Nonlinear Physical Science

Vladimir V. Uchaikin

Fractional Derivatives for Physicists and Engineers

Volume II Applications

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Nonlinear Physical Science

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To my friends who made my life longer

Preface

"God made the integers; all else is the work of man".[1] For centuries, the ancients were satisfied with using natural numbers called simply "numbers". What we call *irrational numbers* was not included into this notion by the Greeks. Not even rational fractions were called numbers.

Hence, *numbers* were conceived as discontinuous, while *magnitudes* were continuous. The two notations appeared, therefore, entirely distinct. The transfer from numbers to magnitudes (to lengths, for example) was a difficult and important step. Perhaps, the most dramatic confrontation of the notions exhibited in *Zeno's paradoxes*. One of them says, "Achilles cannot overtake a tortoise. Why? Achilles must first reach the place from which the tortoise started. By that time, the tortoise will have moved on a little way. Achilles must then traverse that, and still the tortoise will be ahead. He is always nearer, yet never makes up to it". The paradox resolution became possible only after extending the concept "number" over the whole real axis.

The real numbers have formed a basis of classical analysis whose major concept is the *continuity*. In frame of this conception, the set of natural and even rational numbers is vanishing (in cardinality) as compared with the continuum of real numbers. In numerical calculations, we use rational numbers as approximations to real ones, but namely irrational numbers reflect the real world. Nobody can make a rod with rational length and nobody can check that the length is rational.

Of course, when we deal with a set of isolated objects we use the natural numbers for counting the objects. However, if the objects are numbers and we consider the sum of the numbers, the situation may change. Thus, the number n in the expressions

$$y_n = x_1 + x_2 + ... + x_n = \sum_{j=1}^{n} x_j$$

and

[1] I'm very grateful to Prof. V. Kiryakova for her remark that this phrase is due to Leopold Kronecker (1886).

$$z_n = x_1 \cdot x_2 \cdot \ldots \cdot x_n = \prod_{j=1}^{n} x_j$$

needs no comments, it is natural. But if the numbers x_i are identical (and positive), these operations can be easily continued on the whole real axis: $n = \{1, 2, 3, \ldots\} \to \nu \in (-\infty, \infty)$:

$$y_\nu = \nu x, z_\nu = x^\nu, \nu \in \mathbb{R}.$$

The famous Euler invention called Euler's Gamma function

$$\Gamma(\nu) = \int_0^\infty e^{-x} x^{\nu-1} dx, \quad \nu > 0,$$

$$\Gamma(n+1) = n!, \quad n = 1 \cdot 2 \cdot 3 \cdot \ldots$$

has played a crucial role in extending the concept "number of operations" on the noninteger values. With this function, there was made a scientific breakthrough in the differential calculus, which enriched it with differentiation and integration of fractional orders called shortly *fractional calculus*. The foundation of fractional calculus is connected with the names of Riemann, Lioville, Weyl, Grünwald, Letnikov. Though the first works in this direction were made of about two centuries ago, these ideas had not found any practical applications for a long time. However, the situation changed dramatically during a couple of last decades, when about 3000 works were published on the subject.

For better understanding of what the extension of this concept can bring, consider popular differential equations of theoretical physics of the form

$$a \frac{\partial^m f(x,t)}{\partial t^m} + b \frac{\partial^n f(x,t)}{\partial x^n} = F, \tag{1}$$

where x, t are space-time variables, a, b and F are given functions of x and t, and $m, n = 0, 1, 2, \ldots$ are integer numbers. If one of the numbers, say n, is zero, the corresponding variable x becomes a plain parameter. Omitting it, we arrive at the ordinary differential equation:

$$a \frac{d^m f(t)}{dt^m} + bf(t) = F.$$

Putting $n = 1$ in Eq. (1) and interpreting $f(t)$ as the velocity of a material point performing one-dimensional motion under action of the force $F - bf$, we recognize here the simplest version of the Newton equation. If $a, b > 0$, then the term $-bf$

can be interpreted as the friction force, and we meet the relaxation problem. When a denotes mass and f means the coordinate of a particle, we again see the Newton equation describing one-dimensional motion along the x-axis. This time, the term $-bf$ means the elastic force and the equation describes the harmonic oscillator driven by the force F (assuming $b = \text{const}$).

Choosing $m = n = 1$, we obtain a one-dimensional continuity equation. This is the simplest of partial differential equations of mathematical physics. The other popular versions of equations of mathematical physics are represented in Table 1. They are well-known and do not need any comments. In the last line, you can see the equations containing time- and space-derivatives of fractional orders. These operations are significantly less familiar to the majority of physicists and engineers. One could pay no serious attention to such exotic mathematical construction, but only one glance at Fig. 1 may shake the scepticism: we see that the set of well-known and well-investigated differential equations of mathematical physics is represented by only a few points on (μ, ν) diagram, while the continuous set of all other points of the whole (μ, ν)-plane is a *terra incognita*, which can not but attract attention of graduate and postgraduate students, promising scientists and young engineers. This is the readership the book is addressed in the first place. Nevertheless, I hope that it can attract attention of more experienced researches, both physicists and mathematicians, as comparatively new tools for investigation and modelling of complex natural processes.

I was 55 years old when I went on a long-year trip over this "fractional" land and found many treasures there: about three thousand articles using the tools for solving different problems in physics and engineering have been published during the last two decades. They include

Table 1 The other popular versions of equations of mathematical physics

	1D-equations	3D-equations	Phys. sense	Math. type
1,0	$\lvert a\rvert \frac{dv}{dt} + bv = F$	$\lvert a\rvert \frac{d\mathbf{v}}{dt} + b\mathbf{v} = F$	Damped motion	–
2,0	$\lvert a\rvert \frac{d^2x}{dt^2} + bx = F$	$\lvert a\rvert \frac{d^2\mathbf{r}}{dt^2} + b\mathbf{r} = F$	Oscillation	–
1,1	$\lvert a\rvert \frac{\partial f}{\partial t} + \frac{\partial(bf)}{\partial x} = F$	$\lvert a\rvert \frac{\partial f}{\partial t} + \nabla(\mathbf{b}f) = F$	Continuity	–
1,2	$\lvert a\rvert \frac{\partial f}{\partial t} - \lvert b\rvert \frac{\partial^2 f}{\partial x^2} = F$	$\lvert a\rvert \frac{\partial f}{\partial t} - \lvert b\rvert \nabla^2 f = F$	Diffusion	Parabolic
2,2	$\lvert a\rvert \frac{\partial^2 f}{\partial t^2} - \lvert b\rvert \frac{\partial^2 f}{\partial x^2} = F$	$\lvert a\rvert \frac{\partial^2 f}{\partial t^2} - \lvert b\rvert \nabla^2 f = F$	Waves	Hyperbolic
0,2	$af + b\frac{\partial^2 f}{\partial x^2} = F$	$af + b\nabla^2 f = F$	Static fields	Elliptic
μ, ν non-integers	$a\frac{\partial^\mu f}{\partial_t \mu} + b(\frac{\partial^2}{\partial x^2})^{\nu/2} f = F\ t > 0,\ -\infty < x < \infty$	$a\frac{\partial^\mu f}{\partial t^\mu} + b\Delta^{\nu/2} f = F$	?	No classified yet

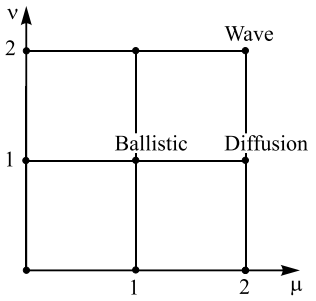


Fig. 1 Continuous manifold of fractional partial equations

- inverse mechanical problems,
- stochastic kinetics and dynamical chaos,
- motion in viscous fluid,
- heat flow spreading,
- electrochemistry of electrodes,
- percolation through porous media,
- rheology of viscoelastic materials,
- electrical and radio engineering,
- plasma physics,
- quantum optics and nanophysics,
- astrophysics and cosmology,
- biophysics and medicine.

Fortunately, our group managed to participate in developing fractional approach to description of anomalous (dispersive) transport in disordered semiconductors, non-Debye relaxation in solid dielectrics, penetration of light beam through a turbulent medium, transport of resonance radiation in plasma, blinking fluorescence of quantum dots, subrecoil laser cooling of atoms, penetration and acceleration of cosmic ray in the Galaxy, large-scale statistical cosmography and solving some other problems. These investigations allowed us to become aware of deep links between fractional calculus, non-Gaussian Lèvy-stable statistics and stochastic fractals. The presence of a time-fractional derivative in the equation is interpreted as a special property of the process under consideration called the *memory*, the *after-effect*, or, if we handle with a stochastic process, the *non-Markovian property*. Fractional derivatives with respect to coordinates reflect a medium with inhomogeneities of some special kind called *self-similar inhomogeneities* or *fractals*. We meet such structures in turbulent flows, plasma, and interstellar media.

As many my colleagues, I'm convinced that the fractional derivative or, as it is often called, the *fractional differintegral* given by the expression

$$ {}_af^{(\nu)}(x) = \frac{1}{\Gamma(n-\nu)} \frac{d^n}{dx^n} \int_0^x (x-\xi)^{n-\nu-1} f(\xi) d\xi $$

with $n-1 \leqslant \nu < n$ if $\nu \geqslant 0$ and $n = 0$ if $\nu < 0$, is much more than merely a sequence of differential and special integral operators.[2] Important properties of Nature underlie this mathematical concept. Starting to write this book, I decided to begin it with discussion of these properties, enveloping, in my opinion, *heredity*, *non-locality*, *self-similarity* and *stochasticity*. This is why the first three chapters of the book united into the first part "Background" contain description of various natural phenomena demonstrating such properties.

Chapter 1 contains a modern exposition of the Volterra heredity concept whose main tool is the integral operation

$$f(x) \mapsto \int_0^x K(x,\xi) f(\xi) d\xi$$

with the kernel $K(x,\xi)$ interpreted as a memory in case x and ξ are time, or as non-locality in case the variables are spatial coordinates. The variety of physical processes including mechanical, molecular, hydrodynamical, thermodynamical phenomena demonstrating hereditary properties are described in this chapter.

In Chap. 2 we review physical processes characterized by power-type memory functions and basic mechanisms generated this property. This list may shake the opinion that the exponential function is the queen-function of theoretical physics and show that without power functions like without maids of honour the queen court would be more tiresome.

Chapter 3 opens a wide panorama of stochastic processes which shows that probabilistic long tails of power type results from self-similarity of the processes and the latter is connected with existence of limit distributions, namely with the Levy-stable laws. The reader will go into the fractional Brownian motion continuous-time random walk processes, fractional Poisson process and walking on fractals.

The second part of the book, "Theory", contains the elements of fractional calculus theory with review of various fractional equations, their analytical and numerical solutions.

Chapter 4 serves as a mathematical introduction to fractional calculus containing basic definitions of fractional operators, their properties and rules of applications. The reader can recognize many of them as corresponding generalization of well-known analogous from integer-order calculus such as the Leibnitz rule or the Taylor formula,

Chapter 5 shows how equations with fractional derivatives are solved. The reader will meet description of some analytical methods of solution and many examples of their applications to ordinary and partial fractional equations. Of course, this review can not pretend on a strict and exhaustive exposition, but it will be useful for physicists and engineers as a first acquaintance.

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[2] Here are two remarkable quotations: "The fractional calculus is the calculus of the XXI century" (K. Nishimoto, 1989) and "We may say that Nature works with fractional time derivatives" (S. Westerlund, 1991).

Chapter 6 contains an introduction to numerical methods of fractional equation solution. Starting with the fractional difference operators and based on this concept the Grünwald-Letnikov definition of fractional derivatives, the reader is acquainted with the finite-difference methods of computing fractional integrals, fractional derivatives, and fractional equations of various kinds. The last section of this chapter is devoted to some aspects of Monte Carlo techniques.

The third part, opening the second volume of the book, exposes a wide field of applications of fractional calculus in modern physics including mechanics, hydrodynamics, viscoelasticity, thermodynamics, electrodynamics, plasma physics, quantum physics, cosmic ray physics, and cosmology. Finally, the fourth, last part of the book contains various auxiliary material (special functions, notation of fractional derivatives, main formulas of fractional calculus, tables and graphs of some functions, playing a special role in solution of fractional equations).

Ulyanovsk, Russia Vladimir V. Uchaikin
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Chapter 1
Mechanics

1.1 Tautochrone Problem

1.1.1 Non-relativistic Case

We start reviewing applications of fractional calculus with the following mechanical problem.

A classical, non-relativistic, point particle of mass m with potential energy $U = mgy$ begins to slide without friction along a curve running through a vertical plane $x - y$ to the origin and reaches it at time τ (Fig. 1.1). The problem is to find the function $\tau(h)$ that specifies the total time of descent from an initial height h. A special case of the problem when $\tau(h) = \text{const}$ is called the *tautochrone problem* (from Greek prefixes *tauto* meaning "same" and *chrono* "time").

The principle of conservation energy says

$$\frac{m}{2}\left(\frac{ds}{dt}\right)^2 = mg(h-y),$$

where s is the length of the curved segment between the origin and a current position, and $h - y$ is the descent of the particle during time t. Hence

$$dt = -\frac{ds}{\sqrt{2g(h-y)}}.$$

After integrating, we directly arrive at the equation

$$\tau(h) = \int_0^h \frac{ds/dy}{\sqrt{2g(h-y)}}dy = \sqrt{\frac{\pi}{2\,g}}\,{}_0^{1/2}\mathsf{D}_h s(h)$$

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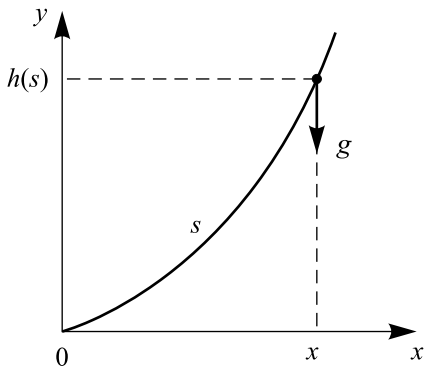


Fig. 1.1 Abel's mechanical problem

called *Abel's integral equation*, solution of which has opened a way to the land of fractional equations [1].

Muñoz and Fernandéz-Anaya [65] apply the fractional approach to investigation of the properties of the tautochrone and brachistochrone curves by introducing a family of curves complying with relations where the time of descent is proportional to a fractional power of the height difference.

1.1.2 Relativistic Case

The relativistic counterpart of this problem has been studied by Kamath [53]. The methods of fractional calculus are shown to be more useful in the derivation of the exact relativistic tautochrone. Relativistic kinematics yields the equation of conservation energy

$$mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}} + Q,$$

where Q is the energy lost by the particle from the gravitation field as it is released from a height h and is given by

$$Q = mc^2 \left\{1 - \exp[g(h - y)/c^2]\right\}$$

(Goldstein and Bender [43]). From two these equations, we find

$$v(y) = c\sqrt{1 - \exp[2g(y - h)/c^2]}.$$

The time of fall is

$$T = \int_0^T dt = -\int_h^0 \frac{ds}{v(y)} = \int_0^h \frac{s'(y)dy}{c\sqrt{1 - \exp[2g(y - h)/c^2]}},$$

with $s'(y) = ds/dy$ being the arclength along the path joining the initial $(x_0, y_0 = h)$ and final $(0, 0)$ end points. By rewriting this equation as

$$cT = i\int_0^h \frac{e^{-gy/c^2}s'(y)dy}{\sqrt{e^{-2gh/c^2} - e^{-2gy/c^2}}} = -\frac{ic^2}{2g}\int_0^h \frac{(-2g/c^2)e^{-gy/c^2}}{\sqrt{e^{-2gh/c^2} - e^{2gy/c^2}}}s'(y)dy,$$

one arrives at the fractional equation for the function $\eta(h) = s'(h)e^{gh/c^2}$ determining the sought curve:

$$c\sqrt{\pi}\ {}_0\mathsf{D}^{-1/2}_{\mu(h)}\eta(h) = 2iTg,$$

with $\mu(h) = e^{2gh/c^2} - 1$. Converting the equation to

$${}_0\mathsf{D}^{1/2}_{\mu(h)}\ {}_0\mathsf{D}^{-1/2}_{\mu(h)}\eta(h) = {}_0\mathsf{D}^{1/2}_{\mu(h)}(2iTg/c\sqrt{\pi}),$$

and using the composition rules, the author reduces it to the form

$$\begin{aligned}\sqrt{\pi}e^{gh/c^2}s'(h) &= \frac{2iTg}{c\sqrt{\pi}}\frac{d}{d\mu(h)}\int_0^h \frac{-2ge^{-2gy/c^2}dy}{c^2\sqrt{e^{-2gh/c^2} - e^{-2gy/c^2}}} \\ &= 2\frac{2iTg}{c\sqrt{\pi}}\frac{d[\mu(h)]^{1/2}}{d\mu(h)} = \frac{2iTg}{c\sqrt{\pi}}[\mu(h)]^{-1/2}.\end{aligned}$$

Solution of this equation leads to the following parametric representation of the sought tautochrone:

$$e^{2gy/c^2} = 1 + \left(\frac{2Tg}{\pi c}\right)^2 \cos^2\theta,$$

$$\frac{gx}{c^2} = \theta - \frac{\pi}{2} + a\left(\frac{\pi}{2} - \arctan\left(\frac{1}{a}\tan\theta\right)\right)$$

with

$$a = 1 + \left(\frac{2Tg}{\pi c}\right)^2.$$

The non-relativistic tautochrone problem was generalized to an arbitrary potential $U(y)$ as well [44]. In this case

$$dt = -\frac{ds}{\sqrt{(2/m)U(h-y)}}$$

and

$$T = -\int_{y_0}^{y} \frac{s'(y)dy}{\sqrt{(2/m)[U(y_0) - U(y)]}}. \tag{1.1}$$

Making $z = U(y)$, one can write

$$\frac{ds}{dy}dy = \frac{ds}{dz}dz$$

with

$$dz = U'dy,$$

so then Eq. (1.1) takes the form

$$T = -\sqrt{\frac{m}{2}}\int_{z_0}^{z}\frac{s'(z)dz}{\sqrt{z_0 - z}}.$$

Solution of this equation

$$x = \int_0^y \sqrt{\frac{2T^2}{m\pi^2}\frac{U'^2}{U} - 1}\, dy$$

was found in Flores and Osler [41].

1.2 Inverse Problems

1.2.1 Finding Potential from a Period-Energy Dependence

Let us look at Sect. 12 of "Mechanics" (Landau and Lifshitz [57]) devoted to finding of potential energy $U(x)$ from the oscillation period T given as a function of the total energy E, $T = T(E)$ (Fig. 1.2). Starting as before from the principle of energy conservation, they obtain the integral equation

$$T(E) = 2\sqrt{2\,m}\int_0^{X(E)}\frac{dx}{\sqrt{E - U(x)}}, \tag{1.2}$$

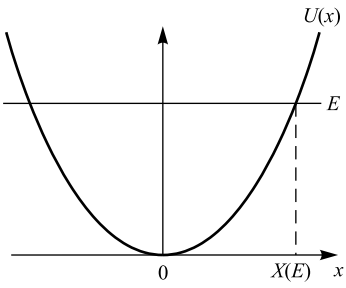


Fig. 1.2 Illustration to Eq. (1.2)

where $X(E)$ is a root of equation $U(x) = E$ (for simplicity let the potential be an even function, monotonically increasing with moving from the origin). Passage to integration variable U leads to the fractional differential equation

$$T(E) = 2\sqrt{2\,m}\int_0^E \frac{dX(U)}{dU}\frac{dU}{\sqrt{E-U}} = 2\sqrt{2\,m\pi}\ {}_0^{1/2}\mathsf{D}_E X(E),$$

the integration of which yields Eq. (12.2) of the mentioned "Mechanics":

$$X(U) = \frac{1}{2\pi\sqrt{2\,m}}\int_0^U \frac{T(E)dE}{\sqrt{U-E}}.$$

It belongs to the class of *inverse problems*[1] formulated as finding properties of a system by observation of its motion. Just as Newton came to differential calculus considering a primal mechanical problem, Abel found a way to fractional differential calculus solving some special inverse problem.

1.2.2 Finding Potential from Scattering Data

One more inverse mechanical problem is to find the interaction potential $U(r)$ from experimental data on scattering of particles with a given energy E. It is related to solving some integral equation (see [30] and Sect. 11.6 of book [70]). The simplest statement of this problem is assuming the potential function $U(r)$ to be monotonically increasing and vanishing at infinity find it form the connection between the scattering angle θ and the impact parameter x is expressed by formula (Landau and Lifshitz [57]) (Fig. 1.3).

$$\theta(x) = \frac{x}{E}\int_x^\infty \frac{dU}{dr}\frac{dr}{\sqrt{r^2-x^2}}. \tag{1.3}$$

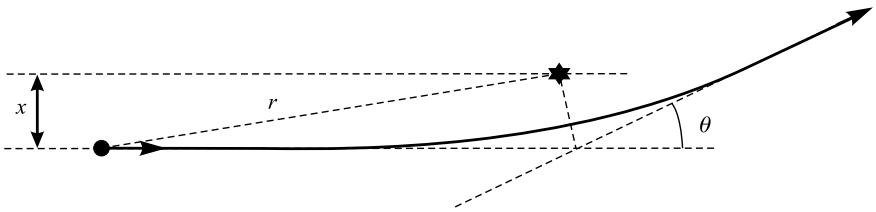


Fig. 1.3 Illustration to Eq. (1.3)

[1] The term "inverse problem" was introduced by Soviet-Armenian astrophysicist Victor Ambartsumian. His paper on the inverse Sturm-Liouville problem (1929) was found by Swedish mathematicians at the end of the Second World War and accepted as a foundation of a new discipline.

The formula represents a right-hand Gerasimov-Caputo semiderivative with respect to x^2:

$$\Theta(x) \equiv \frac{\theta(x)}{x} = \frac{1}{E}\int_{x^2}^{\infty} \frac{dU}{dr^2}\frac{dr^2}{\sqrt{r^2-x^2}} = \frac{\sqrt{\pi}}{E}\,{}^{1/2}_{x^2}\mathsf{D}_{\infty}U(x^2).$$

Inverting this relation gives the result

$$U(r) = \frac{E}{\sqrt{\pi}}\,{}^{r^2}\mathsf{I}^{1/2}_{\infty}\Theta(r^2) = \frac{2E}{\pi}\int_{r}^{\infty}\frac{\theta(x)dx}{\sqrt{x^2-r^2}}, \tag{1.4}$$

coincident with Eq. (11.16) in the book [70]. Finding $x(\theta)$ from the equation

$$\frac{1}{2}\frac{dx^2}{d\cos\theta} = \left[\frac{d\sigma(\theta)}{d\Omega}\right]_{\text{exper}},$$

inverting the obtained function and substituting the result $\theta(x) = \theta_{\text{exper}}(x)$ into Eq. (1.4), we come to the desired function $U(r)$.

1.2.3 Stellar Systems

The construction of self-consistent models for stellar systems is of great interest in astrophysics. The most straightforward way to built such models is to start with an assumed potential defining the mass density ρ and the families of stars orbits with the distribution function f. The integral relation connecting f and ρ is known as the *self-gravitation equation*. Pedraza et al. [71] generalized the classical methods involving the fractional derivatives. We sketch here this approach.

According to Jean's theorem, the phase space distribution function $f(\mathbf{r}, \mathbf{v})$ is a function only of the isolating integrals of motion that are conserved in each orbit. For spherical symmetry these are the energy E and the angular momentum L_z.[2] Assume that Φ is the gravitational potential and define a relative potential $\Psi = -\Phi + \Phi_0$ and a relative energy $\varepsilon = -E + \Phi_0$, in such a way that the system has only stars with energy $\varepsilon > 0$. An axisymmetric system admits two isolated integrals: z-component of the angular momentum about the z-axis, $L_z = Rv_\varphi$ and $\varepsilon = -E + \Phi_0$. For a steady-state axisymmetric stellar system, the even part of the distribution function with respect to L_z, f_+, related to the mass density as

$$\rho(R, \Psi) = \frac{4\pi}{R}\int_0^{\Psi}\int_0^{R\sqrt{2(\Psi-\varepsilon)}} f_+(\varepsilon, L_z)dL_z d\varepsilon. \tag{1.5}$$

[2] As indicated in the cited work, as far back as Eddington showed that it is possible to obtain such distribution functions by first expressing the density as a function of the potential, and then solving an Abel integral equation.

The authors seek the solution of the integral equation in the form

$$f_+(\varepsilon, L_z) = \sum_n L_z^{2\alpha_n} h_n(\varepsilon).$$

Inserting it into Eq. (1.5) and integrating with respect to L_z they obtain

$$\rho(R, \Psi) = \sum_n R^{2\alpha_n} \tilde{\rho}_n(\Psi), \quad \alpha_n > -1/2, \tag{1.6}$$

where

$$\tilde{\rho}_n(\Psi) = \frac{4\pi 2^{\alpha_n+1/2}}{2\alpha_n + 1} \int_0^\Psi h_n(\varepsilon)(\Psi - \varepsilon)^{\alpha_n+1/2} d\varepsilon.$$

This integral equation can easily be inverted by means of fractional derivatives method,

$$h_n(\varepsilon) = \frac{{}_0\mathsf{D}_\Psi^{\alpha_n+3/2} \tilde{\rho}_n(\Psi)\Big|_{\Psi=0}}{\pi 2^{\alpha_n+3/2}\Gamma(\alpha_n + 1/2)},$$

so the result is

$$f_+(\varepsilon, L_z) = \sum_n \frac{{}_0\mathsf{D}_\Psi^{\alpha_n+3/2} \tilde{\rho}_n(\Psi)\Big|_{\Psi=0}}{\pi 2^{\alpha_n+3/2}\Gamma(\alpha_n + 1/2)} L_z^{2\alpha_n}.$$

Here $\tilde{\rho}_n(\Psi)$ is assumed to be known because of Eq. (1.6).

1.3 Motion Through a Viscous Fluid

1.3.1 Entrainment of Fluid by a Moving Wall

We will consider motion of a body in an incompressible viscous Newton's fluid. In absence of external bulk forces, the unsteady flow of such a fluid is governed by the Navier-Stockes (N-S) equations system

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2},$$

where u_i, $i = 1, 2, 3$, are velocity components along x_i-coordinate axis respectively, ρ is the density, p is the pressure, and ν denotes the kinematic viscosity. This system should be supplemented with the incompressibility condition

$$\frac{\partial u_i}{\partial x_i} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} = 0.$$

For specificity of solution, this system must be accompanied by initial and boundary conditions.

Let us apply the N-S equations to the problem of entrainment of fluid by an unbounded (or of large size) plate moving in xOz-plane along x-axis (Fig. 1.4). The plate is at rest until $t = 0$ and then begins to move with a constant velocity V to positive direction of x-axis. The liquid particles velocity will contain only one nonzero component, $u_x = u$, and this component, due to unboundedness of the plate, will depend upon only z-coordinate,

$$u_x = u(z, t), \quad u_y = u_z = 0.$$

The pressure fall is zero too,

$$\frac{\partial p}{\partial x} = \frac{\partial p}{\partial y} = \frac{\partial p}{\partial z} = 0,$$

and we reduce the initial N-S system to the equation

$$\frac{\partial u(z, t)}{\partial t} = \nu \frac{\partial^2 u(z, t)}{\partial z^2} \tag{1.7}$$

of the diffusion type under following initial and boundary conditions:

$$u(z, t) = \begin{cases} 0, & \text{if } t = 0 \text{ and } z \leq 0; \\ V, & \text{if } t \geq 0 \text{ and } z = 0; \\ 0, & \text{if } t \geq 0 \text{ and } z = -\infty. \end{cases} \tag{1.8}$$

For solving the problem, we apply the Laplace transform method with respect to time. Multiplying both sides of Eq. (1.7) by $e^{-\lambda t} dt$, integrating over the positive half-axis,

$$\int_0^\infty e^{-\lambda t} \frac{\partial u(z, t)}{\partial t} dt = \nu \int_0^\infty e^{-\lambda t} \frac{\partial^2 u(z, t)}{\partial z^2} dt,$$

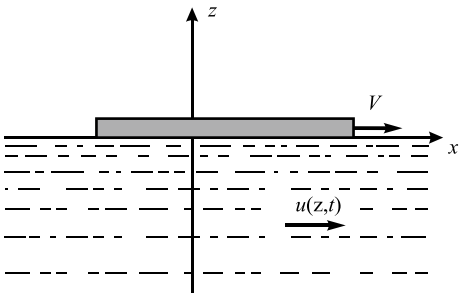


Fig. 1.4 To the problem of a plate on viscous fluid surface

and applying the rule of integration by parts to the left-hand side,

$$\int_0^\infty e^{-\lambda t}\frac{\partial u(z,t)}{\partial t}dt = e^{-\lambda t}u(z,t)\Big|_0^\infty + \nu\int_0^\infty e^{-\lambda t}\frac{\partial^2 u(z,t)}{\partial z^2}dt = \nu\int_0^\infty e^{-\lambda t}\frac{\partial^2 u(z,t)}{\partial z^2}dt,$$

and the rule of commutation of integral and differential operations with respect to independent variables to the right-hand side,

$$\int_0^\infty e^{-\lambda t}\frac{\partial^2 u(z,t)}{\partial z^2}dt = \frac{\partial^2}{\partial z^2}\int_0^\infty e^{-\lambda t}u(z,t)dt,$$

we get the ordinary differential equation

$$\frac{d^2\widehat{u}(z,\lambda)}{dz^2} - \frac{\lambda}{\nu}\widehat{u}(z,\lambda) = 0, \tag{1.9}$$

for the Laplace transform of the solution

$$\widehat{u}(z,\lambda) = \int_0^\infty e^{-\lambda t}u(z,t)dt.$$

The Latter obeys the conditions

$$\widehat{u}(z,\lambda) = \begin{cases} V/\lambda, & z = 0; \\ 0, & z = -\infty; \end{cases} \tag{1.10}$$

following from the original boundary conditions (1.8). The general solution of Eq. (1.9) is represented as

$$\widehat{u}(z,\lambda) = Ae^{z\sqrt{\lambda/\nu}} + Be^{-z\sqrt{\lambda/\nu}}.$$

Upon using Eq. (1.10), we find

$$\widehat{u}(z,\lambda) = (V/\lambda)e^{z\sqrt{\lambda/\nu}}, \quad z < 0.$$

Returning to the original $u(z,t)$ is possible by computing integral

$$u(z,t) = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} e^{\lambda t}\widehat{u}(z,\lambda)d\lambda = \frac{V}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} e^{\lambda t+z\sqrt{\lambda/\nu}}\frac{d\lambda}{\lambda} \tag{1.11}$$

over a straight line in the complex plane parallel to imaginary axis and called the *convergence axis of the Laplace integral*. The integrand in (1.11) has a branchpoint at $\lambda = 0$ so function must consider the infinite sequence of closed loops and use the Jordan lemma for computing the limit of corresponding integrals.

Let us draw a closed contour *ABCDEFA*, consisting of rectilinear segment *AB*, semicircle of a large radius *R*, two cuts *CD* and *EF* and a small circle *DE* of radius *r* around the origin (Fig. 1.5). The integrand has no singularities in the domain bounded by *ABCDEFA*, and according to the Cauchy theorem

$$I_{ABCDEFA} = \oint\limits_{ABCDEFA} e^{\lambda t + z\sqrt{\lambda/\nu}} \frac{d\lambda}{\lambda} = 0$$

whence

$$\int_A^B e^{\lambda t + z\sqrt{\lambda/\nu}} \frac{d\lambda}{\lambda} \equiv I_{AB} = -I_{BC} - I_{CD} - I_{DE} - I_{EF} - I_{FA}.$$

Evidently, unlimited increasing *R* produces

$$I_{AB} \to \int_{\sigma_\infty}^{\sigma+\infty} e^{\lambda t + z\sqrt{\lambda/\nu}} \frac{d\lambda}{\lambda}, \quad I_{BC} \to 0, \quad I_{FA} \to 0,$$

$$I_{DE} = -2\pi i \operatorname*{Res}_{\lambda=0} \left\{ \frac{e^{\lambda t + z\sqrt{\lambda/\nu}}}{\lambda} \right\} = -2\pi i.$$

Taking for cut *CD*

$$\lambda = \xi^2 e^{i\pi} = -\xi^2, \quad \sqrt{\lambda} = \xi e^{i\pi/2} = i\xi, \quad d\lambda = -2\xi d\xi,$$

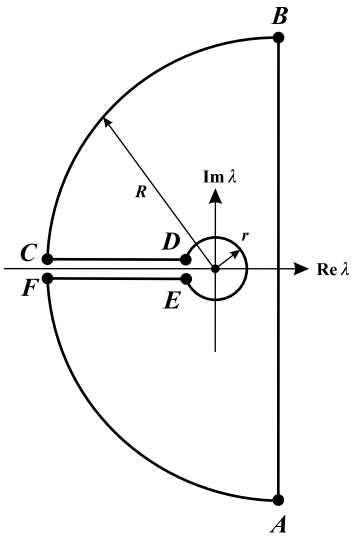


Fig. 1.5 The integration contour *ABCDEFA*

and for cut EF

$$\lambda = \xi^2 e^{-i\pi} = -\xi^2, \quad \sqrt{\lambda} = \xi e^{-i\pi/2} = -i\xi, \quad d\lambda = -2\xi d\xi,$$

we obtain

$$I_{CD} \to \frac{2}{i}\int_0^\infty e^{-\xi^2 t - i(z/\sqrt{\nu})\xi}\frac{d\xi}{\xi},$$

and

$$I_{EF} \to \frac{2}{i}\int_0^\infty e^{-\xi^2 t + i(z/\sqrt{\nu})\xi}\frac{d\xi}{\xi},$$

respectively. As a result, we have

$$g(z,t) := \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} e^{\lambda t + z\sqrt{\lambda/\nu}}\frac{d\lambda}{\lambda} = 1 + \frac{2}{\pi}\int_0^\infty e^{-t\xi^2}\sin\left(\frac{z\xi}{\sqrt{\nu}}\right)\frac{d\xi}{\xi} \qquad (1.12)$$

and consequently

$$u(z,t) = g(z,t)V.$$

Assuming that an acceleration of the plate is not zero after $t = 0$ but can be broken up into a series of small step changes in the velocity, the velocity of entrained fluid can be written in the form of *Duhamel's principle*:

$$u(y,t) = g(y,t)V(0) + \int_0^t g(y,t-t')\dot{V}(t')dt'.$$

Its physical meaning is evident: perturbations added to the velocity field by changes in the plate velocity spread according to diffusion law, and the shift of time in the integral term reflects the retardation effect. From mathematical point of view, $g(z,t)$ is the Green function of the equation under consideration.

If the initial moment is not zero but, say, a, the formula becomes

$$u(y,t) = g(y,t-a)V(a) + \int_a^t g(y,t-t')\dot{V}(t')dt',$$

with admissible value $a = -\infty$. The viscous force per unit surface of the plate (the shear stress) is represented by

$$\tau = \nu\rho\left(\frac{\partial u(y,t)}{\partial y}\right)_{y=0} \equiv \nu\rho u'(0,t)$$

$$= \nu\rho\left[g'(0,t-a)V(a) + \int_a^t g'(0,t-t')\dot{V}(t')dt'\right],$$

where

$$g'(0,t) = \frac{\partial g(y,t)}{\partial y}\bigg|_{y=0} = \left[\frac{\partial}{\partial y}\left(1 - \frac{2}{\pi}\int_0^\infty e^{-(t-a)\xi^2}\sin\left(\frac{y\xi}{\sqrt{\nu}}\right)\frac{d\xi}{\xi}\right)\right]_{y=0}$$
$$= -\rho\sqrt{\frac{\nu}{\pi(t-a)}}.$$

Thus we see that

$$\tau = -\rho\sqrt{\frac{\nu}{\pi}}\left[\frac{V(a)}{\sqrt{t-a}} + \int_a^t \frac{\dot{V}(t')dt'}{\sqrt{t-t'}}\right]$$
$$= -\rho\sqrt{\nu}\left[\frac{V(a)}{\sqrt{\pi(t-a)}} + {}_a^{1/2}\mathsf{D}_t V(t)\right] = -\rho\sqrt{\nu}\,{}_a\mathsf{D}_t^{1/2}V(t). \qquad (1.13)$$

Equation (1.13) shows that the viscous stress at the plate at time t depends on all preceding ($t' < t$) motion of the wall, or, at greater length, the stress observed at time t at a point (x, y) is a result of contributions of liquid particles coming from other points (x', y) of the layer, where they were located, say, at $t' < t$. Because of the translation invariance of the solution relative to x, the same velocity distribution at the moment t' took place also in the observing point (x, y), and the space-shift $u(x', t) \mapsto u(x, t)$ is equivalent to the time-shift $u(x, t') \mapsto u(x, t)$. This is the simplest, mechanism of heredity, the "mechanical" memory. It was discussed in works [56, 72, 82] and others.

1.3.2 Newton's Equation with Fractional Term

As follows from the foregoing, the equation connecting the force F acting on a thin plate of mass m_0 and surface area S immersed into a liquid with density ρ and kinematic viscosity ν, and the corresponding acceleration of this plate has the form

$$m_0\frac{dV}{dt} + 2\rho S\sqrt{\nu}\,{}_a\mathsf{D}_t^{1/2}V(t) = F(t) \qquad (1.14)$$

(the factor 2 accounts that both surfaces of the plate are in contact with the fluid).

In case of a sphere of mass m_0 and of radius R steadily moving through an incompressible viscous fluid under action of an external force $F(t)$, the motion equation has the form

$$m_0\frac{dV}{dt} = F(t) + R(t),$$

where R is the resistance force:

$$R(t) = -6\pi\rho\nu RV(t) - \frac{2}{3}\pi\rho R^3\frac{dV}{dt} - 6\pi\rho R^2\sqrt{\nu}\left[\frac{V(a)}{\sqrt{\pi(t-a)}} + \frac{1}{\sqrt{\pi}}\int_a^t \frac{\dot{V}d\tau}{\sqrt{t-\tau}}\right].$$

The first term represents the *Stokes formula* for the friction force as it appears in hydrodynamic consideration of a sphere motion through a viscous fluid; the second gives the force due to the mass of the liquid displaced by the particle (*associated mass*), and the third term called the *Boussinesq-Basset formula* accounts the retardation effect on the motion of the sphere due to the penetration depth of viscous unsteady flow around the sphere [23, 27]. This formula has been derived on assumption that the sphere starts its motion in a motionless fluid. This means that the body itself has been at rest for a long time before a, so all perturbations excited by its preceding motion have vanished. In other words, the velocity $V(t)$ is 0 for all $t < a$, and the motion equation for a body of mass m_0 inserted into a fluid with kinematic viscosity ν can be written as

$$m\frac{dV}{dt} + b\,{}_a\mathsf{D}_t^{\alpha}V(t) + cV(t) = F(t), \quad V(t) \equiv 0 \quad \text{for} \quad t < a, \tag{1.15}$$

where $\alpha = 1/2$, coefficients

$$m = m_0, \quad b = 2\rho S\sqrt{\nu}, \quad c = 0$$

in case of a thin plate, and

$$m = m_0 + \frac{2}{3}\pi\rho R^3, \quad b = 6\pi\rho\sqrt{\nu}R^2, \quad c = 6\pi\rho\nu R$$

in case of a sphere. Finally, one can suppose that a wide range of problems related to motion of bodies through viscous liquids can be considered on the base of Eq. (1.14), that is the Newton equation with a fractional term, or shorter, the *fractional Newton equation* (FNE). The value $\alpha = 1/2$ means that we deal with the Newtonian fluid. We leave the symbol α in this equation keeping in mind to use it for non-Newtonian fluids when $\alpha \neq 1/2$.

When the driving force F depends on the body coordinate, $x(t)$, the motion is governed by the fractional Newton equation

$$m\frac{dx^2}{dt^2} + b\,{}_a\mathsf{D}_t^{\alpha+1}x(t) + c\frac{dx}{dt} = F(x,t), \quad 0 < \alpha < 1. \tag{1.16}$$

As follows from Tauberian theorem, the main asymptotic (as $t \to \infty$) part of the solution $V^{\text{as}}(t)$ satisfies an equation with a single fractional order derivative,

$$b\,{}_a\mathsf{D}_t^{\alpha}V^{\text{as}}(t) + cV^{\text{as}}(t) = F(t).$$

In what follows, we call it the *reduced fractional Newton's equation* (RFNE) in order to distinguish it from the *total fractional Newton equation* (TFNE) (1.15). The same terms will be used for coordinate representation of the motion Eq. (1.16).

1.3.3 Solution by the Laplace Transform Method

One of the first solutions of FNE, as I know, belongs to Russian mechanician Slyoskin [82]. Let us follow his way. Multiplying both sides of TFNE (1.14), rewritten as

$$m\frac{dV(t)}{dt} + b\left[\frac{V(0)}{\sqrt{\pi t}} + \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{\dot{V}(t')dt'}{\sqrt{t-t'}}\right] = F(t),$$

by $e^{-\lambda t}dt$ and integrating from zero to infinity, he obtained the expression

$$m\lambda\widehat{V}(\lambda) - mV(0) + b\left[\frac{V(0)}{\sqrt{\lambda}} + \frac{1}{\sqrt{\pi}}\int_0^\infty dte^{-\lambda t}\int_0^t \frac{\dot{V}(t')dt'}{\sqrt{t-t'}}\right] = \widehat{F}(\lambda)$$

which after applying the Dirichlet rule to the last term in the left-hand side is reduced to the form

$$m\lambda\widehat{V}(\lambda) - mV(0) + b\left[\frac{V(0)}{\sqrt{\lambda}} + \frac{1}{\sqrt{\pi}}\left(\sqrt{\lambda\pi}\widehat{V}(\lambda) - \sqrt{\frac{\pi}{\lambda}}V(0)\right)\right] = \widehat{F}(\lambda).$$

Solution of this algebraic equation, representing the Laplace transform of the velocity of the plate, immersed into viscous fluid,

$$\widehat{V}(\lambda) = \frac{\widehat{F}(\lambda) + mV(0)}{\lambda(m + b/\sqrt{\lambda})}, \tag{1.17}$$

coincides with that found by Slyoskin (Eq. (2.30) in the cited book). He used this result for analyzing processes of slowing down of a plate in a viscous fluid, generating vortex sheets, moving the fluid between two plates and through a pipe, rotating in a cylinder, and others. Referring to Tauberian theorem, which claims that for any $f(t)$.

$$f(t) \sim At^{-\alpha},\ t \to \infty \quad \Leftrightarrow \quad \widehat{f}(\lambda) \sim A\Gamma(1-\alpha)\lambda^{\alpha-1},\ \lambda \to 0,$$

we bring here two asymptotic results.

1. The plate has the initial velocity V_0 and moves in a fluid without external force, that is $F = 0$. In this case,

$$\widehat{V}(\lambda) = \frac{mV_0}{\lambda(m + b/\sqrt{\lambda})} \sim \frac{mV_0}{b}\lambda^{-1/2}, \quad \lambda \to 0,$$

therefore

$$V_1(t) \sim \frac{mV_0}{b\sqrt{\pi}}\, t^{-1/2}, \quad t \to \infty.$$

2. The plate being at rest till $t = 0$ undergoes action of a constant force F_0, accelerating the plate. Now, $V(0) = 0$, $F(t) = F_0$, and we obtain

$$V_2(t) \sim \frac{2F_0}{b\sqrt{\pi}}\, t^{1/2}, \quad t \to \infty.$$

1.3.4 Solution by the Green Functions Method

Representing Eq. (1.17) in the form

$$\widehat{V}(\lambda) = V(0)\widehat{G}(\lambda) + \frac{1}{m}\hat{F}(\lambda)\widehat{G}(\lambda),$$

with

$$\widehat{G}(\lambda) = \frac{\lambda^{1/2-1}}{\lambda^{1/2} + b/m}.$$

We identify the latter expression with the Laplace transform of the Mittag-Leffler function

$$G(t) = E_{1/2}(-(b/m)\sqrt{t}) \equiv e^{(b/m)^2 t}\left(1 - \frac{2}{\sqrt{\pi}}\int_0^{(b/m)\sqrt{t}} e^{-\xi^2} d\xi\right).$$

Consequently,

$$V(t) = V(0)G(t) + \frac{1}{m}\int_0^t G(t - t')F(t')dt'$$

with Green's function $G(t)$ given by the previous expression. Using this representation, one can investigate short-time behavior of the plate immersed into a viscous incompressible fluid. In particular, for the two problems whose long-time asymptotical solutions are found above, we obtain in a short-time domain ($t \ll (b/m)^2$):

$$V_1(t) \approx V_0\left[1 - \frac{2}{\sqrt{\pi}}\left(\frac{b}{m}\right)t^{1/2}\right]$$

and

$$V_2(t) \approx \frac{F_0 t}{m}\left[1 - \frac{4}{3\sqrt{\pi}}\left(\frac{b}{m}\right)t^{1/2}\right].$$

Solution of a more general case of FNE, relating to a sphere moving through a non-Newtonian fluids,

$$m\frac{dV}{dt} + b\,{}_0\mathsf{D}_t^{\alpha} V(t) + cV(t) = F(t), \quad \alpha \in (0, 1], \tag{1.18}$$

can be represented by means of Green's function found by Podlubny [72] (see Sect. 5.1.9). It is expressed through the two-parameter Mittag-Leffler function and has the form

$$G(t) = \frac{1}{m}\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}\left(\frac{c}{m}\right)^k t^k E^{(k)}_{1-\alpha,1+\alpha k}\left[-(b/m)t^{1-\alpha}\right].$$

1.3.5 Fractionalized Fall Process

Reading numerous articles with various applications of fractional differential calculus, we often meet a simple method of introducing fractional derivative into consideration. Namely, the author of such an article takes a well-known differential equation derived from first physical principles and says: "Now, we replace the first (second) derivative in this equation by its fractional counterpart". After this, he describes a method of solution of the equation, demonstrates numerical results, stress on its difference from the classical analogs and shows that they reduce to classical results when the order of the fractional derivative becomes integer. At the end of the articles, one can read that these results may be applied to some problems arising in viscoelasticity phenomena, physics of polymers or so.

Going this way, we might say, for example: let us take the Newton equation of motion

$$m\dot{V} = F$$

and consider its fractional generalization

$$m\,{}_0\mathsf{D}_t^{\alpha} V = F. \tag{1.19}$$

For notational convenience, we will use for such procedure the term *fractionalization* and call the resulting equation the *fractionalized equation*. Thus, we can say, that Eq. (1.19) is obtained by fractionalization of the classic Newton equation and for this reason may be called the *fractionalized Newton equation*, but we can't use the term for Eq. (1.18), because it is *derived* from the classic mechanic-hydrodynamic system of equations without replacing integer-order derivatives by fractional-order ones.

As an example of fractionalization, one can indicate the work [40] devoted to the problem of a body fall under action of homogeneous gravity and friction force proportional to its velocity. The Newton equation of motion was fractionalized by using the Caputo-Gerasimov time-derivative

$$m_\nu \, {}_0^\nu \mathsf{D}_t V + bV = mg.$$

Beginning with the case when the velocity term is absent,

$$m_\nu \, {}_0^\nu \mathsf{D}_t V = mg,$$

the author comes to the solution

$$V(t) = V_0 + \frac{mgt^\nu}{m_\nu \Gamma(1+\nu)}, \tag{1.20}$$

and obtains by additional integration

$$x(t) = \int_0^t V(\tau) d\tau = x_0 + V_0 t + \frac{mgt^{\nu+1}}{m_\nu \Gamma(2+\nu)}, \tag{1.21}$$

where V_0 is the initial velocity and x_0 the initial coordinate (the x-axis is down-directed). One can see that for short times the fractional derivative gives a greater velocity than the first one ($t^\nu > t$ for $\nu < 1$), while for long times, the velocity is smaller than in a classical case.

Evidently, the parameter m_ν for $\nu \neq 1$ does not have the dimension of mass. In Achar et al. [3], Ryabov and Puzenko [79], the generalized momentum is defined by

$$p_\nu = m_\nu \, {}_0^\alpha \mathsf{D}_t x(t), \quad \alpha = (1+\nu)/2,$$

in order to agree $p_1^2/(2m_1)$ with the dimension of energy. As a result, the total energy of the body is written by

$$E = \frac{p_\nu^2}{2m_\nu} + U(x) = \frac{m_\nu}{2} \left[{}_0^\alpha \mathsf{D}_t x(t) \right]^2 - mgx. \tag{1.22}$$

Observe that the rate of total energy [40] is expressed as

$$\frac{dE}{dt} = \left[\frac{m_\nu \beta V_0 t^{-\nu}}{\Gamma^2(1+\beta)} + \frac{mg}{\Gamma(1+\alpha)\Gamma(1+\beta)} - mg \right] V_0 + \frac{(mg)^2}{m_\nu} \left[\frac{\alpha}{\Gamma^2(1+\alpha)} - \frac{1}{\Gamma(2\alpha)} \right] t^\nu,$$

where

$$\alpha = \frac{1+\nu}{2}, \quad \beta = \frac{1-\nu}{2}.$$

Note that the kinetic energy increases and the velocity of full body's energy increment isn't proportional to the velocity of its motion, except cases when the body was motionless at initial instant

$$\frac{dE}{dt} = \frac{m^2 g^2}{m_\beta}\left[\frac{1}{\alpha[\Gamma(\alpha)]^2} - \frac{1}{\Gamma(2\alpha)}\right] t^\nu, \quad V_0 = 0.$$

If $\nu \to 1$ and $m_1 = m$, formulas (1.20), (1.21), (1.22) turn into well-known equations of classical mechanics:

$$V(t) = V_0 + gt,$$

$$x(t) = x_0 + V_0 t + \frac{gt^2}{2},$$

$$E = \frac{mV^2}{2} - mgx = \text{const}.$$

Accounting of resistive force leads to the next expressions for the velocity and coordinate of the body falling accordingly to fractional differential law

$$V(t) = V_0 E_{\nu,1}[-(b/m_\nu)t^\nu] + \frac{mg}{m_\nu} t^\nu E_{\nu,1+\nu}[-(b/m_\nu)t^\nu],$$

and

$$x(t) = x_0 + V_0 t E_{\nu,2}[-(b/m_\nu)t^\nu] + \frac{mg}{m_\nu} t^{1+\nu} E_{\nu,2+\nu}[-(b/m_\nu)t^\nu].$$

Having analyzed experiments on the six men falling from the height of 31 400 foots, author of the article comes to the conclusion that the fractional differential approach with $\beta = 0.998$ and $m/m_\beta = 1.457$ is not in worse agreement with the experiment dependencies $x(t)$ then the classical one. The author did not comment the cause of the fact.

Baleanu et al. [20] representing a new version of the fractional Newtonian mechanics define the *fractional velocity* and *fractional momentum* for interval $[a, b]$ as

$$V(t) = (1/2)(\rho_1 \, {}^{\alpha}_{a}\mathsf{D}_t + \rho_2 \, {}^{\beta}_{t}\mathsf{D}_b)x(t),$$

and

$$p(t) = (m/2)(\rho_1 \, {}^{\alpha}_{a}\mathsf{D}_t + \rho_2 \, {}^{\beta}_{t}\mathsf{D}_b)x(t) = p_\alpha + p_\beta,$$

respectively, where $0 < \alpha,\ \beta \le 1$, and dimensions of the constants ρ_1 and ρ_2 are $T^{\alpha-1}$ and $T^{\beta-1}$, respectively. The fractional version of Newton's second law is then written as the expression

$$(1/2)(\kappa_1 \, {}_{t}\mathsf{D}^{\alpha}_{b} p_\alpha + \kappa_2 \, {}_{a}\mathsf{D}^{\beta}_{b} p_\beta) = F, \tag{1.23}$$

added by transversality condition

$$\left[{}_{t}\mathsf{D}^{\alpha-1}_{b} p_\alpha - {}_{a}\mathsf{D}^{\beta-1}_{t} p_\beta \right]_a^b = 0. \tag{1.24}$$

The authors note that for $\alpha = 1,\ \beta = 1$ they have

$$ {}_{a}\mathsf{D}_{t}^{\alpha} = {}_{a}^{\alpha}\mathsf{D}_{t} = {}_{t}\mathsf{D}_{b}^{\alpha} = {}_{t}^{\alpha}\mathsf{D}_{b} = \frac{d}{dt}, $$

and Eq. (1.24) reduces to the standard Newtonian equation. This is really so.

However, their next Remark 3.1, "If the generalized force in Eq. (1.23) is zero, then we can generalize Newton's first law as $\kappa_1\, {}_{t}\mathsf{D}_{b}^{\alpha} p_{\alpha} + \kappa_2\, {}_{a}\mathsf{D}_{t}^{\beta} p_{\beta} = 0$" seems to be non quite understable. Really, the first Newtonian law should not be considered as a plain mathematical sequence of the second one (in this case, it would be absolutely unnecessary in the system of mechanical axioms). In classical mechanics, the first law singles out from all possible coordinate systems, the family of inertial systems for which the second law is valid. From this point of view, the concepts of fractional velocity and fractional momentum look too artificial.

1.4 Fractional Oscillations

1.4.1 Fractionalized Harmonic Oscillator

The harmonic oscillator, one of the simplest mechanical systems, whose motion is governed by a second-order being linear differential equation with constant coefficients and whose analogs are ubiquitous in physics, needs no introduction. From mathematical point of view, fractional generalizations of this system have been studied by Bagley and Torvik [14], Gorenflo and Rutman [48], Mainardi [58], Blank [25], Gorenflo and Mainardi [47], Podlubny [72], Achar et al. [3], Trinks and Ruge [89], Stanislavsky [84], and other authors.

Fractionalization of a free oscillation process, governed by differential equation

$$ \ddot{x}(t) + \omega^2 x(t) = 0, \quad \omega^2 = k/m, \quad x(0) = x_0, \quad \dot{x}(0) = V_0, $$

or, equivalently, by integral equation [10]

$$ x(t) = x_0 + V_0 t - \omega^2 \int_0^t d\tau \int_0^{\tau} x(t')dt', $$

can be performed by using both these formulations. The first way leads (for $\alpha \in (1, 2]$) to equation

$$ {}_{0}\mathsf{D}_{t}^{\alpha}[x(t) - x_0 - V_0 t] + \omega^{\alpha} x = 0 $$

with the R-L derivative or to

$$ {}_{0}^{\alpha}\mathsf{D}_{t} x(t) + \omega^{\alpha} x(t) = 0 $$

with the G-C derivative [48]. The second way yields

$$x(t) = x_0 + V_0 t - \omega^\alpha \, {}_0\mathsf{I}_t^\alpha x(t)$$

Achar et al. [3]. The solution, obtained by the Laplace transform method and expressed in terms of two-parameter Mittag-Leffler functions

$$x(t) = x_0 E_{\alpha,1}[-(\omega t)^\alpha] + V_0 t E_{\alpha,2}[-(\omega t)^\alpha]$$

and in case of $V_0 = 0$, has the form

$$x(t) = x_0 E_{\alpha,1}[-(\omega t)^\alpha]. \tag{1.25}$$

According to Gorenflo and Mainardi [47], it can also be written as

$$x(t) = \frac{x_0}{\pi} \int_0^\infty \frac{\xi^{\alpha-1} \sin(\alpha\pi)}{\xi^{2\alpha} + 2\xi^\alpha \cos(\alpha\pi) + 1} e^{-\omega t \xi} d\xi.$$

Corresponding graph is plotted in Fig. 1.6.

As the authors write, their principal result is represented by expressions for the momentum and the total energy of the fractionalized oscillator. Following the fractionalization idea, they introduced these characteristics by generalization of corresponding classical ones:

$$p = m\mathsf{D}_t x(t) \;\mapsto\; p = m \, {}_0^{\alpha/2}\mathsf{D}_t x(t)$$

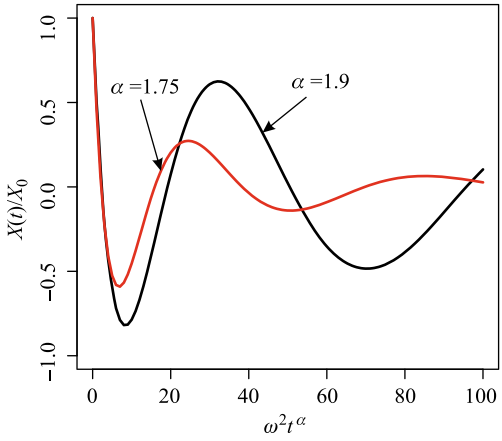


Fig. 1.6 Displacement as a function of time (in reduced variables) as per Eq. (1.25)

and

$$E = \frac{1}{2}kx^2 + \frac{1}{2}m[\mathsf{D}_t x(t)]^2 \mapsto\ E = \frac{1}{2}kx^2 + \frac{1}{2}m[\,{}_0^{\alpha/2}\mathsf{D}_t x(t)]^2,$$

where $1 < \alpha \leq 2$. As a result, in case $V_0 = 0$,

$$p = -mx_0\omega^\alpha t^{\alpha/2} E_{\alpha,1+\alpha/2}[-(\omega t)^\alpha] \tag{1.26}$$

and

$$E = \frac{1}{2}kx_0^2 E^2_{\alpha,1}[-(\omega t)^\alpha] + \frac{1}{2}mx_0^2(\omega^2 t)^\alpha E^2_{\alpha,1+\alpha/2}[-(\omega t)^\alpha].$$

The phase diagram corresponding to Eqs. (1.25) and (1.26) is shown in Fig. 1.7.

In case of a simple harmonic oscillator problem, $\alpha = 2$,

$$x(t) = x_0 \sum_{m=0}^{\infty} \frac{[-(\omega t)^2]^m}{(2\,m)!} = x_0 \cos(\omega t),$$

the motion is periodic, the total energy is constant and the phase plane diagram is a closed elliptic curve. When $\alpha < 2$, the oscillator behavior changes: its response function decays and its diagram in the phase plane (x, p) is not a closed curve anymore but a logarithmic spiral similar to the diagram of the damping classical oscillator (see Figs. 1 and 2 in Achar et al. [3]).

The next article of Achar et al. [2] contains results of studying of the response and resonance characteristics of the fractional oscillator for several cases of forcing function. The inclusion of an external driving force $F(t)$ under zeroth initial conditions

$$x(0) = 0, \quad \dot{x}(0) = 0,$$

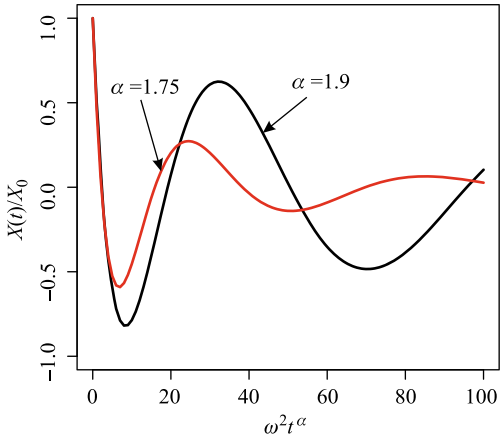


Fig. 1.7 Phase plane diagram (in reduced variables) for $\alpha = 1.9$

Table 1.1 Forcing and response functions

Forcing function $F(t)$	Response function $x(t)$
$\delta(t)$	$t^{\alpha-1}E_{\alpha,\alpha}\,[-(\omega_0 t)^\alpha]$
$1(t)$	$t^{\alpha}E_{\alpha,\alpha+1}\,[-(\omega_0 t)^\alpha]$
$\sin(\omega t)$	$A_1\sin(\omega t-\delta)+A_2\exp(-\beta t)\cos[\omega_0 t\sin(\pi/\alpha)-\phi]+$ $\int_0^\infty \exp(-st)K_\alpha(s)ds$

leads to the integral fractional equation

$$x(t) = {}_0\mathrm{I}_t^\alpha[-\omega_0^\alpha x(t)+F(t)],$$

solution of which for $\alpha\in(1,2)$ is given by

$$x(t)=\int_0^t E_{\alpha,\alpha}[-\omega_0^\alpha(t-\tau)^\alpha](t-\tau)^{\alpha-1}F(\tau)d\tau.$$

Three special results are shown in Table 1.1, where

$$\begin{aligned}
A_1 &= \frac{1}{\sqrt{\omega_0^{2\alpha}+\omega^{2\alpha}+2\omega_0^\alpha\omega^\alpha\cos(\alpha\pi/2)}},\\
\delta &= \arctan\left[\frac{\omega^\alpha\sin(\alpha\pi/2)}{\omega_0^\alpha+\omega^\alpha\cos(\alpha\pi/2)}\right],\\
A_2 &= \frac{2\omega}{\alpha\omega_0^{\alpha-1}\sqrt{\omega_0^4+\omega^4+2\omega_0^2\omega^2\cos(2\pi/\alpha)}},\\
\beta &= -\omega_0\cos(\pi/\alpha),\\
\phi &= \arctan\left[\frac{\omega_0^2\sin((1+\alpha)\pi/\alpha)-\omega^2\sin((1-\alpha)\pi/\alpha)}{\omega_0^2\cos((1+\alpha)\pi/\alpha)+\omega^2\cos((1-\alpha)\pi/\alpha)}\right],
\end{aligned}$$

and

$$K_\alpha(s)=\frac{\omega\sin(\pi\alpha)}{\pi(s^2+\omega^2)(s^{2\alpha}+2\,s^\alpha\omega_0^\alpha\cos\pi\alpha+\omega_0^{2\alpha})}.$$

The response to the δ-function is shown in Fig. 1.8.

The latter response given in Table 1.1 is obtained from the initial expression by using the Mittag-Leffler representation of the sine function

$$\sin(\omega\tau)=\omega\tau E_{2,2}[-(\omega\tau)^2]$$

and evaluating the Laplace transform of the resulting function

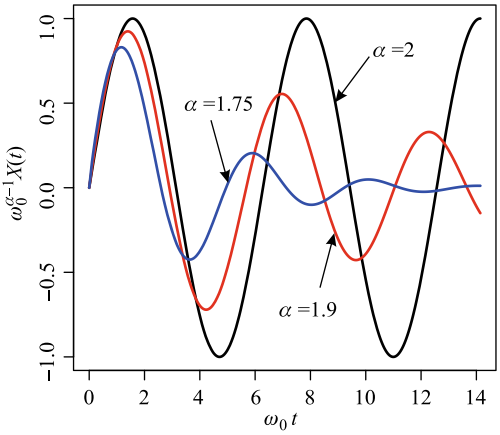


Fig. 1.8 Response function for a δ-forcing function for different values of α

$$\widehat{x}(\lambda) = \omega \frac{\lambda^{-\alpha}}{1 + (\omega_0/\lambda)^{\alpha}} \frac{\lambda^{-2}}{1 + (\omega/\lambda)^2}.$$

Performing inverse transformation represents the responce function as a sum of three terms

$$x(t) = \frac{1}{2\pi i} \int_{\mathrm{Br}} \frac{\omega \exp(\lambda t) d\lambda}{(\lambda^2 + \omega^2)(\lambda^{\alpha} + \omega_0^{\alpha})} = x_1(t) + x_2(t) + x_3(t).$$

The first of them results from calculating the residues of the poles $\lambda = \pm i\omega$:

$$x_1(t) = \left[\frac{\omega_0^{\alpha} \sin(\omega t) + \omega^{\alpha} \sin(\omega t - \alpha\pi/2)}{\omega_0^{2\alpha} + \omega^{2\alpha} + 2\omega_0^{\alpha}\omega^{\alpha} \cos(\alpha\pi/2)} \right].$$

The second is defined by the residues for two other poles $\lambda = \omega_0 \exp(\pm i\pi/\alpha)$:

$$x_2(t) = \left[\frac{\omega e^{\lambda t}}{(\lambda^2 + \omega^2)(d/d\lambda)(\lambda^{\alpha} + \omega_0^{\alpha})} \right]_{\lambda = \omega_0 \exp(\pm i\pi/\alpha)}$$

$$= \frac{2 \exp[\omega_0 t \cos(\pi/\alpha)] C}{\alpha \omega_0^{\alpha - 1} \left[\omega^4 + \omega_0^4 + 2\omega^2 \omega_0^2 \cos(2\pi/\alpha) \right]},$$

with

$$C = \omega \{ \omega_0^2 \cos[\omega_0 t \sin(\pi/\alpha) - (1 + \alpha)\pi/\alpha] + \omega^2 \cos[\omega_0 t \sin(\pi/\alpha) + (1 - \alpha)\pi/\alpha] \}.$$

After some algebra, these terms are reduced to the form given in the Table 1.1:

$$x_1(t) = A_1 \sin(\omega t - \delta),$$

$$x_2(t) = A_2 e^{-\beta t} \cos[\omega_0 t \sin(\pi/\alpha) - \phi].$$

The third integral term originates from the loop consisting of the small circle of radius r and the two lines CD and EF parallel to the negative real axis (see Fig. 1.5)

$$x_3(t) = -\frac{1}{2\pi i}\int_{-\infty}^{0} e^{\lambda t}\widehat{x}_+(\lambda)d\lambda - \frac{1}{2\pi i}\int_{0}^{-\infty} e^{\lambda t}\widehat{x}_-(\lambda)d\lambda.$$

Inserting $\lambda = se^{i\pi}$ into the first integral taken along the upper border and $\lambda = se^{-i\pi}$ into the second integral along the lower border, we get

$$x_3(t) = \int_0^{\infty} e^{-st} K_\alpha(s)ds.$$

Evidently, the term $x_3(t)$ vanishes as $t \to \infty$. When $\alpha \in (1, 2)$ $\cos(\pi/\alpha)$ is negative and $x_2(t)$ becomes also vanishingly small at large time. Both these terms describe a relaxation process in the oscillator, and only the first term defines steady-state oscillation in this system,

$$x_{\rm st}(t) = A\sin(\omega t - \delta),$$

with driving force frequency ω, amplitude

$$A = A_{\rm st} = \frac{1}{\sqrt{\omega^{2\alpha} + \omega_0^{2\alpha} + 2\omega^\alpha \omega_0^\alpha \cos(\alpha\pi/2)}}$$

and phase shift

$$\delta = \arctan\left[\frac{\omega^\alpha \sin(\alpha\pi/2)}{\omega^2 \cos(\alpha\pi/2) + \omega_0^\alpha}\right].$$

Numerical results obstained with the use of these formulas are plotted on Figs. 1.9 and 1.10.

For other treatments of fractional oscillators see, for example, [2, 3, 12, 13, 24, 47, 58] and others.

1.4.2 Linear Chain of Fractional Oscillators

In this subsection, we consider oscillations of a one-dimensional chain of identical point masses each of them interacts with only its nearest neighbors [4].

The ordinary model of a system of masses connected by springs of spring constant k (see Fig. 1.11) is described by equation

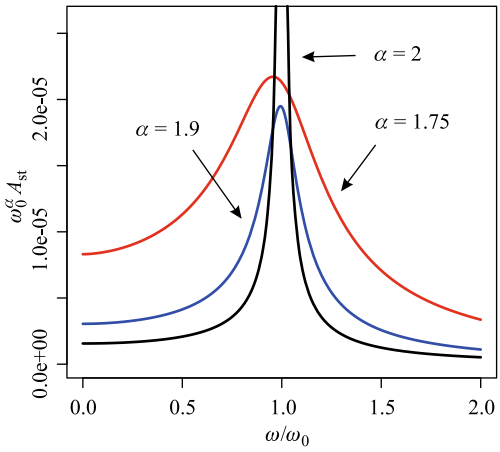


Fig. 1.9 Amplitude response for sinusoidal forcing

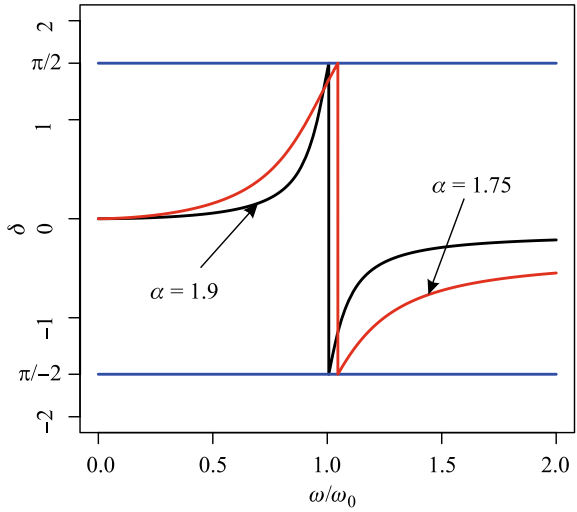


Fig. 1.10 Change in phase angle for sinusoidal forcing

$$m\frac{d^2u_n}{dt^2} = k(u_{n+1} - 2u_n + u_{n-1}),$$

where u_n is the displacement from the equilibrium position of the nth mass ("atom"), $2 \leq n \leq N-1$, and N is the total number of atoms. This model has been generalized to include dissipative effects by incorporating dashpots in parallel to the springs to yield the lattice dynamical version of the Kelvin–Voigt model, or the dashpots in series with the springs to yield the lattice dynamical version of the Maxwell model [11].

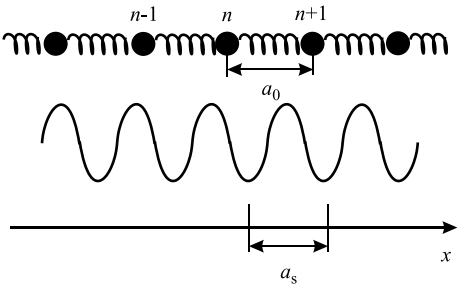


Fig. 1.11 Schematic presentation of the Frenkel-Kontorova model

This system is easily reduced to the integral form

$$u_n(t) = u_n(0) + \dot{u}_n(0)t + \omega_0^2 \int_0^t [u_{n+1}(\tau) - 2u_n(\tau) + u_{n-1}(\tau)](t-\tau)d\tau,$$

$$2 \le n \le N-1,$$

where $\omega_0^2 = k/m$, in the usual notation. For the extremes, we have

$$u_1(t) = u_1(0) + \dot{u}_1(0)t + \omega_0^2 \int_0^t [u_2(\tau) - 2u_1(\tau)](t-\tau)d\tau + \int_0^t F(\tau)(t-\tau)d\tau$$

and

$$u_N(t) = u_N(0) + \dot{u}_N(0)t + \omega_0^2 \int_0^t [u_{N-1}(\tau) - 2u_N(\tau)](t-\tau)d\tau,$$

respectively. An external force $F(t)$ which applied to the end atom numbered by N is assumed to be periodic.

The integrals on the right-hand side of these equations are generalized to fractional integrals of order μ to describe the of motion of a chain of coupled fractional oscillators:

$$u_1(t) = u_1(0) + \dot{u}_1(0)t + \frac{\omega_0^\nu}{\Gamma(\nu)} \int_0^t [u_2(\tau) - 2u_1(\tau)](t-\tau)^{\nu-1}d\tau$$
$$+ \frac{1}{\Gamma(1)} \int_0^t F(\tau)(t-\tau)^{\nu-1}d\tau,$$

$$u_n(t) = u_n(0) + \dot{u}_n(0)t + \frac{\omega_0^\nu}{\Gamma(\nu)} \int_0^t [u_{n+1}(\tau) - 2u_n(\tau) + u_{n-1}(\tau)](t-\tau)^{\nu-1}d\tau,$$

$$2 \le n \le N-1,$$

and

$$u_N(t) = u_N(0) + \dot{u}_N(0)t + \frac{\omega_0^\nu}{\Gamma(\nu)} \int_0^t [u_{N-1}(\tau) - 2u_N(\tau)](t-\tau)^{\nu-1} d\tau.$$

Taking Laplace transform of both sides of the equations yields

$$\widehat{u}_1(\lambda) = u_1(0)\lambda^{-1} + \dot{u}_1(0)\lambda^{-2} + \omega_0^\nu \lambda^{-\nu}[\widehat{u}_2(\lambda) - 2\widehat{u}_1(\lambda)] + \widehat{F}(\lambda)\lambda^{-\nu},$$

$$\widehat{u}_n(\lambda) = u_n(0)\lambda^{-1} + \dot{u}_n(0)\lambda^{-2} + \omega_0^\nu \lambda^{-\nu}[\widehat{u}_{n+1}(\lambda) - 2\widehat{u}_n(\lambda) + \widehat{u}_{n-1}(\lambda)],$$

$$2 \le n \le N-1,$$

and

$$\widehat{u}_N(\lambda) = u_N(0)\lambda^{-1} + \dot{u}_N(0)\lambda^{-2} + \omega_0^\nu \lambda^{-\nu}[\widehat{u}_{N-1}(\lambda) - 2\widehat{u}_N(\lambda)].$$

Then the set of linear equations can be solved numerically and the inverse Laplace transform yields the displacements as functions of time (for details see [5]).

1.4.3 Fractionalized Waves

In the continuum limit when the number of atoms $N \to \infty$, the separation between the atoms $a \to 0$, such that the product $Na \to L$, a finite length,

$$a^2\omega_0^\nu = a^2 k/m = K/\rho,$$

where ρ is the mass density and $K = ka$ is the tension, and

$$u_{n+1}(t) - 2u_n(t) + u_{n-1}(t) = a^2 \frac{[u_{n+1}(t) - u_n(t)]/a - [u_n(t) - u_{n-1}(t)]/a}{a} \to a^2 \frac{\partial^2 u(x,t)}{\partial x^2}.$$

On assumption that at $t = 0$, the displacement of the free end is subject to sinusoidal forcing,

$$u(0,t) = f(t) = A\sin(\omega t),$$

we arrive at the equation

$$u(x,t) = \frac{c_0^\nu}{\Gamma(\nu)} \int_0^t \frac{\partial^2 u(x,\tau)}{\partial x^2} (t-\tau)^{\nu-1} d\tau$$

with the initial conditions $u(x,0)=0$ and $\dot{u}(0,t)=0$ for $x>0$ and $u(0,t)=f(t)$. Observe that c_0 has the dimension of velocity. The Laplace transform with respect to time leads to the equation

$$\frac{\partial^2 \widehat{u}(x,\lambda)}{\partial x^2}-\left(\frac{\lambda}{c_0}\right)^{\nu}\widehat{u}(x,\lambda)=0$$

for $x\neq 0$ and $\widehat{u}(0,\lambda)=\widehat{f}(\lambda)$. Solving this equation,

$$\widehat{u}(x,\lambda)=\widehat{f}(\lambda)\exp\left[-(\lambda/c_0)^{\nu/2}x\right],$$

substituting

$$\widehat{f}(\lambda)=A\omega/(\omega^2+\lambda^2)$$

and performing the inverse transformation by the Bromwich integral, we get

$$u(x,t)=\frac{A\omega}{2\pi i}\int_{\mathrm{Br}}\frac{\exp\left[\lambda t-(\lambda/c_0)^{\nu/2}x\right]d\lambda}{\omega^2+\lambda^2}=u_{\mathrm{tr}}(x,t)+u_{\mathrm{st}}(x,t),$$

where

$$u_{\mathrm{tr}}(x,t)=\int_0^{\infty}ds\frac{A\omega}{\pi(s^2+\omega^2)}\exp\left[-st-\left(\frac{s}{c_0}\right)^{\nu/2}x\cos\left(\frac{\nu\pi}{2}\right)\right]$$
$$\cdot\sin\left[\left(\frac{s}{c_0}\right)^{\nu/2}x\sin\left(\frac{\nu\pi}{2}\right)\right]$$

and

$$u_{\mathrm{st}}(x,t)=A\exp\left[-(\omega/c_0)^{\nu/2}x\cos(\nu\pi/4)\right]\sin\left[\omega t-x(\omega/c_0)^{\nu/2}\sin(\nu\pi/4)\right]$$

represent a transient part and a steady state part of the solution respectively. The first of them arises from the Hankel loop consisting of the small circle and two lines parallel to the negative x-axis and the second from the residues of the poles of the integrand. The transient part decays in time and approaches zero as $t\to\infty$ and vanishes entirely as $\nu=2$. Furthermore, it exhibits attenuation as a function of distance from the end as indicated by the spatial dependence of the integrand. No simple closed form expressions is obtained, the only recourse is through numerical integration (see Achar and Hanneken [4] and references therein).

1.4.4 Fractionalized Frenkel-Kontorova Model

We touch here upon the model suggested by Frenkel and Kontorova [42] to describe in outline the dynamics of a crystal lattice in the vicinity of the dislocation core. The model represents a chain of atoms with the equilibrium distance a_0 (in the absence of on-site potential) interacting with the nearest neighbors via ideal springs characterized by the elastic constant k and subjected to an external harmonic force charaterized by the spatial period a_s and amplitude F_0. The equations of motion are written as

$$m\frac{d^2x_n}{dt^2} = k(x_{n+1} - 2x_n + x_{n-1}) + F_0 \sin Kx_n,$$

where the chain is supposed to be infinite with $a_0 = a_s$, when the ground state of the chain (corresponding to the minimum of the potential energy) is a commensurate structure of the atom. Treating the system as a finite chain of N atoms with periodic boundary conditions, introducing atomic displacements, and passing to the dimensionless variables, in the continuous limit we arrive at the *sine-Gordon* (SG) *equation* (for details see [29])

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = b\sin(\mu f). \tag{1.27}$$

It is worth to note that the equation revealed solitonic properties of its solutions by more than a decade before the Korteweg-de Vries equation. In the sixties, the SG equation appeared in the theory of weak superconductivity and became the main non-linear equation describing the long Josephson junctions [9, 22], pinning in superconductors [49], motion of Bloch magnetic walls in magnetic crystals [39] and some other phenomena.

The physical applications of the nonlinear sine-Gordon equation are related with the description of dislocations in solid state physics [42], motion of Bloch magnetic walls in magnetic crystals [39], magnetic flux propagation in superconductors [22]. Further generalizations of this equation were performed by involving nonlocality both into the dynamic term $b\sin(\mu f)$ and into the kinetic term $a\partial^2 f/\partial x^2$ (see for review [69]).

The first way has led to the equation [35]

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = 2b\cos\left(\frac{f(x,t)}{2}\right)\int_{-\infty}^{\infty} K(x-y)\sin\left(\frac{f(y,t)}{2}\right)dy,$$

which associated with DNA models as well as the *Frenkel-Kontorova model* [42] including long-range iterations.

On the second way, the space derivative operator was replaced by integro-differential one

$$\frac{\partial^2 f}{\partial t^2} - a\mathsf{L}f(x,t) = b\sin(\mu f)$$

and the obtained equation has found application in the nonlocal Josephson electrodynamics.

Equation (1.27) is a nonlinear but local equation. There are derived at least two versions of nonlocal SG equations:

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = 2b\cos\left(\frac{f(x,t)}{2}\right)\int_{-\infty}^{\infty} K(x-x')\sin\left(\frac{f(x',t)}{2}\right)dx',$$

and

$$\frac{\partial^2 f}{\partial t^2} - \mathsf{L}f = b\sin f(x,t),$$

where L is a nonlocal operator. Miskinis [61, 62] considered a special type of the fractionalized SG equation:

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = b\mathbb{D}_x^{-\alpha}\sin(c\mathbb{D}_x^{\alpha} f), \tag{1.28}$$

where a, b and c are constants and $\mathbb{D}_x^{\alpha}$ is the fractional Riesz operator

$$\mathbb{D}_x^{\alpha} = -\frac{{}_{-\infty}\mathsf{D}_x^{\alpha} + {}_x\mathsf{D}_{\infty}^{\alpha}}{2\cos(\alpha\pi/2)}.$$

He obtained it as follows. The first and second partial derivatives with respect to the spatial coordinate are the limits

$$\frac{\partial f(x,t)}{\partial x} = \lim_{a\to 0}\frac{2\pi y_n}{a} = \lim_{a\to 0}\frac{2\pi y_n/x_0}{a/x_0},$$
$$\frac{\partial^2 f(x,t)}{\partial x^2} = \lim_{a\to\infty} x_0^2\frac{2\pi}{a^3}(y_{n+1} - 2y_n + y_{n-1}),$$

where $y_n(t) = x_n(t) - na$, and x_n is the coordinate of the nth atom. Assuming that atoms undergo a chaotic perturbations, so that $\lim_{a\to 0}(2\pi y_n/a)$ could not exist, but $\lim_{a\to 0}(2\pi y_n/a^{\alpha})$ with $\alpha \in (0, 1)$ exists, Miskinis obtains

$$\lim_{a\to 0}\left(x_0^{\alpha-1}\frac{2\pi y_n}{a_n^{\alpha}}\right) = \mathbb{D}_x^{\alpha} f(x,t).$$

Thus, the equation of motion takes the form

$$\frac{\partial^2(\mathbb{D}_x^{\alpha} f)}{\partial t^2} - a\frac{\partial^2(\mathbb{D}_x^{\alpha} f)}{\partial x^2} = b\sin(c\mathbb{D}_x^{\alpha} f),$$

or using properties of the derivative, we obtain

$$\mathbb{D}_x^{\alpha}\left[\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2}\right] = b\sin\left(c\mathbb{D}_x^{\alpha} f\right).$$

Among all possible solutions, a set of functions $f(x, t)$ for which

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = b\mathbb{D}_x^{-\alpha}\sin(c\mathbb{D}_x^{\alpha} f),$$

and the boundary condition obeys the relation $\mathbb{D}_x^{\alpha} f(x, t_0) = 0$.

Let us come back to Eq. (1.28). In case of small values of α, it takes the infinitesimal form

$$\frac{\partial^2 f}{\partial t^2} - a\frac{\partial^2 f}{\partial x^2} = b\sin(cf) + \alpha \text{L} f$$

which turns into the linear Klein-Gordon equation with the mass term bcf if amplitudes $c\mathbb{D}_x^{\alpha} f \ll 1$. Miskinis has found the Lagrangian form for this equation

$$L = \int_{-\infty}^{\infty}\left\{(1/2)\left[(\mathbb{D}_x^{\alpha}\partial f/\partial t)^2 - (\mathbb{D}_x^{\alpha+1} f)^2\right] + (b/c)\left[1 - \cos(c\mathbb{D}_x^{\alpha} f)\right]\right\} dx$$

and two travelling wave solutions—nonlocal generalizations of one-solitonic solutions:

$$f(x, t) = \frac{4}{\lambda}\mathbb{D}_x^{-\alpha}\tan^{-1}\left\{\exp\left[\pm\frac{bc(kx + \mu t + \theta)}{\sqrt{bc(\mu^2 - ak^2)}}\right]\right\} \quad \text{for} \quad bc(\mu^2 - ak^2) > 0$$

and

$$f(x, t) = \frac{4}{\lambda}\mathbb{D}_x^{-\alpha}\tan^{-1}\left\{\exp\left[\pm\frac{bc(kx + \mu t + \theta)}{\sqrt{bc(ak^2 - \mu^2)}}\right]\right\} \quad \text{for} \quad bc(\mu^2 - ak^2) < 0,$$

where μ, k, and θ are arbitrary constants.

1.4.5 Oscillations of Bodies in a Viscous Fluid

Shäfer and Kempfle [81] consider an oscillator governed by the differential equation of the second order with a fractional derivative of order $\alpha \in (0, 2)$. This equation was introduced through its Fourier transform:

$$A(\omega)\widetilde{x}(\omega) \equiv [(i\omega)^2 + a(i\omega)^{\alpha} + b]\widetilde{x}(\omega) = \widetilde{f}(\omega).$$

The solution $x(t)$ is expressed by the convolution

$$x(t) = \int_{-\infty}^{\infty} G(t-\tau) f(\tau) d\tau$$

of the right-hand side with impulse response $G(t)$ obtained as

$$G(t) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{\lambda t} d\lambda}{\lambda^2 + a\lambda^\alpha + b}.$$

When $\alpha = 3/2$, this function describes the response of a thin plate hanging on a spring and immersed into a viscous liquid.

Due to the location of the zeros $\lambda_{1,2} = -\sigma \pm i\Omega$ of the denominator, $G(t)$ consists for $t \geq 0$ of two components:

$$G(t) = G_1(t) + G_2(t),$$

where

$$G_1(t) = Ce^{-\sigma t} \sin(\Omega t + \Phi)$$

with

$$C = \frac{2}{\sqrt{\mu^2 + \nu^2}}, \ \Phi = \arctan\left(\frac{\mu}{\nu}\right), \ \mu = \mathrm{Re}(\lambda_1), \ \nu = \mathrm{Im}(\lambda_1),$$

and

$$G_2(t) = \frac{a \sin(\pi\alpha)}{\pi} \int_0^{\infty} \frac{r^\alpha e^{-rt} dr}{(r^2 + ar^\alpha \cos(\pi\alpha) + b)^2 + (ar^\alpha \sin(\pi\alpha))^2}.$$

Observe that when $a = 0$, the results coincide with those for a pure harmonic oscillator. If $a \neq 0$ but $\alpha = 1$, we meet the case of a damped oscillator characterized by an exponentially decaying amplitude. In the fractional case, G_1 represents a phase-shifted exponentially damped oscillations, whereas G_2 describes an algebraic decaying component with the asymptotic behavior (Fig. 1.12)

$$G_2(t) \sim \frac{a \sin(\pi\alpha)}{\pi b^2} \Gamma(\alpha+1) t^{-\alpha-1}, \qquad t \to \infty.$$

After some time, the latter becomes dominant over G_1 such that $G_1(t)$ perform a finite number of oscillations around $G_2(t)$ and then ends in a monotonic creeping into the rest position from above, if $0 < \alpha < 1$ and from below, if $1 < \alpha < 2$. The maximal decay rate is observed at $\alpha = 1$ (a classical damped oscillator); as α moves away from 1, the rate becomes weaker and at $\alpha = 0$ or 1 the damping completely vanishes: the oscillator becomes harmonic.

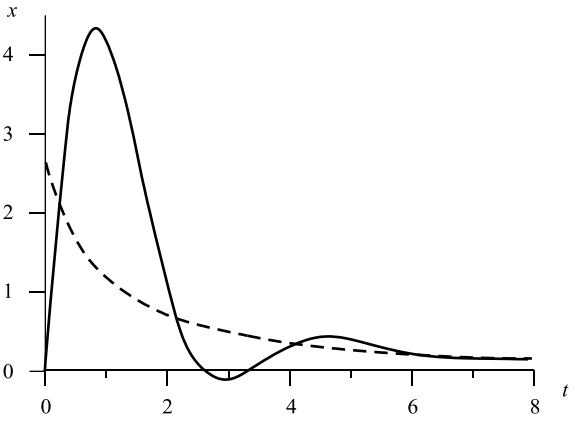


Fig. 1.12 Fractional oscillator motion(solid curve) and its asymptotical behavior (dashed line)

As we have seen above, oscillations of a sphere under the same conditions are described by the equation

$$m\ddot{x}(t) + b_0 \mathsf{D}_t^{1+\alpha} x(t) + c\dot{x}(t) + kx(t) = F(t), \quad 0 < \alpha < 1.$$

A special numerical method for solving this equation is worked out in Trinks and Ruge [89]. We give here a brief account of this method.

Keeping symbol V for the velocity ($V(t) = \dot{x}(t)$), we transform the term with fractional derivative to expression

$$\begin{aligned} {}_0\mathsf{D}_t^{1+\alpha} x(t) &= \frac{1}{\Gamma(1-\alpha)} \frac{d}{dt} \left[\frac{x(0)}{t^\alpha} + \int_0^t \frac{\dot{x}(\tau) d\tau}{(t-\tau)^\alpha} \right] \\ &= \frac{1}{\Gamma(1-\alpha)} \left[\frac{V(0)}{t^\alpha} - \frac{\alpha x(0)}{t^{1+\alpha}} \right] + \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{\dot{V}(\tau) d\tau}{(t-\tau)^\alpha}. \end{aligned}$$

Using integral representation of the gamma function and Euler's reflection formula yields:

$$\begin{aligned} {}_0\mathsf{D}_t^{1+\alpha} x(t) = &\frac{1}{\Gamma(1-\alpha)} \left[\frac{V(0)}{t^\alpha} - \alpha \frac{x(0)}{t^{\alpha+1}} \right] \\ &+ \frac{2 \sin \pi\alpha}{\pi} \int_0^t d\tau \dot{V}(\tau) \int_0^\infty e^{-(t-\tau)y^2} y^{2\alpha-1} dy. \end{aligned}$$

As the authors note, the decisive idea of their method is eliminating the time integral in this definition by introducing a new variable

$$\Theta(y, t) = y^{2\alpha-1} \int_0^t e^{-y^2(t-\tau)} \dot{V}(\tau) d\tau$$

obeying the differential equation

$$\dot{\Theta}(y,t) + y^2\Theta(y,t) = y^{2\alpha-1}\dot{V}(t).$$

Therefore, they can reformulate the problem under consideration by means of the following system of equations

$$m\dot{V}(t) + c\dot{x}(t) + kx(t) + \frac{2b}{\pi}\sin(\pi\alpha)\int_0^\infty \Theta(y,t)dy = F(t) - f(t),$$

$$f(t) = \frac{b}{\Gamma(1-\alpha)}\left[\frac{V(0)}{t^\alpha} - \frac{\alpha x(0)}{t^{1+\alpha}}\right],$$

and

$$\dot{\Theta}(y,t) + y^2\Theta(y,t) = y^{2\alpha-1}\dot{V}(t).$$

Supplementing this system by a finite-difference scheme of the above integral computing, the authors used the Trinks-Ruge numerical method. Some specific computational details can be seen from examples given in the cited paper.

1.5 Dynamical Control Problems

1.5.1 PID Controller and Its Fractional Generalization

Control theory is an interdisciplinary branch of engineering and mathematics that deals with the behavior of dynamical systems. The system has an *input* and an *output*. When the output variable need to follow a certain reference over time, a *controller* manipulates the inputs to the system to obtain the desired effect on the output of the system. The usual aim of the control theory is to calculate solutions for the proper corrective action from the controller. The most commonly used controller is the *proportional-integral-derivative controller* (*PID controller*), involving three separate parameters relating to the proportional, the integral and derivative values, denoted by P, I and D respectively (Fig. 1.13). These three parameters represent information on contributions of the corresponding sources of errors $\varepsilon(t)$ (the difference between a measured process variable and a desired setpoint at time t) into the output $f(t)$:

$$K_p\varepsilon(t) + K_i\int_0^t \varepsilon(\tau)d\tau + K_d\frac{d\varepsilon(t)}{dt} = f(t),$$

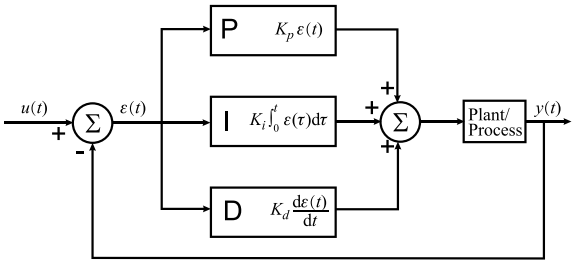


Fig. 1.13 PID controller

where K_p, K_i, and K_d are corresponding tuning parameters (see Fig. 1.13). Under initial zeroth condition, the Laplace transform of this equation can be rewritten as

$$\widehat{\varepsilon}(\lambda) = \widehat{C}(\lambda)\widehat{f}(\lambda)$$

where

$$\widehat{C}(\lambda) = \left[K_p + K_i\lambda^{-1} + K_d\lambda\right]^{-1}$$

is the *PID controller transfer function*.

To make this concept more obvious, we imagine a simple mechanic system consisting of a driving shaft with a motor on one end and a massive disc on the other. The moment of inertia of the motor with an affiliated load (except the disc) is I, damping is represented by the torsional resistance B of the disc. One should ensure matching angular displacements of the long shaft in its beginning (φ_1) and its end (φ_2). The difference $\varphi_2(t) - \varphi_1(t)$ is the same error ε mentioned above. It enters the PID controller which applies a correcting torque

$$\delta\tau(t) = K_p\varepsilon(t) + K_i\int_0^t \varepsilon(\tau)d\tau + K_d\frac{d\epsilon}{dt}.$$

Assuming that the shaft rests until $t = 0$ and then its first end begins to rotate so that the second end rotates according to the linear law $\varphi_2(t) = \Omega t$. The corresponding equations for dependent variables $y(t) \equiv \varphi_1(t)$ and $\varepsilon(t)$ are

$$I\frac{d^2y(t)}{dt^2} = -B\frac{dy(t)}{dt} + \delta\tau(t) \tag{1.29}$$

and

$$\delta\tau(t) = K_p\varepsilon(t) + K_i\int_0^t \varepsilon(\tau)d\tau + K_d\frac{d\varepsilon(t)}{dt} \tag{1.30}$$

with

$$\varepsilon(t) = \varphi_2(t) - \varphi_1(t) = t_+ - y(t).$$

Eliminating $\delta\tau(t)$ from Eqs. (1.29) and (1.30), we arrive at the system

$$I\frac{d^2y}{dt^2} + B\frac{dy}{dt} = K_p\varepsilon(t) + K_i\int_0^t \varepsilon(\tau)d\tau + K_d\frac{d\varepsilon(t)}{dt},$$

$$\varepsilon(t) = t_+ - y(t),$$

the Laplace transform of which is

$$\left[I\lambda^2 + B\lambda\right]\widehat{y}(\lambda) = \left[K_p + K_i\lambda^{-1} + K_d\lambda\right]\widehat{\varepsilon}(\lambda),$$

$$\widehat{\varepsilon}(\lambda) = \lambda^{-2} - \widehat{y}(\lambda).$$

Finally, we have

$$\widehat{\varepsilon}(\lambda) = \widehat{H}(\lambda).$$

As a new generalization of the classical PID-controller, the idea of $PI^{\mu}D^{\nu}$-controller, involving fractional-order integrator and fractional-order differentiator, has been found to be a more efficient control of fractional-order dynamical systems. Such a system is described by the fractional differential equation

$$\left[K_p + K_i\,{}_0\mathsf{D}_t^{-\mu} + K_d\,{}_0\mathsf{D}_t^{\nu}\right]\varepsilon(t) = f(t),$$

or in a more general case,

$$\left[\sum_{k=0}^{n} a_k\,{}_0\mathsf{D}_t^{\alpha_k}\right]\varepsilon(t) = f(t),$$

where $\alpha_k > \alpha_{k-1}$ are arbitrary real numbers and a_k are arbitrary constants. The respective *fractional transfer function* (FTF) is given by

$$\widehat{C}(\lambda) = \left[\sum_{k=0}^{n} a_k\lambda^{\alpha_k}\right]^{-1}.$$

The unit-impulse response

$$C(t) = \mathscr{L}^{-1}\{\widehat{C}(\lambda)\}(t)$$

can be represented by means of functions

$$\mathscr{E}_k(t, y; \alpha, \beta) = t^{\alpha k+\beta-1}E^{(k)}_{\alpha,\beta}(yt^{\alpha}), \quad k = 0, 1, 2, \ldots$$

introduced in Podlubny [72]. They have the Laplace transforms given by

$$\widehat{\mathscr{E}}_k(\lambda, y; \alpha, \beta) = \frac{k!\lambda^{\alpha-\beta}}{(\lambda^\alpha + y)^{k+1}},$$

and also possess the useful property

$${}_0\mathsf{D}_t^\nu \mathscr{E}_k(t, y; \alpha, \beta) = \mathscr{E}_k(t, y; \alpha, \beta - \nu), \quad \nu < \beta.$$

I. Podlubny found that

$$C(t) = \frac{1}{a_n} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \sum_{k_0>0,\ldots,k_{n-2}>0} \delta_{k_0+\cdots+k_{n-2},m} \binom{m}{k_0, \cdots, k_{n-2}} \times \prod_{i=0}^{n-2} \left(\frac{a_i}{a_n}\right)^{k_i} \mathscr{E}_m\left(t, -a_{n-1}/a_n; \beta_n - \beta_{n-1}, \beta_n + \sum_{j=0}^{n-2}(\beta_{n-1} - \beta_j)k_j\right).$$

For example, in case of the FTF

$$\widehat{C}(\lambda) = \left[b\lambda^\beta + a\lambda^\alpha + 1\right]^{-1}$$

with $\beta > \alpha$ we obtain

$$\varepsilon(t) = \frac{1}{b} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left(\frac{1}{b}\right)^k \mathscr{E}_k(t, -a/b; \beta - \alpha, \beta + \alpha k + 1).$$

1.5.2 Fractional Transfer Functions

A single input-single output linear time invariant system, relaxed at $t = 0$ may be described by the following differential equation

$$\left[\sum_{k=0}^{n} a_k \, {}_0^{\alpha_k}\mathsf{D}_t\right] y(t) = \left[\sum_{j=0}^{m} b_k \, {}_0^{\beta_k}\mathsf{D}_t\right] u(t).$$

In the Laplace domain, the differential equation leads us to fractional transfer functions such as

$$\widehat{H}(\lambda) = \frac{\widehat{y}(\lambda)}{\widehat{u}(\lambda)} = \frac{b_m\lambda^{\beta_m} + b_{m-1}\lambda^{\beta_{m-1}} + \cdots + b_0\lambda^{\beta_0}}{a_n\lambda^{\alpha_n} + a_{n-1}\lambda^{\alpha_{n-1}} + \cdots + a_0\lambda^{\alpha_0}}.$$

Those where all orders β_k and α_k are integer multipliers of a least common devisor γ called commensurable, these functions are the ratio of two polynomials in λ^γ,

$$\widehat{H}(\lambda) = \frac{\widehat{y}(\lambda)}{\widehat{u}(\lambda)} = \frac{b_m\lambda^{m\gamma} + b_{m-1}\lambda^{(m-1)\gamma} + \cdots + b_0}{a_n\lambda^{n\gamma} + a_{n-1}\lambda^{(n-1)\gamma} + \cdots + a_0}.$$

According to the *Matignon theorem* [59], commensurable transfer functions are stable if and only if the roots $\sigma_k,\ k = 1, 2, \ldots, n$ of polynomial $A(\sigma) = \sum_{k=0}^{n} a_k\sigma^k$ satisfy the condition $|\arg\sigma_k| > \alpha\pi/2$.

It is possible to find usual transfer functions (with $\gamma = 1$) that approximate the behaviour of fractional transfer functions. The most widely used approximation is the CRONE (Commande Robuste d'Ordre Non-Entier) approximation. It has N poles and N zeros within a frequency range $[\omega_l, \omega_h]$ and is given by

$$\lambda^{\gamma} \approx C\prod_{m=1}^{N}\frac{1+\lambda/\omega_{\zeta,m}}{1+\lambda/\omega_{\pi,m}},$$

$$\omega_{\zeta,m} = \omega_l\left(\frac{\omega_h}{\omega_l}\right)^{(2m-1-\gamma)/(2N)}, \quad \omega_{\pi,m} = \omega_l\left(\frac{\omega_h}{\omega_l}\right)^{(2m-1+\gamma)/(2N)}.$$

As is clear from these formulas, if γ is real then poles and zeros are real too and recursively placed alternating on the negative real semi-axis:

$$\frac{\omega_{\zeta,m+1}}{\omega_{\zeta,m}} = \frac{\omega_{\pi,m+1}}{\omega_{\pi,m}} = \left(\frac{\omega_h}{\omega_l}\right)^{1/N}.$$

This constant ratio implies that both poles and zeros are equidistant in a logarithmic scale of frequencies.

Fractional transfer functions can be employed to model dynamic systems and to devise controllers for various dynamic systems. The plants to which fractional controllers are applied may or may not be fractional themselves. One can find among them such processes as heat fluxes in the soil, found from surface soil temperature measurements, the propagation of sound in porous materials, the transport of substances by water in soils, diffusion phenomena in plasmas, viscoelasticity phenomena, optical systems, and so on.

1.5.3 Fractional Optimal Control Problem

In conclusion of this section, we consider the optimization problem [8]. In the simplest case the *fractional optimal control problem* (FOCP) is formulated as follows. Find the optimal control function $u(t),\ 0 < t < 1$, that minimizes the functional

$$J[u(\cdot)] = \int_0^1 F(x(t), u(t), t)dt$$

subject to the system constraints

$$ {}_0\mathsf{D}_t^{\alpha} x(t) = G(x(t), u(t), t). $$

Here F and G are assumed to be arbitrary x- and u-differentiable functions, $\alpha \in (0, 1)$ and the initial condition $x(0) = 0$. The modified functional, taking above mentioned constraints into account, is written as

$$ \bar{J}(u(\cdot)) = \int_0^1 \left\{ F(x, u, t) + \lambda \left[G(x, u, t) - {}_0\mathsf{D}_t^{\alpha} x \right] \right\} dt, $$

where λ is a Lagrange multiplier. Taking variation of this equation, we obtain

$$ \begin{aligned} \delta\bar{J}(u(\cdot)) = \int_0^1 \Bigg\{ & \frac{\partial F}{\partial x}\delta x + \frac{\partial F}{\partial u}\delta u + \delta\lambda \left[G(x, u, t) - {}_0\mathsf{D}_t^{\alpha} x \right] \\ & + \lambda \left[\frac{\partial G}{\partial x}\delta x + \frac{\partial G}{\partial u}\delta u - \delta \left({}_0\mathsf{D}_t^{\alpha} x \right) \right] \Bigg\} dt, \end{aligned} $$

where the variations δx, δu, and $\delta\lambda$ are consistent with a special condition. According to the formula of fractional integration by parts,

$$ \int_0^1 f(t)\, {}_0\mathsf{D}_t^{\alpha} g(t) dt = \int_0^1 g(t)\, {}_t^{\alpha}\mathsf{D}_1 f(t) dt = \int_0^1 g(t)\, {}_t\mathsf{D}_1^{\alpha} f(t) dt, $$

so one can rewrite the variation as

$$ \begin{aligned} \delta\bar{J}(u(\cdot)) = \int_0^1 \Bigg\{ & \left(\frac{\partial F}{\partial x} + \lambda\frac{\partial G}{\partial x} - {}_t\mathsf{D}_1^{\alpha}\lambda \right) \delta x + \left(\frac{\partial F}{\partial u} + \lambda\frac{\partial G}{\partial u} \right) \delta u \\ & + \left[G(x, u, t) - {}_0\mathsf{D}_t^{\alpha} x \right] \delta\lambda \Bigg\} dt, \end{aligned} $$

Equating the coefficients in front of δx, δu, and $\delta\lambda$ to zero, we find that the $\bar{J}$ reaches the minimum under conditions

$$ {}_0\mathsf{D}_t^{\alpha} x(t) = G(x, u, t), $$

$$ {}_t\mathsf{D}_1^{\alpha}\lambda = \frac{\partial F}{\partial x} + \lambda\frac{\partial G}{\partial x}, $$

$$ \frac{\partial F}{\partial u} + \lambda\frac{\partial G}{\partial u} = 0, \tag{1.31} $$

and

$$ x(0) = 0, \quad \lambda(1) = 0. $$

These equations, giving the necessary conditions for the optimality of the control function, are similar to the Euler-Lagrange equations for classical case except that the resulting differential equations contain the left and the right fractional derivatives.

O. P. Agrawal, whose work [8] is cited here, considers a case with the Lagrangian of quadratic form,

$$J(u(\cdot)) = \frac{1}{2}\int_0^1 \left[q(t)x^2(t) + r(t)u^2(t)\right]dt, \quad q(t) \geq 0, \quad r(t) > 0,$$

and the dynamics of the system is described by the fractional equations

$${}_0\mathsf{D}_t^{\alpha} x = a(t)x(t) + b(t)x(t)$$

and

$${}_t\mathsf{D}_1^{\alpha} \lambda = q(t)x(t) + a(t)\lambda(t) \tag{1.32}$$

complemented by the constraint condition

$$r(t)u(t) + b(t)\lambda(t) = 0. \tag{1.33}$$

As a result, we have

$${}_0\mathsf{D}_t^{\alpha} x(t) = a(t)x(t) - r^{-1}(t)b^2(t)\lambda(t). \tag{1.34}$$

The state $x(t)$ and the *costate* $\lambda(t)$ are obtained by solving Eqs. (1.32) and (1.34) subject to the terminal conditions given by Eq. (1.31). Once $\lambda(t)$ is known, the control variable $u(t)$ can be found using Eq. (1.33).

For a more detailed acquaintance of the reader with application of fractional methods to control problems, I recommend the book of Das [36] and other special articles (see References).

1.6 Analytical Fractional Dynamics

1.6.1 Euler-Lagrange Equation

It is well know that variational principles play a great role in modern theoretical physics. Introducing fractional derivatives allows to expand the area of applications of these principles by involving nonconservative systems. It is especially necessary within the problem of interpretation of dynamical variables in fractional differential mechanics [77, 78].

The simplest variational problem looks as follows [7]. Let $L(q, u, v, t)$ be a function with continuous first and second partial derivatives with respect to all its arguments, and let $\{q(t)\}$ be a set of functions on $[a, b]$ with continuous left and right derivatives of orders $\alpha \in (0, 1]$ and $\beta \in (0, 1]$ satisfying the boundary conditions

$$q(a) = q_a, \ q(b) = q_b.$$

It is asked to find such a function of this class that makes the function

$$S[q(\cdot)] = \int_a^b L(q(t), \ {}_aq_t^{(\alpha)}, \ {}_tq_b^{(\beta)}, t)dt$$

attain an extremum.

Denote the unknown function by $\overline{q}(t)$ and represent a trial function as

$$q(t) = \overline{q}(t) + \epsilon\eta(t), \tag{1.35}$$

where ϵ is a constant parameter corresponding to the chosen function, $\eta(t)$ is a function which meets the same requirements as $q(t)$ does but obeys the zeroth boundary conditions, $\eta(a) = \eta(b) = 0$. Performing a fractional differentiation of the function (1.35) and substituting the result into the functional, we then make it to be a function of the parameter ϵ:

$$S(\epsilon) = \int_a^b L(\overline{q} + \epsilon\eta, \ {}_a\overline{q}_t^{(\alpha)} + \epsilon \ {}_a\eta_t^{(\alpha)}, \ {}_t\overline{q}_b^{(\beta)} + \epsilon \ {}_t\eta_b^{(\beta)}, t)dt.$$

A necessary condition for an extremum of the function is

$$\frac{dS(\epsilon)}{d\epsilon} = \int_a^b \left[\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \ {}_aq_t^{(\alpha)}} \ {}_a\eta_t^{(\alpha)} + \frac{\partial L}{\partial \ {}_tq_b^{(\beta)}} \ {}_t\eta_b^{(\beta)} \right] dt = 0. \tag{1.36}$$

Integrating by parts, we transform the second and third terms in the right-hand side to the forms

$$\int_a^b \frac{\partial L}{\partial \ {}_aq_t^{(\alpha)}} \ {}_a\eta_t^{(\alpha)}dt = \int_a^b \left(\frac{\partial L}{\partial \ {}_aq_t^{(\alpha)}} \right) \ {}_a\mathsf{D}_t^{\alpha}\eta dt = \int_a^b \ {}_t\mathsf{D}_b^{\alpha} \left(\frac{\partial L}{\partial \ {}_aq_t^{(\alpha)}} \right) \eta dt$$

and

$$\int_a^b \frac{\partial L}{\partial \ {}_tq_b^{(\beta)}} \ {}_t\eta_b^{(\beta)}dt = \int_a^b \left(\frac{\partial L}{\partial \ {}_tq_b^{(\beta)}} \right) \ {}_t\mathsf{D}_b^{\beta}\eta dt = \int_a^b \ {}_a\mathsf{D}_t^{\beta} \left(\frac{\partial L}{\partial \ {}_tq_b^{(\beta)}} \right) \eta dt.$$

Substituting the resulting equation into Eq. (1.36),

$$\int_a^b \left[\frac{\partial L}{\partial q} + {}_t\mathsf{D}_b^{\alpha} \left(\frac{\partial L}{\partial\, {}_a q_t^{(\alpha)}} \right) + {}_a\mathsf{D}_t^{\beta} \left(\frac{\partial L}{\partial\, {}_t q_b^{(\beta)}} \right) \right] \eta dt = 0,$$

and taking into account that $\eta(t)$ is an arbitrary function of time, we come to a fractional analog of the Euler-Lagrange equation

$$\frac{\partial L}{\partial q} + {}_t\mathsf{D}_b^{\alpha} \left(\frac{\partial L}{\partial\, {}_a q_t^{(\alpha)}} \right) + {}_a\mathsf{D}_t^{\beta} \left(\frac{\partial L}{\partial\, {}_t q_b^{(\beta)}} \right) = 0. \tag{1.37}$$

When $\alpha = \beta = 1$, it turns into an ordinary Euler-Lagrange equation (remember that ${}_a\mathsf{D}_t^1 = d/dt$, but ${}_t\mathsf{D}_b^1 = -d/dt$).

Equation (1.37) is easily generalized for the case when the Lagrange function depends on several coordinate derivatives,

$$\frac{\partial L}{\partial q} + \sum_{j=1}^{n} {}_t\mathsf{D}_b^{\alpha_j} \left(\frac{\partial L}{\partial\, {}_a q_t^{(\alpha_j)}} \right) + \sum_{k=1}^{m} {}_a\mathsf{D}_t^{\beta_k} \left(\frac{\partial L}{\partial\, {}_t q_b^{(\beta_k)}} \right) = 0.$$

In case of several degrees of freedom

$$S[q_j(\cdot)] = \int_a^b L \left(q_1, \ldots, q_n;\ {}_a q_{1,t}^{(\alpha)}, \ldots,\ {}_a q_{n,t}^{(\alpha)},\ {}_t q_{1,b}^{(\beta)}, \ldots,\ {}_t q_{n,b}^{(\beta)} \right) dt.$$

The Euler-Lagrange equation has the form of the system of equations [7]:

$$\frac{\partial L}{\partial q_j} + {}_t\mathsf{D}_b^{\alpha} \left(\frac{\partial L}{\partial\, {}_a q_{j,t}^{(\alpha)}} \right) + {}_a\mathsf{D}_t^{\beta} \left(\frac{\partial L}{\partial\, {}_t q_{j,b}^{(\beta)}} \right) = 0, \quad j = 1, 2, \ldots, n. \tag{1.38}$$

1.6.2 Discrete System Hamiltonian

Following the work [76], we define generalized impulses for a one-dimensional system with the Lagrangian matching Eq. (1.38) via the relations

$$p_\alpha = \frac{\partial L}{\partial_a q_t^{(\alpha)}}, \quad p_\beta = \frac{\partial L}{\partial_t q_b^{(\beta)}} \tag{1.39}$$

and introduce a Hamiltonian

$$H = p_\alpha\, {}_a q_t^{(\alpha)} + p_\beta\, {}_t q_b^{(\beta)} - L.$$

Its total differential has the form

$$dH = p_\alpha d_a q_t^{(\alpha)} + {}_a q_t^{(\alpha)} dp_\alpha + p_\beta d_t q_b^{(\beta)} + {}_t q_b^{(\beta)} dp_\beta - \frac{\partial L}{\partial_a q_t^{(\alpha)}} {}_a q_t^{(\alpha)} - \frac{\partial L}{\partial_t q_b^{(\beta)}} {}_t q_b^{(\beta)} - \frac{\partial L}{\partial q} dq - \frac{\partial L}{\partial t} dt.$$

Substituting here momenta from Eq. (1.39) and using the Euler-Lagrange equation (1.38) from the previous section, we get

$$dH = {}_a q_t^{(\alpha)} dp_\alpha + {}_t q_b^{(\beta)} dp_\beta + \left[{}_a p_{\beta,t}^{(\beta)} + {}_t p_{\alpha,b}^{(\alpha)} \right] dq - \frac{\partial L}{\partial t} dt. \tag{1.40}$$

Comparison of this result with the Hamiltonian differential being a function of variables q, p_α, p_β, t,

$$dH = \frac{\partial H}{\partial q} dq + \frac{\partial H}{\partial p_\alpha} dp_\alpha + \frac{\partial H}{\partial p_\beta} dp_\beta + \frac{\partial H}{\partial t} dt,$$

leads to the next system of Hamiltonian equations

$${}_a q_t^{(\alpha)} = \frac{\partial H}{\partial p_\alpha}, \quad {}_t q_b^{(\beta)} = \frac{\partial H}{\partial p_\beta},$$

$${}_a p_{\beta,t}^{(\beta)} + {}_t p_{\alpha,b}^{(\alpha)} = \frac{\partial H}{\partial q}, \quad \frac{\partial L}{\partial t} = -\frac{\partial H}{\partial t},$$

describing a one-dimensional fracttional motion. As it is seen from Eq. (1.40), the fractional differential Hamoltonian is not an integral of motion even when Lagrangian doesn't explicitly depends on time.

1.6.3 Potentials of Non-concervative Forces

Involving fractional derivatives into the analytical mechanics opens up new possibilities for description of non-conservative forces in terms of potentials. Rabei et al. [75] have derived a general formula for fractional potentials U corresponding to arbitrary forces, both conservative and non-conservative, so that for all cases the Lagrangian reads as difference between kinetic and potential energies,

$$L = T - U.$$

Starting from the definition of conservative forces,

$$F(x) = -\frac{\partial U}{\partial x},$$

they introduced the generalized definition,

$$F(q^{(\beta)}) = (-1)^{\alpha+1}\, {}_a\mathsf{D}_t^{\alpha} \frac{\partial U}{\partial q^{(\alpha)}}, \quad \alpha,\ \beta \geq 0,$$

where $q^{(\beta)} = {}_b\mathsf{D}_t^{\beta} x(t), \quad \alpha = \beta/2 \in (m-1, m]$. This equation can be solved by applying the Laplace transform with respect to time. Indeed,

$$\mathscr{L}\{F(q^{(\beta)})\}(\lambda) = (-1)^{\alpha+1}\lambda^{\alpha}\mathscr{L}\{\partial U/\partial q^{(\alpha)}\}(\lambda)$$

(under condition that $\partial U/\partial q^{(\alpha)}$ and its $m-1$ derivatives vanish at $t = 0^+$),

$$\mathscr{L}\{\partial U/\partial q^{(\alpha)}\}(\lambda) = (-1)^{-\alpha-1}\lambda^{-\alpha}\mathscr{L}\{F(q^{(\beta)})\}(\lambda).$$

Taking the inverse transform for both sides of this equation,

$$\partial U/\partial q^{(\alpha)} = (-1)^{-\alpha-1}\mathscr{L}^{-1}\{\lambda^{-\alpha}\mathscr{L}\{F(q^{(\beta)})\}(\lambda)\}(q^{(\alpha)}),$$

and integrating both sides with respect to q_α leads to the result

$$U = (-1)^{-\alpha-1}\int \mathscr{L}^{-1}\{\lambda^{-\alpha}\mathscr{L}\{F(q^{(\beta)})\}(\lambda)\}(q^{(\alpha)})dq^{(\alpha)}.$$

Let, for example,

$$F(q^{(\beta)}) = -\gamma q^{(\beta)}, \quad \beta > 0,$$

then

$$\mathscr{L}[F(q^{(\beta)})\}(\lambda) = \gamma\lambda^{\beta}\widehat{q}(\lambda),$$

and

$$U = \frac{\gamma}{2(-1)^{\beta/2}}\left[q^{(\beta/2)}\right]^2.$$

1.6.4 Hamilton-Jacobi Mechanics

Having Hamiltonian of a system in the form

$$H(q, p_\alpha, p_\beta, t) = p_\alpha\, {}_aq_t^{(\alpha)} + p_\beta\, {}_tq_b^{(\beta)} - L(q,\ {}_aq_t^{(\alpha)}, {}_tq_b^{(\beta)}, t),$$

consider a canonical transformation with the generating function $S\left({}_aq_t^{(\alpha-1)}, {}_tq_b^{(\beta-1)}, P_\alpha, P_\beta, t\right)$ called the *Hamilton principal function*. Under this transformation, the Hamiltonian comes to the form

$$K(Q, P_\alpha, P_\beta, t) = P_\alpha \, {}_aQ_t^{(\alpha)} + P_\beta \, {}_tQ_b^{(\alpha)} - L'(Q, \, {}_aQ_t^{(\alpha)}, \, {}_tQ_b^{(\beta)}, t).$$

The old and new canonical variables satisfy the correspondent forms of the fractional Hamiltonian principle, that is

$$\delta \int_{t_1}^{t_2} \left(p_\alpha \, {}_aq_t^{(\alpha)} + p_\beta \, {}_tq_b^{(\beta)} - H \right) dt = 0$$

and

$$\delta \int_{t_1}^{t_2} \left(P_\alpha \, {}_aQ_t^{(\alpha)} + P_\beta \, {}_tQ_b^{(\beta)} - K \right) dt = 0$$

respectively. As in classical (integer-order) mechanics, both statements are satisfied if the integrand are connected via interrelation

$$p_\alpha \, {}_aq_t^{(\alpha)} + p_\beta \, {}_tq_b^{(\beta)} - H = P_\alpha \, {}_aQ_t^{(\alpha)} + P_\beta \, {}_tQ_b^{(\beta)} - K + dF/dt,$$

where

$$F = S \left({}_aq_t^{(\alpha-1)}, \, {}_tq_b^{(\beta-1)}, P_\alpha, P_\beta, t \right) - P_\alpha \, {}_aQ_t^{(\alpha-1)} - P_\beta \, {}_tQ_b^{(\beta-1)}.$$

Thus,

$$\begin{aligned} \frac{dF}{dt} &= \frac{dS}{dt} - \frac{dP_\alpha}{dt} \, {}_aQ_t^{(\alpha-1)} - P_\alpha \frac{d \, {}_aQ_t^{(\alpha-1)}}{dt} - \frac{dP_\beta}{dt} \, {}_tQ_b^{(\beta-1)} - P_\beta \frac{d \, {}_tQ_b^{(\beta-1)}}{dt} \\ &= \frac{dS}{dt} - \frac{dP_\alpha}{dt} \, {}_aQ_t^{(\alpha-1)} - P_\alpha \, {}_aQ_t^{(\alpha)} - \frac{dP_\beta}{dt} \, {}_tQ_b^{(\beta-1)} - P_\beta \, {}_tQ_b^{(\beta)}. \end{aligned}$$

Continuing to follow classical way of mathematical transformations, [74] get

$${}_aQ_t^{(\alpha-1)} = \frac{\partial S}{\partial P_a}, \quad {}_tQ_b^{(\beta-1)} = \frac{\partial S}{\partial P_b}, \quad p_\alpha = \frac{\partial S}{\partial \, {}_aq_t^{(\alpha-1)}}, \quad p_\beta = \frac{\partial S}{\partial \, {}_aq_t^{(\beta-1)}},$$

and

$$H + \frac{\partial S}{\partial t} = K.$$

Putting transformed Hamiltonian $K = 0$ makes the new variables $Q, \, P_\alpha, \, P_\beta$ constant and leads to the Hamilton-Jacobi equation

$$\frac{\partial S}{\partial t} = -H.$$

1.6.5 Hamiltonian Formalism for Field Theory

Rabei et al. [76] have applied the fractional approach to analytical dynamics of continuous medium represented by a classical field $\phi(x,t),\ a \leqslant t \leqslant b,\ c \leqslant x \leqslant d$. Its Lagrangian (Lagrangian density, to be more precisely, but we omit the word "density" for brevity) includes as arguments partial derivatives of ϕ as by coordinated so by time:

$$L = L(\phi,\ {}_a\phi_t^{(\alpha)},\ {}_t\phi_b^{(\beta)},\ {}_c\phi_x^{(\alpha)},\ {}_x\phi_d^{(\alpha)}).$$

Correspondent Hamiltonian has the form

$$H = \pi_\alpha\ {}_a\phi_t^{(\alpha)} + \pi_\beta\ {}_t\phi_b^{(\beta)} - L, \quad \pi_\alpha = \frac{\partial L}{\partial_a \phi_t^{(\alpha)}},\ \pi_\beta = \frac{\partial L}{\partial_t \phi_b^{(\beta)}}.$$

Taking a total differential, substituting impulse densities and using the Euler-Lagrange equation we come to the next system of canonical equations for a classical field of fractional differential type:

$${}_a\phi_t^{(\alpha)} = \frac{\partial H}{\partial \pi_\alpha}, \quad {}_t\phi_b^{(\beta)} = \frac{\partial H}{\partial \pi_\beta}, \quad \frac{\partial L}{\partial t} = -\frac{\partial H}{\partial t},$$

$$\frac{\partial L}{\partial_c \phi_x^{(\alpha)}} = -\frac{\partial H}{\partial_c \phi_x^{(\alpha)}}, \quad \frac{\partial L}{\partial_x \phi_d^{(\beta)}} = -\frac{\partial H}{\partial_x \phi_d^{(\beta)}},$$

$${}_a\pi_{\beta,t}^{(\beta)} + {}_t\pi_{\alpha,b}^{(\alpha)} = \frac{\partial H}{\partial \phi} + {}_c\mathsf{D}_x^\alpha \frac{\partial H}{\partial\ {}_x\phi_d^{(\alpha)}} + {}_x\mathsf{D}_d^\beta \frac{\partial H}{\partial\ {}_c\phi_x^{(\beta)}}.$$

Taking into the Lagrangian in the form

$$L = {}_a\phi_t^{(\alpha)}\ {}_a\phi_t^{*(\alpha)} - c_0^2\ {}_c\phi_x^\alpha\ {}_c\phi_x^{*\alpha} - \mu_0^2 c_0^2 \phi\phi^*,$$

and having found the canonically conjugated impulses

$$\pi_a = {}_a\phi_t^{*(\alpha)}, \quad \pi_\beta = 0, \quad \pi_a^* = {}_a\phi_t^{(\alpha)},\ \pi_\beta^* = 0,$$

we arrive at the Hamiltonian

$$H = \pi_\alpha\ {}_a\phi_t^{(\alpha)} + \pi_\alpha^*\ {}_a\phi_t^{(\alpha)} - L = \pi_a \pi_a^* + c_0^2\ {}_a\phi_x^{(\alpha)}\ {}_a\phi_x^{*(\alpha)} + \mu_0^2 c_0^2 \phi\phi^*.$$

Its insertion into the canonical equations leads to the equation of motion

$$\pi_\alpha^* = {}_a\phi_t^{(\alpha)}, \quad \frac{\partial H}{\partial \pi_\beta} = 0, \quad \pi_\alpha = {}_a\phi_t^{*(\alpha)}, \quad \frac{\partial H}{\partial \pi_\beta^*} = 0,$$

$$ {}_t\mathsf{D}_b^{\alpha}\ {}_a\phi_t^{*(\alpha)} - c_0^2\ {}_x\mathsf{D}_d^{\beta}\ {}_a\phi_x^{*(\beta)} = \mu_0^2 c_0^2 \phi^*, $$

and

$$ {}_t\mathsf{D}_b^{\alpha}\ {}_a\phi_t^{(\alpha)} - c_0^2\ {}_x\mathsf{D}_d^{\beta}\ {}_a\phi_x^{(\beta)} = \mu_0^2 c_0^2 \phi. $$

Last two equations can be considered as fractional generalizations of the *Klein-Gordon equations* [76].

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Chapter 2
Continuum Mechanics



2.1 Classical Hydrodynamics

2.1.1 A Simple Hydraulic Problem

Considering the motion of rigid bodies in an ordinary (Newtonian) viscous fluid, we saw in the preceding chapter, how naturally fractional derivatives (semiderivatives) arise in framework of classical hydrodynamics. Here, we bring four more such examples. One of them is linked with the design of a *weir notch* [92, 98, 113]. A weir is a small overflow dam used to alter the flow characteristics of a river or stream. In most cases weirs take the form of a barrier across the river that causes water to pool behind the structure, but allows water to split through a weir notch over the dam. Weirs are commonly used to alter the flow regime of the river, prevent flooding, measure discharge and to help render a river navigable. Schematically, the dam element is depicted in Fig. 2.1.

The problem is to find the shape $y = y(x)$ of a notch cut into the weir such that the rate of flow of the water through the notch, Q is a specified function of the hight h of the opening. In order to solve it, we write the Bernoulli theorem connecting the water characteristics at two points 1 and 2 taken on the same "tube of flow" on the level y:

$$\frac{p_1}{\rho} + gy + \frac{u_1^2}{2} = \frac{p_2}{\rho} + gy + \frac{u_2^2}{2}.$$

Because

$$p_2 = p_1 - \rho g(h - y) = p_{\text{atm}}$$

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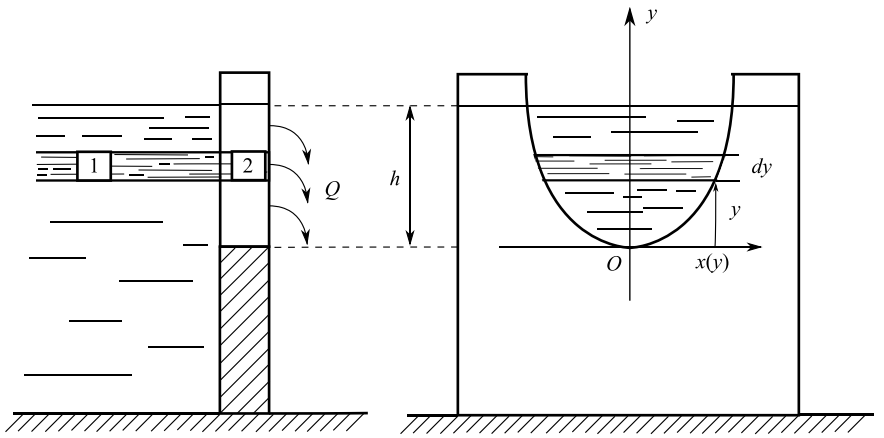


Fig. 2.1 A cross section of the dam (left panel) and its front view (right panel)

and

$$u_1 \approx 0,$$

we obtain $u \equiv u_2$ as the following function of y:

$$u(y) = \sqrt{2g}(h - y)^{1/2}.$$

Assuming the shape of the notch to be symmetrical about the y-axis, we represent the total flow rate Q (say, in cubic meters per second) as a function of the opening height in the form of integral

$$Q(h) = q \int_0^h x(y)u(y)dy = 2\sqrt{2g} \int_0^h x(y)(h - y)^{1/2}dy = \sqrt{2\pi g}\ {}_0\mathsf{D}_h^{-3/2}x(h).$$

Consequently, the desired function $x(y)$ characterizing the notch shape is given by expression

$$x(y) = \frac{1}{\sqrt{2\pi g}}{}_0\mathsf{D}_y^{3/2}Q(y).$$

2.1.2 Liquid Drop Oscillations

The second example we take from the work [82], where the fractional equation describes oscillations of a drop suspended in the field of an acoustic wave. The authors notice that neglecting the terms related to the motion's prehistory for the sake of simplification of the problem leads to discrepancy between theoretical results and experimental data. Taking into account the prehistory effect and passing from

the velocity $V(t)$ to the acceleration $W(t) = \dot{V}(t)$, Kutz et al. reduce the integro-differential equation of the drop motion to the integral equation

$$W(t) + \alpha^2 \int_0^t W(\tau)d\tau + \frac{2\beta}{\sqrt{\pi}} \int_0^t \frac{W(\tau)d\tau}{\sqrt{t-\tau}} = f(t).$$

On assumption that the gas velocity is

$$v(t) = v_0 + A\omega \sin \omega t,$$

the right-hand side can be written as

$$f(t) = \alpha^2(v_0 + A\omega \sin \omega t) + A\omega^2 \cos \omega t + \frac{2\beta}{\sqrt{\pi}} \int_0^t \frac{A\omega^2 \cos \omega\tau d\tau}{\sqrt{t-\tau}}.$$

Constants α and β are connected with the radius a and the density ρ of the particle via relations

$$\alpha^2 = \frac{9\eta}{2a^2(\rho + \rho_0/2)}, \quad \beta = \frac{9\sqrt{\eta\rho_0}}{4a(\rho + \rho_0/2)},$$

where ρ_0 and η are the density and the viscosity of the gas respectively.

The Laplace transform

$$\hat{W}(\lambda) = \int_0^\infty e^{-\lambda t} W(t)dt$$

takes the following form

$$\hat{W}(\lambda) = \frac{\lambda \hat{f}(\lambda)}{\lambda + 2\beta\sqrt{\lambda} + \alpha^2} = A(\lambda)\hat{f}(\lambda),$$

where

$$A(\lambda) = \frac{\lambda_1 + \alpha^2}{\lambda_1 - \lambda_2} \frac{\lambda}{\lambda - \lambda_1} - \frac{\lambda + \alpha^2}{\lambda - \lambda_1} \frac{\lambda}{\lambda - \lambda_1} - \frac{2\beta}{\lambda_1 - \lambda_2} \left(\frac{\lambda_1\sqrt{\lambda}}{\lambda - \lambda_1} - \frac{\lambda_2\sqrt{\lambda}}{\lambda - \lambda_2} \right)$$

$$\lambda_{1,2} = [\beta \pm \sqrt{\beta^2 - \alpha^2}]^2.$$

The final expression for the velocity of the particle, being motionless at the initial instant has the form

$$V(t) = \int_0^t W(\tau)d\tau = \left[\frac{1}{2\sqrt{\beta^2 - \alpha^2}} \left(\frac{e^{\lambda_1 t} \mathrm{erf}(\sqrt{\lambda_1 t})}{\beta + \sqrt{\beta^2 - \alpha^2}} - \frac{e^{\lambda_2 t} \mathrm{erf}(\sqrt{\lambda_2 t})}{\beta - \sqrt{\beta^2 - \alpha^2}} \right) + \frac{1}{\alpha^2} \right] f(t).$$

The drop preserves its density due to the internal molecular forces. Internal pressure p in a drop is proportional to the surface strain σ:

$$p = 2\sigma/a.$$

Owing to resistive forces a drop crumbles while moving through the gaseous medium. Supposing that the resistive force acts only on a half of the surface and its numerical value is defined by maximal values of overpressure, it follows from the condition of equality of forces acting on the drop that the critical radius $a_{\rm cr}$ is given with the formula

$$a_{\rm cr} = \sqrt[4]{\frac{27}{8}\frac{\eta\sigma}{A\omega^2\rho^2}}.$$

This estimation is in a rough agreement with experimental data. Thus, at frequency 12 kHz and sound pressure 154 dB, this formula produces $a_{\rm cr} = 40$ μm, whereas experiments give 55 μm.

2.1.3 Sound Radiation

Let us look at Sect. 74 of the book (Landau and Lifshitz, 1981), where the authors deal with sound radiation by an infinite round cylinder with the variable cross-section area $S(t)$. At a distance r from the axis much greater than the cylinder diameter and much smaller than wave-length, the wave potential in homogeneous medium is determined as

$$\varphi(r,t) = -\frac{c}{2\pi}\int_{-\infty}^{t-r/c}\frac{\dot S(t')dt'}{\sqrt{c^2(t-t')^2 - r^2}},$$

where c is the sound speed and $\dot S = dS/dt$. The region of values $t - t' \sim r/c$ plays a general role in the wave domain. That is why one can accept

$$(t-t')^2 - \frac{r^2}{c^2} \simeq 2\frac{r}{c}\left(t - t' - \frac{r}{c}\right).$$

As a result, we obtain the potential

$$\varphi(r,t) = -\frac{1}{2\pi}\sqrt{\frac{c}{2r}}\int_{-\infty}^{t-r/c}\frac{\dot S(t')dt'}{\sqrt{t-r/c-t'}} = -\frac{1}{2}\sqrt{\frac{c}{2\pi r}}\,{}_{-\infty}^{1/2}\mathsf{D}S(t-r/c)$$

and the speed

$$v(r,t) = \frac{\partial\varphi}{\partial r} \simeq \frac{1}{2\pi\sqrt{2cr}}\int_{-\infty}^{t-r/c}\frac{\ddot S(t')dt'}{\sqrt{t-r/c-t'}} = \frac{1}{2\sqrt{2\pi cr}}\,{}_{-\infty}^{3/2}\mathsf{D}S(t-r/c).$$

2.1.4 Deep Water Waves

Matsuuchi [91] arrives at a fractional differential equation when investigating weakly nonlinear gravity waves traveling on deep water. When the flow is irrotational, the velocity potential ϕ is governed by the two-dimensional Laplace equation in Cartesian coordinates (x, z):

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0, \tag{2.1}$$

where x is measured horizontally to the right and z vertically upward. At the free surface, given by infinitely small $z = \zeta(x, t)$, the velocity potential must satisfy the boundary condition

$$\left(\frac{\partial \phi}{\partial z} + \frac{1}{g} \frac{\partial^2 \phi}{\partial t^2} \right)_{z=0} = 0 \tag{2.2}$$

with g being the acceleration due to gravity. For the perturbed quantities ϕ_1 and ζ_1 to ϕ and ζ, respectively, Eqs. (2.1) and (2.2) yield

$$\zeta_1(x, t) = a e^{i(kx - \omega t)}$$

and

$$\phi_1(x, z, t) = -i(\omega/k) a e^{kz} e^{i(kx - \omega t)}.$$

In the above expression, a is the complex amplitude, k the wave number, and ω the frequency. The latter two quantities must satisfy the dispersion relation

$$\omega^2 = gk. \tag{2.3}$$

Any waves of arbitrary shape can be expressed by superposition of the above monochromatic waves as

$$\zeta_1(x, t) = \int_{-\infty}^{\infty} a(k) e^{i(kx - \omega t)} dk. \tag{2.4}$$

Thus far we have assumed that k is positive: for the purpose of using Eq. (2.4), k should be permitted to have a negative sign. Confining ourselves to waves traveling only in the positive x-direction, instead of the dispersion relation (2.3) it should be written for both signs of k,

$$\omega = |gk|^{1/2} \text{sign } k. \tag{2.5}$$

Similarly, for the perturbed potential, we obtain

$$\phi_1(x, z, t) = -i \int_{-\infty}^{\infty} \frac{\omega}{|k|} a(k) e^{|k|z} e^{i(kx - \omega t)} dk$$

$$= -i \int_{-\infty}^{\infty} \sqrt{g} a(k) |k|^{-1/2} \text{sign}\, k \; e^{|k|z} e^{i(kx-\omega t)} dk. \tag{2.6}$$

Differentiating Eq. (2.4) with respect to time and using relation (2.5), we have

$$\frac{\partial \zeta_1}{\partial t} + i \int_{-\infty}^{\infty} \sqrt{g} |k|^{1/2} \text{sign}\, k \; a(k) e^{i(kx-\omega t)} dk = 0. \tag{2.7}$$

Taking into account formulas

$$\int_{-\infty}^{\infty} \widetilde{f}(k) |k|^{-\alpha-1} e^{ikx} dk = \frac{1}{2\cos[\pi(\alpha+1)/2]} \left({}_{-\infty}\mathsf{D}_x^{-\alpha-1} + {}_x\,\mathsf{D}_{\infty}^{-\alpha-1} \right) f(x), \tag{2.8}$$

$$\int_{-\infty}^{\infty} \widetilde{f}(k) |k|^{-\alpha-1} \text{sign}\, k e^{ikx} dk = \frac{i}{2\sin[\pi(\alpha+1)/2]} \left({}_{-\infty}\mathsf{D}_x^{-\alpha-1} - {}_x\,\mathsf{D}_{\infty}^{-\alpha-1} \right) f(x), \tag{2.9}$$

we arrive at the Riemann-Liouville integral and the Weyl integral of fractional order respectively. With the use of Eqs. (2.9), (2.7) can be rewritten as

$$\frac{\partial \zeta_1}{\partial t} + \sqrt{\frac{g}{2}} \left({}_{-\infty}\mathsf{D}_x^{1/2} - {}_x\mathsf{D}_{\infty}^{1/2} \right) \zeta_1(x,t) = 0. \tag{2.10}$$

Equation (2.10) including the fractional derivatives of order 1/2 can describe any waves of infinitesimal amplitude traveling in the positive x-direction.

Under the assumption of weak nonlinearity, K. Matsuuchi has derived an exact equation governing one-dimensional waves of arbitrary shape traveling on deep water under the assumption of weak nonlinearity.

2.2 Turbulent Motion

2.2.1 Kolmogorov's Model of Turbulence

According to Kolmogorov's model of homogeneous isotropic turbulence (for details see [53]), in the inertial domain, a velocity of relative motion of turbulent medium points, located at a distance r from each other,

$$\Delta \mathbf{u} = \mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x}), \tag{2.11}$$

is considered as a random constant, and mean square of which increases with distance proportionally to $r^{2/3}$:

$$\langle (\Delta \mathbf{u})^2 \rangle = C_0 \varepsilon^{2/3} r^{2/3}, \quad l < r < L. \tag{2.12}$$

Here C_0 is a dimensionless variable, ε is a rate of dissipation of kinetic energy per a mass unit, $l = (\varepsilon^3/v)^{1/4}$ is the Kolmogorov length. Corresponding to this model energy spectrum is given by *the five-thirds power law*

$$E(k) = C\varepsilon^{2/3}k^{-5/3},$$

where k is the wave number, and C is the Kolmogorov constant.

From another site, it is known that relative diffusion (diffusion of distance vector between two indicated particles) satisfies the *Richardson law* (the three-halves power law)

$$\langle r^2 \rangle = C_1 \varepsilon (\Delta t)^3, \tag{2.13}$$

i.e. essentially faster with respect to normal diffusion. If one implies in Eq. (2.11) the difference of point velocities, the variable r becomes random and formula (2.12) can be interpreted as

$$\langle (\Delta \mathbf{u})^2 \rangle = C_0 \varepsilon^{2/3} \langle r^2 \rangle^{1/3}.$$

Substituting here the mean square of relative distance from (2.13), we find

$$\langle (\Delta \mathbf{u})^2 \rangle = C_2 \varepsilon \Delta t, \;\; C_2 = C_0 C_1^{1/3}. \tag{2.14}$$

2.2.2 *From Kolmogorov's Hypothesis to the Space-Fractional Equation*

Fractional derivatives, namely fractional Laplacians entered the turbulence through the problem of diffusion of a passive scalar[1] in a turbulent environment, called for brevity, *turbulent diffusion* (TD). It is caused by influence on the particle of vortices of different sizes that exist in turbulent medium. The distance between two test particles may change significantly over short time by the action vortex of size comparable with this distance. If these particles are farther from each other, the greater the size of the vortices which spread them from each other, the greater the speed increasing the distance l between them. In frames of the classical theory of diffusion, such effect can be achieved by introducing dependence of the diffusion coefficient D on relative coordinates, i.e. distance, $D = D(r)$. This approach has been used in the pioneering work of Richardson [112] who wrote the equation for the density distribution $p(r, t)$ of random distance between a pair of particles of impurities that were at the moment $t = 0$ in one point. This equation has the form

[1] That is a particle that does not modify the velocity field of the fluid while being transported.

$$\frac{\partial p}{\partial t} = \frac{\partial}{\partial r}\left[D\,(r)\,\frac{\partial p}{\partial r}\right]$$

with the diffusion coefficient $D\,(r) \propto r^{4/3}$, corresponding to increase of the diffusion packet width $\Delta(t)$ according to the law

$$\Delta(t) \propto t^{3/2},$$

that sufficiently differ from normal diffusion law $\Delta(t) \propto t^{1/2}$. Theoretically such behavior has found support in known papers of Kolmogorov [78] and Obukhov [101, 102] as a consequence of the self-similarity hypothesis for locally isotropic turbulence, determined by the single dimensional parameter, which is the rate of dissipation of turbulent energy ε.

The relations (2.13) and (2.14) show that when behavior of relative velocity of two particles is compatible with the Brownian model of normal diffusion in a homogeneous stationary medium (laboratory studies in [69] confirm this), the distance between them behave differently, its mean square is proportional to the third power. If we say about linear size r of cloud diffusing in turbulent medium, it is proportional to $t^{3/2}$, whereas in the normal case, this size increases in proportion to $t^{1/2}$: TD is an enhanced diffusion (superdiffusion).

Qualitatively, the above results agree with experiment, but the fact that in a homogeneous environment, the average diffusion coefficient has to depend on the distance between particles, created some problems in the physical interpretation of the results.

Monin [95, 96] found the method to combine the rapid character of turbulent diffusion with constancy of the coefficient characterizing the medium. With his papers, applications of non-Gaussian stable distributions and equations with derivatives of fractional orders begin in the theory of turbulent diffusion (however he didn't use this terminology). In form slightly changed with respect to the original work, this method justified as follows.

Let's take as a basis a Markovian jump process approximating the turbulent diffusion of a particle located at the point $\mathbf{x} = 0$ at the moment $t = 0$. Random time between two successful jumps is distributed with exponential density

$$q(t) = \mu e^{-\mu t}, \quad t > 0.$$

The probability distribution function $p(\mathbf{x}, t)$ of random coordinate of the particle satisfies the Einstein-Smoluchowski equation

$$\frac{\partial p(\mathbf{x}, t)}{\partial t} = \mu \int K(\mathbf{x} - \mathbf{x}')\left[p(\mathbf{x}', t) - p(\mathbf{x}, t)\right] d\mathbf{x}',$$

with the initial condition $p(\mathbf{x}, 0) = \delta(\mathbf{x})$, that can be included in the right side of the equation with the condition $p(\mathbf{x}, t) = 0$ at $t < 0$:

$$\frac{\partial p(\mathbf{x},t)}{\partial t} = \mu \int K(\mathbf{x}-\mathbf{x}')\left[p(\mathbf{x}',t) - p(\mathbf{x},t)\right] d\mathbf{x}' + \delta(\mathbf{x})\delta(t).$$

Corresponding equation for the Fourier transform, i.e. for the characteristic function (CF)

$$\widetilde{p}(\mathbf{k},t) = \int e^{i\mathbf{k}\mathbf{x}} p(\mathbf{x},t) d\mathbf{x}$$

has the solution

$$\widetilde{p}(\mathbf{k},t) = \exp\left\{-\mu\left[1 - \widetilde{K}(\mathbf{k})\right]t\right\}.$$

Introducing new variables through the relations $\mathbf{z} = \mathbf{k}t^{1/\alpha}$ è $\widetilde{P}(\mathbf{z},t) = \widetilde{p}(\mathbf{z}t^{-1/\alpha},t)$, one can certain that asymptotic self-similarity of the solution, i.e. existence of a non-zero limit, $\widetilde{P}(\mathbf{z},\infty) = \lim\limits_{t\to\infty} \widetilde{P}(\mathbf{z},t)$ requires fulfillment of the conditions

$$|1 - \widetilde{K}(\mathbf{k})| \propto |\mathbf{k}|^{\alpha},\ \mathbf{k} \to 0.$$

$\widetilde{K}(\mathbf{k})$ will be the characteristic function at $\alpha \in (0,2]$ (out of these limits, the one of the compulsory conditions for CF $\widetilde{K}(0) = 1$ and $\int e^{-i\mathbf{k}\mathbf{x}} \widetilde{K}(\mathbf{k}) d\mathbf{k} \geqslant 0$ is broken).

In the case of isotropic diffusion

$$1 - \widetilde{K}(\mathbf{k}) \sim B|\mathbf{k}|^{\alpha},\ \mathbf{k} \to 0,$$

and the main asymptotical term

$$\widetilde{f}(\mathbf{k},t) \equiv \widetilde{P}(\mathbf{k}t^{\nu},\infty)$$

satisfies the equation

$$\frac{d\widetilde{f}(\mathbf{k},t)}{dt} = -B|\mathbf{k}|^{\alpha}\widetilde{f}(\mathbf{k},t) + \delta(t).$$

Its solution

$$\widetilde{f}(\mathbf{k},t) = \exp\{-B|\mathbf{k}|^{\alpha}t\}$$

represents the characteristic function of isotropic stable distribution with the exponent α.

At $\alpha = 2$, the inverse transformation of this equation has the form of a usual diffusion equation ($-|\mathbf{k}|^2$ is the Fourier transformation of the Laplace operator Δ_3), at $\alpha \neq 2$, the Laplacian of fractional order arises

$$\frac{\partial f(\mathbf{x},t)}{\partial t} = -B(-\Delta)^{\alpha/2} f(\mathbf{x},t) + \delta(\mathbf{x})\delta(t). \tag{2.15}$$

Its solution is expressed through three-dimensional density of isotropic stable distribution (the Lévy-Feldgheim distribution) $g_3(\mathbf{x};\alpha)$ by the relation

$$f(\mathbf{x},t) = (Bt)^{-3/\alpha} g_3((Bt)^{-1/\alpha}\mathbf{x};\alpha), \quad \alpha \in (0,2].$$

The variance of this distribution:

$$\langle \mathbf{X}^2 \rangle = \infty,$$

and one should choose another measure of width for characterization of spreading of diffusion packet with time, for example, the width on a height h, Δ_h, or radius of the sphere R_p, containing specified probability p. These measures are proportional to each other and grow with a time as $t^{1/\alpha}$, which is in accordance with the Richardson law at $\alpha = 2/3$ and with the normal diffusion case when $\alpha = 2$.[2]

The physical interpretation of a role of fractional power of the Laplacian in these equations reduces to the fact that random particle trajectories corresponding to these processes are not continous in contrast to the Brownian ones ($\alpha = 2$). They represent clots (clusters) chaotically distributed in the space, i.e. they are random fractals. A particle moving in such way "marks time" in volumes of relatively small sizes, then it suddenly "flies away" on large distance and becomes "to mark time" there. Such type of motion is called *Lévy flights* (see also the discussion in [31]).

2.2.3 From Boltzmann's Equation to the Time-Fractional Telegraph One

The transport problem with account of the finite free motion speed was discussed in Sect. 3.6. For the sake of simplicity, we consider here a one-dimensional version of the process with exponential distribution of free motion times without trapping. In this case, instead of general Eqs. (3.60)–(3.62) we have

$$\frac{df_+}{dt} \equiv \frac{\partial f_+}{\partial t} + V\frac{\partial f_+}{\partial x} = \mu(f_- - f_+) \tag{2.16}$$

and

$$\frac{df_-}{dt} \equiv \frac{\partial f_-}{\partial t} - V\frac{\partial f_-}{\partial x} = \mu(f_+ - f_-), \tag{2.17}$$

where $f_+(x,t)$ and $f_-(x,t)$ are concentrations of particles moving to the right and to the left respectively, μ is the rate of direction changes, V is a constant speed of

[2] Thereby, to classify different types of diffusion with respect to rates of diffusion packet spreading one should use not a variance that does not always exist, but the one of measures indicated above. They always exist and can be chosen in such way to coincide with the variance at the limit $\alpha \to 2$. This concerns of $\langle r^2 \rangle$, $\langle (\Delta \mathbf{u})^2 \rangle$ and other characteristics.

the walker. This is nothing but a special case of the one-velocity Boltzmann equation (see Uchaikin. 1998). Summing these equations with the use of designations

$$f = f_+ - f_-,$$

for the total concentration and

$$j = V(f_+ - f_-)$$

for the particles current along the x-axis, we obtain

$$\frac{\partial f}{\partial t} = -\frac{\partial j}{\partial x}. \tag{2.18}$$

This is merely the particles conservation law. After complementing it with the phenomenological Fick law

$$j(x,t) = -K\frac{\partial f(x,t)}{\partial x}, \tag{2.19}$$

we obtain an ordinary diffusion equation

$$\frac{\partial f(x,t)}{\partial t} = K\frac{\partial^2 f(x,t)}{\partial x^2}. \tag{2.20}$$

However, this model allows us to get the current-concentration relation in a strict way, namely, by subtracting Eq. (2.17) from Eq. (2.16), and the result varies from Eq. (2.19):

$$j = -\frac{V^2}{\mu}\frac{\partial f}{\partial x} - \frac{1}{\mu}\frac{\partial j}{\partial t}.$$

In contrast to Fick's law, this expression known as the *Maxwell-Cattaneo equation* takes into account a finite speed of moving particles. In order to see this, we transform this equation as follows:

$$\frac{\partial j(x,t)}{\partial t} + \mu j(x,t) = \mu j_0(x,t), \quad j_0(x,t) = -\frac{V^2}{\mu}\frac{\partial f(x,t)}{\partial x},$$

$$j(x,t) = \int_0^t j_0(x,t-t')\psi_1(t')dt' = -\left\langle \frac{V^2}{\mu}\frac{\partial f(x,t-T)}{\partial x}\right\rangle_1, \tag{2.21}$$

where

$$\psi_1(t) = \mu e^{-\mu t}$$

and the angular bracket with the subscript 1 denotes averaging over the retardation time T distributed with this probability density function. Observe that the pair of Eqs. (2.18) and (2.21) strictly describes the one-dimensional walk with a finite speed V.

Now, we replace exponential density $\psi_1(t)$ by its fractional counterpart $\psi_\nu(t)$ in these equations (see for details Sect. 3.5). As a result, we obtain an integro-differential generalization of the diffusion equation:

$$\frac{\partial f_\nu(x,t)}{\partial t} = \frac{V^2}{\mu}\int_0^t \frac{\partial^2 f_\nu(x,t-t')}{\partial x^2}\psi_\nu(t')dt'. \tag{2.22}$$

Recall that

$$\psi_0(t) = \delta(t),$$

$$\psi_1(t) = \mu e^{-\mu t},$$

and

$$\psi_\nu(t) = \mu x^{\nu-1} E_{\nu,\nu}\left(-\mu x^\nu\right)$$

(see Sect. 3.5.3). Inserting ψ_0 in Eq. (2.22) returns us to the ordinary diffusion equation

$$\frac{\partial f_0(x,t)}{\partial t} = \frac{V^2}{\nu}\frac{\partial^2 f_1(x,t)}{\partial x^2}.$$

Using the density ψ_1 and changing integration variable, we bring the integral in the right side of Eq. (2.22) to the form

$$\int_0^t \frac{\partial^2 f_1(x,t-t')}{\partial x^2}\psi_1(t')dt' = e^{-\mu t}\int_0^t \frac{\partial^2 f_1(x,t')}{\partial x^2}\mu e^{\mu t'}dt'.$$

Inserting it into Eq. (2.22), differentiating the result with respect to time and using the evident equality

$$e^{-\mu t}\int_0^t \frac{\partial^2 f_1(x,t')}{\partial x^2}\mu e^{\mu t'}dt' = \frac{\mu}{v^2}\frac{\partial f(x,t)}{\partial t},$$

we arrive at the *telegraph equation*

$$\frac{\partial^2 f_1}{\partial t^2} + \mu\frac{\partial f_1}{\partial t} = V^2\frac{\partial^2 f_1}{\partial x^2}. \tag{2.23}$$

In the general case, we apply the Laplace transform to Eq. (2.22). Taking into account that

$$\widehat{\psi}_\nu(\lambda) \equiv \int_0^\infty e^{-\lambda t}\psi_\nu(t)dt = \frac{\mu}{\mu+\lambda^\nu}, \quad 0 < \nu \leqslant 1,$$

we obtain

$$\lambda^{\nu+1}\widehat{f_\nu}(x,\lambda)+\mu[\lambda\widehat{f_\nu}(x,\lambda)-f_\nu(x,0)]=V^2\frac{\partial^2\widehat{f_\nu}(x,\lambda)}{\partial x^2}+\lambda^\nu f_\nu(x,0).$$

The inverse Laplace transform yields a fractional generalization of the telegraph equation

$${}_0\mathsf{D}_t^{\nu+1}f_\nu(x,t)+\mu\frac{\partial f_\nu(x,t)}{\partial t}=V^2\frac{\partial^2 f_\nu(x,t)}{\partial x^2}+\delta_\nu(t)f_\nu(x,0),\quad 0<\nu<1, \tag{2.24}$$

where

$$\delta_\nu(t)\equiv{}_0\mathsf{D}_t^\nu 1_+(t)=\frac{t^{-\nu}}{\Gamma(1-\nu)}$$

is the fractional counterpart of delta-function. Note that Eq. (2.23) can also be represented in the form of inhomogeneous equation, containing an initial condition in its right-hand side:

$$\frac{\partial^2 f_1(x,t)}{\partial t^2}+\mu\frac{\partial f_1(x,t)}{\partial t}=V^2\frac{\partial^2 f_1(x,t)}{\partial x^2}+\delta(t)f_1(x,0). \tag{2.25}$$

Solutions of the latter equation and its modifications are investigated by many authors (including our works [145–149]). It is worth to notice that the solution of Eq. (10) differs from zero if only $|x|\leqslant Vt$. Both equations can be extended to the three-dimensional case by the replacement of operators $\partial^2 f/\partial x^2\mapsto\Delta$ in their right-hand sides:

$${}_0\mathsf{D}_t^{\nu+1}f_\nu(\mathbf{x},t)+\mu\frac{\partial f_\nu(\mathbf{x},t)}{\partial t}=V^2\Delta f_\nu(\mathbf{x},t)+\delta_\nu(t)f_\nu(\mathbf{x},0),\quad 0<\nu<1. \tag{2.26}$$

2.2.4 Turbulent Diffusion in a Viscous Fluid

In the preceding subsection, we have adjusted the equation in accordance with the principle of a limited speed of the particles. However, another problem remains open: numerical simulations show that the walker trajectory contains long rectangular segments, which are not typical for tracers trajectories in a turbulent flow. The situation may be improved by involving viscosity forces. The relationships (2.12) and (2.13) lying in the base of the fractional differential equation of turbulent diffusion (2.15) relate to homogeneous turbulence at high enough (formally infinite) Reynolds numbers and reflect statistical self-similarity typical for accelerative interval of frequencies. Due to this the Eq. (2.15) represents "coarse-grained" description of diffusion process. In the domain of small distances, deviations from the model of Lévy motion conditioned by influence of viscosity take place. Laboratory studies

and numerical simulation [106, 137] testify the influence of viscosity. Inclusion of the component with viscosity into the equation[3]

$$\frac{\partial f}{\partial t} + B(-\Delta)^{1/3} f(\mathbf{x}, t) - \eta \Delta f(\mathbf{x}, t) = 0 \tag{2.27}$$

takes into account molecular diffusion responsible for intermittency at finite Reynolds numbers.

Supplementing it with the initial condition

$$f(\mathbf{x}, 0) = \delta(\mathbf{x}),$$

we perform Fourier transformation on space variable

$$\frac{d\tilde{f}}{dt} + [B|\mathbf{k}|^{2/3} + \eta k^2]\tilde{f} = 0, \quad \tilde{f}(\mathbf{k}, 0) = 1. \tag{2.28}$$

The solution of the transformed equation

$$\tilde{f}(\mathbf{k}, t) = e^{-B|\mathbf{k}|^{2/3}t} \cdot e^{-\eta k^2 t}$$

has the form of product of two characteristic function, the Lévy-Feldgheim function with $\alpha = 2/3$ and the Gaussian one. At small k (large distances) the component corresponding to turbulent superdiffusion with $\alpha = 2/3$ will play the main role in Eq. (2.29a), at large k (small distances) the Gaussian component describing molecular diffusion is main. A general solution has the form of convolution of two distributions

$$\psi(\mathbf{x}, t) = (Kt)^{-9/2} g_3\left((Kt)^{-3/2}\mathbf{x}; 2/3\right)$$

and

$$\chi(\mathbf{x}, t) = (\eta t)^{-3/2} g_3\left((\eta t)^{-1/2}\mathbf{x}; 2\right),$$

$$f(\mathbf{x}, t) = \int_{\mathbb{R}^3} \psi(\mathbf{x} - \mathbf{x}', t)\chi(\mathbf{x}', t)d\mathbf{x}'.$$

Qualitatively similar estimation of the role of molecular diffusion in turbulence is obtained by Kraichnan [80] in frames of formalism that is sufficiently more complicated in mathematical aspects. The present approach developed in [28, 29] based on the use of fractional derivatives is clearer in stochastic sense and easier in computing aspects.

[3] Concentrating attention of the reader on the viscosity influence, we omit here the term with the time-derivative of fractional order discussed above. It may be included at any moment.

2.2.5 Navier-Stokes Equation

In essence, the Kolmogorov five-thirds power law characterizing statistical self-similarity of turbulent motion at small scales is based on the locally homogeneous isotropic model. To a certain extent this law is confirmed by numerous experiments and numerical simulation [97] at quite high Reynolds numbers. However experiments [106] performed with use of high-speed optical technics found out that statistics of Lagrangian accelerations is characterized by distributions with long tails of power law type. Non-Gaussian character of these distributions contradicts to the assumption about Gaussian character of the turbulence shared by many authors (see [133]).

The Gaussian distribution belongs to the class of stable distributions occupying the central position in modern probability theory due to their property of being unique probable limits of distributions of normalized sums of independent random variables. The Gaussian is the only member of this class having a finite variance. Variances of other members of the stable family are infinite, and tails of their distributions (at least one of them) characterize by power law asymptotics. One can look for a generalization of the turbulence model on the base of all class of stable distributions. Distributions of this class differing from the normal one satisfy equations with fractional derivatives, and multidimensional isotropic stable distributions satisfy equations with the Laplacian of fractional order.

Distributions of this class different from the Gaussian satisfy equations with fractional derivatives, and multidimensional isotropic stable distributions satisfy equations with the Laplacian of fractional order. The logic of development of fractional turbulence models lies indeed in these facts. Two such generalizations were considered by us. The third one would be the fractional Navier-Stokes' equation.

The fractional Navier-Stokes equation is written down in the form [29]:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p - \frac{1}{\mathsf{Re}}(-\Delta)^{\alpha/2}\mathbf{u}, \quad \alpha \in (0, 2], \tag{2.29}$$

where Re is an analogue of the Reynolds number that is equal to it at $\alpha = 2$ (note that dimension of the operator $\Delta^{\alpha/2}$ is the length to the $-\alpha$). A distinction of the exponent α from classical value 2 denotes that a determining (constitutive) equation of turbulent medium differs from classical Newtonian gradient law.

Dimension analysis of Eq. (2.29) leads to the following energy spectrum

$$E(k) = \bar{C}\bar{\varepsilon}^{2/3}k^{-\gamma},$$

with the exponent

$$\gamma = (9 - 2\alpha)/3$$

lying in the interval [5/3,3).[4] The left boundary corresponds to the standard Kolmogorov model of turbulence, but it should be noted that values closed to the right boundary were also observed in the experiments with soap films [89] and plasma [39]. Moreover, the ordinary Newtonian fluid ($\alpha = 2, \gamma = 5/3$) in a turbulence wall layer displays sufficient distinction from the standard model and characterized by the values $\alpha = 1/2, \gamma = 8/3$ [18].

In the paper [28] there is another way of generalization of the Kolmogorov five-thirds power law. This way is based on an introduction into the Navier-Stokes equation of fractional time derivative,

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \frac{1}{\mathrm{Re}}{}_0\mathrm{D}_t^{\mu}\Delta \mathbf{u}, \quad \mu \in [0, 1). \tag{2.29a}$$

It leads to the spectrum

$$E(k) = \bar{C}\bar{\varepsilon}^{2/3}k^{-\gamma}$$

with the exponent

$$\gamma = (5 - 3\mu)/(3 - \mu),$$

value area of which (1, 5/3] lies on the left of the standard value 5/3.

Another way to this equation is used by Cavallini [27]. He proposes to consider the fractional equation as an interpolation of the sequence of constitutive equations relating the stress T to the pressure $p = -\mathrm{tr}\mathsf{T}$ and the strain E

$$\mathsf{T} = -p\mathsf{I} + 2\mu_0\mathrm{D}_t^0\left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right), \qquad \text{(solids)}$$

$$\mathsf{T} = -p\mathsf{I} + 2\mu_1\mathrm{D}_t\left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right), \qquad \text{(viscous fluid)}$$

$$\mathsf{T} = -p\mathsf{I} + 2\mu_2\mathrm{D}_t^2\left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right). \qquad \text{(elastic fluid)}$$

Passing to the generalized fluid by involving a memory kernel $K(t)$,

$$\mathsf{T} = -p\mathsf{I} + 2\mu K \star \mathrm{D}_t^n\left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right), \qquad \text{(gen. fluid)}$$

and choosing the kernel being proportional to the power function $\Phi_{\beta-n}(t)$, one arrives at the fractional constitutive equation

[4] Mandelbrot indicated that non-Gaussian property of turbulence, intermittency, is caused by the spectrum, the slope of which exceeds 5/3.

$$\begin{aligned}\mathsf{T} = & -p\mathsf{I} + 2\eta_\beta \Phi_{\beta-n} \star \mathsf{D}_t^n \left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right) \\ = & -p\mathsf{I} + 2\eta_{\beta_0}{}^{\beta}\mathsf{D}_t \left(\mathsf{E} - \frac{1}{3}(\mathrm{tr}\,\mathsf{E})\mathsf{I}\right), \qquad \text{(frac. fluid)}\end{aligned}$$

Inserting this equation into the momentum balance equation

$$\rho \frac{d\mathbf{u}}{dt} \equiv \rho \left[\frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla\right]\mathbf{u} = \mathrm{div}\mathsf{T} + \mathbf{f}$$

and using the incomressibility condition div $\mathbf{v} = 0$ leads to a fractional version of the Navier-Stokes equation

$$\rho \left[\frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla\right]\mathbf{u} = \mathbf{f} - \nabla p + \eta_{\beta_0}{}^{\beta}\mathsf{D}_t \Delta\mathbf{v}.$$

Cavallini concludes that this equation may describe turbulence more accurately than the classical Navier-Stokes equation, and constitute a starting point for developing a more reliable formulation of geophysical fluid dynamics.

Combining the space- and time-fractionalization leads to the bifractional version of the Navier-Stokes equation [28],

$$\left[\frac{\partial}{\partial t} + \mathbf{u}\cdot\nabla\right]\mathbf{u} = -\frac{1}{\rho}\nabla p - \frac{1}{\mathsf{Re}}\,{}_0\mathsf{D}_t^{\mu}(-\Delta)^{\alpha/2}\mathbf{u}, \tag{2.30}$$

where $\alpha \in (0, 2]$ at $\mu = 0$, $\mu \in [0, 1)$ at $\alpha = 2$ and Re corresponds to the Reynolds number renormalized on fractional dimensions of time and length. In this model, energy transfer from large scales to small ones is split into three phases, slow transfer (subdiffusion) in the domain of values before Kolmogorov's one ($\gamma < 5/3$), normal transfer ($\gamma = 5/3$), and rapid transfer in the Kolmogorov-Lévy domain ($\gamma > 5/3$).

To mention an example of a system described by Eq. (2.30), the author of quoted work refers to the article (Croisman and Steinberg 2000) devoted to turbulent motion of viscoelastic medium produced by polymer solution. This motion differs from ordinary a turbulent motion of Newtonian fluid by non-local character of interactions producing a fractional power of the Laplacian and by presence of memory effect producing fractional time derivative. Both these effects are the consequences of the dynamics of long polymer molecules in the solution. Author notes that at $\alpha = 2$ and $\mu \neq 0$, the exponent γ of such medium does not coincide with Kolmogorov's value 5/3 even in the limit of the infinite Reynolds number.

2.2.6 Reynolds' Equation

Recall that in the classical theory of turbulence, the Reynolds equation arises from the Navier-Stokes equation

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \Delta \mathbf{u}$$

and the condition of fluid incompressibility

$$\nabla \mathbf{u} = 0$$

through splitting the fields $\mathbf{u}$ and p into mean component and rippling (fluctuations)

$$\mathbf{u} = \bar{\mathbf{u}} + \tilde{\mathbf{u}}, \quad p = \bar{p} + \tilde{p}.$$

The first of them obeys equation

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \nu \Delta \bar{u}_i - \frac{\partial}{\partial x_j}\langle \tilde{u}_i \tilde{u}_j \rangle, \tag{2.31}$$

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0.$$

A spread of ripples $\tilde{u}_i$ arisen in some small domain of medium is described by diffusion equation

$$\frac{\partial \tilde{u}_i}{\partial t} = K \Delta(\tilde{u}_i + \bar{u}_i). \tag{2.32}$$

Comparing it with the continuity equation

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial}{\partial x_j}(\tilde{u}_i \tilde{u}_j) = 0,$$

we obtain

$$\frac{\partial}{\partial x_j}(\tilde{u}_i \tilde{u}_j) = -K \Delta(\tilde{u}_i + \bar{u}_i). \tag{2.33}$$

Generalizing Eq. (2.32) on turbulent superdiffusion,

$$\frac{\partial \tilde{u}_i}{\partial t} = -K(-\Delta)^{\alpha/2}(\tilde{u}_i + \bar{u}_i),$$

we obtain instead Eq. (2.33)

$$\frac{\partial}{\partial x_j}(\tilde{u}_i\tilde{u}_j) = K(-\Delta)^{\alpha/2}(\tilde{u}_i + \bar{u}_i),$$

and after averaging over rippling

$$\frac{\partial}{\partial x_j}\langle\tilde{u}_i\tilde{u}_j\rangle = K(-\Delta)^{\alpha/2}\bar{u}_i.$$

As a result instead Eq. (2.31) we obtain fractional differential analogue of the Reynolds equation

$$\frac{\partial\bar{\mathbf{u}}}{\partial t} + (\bar{\mathbf{u}}\cdot\nabla)\bar{\mathbf{u}} = -\frac{1}{\rho}\nabla\bar{p} + \nu\Delta\bar{\mathbf{u}} - K(-\Delta)^{\alpha/2}\bar{\mathbf{u}}. \tag{2.34}$$

The next to last component in this equation is related to molecular diffusion reflecting physical properties of a fluid, at the same time the last component containing the fractional Laplacian (remind that $\alpha = 2/3$) characterizes properties of inertial diffusion of a non-Newtonian fluid, long-range correlations arising spontaneously in its irregular turbulent motion. The numeric example of [28] shows that coefficient of inertial diffusion can be large in three times than molecular diffusion coefficient.

2.2.7 *Diffusion in Lane Flows*

In some cases, it is convenient to present a complicated process of non-regular turbulent motion in the form of diffusion motion in deterministic medium with the velocity depending on coordinates. The *Dreizin-Dyhne model* (Dreizin-Dyhne 1972), cited in the reviews [12, 15] is a namely such case. Following these authors, we consider the space filled up by plane layers with identical thickness a moving in opposite directions of the x-axis with constant absolute value velocity (Fig. 2.2). A probe particle

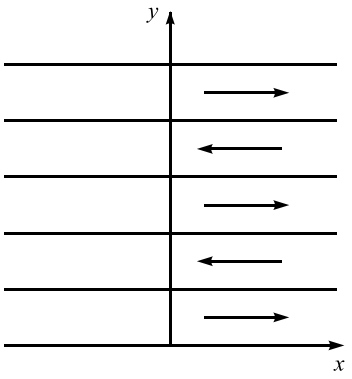


Fig. 2.2 Lane flows

diffuses in this medium. Medium velocity is large enough and longitudinal (along x) displacements of the particle are simply determined by its shifts by moving layers, whereas in cross direction (along y) it perform ordinary diffusion (with a coefficient K_y) and changing velocity x-projection to opposite value it goes from one layer to other. Making the following assumptions:

(1) the mean number of layers traversed by the particle during time t is equal to

$$\langle N \rangle = \sqrt{2K_y t}/a;$$

(2) the particle displacement Δx along the x-axis is determined by a fraction of non-compensated rippling $P \sim \delta N/N$:

$$K_x \approx \frac{(\Delta x)^2}{t} = \frac{(V_0 t)^2 P^2}{t};$$

(3) the random variable δN obeys Gaussian statistics

$$\left\langle \left(\frac{\delta N}{N} \right)^2 \right\rangle \sim \frac{1}{\langle N \rangle},$$

authors arrive at the conclusion that the x-axis diffusion coefficient determined in such way

$$K_x = \frac{V_0^2 t}{\langle N \rangle} = \frac{V_0^2 t a}{\sqrt{2K_y t}} = \frac{V_0^2 a}{\sqrt{2K_y}} t^{1/2}$$

grows with time as $\sqrt{t}$, in other words superdiffusion regime takes place.

This assumptions can be easily criticized however. If τ is a mean time that the particle spends in a layer, then number of layers visited by it during time t

$$\langle N(t) \rangle = t/\tau$$

must be proportional to t. The formula

$$\langle N(t) \rangle \propto t^{1/2}$$

takes place only in the case, when $\tau = \infty$ that is possible only for infinite thickness of layers. Further, in the expression

$$X = \sum_{i=1}^{N} \Delta X_i$$

successive displacements ΔX_i are not independent due to their sign alternating. To turn to statistics of independent events one should combine successful displacements ΔX, $\Delta X'$ into one $\Delta Z = \Delta X + \Delta X'$. Then

$$X \approx \sum_{i=1}^{N/2} \Delta Z_i,$$

and

$$K_x = \frac{\langle X^2 \rangle}{t} = \frac{\langle N \rangle}{2} \frac{\langle (\Delta Z)^2 \rangle}{t} = \frac{\langle (\Delta Z)^2 \rangle}{2\tau}.$$

Thus we come to constant diffusion coefficient along x that is evidence of normal diffusion in this model.

Let us consider the case of two semi-infinite (in the y direction) layers moving in opposite directions. The time T between changing of particle motion direction (return time to layer boundary) is distributed according to the power law

$$p_T(t) \propto At^{-3/2}, \quad t \to \infty.$$

The mean value does not exist, the process is superdiffusive and it obeys fractional differential equation in the asymptotics.[5]

The interesting model was proposed by Chukbar [32]. A fluid flow and flow velocity in narrow layer along the x-axis is large. An impurity can escape this layer and perform ordinary diffusion with a coefficient k out of it. The diffusion equation with drift in incompressible fluid flow

$$\frac{\partial n}{\partial t} + \mathbf{v}\nabla n = K\Delta n, \quad \Delta \mathbf{v} = 0,$$

out of the layer has the form

$$\frac{\partial n}{\partial t} = K \frac{\partial^2 n}{\partial y^2},$$

and all influence of the flow is reduced to the boundary condition

$$\left(\frac{\mathsf{P}}{2} \frac{\partial n}{\partial x} - \frac{\partial n}{\partial y} \right)_{y=0} = 0, \tag{2.35}$$

where P is the Peclet number being equal to the ratio of characteristic value of the flow current function to the diffusion coefficient K. This problem arises at description of fast penetration of magnetic field into magnetized plasma along a well conductive electrode [57].

[5] The general solution of the one-dimensional random walk problem with given path length distribution can be found in [146].

The Laplace transformation with respect to time turns Eq. (2.35) to the following one

$$\lambda\hat{n} = K\frac{d^2\hat{n}}{dy^2} + n_0(x, y), \tag{2.36}$$

where n_0 is an initial impurity distribution. Taking into account the boundary condition

$$\hat{n}|_{y\to\infty} = 0,$$

we can rewrite the solution (2.36) in the form

$$\begin{aligned}\hat{n}(x, y, \lambda) = \; & C\exp\left(-\sqrt{\lambda/K}\,y\right) + \exp\left(-\sqrt{\lambda/K}\,y\right)\int_0^y \exp\left(\sqrt{\lambda/K}\,y'\right)\frac{n_0(x, y')}{2\sqrt{K\lambda}}dy' \\ & + \exp\left(\sqrt{\lambda/K}\,y\right)\int_y^\infty \exp\left(\sqrt{\lambda/K}\,y'\right)\frac{n_0(x, y')}{2\sqrt{K\lambda}}dy',\end{aligned} \tag{2.37}$$

where $\sqrt{\lambda}$ is positive and real for positive and real λ, and C is an arbitrary constant determined by the boundary condition at $y = 0$. Finding from Eq. (2.37) that $\partial\hat{n}/\partial y|_{y=0}$ and substituting the result in the transformant of the condition (2.35), we obtain for $n_1 = n(x, 0, t)$

$$\sqrt{\lambda/K}\hat{n}_1 + \frac{\mathbf{P}}{2}\frac{d\hat{n}_1}{dx} = \frac{1}{K}\int_0^\infty \exp\left(-\sqrt{\lambda/K}\,y\right)n_0(x, y)dy.$$

In coordinate-time variables the equation has the form

$$\frac{\partial}{\partial t}\int_0^t \frac{n_1(x, t')}{\sqrt{\pi K(t - t')}}dt' + \frac{\mathbf{P}}{2}\frac{\partial n_1}{\partial x} = -\int_0^\infty \frac{n_0(x, y)}{\sqrt{\pi Kt}}\frac{d}{dy}\exp\left(-\frac{y^2}{4\,Kt}\right)dy. \tag{2.38}$$

We see the fractional derivative in the left part.

The general solution of Eq. (2.38) describing impurity diffusion in a narrow flow is expressed through its Green's function

$$G(x, y, t) = \frac{l(x)}{\mathbf{P}Kt\sqrt{\pi Kt}}\left(\frac{2x}{\mathbf{P}} + y\right)\exp\left(-\frac{(2x\mathbf{P} + y)^2}{4Kt}\right)$$

by means of the inegral

$$n_1(x, t) = \int_{-\infty}^\infty dy\int_0^\infty dx' n_0(x', y)G(x - x', y, t).$$

2.2.8 Subdiffusion in a Random Compressible Flow

Chukbar and Zaburdaev [33] consider one-dimensional advection-diffusion process in a static random velocity field $u(x) = -d\varphi(x)/dx$ with potential $\varphi(x)$ having the form of a periodic sequence of symmetric triangular wells of equal width $2a$ but random depths b separated by equal distances l (see Fig. 2.3).

The essential point here is the separation of the particles into two groups, diffusing between the wells and wondering inside them. The particles being in the left half of a well diffuse in the fluid flowing to the right with velocity $u \equiv u_x = b/a$ whereas the particles in the right half of the well diffuse in the flow with velocity $u = -b/a$ directed to the left (the fluid is supposed to be compressible). Drift to the center and isotropic diffusion are two competitors which create the trap effect with a long trapping time. Under condition $2a \ll l$, these wells can be considered as point traps uniformly distributed along the axis. A special choice of their random depths distribution,

$$f(b)db = \frac{A}{D^2}\frac{be^{-\alpha b/D}}{(1+b/D)^{2(1-\alpha)}}db,$$

(where A is the normalizing factor, D the diffusion coefficient, and α a positive constant belonging to the interval $(0, 1)$) produces the asymptotically power type of the trapping time distribution. As a result, the authors come to the subdiffusion time-fractional equation:

$${}_0\mathrm{D}_t^{\alpha} n(x,t) = D_\alpha \frac{\partial^2 n(x,t)}{\partial x^2} + \frac{n_0(x)}{\Gamma(1-\alpha)t^{\alpha}}$$

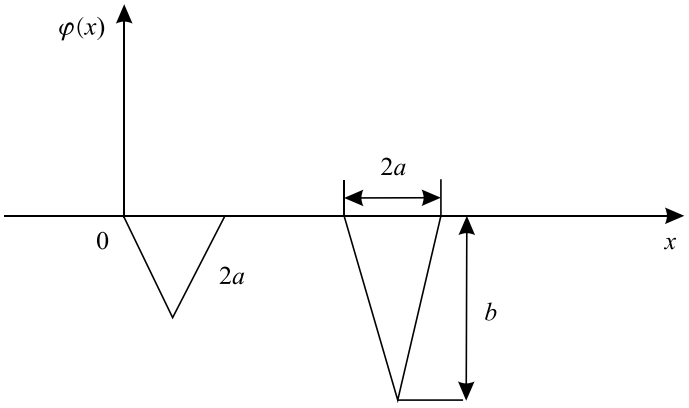


Fig. 2.3 Triangular wells potential

with the diffusivity

$$D_\alpha = la^{2(1-\alpha)} \sin[\pi(1-\alpha)]D^\alpha/(2a\pi).$$

The authors end their paper by pointing to close connection between this model and comb-model leading to the same equation.

2.3 Fractional Models of Viscoelasticity

2.3.1 Two First Models of Fractional Viscoelasticity

Among the first authors employing fractional calculus to describe wrote up viscoelasticity, we meet names of Gemant [55] and Gerasimov [56]. The first of them started with the complex viscosity

$$V = (1 - i(\omega\tau)^{1/2})/(1 + \omega\tau)$$

as the best in agreement with experiments on steady vibrations. Applying the Heaviside method, that is performing replacement

$$i\omega \to p \to d/dt,$$

Gemant wrote up the fractional interrelation

$$\frac{d\varepsilon}{dt} = \frac{\sigma}{\eta} + \frac{1}{(\eta E)^{1/2}} \frac{d^{1/2}\sigma}{dt^{1/2}}$$

between the stress σ and the strain ε (η stands for viscosity and E for shear modulus).

Gerasimov [56] began with the discussion of the Boltzmann equation and then wrote.

"Experimental data show that the particular form of this equation, corresponding to the hereditary part of stress only,

$$\sigma(t) = \int_0^\infty G(\tau)\varepsilon(t-\tau)d\tau,$$

requires special attention. Of no less interest is the case when the stress $\sigma(t)$ depends on all previous properly weighed strain rates, but not on values of the strain. For such deformation processes, the relationship between σ and ε has the form

$$\sigma(t) = \int_0^\infty K(\tau)\dot{\varepsilon}(t-\tau)d\tau,$$

if this does not go beyond the hereditary part of stress."

Note that this equation makes it possible to visualize[6] the relationship between stress and strain if the kernel $K(\tau)$ is interpreted as reaction (response) of stress at the moment t to the unit step strain $1(t-\tau)$ at the moment $t-\tau$. Approximating the smooth dependence $\varepsilon(t)$ by the step function determined by jump increments $\Delta\varepsilon(t_j)$ at times t_j and using the principle of linearity, we obtain the integral sum

$$\sigma(t) = \sum_{t_j<t} K(t-t_j)\Delta\varepsilon(t_j).$$

When $\Delta t_j \to 0$, this sum takes the form

$$\sigma(t) = \int_0^t K(t-t')d\varepsilon(t') = \int_0^t K(\tau)\dot{\varepsilon}(t-\tau)d\tau.$$

Let us continue quoting Gerasimov's article (I think it is important, because the priority in the creation and development of the fractional differential model of viscoelasticity is usually attributed to later publications).

"A hereditary function for some materials (of fiber structure) must have the form

$$K(\tau) = \frac{A}{\tau^\alpha},$$

where A and α are positive constants. Using the representation

$$A = \frac{\kappa_\alpha}{\Gamma(1-\alpha)},$$

we obtain

$$\sigma(t) = \frac{\kappa_\alpha}{\Gamma(1-\alpha)} \int_0^\infty \frac{\dot{\varepsilon}(t-\tau)}{\tau^\alpha} = \kappa_\alpha \, {}_0^\alpha \mathsf{D}_t \varepsilon(t), \quad 0 \leqslant \alpha \leqslant 1.$$

Here the constant $\kappa_\alpha > 0$ depends on substance properties, and Γ stands for the Euler integral of the second kind. This linear interrelation between ε and σ reduces to the Hooke law when $\alpha = 0$, and to the Newton law of internal friction when $\alpha = 1$. We are going to consider this dependence with any α between 0 and 1."

Thus, the viscoelastic medium in some respects occupies an intermediate position between viscous inelastic matter and nonviscous, perfectly elastic matter,

$$\sigma(t) = \begin{cases} E\varepsilon(t), & \text{(Hooke's law);} \\ \eta d\varepsilon(t)/dt, & \text{(Newton's law).} \end{cases} \tag{2.39}$$

The formula

$$\sigma(t) = \kappa_\alpha \, {}_0^\alpha \mathsf{D}_t \varepsilon(t) \tag{2.40}$$

[6] Here $\dot{\varepsilon}$ denotes $d\varepsilon/dt$.

can be considered as a possible way of interpolation between these laws: when $\alpha = 0$, it produces the Hooke law (the corresponding elastic element is symbolically pictured by spring), and when $\alpha = 1$, we recognize it as the Newton law for motion in a viscous medium (the viscous element is pictured by dashpot). Continuing this graphical tradition, Koeller [75] has introduced the term "spring-pot" for the element representing an intermediate case with the constitutive Eq. (2.40). Of course, Eq. (2.40) is not the only way to do the interpolation, but the fact that the Nutting power law is related to a solution of an equation with fractional derivative plays a crucial role. Nevertheless, some investigators consider one-term fractional interpolation as a crude model. Heymans and Podlubny [65] write: "a spring-pot is a particularly crude model, which has several unrealistic and unphysical characteristics. As pointed out above, it has a vanishing initial compliance or an infinite initial modulus. Viscoelastic solids, on the contrary, have a well-defined instantaneous modulus. At long times there is no limit to anelastic strain of a spring-pot: creep continues indefinitely. Also, stress relaxes to vanishingly small values. The increase of stress in constant strain-rate conditions means that if attempting to describe a viscoelastic fluid, steady state flow is never attained. When describing a viscoelastic solid, again we find an unbounded modulus at $t = 0$."

The development of the fractional differential approach in rheology is associated with such names as Scott-Blair, Bagley and Torvik, Gorenflo and Mainardi, Friedrich and Schiessel, Blumen and Metzler, Nonnenmacher, Glöckle, Klafter and Schlesinger, Podlubny and Heymans, and others. This approach has been successfully applied to describe rheological behavior of organic glasses, elastomers, polyurethane, polyisobutylene, monodisperse polybutadiene, and solid amorphous polymers in a wide temperature range (see for example [2] and references therein). Vandamme and Ulm [151] and Shahsavari and Ulm [130] derived equations governing the time-dependent indentation response for axisymmetric indenters into a fractional viscoelastic half-space and have proposed an original method for the inverse analysis of fractional viscoelastic properties and applied to experimental indentation creep data of polystyrene. The method is based on fitting the time-dependent indentation data (in the Laplace domain) to the fractional viscoelastic model response. It is shown that the particular time-dependent response of polystyrene is best captured by a bulk-and-deviator fractional viscoelastic model of the Zener type. We shall dwell in details on fractional differential models of viscoelasticity and then consider a few standard hydrodynamic problems in the simplest model of this type.

2.3.2 Fractionalized Maxwell Model

Mechanical models composed of elements of two types, an elastic element (spring) and viscous element (damper), have played an important heuristic role in rheology since the times of Maxwell and Kelvin. By combining these elements in different orders, one can get various schemes of viscoelastic materials just as electric and radio engineers compose various electric circuits by using resistors, capacitors, and

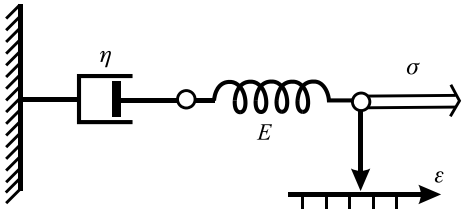


Fig. 2.4 The Maxwell model

inductors. Hooke's law for an elastic body and Newton's one for a viscous liquid were combined into a single constitutive equation of a viscoelastic medium by Maxwell,

$$\sigma(t) + \tau \mathsf{D}_t \sigma(t) = E\tau \mathsf{D}_t \varepsilon(t), \quad \tau > 0.$$

This equation started contemporary phenomenology of viscoelasticity. It describes behavior of a viscoelastic body in terms of elastic (with a modulus of elasticity E) and viscous (with viscosity $\eta = E\tau$) elements (spring and damper) connected in series (Fig. 2.4). A formal generalization of this model to the fractional differential one [48, 100] can be obtained by substitutions

$$\tau \mathsf{D}_t \sigma \mapsto \tau^{\alpha}{}_0\mathsf{D}_t^{\alpha}\sigma(t), \quad \tau \mathsf{D}_t \varepsilon \mapsto \tau^{\beta}{}_0\mathsf{D}_t^{\beta}\varepsilon(t), \quad 0 < \alpha, \beta < 1;$$

$$\sigma(t) + \tau_0^{\alpha}\mathsf{D}_t^{\alpha}\sigma(t) = E\tau_0^{\beta}\mathsf{D}_t^{\beta}\varepsilon(t). \tag{2.41}$$

For the prescribed time-dependence of the strain $\varepsilon(t)$ ($\varepsilon(t) = 0$ for $t < 0$), the solution of this equation for the stress $\sigma(t)$ has the form

$$\sigma(t) = cG(t) + E\tau^{\beta}\int_0^t G(t-z)_0\mathsf{D}_z^{\beta}\varepsilon(z)dz,$$

where

$$G(t) = (t^{\alpha-1}/\tau^{\alpha})E_{\alpha,\alpha}(-t^{\alpha}/\tau^{\alpha}).$$

In the case $\alpha = \beta = 1$,

$$G(t) = (1/\tau)E_{1,1}(-t/\tau) = (1/\tau)e^{-t/\tau}. \tag{2.42}$$

If the strain $\varepsilon(t)$ instantly changes its value from 0 to ε_0 at the moment $t = 0$ and then remains constant, the Green function (2.42) provides an ordinary (Debye) stress relaxation law,

$$\sigma(t) = \varepsilon_0 E e^{-t/\tau}.$$

2.3.3 Fractionalized Kelvin-Voigt Model

Another popular classical model, the Kelvin-Voigt model, is based on the parallel connection of a spring and a damper (Fig. 2.5). Its constitutive equation has the form

$$\sigma(t) = E\left[\varepsilon(t) + \tau \mathsf{D}_t \varepsilon(t)\right]$$

and was generalized by Slonimsky [136] who introduced fractional derivatives into this model to describe the relaxation processes in polymers. The fractional model is written as

$$\sigma(t) = E\left[\varepsilon(t) + \tau^{\beta}\, {}_0\mathsf{D}_t^{\beta}\varepsilon(t)\right],\ 0 \leqslant \alpha < \beta \leqslant 1.$$

or even as

$$\sigma(t) = E\left[\tau^{\alpha}\, {}_0\mathsf{D}_t^{\alpha}\varepsilon(t) + \tau^{\beta}\, {}_0\mathsf{D}_t^{\beta}\varepsilon(t)\right],\ 0 \leqslant \alpha < \beta \leqslant 1$$

(see for details [65, 128]). The particular solution of this equation is

$$\varepsilon(t) = \frac{1}{E\tau^{\beta}} \int_0^t (t-z)^{\beta-1} E_{\beta-\alpha,\beta}\left[-\tau^{\alpha-\beta}(t-z)^{\beta-\alpha}\right]\sigma(z)dz.$$

The substitution of $\alpha = 0,\ \beta = 1$ brings us back into the frames of the standard Kelvin model, in which the step stress $\sigma_0 1(t)$ at the moment $t = 0$ produces the strain

$$\varepsilon(t) = \frac{\sigma_0}{E}\left(1 - e^{-t/\tau}\right),$$

which exponentially approaches Hooke's limit σ_0/E at $t \to \infty$.

2.3.4 Standard Model and Its Generalization

Both these classical models are successfully combined in the *standard model* [67]: [157], in which the constitutive equation has a symmetrical form,

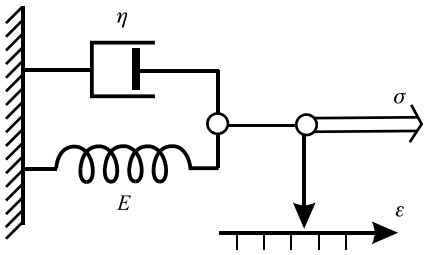


Fig. 2.5 The Kelvin-Voigt model

$$\sigma(t) + \tau \mathsf{D}_t \sigma(t) = E\left[\varepsilon(t) + \theta \mathsf{D}_t \varepsilon(t)\right]. \tag{2.43}$$

This three-parameter model is called the *standard model*. Its fractional generalization, the four-parameter model, proposed by Caputo and Mainardi [19], is described by the constitutive equation with fractional derivatives of the same order α (fourth parameter):

$$\sigma(t) + \tau^{\alpha}\, {}_0\mathsf{D}_t^{\alpha} \sigma(t) = E\left[\varepsilon(t) + \theta^{\alpha}\, {}_0\mathsf{D}_t^{\alpha} \varepsilon(t)\right], \quad 0 < \alpha < 1. \tag{2.44}$$

Interpreting parameters τ and θ as temperature relaxation times at constant stress and constant strain, respectively, and E as an isothermal modulus, Zener had given a thermodynamic basis for the standard model in frames of thermoelasticity theory of solids (metals) [157]. Two basic equations of thermoelasticity connect the temperature deviation δT from its equilibrium value with stress and strain. The first equation is related to thermal expansion of solid

$$\varepsilon = \frac{1}{E}\sigma + \lambda \delta T \tag{2.45}$$

(λ is a linear coefficient of thermal expansion). The second one reflects two causes of temperature change, thermodiffusion (temperature relaxation)

$$\left(\frac{d\delta T}{dt}\right)_{\text{diff}} = -\frac{1}{\tau_\varepsilon}\delta T, \tag{2.46}$$

and adiabatic change of strain

$$\left(\frac{d\delta T}{dt}\right)_{\text{adiab}} = -\gamma \frac{d\varepsilon}{dt},$$

where $\gamma = (\partial T/\partial \varepsilon)_{\text{adiab}}$. As a result, we obtain

$$\left(\frac{d\delta T}{dt}\right) = \left(\frac{d\delta T}{dt}\right)_{\text{diff}} + \left(\frac{d\delta T}{dt}\right)_{\text{adiab}} = -\frac{1}{\tau_\varepsilon}\delta T - \gamma \frac{d\varepsilon}{dt}. \tag{2.47}$$

Assuming that $1 + \lambda\gamma = \tau_\sigma/\tau_\varepsilon$ and excluding from Eqs. (2.45) and (2.46) the member with temperature change δT, Zener arrived at Eq. (2.44).

Replacing Eq. (2.46) by the fractional one for subdiffusive relaxation

$${}_0\mathsf{D}_t^{\alpha}\delta T = -\frac{1}{\tau^{\alpha}}\delta T, \quad 0 < \alpha < 1,$$

F. Mainardi obtained the following equation

$${}_0\mathsf{D}_t^{\alpha}\delta T = -\frac{1}{\tau^{\alpha}}\delta T - \gamma_0 \mathsf{D}_t^{\alpha}\varepsilon,$$

instead of Eq. (2.47) which has led to the fractional generalization of the standard model (2.44). Here the following relation

$$1 + \lambda\gamma = (\theta/\tau)^{\alpha}$$

has been used.

An equation of the type (2.44) was proposed by Caputo and Mainardi [19, 20] as an empirical model allowing a more accurate description of experimental data. The use of Mittag-Leffler function's was proposed by Gross [59] to give a more precise tool than a power law function for approximating experimental data [59].

2.3.5 *Bagley-Torvik Model*

Continuing the development of fractional differential phenomenology beyond the generalized standard model considered above, Torvik [11] took an equation with fractional derivatives of different orders for strain and stress,

$$\sigma(t) + \tau^{\beta}\,{}_0\mathsf{D}_t^{\beta}\sigma(t) = E\left[\varepsilon(t) + \theta^{\alpha}\,{}_0\mathsf{D}_t^{\alpha}\varepsilon(t)\right]. \tag{2.48}$$

Following their article, let us use the Laplace transformation with the imaginary parameter $\lambda = i\omega$,

$$\widehat{f}(i\omega) = \int_0^{\infty} e^{-i\omega t} f(t)dt.$$

As a result of this transformation, we obtain

$$\widehat{\sigma}(i\omega) = \widehat{E}(i\omega)\widehat{\varepsilon}(i\omega),$$

where

$$\widehat{E}(i\omega) = \frac{1 + (i\omega\theta)^{\alpha}}{1 + (i\omega\tau)^{\beta}}E$$

is a complex module. At very low frequencies $\widehat{E}(i\omega) \to E$, the material behaves itself as absolutely elastic ideal rubber. At high frequencies, the transform $\widehat{E}(i\omega) \sim (\theta^{\alpha}/\tau^{\beta})(i\omega)^{\alpha-\beta}$ has a finite limit only if $\alpha = \beta$, and the same material behaves like a glass. The presence of such a transition region is confirmed by the experimental data presented in [11].

The parameters of the model must be real, but not arbitrary, because calculations based on this constitutive equation must give non-negative internal work and rate of energy dissipation. In order to find thermodynamic limitations of the model parameters following from the last statement, Bagley and Torvik considered the sinusoidal strain

$$\varepsilon(t) = \sin(\omega_0 t)$$

producing the stress

$$\sigma(t) = A\sin(\omega_0 t) + B\cos(\omega_0 t).$$

The internal work produced in a unit volume during a unit time is given by the expression

$$\sigma(t)\dot{\varepsilon}(t) = A\omega_0\cos(\omega_0 t)\sin(\omega_0 t) + B\omega_0\cos^2(\omega_0 t). \tag{2.49}$$

At a constant temperature this expression is equivalent to the thermodynamic relation

$$\dot{p} = \dot{\Psi} + \dot{\epsilon} \tag{2.50}$$

asserting that the rate of mechanical work production by internal forces $\dot{p}$ is equal to the rate of free energy increase $\dot{\Psi}$ plus the rate of energy dissipation $\dot{\epsilon}$. The comparison of these two equations allows to clarify the situation with a thermodynamic restriction on the model parameters. Indeed, to guarantee non-negativity of the energy dissipation rate $\dot{\epsilon} \geqslant 0$, non-negativity of B is required, the inequality

$$\operatorname{Im}\widehat{E}(i\omega) \geqslant 0, \quad 0 < \omega < \infty$$

should takes place for all frequencies.

Similarly, a limitation of non-negative internal work is constrained. Integrating Eqs. (2.49) and (2.50) on time and taking into account non-negativity of B, it is enough to demand non-negativity of A to guarantee non-negativity of work of internal forces. Non-negativity of A is achieved by the following condition:

$$\operatorname{Re}\widehat{E}(i\omega) \geqslant 0, \quad 0 < \omega < \infty.$$

Separating real and imaginary parts of the complex module and using the restrictions listed above, Bagley and Torvik [11] have obtained the following restrictions on the model parameters:

$$(1)\ E \geqslant 0, \quad (2)\ \alpha = \beta, \quad (3)\ \theta \geqslant \tau, \tag{2.51}$$

which guarantee thermodynamic consistency of the fractional model of viscoelasticity and form the basis for further development of the fractional phenomenology.

2.3.6 Hysteresis Loop

We have seen above that thermodynamic restrictions reduce the 5-parameter model to the 4-parameter one with the constitutive equation

$$\sigma(t) + \tau_0^{\alpha}\mathsf{D}_t^{\alpha}\sigma(t) = E\left[\varepsilon(t) + \theta^{\alpha}{}_0\mathsf{D}_t^{\alpha}\varepsilon(t)\right]. \tag{2.52}$$

Let's devide the stress $\sigma(t)$ into two components

$$\sigma(t) = \sigma_0(t) + \sigma_1(t),$$

determined by the equations

$$\sigma_0(t) + \tau_0^\alpha \mathsf{D}_t^\alpha \sigma_0(t) = E\varepsilon(t) \tag{2.53}$$

and

$$\sigma_1(t) + \tau^\alpha \, {}_0\mathsf{D}_t^\alpha \sigma_1(t) = E\theta^\alpha \, {}_0\mathsf{D}_t^\alpha \varepsilon(t). \tag{2.54}$$

Applying the operator $\theta^\alpha {}_0\mathsf{D}_t^\alpha$ to both sides of Eq. (2.53) and comparing the result with Eq. (2.54), one can see that the components introduced above are connected by the relationship

$$\sigma_1(t) = \theta_0^\alpha \mathsf{D}_t^\alpha \sigma_0(t),$$

and the resultant stress is completely determined by the initiating component $\sigma_0(t)$,

$$\sigma(t) = \sigma_0(t) + \theta_0^\alpha \mathsf{D}_t^\alpha \sigma_0(t). \tag{2.55}$$

Let this component be 0 till the moment $t = 0$, then it changes according to the sine law with the frequency ω_0,

$$\sigma_0(t) = 1(t)\sigma_0 \sin(\omega_0 t).$$

Substituting this expression into Eq. (2.55), we obtain

$$\sigma(t) = \sigma_0 \sin(\omega_0 t) + \frac{\sigma_0 \omega_0 \theta^\alpha}{\Gamma(1-\alpha)} \left(\cos(\omega_0 t) \int_0^t \cos(\omega_0 z) z^{-\alpha} dz + \sin(\omega_0 t) \int_0^t \sin(\omega_0 z) z^{-\alpha} dz \right).$$

In an analogous way, we find from Eq. (2.53)

$$\varepsilon(t) = \frac{\sigma_0}{E} \sin(\omega_0 t) + \frac{\sigma_0 \omega_0 \tau^\alpha}{\Gamma(1-\alpha) E_0} \left(\cos(\omega_0 t) \int_0^t \cos(\omega_0 z) z^{-\alpha} dz + \sin(\omega_0 t) \int_0^t \sin(\omega_0 z) z^{-\alpha} dz \right).$$

In the short times region, these functions increase according to the inverse power law with the exponent $1 - \alpha$,

$$\sigma(t) \sim \frac{\sigma_0 \omega_0 \theta^\alpha}{\Gamma(2-\alpha) E} t^{1-\alpha}, \quad \varepsilon(t) \sim \frac{\sigma_0 \omega_0 \tau^\alpha}{\Gamma(2-\alpha) E} t^{1-\alpha}, \quad t \to 0,$$

but at long times they turn into harmonic oscillations with the determined frequency ω_0,

$$\sigma(t) = \sigma_0 \left\{ \left[1 + (\omega_0\theta)^\alpha \cos\left(\frac{\alpha\pi}{2}\right) \right] \sin(\omega_0 t) + \left[(\omega_0\theta)^\alpha \sin\left(\frac{\alpha\pi}{2}\right) \right] \cos(\omega_0 t) \right\},$$

$$\varepsilon(t) = \frac{\sigma_0}{E_0} \left\{ \left[1 + (\omega_0\tau)^\alpha \cos\left(\frac{\alpha\pi}{2}\right) \right] \sin(\omega_0 t) + \left[(\omega_0\tau)^\alpha \sin\left(\frac{\alpha\pi}{2}\right) \right] \cos(\omega_0 t) \right\}.$$

The hysteresis loop becomes elliptic, and the dissipation factor is given by the expression

$$\eta = \mathrm{tg}(\varphi_\sigma - \varphi_\varepsilon),$$

where

$$\varphi_\sigma = \mathrm{arctg}\left(\frac{(\omega_0\theta)^\alpha \sin(\alpha\pi/2)}{1 + (\omega_0\theta)^\alpha \cos(\alpha\pi/2)} \right)$$

and

$$\varphi_\varepsilon = \mathrm{arctg}\left(\frac{(\omega_0\tau)^\alpha \sin(\alpha\pi/2)}{1 + (\omega_0\tau)^\alpha \cos(\alpha\pi/2)} \right).$$

Using the known trigonometric formula, we obtain

$$\eta = \frac{\omega_0^\alpha(\theta^\alpha - \tau^\alpha)\sin(\alpha\pi/2)}{1 + \omega_0^\alpha(\theta^\alpha + \tau^\alpha)\cos(\alpha\pi/2) + \omega_0^{2\alpha}(\tau\theta)^\alpha}.$$

From the thermodynamic restriction, it follows that losses are positive for all frequencies.

A further development of the fractional differentiation method in rheology is related to studies of many-parameter models with constitutive equations of the form

$$\sigma(t) + \sum_{j=1}^{m} \tau_j^{\nu_j} {}_0\mathsf{D}_t^{\nu_j} \sigma(t) = E\left[\varepsilon(t) + \sum_{k=1}^{m} \theta_k^{\mu_k} {}_0\mathsf{D}_t^{\mu_k} \varepsilon(t) \right].$$

2.3.7 *Rabotnov's Model*

Rabotnov [110] put the following equation

$$\sigma(t) = E\left[\varepsilon(t) - \beta \int_0^t \phi_\alpha(-\beta, t-\tau)\varepsilon(\tau)d\tau \right] \tag{2.56}$$

into the base of his model of the hereditary mechanics of solids. Here $\alpha \in (-1, 0]$, $\beta \neq 0$ and the function ϕ_α is given by the following power series

$$\phi_\alpha(\beta, x) = x^\alpha \sum_{n=0}^{\infty} \frac{\beta^n x^{n(\alpha+1)}}{\Gamma((n+1)(1+\alpha))}. \tag{2.57}$$

Substituting Eq. (2.57) into Eq. (2.56) and using the formula of fractional integration, we obtain

$$\begin{aligned}
\sigma(t) = & E\left[\varepsilon(t) + \sum_{n=0}^{\infty} \frac{(-\beta)^{n+1}}{\Gamma[(\alpha+1)(n+1)]} \int_0^t \frac{\varepsilon(\tau)d\tau}{(t-\tau)^{1-(\alpha+1)(n+1)}}\right] \\
= & E\left[\varepsilon(t) + \sum_{n=0}^{\infty} (-\beta)^{n+1}\, {}_0\mathsf{D}_t^{-(\alpha+1)(n+1)}\varepsilon(t)\right] \\
= & E\left[\varepsilon(t) + {}_0\mathsf{D}_t^{-(\alpha+1)} \sum_{n=0}^{\infty} (-\beta)^{n+1}\, {}_0\mathsf{D}_t^{-(\alpha+1)n}\varepsilon(t)\right].
\end{aligned} \tag{2.58}$$

Following Nakhushev's book [98], let us apply the operator ${}_0\mathsf{D}_t^{\alpha+1}$ to both sides of the last equation,

$$\begin{aligned}
{}_0\mathsf{D}_t^{\alpha+1}\sigma(t) = & E\left[{}_0\mathsf{D}_t^{\alpha+1}\varepsilon(t) + \sum_{n=0}^{\infty}(-\beta)^{n+1}\, {}_0\mathsf{D}_t^{-(\alpha+1)n}\varepsilon(t)\right] \\
= & E\left[{}_0\mathsf{D}_t^{\alpha+1}\varepsilon(t) - \beta\varepsilon(t) - \beta\sum_{n=1}^{\infty}(-\beta)^{n}\, {}_0\mathsf{D}_t^{-(\alpha+1)n}\varepsilon(t)\right] \\
= & E\left[{}_0\mathsf{D}_t^{\alpha+1}\varepsilon(t) - \beta\varepsilon(t) - \beta\sum_{n=0}^{\infty}(-\beta)^{n+1}\, {}_0\mathsf{D}_t^{-(\alpha+1)(n+1)}\varepsilon(t)\right].
\end{aligned}$$

Finding from the last relation the following expression

$$E\beta\sum_{n=0}^{\infty}(-\beta)^{n+1}\, {}_0\mathsf{D}_t^{-(\alpha+1)(n+1)}\varepsilon(t) = E[{}_0\mathsf{D}_t^{\alpha+1}\varepsilon(t) - \beta\varepsilon(t)] - {}_0\mathsf{D}_t^{\alpha+1}$$

and substituting it instead of the last component in Eq. (2.58), we arrive at

$$\sigma(t) = E\varepsilon(t) + (E/\beta)[{}_0\mathsf{D}_t^{\alpha+1}\varepsilon(t) - \beta\varepsilon(t)] - (1/\beta)\, {}_0\mathsf{D}_t^{\alpha+1}\sigma(t)$$

that takes the form of

$$\sigma(t) + \tau\ {}_0\mathsf{D}_t^{\rho}\sigma(t) = \tau E\ {}_0\mathsf{D}_t^{\rho}\varepsilon(t),$$

after simple transformations. Here $\tau = 1/\beta$ and $\rho = \alpha + 1$. Thus, the Rabotnov model is the particular case of the 4-parameter Bagley-Torvik model. At $\alpha = 0$ it coincides with the defining Maxwell equation.

2.3.8 Compound Mechanical Models

It is natural to put a question: can one construct the fractional model of viscoelasticity with the help of standard elements of two types (elastic and viscous)?

Such a model was built by Gross and Marvin for $\alpha = 1/2$ [144] by analogy with its electrotechnical prototype stated in the book [103], and it was generalized by Schiessel and Blumen [125] to an arbitrary $\alpha \in (0, 1)$. It represents a stair type structure composed of an infinite sequence of elements with parameters chosen in a certain way (see Fig. 2.6).

Let us denote the extension of kth spring (with the elasticity module E_k) by ε_k^e, the displacement of kth damper (with the viscosity η_k) by ε_k^d and introduce the analogous notation for the corresponding components of stress. Similarly to electrotechnical circuits, mechanical ones are calculated according to a special set of rules (these methodology is well described in the book [54].

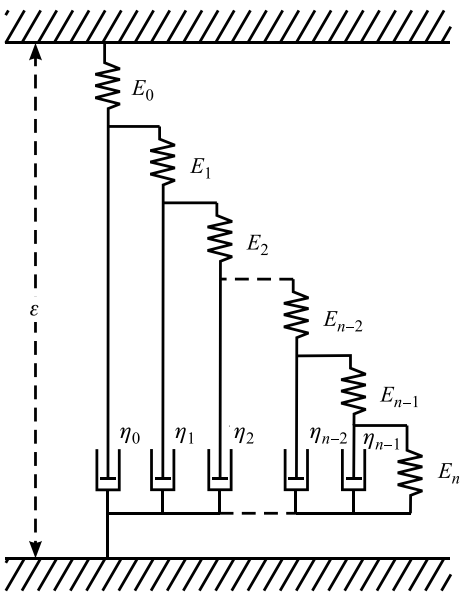


Fig. 2.6 The Gross-Marvin model [144]

1. Additivity of serial strains

$$\varepsilon_k^d = \varepsilon_{k+1}^e + \varepsilon_{k+1}^d, \quad k = 0, 1, \ldots, n-2. \tag{2.59}$$

At the upper end

$$\varepsilon = \varepsilon_0^e + \varepsilon_0^d,$$

at the lower one

$$\varepsilon_{n-1}^d = \varepsilon_n^e.$$

2. Additivity of parallel stresses

$$\sigma_k^e = \sigma_{k+1}^e + \sigma_k^d, \quad k = 0, 1, \ldots, n-1. \tag{2.60}$$

At the end of the stairs

$$\sigma = \sigma_0^e.$$

3. Components of strains and stresses are related by the following relationships

$$\varepsilon_k^e = \frac{1}{E_k}\sigma_k^e \tag{2.61}$$

and

$$\sigma_k^d = \eta_k \frac{d\varepsilon_k^d}{dt}. \tag{2.62}$$

4. It is assumed that

$$\varepsilon(t) = 0, \quad t \leqslant 0,$$

and

$$\sigma(t) = 0, \quad t \leqslant 0.$$

Substituting Eq. (2.61) into Eq. (2.59) and turning to the Laplace images, we obtain

$$E_{k+1}\widehat{\varepsilon}_k^d(\lambda) = \widehat{\sigma}_{k+1}^e(\lambda) + E_{k+1}\widehat{\varepsilon}_{k+1}^d(\lambda), \quad k = 0, 1, \ldots, n-2. \tag{2.63}$$

Similarly, substituting Eq. (2.62) into Eq. (2.60), we arrive at

$$\widehat{\sigma}_k^e(\lambda) = \widehat{\sigma}_{k+1}^e(\lambda) + \lambda\eta_k\widehat{\varepsilon}_k^d(\lambda), \quad k = 0, 1, \ldots, n-1. \tag{2.64}$$

Dividing two parts of Eq. (2.63) by $\widehat{\sigma}_{k+1}^e(\lambda)$ and substituting the stress found from Eq. (2.64) in the obtained equation

$$\widehat{\sigma}_{k+1}^e(\lambda) = \widehat{\sigma}_{k+2}^e(\lambda) + \lambda\eta_{k+1}\widehat{\varepsilon}_{k+1}^d(\lambda)$$

and producing some reduction, we represent the result in the form

$$E_{k+1}\frac{\widehat{\varepsilon}_k^d(\lambda)}{\widehat{\sigma}_{k+1}^e(\lambda)} = 1 + \frac{E_{k+1}}{\eta_{k+1}}\frac{1}{\lambda + \frac{1}{\eta_{k+1}}\frac{\widehat{\sigma}_{k+2}^e(\lambda)}{\widehat{\varepsilon}_{k+1}^d(\lambda)}},$$

$$E_{n-1}\frac{\widehat{\varepsilon}_{n-2}^d(\lambda)}{\widehat{\sigma}_{n-1}^e(\lambda)} = 1 + \frac{E_{n-1}}{\eta_{n-1}}\frac{1}{\lambda + \frac{E_n}{\eta_{n-1}}}.$$

A multiple application of this procedure leads to the representation of the ratio $\widehat{\varepsilon}(\lambda)/\widehat{\sigma}(\lambda)$ in the form of a chain fraction

$$E_0\frac{\widehat{\varepsilon}(\lambda)}{\widehat{\sigma}(\lambda)} = 1 + \frac{\lambda^{-1}E_0/\eta_0}{1+}\,\frac{\lambda^{-1}E_1/\eta_0}{1+}\cdots\frac{\lambda^{-1}E_{n-1}/\eta_{n-1}}{1+}\,\frac{\lambda^{-1}E_n/\eta_{n-1}}{1}.$$

Comparing the obtained expression with the expansion

$$x(x+1)^{\alpha-1} = \frac{x}{1+}\,\frac{(1-\alpha)x}{1+}\,\frac{\frac{1\cdot(0+\alpha)}{1\cdot 2}}{1+}\,\frac{\frac{1\cdot(2-\alpha)}{2\cdot 3}}{1+}\,\frac{\frac{1\cdot(1+\alpha)}{3\cdot 4}}{1+}\,\frac{\frac{1\cdot(3-\alpha)}{4\cdot 5}}{1+}\cdots,$$

one can see that with an appropriate choice of parameters

$$E_1/\eta_0 = (1-\alpha)c_0, \quad E_1/\eta_1 = \frac{1\cdot(0+\alpha)}{1\cdot 2}c_0, \quad \cdots,$$

the relation under investigation has the limit

$$E_0\frac{\varepsilon(\lambda)}{\sigma(\lambda)}\bigg|_{n\to\infty} = 1 + (c_0/\lambda)(c_0/\lambda+1)^{\alpha-1}.$$

In the asymptotics of small λ corresponding to large times t,

$$E_0\frac{\varepsilon(\lambda)}{\sigma(\lambda)} \sim (c_0/\lambda)^\alpha.$$

It corresponds to the expression with a fractional derivative in the time domain

$$\sigma(t) = \eta_0^\alpha E_0^{1-\alpha}\,{}_0\mathsf{D}_t^\alpha\varepsilon(t).$$

A more complicated construction of the model leads to the fractional generalization of the Maxwell model

$$\sigma(t) + \tau^\alpha\,{}_0\mathsf{D}_t^\alpha\sigma(t) = \tau^\beta E\,{}_0\mathsf{D}_t^\beta\varepsilon(t).$$

Constructions of this sort are very visual and they can be found in a number of works, although their explicit formalism and detachment toward physical essence of phenomena do not allow to take them as a physical interpretation of the fractional model.

2.3.9 The Rouse Model of Polymers

The physical explanation of the fractional relation between strain and stress was given on the basis of Rouse's statistical theory for viscoelasticity of polymer solutions in Newtonian fluid. Separating the velocity of each molecule of a polymer chain into the velocity of the chain center of mass, coinciding with a velocity of fluid flow, and into the relative velocity describing the return of a deformed chain to an equilibrium shape, and performing necessary calculations, Rouse has obtained the following expression for the shift module $\widetilde{G}(\omega)$:

$$\widetilde{G}(\omega) = nkT \sum_{j=1}^{N} \frac{\tau_j^2 \omega^2}{1 + \tau_j^2 \omega^2} + i \left\{ \mu_s \omega + nkT \sum_{j=1}^{N} \frac{\tau_j \omega}{1 + \tau_j^2 \omega^2} \right\}. \tag{2.65}$$

Here T is an absolute temperature, k is the Boltzmann constant, n is a number of molecules in the unit volume of the solution, and τ_j are characteristic relaxation times having (at $j < N/5$ and $\omega\tau_1 < N^2/250$) the form

$$\tau_j \simeq \frac{\tau_1}{j^2}, \tag{2.66}$$

$$\tau_1 \simeq \frac{6(\mu_0 - \mu_s)}{n\pi^2 kT}, \tag{2.67}$$

where μ_0 and μ_s are stationary viscosities of a solution and a solvent, respectively. Substituting Eq. (2.66) into Eq. (2.65), we arrive at the expression

$$\widetilde{G}(\omega) \simeq i\mu_s\omega + nkT \sum_{j=1}^{[N/2]} [1 - ij^2/(\tau_1\omega)]^{-1}.$$

Passing from summation to integration with the use of the formula

$$\lim_{\Delta x \to 0} \sum_{j=1}^{\infty} [1 - i(j\Delta x)^2]\Delta x = \int_0^{\infty} (1 - i\xi^2)^{-1} d\xi = i^{1/2}\pi/2$$

at $\Delta x = (\tau_1\omega)^{-1/2}$ and $[N/5] \gg \tau_1\omega \geqslant 25$, we obtain

$$\widetilde{G}(\omega) \simeq i\mu_s\omega + nkT(\pi/2)\sqrt{i\tau_1\omega} = i\mu_s\omega + \sqrt{(3/2)(\mu_0 - \mu_s)nkT}\sqrt{i\omega}.$$

Inverse Fourier transformation of the relation

$$\widetilde{\sigma}(\omega) = \widetilde{G}(\omega)\widetilde{\varepsilon}(\omega)$$

leads to the desired result

$$\sigma(t) = \mu_s \mathsf{D}_t \varepsilon(t) + \sqrt{(3/2)(\mu_0 - \mu_s)nkT}\ {}_0\mathsf{D}_t^{1/2}\varepsilon(t).$$

From the found expression one can see that the stress in the polymer solution consists of two parts, one of which characterizes a solvent (Newtonian fluid), and the second one gives contribution of polymer chains, which depends on their prehistory, and therefore, on the prehistory of the flow. The Rouse theory gives the form of this dependence, providing the physical support of the application of fractional derivatives to describe the process. Observe that in case of a stepwise sress

$$\varepsilon(t) = \varepsilon_0 1(t) = \begin{cases} 0, & \text{t<0,} \\ \varepsilon_0, & \text{t>0,} \end{cases}$$

its derivative is equal to 0 for $t > 0$, whereas the semi-derivative differs from 0 and decreases with time according to the power law,

$$\sigma(t) = \sqrt{(3/2\sqrt{\pi})(\mu_0 - \mu_s)nkT}\ \varepsilon_0 t^{-1/2}.$$

Thus, the Rouse calculation leads to the constitutive equation with the half-derivative,

$$\sigma(t) = E_0\varepsilon(t) + E_{10}\mathsf{D}_t^{1/2}\varepsilon(t).$$

This equation represents a special case of the Bagley-Torvik model, and experiments with numerous solid polymers have shown that α can take different values from the interval (0, 1) for various materials. The explanation of such data requires a more general model than before. To give an example of such approach, we refer to the article [125].

2.3.10 Hamiltonian Dynamic Approach

Let A be a dynamic variable of a homogeneous dynamical system and $V(t)$ is an external field perturbing the equilibrium state of the system. After the activation of this field, the system begins to relax and tends to the equilibrium state corresponding to new conditions. The Hamiltonian of such a system is the sum of the unperturbed Hamiltonian and the Hamiltonian of perturbation related to the potential of the external field,

$$H = H_0 + H_{\text{int}}.$$

This relationship is assumed to be linear with a proportionality coefficient b that depends on time through generalized coordinates and momentums (as well as the dynamic variable A itself),

$$H_{\text{int}} = -b(p(t), q(t))V(t).$$

The evolution of the perturbed system is described by the well-known *Kubo formula*,

$$\langle A(t)\rangle = \langle A(0)\rangle + \frac{1}{2\pi}\int_{-\infty}^{\infty} \widetilde{V}(\omega)\widetilde{G}(\omega)e^{-i\omega t}d\omega, \tag{2.68}$$

where G is the retarded Green function (thermodynamic correlation function) of the system,

$$G(t-t') = -1(t-t')\langle\{b(t'), A(t)\}\rangle_{\text{eq}} \equiv \langle\langle b(t')A(t)\rangle\rangle,$$

and $\{\dots\}$ are the Poisson brackets.

On the basis of this formalism we consider the system of entangled polymer chains (glomeruli) of different lengths. Their sizes are assumed to be much smaller than the macroscopic scale, but much larger than the size of the monomer. Under the influence of an external field, these glomeruli can deform, slip, break, and recombine. Let x_1 and x_2 be two points of the chain that are taken in the direction of tension. The distance between them is much shorter than the length of the chain part connecting these points. The difference between these lengths is called the *stored length*. If one of these points starts to move, the section $[x_1, x_2]$ is firstly straightened, and only after that does the second point x_2 begin to move. The energy dissipation in neighboring domains occurs with delay, and this delay generates the damping law containing a fractional derivative. The perturbation of the system lies in its deformation, $V(t) = \varepsilon(t)$, and the response of the system, the stress $\sigma(t)$, is represented as a sum of the elastic (reversible) component, $\sigma_e = E\varepsilon$, and the non-elastic (dissipative, irreversible) one, $\sigma_d = A$.

Schiessel and Blumen [125] write: "Direct calculation of the Green function for a random potential is very difficult. Therefore, we replace the Hamiltonian of interaction by its average value",

$$\overline{H}_{\text{int}}(p, q, t) = -\overline{b}(p, q)V(t).$$

Averaging is carried out on a discretized time delay of the nth monomer nt_0 with probability distribution

$$P(nt_0) = \frac{t_0 N_0}{(nt_0)^{\nu+1}},\ n > N = \theta/t_0,\ N_0 = (\nu+2)\theta^{\nu+2},\ 0 < \nu < 1.$$

and it leads to the expression

$$G(t-t') = \left\langle\left\langle \sum_{n=N}^{\infty} P(nt_0) b(t'-nt_0)\sigma_d(t) \right\rangle\right\rangle = N_0 \sum_{n=N}^{\infty} \frac{\langle\langle b(t'-nt_0)\sigma_d(t)\rangle\rangle}{t_0^{\nu} n^{\nu+1}}.$$

Further, the authors neglect the thermodynamic correlations between $b(t')$ and $\sigma_d(t)$,

$$G(t-t') = \begin{cases} -N_0 \sum_{n=N}^{\infty} t_0^{-\nu} n^{-\nu-1} \delta(t-t'-nt_0), & t > t', \\ 0, & t \leqslant t' \end{cases}$$

and find the Fourier transform of the Green function,

$$\begin{aligned} \widetilde{G}(\omega) = & -N_0 \sum_{n=N}^{\infty} t_0^{-\nu} n^{-\nu-1} \exp(-i\omega t_0 n) \\ = & -N_0 \left[\sum_{n=1}^{\infty} t_0^{-\nu} n^{-\nu-1} \exp(-i\omega t_0 n) - \sum_{n=1}^{N-1} t_0^{-\nu} n^{-\nu-1} \exp(-i\omega t_0 n) \right]. \quad (2.69) \end{aligned}$$

Carrying out the direct and inverse Mellin transformations of the function $\exp(-i\omega t_0 n),\ \omega > 0$, one can check the identity

$$\exp(-i\omega t_0 n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(s)(\omega t_0 n)^{-s} \exp(-is\pi/2) ds,\ 0 < c < 1.$$

Inserting it into Eq. (2.66) and applying the residue theorem lead to the expression

$$\begin{aligned} \widetilde{G}(\omega) = & -N_0 \Gamma(-\nu)\omega^{\nu} e^{i\nu\pi/2} \\ & -N_0 \sum_{n=0}^{\infty} t_0^{n-\nu} \left[\zeta(1+\nu-n) - \sum_{m=1}^{N-1} m^{-nu-1+n} \right] [(-1)^n/n!] \omega^n e^{i\nu\pi/2}, \end{aligned}$$

where $\zeta(z)$ is the Riemann zeta function,

$$\mathrm{res}_{z=-\nu} \zeta(1+z+\nu) = 1.$$

Passing to $\omega < 0$ changes the sign in front of the imaginary unit.

In the limit $t_0 \to 0$, all the terms in square brackets vanish except for the first one (with $n = 0$) that can be found by using the Hermitian representation of the generalized zeta-function $\zeta(\nu+1, \theta/t_0)$,

$$\lim_{t_0 \to 0} \widetilde{G}(\omega) = -N_0 \left[\Gamma(-\nu)(i\omega)^{\nu} + (\nu\theta^{\nu})^{-1} \right].$$

Substituting this result in the Kubo formula (2.68), we arrive at the constitutive equation

$$\begin{aligned}\langle\sigma_d\rangle &= -N_0\Gamma(-\nu)\frac{1}{2\pi}\int_{-\infty}^{\infty}(i\omega)^{\nu}\widetilde{\varepsilon}(\omega)e^{-i\omega t}d\omega - N_0(\nu\theta^{\nu})\varepsilon(t)\\ &= \eta_{\nu a}\mathsf{D}^{\nu}\varepsilon(t)+\xi_{\nu}\varepsilon(t),\\ &\eta_{\nu} = -N_0\Gamma(-\nu), \quad \xi_{\nu} = -N_0(\nu\theta^{\nu})^{-1},\end{aligned}$$

containing the fractional derivative of order ν.

2.4 Viscoelastic Fluids Motion

Comprehensive reviews on applications of fractional calculus to dynamic problems of viscoelastic material have been given by Shimizu and Zhang [114, 132]. As noticed in Trinks and Ruge [143], there exist three different methodologies to tackle the above problem: the Laplace transformation method; the application of the correspondence principle, the direct numerical integration of the respective equations. Several authors [42, 104] base their efforts on the Grünwald-Letnikov definition of the fractional operator and use a truncated numerical representation. Other authors (Zhang and Shimizu [76, 134, 143, 158]) use the Riemann-Liouville definitions as the starting point of the derivation of numerical schemes.

2.4.1 Gerasimov's Results

2.4.1.1 Plain Flow

The first fractional generalization of a simplest model of fluid, Newtonian model, with the constitutive equation

$$\sigma(t) = \kappa_1 \mathsf{D}_t\varepsilon(t)$$

was performed by Gerasimov (see Fig. 2.7) who represented it in the form

$$\sigma(t) = \kappa_{\alpha}\,{}_0^{\alpha}\mathsf{D}_t\varepsilon(t),$$

and solved two problems.

In the first of them, the fluid is assumed to be enclosed between two parallel planes. The lower plane ($z = 0$) is fixed, the upper one ($z = a$) moves in the direction of OX-axis according to the given law $X(t)$, $X(0) = 0$, $\dot{X}(0) = 0$. The initial and boundary conditions for the x-coordinate of a fluid particle $x(z, t)$ have the form

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ОБОБЩЕНИЕ ЛИНЕЙНЫХ ЗАКОНОВ ДЕФОРМИРОВАНИЯ И ЕГО ПРИМЕНЕНИЕ К ЗАДАЧАМ ВНУТРЕННЕГО ТРЕНИЯ [1]

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Явления упругого последействия в свете теории наследственности описываются линейным интегральным уравнением Больцмана [1]

$$\sigma(t) = E\varepsilon(t) + \int_0^\infty G(\tau)\,\varepsilon(t-\tau)\,d\tau$$

где E — упругая постоянная и $G(\tau)$ — наследственная функция, определяемая из опыта. Экспериментальные исследования показывают, что особого внимания заслуживает частный вид этого уравнения, соответствующий только наследственной части напряжения,

$$\sigma(t) = \int_0^\infty G(\tau)\,\varepsilon(t-\tau)\,d\tau$$

Не меньший интерес представляет также тот случай, когда напряжение $\sigma(t)$ зависит от всех предшествующих, надлежащим образом взвешенных значений скоростей деформации, но не деформаций.

Для таких процессов деформирования зависимость между σ и ε

$$\sigma(t) = \int_0^\infty K(\tau)\,\dot{\varepsilon}(t-\tau)\,d\tau$$

если ограничиваться только наследственной частью напряжения.

Наследственная функция для некоторых материалов (волокнистой структуры) должна иметь вид

$$K(\tau) = \frac{A}{\tau^\alpha}$$

где постоянная $A > 0$ и α лежит между нулем и единицей. Положив еще

$$A = \frac{\varkappa}{\Gamma(1-\alpha)}$$

где постоянная $\varkappa > 0$ зависит от свойств вещества и Γ есть эйлеров интеграл второго рода, получим

$$\sigma(t) = \varkappa \frac{1}{\Gamma(1-\alpha)} \int_0^\infty \frac{\dot{\varepsilon}(t-\tau)\,d\tau}{\tau^\alpha} = \varkappa \frac{\partial^\alpha \varepsilon(t)}{\partial t^\alpha} \qquad (0 < \alpha < 1) \tag{1}$$

так как производная от $\varepsilon(t)$ по t порядка α будет

$$\frac{1}{\Gamma(1-\alpha)} \int_0^\infty \frac{\dot{\varepsilon}(t-\tau)\,d\tau}{\tau^\alpha}$$

[1] Доложено в Институте механики АН СССР 29 мая 1947 г.

Fig. 2.7 The first page of the Gerasimov article [56]

$$x(z,0)=0,\ \left.\frac{\partial x(z,t)}{\partial t}\right|_{t=0}=0,\ x(0,t)=0,\ x(a,t)=X(t).$$

Let us choose an elementary fluid parallelepiped, bounded by two horizontal edges with the unit area and the distance dz between them. The equation of motion of the center of mass has the form

$$\rho\frac{\partial^2 x}{\partial t^2}=\frac{\partial\sigma}{\partial z},$$

where $\rho=\text{const}$ is the fluid density. Taking into account that

$$\varepsilon=\frac{\partial x}{\partial z},\quad \sigma=\kappa_\alpha\,{}_0^\alpha\mathsf{D}_t\varepsilon=\kappa_\alpha\,{}_0^\alpha\mathsf{D}_t\frac{\partial x}{\partial z}$$

and

$$\frac{\partial\sigma}{\partial z}=\frac{\partial}{\partial z}\left(\kappa_\alpha\,{}_0^\alpha\mathsf{D}_t\varepsilon\right)=\kappa_\alpha\,{}_0^\alpha\mathsf{D}_t\frac{\partial^2 x}{\partial z^2},$$

we obtain the equation of motion in the form

$$c^2\frac{\partial^2 x}{\partial t^2}={}_0^\alpha\mathsf{D}_t\frac{\partial^2 x}{\partial z^2},\qquad c_\alpha^2=\frac{\rho}{\kappa_\alpha}.$$

Gerasimov used the operational Heaviside method to obtain the solution of this problem. Passing from the function $x(z,t)$ to its image $\hat{x}(z,p)$ leads to the ordinary differential equation of the 2nd order

$$\frac{d^2\hat{x}}{dz^2}=c_\alpha^2 p^{2-\alpha}\hat{x}$$

with the boundary conditions

$$\hat{x}(0,p)=0,\quad \hat{x}(a,p)=\hat{X}(p).$$

Its solution has the form

$$\hat{x}(z,p)=\hat{K}(z,p)\hat{X}(p),$$

where

$$\hat{K}(z,p)=\frac{\operatorname{sh}(c_\alpha p^{1-\alpha/2}z)}{\operatorname{sh}(c_\alpha p^{1-\alpha/2}a)}.$$

The inverse transformation was fulfilled for three cases.

1. The viscose (Newtonian) fluid ($\alpha=1$, that is $m=1/2$):

$$x(z,t)=\frac{2}{(\pi c_1)^2}\sum_{n=1}^{\infty}(-1)^{n-1}n\sin\left(\frac{n\pi z}{a}\right)\int_0^t\exp\left\{-\left(\frac{n\pi}{c_1 a}\right)^2(t-\tau)\right\}X(\tau)d\tau,$$

$$0 \leqslant z < a.$$

2. The elastic medium ($\alpha = 0$, that is $m = 1$):

$$x(z,t) = \frac{2}{c_0 a}\sum_{n=1}^{\infty}(-1)^{n-1}\sin\left(\frac{n\pi z}{a}\right)\int_0^t \sin\left\{\left(\frac{n\pi}{c_0 a}\right)(t-\tau)\right\}X(\tau)d\tau,$$

$$0 \leqslant z < a.$$

3. The intermediate case ($\alpha = 1/2$, that is $m = 3/4$):

$$x(z,t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin\left(\frac{n\pi z}{a}\right)\int_0^t \sum_{k=1}^{\infty}\left(\frac{n\pi}{c_{1/2}a}\right)^{2k}\frac{(-1)^k(t-\tau)^{3k/2-1}}{\Gamma(3k/2)}X(\tau)d\tau,$$

$$0 \leqslant z < a.$$

For uniform motion of the upper plane

$$X(t) = Vt, \quad V = \text{const} > 0,$$

Gerasimov has also found the stress $\sigma(z,t)$ on the upper surface $z = a$ for an arbitrary α in the form of double series, $\sigma(a,t)$

$$= \kappa_\alpha c_\alpha V\left\{\frac{1}{\Gamma(1-\alpha/2)}t^{-\alpha/2} + 2\sum_{k=0}^{\infty}\sum_{j=0}^{\infty}\frac{[-(2k+2)c_\alpha a]^j}{j!\Gamma((1-j)(1-\alpha/2))}t^{-j(1-\alpha/2)-\alpha/2}\right\}.$$

For $t = 0$ it becomes infinite of order $\alpha/2$, and it is understandable because a plate immersed in a viscous fluid can not be instantly accelerated up to a finite velocity by a finite effort.

2.4.1.2 Rotating Flow

The second problem solved by Gerasimov is the motion of a viscoelastic fluid between coaxial cylindrical surfaces $r_1 < r < r_2$ rotating according to the given laws $\Phi_1(t)$ and $\Phi_2(t)$. Evidently,

$$\varepsilon = r\frac{\partial\varphi}{\partial r},$$

so the resulting moment of inertial force acting on the layer $(r, r + dr)$ of the unit length (the left-hand side of equation) is

$$dM = 2\pi\rho\frac{\partial^2\varphi}{\partial t^2}r^3 dr,$$

while the moment of stress force applied to it (the right-hand side of equation) reads

$$dM = 2\pi \frac{\partial(r^2\sigma)}{\partial r} dr.$$

Equating these values and inserting the constitutive equation yield

$$\rho r^3 \frac{\partial^2 \varphi(r,t)}{\partial t^2} = \kappa_\alpha \frac{\partial}{\partial r}\left[r^3 \frac{\partial}{\partial r}\left({}_0^\alpha \mathsf{D}_t \varphi(r,t) \right)\right], \quad r_1 \leqslant r \leqslant r_2.$$

Its solution must satisfy the initial conditions

$$\varphi(r,0) = 0, \quad \left.\frac{\partial \varphi}{\partial t}\right|_{t=0} = 0$$

and the boundary conditions

$$\varphi(r_1,t) = \Phi_1(t), \quad \varphi(r_2,t) = \Phi_2(t),$$

expressing adhesion of particles on the walls. The functions $\Phi_1(t)$ and $\Phi_2(t)$ with their derivatives are assumed to be 0 at the initial time moment $t = 0$.

Applying again the Heaviside method, Gerasimov arrives at the equation

$$c^2 r^3 p^{2m} = \frac{\partial}{\partial r}\left[r^3 \frac{\partial \hat{\varphi}}{\partial r}\right]$$

with constants

$$c^2 = \rho/\kappa_\alpha, \quad m = (2-\alpha)/2, \quad 1/2 < m < 1$$

and conditions

$$\hat{\varphi}(r_1,p) = \hat{\Phi}_1(p), \quad \hat{\varphi}(r_2,p) = \hat{\Phi}_2(p).$$

Introducing the arc length

$$l(r,t) = r\varphi(r,t)$$

leads to equation of the form

$$\frac{d^2\hat{l}}{dr^2} + \frac{1}{r}\frac{d\hat{l}}{dr} - \left(c^2 p^{2m} + \frac{1}{r^2}\right)\hat{l}(r,p) = 0$$

with corresponding boundary conditions. Its general integral is expressed through the superposition of Bessel's functions $J \equiv J_1$ and $N \equiv N_1$,

$$\hat{l}(r,p) = AJ(ibr) + BN(ibr), \quad b = cp^m,$$

where constants A and B must obey the conditions

$$AJ(ibr_1) + BN(ibr_1) = r_1\hat{\Phi}_1(p),$$

$$AJ(ibr_2) + BN(ibr_2) = r_2\hat{\Phi}_2(p).$$

As a result, we obtain

$$\begin{aligned}\hat{\varphi}(r,\,p) = \; & C\left\{\left[r_1\hat{\Phi}_1(p)J(ibr)N(ibr_2) - r_2\hat{\Phi}_2(p)J(ibr)N(ibr_1)\right]\right. \\ & \left.+\left[r_2\hat{\Phi}_2(p)J(ibr_1)N(ibr) - r_1\hat{\Phi}_1(p)J(ibr_2)N(ibr)\right]\right\},\end{aligned}$$

where

$$C = \frac{1}{J(ibr_1)N(ibr_2) - J(ibr_2)N(ibr_1)}$$

(on assumption that the denominator is not equal to 0).

In a special case, when $r_2 - r_1 = \delta_0$ is small with respect to r_1, so that $J(ibr)$ and $N(ibr)$ can be approximated by $J(ibr_1) + J'(ibr_1)ib(r - r_1)$ and $N(ibr_1) + N'(ibr_1)ib(r - r_1)$ respectively, we arrive at the image

$$\hat{\varphi}(r,\,p) = \frac{r_1(r_2 - r)}{\delta_0 r}\hat{\Phi}_1(p) + \frac{r_2(r - r_1)}{\delta_0 r}\hat{\Phi}_2(p),$$

which can be immediately inverted:

$$\varphi(r,\,t) = \frac{r_1(r_2 - r)}{\delta_0 r}\Phi_1(t) + \frac{r_2(r - r_1)}{\delta_0 r}\Phi_2(t), \quad r_1 \leqslant r \leqslant r_1 + \delta_0.$$

For the strain and stress we obtain

$$\varepsilon(r,\,t) = r\frac{\partial\varphi}{\partial r} = \frac{r_1 r_2}{\delta_0 r}\Psi(t)$$

and

$$\sigma(r,\,t) = \kappa_\alpha\, {}_0^\alpha\mathsf{D}_t\epsilon(r,\,t) = \frac{\kappa_\alpha r_1 r_2}{r\delta_0}\,{}_0^\alpha\mathsf{D}_t\Psi(t),$$

with

$$\Psi(t) = \Phi_2(t) - \Phi_1(t).$$

In particular, on the surfaces

$$\sigma(r_1,\,t) = \frac{\kappa_\alpha r_2}{\delta_0}\,{}_0^\alpha\mathsf{D}_t\psi(t), \quad \sigma(r_2,\,t) = \frac{\kappa_\alpha r_1}{\delta_0}\,{}_0^\alpha\mathsf{D}_t\psi(t). \tag{2.70}$$

Note that the stress on the external surface is r_2/r_1 times less than the stress on the internal one. The cause of this is purely geometrical: according to the law of action and reaction, forces must be equal to each other, but they are distributed over different areas, and the concentration of the force (stress) on the surface with a smaller area must be greater. If $\alpha = 1$, Eq. (2.70) reveals the *Coulomb law*: the stress is proportional to the relative speed of rotation of the surfaces. When $\alpha = 0$, it turns into the *Hooke law*: the stress is proportional to the relative displacement of the surfaces. In case of sinusoidal vibrations of the surfaces,

$$\Psi(t) = A\sin(\omega t),$$

the stress repeats these vibrations with the same frequency but shifted in phase:

$$\sigma(r_1, t) = \frac{\kappa_\alpha r_2}{\delta_0} A \sin\left(\omega t + \frac{\alpha\pi}{2}\right).$$

In a purely elastic medium, stresses oscillate in-phase with the surfaces, in a purely viscous medium the phase shift becomes $\pi/2$.

2.4.2 El-Shahed-Salem Solutions

El-Shahed and Salem [41] solved the time-fractional generalization of the Navier-Stokes equation:

$$ {}_0^{\beta}\mathsf{D}_t\mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -(1/\rho)\nabla p + \nu\Delta\mathbf{u},$$

referring to problems with flows in porous media. We shall discuss these problems in the next chapter, but here we bring two simplest examples from the paper of El-Shahed and Salem, because they contain nothing special connected with porosity and can be directly compared with the Gerasimov results.

2.4.2.1 The First Stokes Problem

An infinite plate lying on the fluid surface suddenly (at $t = 0$) starts to move in its own plane with a constant velocity U_0 directed along the x-axis. The fluid velocity $u_x = u(y, t)$ obeys the equation,

$$ {}_0^{\beta}\mathsf{D}_t u = \nu\frac{\partial^2 u}{\partial y^2}, \quad 0 < \beta < 1,$$

with initial and boundary conditions

$$u(y, 0) = 0; \quad u(0, t) = U_0, t > 0; \quad u(\infty, t) = 0.$$

Using the Fourier sine-transform

$$\widetilde{u}(n,t) = \sqrt{2/\pi}\int_0^\infty u(y,t)\sin(ny)dy,$$

we obtain the equation

$${}_0^\beta \mathsf{D}_t \widetilde{u}(n,t) = \nu U_0\sqrt{2/\pi} - \nu n^2 \widetilde{u}(n,t),$$

the Laplace transform of which with respect to time leads to an algebraic equation having the solution

$$\widetilde{\widetilde{u}}(n,\lambda) = \sqrt{2/\pi}\,\nu n U_0 \frac{\lambda^{-1}}{\lambda^\beta + \nu n^2}.$$

Returning to the original variables yields

$$u(y,t) = (2/\pi)\nu U_0 t^\beta \sum_{n=1}^{\infty} n\sin(ny)E_{\beta,\beta+1}(-\nu n^2 t^\beta).$$

In terms of the Wright function, it is written as

$$u(y,t) = U_0 W(-y/\sqrt{\nu t^\beta}, -\alpha/2, 1).$$

In case $\beta \to 1$, we recognize a known result:

$$u(y,t) = U_0 \mathrm{erfc}\left(\frac{y}{2\sqrt{\nu t}}\right).$$

2.4.2.2 The Couette Flow

Now, the fluid is bounded from below by another plate being parallel to the first one but motionless. In other words, this is a bottom of the vessel, $z = h$. In this case, $u(y,t)$ satisfies the same equation

$${}_0^\beta \mathsf{D}_t u = \nu \frac{\partial^2 u}{\partial y^2}, \quad 0 < \beta < 1,$$

but changed boundary conditions:

$$u(y,0) = 0; \quad u(0,t) = U_0, \quad t > 0; \quad u(h,t) = 0.$$

The solution to the problem is

$$u(y,t) = (2\nu U_0/h)t^{\beta}\sum_{n=1}^{\infty} E_{\beta,\beta+1}(-n(\pi/h)^2 t^{\beta})\sin(n\pi y/h)$$

and turns into classical result when $\beta \to 1$:

$$u(y,t) = (2\nu h U_0/\pi^2)\sum_{n=1}^{\infty}\frac{1}{n}\left[1-\exp(-n(\pi/h)^2 t)\right]\sin(n\pi y/h),$$

Because (see A.2.1)

$$tE_{1,2}(mt) = (1/m)(e^{mt}-1).$$

2.4.2.3 Poiseuille Flow

In case of irrotational motion of the fluid in a long circular pipe of the radius R, the equation for longitudinal velocity $u_z = u(r,t)$ reads

$$\rho_0^{\beta}\mathsf{D}_t u = P + \mu\left(\frac{\partial^2 u}{\partial r^2}+\frac{1}{r}\frac{\partial u}{\partial r}\right). \tag{2.71}$$

The flow is considered as axially symmetric and fully developed. If the fluid is initially at rest and a constant pressure gradient P_0 along the axis of the pipe is suddenly imposed at $t = 0$, the initial and boundary conditions are

$$u(r,0) = 0,\ u(R,t) = 0,\ u(0,t) < \infty.$$

For solving this equation, the authors use the finite Hankel transform

$$\overline{u}(\xi_n) \equiv \mathscr{H}_0\{u(r)\}(n) = \int_0^R f(r)J_0(\xi_n r)r\,dr,\quad u(r) = \frac{2}{R^2}\sum_{n=1}^{\infty}\overline{u}_n J_0(\xi_n r)/J_1^2(\xi_n R),$$

where ξ_n are the positive roots of the equation $J_0(R\xi) = 0$. Taking into account the known formulas

$$\mathscr{H}_0\left\{\frac{d^2 f}{dr^2}+\frac{1}{r}\frac{df}{dr}\right\} = -\xi_n^2\overline{f}(\xi_n) + R\xi_n J_1(\xi_n R)f(R),$$

$$R^2 - r^2 = \frac{8}{R}\sum_{n=1}^{\infty}\frac{J_0(\xi_n r)}{\xi_n^3 J_1(\xi_n R)},$$

and applying the Laplace transform with respect to time variable, we obtain sequentially

$$\rho_0^{\beta} \mathsf{D}_t \overline{u}(\xi_n) = PRJ_1(\xi_n R)/\xi_n - \mu \xi_n^2 \overline{u}(\xi_n)$$

and

$$\rho \xi^{\beta} \widehat{\overline{u}}(\xi_n) = PRJ_1(\xi_n R)/(\xi_n \lambda) - \mu \xi_n^2 \widehat{\overline{u}}(\xi_n).$$

The last equation has the solution

$$\widehat{\overline{u}}(\xi_n) = \frac{PRJ_1(\xi_n R)}{\xi_n \lambda(\mu \xi_n^2 + \rho \lambda^{\beta})}.$$

Performing the inverse Hankel-Laplace transform yields

$$u(r,t) = \frac{2Pt^{\beta}}{\rho R} \sum_{n=1}^{\infty} \frac{J_0(\xi_n r)}{\xi_n J_1(\xi_n R)} E_{\beta,\beta+1}(-\nu \xi_n^2 t^{\beta}).$$

If $\beta = 1$, this result coincides with the classical one:

$$\begin{aligned} u(r,t) &= \frac{2Pt}{\rho R} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n r)}{\lambda_n J_1(\lambda_n R)} E_{1,2}(-\nu \lambda_n^2 t) \\ &= \frac{P}{4\mu}(R^2 - r^2) - \frac{2P}{\rho R} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n r)}{\lambda_n^3 J_1(\lambda_n R)} e^{-\nu \lambda_n^2 t}. \end{aligned}$$

When $\beta \to 0$,

$$u(r,t) = \frac{2P}{R} \sum_{n=1}^{\infty} \frac{J_0(\xi_n r)}{\xi_n(\rho + \mu \xi_n^2) J_1(\xi_n R)}.$$

2.4.3 *Fractional Maxwell Fluid: Plain Flow*

Tan and Xu [141], Tan et al. [140] consider the flow of a viscoelastic fluid of the Maxwell type over an infinite flat plate, which is driven by the impulsive motion of the flat plate with constant velocity V along x-axis. Selecting the y-axis perpendicularly to the wall so that $\mathbf{v} = u(y,t)\mathbf{e}_x$ and representing the Maxwell constitutive equation as

$$\sigma_{yx} + \tau^{\alpha}\, {}_0\mathsf{D}_t^{\alpha} = G\tau^{\beta}\, {}_0\mathsf{D}_t^{\beta-1} \dot{\varepsilon}_{yx} = G\tau^{\beta}\, {}_0\mathsf{D}_t^{\beta-1} \left(\frac{\partial u}{\partial y} \right),$$

the authors arrive at the basic equation

$$\rho\frac{\partial u}{\partial t}=\frac{\partial\sigma_{yx}}{\partial y}=-\rho\tau^{\alpha}\,{}_0\mathsf{D}_t^{\alpha+1}u+G\tau^{\beta}\,{}_0\mathsf{D}_t^{\beta-1}\left(\frac{\partial^2 u}{\partial y^2}\right).$$

Passing to dimensionless variables $u^*=u/V,\ y^*=y\rho V/\mu,\ t^*=tV^2\rho/\mu,\ \eta=V^2\rho/G$ and omitting mark $*$ lead to the equation

$$\frac{\partial u}{\partial t}+\eta^{\alpha}\,{}_0\mathsf{D}_t^{\alpha+1}u=\eta^{\beta-1}\,{}_0\mathsf{D}_t^{\beta-1}\left(\frac{\partial^2 u}{\partial y^2}\right).$$

Adding the initial and boundary conditions

$$u(x,0)=0,\ y>0;\quad u(0,t)=1,\ t>0;\quad u\to 0,\ y\to\infty,$$

and applying the Laplace transform with respect to time yield the ordinary differential equation

$$\frac{d^2\widehat{u}}{dy^2}=\frac{1+\eta^{\alpha}\lambda^{\alpha+1}}{\eta^{\beta-1}\lambda^{\beta-1}}\widehat{u}.$$

Solving the equation under conditions

$$\widehat{u}(0,\lambda)=1/\lambda;\quad \widehat{u}(y,\lambda)\to 0,\ y\to\infty$$

and applying the discrete inverse Laplace transform to the series representation of the solution,

$$\widehat{u}(y,\lambda)=\frac{1}{\lambda}\exp\left[-\left(\frac{1+\eta^{\alpha}\lambda^{\alpha}}{\eta^{\beta-1}\lambda^{\beta}}\right)^{1/2}y\lambda\right]$$

$$=\frac{1}{\lambda}+\sum_{n=1}^{\infty}\frac{(-y)^n\eta^{n(1+\alpha-\beta)/2}}{n!}\sum_{m=0}^{\infty}\frac{\Gamma(n/2+1)\eta^{-\alpha m}}{m!\Gamma(n/2-m+1)}\lambda^{n-n(\beta-\alpha)/2-m\alpha-1},$$

Tan and Xu [141] express the result though the Fox function:

$$u(y,t)=1+\sum_{n=1}^{\infty}\frac{(-y)^n}{n!}\eta^{n(1+\alpha-\beta)/2}t^{n(\beta-\alpha)/2-n}$$

$$\times H_{13}^{11}\left(-\frac{t^{\alpha}}{\eta^{\alpha}}\,\middle|\,\begin{matrix}(-n/2,\ 0)\\ (\,0,\ 1\,)\,(-n/2,\ -1)\,(n-n(\beta-\alpha)/2,\ \alpha)\end{matrix}\right).$$

The dimensionless stress field

$$F(y,t)=\sigma_{yx}/(\rho V^2)$$

is obtained from the equation

$$F + \eta_0^{\alpha} \mathsf{D}_t^{\alpha} F = \eta_0^{\beta-1} \mathsf{D}_t^{\beta-1} \left(\frac{\partial u}{\partial y} \right)$$

and has the form

$$F(y,t) = -\sum_{n=0}^{\infty} \frac{(-y)^n}{n!} \eta^{(n-1)(1-\beta)/2} t^{(n-1)(\beta-\alpha)/2} \times H_{13}^{11} \left(-\frac{t^{\alpha}}{\eta^{\alpha}} \middle| \begin{matrix} (1/2 - n/2,\ 0) \\ (0,\ 1)\ (1/2 - n/2,\ -1)\ (n - (n-1)(\beta-\alpha)/2,\ \alpha) \end{matrix} \right).$$

Numerical analysis performed by Tan and Xu [141] has shown that the greater the α is, the more slowly the fluid velocity changes near the plate, the effect of β on the velocity is opposite to that of α, but in a critical point their effects on velocity change direction. For large times, the viscoelastic effects become weak.

2.4.4 *Fractional Maxwell Fluid: Longitudinal Flow in a Cylinder*

The article [135] deals with the longitudinal flow of a generalized Maxwell fluid in an infinite circular cylinder of radius R. At the initial moment, the cylinder is pulled by a time-dependent shear stress along its axis Oz. The constitutive equation for this fluid is taken as

$$\tau(r,t) + \vartheta_0 \mathsf{D}_t^{\alpha} \tau(r,t) = \mu \frac{\partial u(r,t)}{\partial r},$$

where $\tau(r,t)$ is the shear stress and $u(r,t)$ is z-component of the velocity. Neglecting the body forces and the pressure gradient in the axial direction, one can write the balance of linear momentum by means of the partial differential equation

$$\rho \frac{\partial u(r,t)}{\partial t} = \left(\frac{\partial}{\partial t} + \frac{1}{r} \right) v(r,t),$$

where ρ is a constant density of the fluid. Eliminating τ between these equations, Siddique [135] acquires the following motion equation for fractional Maxwell fluid:

$$\left(1 + \vartheta_0 \mathsf{D}_t^{\alpha}\right) \frac{\partial u(r,t)}{\partial t} = u \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} \right) u(r,t), \quad \nu = \mu/\rho,\ r \in (0,R),\ t > 0.$$

Under initial and boundary conditions

$$u(r,0)=\frac{\partial u(r,0)}{\partial t}=0,\quad \tau(r,0)=0,\quad r\in[0,R),$$

$$\left(1+\vartheta_0\mathsf{D}_t^{\alpha}\right)\tau(R,t)=\mu\frac{\partial u(r,t)}{\partial r}\bigg|_{r=R}=ft^a,\quad t>0,$$

with f and $a>0$ being constants, this equation was sequentially subjected to the Laplace transform

$$(\lambda+\vartheta\lambda^{\beta+1})\widehat{u}(r,\lambda)=\nu\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\frac{\partial}{\partial r}\right)\widehat{u}(r,\lambda),\quad \widehat{u}(r,\lambda)=\int_0^{\infty}u(r,t)e^{-\lambda t}dt,$$

and the finite Hankel transform

$$\bar{u}(r_n,\lambda)\equiv\int_0^R\widehat{u}(r,\lambda)J_0(rr_n)rdr=\frac{RfJ_0(Rr_n)\Gamma(a+1)}{\rho\lambda^{a+1}(\lambda+\vartheta\lambda^{\beta+1}+\nu r_n^2)},$$

where r_n are positive roots of the transcendental equation $J_1(Rr_n)=0$. In case $a=0$, the inverse Hankel transform of $\bar{v}$ is

$$\widehat{u}(r,\lambda)=\frac{fr^2}{2\mu R}-\frac{2f}{\mu R}\sum_{n=1}^{\infty}\frac{g(r_n,t)J_0(rr_n)}{r_n^2J_0(Rr_n)},$$

where $g(r_n,t)$ is expressed through the generalized G-function

$$G_{bcd}(p,t)=\sum_{j=0}^{\infty}\frac{\Gamma(d+j)p^j}{\Gamma(d)\Gamma(j+1)\Gamma((d+j)b-c)}t^{[(d+j)b-c]}$$

as follows:

$$g(r_n,t)=\sum_{k=0}^{\infty}\left(-\frac{ur_n^2}{\vartheta}\right)^k\left[G_{\beta,\beta-k-1,k+1}\left(-\frac{1}{\vartheta},t\right)+\frac{1}{\vartheta}G_{\beta,-k-1,k+1}\left(-\frac{1}{\vartheta},t\right)\right].$$

The corresponding shear stress is written in the form

$$\tau(r,t)=\frac{fr}{\vartheta R}G_{\beta,-1,1}\left(-\frac{1}{\vartheta},t\right)+\frac{2f}{\vartheta R}\sum_{n=1}^{\infty}\frac{J_1(rr_n)}{r_nJ_0(Rr_n)}\sum_{k=0}^{\infty}\left(-\frac{ur_n^2}{\vartheta}\right)^kG_{\beta,-k-1,k+1}\left(-\frac{1}{\vartheta},t\right).$$

Analyzing numerical results, the authors conclude that in the central area of the channel, the Maxwell fluid (the case $\beta = 1$) has lower velocity and the velocity of the generalized Maxwell fluid increases if the fractional order decreases. For long times, the differences between Maxwell, generalized Maxwell and Newtonian ($\beta = 1$ and $\vartheta \to 0$) fluids disappear.

2.4.5 Magnetohydrodynamic Flow

Viscoelastic fluid flows important objects for study because of their numerous applications in modern engineering. Typical examples are enhanced oil recovery, cooling of a metallic plate in a bath, artificial and natural gels, paper and textile coating and so on. Fractional models become more and more popular in magnetohydrodynamics [61–63, 72].

A number of works are devoted to fractional generalization of so-called Oldroyd-B model as a special viscoelastic non-Newtonian fluid successfully describing polymeric liquids and being more amenable to analysis and more importantly experimental. Let us see one of such articles [84].

The Cauchy stress tensor of an incompressible and unsteady Oldroyd-B fluid is written in the form

$$\mathsf{T} = -p\mathsf{I} + \mathsf{S},$$

where $-p\mathsf{I}$ denotes the indeterminate spherical stress and S is the extra-stress tensor satisfying the constitutive equation of the fractional type:

$$\left(1 + \tau^{\alpha} \mathscr{D}_t^{\alpha}\right) \mathsf{S} = \mu \left(1 + \tau_r^{\beta} \mathscr{D}_t^{\beta}\right) \mathsf{A}. \tag{2.72}$$

The *first Rivlin-Ericksen tensor* is expressed through the velocity gradient L as $\mathsf{A} = \mathsf{L} + \mathsf{L}^{\mathrm{T}}$, μ, θ, θ_r are material constants,

$$\mathscr{D}_t^{\alpha} \mathsf{S} =_0 \mathsf{D}_t^{\alpha} \mathsf{S} + \mathbf{V} \cdot \nabla \mathsf{S} - \mathsf{L}\mathsf{S} - \mathsf{S}\mathsf{L}^{\mathrm{T}},$$

and the similar representation takes place for $\mathscr{D}_t^{\beta} \mathsf{A}$. On assumption that the velocity field $\mathbf{V} = u(y, t)\mathbf{e}_x$ and the stress field

$$S_{yy} = S_{zz} = S_{xz} = S_{yz} = 0, \quad S_{xy} = S_{yx} = S(y, t),$$

Equation (2.72) is then reduced to

$$(1 + \tau^{\alpha}\, {}_0\mathsf{D}_t^{\alpha}) S = \mu (1 + \tau_r^{\beta}\, {}_0\mathsf{D}_t^{\beta}) \frac{\partial u(y, t)}{\partial y}. \tag{2.73}$$

Liu et al. consider the fluid permeated by an imposed magnetic field B_0 which acts in the y-direction, so that in low-magnetic Reynolds approximation, the magnetic body force is $\sigma B_0^2 u$. In the absence of a pressure gradient in the x-direction, the motion equation is written as

$$\rho \frac{\partial u}{\partial t} = \frac{\partial S}{\partial y} - \sigma B_0^2 u. \tag{2.74}$$

Eliminating S between Eqs. (2.73) and (2.74) yields

$$\left(1 + \tau^\alpha \, {}_0\mathsf{D}_t^\alpha\right) \frac{\partial u(y,t)}{\partial t} = \nu \left(1 + \tau_r^\beta \, {}_0\mathsf{D}_t^\beta\right) \frac{\partial u(y,t)}{\partial y} - M \left(1 + \tau^\alpha \, {}_0\mathsf{D}_t^\alpha\right) u(y,t), \tag{2.75}$$

where $\nu = \mu/\rho$ is the kinematic viscosity and $M = \sigma B_0^2/\rho$.

The problem is stated as follows. The fluid occupies the space above a flat plate and initially both of them are at rest. At $t = 0$, the plat begins harmonic oscillations with the amplitude V_0 and the frequency ω, $V(t) = V_0 \sin \omega t$. Due to the shear, the fluid is moved gradually with the velocity $u(x,t)$ obeying Eq. (2.75). Applying the Fourier sine transform with respect to x and the Laplace transform with respect to t, the authors find (in dimensionless variables such that $\nu = 1$ and $V_0 = 1$)

$$\bar{u}(\xi, \lambda) = \sqrt{\frac{2}{\pi}} \frac{\omega \xi (1 + \tau_r^\beta \lambda^\beta)}{(\lambda^2 + \omega^2)[(\lambda + M)(1 + \tau^\alpha \lambda^\alpha) + \xi^2 (1 + \tau_r^\beta \lambda^\beta)]}.$$

The authors give also results of inverse transform, $\bar{u}(\xi, \lambda) \mapsto u(x,t)$, in an analytical form, but the formula with a double integration, triple summation and infinite number of special functions including in their turn gamma functions and Pochhammer polynomials, is so cumbersome, that I have to restrict myself only by the Fourier-sin-Laplace transform written above. Numerical results ending the cited article demonstrate that the MHD fluid velocity is more steady than hydrodynamic, and the magnetic body force is favorable to decay of the velocity. The non-Newtonian effects are stronger at large values of α; the smaller α, the more steady of the velocity field.

2.4.6 Burgers' Equation

2.4.6.1 Classical Burgers' Equation

The non-linear Burgers' equation for the velocity $u = u(x,t)$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

is a fundamental partial differential equation from fluid mechanics, gas dynamics, plasma kinetics and so on. When viscosity coefficient $\nu = 0$, the equation takes the conservation form

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} = 0.$$

The viscous Burgers' equation can be linearized by the *Cole-Hope ansatz*

$$u = -\frac{2\nu}{\phi}\frac{\partial \phi}{\partial x},$$

which turns it into diffusion equation

$$\frac{\partial \phi}{\partial t} = \nu \frac{\partial^2 \phi}{\partial x^2}.$$

This gives one the solution of an initial value problem for Burgers' equation in the form

$$u(x,t) = -2\nu \frac{\partial}{\partial x} \ln \left\{ \frac{1}{2\sqrt{\pi \nu t}} \exp \left[-\frac{(x-x')^2}{4\nu t} - \frac{1}{2\nu} \int_0^{x'} u(x'',0) dx'' \right] dx' \right\}.$$

Below, we consider solving fractional generalizations of Burgers' equations, following [81, 156].

2.4.6.2 Time-Fractional Burgers' Equation

Let us start with the time-fractional equation

$${}_0^{\alpha}\mathsf{D}_t u + u\frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0, \quad 0 < \alpha \leqslant 1, \quad t > 0,$$

and subject to the initial condition

$$u(x,0) = g(x) = \frac{\mu + \sigma + (\sigma - \mu)e^{\xi}}{1 + e^{\xi}},$$

where $\xi = (\mu/\nu)(x - \lambda)$ and μ, ν, σ, and λ are constant. According to the HAM (homotopy analysis method, see Sect. 6.4.2), the zeroth-order deformation equation can be written as

$$(1-q)\mathsf{L}[\phi(x,t;q) - u_0(x,t)] = qhH(x,t)\left({}_0^{\alpha}\mathsf{D}_t \phi + \phi \frac{\partial \phi}{\partial x} - \nu \frac{\partial^2 \phi}{\partial x^2} \right)$$

with ${}_0^{\alpha}\mathsf{D}_t$ taken as an auxiliary linear operator L, auxiliary function $H(x,t)=1$, and the initial approximation $u_0(x)=g(x)$. Then, the mth-order deformation is given by

$$\mathsf{L}[u_m(x,t)-1_m u_{m-1}(x,t)]=hH(x,t)R_m(u_0,\ldots,u_{m-1}), \tag{2.76}$$

where

$$R_m(u_0,\ldots,u_{m-1})={}_0^{\alpha}\mathsf{D}_t u_{m-1}+\sum_{j=0}^{m-1}\frac{\partial u_{m-j-1}}{\partial x}-\nu\frac{\partial^2 u_{m-1}}{\partial x^2}.$$

The solution of Eq. (2.76) is expressed as

$$u_m(x,t)=1_m u_{m-1}(x,t)+h\mathsf{L}^{-1}[R_m(u_0,\ldots,u_{m-1})],$$

or, in an explicit form,

$$u_0(x,t)=\frac{\mu+\sigma+(\sigma-\mu)e^{\xi}}{1+e^{\xi}},$$

$$u_1(x,t)=-h\frac{2\mu\sigma^2e^{\xi}}{\nu(1+e^{\xi})^2}\frac{t^{\alpha}}{\Gamma(\alpha+1)},$$

$$u_2(x,t)=(1+h)u_1(x,t)+h^2\frac{\mu^3\sigma^2e^{\xi}(e^{\xi}-1)}{\nu^2(1+e^{\xi})^3}\frac{t^{2\alpha}}{\Gamma(2\alpha+1)}$$

and so on. Hence, for $h=-1$

$$u(x,t)=\frac{\mu+\sigma+(\sigma-\mu)e^{\xi}}{1+e^{\xi}}+\frac{2\mu\sigma^2e^{\xi}}{\nu(1+e^{\xi})^2}\frac{t^{\alpha}}{\Gamma(\alpha+1)}$$
$$+\frac{\mu^3\sigma^2e^{\xi}(e^{\xi}-1)}{\nu^2(1+e^{\xi})^3}\frac{t^{2\alpha}}{\Gamma(2\alpha+1)}+\cdots.$$

In case $\alpha=1$ this solution turns into the classic result

$$u(x,t)=\frac{\mu+\sigma+(\sigma-\mu)\exp[(\mu/\nu)(x-\sigma t-\lambda)]}{1+\exp[(\mu/\nu)(x-\sigma t-\lambda)]}.$$

2.4.6.3 Space-Fractional Burgers' Equation

Now, we solve the equation

$$\frac{1}{\nu}\frac{\partial u}{\partial t}+\frac{u}{\nu}\frac{\partial u}{\partial x}-\frac{\partial^2 u}{\partial x^2}+\frac{\eta}{\nu}{}_0^{\alpha}\mathsf{D}_x=0,\quad x,t>0,\quad 0<\alpha\leqslant 1,$$

under conditions

$$u(0,t)=0,\quad \left.\frac{\partial u(x,t)}{\partial x}\right|_{x=0}=\frac{1}{t}-\frac{\pi^2}{2\nu t^2}.$$

The zeroth-order deformation can be taken as

$$(1-q)\mathsf{L}[\phi(x,t;q)-u_0(x,t)]=qhH(x,t)\left(\frac{\partial^2\phi}{\partial x^2}-\frac{\eta}{\nu}\,{}_0^{\alpha}\mathsf{D}_t\phi-\frac{1}{\nu}\frac{\partial\phi}{\partial t}-\frac{\phi}{\nu}\frac{\partial\phi}{\partial x}\right)$$

with $\mathsf{L}=\partial^2/\partial x^2$. The higher-order deformation equations are given by

$$\mathsf{L}[u_m(x,t)-1_m u_{m-1}(x,t)]=hH(x,t)R_m(u_0,\ldots,u_{m-1}),$$

where

$$R_m(u_0,\ldots,u_{m-1})=\frac{\partial^2 u_{m-1}}{\partial x^2}-\frac{\eta}{\nu}\,{}_0^{\alpha}\mathsf{D}_x u_{m-1}-\frac{1}{\nu}\frac{\partial u_{m-1}}{\partial t}-\frac{1}{\nu}\sum_{j=0}^{m-1}u_j\frac{\partial u_{m-j-1}}{\partial x}.$$

The solution of the system is

$$u_m(x,t)=1_m u_{m-1}(x,t)+h\mathsf{L}^{-1}[R_m(u_0,\ldots,u_{m-1})],\quad m\geqslant 1.$$

Taking the zeroth approximation in the form

$$u_0(x,t)=\left(\frac{1}{t}-\frac{\pi^2}{2\nu t^2}\right)x,$$

for the next terms of the series we get

$$u_1(x,t)=\ h\left[-\frac{\pi^4x^3}{24\nu^3t^4}+\frac{\eta x^{3-\alpha}}{(3-\alpha)(-2+\alpha)\Gamma(2-\alpha)\nu}\right],$$

$$u_2(x,t)=\ (1+h)u_1(x,t)-h^2\left[\frac{\pi^6x^5}{240\nu^5t^6}+\frac{\eta(2\nu t-\pi^2)x^{5-\alpha}}{2(5-2\alpha)(-3+\alpha)\Gamma(2-\alpha)\nu^3t^2}\right]$$

$$+h^2\frac{\eta\Gamma(4-\alpha)x^{5-2\alpha}}{2(5-2\alpha)(-3+\alpha)(-2+\alpha)^2\Gamma(2-\alpha)\Gamma(3-2\alpha)\nu},$$

and so on. Finally, the HAM series (for $h=-1$) is

$$u(x,t)=\ \left(\frac{1}{t}-\frac{\pi^2}{2\nu t^2}\right)x+\left[\frac{\pi^4x^3}{24\nu^3t^4}-\frac{\eta x^{3-\alpha}}{(3-\alpha)(-2+\alpha)\Gamma(2-\alpha)\nu}\right]$$

$$-\left[\frac{\pi^6 x^5}{240\nu^5 t^6}+\frac{\eta(2\nu t-\pi^2)x^{5-\alpha}}{2(5-2\alpha)(-3+\alpha)\Gamma(2-\alpha)\nu^3 t^2}\right]$$

$$+\frac{\eta\Gamma(4-\alpha)x^{5-2\alpha}}{2(5-2\alpha)(-3+\alpha)(-2+\alpha)^2\Gamma(2-\alpha)\Gamma(3-2\alpha)\nu}+\cdots.$$

The solution becomes significantly simpler in case $\alpha = 1/2$,

$$u(x,t)=\left(\frac{1}{t}-\frac{\pi^2}{2\nu t^2}\right)x+\frac{\pi^4 x^3}{24\nu^3 t^4}-\frac{\pi^6 x^5}{240\nu^5 t^6}+\frac{8\eta x^{5/2}}{15\nu\sqrt{\pi}}+\cdots,$$

and $\eta = 0$:

$$u(x,t)=\left(\frac{1}{t}-\frac{\pi^2}{2\nu t^2}\right)x+\frac{\pi^4 x^3}{24\nu^3 t^4}-\frac{\pi^6 x^5}{240\nu^5 t^6}+\cdots.$$

Some other methods of numerical solution of Burgers' equation and their results can be found in [4, 66, 94, 94, 138, 152, 153].

2.5 Solid Bodies

2.5.1 Viscoelastic Rods

Kempfle et al. [71] consider the longitudinal displacement $u(x,t)$ of a one-dimensional viscoelastic rod. Within the framework of the ordinary Kelvin-Voigt model, it is described by the equation

$$\rho u_{tt}(x,t)=Eu_{xx}(x,t)+\eta u_{xxt}(x,t),$$

where η stands for viscosity. The fractional model is obtained by substituting the order of time derivative in the damping term by a fractional number α:

$$\rho u_{tt}(x,t)=Eu_{xx}(x,t)+\eta u_{xxt^\alpha}(x,t).$$

The boundary conditions contain information about external forces $F(x,t)$ acting on both ends of the rod:

$$F(0,t)=AEu_x(0,t)+\eta Au_{x,t^\alpha}(0,t)$$

and

$$F(b,t)=B\cos(\Omega t)=AEu_x(b,t)+A\eta u_{x,t^\alpha}.$$

The solution of the ordinary differential equation resulting after the Fourier transform with respect to time reads

$$\tilde{u}(x,\omega) = \pi \frac{[\delta(\omega-\Omega)+\delta(\omega+\Omega)]\cosh\left(x\sqrt{-\rho\omega^2/[E+\eta(i\omega)^\alpha]}\right)}{Ai\omega\sqrt{\rho[E+\eta(i\omega)^\alpha]}\sinh\left(b\sqrt{-\rho\omega^2/[E+\eta(i\omega)^\alpha]}\right)}\tilde{F}(\omega).$$

In the time domain this is a phase shifted stationary oscillation of the form

$$u(x,t) = U(x,\Omega)\cos[\Omega t + \varphi(x,\Omega)].$$

To verify this result and receive evidence that the fractional phenomenon takes place and can be observed, Kempfle et al. [71] carried out the following test. They took rods from teflon ($\rho = 3850$ kg/m^3, $E = 0.12\cdot 10^{10}$ N/m^2) of length 1 m and placed a force transducer linked with a shaker at one end, and an acceleration transducer at the other (free) end. Fitting the parameters $\alpha = 0.086$ and $\eta = 0.44\cdot 10^9$ Ns$^{\alpha/m^2}$, they observed a very good agreement of the fractional model with experimental data. More details can be found in (Shäfer 2000).

Atanackovic and Stankovic [5, 6] studied the lateral motion of an elastic horizontal column on a viscoelastic foundation. It is fixed at one end and loaded by a concentrated force F_0 at the other (free) end. Assuming that the column is positioned on a viscoelastic foundation described by a constitutive equation of fractional derivative type they derived a system of fractional equations for the lateral displacement $y(s)$ as a function of the arc-length of the rod axis $s \in [0, L]$. The system consists of the classical motion equation

$$EI\frac{\partial^4 y}{\partial s^4} + F_0\frac{\partial^2 y}{\partial s^2} + \rho\frac{\partial^2 u}{\partial t^2} + F = 0$$

and the fractional constitutive equation for a viscoelastic material of the foundation

$$F + \tau_{F0}\mathsf{D}_t^\alpha F = E_p(y + \tau_y\,{}_0\mathsf{D}_t^\alpha y)$$

with $0 < \alpha \leqslant 1$. In these equations, EI is the bending stiffness of the column, ρ is its line density, F is the force (per unit length) of the foundation on which the column is lying, the constants E_p, τ_F, and τ_y are instantaneous moduli of the foundation and the relaxation times, respectively. By solving this system, the stability boundary is determined. It is shown in the work cited that as in the case of an elastic (Winkler) type of foundation the stability boundary remains the same as for the column without a foundation (the column exhibits the so-called *Hermann-Smith paradox*).

As the authors assert, this model of viscoelastic foundation can be used, for example, to model railpads in a study of the stability of railway track.

In paper [1] the dynamics of the rigid body with memory is presented. The memory was described by the variables with distributed delay and by the CG fractional derivative.

2.5.2 Local Fractional Approach

The actual aim of the modern damage mechanics is to develop such models which are capable of describing microstructural effects. The passage to the microscales involves the fractality concept. Investigations of damage in materials, especially with porous concrete microstructures often exhibit a fractal-like structure at the mesolevel [22]. It makes use of the power kind of functions which leads directly to fractional integrals and derivatives. First attempts to apply the fractional calculus to the problem of fractal media, i.e. solids where the deformation is localized on a fractal subset, were undertaken on the local Kolwankar-Gangal (KG) fractional calculus [21, 23, 24, 26]. Recall that by using the local fractional KG operator D_x^α, one can represent the beginning of the Taylor expansion as

$$f(x) - f(y) \sim \frac{D_x^\alpha f(y)}{\Gamma(1+\alpha)}(x-y)^\alpha, \quad x \to y. \tag{2.77}$$

Thus, Carpinteri et al. [21, 24] modeling a specimen of disordered material by the *fractal Cantor bar* [43] define analytically the fractal strain ε^* as a local KG-derivative of the displacement $w(z)$:

$$\varepsilon^*(x) = D_x^\alpha w(x). \tag{2.78}$$

The explanation sounds as follows. As mentioned above, experimental results led ones to believe that a consistent modelling of damage in concrete and some other structures can be achieved by assuming that the rarefied resisting sections in correspondence of the critical load can be represented by stochastic lacunar fractal sets. Suppose to apply a tensile load in the x (axial) direction. Because of the fractional localization of strain, the plot of $w(z)$ is a Cantor staircase. This strain field is zero everywhere except in an infinite number of crack points where it is singular. The displacement singularities can be characterized by KG derivatives of order equal to the fractal dimension $\alpha = 1 - d_\varepsilon$ of the singularities. The fractional exponent $d_\varepsilon \in (0, 1)$ is intimately related to the degree of disorder in the mesoscopic damage process.

Assuming that the body is in equilibrium, Carpinteri et al. [21] consider a bar with $x = 0$ and $x = b$ being its extreme cross sections. Indicating with $p(x)$ the axial load per unit of fractal length acting upon the fractal bar, with $N(z)$ the axial force acting on the generic cross section, and using the rule of fractal integration by parts:

$${}_a I_b^\alpha [f(x) D_x^\alpha g(x)] = [f(x)g(x)]_a^b - {}_a I_b^\alpha [g(x) D_x^\alpha f(x)],$$

these authors obtain

$${}_a I_b^\alpha [N(x)\varepsilon^*(x)] =_a I_b^\alpha [N(x) D_x^\alpha w(x)] = [N(x)w(x)]_0^b - {}_a I_b^\alpha [w(x) D_x^\alpha N(x)].$$

According to the virtual work principle, this equality is true if and only if the static axial equation

$$D_x^{\alpha} N(x) + p(x) = 0$$

holds. The internal axial force in the equilibrium state is constant and equal to N throughout the bar. Kinematic equation becomes

$$D_x^{\alpha} w(x) = \frac{N}{EA} 1(x; C_{\alpha}^{[0,b]})$$

where E is the Young modulus, $C_{\alpha}^{[0,b]}$ is the triadic Cantor set built on $[0, b]$, A is a constant in the expression for the fractal strain $\varepsilon^* = N/EA$. Using the scaling property

$$D_x^{\alpha} w(x) = b^{1-\alpha} D_{\tilde{x}} \tilde{w}(\tilde{x}), \quad \tilde{x} = x/b, \quad \tilde{w} = w/b,$$

allows to present the equation in dimensionless form as follows:

$$D_{\tilde{x}}^{\alpha} \tilde{w}(\tilde{x}) = \frac{N}{EAb^{1-\alpha}} 1(\tilde{x}; C_{\alpha}^{[0,1]}).$$

Solving this equation and recovering the initial quantities yield

$$w(x) = \frac{Nb^{\alpha}}{EA\Gamma(1+\alpha)} S_C\left(\frac{z}{b}\right),$$

where $S_C(x)$ is the Cantor staircase built on the segment [0, 1]. As noted in [25], this equation provides important information about the size effect. It shows in particular that the displacement increases less than linearly with the bar length, as occurs with classical elastic bodies.

2.5.3 Nonlocal Approach

Another model based on non-local RL derivatives is suggested in [36] and described in [25] as well. Discretizing an elastic bar into m equal elements $V_j = A\delta x$, $j = 1, \ldots, m$, $\delta x = b/m$, and introducing long-range forces $Q^{(h,j)}$, $h \neq j$ that surrounding volume elements V_h apply on element V_j, the authors of this approach model these forces as

$$Q^{(h,j)} = \text{sign}(x_h - x_j)[w(x_h) - w(x_j)]g(x_h, x_j)V_j V_h,$$

with $g(x_h, x_j)$ being a power type decreasing function

$$g(x_h, x_j) = \frac{E c_\alpha \alpha}{A\Gamma(1-\alpha)|x_h - z_j|^{1+\alpha}}, \quad 0 \leqslant \alpha \leqslant 1.$$

Taking into account that $\Delta N_j \equiv N_{j+1} - N_j$ obeys the equation

$$\Delta N_j + \sum_{h=1}^{j-1} Q^{(h,j)} - \sum_{h=j+1}^{m} Q^{(h,j)} = -f_j A \Delta x$$

with $f_j \equiv f(x_j)$ being a longitudinal force per unit volume, yields

$$\Delta N_j - \frac{\alpha c_\alpha AE}{\Gamma(1-\alpha)} \left[\sum_{h=1}^{j-1} \frac{w(x_h) - w(x_j)}{(x_j - x_h)^{1+\alpha}} \Delta x - \sum_{h=j+1}^{m} \frac{w(x_j) - w(x_h)}{(x_j - x_h)^{1+\alpha}} \Delta x \right] = -f_j A \Delta x.$$

Dividing this equation by $EA\Delta x$ and assuming $\sigma(x) = E\varepsilon(x) = dw/dx$, the following equation is obtained by Di Paola and Zingales [36]:

$$\frac{d^2 w}{dx^2} - c_\alpha AE \left[\frac{\alpha}{\Gamma(1-\alpha)} \int_0^x \frac{w(x) - w(\xi)}{(x-\xi)^{1+\alpha}} d\xi \right.$$

$$\left. + \frac{\alpha}{\Gamma(1-\alpha)} \int_x^b \frac{w(x) - w(\xi)}{(\xi - x)^{1+\alpha}} d\xi \right] = -\frac{f(x)}{E}.$$

The square brackets contain integral terms in the Marchaud fractional derivatives.

Comparing these approaches, one of which deals with fractal media, i.e. with solids whose microstructure is such that strain localizes onto fractal subsets, and the other approach dealing with solids characterized by nonlocal long-range interaction, [25] note that the local approach is based on kinematic arguments, while the non-local one stems from a static analysis. Both the models introduce a displacement derivative of order lower than the classical one, but in the local model the fractional derivative replaces the classical counterpart, whereas in the nonlocal model it represents a correction to be added to the classical equilibrium equation. Finally, the first approach assumes that microstructure of the medium is fractal, but this aspect remains hidden in the second approach.

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Chapter 3
Porous Media



3.1 Diffusion

3.1.1 Main Concepts of Anomalous Diffusion

We start discussion of the problems related to discontinuous (porous, fractal, tortuous, boiling, bubbling and so on) media with diffusion (percolation) phenomena.

Diffusion theory is widely applied in geological and geophysical problems where the model objects are the processes of motion of groundwater aquifers, gas and oil in petroliferous stratums, and radioactive and toxic industrial waste. Consider the persisting chemical substance propagation through geological formations fully or partially saturated with water. Formations' heterogeneity reveals itself in breaks and cracks, lenticles and pores filled with other minerals (quartz, clay, and carbonates). This medium with all its inhomogeneities represents a realization of some statistical ensemble of the random media. The diffuse substance moves through the media due to the gravitation force and the density gradient. Its particles bypass impenetrable inclusions, they are transferred by underground flows and trapped in pores and penetrable inclusions. Every trajectory looks as a very irregular line, like a fibre with a lot of nodes in the places where the particle has been trapped for a long time, and long smooth sections ("free paths") where it was carried by an underground flow. Strictly speaking, neither the paths are rectilinear nor the nodes are point, but these differences can be neglected in the first approximation.

Accepting this approximation and complementing them by assumption on independency of path lengths and trapping times lead to the CTRW or even CTRF model. In the framework of the latter model, every trajectory of the walker is represented on the space-time diagram by a broken line. Each of its segments is parallel either to the coordinate axis (flying with an infinite speed) or the time axis (staying in a trap). Random directions of flights are usually assumed to be equiprobably, so only two functions are sufficient to statistically describe such motion: trapping times and flight lengths distributions.

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Discussing the distributions $q(t)$ of random trapping times T, Margolin and Berkowitz [81] firstly stated that there exist only two simple formulas for such distributions: the exponential formula

$$q(t) = \mu e^{-\mu t}, \quad t > 0$$

and the power one

$$q(t) = \text{const} \cdot t^{-\nu-1}, \quad t > t_1. \tag{3.1}$$

Of course, the fact that these functions are easy to differentiate and integrate can not serve as a physical foundation of the theory. The more essential fact is that the asymptotics of the first passage distribution of the particle has namely the power form with $\nu = 1/2$. The same result was obtained for the trapping time in a cubic Brownian trap [117]. But the power type distribution (3.1) is originated not only from diffusion processes, though the last formed a basis for non-Debye relaxation theory. A simple example given in Margolin and Berkowitz [81] shows that this type of distribution appears in the simplest hydrodynamic problem—the Poiseuille flow in a round tube. Consider a tube section of radius R and length L. The speed profile has a parabolic form

$$V(r) = V_0(1 - r^2/R^2), \quad 0 < r < R.$$

A liquid particle with such velocity passes the section during time

$$T = \frac{L}{V(r)} = \frac{L}{V_0}(1 - r^2/R^2)^{-1}.$$

Therefore,

$$\mathsf{P}(T > t) = \mathsf{P}\left(\frac{r^2}{R^2} > 1 - \frac{L}{V_0 t}\right) = \frac{L}{V_0 t},$$

and consequently

$$q(t) = -\frac{d\mathsf{P}(T > t)}{dt} = \frac{L}{V_0} t^{-1-\omega}, \quad \nu = 1.$$

However, examples cannot substitute a proof, which indeed can hardly be derived for such systems as geological structures. But what I find to be important here is that since within the CTRF model there are only two functions $q(r)$ and $q(t)$ completely determining the process,[1] then the asymptotical self-similar behavior is possible if one of the four conditions holds:

[1] We use this abbreviation for the *Continuous Time Radom Flight* that is a decoupled CTRW (*Continuous Time Radom Walk*) (See Vol. I, Sect 3.6.4).

$$\int_0^\infty rq(r)dr < \infty, \quad \int_0^\infty tq(t)dt < \infty;$$

$$\int_r^\infty p(r)dr \propto r^{-\alpha}, \quad \int_0^\infty tq(t)dt < \infty;$$

$$\int_0^\infty rp(r)dr < \infty, \quad \int_t^\infty q(t)dt \propto t^{-\nu};$$

$$\int_r^\infty p(r)dr \propto r^{-\alpha}, \quad \int_t^\infty q(t)dt \propto t^{-\nu}.$$

In the first case, the asymptotic part of CTRF solution satisfies the ordinary diffusion equation, the second condition yields superdiffusion with the fractional Laplace power, in the third case subdiffusion with fractional time derivative takes place, and the fourth condition leads to the space-time-fractional equation of anomalous diffusion. The problem is finally reduced to the question: if diffusion through the porous material under investigation really satisfies the self-similarity condition, at least as an intermediate asymptotics. The positive answer the question complemented by numerical values of α and ν, allows us to perform the numerical analysis based on solution of the fractional equation

$${}_0\mathsf{D}_t^\nu f(\mathbf{r}, t) = -(-\Delta)^{\alpha/2} f(\mathbf{r}, t) + S(\mathbf{r}, t).$$

This *first way* of solving the problem of diffusion in a porous media can be considered as phenomenological one. Following this way, we mentally deal with a homogeneous media, and all its specific property transfer to diffuse particles by prescribing them α, ν, and corresponding scale factors.

In the framework of the *second* approach, a model of porosity, or, more generally, a random medium model is constructed and motion of fluid through this model medium is considered. With a few exceptions any transport model can be escribed to one of these types. The concretization of the model is performed by properties and parameters of the walk in the first case, and statistical ensemble properties of a randomly inhomogeneous medium in the second one. We start with consideration of some examples of solution belonging to the first type.

3.1.2 Granular Porosity

Erochenkova and Lima [40] introduced a method, allowing (in statistical sense) to connect these two approaches. Consider a half-infinite cylinder filled with granulas. Voids between granulas represent a porous effect: a liquid percolates through the voids. Filling the cylinder sector with granulas again, we get different spatial distributions ω, each of them being described by a *porosity function*

$$\varepsilon_\omega(x) = [S(x)/S]_\omega,$$

which is a fraction of the area of void part of a cylinder section over the total section area (Fig. 3.1). The set $\Omega = \{\omega\}$ forms a statistical ensemble of such media, and $\varepsilon_\omega(x) \in [0, 1]$ can be considered as a random realization of some process.

The corresponding concentration $u_\omega(x, t)$ obey the diffusion equation

$$\varepsilon_\omega(x)\frac{\partial u_\omega}{\partial t} = K\frac{\partial^2 u_\omega}{\partial x^2}, \quad t > 0, \quad 0 \leqslant x < \infty, \tag{3.2}$$

with boundary conditions

$$u_\omega(0, t) = u_0 = \text{const}, \quad u_\omega(x, 0) = 0, \quad u_\omega(x, \infty) = 0.$$

The derivation of this equation is based on the following concepts. The amount of the liquid, flowing through the cylinder cross-sections $x = x_1$ and $x = x_2$ during time dt, equals correspondingly

$$dQ_1 = -K_\omega(x_1)\frac{\partial u_\omega(x_1, t)}{\partial x_1}Sdt,$$

$$dQ_2 = -K_\omega(x_2)\frac{\partial u_\omega(x_2, t)}{\partial x_2}Sdt.$$

The variation of the liquid amount in the layer (x_1, x_2) during time dt is given with the difference

$$dQ_1 - dQ_2 = \left[K_\omega(x_2)\frac{\partial u_\omega(x_2, t)}{\partial x_2} - K_\omega(x_1)\frac{\partial u_\omega(x_1, t)}{\partial x_1}\right]Sdt,$$

and during time (t_1, t_2)—by the integral

$$\Delta Q(t_1, t_2) = \int_{t_1}^{t_2}\left[K_\omega(x_2)\frac{\partial u_\omega(x_2, t)}{\partial x_2} - K_\omega(x_1)\frac{\partial u_\omega(x_1, t)}{\partial x_1}\right]Sdt.$$

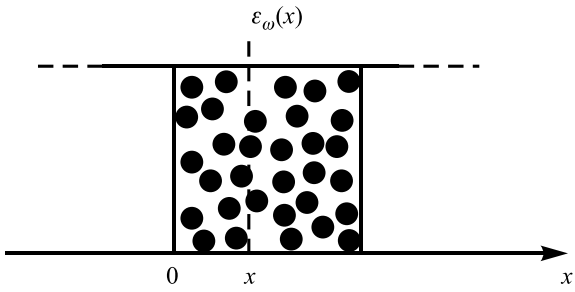


Fig. 3.1 A granular porosity model

Following the law of mass conservation (continuity theorem),

$$\Delta Q(t_1, t_2) = \int_{x_1}^{x_2} [u_\omega(x, t_2) - u_\omega(x, t_1)]\varepsilon_\omega(x) S dx.$$

By equating right-hand sides of these equations and assuming $(t_1, t_2) = (t, t + \delta t)$, $(x_1, x_2) = (x, x + \delta x)$ with $\delta t \to 0$, $\delta x \to 0$, we come to the equation

$$\varepsilon_\omega(x) \frac{\partial u_\omega}{\partial t} = \frac{\partial}{\partial x} K_\omega(x) \frac{\partial u_\omega}{\partial x}.$$

Within small liquid concentrations, K_ω can be replaced by the effective constant diffusion coefficient K, leading to the Eq. (3.2).

Let us write a random porosity $\varepsilon_\omega(x)$ as a sum of the coordinate independent average porosity $\overline{\varepsilon}$ and the random porosity fluctuation $\tilde{\eta}_\omega(x)$:

$$\varepsilon_\omega(x) = \overline{\varepsilon} + \eta_\omega(x). \tag{3.3}$$

Then, Eq. (3.2) becomes

$$\overline{\varepsilon} \frac{\partial u_\omega}{\partial t} + \eta_\omega \frac{\partial u_\omega}{\partial t} = K \frac{\partial^2 u_\omega}{\partial x^2}.$$

The percolation through a porous medium is accompanied by mixing trajectories of liquid particles, and this process produces self-averaging effect: the experimenter observes the average concentration $\langle u_\omega \rangle$ at the outlet of the cylinder (if the length of the cylinder L is much larger than the granule size l) [39]. This quantity satisfies the equation

$$\overline{\varepsilon} \frac{\partial \langle u_\omega \rangle}{\partial t} + \frac{\partial \langle u_\omega \eta_\omega \rangle}{\partial t} = K \frac{\partial^2 \langle u_\omega \rangle}{\partial x^2}. \tag{3.4}$$

Let us consider the fluctuating component of porosity $\eta_\omega(x)$ in the form of

$$\eta_\omega(x) = (-1)^{N(x)} A,$$

where $N(x)$ is a Poisson process,

$$\mathsf{P}(N(x) = k) = \frac{(\mu x)^k}{k!} e^{-\mu x}, \quad k = 0, 1, 2, \ldots,$$

and A is a random variable independent of it and taking two equally probable values $\pm a$ with equal probabilities:

$$\langle A \rangle = 0, \quad \langle A^2 \rangle = a^2.$$

Evidently, conditional and unconditional averaging respectively yield

$$\langle \eta_\omega(x)|A=a\rangle = a\sum_{k=0}^{\infty}(-1)^k\frac{(\mu x)^k}{k!}e^{-\mu x} = ae^{-2\mu x},$$

$$\langle \eta_\omega(x)\rangle = 0, \quad \langle \eta_\omega^2(x)\rangle = a^2.$$

Similarly, for the correlation function within $x_2 > x_1$, we have

$$\begin{aligned}&\langle \eta_\omega(x_1)\eta_\omega(x_2)\rangle = \langle a(-1)^{N(x_1)}a(-1)^{N(x_2)}\rangle\\ &= a^2\langle(-1)^{2N(x_1)+N(x_2-x_1)}\rangle = a^2\langle(-1)^{N(x_2-x_1)}\rangle\\ &= a^2\sum_{k=0}^{\infty}(-1)^k\frac{[\mu(x_2-x_1)]^k}{k!}e^{-\mu(x_2-x_1)} = a^2e^{-2\mu(x_2-x_1)}.\end{aligned}$$

The following differentiation theorem takes place [107].

If $\phi_x[\eta_\omega(\cdot)]$ is some linear functional of the Markovian process $\eta_\omega(x')$, $x' < x$, with the zero mean $\langle\eta_\omega\rangle = 0$, then

$$\frac{\partial}{\partial x}\langle \eta_\omega(x)\phi_x[\eta_\omega(\cdot)]\rangle = \left\langle \eta_\omega(x)\,\frac{\partial}{\partial x}\phi_x[\eta_\omega(\cdot)]\right\rangle - 2\mu\langle \eta_\omega(x)\,\phi_x[\eta_\omega(\cdot)]\rangle. \tag{3.5}$$

Now let us come back to Eq. (3.4) and perform the Laplace transform with respect to time:

$$\overline{\varepsilon}\lambda\langle\hat{u}_\omega\rangle + \lambda\langle\eta_\omega\hat{u}_\omega\rangle = K\frac{\partial^2\langle\hat{u}_\omega\rangle}{\partial x^2}. \tag{3.6}$$

Using differentiation theorem (3.5), we have

$$\frac{\partial}{\partial x}\langle\eta_\omega\hat{u}_\omega\rangle = -2\mu\langle\eta_\omega\hat{u}_\omega\rangle + \left\langle\eta_\omega\frac{\partial\hat{u}_\omega}{\partial x}\right\rangle, \tag{3.7}$$

$$\frac{\partial}{\partial x}\left\langle\eta_\omega\frac{\partial\hat{u}_\omega}{\partial x}\right\rangle = -2\mu\left\langle\eta_\omega\frac{\partial\hat{u}_\omega}{\partial x}\right\rangle + \left\langle\eta_\omega\frac{\partial^2\hat{u}_\omega}{\partial x^2}\right\rangle. \tag{3.8}$$

On substituting the second derivative found from (3.4) into (3.8) and performing simple transformations, we see that the function

$$\hat{W}(x,\lambda) = \langle\eta_\omega\hat{u}_\omega\rangle$$

satisfies the equation

$$\frac{\partial^2\hat{W}}{\partial x^2} + 4\mu\frac{\partial\hat{W}}{\partial x} + \left(4\mu^2 - \frac{\overline{\varepsilon}\lambda}{K}\right)\hat{W} = \frac{a^2\lambda}{K}\langle\hat{u}_\omega\rangle \tag{3.9}$$

with the initial condition

$$\hat{W}(0,\lambda)=0.$$

The large x asymptotics of the solution to Eq. (3.9) is

$$\widehat{W}(x,\lambda)\approx-\frac{a^2\lambda}{2K}\int_0^x d\xi\sqrt{K/\bar{\varepsilon}\lambda}\langle\hat{u}(\xi,\lambda)\rangle\exp\left[-\left(2\mu+\sqrt{\bar{\varepsilon}\lambda/K}\right)(x-\xi)\right]$$
$$\sim\frac{a^2}{4\mu\sqrt{\pi\bar{\varepsilon}K}}\lambda^{1/2},\quad\lambda\to 0.$$

Using the inverse Laplace transformation yields the correlations

$$\langle\tilde{\varepsilon}u\rangle=\frac{1}{2\pi i}\int_C e^{\lambda t}\hat{W}(x,\lambda)d\lambda,$$

and inserting them into (3.4) leads to the equation

$$\bar{\varepsilon}\frac{\partial\langle u_\omega\rangle}{\partial t}=K\frac{\partial^2\langle u_\omega\rangle}{\partial x^2}+\frac{a^2}{4\mu\sqrt{\pi\bar{\varepsilon}K}}\ {}_0\mathsf{D}_t^{3/2}\langle u_\omega\rangle,$$

obtained in Erochenkova and Lima [40].

3.1.3 Fiber Porosity

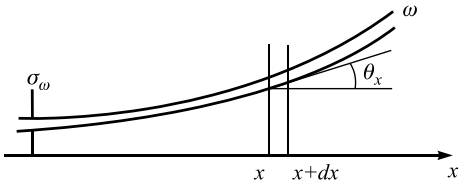
In the work [61] developing the same approach, a bit different structure is under consideration: trivariate 3d-space is filled with a chaotic set of convoluted flexible fibers (tubes), through which the process of normal diffusion proceeds. Generally, these tubes tend to be located along the x-axis. Let ω be one of such tubes (Fig. 3.2), σ_ω be its cross-section area, $\theta_\omega(x)$ be the angle between x-axis direction and the tube element direction. Suppose, the tubes are filled with steady fluid. At the instant $t=0$ in the plane $x=0$ one injects into all tubes a solution which diffuses along them according to Fick's law. Then

$$\frac{\partial u_\omega(x,t)\sigma_\omega dx}{\partial t}=q_\omega(x,t)\sigma_\omega\cos\left[\theta_\omega(x)\right]-q_\omega(x+dx,t)\sigma_\omega\cos\left[\theta_\omega(x+dx)\right]$$

and the continuity relation between the concentration of the diffunding matter and its current density $q_\omega(x,t)$ in the given tube ω is written in the form

$$\frac{\partial u_\omega(x,t)}{\partial t}=-\frac{\partial[q_\omega(x,t)\cos\theta_\omega(x)]}{\partial x}.$$

Fig. 3.2 Diffusion tube



Involving the Fick law

$$q_\omega(x,t) = -K_0 \frac{\partial u_\omega(x,t)}{\partial l} = -K_0 \cos[\theta_\omega(x)] \frac{\partial u_\omega(x,t)}{\partial x}$$

yields the equation of normal diffusion along the tube:

$$\frac{\partial u_\omega(x,t)}{\partial t} = \frac{\partial}{\partial x} \left\{ K_0 \cos^2[\theta_\omega(x)] \, \frac{\partial u_\omega(x,t)}{\partial x} \right\}. \tag{3.10}$$

Following the cited work, we suppose that

$$\sqrt{K_0} \cos[\theta_\omega(x)] = \bar{\varepsilon} + \eta_\omega(x)$$

is a random process, for example, the stationary Gaussian process with the mean $\bar{\varepsilon}$ and the correlation function

$$\langle \eta_\omega(x_1) \eta_\omega(x_2) \rangle = a^2 e^{-\mu |x_2 - x_1|}.$$

Here a and $1/\mu$ are supposed to be small.

We now average (3.10) over ω, assuming σ_ω statistically independent on $\theta_\omega(x)$:

$$\frac{\partial \langle u \rangle}{\partial t} - \overline{\varepsilon^2} \frac{\partial^2 \langle u \rangle}{\partial x^2} = 2\bar{\varepsilon} \frac{\partial}{\partial x} \langle \eta_\omega(x) \, \partial u(x,t)/\partial x \rangle + \frac{\partial}{\partial x} \langle \eta_\omega^2(x) \, \partial u(x,t)/\partial x \rangle.$$

Considering concentration $u_\omega(x,t)$ as a functional of $\eta_\omega(x)$ and applying Novikov-Furutsu formula [56] we get

$$\left\langle \eta_\omega \frac{\partial u}{\partial x} \right\rangle = \sum_{k=0}^{\infty} \frac{1}{k!} \int_{\mathbb{R}} dx_1 \cdots \int_{\mathbb{R}} dx_k \left\langle \frac{\delta^k [\partial u(x,t)/\partial x]}{\delta \eta_\omega(x_1) \cdots \delta \eta_\omega(x_k)} \right\rangle \Psi_k(x, x_1, \ldots, x_k),$$

where $\Psi_k(x, x_1, \ldots, x_k)$ is the density of $k+1$ point cumulant density of the random field $\eta_\omega(x)$. For given $\eta_\omega(x)$,

$$\Psi_1(x, x_1) = \langle \eta_\omega(x) \eta_\omega(x_1) \rangle = a^2 e^{-\mu |x - x_1|}$$

and

$$\Psi_k = 0, \quad k \neq 1.$$

Finally, we have

$$\langle \eta_\omega \, \partial u(x,t)/\partial x \rangle = a^2 \int_{\mathbb{R}} \frac{\partial \langle f(x,x',t) \rangle}{\partial x} e^{-\mu|x-x'|} dx',$$

where

$$f(x,x',t) = \frac{\delta u_\omega(x,t)}{\delta \eta_\omega(x')}.$$

Further,

$$\left\langle \eta_\omega^2 \frac{\partial u(x,t)}{\partial x} \right\rangle = \left\langle \eta_\omega (\eta_\omega \, \frac{\partial u(x,t)}{\partial x}) \right\rangle = a^2 \int_{\mathbb{R}} \left\langle \frac{\delta(\eta_\omega \partial u(x,t)/\partial x)}{\delta \eta_\omega(x_1)} \right\rangle e^{-\mu|x-x_1|} dx_1,$$

hence,

$$\left\langle \frac{\delta(\eta_\omega \partial u(x,t)/\partial x)}{\delta \eta_\omega(x_1)} \right\rangle = \delta(x-x_1) \frac{\partial u(x,t)}{\partial x} + a^2 \int_{\mathbb{R}} \left\langle \frac{\delta^2 \partial u(x,t)/\partial x}{\delta \eta_\omega(x_1) \delta \eta_\omega(x_2)} \right\rangle e^{-\mu|x-x_2|} dx_2,$$

and finally

$$\left\langle \tilde{\varepsilon}^2 \frac{\partial u(x,t)}{\partial x} \right\rangle = a^2 \frac{\partial \langle u(x,t) \rangle}{\partial x} + a^4 \int_{\mathbb{R}}^2 \left\langle \frac{\delta^2 \partial u(x,t)/\partial x}{\delta \tilde{\varepsilon}(x_1) \delta \tilde{\varepsilon}(x_2)} \right\rangle e^{-\mu(|x-x_2|+|x-x_1|)} dx_1 dx_2.$$

Applying functional differentiation to Eq. (3.10)

$$\frac{\partial f_\omega(x,x',t)}{\partial t} - \frac{\partial}{\partial x} \left[(\bar{\varepsilon} + \eta_\omega)^2 \frac{\partial f_\omega(x,x',t)}{\partial x} \right] = 2 \frac{\partial}{\partial x} \left[(\bar{\varepsilon} + \eta_\omega) \delta(x-x') \frac{\partial u_\omega(x,t)}{\partial x} \right]$$

and neglecting terms that include a^2 and a^4, we get

$$\frac{\partial \langle f(x,x',t) \rangle}{\partial t} - \bar{\varepsilon}^2 \, \frac{\partial^2 \langle f(x,x',t) \rangle}{\partial x^2} = 2\bar{\varepsilon} \frac{\partial [\delta(x-x') \partial \langle u(x,t) \rangle / \partial x]}{\partial x} \tag{3.11}$$

and

$$\frac{\partial \langle u(x,t) \rangle}{\partial t} - \bar{\varepsilon}^2 \frac{\partial^2 \langle u(x,t) \rangle}{\partial x^2} + 2\bar{\varepsilon} a^2 \frac{\partial}{\partial x} \int_{\mathbb{R}} \frac{\partial \langle f(x,x',t) \rangle}{\partial x} e^{-\mu|x-x'|} dx'. \tag{3.12}$$

After some bulky calculations, the system of Eqs. (3.11) and (3.12) is reduced to fractional differential equation

$$\frac{\partial \langle u(x,t)\rangle}{\partial t} = K' \frac{\partial^2 \langle u(x,t)\rangle}{\partial x^2} + \eta\, {}_0\mathsf{D}_t^{1/2} \frac{\partial^2 \langle u(x,t)\rangle}{\partial x^2},$$

obtained by Logvinova and Neel [61].

3.1.4 Filtration

One of the methods to describe a process of a non-stationary filtration of incompressible liquid in a half-infinite porous medium with its peculiar frow-through zones and stagnation regions (labelled by 1 and 0 respectively) is based on the system of equations for the corresponding pressures $p_1(x,t)$ and $p_0(x,t)$ $(x \geq 0,\ t \geq 0)$ [4]

$$\epsilon \frac{\partial p_1(x,t)}{\partial t} + (1-\epsilon)\frac{\partial p_0(x,t)}{\partial t} = K \frac{\partial^2 p_1(x,t)}{\partial x^2},$$

$$\frac{\partial p_0(x,t)}{\partial t} = \gamma(p_1 - p_0)$$

with boundary conditions

$$p_1(+0,t) = P_s(t),\ \ p_1(x,0) = p_1(\infty,t) = p_0(x,0) = p_0(\infty,t) = 0.$$

In these equations ϵ is the volume part of the flowing zones, γ is the coefficient of heat transfer between the zones, K is a diffusion coefficient, $P_s(t)$ is a given time-dependence of the surface pressure. There is required to find a pressure gradient on this surface $p'(t) \equiv \partial p_1(x,t)/\partial x|_{x=0}$.

Solving the second equation of the system and substituting the result into the first one, we get

$$(\mathsf{L}^2 - K\mathsf{D}_x^2)p_1(x,t) = 0, \tag{3.13}$$

where the operator L^2 is defined with the formula

$$\mathsf{L}^2 p_1(x,t) \equiv [\epsilon \mathsf{D}_t + (1-\epsilon)\gamma - (1-\epsilon)\gamma^2 e^{-\gamma t}\, {}_0\mathsf{D}_t^{-1}]p_1(x,t).$$

Performing direct calculations, one can make sure in validity of the expressions

$$\mathsf{L}^2 p_1(x,t) \equiv \left\{e^{-\gamma t}[\epsilon \mathsf{D}_t + (1-2\epsilon)\gamma - (1-\epsilon)\gamma^2\, {}_0\mathsf{D}_t^{-1}]e^{\gamma t}\right\} p_1(x,t)$$

and

$$\mathsf{L}p_1(x,t) = \left\{e^{-\gamma t}\sqrt{\epsilon \mathsf{D}_t + (1-2\epsilon)\gamma - (1-\epsilon)\gamma^2\, {}_0\mathsf{D}_t^{-1}}\, e^{\gamma t}\right\} p_1(x,t). \tag{3.14}$$

Splitting Eq. (3.13) results in

$$(\mathsf{L} - \sqrt{K}\mathsf{D}_x)(\mathsf{L} + \sqrt{K}\mathsf{D}_x)p_1(x,t) = 0,$$

which leads to the required gradient

$$\begin{aligned} p'(t) &= -\sqrt{1/K}\ \mathsf{L}p_1(0,t) \\ &= -\sqrt{1/K}e^{-\gamma t}\sqrt{\epsilon\mathsf{D}_t + (1-2\epsilon)\gamma - (1-\epsilon)\gamma^2\ {}_0\mathsf{D}_t^{-1}}e^{\gamma t}P_s(t). \end{aligned}$$

In the extreme case of continuous flow-through zone ($\epsilon = 1$), we find from above it follows the usual solution for diffusion in a viscous liquid

$$p'(t) = -\sqrt{1/K}\ {}_0\mathsf{D}_t^{1/2}P_s(t).$$

In another extreme case of continuous stagnant zone, we have $\epsilon = 0$,

$$\begin{aligned} p'(t) &= -\sqrt{1/K}e^{-\gamma t}\sqrt{\gamma - \gamma^2\ {}_0\mathsf{D}_t^{-1}}e^{\gamma t}P_s(t) \\ &= -\sqrt{\gamma/K}e^{-\gamma t}\ {}_0\mathsf{D}_t^{-1/2}\sqrt{\mathsf{D}_t - \gamma}e^{\gamma t}P_s(t) \\ &= -\sqrt{\gamma/K}e^{-\gamma t}\ {}_0\mathsf{D}_t^{-1/2}e^{\gamma t}\ {}_0\mathsf{D}_t^{1/2}P_s(t). \end{aligned}$$

And at last, in the case when mass transfer between the zones is absent ($\gamma = 0$) and

$$p'(t) = -\sqrt{1/K}\sqrt{\epsilon}\ {}_0\mathsf{D}_t^{1/2}P_s(t),$$

we have the same result as in the first case, but with the larger diffusion coefficient $K' = K/\epsilon$.

3.1.5 MHD Flow in Porous Media

In their recent papers, El-Shahed and Salem [38] and Odibat and Momani [91] have generalized the classical Navier-Stokes equations by replacing the first time derivative by a fractional derivative of order α, $0 < \alpha < 1$. They use the Laplace, Fourier sine, and finite Hankel transforms to obtain exact solution for the time fractional Navier-Stokes equations.

Chaurasia and Kumar [20] consider the time-fractional Navier-Stokes equation for the unsteady magnetohydrodynamic (*MHD*) flow of a viscous fluid in a porous medium past a circular cylinder of radius R and length L:

$${}_0^{\alpha}\mathsf{D}_t u(r,t) = P/\rho + \nu\left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r}\right) - Au \tag{3.15}$$

where

$$P = -\frac{\partial p}{\partial z}$$

is the pressure drop along z-axis coinciding with the cylinder axis. The coefficient

$$A = \frac{\sigma}{\rho} B_0^2 + \frac{\nu}{K}$$

contains the fluid electrical conductivity σ, the magnetic field intensity B_0, and the porous medium permeability K. The initial and boundary conditions for $u = u(t, t)$ are

$$u(r, 0) = u_0(r), \quad u(R, t) = 0, \quad u(0, t) \text{ is finite.}$$

Applying the *Hankel transform*

$$\breve{f}(\xi_n) = \mathcal{H}\{f(r)\}(\xi_n) = \int_0^R J_0(\xi_n r) f(r) r dr,$$

$$f(r) = \frac{2}{R^2} \sum_{n=1}^{\infty} \frac{J_0(r\xi_n)}{J_1(R\xi_n)} \breve{f}(\xi_n)$$

(with ξ_n being the roots of the equation $J_0(\xi) = 0$) reduces the partial differential equation to

$${}_0^{\alpha}\mathsf{D}_t \breve{u}(\xi_n, t) = (P/\rho\xi_n) R J_1(\xi_n R) - \nu \xi_n^2 \breve{u}(\xi_n, t) - A\breve{u}(\xi_n, t).$$

Using the Laplace transform with respect to time t yields an algebraic equation solution of which is

$$\bar{\breve{u}}(\xi_n, \lambda) = \frac{P R J_1(\xi_n R)}{\rho \xi_n} \frac{\lambda^{-1}}{\lambda^{-\alpha} + \nu \xi_n^2 + A} + \frac{\breve{U}(\xi_n) \lambda^{-\alpha}}{\lambda^{-\alpha} + \nu \xi_n^2 + A}.$$

Returning to the origin space-time variable by inverse transformation, one obtains the following exact solution:

$$u(r, t) = \frac{2Pt^{\alpha}}{\rho R} \sum_{n=1}^{\infty} \frac{J_0(\xi_n r)}{\xi_n J_1(\xi_n R)} E_{\alpha,\alpha+1}\left[-(\nu \xi_n^2 + A) t^{\alpha}\right] + \frac{2}{R^2} \sum_{n=1}^{\infty} \frac{J_0(r\xi_n)}{J_1(R\xi_n)} U(\xi_n) E_{\alpha,1}\left[-(\nu \xi_n^2 + A) t^{\alpha}\right]. \tag{3.16}$$

When $\alpha = 1$, the solution becomes

$$u(r,t) = \frac{2P}{\rho R}\sum_{n=1}^{\infty}\frac{J_0(r\xi_n)}{\xi_n J_1(R\xi_n)}\frac{1-e^{-(\nu\xi_n^2+A)t}}{\nu\xi_n^2+A} + \frac{2}{R^2}\sum_{n=1}^{\infty}\frac{J_0(r\xi_n)}{J_1(R\xi_n)}U(\xi_n)e^{-(\nu\xi_n^2+A)t}. \tag{3.17}$$

The result (3.16) may be obtained in a different way: one can solve Eq. (3.15) with $\alpha = 1$ and then pass from solution (3.16) to (3.17) by using the interrelation between this two solutions considered in Sect. 5.2.16 of Vol. I.

3.1.6 *Advection-Diffusion Model*

Dentz and Tartakovsky [31] presented a macroscopic model of non-Fickian diffusion in porous media, that postulates a time delay τ_d between the cause and effect in Fick's law,

$$\mathbf{J}_d(\mathbf{r},t) = -D\nabla c(\mathbf{r}, t-\tau_d), \tag{3.18}$$

and a time delay τ_a in the advective flux,

$$\mathbf{J}_a(\mathbf{r},t) = \mathbf{u}c(\mathbf{r}, t-\tau_a).$$

This time delay can be caused by variable porosity and adsorption properties of the porous medium, as well as by trapping in low flow zones and in closed streamlines. Inserting these terms into the mass balance equation

$$\frac{\partial c}{\partial t} = \nabla\cdot\mathbf{J}_a(\mathbf{r},t) + \nabla\cdot\mathbf{J}_d(\mathbf{r},t)$$

leads to a delayed advection-diffusion equation

$$\frac{\partial c}{\partial t} = \nabla\cdot[\mathbf{u}c(\mathbf{r}, t-\tau_a)] - D\Delta c(\mathbf{r}, t-\tau_d) = 0.$$

The authors assumed that the transport described by the equation with deterministic delay times becomes Fickian at long times, that is

$$c(\mathbf{r}, t-\tau) = c(\mathbf{r},t) - \frac{t\partial c(\mathbf{r},t)}{\partial t}\frac{\tau}{t} + \cdots \sim c(\mathbf{r},t), \quad t \gg \tau,$$

and, moreover, this statement remains valid for the average concentration in case of random τ, but only if its expectation is finite. In other words, they suppose that

$$\langle c(\mathbf{r}, t-\tau)\rangle = \langle c(\mathbf{r},t)\rangle - \frac{t\partial\langle c(\mathbf{r},t)\rangle}{\partial t}\frac{\langle\tau\rangle}{t} + \cdots \sim \langle c(\mathbf{r},t)\rangle, \quad t \gg \langle\tau\rangle.$$

If $\langle \tau_{a,d} \rangle = \infty$ but the delay times themselves have power type tails, $\mathsf{P}(\tau_a > t) \propto t^{-\alpha}$ and $\mathsf{P}(\tau_d > t) \propto t^{-\delta}$, then the advection-diffusion equation for the concentration averaged over random delay times takes the fractional form

$$\frac{\partial \langle c \rangle}{\partial t} + \theta_a^{-\alpha} \, {}_0\mathsf{D}_t^{-\alpha} \nabla \cdot \langle \mathbf{u} c \rangle = D \theta_d^{-\delta} \, {}_0\mathsf{D}_t^{-\delta} \Delta \langle c \rangle + S(\mathbf{r}, t),$$

where $\theta_{a,d}$ are time-dimension constants, and S is the source density. When $\alpha = \delta = 0$, we recognize here an ordinary advection-diffusion equation for $\rho \equiv \langle c \rangle$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\mathbf{u} \rho) = D \Delta \rho + S(\mathbf{r}, t),$$

which for an incompressible fluid with space-dependent diffusivity D becomes

$$\frac{\partial \rho}{\partial t} + \mathbf{u} \cdot \nabla \rho = \nabla \cdot D \nabla \rho + S(\mathbf{r}, t).$$

These equations model the flow and spread of contaminant particles carried by a fluid through a porous medium. The spreading of a contaminant plume in such model is due to mechanical dispersion, the velocity variations imposed by the tortuosity of paths the particles must take to navigate around obstacles in the porous medium. This is why the equations are also called the *advection-dispersion equations*.

Another, space-fractional generalization of these equations was introduced by Meerschaert et al. [85] (see also Meerschaert et al. [84]) to model ground water flow:

$$\frac{\partial \rho}{\partial t} = -\mathbf{u} \cdot \nabla \rho(\mathbf{r}, t) + K \mathbb{D}_M^{\alpha} \rho(\mathbf{r}, t)$$

where $\alpha \in (1, 2]$, $M(d\Omega)$ is the mixing measure, K is the diffusivity constant, and the fractional differential operator $\mathbb{D}_M^{\alpha}$ is defined in terms of its Fourier transform

$$\int e^{-i\mathbf{k}\cdot\mathbf{r}} \mathbb{D}_M^{\alpha} \rho(\mathbf{r}, t) d\mathbf{r} = \int_{||\Omega||=1} (i\mathbf{k} \cdot \mathbf{r})^{\alpha} \tilde{\rho}(\mathbf{r}, t) M(d\Omega).$$

The operator can also be represented as a mixture of fractional directional derivatives,

$$\mathbb{D}_M^{\alpha} \rho(\mathbf{r}, t) = \int_{||\Omega||=1} {}_0\mathsf{D}_{\Omega}^{\alpha} \rho(\mathbf{r}, t) M(d\Omega),$$

and, in case of choice of the uniform mixing measure ($M = M_0$), be reduced to the Riesz fractional operator:

$$\mathbb{D}_{M_0}^{\alpha} = -c(-\Delta)^{\alpha/2}.$$

One can also easily check by means of the Fourier transform, that

$$\nabla \cdot \nabla_M^{\alpha-1} \rho = \mathbb{D}_M^{\alpha} \rho.$$

A physical derivation of the fractional advection-dispersion equation was developed in Schumer et al. [105] by introducing the fractional Fick law as the sum of two terms representing the advective flux and the dispersive flux respectively:

$$\mathbf{J} = \mathbf{u}\rho - K\nabla_M^{\alpha-1}\rho.$$

Combining it with the classical mass-balance equation gives

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\mathbf{u}\rho - K\nabla_M^{\alpha-1}\rho) = -\nabla \cdot \mathbf{u}\rho + \mathbb{D}_M^{\alpha}\rho(\mathbf{r}, t).$$

Explaining the physical meaning of the fractional dispersive flux, Meerschaert et al. [85] write that it "is revealed by decomposing the fractional gradient into its radial components. The fractional gradient is a weighted sum of the fractional directional derivatives $\mathsf{D}_{\boldsymbol{\Omega}}^{\beta}\rho$ of the contaminant particle density in each direction, laid out along those directions. If the mixing measure M is a point mass at each coordinate vector $\mathbf{e}_j$, then $\mathsf{D}_{\mathbf{e}_j}^{\beta}\rho = \partial^{\beta}\rho/\partial x_j^{\beta}$ for $j = 1, 2, \ldots, d$ and then we get a simplified form of the fractional gradient in explicit coordinate form

$$\nabla_M^{\beta}\rho = \frac{\partial^{\beta}\rho}{\partial x_1^{\beta}}\mathbf{e}_1 + \cdots + \frac{\partial^{\beta}\rho}{\partial x_d^{\beta}}\mathbf{e}_d$$

that reduced to the usual gradient when $\beta = 1$ [105].

3.1.7 Reaction-Diffusion Equations

Many works using fractional diffusion equations of various form are devoted to chemical reactions running in nonhomogeneous say porous media. In a homogeneous case, the process is traditionally described by the reaction-diffusion equation

$$\frac{\partial c}{\partial t} = D\Delta c + R(c),$$

where c is a given particle concentration, D is the diffusion coefficient, and $R(c)$ is a reaction rate. This equation is based on assumption that the transport process is adequately represented by the standard classical diffusion model, known also as Brownian motion, at any spatial and time scales.

However, dealing with porous materials leads us to another kind of structures, which is rather fractal then homogeneous one. This suggests to involve into the equation fractional derivatives instead of integer-order ones. Keeping in mind, that the normal diffusion equation

$$\frac{\partial c}{\partial t} = D\Delta c + S$$

is derived from two first-order equations: the continuity equation

$$\frac{\partial c}{\partial t} = -\nabla \cdot \mathbf{J}$$

and the Fick's law

$$\mathbf{J} = -D\nabla c.$$

We find ourselves in front of the question: which of these two equations should be fractionalized? Majority of authors resolve the question in favour of the second equation. Firstly, the continuity equation can be considered as a fundamental equation following directly from the mass-balance law whereas the Fick's equation is a more or less appropriate approximation to the real process in a homogeneous medium. Secondly, many investigations confirm this choice: Chaves [21] showed that fractional Fick's law can recover Lévy statistics, Schumer et al. [106] derived a simple fractional Fick's law in the context of hydrological models and showed that it was able to resemble solute plumes in granular aquifers, Paradisi et al. [93] obtained a fractional Fick's law generating the Lévy Feller statistics. Del Rio and Lopez de Haro [30] obtained approximate time evolution equations for fluxes predicting a finite propagation velocity within the porous medium.

Compte and Metzler [25] have proposed fractional modifications to the Maxwell-Cattaneo constitutive equation, which is used by Valdes-Parada et al. [121] for description of reactions in disordered porous media. They considered the process of diffusion and reaction in a porous catalytic particle assuming that only two phases are presented in it: the solid (κ) and fluid phase (γ). The chemical reaction takes place only on the surface of the solid phase. The transport process in the γ-phase satisfies the conservation equation

$$\frac{\partial c_\gamma}{\partial t} + \nabla \cdot \mathbf{J}_\gamma = 0, \quad \mathbf{r} \in \gamma. \tag{3.19}$$

The boundary condition at the $\gamma\kappa$-interface with the inward normal surface vector $\mathbf{n}_{\gamma\kappa}$

$$\mathbf{n}_{\gamma\kappa} \cdot \mathbf{J}_\gamma = k c_\gamma, \quad \mathbf{r} \in \gamma\kappa. \tag{3.20}$$

Instead of the classical Maxwell-Cattaneo constitutive equation

$$\tau \frac{\partial \mathbf{J}_\gamma}{\partial t} + \mathbf{J}_\gamma = -D_\gamma \nabla c_\gamma,$$

Valdes-Parada et al. [121] use its fractional counterpart

$$\tau^\alpha \, {}_0\mathrm{D}_t^\alpha \frac{\partial \mathbf{J}_\gamma}{\partial t} + \mathbf{J}_\gamma = -D_\gamma \nabla c_\gamma, \quad \alpha \in [0, 1]. \tag{3.21}$$

Combining Eqs. (3.19) and (3.21) yields

$$\tau^{\alpha}\,{}_0\mathsf{D}_t^{\alpha+1}c_\gamma + \frac{\partial c_\gamma}{\partial t} = D_\gamma \Delta c_\gamma, \quad \mathbf{r} \in \gamma.$$

Passing to the microscopic reaction-diffusion model for catalyst reactive systems is realized by combination of the mass-balance averaged within a spatial scale,

$$\frac{\partial \langle c\rangle}{\partial t} + \nabla \cdot \langle \mathbf{J}\rangle = k_{\text{eff}}\langle c\rangle, \tag{3.22}$$

and the corresponding fractional constitutive equation

$$\tau^{\alpha}\,{}_0\mathsf{D}_t^{\alpha}\langle \mathbf{J}\rangle + \langle \mathbf{J}\rangle = -D_{\text{eff}}\nabla\langle c\rangle, \quad \alpha \in [0,1]$$

brings to the time-fractional equation

$$\tau^{\alpha}\,{}_0\mathsf{D}_t^{\alpha+1}\langle c\rangle + \tau^{\alpha}k_{\text{eff}}\,{}_0\mathsf{D}_t^{\alpha}\langle c\rangle + \frac{\langle \partial c\rangle}{\partial t} = D_{\text{eff}}\Delta\langle c\rangle + k_{\text{eff}}\langle c\rangle, \quad \alpha \in [0,1],$$

which governs the dynamic of the average concentration $\langle c\rangle$.

In their next work (Valdes-Palada et al. [121]), the authors use the other version of the fractionalized Fick law,

$$\langle \mathbf{J}\rangle = -D_{\text{eff}}\nabla^{\alpha}\langle c\rangle, \quad \alpha \in (0,1],$$

combination of which with Eq. (3.22) yields a space-fractional reaction-diffusion equation

$$\frac{\partial \langle c\rangle}{\partial t} = \nabla \cdot (D_{\text{eff}}\nabla^{\alpha}\langle c\rangle) + k_{\text{eff}}\langle c\rangle. \tag{3.23}$$

The main aim of the cited paper declared by its authors is to derive a fractional spatial averaging theorem and demonstrate the consistency of the *pseudo-homogeneous model* represented by Eq. (3.23) relative to the corresponding heterogeneous model. It seems a bit puzzling that the authors of this article do not refer to their previous article cited above. The absence of numerical results disables us to compare these two approaches and make necessary conclusion. One can say only, that the last equation represents a Markovian process, but interpretation of non-locality in terms of Fick's law seems to be more difficult problem than the memory phenomena in non-Markovian processes. Park et al. [95] investigated the behavior of motile particles in porous systems.

3.2 Fractional Acoustics

3.2.1 Lokshin-Suvorova Equation

We start the discussion of the fractional wave equations by citing the results of Russian authors [63]. Their approach can be considered as a further developing the Rabotnov rheological concept (Sect. 2.3.7). Writing the dynamical stress-strain relation in a viscoelastic medium as

$$E\epsilon(t) = \left[\sigma(t) + \int_{-\infty}^{t} K(t-\tau)\sigma(\tau)d\tau\right]$$

and considering it as an integral equation in $\sigma(t)$,

$$\sigma(t) = -\int_{-\infty}^{t} K(t-\tau)\sigma(\tau)d\tau + E\epsilon(t),$$

one can represent its solution by the Neumann series as

$$\sigma(t) = E\left[\epsilon(t) - \int_{-\infty}^{t} K(t-\tau)\sigma(\tau)d\tau + \int_{-\infty}^{t} K(t-\tau)\int_{-\infty}^{\tau'} K(\tau-\tau')\sigma(\tau')d\tau d\tau' + \ldots\right]$$

$$= E\left[\epsilon(t) - \int_{-\infty}^{t} R(t-\tau)\epsilon(\tau)\,d\tau\right],$$

where

$$R(t) = \sum_{n=1}^{\infty}(-1)^{n-1}K^{(n)}(t), \quad K^{(1)}(t) = K(t) \tag{3.24}$$

and

$$K^{(n)}(t) = \int_{0}^{t} K(t-\tau)K^{(n-1)}(\tau)d\tau \equiv K(t) \star K^{(n-1)}(t).$$

The functions $K(t)$ and $R(t)$ defined for $t > 0$ are referred to as the *creepage kernel* and the *relaxation kernel* respectively. Experiments demonstrate that, as $t \to 0$, these functions rapidly increase, which gives us grounds to make use of singular (i.e., growing without bounds as $t \to 0$) kernels that satisfactorily simulate the experimental data in a certain time interval $0 < a < t < b$.

Let $u(x,t)$ describe the strain of the one-dimensional medium, ρ be its density and $S(x,t)$ be the volume stress density. Then the behavior of a residual elastic medium is determined by the known simultaneous equations

$$\rho\frac{\partial^2 u(x,t)}{\partial t^2} = \frac{\partial\sigma(x,t)}{\partial x} + S(x,t), \tag{3.25}$$

$$\frac{\partial u(x,t)}{\partial x} = \epsilon(x,t), \tag{3.26}$$

complemented by the constitutive equation

$$\begin{aligned}\sigma(x,t) &= E\left[\epsilon(x,t) - \int_{-\infty}^{t} R(t-\tau)\epsilon(\tau,x)d\tau\right] \\ &= E\left[\epsilon(x,t) - R(t) \star \epsilon(x,t)\right].\end{aligned} \tag{3.27}$$

Substituting Eq. (3.27) into Eq. (3.25) and eliminating $\epsilon(x,t)$ from the resulting relation with the use of Eq. (3.26) yields the equation

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + R \star \frac{\partial^2 u}{\partial x^2} = \frac{S}{\rho},$$

where E/ρ is taken to be equal to one and all independent variables are dimensionless. Choosing

$$K(t) = \Phi_{1-\nu}(t) = \frac{t_+^{-\nu}}{\Gamma(1-\nu)},$$

Lokshin and Suvorova arrived at the equation

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + \Phi_{1-\nu} \star \frac{\partial^2 u}{\partial x^2} + \Phi_{2(1-\nu)} \star \frac{\partial^2 u}{\partial x^2} - \cdots = \frac{S}{\rho},$$

or in terms of fractional derivatives

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = \frac{S}{\rho} - {}_0\mathsf{D}_t^{\nu-1}\frac{\partial^2 u}{\partial x^2} - {}_0\mathsf{D}_t^{2(\nu-1)}\frac{\partial^2 u}{\partial x^2} + \cdots.$$

3.2.2 *Schneider-Wyss Equation*

Spreading of a stress perturbation in viscoelastic media is accompanied by a viscoelastic damping and viscothermal losses. In some sense, these processes turn out to be intermediate between the ordinary diffusion governed by the equation

$$\frac{\partial f(x,t)}{\partial t} = K_1 \frac{\partial^2 f(x,t)}{\partial x^2} + S_1(x,t),$$

and the wave propagation in a pure elastic medium described by

$$\frac{\partial^2 f(x,t)}{\partial t^2} = K_2 \frac{\partial^2 f(x,t)}{\partial x^2} + S_2(x,t),$$

where K_j, $j = 1, 2$ are constants and S_j play a role of sources. It is reasonable to suppose that these equations are only two terms of the whole family

$$ {}_0\mathsf{D}_t^{\nu} f(x,t) = K_{\nu} \frac{\partial^2 f(x,t)}{\partial x^2} + S_{\nu}(x,t).$$

A part of this family has already been discussed in connection with the subdiffusion problem (see Sect. 5.2 of Vol. I). This subfamily includes equations with $\nu \in (0, 1]$, the exponent ν must not exceed unity to preserve the property of non-negativity of the solution in accordance with its probabilistic sense. But now we talk about wave processes described by alternating solution producing by another subfamily. In order to get to this domain with minimal efforts, one should only cross the point $\nu = 1$: the region $\nu \in (1, 2]$ contains fractional diffusive waves of the simplest structure (the Schneider-Wyss equation),

$$ {}_0\mathsf{D}_t^{\nu} f(x,t) = K_{\nu} \Delta f(x,t) + S_{\nu}(x,t), \quad \nu \in (1,2], \tag{3.28}$$

solution of which was discussed in Sect. 5.2.8 of Vol. I.

3.2.3 Matignon et al. Equation

Investigating the propagation of pressure waves in the air considered as a real medium with thermal and viscose losses, Matignon et al. [82] used the fractional equation

$$\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = -\varepsilon \, {}_0\mathsf{D}_t^{3/2} f(x,t) - \varepsilon^2 \frac{\partial f}{\partial t},$$

and in the next their work [83] the authors introduced the equation

$$\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = -\varepsilon \, {}_0\mathsf{D}_t^{1+\nu} f(x,t)$$

referring to the case with $\nu = 1/2$ as the Lokshin model.

In some sense different approach has been developed in Fellah and Depollier [43], Depollier et al. [32], Fellah et al. [45]. The authors consider acoustic waves in a homogeneous isotropic porous material saturated with a compressible and viscous fluid of density ρ, compressibility modulus K_a, and viscosity η. The wavelength of sound is assumed to be much larger than the sizes of the pores, so the porous material can be viewed as an *equivalent fluid*. Its wave motion is described by the Euler equation and the constitutive equations having the following frequency representation

(for a wave propagating along the x-axis):

$$-\rho\widetilde{a}(\omega)i\omega\widetilde{u}(x,\omega) = -\frac{\partial\widetilde{p}(x,\omega)}{\partial x},$$

$$-\widetilde{b}(\omega)i\omega\widetilde{p}(x,\omega) = -K_a\frac{\partial\widetilde{u}(x,\omega)}{\partial x},$$

$\widetilde{u}$, $\widetilde{p}$, $\widetilde{a}$, and $\widetilde{b}$ are Fourier images of the liquid particle velocity, the acoustic pressure, the dynamic tortuosity and the dynamic compressibility of the fluid (air) included in the porous material, respectively. Recall that a describes the inertial and viscous losses whereas b represents the thermal exchanges. Further, v is the particle velocity, p is the acoustic pressure, and K_a is the compressibility modulus of the fluid. Passing to the time domain transform these equation into integro-differential form:

$$\rho a \star \frac{\partial u}{\partial t} \equiv \rho\int_0^t a(t-t')\frac{\partial u}{\partial t'}dt' = -\frac{\partial p}{\partial x}, \tag{3.29}$$

$$b \star \frac{\partial p}{\partial t} \equiv \int_0^t b(t-t')\frac{\partial p}{\partial t'}dt' = -K_a\frac{\partial u}{\partial x}. \tag{3.30}$$

As known from theoretical and experimental investigations, the response factors $\widetilde{a}$ and $\widetilde{b}$ have the following high-frequency ($\omega \to \infty$) asymptotics

$$\tilde{a}(\omega) \sim a_\infty[1 + (i\omega\tau_a)^{-1/2}],$$

$$\tilde{b}(\omega) \sim 1 + (i\omega\tau_b)^{-1/2},$$

where τ_a and τ_b are physical coefficients, defining the model properties. In the time domain, their expressions are given by

$$a(t) = a_\infty\left[\delta(t) + (\pi t/\tau_a)^{-1/2}\right]$$

and

$$b(t) = \delta(t) + (\pi t/\tau_b)^{-1/2}.$$

The first term in the right-hand part of every above expression gives the instantaneous response of the medium while the second one reveals its memory.

Inserting this asymptotic expression into the system (3.29)–(3.30) transforms it to

$$A\frac{\partial u}{\partial t} + B_1\,{}_0^{1/2}\mathsf{D}_t u = -K\frac{\partial p}{\partial x},$$

$$\frac{\partial p}{\partial t} + B_2\,{}_0^{1/2}\mathsf{D}_t p = -K\frac{\partial u}{\partial x},$$

where A, B_1, and B_2 are specified constants. Differentiating the first equation with respect to x, the second one with respect to t and eliminating the mixed derivative $\partial^2 u/\partial x \partial t$ brings the system to a single equation:

$$\frac{\partial^2 p}{\partial x^2} - A\frac{\partial^2 p}{\partial t^2} - B\,{}_0^{3/2}\mathsf{D}_t\, p - C\frac{\partial p}{\partial t} = 0,$$

where

$$B = B_1/K^2 + AB_2/K^2,$$

and

$$C = B_1 B_2/K^2.$$

The equation belongs to the family of fractional generalizations of the classical telegraph equation, the solution of which corresponds to a normal diffusion at long times and a pure wave at short times.

A bit other way of thinking was chosen in our work [114]. It is based on the known law of attenuation of a plane wave propagating along z-axis in a lossy media

$$A(z) = A(0)e^{-\gamma z},$$

where the absorption coefficient γ depends on sound wave frequency, $\gamma = \gamma(\omega)$. The standard form of the relationship between wave number (length of wave vector) k, the frequency, and the absorption coefficient is as follows:

$$k = \frac{\omega}{c_0} + i\gamma(\omega),$$

where $\gamma(\omega) = a_2\omega^2$ [59]. This quadratic dependence takes place only under condition of small absorbtion which is almost always valid for gases, but for condensed media, the dependence $\gamma(\omega)$ can differ from quadratic. Particularly in a viscous medium with very high heat-conductivity, the coefficient of absorption increases quadratically at low frequencies and approaches a constant value at high frequencies:

$$\gamma(\omega) \sim \begin{cases} a_2\omega^2, & \omega \to 0, \\ a_0, & \omega \to \infty. \end{cases}$$

Uchaikin [114] uses the combined representation

$$\gamma(\omega) = a_0 + a_\nu|\omega|^\nu \tag{3.31}$$

with the exponent ν belonging to the interval $(0, 2)$ but not coinciding with its boundary values 0 and 2 [7, 110, 126]. Multiplying both sides of Eq. (3.31) by $-i\,\widetilde{p}_\nu(k, \omega)$,

$$k = \frac{\omega}{c_0} + i[a_0 + a_\nu |\omega|^\nu]$$

where

$$\widetilde{p}_\nu(k, \omega) = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dt e^{i(kx-\omega t)} p_\nu(x, t)$$

is the Fourier transform (with respect to space and time) of pressure in a sound wave, we obtain the Fourier image of the partial differential equation

$$\frac{\partial p_\nu}{\partial x} = -\frac{1}{c_0}\frac{\partial p_\nu}{\partial t} - \left[a_0 + a_\nu \left(-\frac{\partial^2}{\partial t^2}\right)^{\nu/2}\right] p_\nu(x, t).$$

Introducing the new variables $\tau = t - x/c_0$ and $f_\nu(x, \tau) = e^{a_0 x} p_\nu(x, \tau)$, we reduce the equation to a shorter form:

$$\left[\frac{\partial}{\partial x} + a_\nu \left(-\frac{\partial^2}{\partial \tau^2}\right)^{\nu/2}\right] f_\nu(x, \tau) = 0. \tag{3.32}$$

Assuming that a sound source is situated in the plane $x = 0$ we write a boundary condition in the form $f_\nu(0, \tau) = p_\nu(0, \tau) = p_0(\tau)$, then the corresponding solution is expressed through the one-dimensional symmetric stable density $g(\tau; \nu, 0)$ by the integral

$$p_\nu(x, \tau) = (a_\nu x)^{-1/\nu} e^{-a_0 x} \int_{-\infty}^{\tau} g\left((\tau - \tau')(a_\nu x)^{-1/\nu}; \nu, 0\right) p_0(\tau') d\tau', \ x > 0.$$

In case of an impulsive source, or at large distances from a short-time source, the solution takes the form

$$p_\nu(x, \tau) = (a_\nu x)^{-1/\nu} e^{-a_0 x} g\left(\tau (a_\nu x)^{-1/\nu}; \nu, 0\right) S_0, \quad S_0 = \int_{-\infty}^{\infty} p_0(\tau') d\tau'.$$

Observe that the pulse profile width increases with the distance x traversed by the wave as $x^{1/\nu}$, whereas its amplitude and consequently the total energy decreases as $x^{-1/\nu}$. In case $\nu = 2$, the formulas coincide with their classical counterparts

$$p_2(x, \tau) = \frac{1}{2\sqrt{\pi a_\nu x}} e^{-a_0 x} \int_{-\infty}^{\tau} \exp\left(-\frac{(\tau - \tau')^2}{4 a_\nu x}\right) p_0(\tau') d\tau', \ x > 0,$$

and

$$p_2(x, \tau) = \frac{1}{2\sqrt{\pi a_\nu x}} e^{-a_0 x} \exp\left(-\frac{\tau^2}{4 a_\nu x}\right) S_0$$

(see Eqs. (79.10) and (79.11) in Landau and Lifshitz [59]).

One should notice that Eq. (3.32) after applying the operator

$$\left[\frac{\partial}{\partial x} - a_\nu \left(-\frac{\partial^2}{\partial \tau^2}\right)^{\nu/2}\right]$$

becomes

$$\frac{\partial^2 f_\nu(x,\tau)}{\partial x^2} = a_\nu^2 \left(-\frac{\partial^2}{\partial \tau^2}\right)^{\nu} f_\nu(x,\tau) = 0. \tag{3.33}$$

3.2.4 Viscoelastic Loss Operators

The wave equations written above can be presented in the common form

$$\frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial x^2} = \mathsf{L} f,$$

with a *loss operator* L in their right hand sides. Initially, this representation was used by Szabo [109] who expressed L only through integer-order derivatives. To reconcile this approach with the classical viscoelastic wave equation, one should take

$$\mathsf{L} = \tau \mathsf{D}_t \Delta, \tag{3.34}$$

where τ is a relaxation time characterizing the medium. As indicated in Holm and Sinkus [52], a wave equation with such a loss operator represents a spring and dashpot in parallel, which corresponds to the classical Kelvin-Voigt model. Another physics-based loss operator,

$$\mathsf{L} = -\theta \mathsf{D}_t,$$

relates to a medium which in the high-frequency limit has a constant loss, independent of frequency.

Chen and Holm [24] involved in L the R-L time derivatives of an arbitrary non-integer order, and then C-G fractional derivatives as having better numerical properties:

$$\mathsf{L} = -\theta \, {}_0^{\gamma+1}\mathsf{D}_t. \tag{3.35}$$

Later, Chen and Holm reformulated the loss wave equation by using fractional Laplacian,

$$\mathsf{L} = -\theta \mathsf{D}_t (-\Delta)^{\gamma/2}. \tag{3.36}$$

When γ is taken 2, this operator becomes its classical counterpart (3.34). The next modification of this operator

$$\mathsf{L} = \theta\ {}_0\mathsf{D}_t^{\gamma-1}\Delta \tag{3.37}$$

with $\gamma \in (1, 2)$. As shown by Wismer [127], this form gives a causal wave equation. It was first derived by Caputo [16] on the base of a fractional generalization of the loss Hooke's law:

$$\sigma = c\epsilon + \eta\ {}_0^{\gamma}\mathsf{D}_t\epsilon,$$

where c is the stiffness and η is viscosity. The exponent γ covers the range from $\gamma = 0$ (no viscosity) to $\gamma = 1$ (classical viscosity). Bagley and Torvik [6] traced the origins of this relationship back to 1921 and motivated it by reference to molecular theories. Combining this equation with Newton's law for displacement u

$$\rho\frac{\partial^2 u}{\partial t^2} = \frac{\partial \sigma}{\partial x}$$

and taking into account the strain-displacement interrelation

$$\epsilon = \frac{\partial u}{\partial x}$$

yield a wave equation

$$\frac{1}{c_0^2}\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = \mathsf{L}u$$

with the loss operator

$$\mathsf{L} = \tau^{\gamma}\ {}_0^{\gamma}\mathsf{D}_t\left(\frac{\partial}{\partial x}\right)$$

being in accordance with Eq. (3.37).

Many other interesting details can be found by the reader in the book [101] and the review article [52].

3.3 Geophysical Applications

3.3.1 Water Transport in Unsaturated Soils

Transport of soil water has an influence on heat and solute transport in soils, on rates of biological processes in plants, transpiration and ground water replenishment, and performs many other functions. This knowledge finds applications in agronomy, hydrology, meteorology, and environmental protection. Therefore, simulations of water transport in soil have many applications in hydrology, meteorology, agronomy, environmental protection, and other soil-related disciplines. Success of a multitude of projects depends on the correctness of the model of soil water transport.

Usually, this process is simulated on the base of Richards' equation [98] derived on assumption that the Darcy law originally devised for saturated flow in porous

media is also applicable to unsaturated flow in porous media. For a one-dimensional horizontal soil column, this equation takes the form

$$\frac{\partial\theta}{\partial t} = \frac{\partial}{\partial x}\left[K(\theta)\frac{\partial\theta}{\partial x}\right],$$

where θ is the volumetric soil water content, K is the soil water diffusivity, x is the distance. Passage to the scaling variable $\xi = x/\sqrt{t}$ reduces the partial differential equation to an ordinary one

$$-\frac{\xi}{2}\frac{d\theta}{d\xi} = \frac{d}{d\xi}\left[K(\theta)\frac{d\theta}{d\xi}\right].$$

However, Pachepsky et al. [92] reviewing numerous experimental data have shown that the Richards equation is not general enough to simulate transport in various soils, and suggested to use its fractional generalization

$${}_0\mathrm{D}_t^{\gamma} = \frac{\partial}{\partial x}\left[K_\gamma(\theta)\frac{\partial\theta}{\partial x}\right]$$

becoming the classical Richards' equation as $\gamma = 1$. In this fractional model, the water flux Q is governed by the *fractionalized Darcy law*

$$Q = -\,{}_0\mathrm{D}_t^{1-\gamma}K_\gamma(\theta)\frac{\partial\theta}{\partial x} = \frac{1}{\gamma}\frac{\partial}{\partial t}\int_0^t \frac{K_\gamma(\theta)}{(t-\tau)^{1-\gamma}}\frac{\partial\theta(x,\tau)}{\partial x}d\tau.$$

simulation of this process Pachepsky et al. [92] simulate water transport in unsaturated soils by using the fractionalized Richards' equation.

3.3.2 Seepage Flow

It is known that in the seepage flow dynamics heavy oil with relaxation properties reveals non-Newtonian behavior. The equation of this fluid motion in porous media is written as [1]

$$\mathbf{v} + \tau_v\frac{\partial\mathbf{v}}{\partial t} = -\frac{k}{\mu}\left(1 + \tau_p\frac{\partial}{\partial t}\right)\nabla p,$$

where τ_v and τ_p is velocity relaxation and pressure retardation times. But subsequent works dealing with the fractional approach to the viscoelastic fluids motion (see Sect. 2.2.4 and References therein) Ge and Tong [46] have written the following equation

for the flow of viscoelastic fluid through porous media:

$$\mathbf{v} + \tau_v^{\alpha}\, {}_0\mathsf{D}_t^{\alpha}\mathbf{v} = -\frac{k}{\mu}\left(1 + \tau_p^{\beta}\, {}_0\mathsf{D}_t^{\beta}\right)\nabla p. \tag{3.38}$$

On the base of this equation, Wang and Tong [125] study the flow of a viscoelastic fluid in a horisontal well. The $2L$-long horizontal well is parallel to the impermeable boundaries of a reservoir of given thickness, conductivity, and capacity ϕc_{t}. The continuity equation is approximated by

$$\nabla \cdot \mathbf{v} = -\phi c_{\mathrm{t}} \frac{\partial p}{\partial t}. \tag{3.39}$$

Inserting Eq. (3.38) into Eq. (3.39) and imposing corresponding initial and boundary conditions, the authors obtain the starting formulation of the problem under consideration.

As show analysis of numerical calculation results, the pressure in the flow are influenced by the α, β but mainly in the initial stages of the process. It is smaller for the larger order of corresponding fractional derivative. Discussion of other details can be found by the reader at the end of the cited paper.

3.3.3 Foam Drainage Equation

Foams are of great importance in many technological processes and applications, and their properties are subject of intensive studies from both practical and scientific points of view [96]. The study of foam drainage equation is very significant for which the equation is a simple model of the flow of liquid through channels and nodes (intersection of four channels) between the bubbles, driven by gravity and capillarity . It has been studied by many authors [49, 51, 124]. The classical equation for description of the process is of the form

$$\frac{\partial u}{\partial t} = \frac{u}{2}\frac{\partial^2 u}{\partial x^2} - 2u^2\frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x}\right)^2.$$

Its fractional counterpart

$${}_0^{\alpha}\mathsf{D}_t u = \frac{u}{2}\frac{\partial^2 u}{\partial x^2} - 2u^2\, {}_0^{\beta}\mathsf{D}_x u + \left({}_0^{\beta}\mathsf{D}_x u\right)^2, \quad 0 < \alpha \le 1, \quad 0 < \beta \le 1$$

has been proposed in Dahmani et al. [28] and solved initially by the Adomian decomposition method and later by the VI method [27]. We discuss here the latter solution.

The simplest version of the fractional equation is the time-fractional equation

$$ {}_0^{\alpha}\mathsf{D}_t u = \frac{u}{2}\frac{\partial^2 u}{\partial x^2} - 2u^2\frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x}\right)^2, \quad 0 < \alpha \leq 1. $$

Its solution subject initial condition

$$ u(x,0) = f(x) = \begin{cases} -\sqrt{c}\tanh[\sqrt{c}(x-ct)], & x \leq ct; \\ 0, & x > ct \end{cases} $$

with c being the velocity of wavefront is considered in Dahmani and Anber [27].

The correction functional can be approximately written as follows:

$$ u_{n+1}(x,t) = u_n(x,t) + \int_0^t \lambda(\tau)\left[{}_0^{\alpha}\mathsf{D}_\tau u_n(x,\tau) - \frac{\tilde{u}_n}{2}\frac{\partial^2 \bar{u}_n}{\partial x^2} + 2\bar{u}_n^2\frac{\partial \bar{u}_n}{\partial x} - \left(\frac{\partial \bar{u}_n}{\partial x}\right)^2\right] d\tau. $$

Making this correction functional stationary and noticing that $\delta\bar{u}_n = 0$, one obtains

$$ \begin{aligned} \delta u_{n+1}(x,t) &= \delta u_n(x,t) + \int_0^t \lambda(\tau)\left[{}_0^{\alpha}\mathsf{D}_\tau u_n(x,\tau) + \mathsf{N}\bar{u}_n(x,\tau)\right] d\tau \\ &= \delta u_n(x,t) + \int_0^t \lambda(\tau)\, {}_0^{\alpha}\mathsf{D}_\tau u_n(x,\tau) d\tau \\ &= [1+\lambda(t)]\delta u_n(x,t) + \int_0^t \delta u_n(x,\tau)\lambda'(\tau) d\tau = 0. \end{aligned} $$

Consequently, the stationary conditions are

$$ 1 + \lambda(t) = 0 $$

and

$$ \lambda'(\tau) = 0, \quad 0 < \tau < t. $$

Therefore, the Lagrange multiplier can be identified as

$$ \lambda(\tau) = -1, \quad 0 < \tau < t, $$

and the following variation iteration formula has resulted:

$$ u_{n+1}(x,t) = u_n(x,t) - \int_0^t \left[{}_0^{\alpha}\mathsf{D}_\tau u_n - \frac{\bar{u}_n}{2}\frac{\partial^2 \bar{u}_n}{\partial x^2} + 2\bar{u}_n^2\frac{\partial \bar{u}_n}{\partial x} - \left(\frac{\partial \bar{u}_n}{\partial x}\right)^2\right] d\tau. $$

The follow-up calculations are evident:

$$u_0(x,t) = f(x),$$

$$u_1(x,t) = u_0(x,t) - \int_0^t \left[{}_0^{\alpha}\mathsf{D}_\tau u_0 - \frac{\bar{u}_0}{2}\frac{\partial^2 \bar{u}_0}{\partial x^2} + 2\bar{u}_0^2 \frac{\partial \bar{u}_0}{\partial x} - \left(\frac{\partial \bar{u}_0}{\partial x}\right)^2 \right] d\tau = f(x) + t f_1(x);$$

$$\begin{aligned} u_2(x,t) &= u_1(x,t) - \int_0^t \left[{}_0^{\alpha}\mathsf{D}_\tau u_1 - \frac{\bar{u}_1}{2}\frac{\partial^2 \bar{u}_1}{\partial x^2} + 2\bar{u}_1^2 \frac{\partial \bar{u}_1}{\partial x} - \left(\frac{\partial \bar{u}_1}{\partial x}\right)^2 \right] d\tau \\ &= f - f_2 t^{2-\alpha} + f_3 t + f_4 t^2/2 + f_5 t^3/3 - f_6 t^4/4, \end{aligned}$$

and so on. Here

$$\begin{aligned} f_1 &= (1/2) f f_{xx} - 2f^2 f_x + f_x^2, \\ f_2 &= \frac{f_1}{(2-\alpha)\Gamma(2-\alpha)}, \\ f_3 &= f_1 + (1/2) f_x f_{xx} - 2 f_x f^2 + f_x^2, \\ f_4 &= (1/2) f_x f_{1xx} + f_{xx} f_{1x} - 2(f^2 f_{1x} + 2 f f_1 f_x) + 2 f_x f_{1x}, \\ f_5 &= (1/2) f_{1x} f_{1xx} - 2(2 f f_1 f_{1x} + f_x f_1^2) + f_{1x}^2, \\ f_6 &= 2 f_1^2 f_{1x}. \end{aligned}$$

The authors of the cited work gives an algorithm for solution of the space-fractional foam-drainage equation

$$\frac{\partial u}{\partial t} - \frac{u}{2}\frac{\partial^2 u}{\partial x^2} + 2u^2 \, {}_0^{\beta}\mathsf{D}_x u - ({}_0^{\beta}\mathsf{D}_x u)^2 = 0, \quad 0 < \beta \le 1.$$

3.3.4 Seismic Waves

One of the first applications of the fractional diffusive-wave equation was found in frame of so-called *constant-Q model* providing a good parametrization of seismic attenuation in rocks, in oil exploration and seismology. Mainardi and Tomirotti [72] reformulated the model in terms of fractional derivatives. Their one-dimensional version is based on the following starting line: the motion equation

$$\rho \frac{\partial^2 u(x,t)}{\partial t^2} = \frac{\sigma(x,t)}{\partial x},$$

$$\varepsilon(x,t) = \frac{\partial u(x,t)}{\partial x}$$

and the constitutive equation

$$\varepsilon(x,t) = J_0\sigma(x,t) + J(t) \star \sigma(x,t).$$

The medium is supposed to be time-independent and homogeneous ($\rho = \text{const}$) and undisturbed for $t < 0$, the star denotes the time-convolution from $0+$ to t, $J(t)$ stands for creep compliance with instantaneous value $J_0 = J(0+) \geq 0$. The Laplace transform with respect to time reduces the equation for the response variable $w(x,t)$ (chosen among u, $v = \partial u/\partial t$, σ, or ε) to the second-order ordinary form

$$\frac{d^2\widehat{w}}{dx^2} = \mu^2(\lambda)\widehat{w}(x,\lambda),$$

where $\mu(\lambda) = \lambda[\rho\lambda\widehat{J}(\lambda)]^{1/2}$. The generalized wave equation was derived in the cited article on assumptions

$$J(t) - J_0 \propto t^{\nu}, \quad t \to 0+, \ \nu \in (0,1]$$

and

$$\lim \frac{\mu(\lambda)}{\lambda} = \sqrt{\rho J_0} := \frac{1}{c}.$$

Introducing the dimensionless rate of creep

$$\psi(t) := \frac{1}{J_0}\frac{dJ(t)}{dt} \geq 0 \quad t > 0$$

yields

$$\mu^2(\lambda) = (\lambda/c)^2[1 + \widehat{\psi}(\lambda)]$$

which leads to the generalized integro-differential wave equation

$$\frac{\partial^2 w}{\partial t^2} + \psi \star \frac{\partial^2 w}{\partial t^2} = c^2\frac{\partial^2}{\partial x^2}.$$

When the creep compliance satisfies the simple power law

$$J(t) = \frac{1}{\rho D}\frac{t^{\nu}}{\Gamma(\nu+1)},$$

then $\sigma = \rho \mathsf{D}\,{}_0^{\nu}D_t\varepsilon$, $0 < \nu < 1$, the equation becomes the fractional diffusive-wave equation:

$$ {}_0^{2-\nu}\mathsf{D}_t w(x,t) = D\frac{\partial^2 w}{\partial x^2}. $$

This equation was investigated in the works [65, 68, 70] and applied to the seismic pulse propagation in Mainardi and Tomirotty [72]. The fractional equation or its modifications are used in application to geophysical problems by many other authors.

In a later version [17], the seismic wave propagation in rocks is described by the 2D wave equation

$$ {}_0\mathsf{D}_t^{\beta}\epsilon = \Delta_\rho(\rho K\epsilon) + S, $$

where $\beta = 2(1-\gamma)$,

$$ \epsilon(x,z,t) = \frac{\partial u_x}{\partial x} + \frac{\partial u_z}{\partial z}, $$

u_x and u_z are the displacement components,

$$ \Delta_\rho = \frac{\partial}{\partial x}\frac{1}{\rho}\frac{\partial}{\partial x} + \frac{\partial}{\partial z}\frac{1}{\rho}\frac{\partial}{\partial z}, $$

ρ is assumed to be space-dependent function and S is the seismic waves source.

The authors of the cited paper show that the fractional wave equation is in agreement with the above propositions. Having performed the numerical modeling of a cross-well seismic experiment experiment using a Kjartansson's attenuation model, they write that fractional approach can provide important applications for porous media well, since fractional derivatives appear in Biot theory [12], related the strain-like quantity ϵ_z to the stress-like quantity τ_{xy} in the frequency domain through the equation

$$ \epsilon_z = \frac{1}{A}(i\omega)^{-\alpha}\tau_{xy}, \quad A = \text{const} $$

whose time-domain counterpart is

$$ \tau_{xy} = a\ {}_a\mathsf{D}_t^{\alpha}\epsilon(t). $$

It is worth to note that instead of direct numerical solution of the fractional differential equation undertaken by the authors the interrelation between solutions of equations with fractional order and first-order derivatives could be used (Sect. 5.2.16 of Vol. I).

Mori and Kawamura [88], El-Misiery and Ahmed [36] develop another fractional version of the earthquake model based on the spring-block Barridge-Knopoff (BK) model. Mori and Kawamura [88] investigated spatio-temporal correlations by simulating the one-dimensional version of the BK model consisting of N identical blocks of mass m, which are mutually connected with the two neighboring blocks via the elastic springs of the elastic constant k_c, and are also connected to the moving plate via the springs of the elastic constant k_p. Measuring the time t in inverse units of the characteristic frequency $\sqrt{k_p/m}$, Mori and Kawamura write the motion equation for

the displacement of the jth block as

$$\ddot{u}_j = Vt - u_j + l^2(u_{i+1} - 2u_j + u_{j-1}) - \phi(\dot{u}_j),$$

where $l = \sqrt{k_c/k_p}$ is the stiffness parameter, V is the speed of plate, and ϕ is the friction force (all these values are dimensionless). El-Misiery and Ahmed [36] reduced the system to the differential equation

$$\frac{\partial^2 u}{\partial t^2} = l^2 \frac{\partial^2 u}{\partial x^2} - u + Vt - \phi\left(\frac{\partial u}{\partial t}\right)$$

and then replaced the second order derivatives by fractional order ones:

$${}_0\mathsf{D}_t^{\beta} \frac{\partial u}{\partial t} = l^2 \, {}_0\mathsf{D}_t^{\alpha} \frac{\partial u}{\partial x} - u + Vt - \phi\left(\frac{\partial u}{\partial t}\right).$$

This equation was supplied with the boundary initial conditions

$$u(0,t) = 0, \quad u(1,t) = 0, \quad u(x,0) = 0, \quad u_t(x,0) = 0.$$

Numerical solving this problem for various values of α, β, t, and x shows earthquakes with very clear aftershocks, which gave the grounds to claim this model is very efficient for earthquake modeling.

3.3.5 Multi-degree-of-Freedom System of Devices

To obtain more information about seismic processes, various kinds of devices are used: viscous fluid dampers, solid viscoelastic dampers, friction dampers, and yielding metallic dampers. For the design of structures with such devices, it is important to develop a corresponding model to calculate seismic response for different installation options. The paper [19] presents such a multi-degree-of-freedom analysis approach. In some sense, it is similar to that used for the analysis of linear systems, for solving the equations of motion with fractional derivative terms for arbitrary forcing functions (see Sect. 1.5).

For a multi-degree of freedom structure installed with added viscoelastic dampers, the equations of motion caused by earthquake induced base acceleration are written in the cited paper as

$$\mathsf{M}\ddot{\mathbf{x}} + \mathsf{C}\dot{\mathbf{x}} + \overline{\mathsf{C}}\mathbf{x}^{(\alpha)} + \mathsf{K}\mathbf{x} = -\mathsf{M}\mathbf{r}\ddot{x}_g(t),$$

where $\mathbf{x}^{(\alpha)} = {}_0\mathsf{D}_t^{\alpha}\mathbf{x}(t)$, $\alpha \in (0,1)$, M is the mass matrix, C is the damping matrix, $\overline{\mathsf{C}}$ is the fractional damping matrix associated with the viscoelastic dampers, and K represents the total stiffness including the contributions from the structure as well as

viscoelastic dampers. The vector $\mathbf{x}$ denotes the relative displacement vector of the structure, $\ddot{x}_g$ is the ground acceleration, and $\mathbf{r}$ is the influence coefficient vector of size N where N is the system degree of freedom. Passage to the principal coordinates q_j reduces the equation to a set of uncoupled modal equations

$$ {}_0\mathsf{D}_t^{\nu} q_j(t) - \lambda_j q_j(t) = Q_j(t), \tag{3.40} $$

where $\nu = \alpha/l$ arises from the rational representation $\alpha = l/m$, λ_j are eigenvalues of the matrix $\mathsf{A}^{-1} \cdot \mathsf{B}$, $Q_j(t) = \rho_j \ddot{x}_g(t)$ is the modal forcing function. The solution to this equation is expressed through the impulse response function

$$ K_j(t) = \mathscr{L}^{-1}\left\{\frac{1}{\lambda^{\nu} - \lambda_j}\right\}(t) = \sum_{k=1}^{m} \lambda_j^{k-1} \mathscr{L}^{-1}\left\{\frac{\lambda^{1-k\nu}}{\lambda - \lambda_j^m}\right\}(t). $$

Taking into account that

$$ \frac{\lambda^{\alpha}}{\lambda - c} = \mathscr{L}\left\{ {}_0\mathsf{D}_t^{\alpha} e^{ct} \right\}(\lambda), $$

the authors write down the desired solution as

$$ q_j(t) = \sum_{k=1}^{m} \int_0^t {}_0\mathsf{D}_t^{1-k\nu} \exp[\lambda_j^m (t-\tau)] Q_j(\tau) d\tau. $$

The article [19] contains description of the numerical procedure and numerical results obtained with its application. It is ended with the conclusion that the proposed approach can be effectively used to predict such response reductions as the force response, floor acceleration response, and some others caused by viscoelastic dampers.

Dickmen [34] descries the seismic waves in soil structures using the finite element method, in which the displacement type formulation is given as

$$ u = N\check{u}, \tag{3.41} $$

where u, N, and $\check{u}$ stand for displacement, shape function, and displacement at nodal points of finite element. For plane strain condition, strain and nodal displacement are linked by the following equation

$$ \varepsilon = B\check{u}, \tag{3.42} $$

where B is the *kinematics matrix* [129].

The stress—nodal displacement interrelation in framework of the Kelvin-Voigt model takes the form

$$ \sigma = EB\check{u} + \eta B\, {}_0\mathsf{D}_t^{\alpha} \check{u}. \tag{3.43} $$

Denoting body and external forces by F_B and F_S respectively, Dikmen writes the principle of virtual work in the form of

$$\int_{\Delta V} \varepsilon^{\mathrm{T}} \sigma dV + \int_{\Delta V} F u^{\mathrm{T}} dV + \int_{S(\Delta V)} f u^{\mathrm{T}} dS = 0. \tag{3.44}$$

Substituting Eqs. (3.42)–(3.43) into Eq. (3.44) leads to the following equation of motion for multi degrees of freedom systems:

$$m\ddot{\breve{u}}(t) + c\ {}_0\mathsf{D}_t^{\alpha}\breve{u}(t) + k\breve{u}(t) = f(t),$$

where

$$m = \int_{\Delta V} \mathsf{N}^{\mathrm{T}} \rho \mathsf{N} dV,$$

$$c = \int_{\Delta V} \mathsf{B}^{\mathrm{T}} \eta \mathsf{B} dV,$$

and

$$k = \int_{\Delta V} \mathsf{B}^{\mathrm{T}} E \mathsf{B} dV$$

are consistent mass, damping, and stiffness respectively. This equation has numerically been solved by using the Newmark-Beta method. At the end of the article its author remarks that the main disadvantages of fractional order damping scheme is the increasing numerical effort and the storages requirements due to local operators. But, the fractional order damping scheme permits for the continuous transition from the fluid to the solid state to represent damping behavior in soil materials.

3.3.6 *Spatial-Temporal Distribution of Aftershocks*

As was noted in Sect. 2.7.3 of Vol. I, the first shock of an earthquake is normally accompanied with a sequence of seconds earthquake shocks: *aftershocks*. The aftershocks follow one after another in random time intervals at random points. Such a random distribution of aftershocks suggests to consider the sequence $\{\mathbf{x}_i, t_i, \}_{i=1,2,3,\ldots}$ as a trajectory of some Markovian chain in the coordinate-time phase space. This idea is realized in the series the Helmstetter and Sornette works (see the article [50] and References therein). In the framework of this model, the average density of shocks $n(\mathbf{x}, t)$ in a homogeneous boundless seismic medium obeys the integral equation

$$n(\mathbf{x}, t) = \int_{\mathbb{R}^3} \int_0^t K(\mathbf{x} - \mathbf{x}', t - t') n(\mathbf{x}', t') d\mathbf{x}' dt' + \delta(\mathbf{x})\delta(t),$$

where $K(\mathbf{x}-\mathbf{x}', t-t')d\mathbf{x}dt$ is the probability that the next shock will occur in the element $d\mathbf{x} \times dt$ around $(\mathbf{x}, t)$ under the condition that the preceding one happened at $(\mathbf{x}', t')$, and the last term in the equation relates to the first shock. Further development of the model is based on the *factorization hypothesis* assuming separation of spatial and temporal variables in the transition probability

$$K(\mathbf{x}, t) = \psi(\mathbf{x})\phi(t).$$

As a result, we obtain the factorized solution

$$n(\mathbf{x}, t) = f(\mathbf{x})g(t)$$

with multiplicands f and g obeying the corresponding equations: the spatial equation

$$f(\mathbf{x}) = \int \psi(\mathbf{x}-\mathbf{x}')f(\mathbf{x}')d\mathbf{x}' + \delta(\mathbf{x})$$

and the temporal equation

$$g(t) = \int \phi(t-t')g(t')dt' + \delta(t).$$

The second hypothesis lying in the base of the model is formulated by including the fractal assumption for spatial and temporal interval between consecutive aftershocks (see Sect. 2.7.3 of Vol. 1):

$$\int_{|\mathbf{x}|>r} \psi(\mathbf{x})d\mathbf{x} \propto r^{-\alpha},\ \alpha \in (0, 2], \quad \int_t^\infty \xi(t)dt \propto t^{-\beta},\ \beta \in (0, 1].$$

The Fourier-Laplace transform

$$N(\mathbf{x}, t) \ \mapsto \ \widetilde{N}(\mathbf{k}, \lambda) = \int_{\mathbb{R}^3}\int_0^\infty N(\mathbf{x}, t)e^{i\mathbf{k}\mathbf{x}-\lambda t}d\mathbf{x}dt$$

reduces the integral equation to the algebraic one which in the diffusion limit ($\mathbf{k} \to 0,\ 1-\widetilde{\psi}(\mathbf{k}) \sim |a\mathbf{k}|^\alpha, \quad \lambda \to 0,\ 1-\widetilde{\xi}(\lambda) \sim (b\lambda)^\beta$) becomes

$$(b\lambda)^\beta \widetilde{n}(\mathbf{k}, \lambda) = -|a\mathbf{k}|^\alpha \widetilde{n}(\mathbf{k}, \lambda) + 1.$$

Returning to the initial space-time variables, we obtain the differential equation with partial derivatives of fractional orders:

$$(b_0\mathsf{D}_t)^\beta n(\mathbf{x}, t) = -(-a\Delta)^{\alpha/2}n(\mathbf{x}, t) + \delta(\mathbf{x})\delta(t)$$

(see Uchaikin and Uchaikin [118]).

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Chapter 4
Thermodynamics

4.1 Classical Heat Transfer Theory

4.1.1 Heat Flux Through Boundaries

In thermodynamics, fractional differential operators are used for numerical solving some problems in the classical heat transfer theory, for modeling heat transfer processes in porous, thermoelastic, and other special materials with complex properties, and for attempts to generalize the classical thermodynamics to non-classical systems (systems with memory, nanosystems, etc.).

We begin with classical problems. The simplest problem of such kind is as follows. There is a semi-infinite rod ($x > 0$) with an isolated lateral surface having the same zero temperature everywhere till the moment ($t = 0$). Find a thermal current (more strictly, its x-projection) $q_0(t) \equiv q_x(0, t)$, with which the rod must be supplied through its end to provide a given temperature dependence $T_0(t) = T(0, t)$ of the end (Fig. 4.1).

To answer this question, we have to resolve the heat conduction equation for the temperature function $T(x, t)$,

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}, \quad x \geqslant 0, \ t \geqslant 0, \tag{4.1}$$

under conditions

$$T(0, t) = T_0(t), \ t > 0; \quad T(x, t) = 0, \ t \leqslant 0; \quad T(\infty, t) = 0, \ -\infty < t < \infty,$$

to calculate its gradient at $x = 0$,

$$T_0'(t) = \left(\frac{\partial T(x, t)}{\partial x} \right)_{x=0}$$

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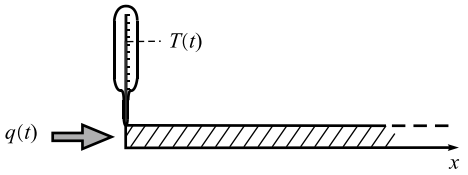


Fig. 4.1 To the problem of a rod heating

and then to make use of the Fourier law

$$q_0(t) = -\kappa T_0'(t).$$

Direct solving Eq. (4.1) by standard methods yields the temperature function

$$T(x,t) = \int_0^t \frac{1}{\sqrt{\kappa\pi(t-t')}} \exp\left[-\frac{x^2}{4\kappa(t-t')}\right] q_0(t')dt',$$

which at the end of the rode becomes

$$T_0(t) = \int_0^t \frac{q_0(t')dt'}{\sqrt{\kappa\pi(t-t')}},$$

or, in terms of fractional derivatives,

$$T_0(t) = \frac{1}{\sqrt{\kappa}} {}_0\mathsf{D}_t^{-1/2} q_0(t). \tag{4.2}$$

This result can be obtained in other way. Rewriting Eq. (4.1) as

$$\left(\frac{\partial}{\partial t} - \kappa\frac{\partial^2}{\partial x^2}\right) T(x,t) = \left({}_0\mathsf{D}_t^{1/2} - \sqrt{\kappa}\frac{\partial}{\partial x}\right)\left({}_0\mathsf{D}_t^{1/2} + \sqrt{\kappa}\frac{\partial}{\partial x}\right) T(x,t) = 0 \tag{4.3}$$

and taking into account that the solution of the equation

$$\left({}_0\mathsf{D}_t^{1/2} + \sqrt{\kappa}\frac{\partial}{\partial x}\right) T(x,t) = 0$$

satisfies Eq. (4.1), we obtain

$$\frac{\partial T(x,t)}{\partial x} = -\frac{1}{\sqrt{\kappa}} {}_0\mathsf{D}_t^{1/2} T(x,t).$$

Multiplying this equation by $-\kappa$ and assuming $x = 0$, we arrive at the Eq. (4.2)

$$q_0(t) = \sqrt{\kappa}\, {}_0\mathsf{D}_t^{1/2} T_0(t) \tag{4.4}$$

being inverse to Eq. (4.2). While acknowledging the authors (Gerasimov, Babenko, Oldham and Spanier, Caputo and Mainardi) whose theoretical works stimulated the penetration of fractional calculus in applied fields, it should be noted that the semi-derivatives occur as early as the first editions of Mechanics of Continuous Media by Landau and Lifshitz, that is in 1944–1953.

Later, Babenko [4] generalized this method to the process described by the equation

$$[\mathsf{D}_t - \alpha(x,t)\mathsf{D}_x^2 - \beta(x,t)\mathsf{D}_x + \gamma(x,t)]T(x,t) = 0.$$

Splitting this equation led to the following representation:

$$\left[\sum_{j=0}^{\infty} b_j(x,t)\,{}_0\mathsf{D}_t^{(1-j)/2} - B(x,t)\frac{\partial}{\partial x}\right]\left[\sum_{j=0}^{\infty} a_j(x,t)\,{}_0\mathsf{D}_t^{(1-j)/2} + A(x,t)\frac{\partial}{\partial x}\right]T(x,t) = 0,$$

where the new coefficients a_j, b_j, A, and B are determined by the original ones α, β and γ. Separating the equation

$$\left[\sum_{j=0}^{\infty} a_j(x,t)\,{}_0\mathsf{D}_t^{(1-j)/2} + A(x,t)\frac{\partial}{\partial x}\right]T(x,t) = 0$$

and substituting $x = 0$ yield directly the desired solution:

$$A(0,t)T_0'(t) = -\sum_{j=0}^{\infty} a_j(0,t)\,{}_0\mathsf{D}_t^{(1-j)/2}T_0(t), \tag{4.5}$$

where

$$T_0'(t) = \left(\frac{\partial T(x,t)}{\partial x}\right)_{x=0},$$

as above.

In particular, for constant coefficients α, β, γ, and $T_0 = \text{const}$, we obtain

$$\begin{aligned} T_0'(t) &= (1/\sqrt{\alpha})\left({}_0\mathsf{D}_t^{1/2} + \frac{\beta}{2\sqrt{\alpha}} + \frac{\delta}{2}\,{}_0\mathsf{D}_t^{-1/2} - \frac{\delta^2}{8}\,{}_0\mathsf{D}_t^{-3/2} + \cdots\right)T_0(t) \\ &= \left(\frac{1}{\sqrt{\pi t}} + \frac{\beta}{2\sqrt{\alpha}} + \frac{\delta}{\sqrt{\pi}}\sqrt{t} - \frac{\delta^2}{6\sqrt{\pi}}t\sqrt{t} + \cdots\right)T_0(t). \end{aligned} \tag{4.6}$$

Equation (4.6) is the basic formula of the splitting method for finding the heat and diffusion fluxes at the boundary of a semi-infinite domain for processes described by

a linear equation with arbitrary coefficients. The solution procedure reduces to the calculation of new coefficients using simple algebraic operations and differentiation. For example, if the coefficients do not depend on time, we have:

$$A = B = \sqrt{\alpha},$$

$$a_0 = 1, \quad b_0 = 1,$$

$$\begin{aligned}
a_1 &= -b_1 = \frac{\beta}{2\sqrt{\alpha}} - \frac{1}{4\sqrt{\alpha}}\frac{d\alpha}{dx}, \\
a_2 &= b_2 = \frac{1}{2}\left(\sqrt{\alpha}\frac{da_1}{dx} + a_1^2 + \gamma\right), \\
a_3 &= b_3 = \frac{1}{2}\left(\sqrt{\alpha}\frac{da_2}{dx}\right), \\
a_4 &= b_4 = \frac{1}{2}\left(\sqrt{\alpha}\frac{da_3}{dx} - a_2^2\right), \\
a_n &= b_n = \frac{1}{2}\left(\sqrt{\alpha}\frac{da_{n-1}}{dx} - \sum_{j=2}^{n-2} a_{n-j}a_j\right) \quad \text{for } n > 4.
\end{aligned} \tag{4.7}$$

A practical realization of this approach based on approximation of the temperature history at the point using a series containing fractional power terms whose coefficients are determined by using the Least Square Method is considered by Agrawal [1].

4.1.2 Flux Through a Spherical Surface

A bit more complicated problem is to determine a thermal or diffusion flux through a spherical boundary. This process is governed by equation

$$\left[\frac{\partial}{\partial t} - \kappa\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right)\right] T(r,t) = 0, \; r > 0, \; t \geqslant 0,$$

and the solution of the problem has the form

$$q_r(R,t) = \sqrt{\kappa}({}_0\mathrm{D}_t^{1/2} + \sqrt{\kappa}/R)T(R,t).$$

Among the problems, arising in the polarization theory, the diffusion of substance from a spherical liquid drop into the environment is considered. The environment is filled with some other field, the drop radius increases with time according to a known law $R = R(t)$, and a substance concentration $C_s(t)$ on the drop surface is given. The concentration inside the drop $c(r, t)$ is described by the equation

$$\left[\frac{\partial}{\partial t} - K\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right) + \frac{R^2\dot{R}}{r^2}\frac{\partial}{\partial r}\right] c(r,t) = 0,\ r > 0,\ t \geqslant 0$$

with the boundary conditions

$$c(R,t) = C_s(t); \quad c(\infty, t) = 0; \quad c(r,0) = 0, \text{ for } r > R(0).$$

Using the splitting method, Babenko [4] expresses the flow density on the drop surface in the form of the series:

$$j_r(R(t),t) = \sqrt{K}\left[{}_0\mathsf{D}_t^{1/2} + \frac{\sqrt{K}}{R(t)} + \frac{\dot{R}(t)}{2R(t)}{}_0\mathsf{D}_t^{-1/2} + \left(\frac{3\dot{R}^2(R)}{8R^2(t)} - \frac{\ddot{R}(t)}{4R(t)}\right){}_0\mathsf{D}_t^{-3/2} + \cdots\right] C_s(t).$$

With the constant concentration on the surface:

$$j_r(R(t),t) = \sqrt{K}\left[\frac{1}{\sqrt{\pi t}} + \frac{\sqrt{K}}{R(t)} + \frac{\dot{R}(t)}{2R(t)}\frac{2\sqrt{t}}{\pi} + \left(\frac{3\dot{R}^2(R)}{8R^2(t)} - \frac{\ddot{R}(t)}{4R(t)}\right)\frac{4t^{3/2}}{3\sqrt{\pi}} + \cdots\right] C_s(t).$$

The first two members of the obtained expansion describe the process in the case of a fixed boundary, and the other members represent corrections taking into account increase of the drop size.

In a certain sense, the process of gas dissolving in a fluid at its compression represents a process inverse to the considered one: the spherical volume filled with gas is surrounded by the fluid, and the pressure of the fluid slowly increases. As a result, the volume and mass of the gas decrease. The latter decreases due to dissolving of the gas in the fluid. This dissolving can be considered as diffusion of molecules through the drop surface. As it is shown in Babenko [4], the pressure in a cavity also satisfies the equation with fractional derivatives.

4.1.3 Splitting Inhomogeneous Equations

Kholpanov and Zakiev [41] have proved the following theorem extending the application of this method to inhomogeneous equations.

The solution $T = T(x,t)$ of the problem

$$\left(\frac{\partial}{\partial t} - \kappa\frac{\partial^2}{\partial t^2}\right) T = S(x,t), \quad t > 0, \quad x > 0;$$

$$T(x,0) = \psi(x), \quad T(0,t) = \varphi(t) + M(t) + F(t), \quad t > 0;$$

$$M(t) = \frac{1}{2\sqrt{\pi\kappa t}} \int_0^\infty d\tau \exp\left(-\frac{\tau^2}{4\kappa t}\right) \psi(\tau);$$

$$F(t) = \frac{1}{\sqrt{\pi}} \int_0^t d\tau \int_0^\infty d\xi \exp\left(-\frac{\xi^2}{4\kappa(t-\tau)}\right) S(\xi,\tau)$$

can be represented in the form of the sum

$$T = u_0 + u_1 + u_2,$$

whose terms are the solutions of the following problems:

$$\left({}_0\mathsf{D}_t^{1/2} + \sqrt{\kappa}\frac{\partial}{\partial x}\right) u_0 = 0, \quad u_0(x,0) = 0, \quad u_0(0,t) = \varphi(t);$$

$$\left({}_0\mathsf{D}_t^{1/2} + \sqrt{\kappa}\frac{\partial}{\partial x}\right) u_1 = \frac{1}{2}\left({}_0\mathsf{D}_t^{-1/2} S(x,t) + \frac{\psi(x)}{\sqrt{\pi t}}\right),$$

$$u_1(x,0) = \frac{\psi(x)}{2}, \quad u_1(0,t) = 0, \quad t > 0;$$

$$\left({}_0\mathsf{D}_t^{1/2} - \sqrt{\kappa}\frac{\partial}{\partial x}\right) u_2 = \frac{1}{2}\left({}_0\mathsf{D}_t^{-1/2} S(x,t) + \frac{\psi(x)}{\sqrt{\pi t}}\right),$$

$$u_2(x,0) = \frac{\psi(x)}{2}, \quad u_2(0,t) = M(t) + F(t), \quad t > 0.$$

4.1.4 Heat Transfer in Porous Media

From mathematical point of view, the heat transfer is similar to the mass transfer, they are governed by the same equations. So, all models and methods which are

developed for modeling diffusion in porous media (Sect. 3.1) are applicable to the problem of the heat transfer in such media. Here, we add the list of models considered above by one more introduced by [5]. Formally, it is based on the comb-model of an inhomogeneous medium (Sect. 3.7.3 of Vol. I). Heat or mass transfer in the porous medium is considered as the classical diffusion in a narrow channel with branches ("teeth"). Moving along these teeth, the particles leave the main channel and some time later return to it. These processes slow down the motion along the main channel simulating restrictions created by pores.

For description of diffusion in a semi-infinite channel, Babenko uses the equation

$$\left(\mathsf{D}_t + k_0 \mathsf{D}_t^{1/2} - \frac{\partial^2}{\partial x^2}\right) T(x,t) = 0, \quad x \geqslant 0, \quad t > 0,$$

with boundary-initial conditions for the the temperature (or concentration) $T(r,t)$

$$T(0,t) = T_0(t), \quad T(\infty,t) = T(x,0) = 0, \quad k > 0.$$

The origin of the term with the semiderivative can be explained by referring to Eq. (4.4): the flux from the main channel into side ones is proportional to the semiderivative of the temperature at their common point. The parameter k characterizes the "relative population" of the secondary channels (the density of teeth). Splitting this equation,

$$\left(\sqrt{\mathsf{D}_t + k_0 \mathsf{D}_t^{1/2}} - \frac{\partial}{\partial x}\right)\left(\sqrt{\mathsf{D}_t + k_0 \mathsf{D}_t^{1/2}} + \frac{\partial}{\partial x}\right) T(x,t) = 0,$$

Babenko wrote down a symbolic solution for the boundary flux as

$$q_0(t) = -\frac{\partial T}{\partial x}\bigg|_{x=0} = \sqrt{\mathsf{D}_t + k_0 \mathsf{D}_t^{1/2}}\, T_0(t) = {}_0\mathsf{D}_t^{1/2} \sqrt{1 + k_0 \mathsf{D}_t^{-1/2}}\, T_0(t). \tag{4.8}$$

For sufficiently short times, this solution can be calculated as

$$q_0(t) = \sum_{n=0}^{\infty} \binom{1/2}{n} k^n {}_0\mathsf{D}_t^{(1-n)/2} T_0(t), \quad t \simeq 0, \tag{4.9}$$

but the long-time behaviour of the solution is better described by its asymptotical series

$$q_0(t) = \sqrt{k_0}\, \mathsf{D}_t^{1/4} \sum_{n=0}^{\infty} \binom{1/2}{n} k^{-n} {}_0\mathsf{D}_t^{n/2} T_0(t), \quad k^2 t \simeq \infty. \tag{4.10}$$

Babenko went further and suggested that each second branch in its turn intensively branches and so on. In his opinion, this construction can be considered as a sort of fractal systems and described by the equation

$$\left(\mathsf{S}_\epsilon^2(k) - \frac{\partial^2}{\partial x^2}\right) T(x,t) = 0,$$

where

$$\mathsf{S}_\epsilon(k) = \sqrt{\mathsf{D}_t + k\sqrt{\mathsf{D}_t + \epsilon k\sqrt{\mathsf{D}_t + \epsilon^2 k\sqrt{\mathsf{D}_t + \cdots}}}}$$

and ϵ reflects different contributions into the process of branches of different levels.

As one can immediately see, the S-operator obeys the equation

$$\mathsf{S}_\epsilon^2(k) = \mathsf{D}_t + k\mathsf{S}_\epsilon(\epsilon k),$$

solutions of which for three special values of ϵ are

$$\mathsf{S}_0(k) = \sqrt{\mathsf{D}_t + {}_0\mathsf{D}_t^{1/2}},$$

$$\mathsf{S}_{1/2}(k) = {}_0\mathsf{D}_t^{1/2} + k/2,$$

and

$$\mathsf{S}_1(k) = \sqrt{\mathsf{D}_t + k^2/4} + k/2$$

(this statement can be verified by direct substitutions). The solution relating to the first case is considered above (Eq. 4.8). In two last cases, the solutions have the form

$$[q_0(t)]_{\epsilon=1/2} = \left({}_0\mathsf{D}_t^{1/2} + k/2\right) T_0(t)$$

and

$$[q_0(t)]_{\epsilon=1} = \left(\sqrt{\mathsf{D}_t + k^2/4} + k/2\right) T_0(t) = \left(e^{-k^2 t/4}\, {}_0\mathsf{D}_t^{1/2} e^{k^2 t/4} + k/2\right) T_0(T).$$

4.1.5 Hyperbolic Heat Conduction Equation

In Sect. 3.6.3 of the book [5], the method of splitting is applied to the hyperbolic equation

$$\left(\tau\mathsf{D}_t^2 + \mathsf{D}_t - \kappa\frac{\partial^2}{\partial x^2}\right) T(x,t) = 0$$

describing the heat propagation with a finite speed. Splitting the operator as

$$\left(\tau\mathsf{D}_t^2 + \mathsf{D}_t - \kappa\frac{\partial^2}{\partial x^2}\right) = \left(\sqrt{\tau\mathsf{D}_t^2 + \mathsf{D}_t} - \sqrt{\kappa}\frac{\partial}{\partial x}\right)\left(\sqrt{\tau\mathsf{D}_t^2 + \mathsf{D}_t} + \sqrt{\kappa}\frac{\partial}{\partial x}\right)$$

and repeating the arguments presented in the above subsection, we obtain

$$\sqrt{\kappa}\left(\frac{\partial T}{\partial x}\right)_{x=0} = -\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t}\, T_0(t). \tag{4.11}$$

Babenko gives three equivalent realization for the "root operator":

$$\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t} = \sqrt{\tau} \sum_{n=0}^{\infty} \binom{1/2}{n} \tau^{-n}\, {}_0\mathsf{D}_t^{1-n}, \tag{4.12}$$

$$\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t} = \sqrt{\tau}\, {}_0\mathsf{D}_t^{1/2} \exp(-t/\tau)\, {}_0\mathsf{D}_t^{1/2} \exp(t/\tau), \tag{4.13}$$

and

$$\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t} = \sqrt{\tau} \exp(-t/\tau)\, {}_0\mathsf{D}_t^{1/2} \exp(t/\tau)\, {}_0\mathsf{D}_t^{1/2}. \tag{4.14}$$

Validity of each this realization verifies by its direct substitution into the defining expression

$$\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t}\left[\sqrt{\tau \mathsf{D}_t^2 + \mathsf{D}_t} f(t)\right] = \left[\tau \mathsf{D}_t^2 + \mathsf{D}_t\right] f(t).$$

According to Maxwell-Cattaneo model, the heat flow is linked to the temperature gradient as

$$\left[\tau \mathsf{D}_t + 1\right] q(x,t) = -\kappa \frac{\partial T(x,t)}{\partial x}.$$

Acting on both sides of the expression by the operator $[\tau \mathsf{D}_t + 1]^{-1}$ and using Eq. (4.11), we represent the desired solution as

$$q_0(t) = -\frac{\kappa}{\tau} \frac{1}{\mathsf{D}_t + 1/\tau} \left(\frac{\partial T}{\partial x}\right)_{x=0} = \sqrt{\frac{\kappa}{\tau}} \frac{{}_0\mathsf{D}_t^{1/2}}{\sqrt{\mathsf{D}_t + 1/\tau}} T_0(t).$$

According to formulas (4.12)–(4.14), it can be rewritten in three equivalent forms:

$$q_0(t) = \sqrt{\frac{\kappa}{\tau}} \sum_{n=0}^{\infty} \binom{-1/2}{n} \tau^{-n}\, {}_0\mathsf{D}_t^{-n} T_0(t),$$

$$q_0(t) = \sqrt{\kappa/\tau}\, {}_0\mathsf{D}_t^{1/2} \exp(-t/\tau)\, {}_0\mathsf{D}_t^{-1/2} \exp(t/\tau) T_0(t),$$

and

$$q_0(t) = \sqrt{\kappa/\tau} \exp(-t/\tau)\, {}_0\mathsf{D}_t^{-1/2} \exp(t/\tau)\, {}_0\mathsf{D}_t^{1/2} T_0(t).$$

Choosing one of these realizations is due to considerations of convenience in relation to a particular function $T_0(t)$.

4.1.6 Inverse Problems

Equation (4.5) establishing a functional relationship between temperature and its gradient at a boundary allows us to solve the inverse problems of certain type. Let us come back to the parabolic equation with which we began to discuss the heat transfer, but this time we assume that $\kappa = \kappa(x, t)$:

$$q(x,t) = -\kappa(x,t)\frac{\partial T}{\partial x},$$

$$\frac{\partial T(x,t)}{\partial t} = \frac{\partial}{\partial x}\left[\kappa(x,t)\frac{\partial T(x,t)}{\partial x}\right] \equiv \frac{\partial \kappa(x,t)}{\partial x}\frac{\partial T(x,t)}{\partial x} + \kappa(x,t)\frac{\partial^2 T(x,t)}{\partial x^2}. \tag{4.15}$$

In a general statement, the inverse problem is formulated as follows: find the thermal conductivity $\kappa = \kappa(x, t)$ given a set of functionals of the solution $T(x, t)$.

For the sake of illustration, Babenko [5] begins with the equation

$$\left[\frac{\partial}{\partial t} - \alpha(x)\frac{\partial^2}{\partial x^2}\right] T(x,t) = 0,$$

$$x \in [0,\infty), \quad t \in (0,\infty), \quad T(\infty,t) = 0, \quad T(x,0) = 0,$$

containing only one variable coefficient which depends only on the coordinate. The problem is to find $\kappa(x)$ given $T_0(t)$ and $T_0'(t)$. According to Eq. (4.5),

$$-\sqrt{\alpha(0)}T_0'(t) = \left[{}_0\mathsf{D}_t^{1/2} + a_1(0) + a_2(0)\,{}_0\mathsf{D}_t^{-1/2} + a_3(0)\,{}_0\mathsf{D}_t^{-1} + \cdots\right] T_0(t), \tag{4.16}$$

where constants $a_n(0) = a_n(x)|_{x=0}$ are determined by Eq. (4.7):

$$a_1(0) = -\frac{\alpha'(0)}{4\sqrt{\alpha(0)}},$$

$$a_2(0) = \frac{3}{32}\frac{[\alpha'(0)]^2}{\alpha(0)} - \frac{\alpha''(0)}{8},$$

$$a_3(0) = \frac{3}{32}\frac{\alpha'(0)\alpha''(0)}{\sqrt{\alpha(0)}} - \frac{3}{64}\frac{[\alpha'(0)]^3}{[\alpha(0)]^{3/2}} - \frac{\sqrt{\alpha(0)}\alpha'''(0)}{16} + \cdots.$$

In order to continue the calculations, one has to represent experimental data obtained by measurements at the end point $x = 0$ as series[1]

$$T_0(t) = \sum_{n=0}^{\infty} \frac{c_n}{\Gamma(1+n/2)} t^{n/2}, \quad -T_0'(t) = \sum_{n=0}^{\infty} \frac{e_n}{\Gamma((1+n)/2)} t^{(n-1)/2}. \tag{4.17}$$

Inserting the series (4.17) into Eq. (4.16), computing the fractional derivatives and equating coefficients in front of equal powers of time, we get:

in front of $t^{-1/2}$,

$$\sqrt{\alpha(0)} e_0 = c_0,$$

in front of t^0,

$$\sqrt{\alpha(0)} e_1 = c_1 + a_1(0) c_0,$$

in front of $t^{1/2}$,

$$\sqrt{\alpha(0)} e_2 = c_2 + a_1(0) c_1 + a_2(0) c_0,$$

in front of t,

$$\sqrt{\alpha(0)} e_3 = c_3 + a_1(0) c_2 + a_2(0) c_1 + a_3(0) c_0,$$

in front of $t^{(n-1)/2}$, $n > 3$,

$$\sqrt{\alpha(0)} e_n = \sum_{j=0}^{n} a_j c_{n-j}.$$

Solving the system yields the solution of the problem:

$$\alpha(x) = \alpha(0) + \alpha'(0) x + \alpha''(0) \frac{x^2}{2} + \cdots,$$

$$\alpha(0) = c_0^2 / e_0^2,$$

$$\alpha'(0) = 4\sqrt{\alpha(0)} [c_1 - \sqrt{\alpha(0)} e_1] / c_0,$$

$$\alpha''(0) = 8 \left[c_2 - \sqrt{\alpha(0)} e_2 \right] / c_0 + (3/4) [\alpha'(0)]^2 / \alpha(0) - 2\alpha'(0) c_1 / \left[\sqrt{\alpha(0)} c_0 \right],$$

and so on.

Following this way, Babenko considers a more realistic problem of determining the space-dependent heat conductivity $\kappa = \kappa(x)$,

[1] This is an essential disadvantage of the method.

$$\frac{\partial T(x,t)}{\partial t} = \frac{\partial}{\partial x}\left[\kappa(x,t)\frac{\partial T(x,t)}{\partial x}\right], \tag{4.18}$$

under conditions

$$T_0(t) = 1, \quad -T_0'(0) = \frac{1}{\sqrt{\pi t}} + \frac{1+g}{3}.$$

In this case

$$c_0 = 1, \quad e_1 = \frac{1+g}{3}, \quad e_n = 0, \quad n \geqslant 2,$$

and the solution has the form

$$\kappa(x) = 1 + \frac{4(1+g)}{3}x + \frac{2(1+g)^2}{9}x^2 - \frac{4(1+g)^3}{81}x^3 + \cdots = [1+(1+g)x]^{4/3}.$$

4.2 Fractional Heat Transfer Models

4.2.1 Fractional Heat Conduction Laws

Classical theory of heat conduction is based on the Fourier law, relating the heat flux components q_i to the temperature gradient by

$$q = -\kappa\frac{\partial T}{\partial x}, \tag{4.19}$$

or, in a 3D vector form,

$$\mathbf{q} = -\kappa\nabla T.$$

Having the same form as the flux-concentration interrelation in the mass diffusion theory, this law implies the ability to heat instantaneously spread far and wide, and slowe heat transfer, which is observed on a macroscopic scale, due to the large variability routes of heat transmission.

As is reported by Joseph and Preziosi [38] with reference to I. Müller's report, Cattaneo first proposes that the heat flux depends on the history of the temperature gradient, writing

$$q = -\kappa_1\frac{\partial T}{\partial x} + \kappa_2\frac{\partial}{\partial t}\left(\frac{\partial T}{\partial x}\right). \tag{4.20}$$

The second term takes into account that the flux $q(x,t)$ remembers the temperature gradient at an earlier time. Combining Eq. (4.20) with the energy equation

$$\rho c\frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0$$

yields a parabolic equation

$$\frac{\partial T}{\partial t} = \frac{\kappa_1}{\rho c}\frac{\partial^2 T}{\partial x^2} - \frac{\kappa_2}{\rho c}\frac{\partial}{\partial t}\left(\frac{\partial^2 T}{\partial x^2}\right),$$

admitting the spreading of pulses with infinite speed. Müller believes that Cattaneo must have noticed this, because he proceeded to another equation through the following sequence of steps:

$$q = -\kappa_1\left(1 - \sigma\frac{\partial}{\partial t}\right)\frac{\partial T}{\partial x}, \quad \sigma = \kappa_1/\kappa_2,$$

$$\left(1 - \sigma\frac{\partial}{\partial t}\right)^{-1} q = -\kappa_1\frac{\partial T}{\partial x} \quad \mapsto \quad \left(1 + \sigma\frac{\partial}{\partial t}\right) q = -\kappa_1\frac{\partial T}{\partial x}.$$

Evidently, this passage can be considered as a rough approximation being valid only if $\sigma\partial q/\partial t \ll q$. On the contrary to this way, the derivation which is performed in Sect. 2.2.3 looks quite acceptable, because the system (Sect. 2.2.3) takes the thermodynamical form

$$\left(1 + \sigma\frac{\partial}{\partial t}\right) q = -\kappa\frac{\partial T}{\partial x} \tag{4.21}$$

without any additional assumptions. The equation can also be rewritten in a non-local form,

$$q(x,t) = -\kappa\int_0^t \psi_1(t-t')\frac{\partial T(x,t')}{\partial x}dt', \quad \psi_1(t) = 1/\sigma\exp(-t/\sigma), \tag{4.22}$$

which expresses explicitly the heredity effect. Fractionalizing Eq. (4.21) yields the following generalizations of the Fourier law:

$$\left(1 + \tau_0{}_0^{\alpha}\mathsf{D}_t\right)\mathbf{q} = -\kappa\nabla T, \quad 0 < \alpha < 1 \tag{4.23}$$

Sherief et al. [66] and

$$\left(1 + \tau_0\frac{\partial}{\partial t}\right)\mathbf{q} = -\kappa_0\mathsf{D}_t^{1-\alpha}\nabla T, \quad 1 < \alpha \leqslant 2 \tag{4.24}$$

Youssef [84].

4.2.2 Fractional Equations for Heat Transport

In the phenomenological heat conduction theory, the speed of heat propagation is assumed to be infinitely large. Indeed, in order to get Eq. (4.19) from Eq. (4.21), we must replace $\psi_1(t)$ by $\delta(t)$. This assumption is supported by the results of calculation of temperature fields in different bodies under normal conditions encountered in practice. However, in rarefied gases under high-intensity transient heat transfer processes must be understood that the heat does not extend infinitely fast, but with a finite velocity V [83]. According to molecular-kinetic theory of the heat transfer,

$$V = \sqrt{\frac{\kappa'}{c\gamma\tau_r}},$$

where γ is the mass density of the matter, $c = c_p = c_V$ specific heat, κ' is proportional to κ, and τ_r is the relaxation time.

As discussed in Sect. 2.2.3, the Fourier heat conduction law, leading to the standard parabolic equation

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2},$$

should be replaced by more exact law (4.21) taking into account a finite speed of heat spreading. This replacement leads to the hyperbolic form of the heat conduction equation

$$\frac{\partial T}{\partial t} + \tau \frac{\partial^2 T}{\partial t^2} = \kappa \frac{\partial^2 T}{\partial x^2}.$$

Some authors call it the *Maxwell-Cattaneo-Lykov* (*MCL*) equation, because the hypothesis about limited velocity of heat and mass transfer in porous media was proposed in Lykov [47]. Three-dimensional version of this equation is written by means of the Laplace operator:

$$\frac{\partial T}{\partial t} + \tau \frac{\partial^2 T}{\partial t^2} = \kappa \Delta T. \tag{4.25}$$

Considering two asymptotical domains for the relaxation time τ, we arrive at two extreme forms of this equation:

$$\frac{\partial T}{\partial t} + \tau \frac{\partial^2 T}{\partial t^2} = \kappa \Delta T \quad \mapsto \quad \begin{cases} \dfrac{\partial T}{\partial t} = \kappa \Delta T, & \tau \to 0 \text{ (diffusion equation);} \\ \tau \dfrac{\partial^2 T}{\partial t^2} = \kappa \Delta T, & \tau \to \infty \text{ (wave equation).} \end{cases}$$

Writing both asymptotical equations as

$$\tau^{\nu-1}\frac{\partial^{\nu}T}{\partial t^{\nu}} = \kappa\Delta T, \quad \nu = 1, 2,$$

and then allowing ν to take all values inside the interval $(0, 2]$, we obtain the time-fractional generalization of the heat transfer equation

$$\tau^{\nu-1}{}_0\mathsf{D}_t^{\nu}T = \kappa\Delta T, \quad \nu \in (0, 2], \tag{4.26}$$

named also wave-diffusion equation solution of which is considered in Sect. 5.2 of Vol. I.

During last decades, the MCL equation is used as an important component of the basis of *Extended Irreversible Thermodynamics* (*EIT*), providing generalized transport equations [39, 40]. These equations should incorporate memory and non-local effect, which reduce to the classical transport equations for low frequency and short mean free-paths, and at the same time, they must be able to describe the high-frequency behavior signals with accounting a finite speed of propagation of heat pulses. Such equations may be actively explored in several modern technologies: in miniaturized electronic devices, in nanotubes and nanowires, in description of sophisticated experiments at micro- and nanoscales. For these aims, embedding fractional operators into EIT mathematics seems to be very perspective.

To get fractional generalizations of the MCL equation, we take gradient from both sides of Eqs. (4.23)–(4.24) and then apply the heat balance equation

$$\nabla \cdot \mathbf{q} = -\frac{\partial T}{\partial t}.$$

As a result, we have

$$\frac{\partial T}{\partial t} + \tau_0\mathsf{D}_t^{1+\alpha}T = \kappa\Delta T, \quad 0 < \alpha < 1 \tag{4.27}$$

and

$$\frac{\partial T}{\partial t} + \tau\frac{\partial^2 T}{\partial t^2} = \kappa_0\mathsf{D}_t^{1-\alpha}\Delta T, \quad 1 < \alpha < 2, \tag{4.28}$$

respectively.

We consider below some applications of these equations.

4.2.3 Application to Thermoelasticity

The theory of thermoelasticity deals with mechanical and thermal effects in an elastic body. Introducing the heat flux—temperature gradient relation in the integral (hereditary) form into the thermoelastic theory was performed by Chen and Curtin [10]. Among a few works devoted to applications of fractional calculus to thermoelasticity, I'd like to refer to the works of Povstenko [60–62], who introduced a fractional heat

conduction law, found the associated thermal stresses, established the fundamental solutions to the Cauchy problem in the case of spherical symmetry, and investigated stresses due to the fractional heat conduction law in an infinite body with a circular cylindrical hole.

In Povstenko [60], the effect of heat on the deformation of a linear isotropic elastic solid is analyzed by using the fractional heat conduction Eq. (4.26) with Gerasimov-Caputo form derivatives:

$$ {}_0^{\alpha}\mathsf{D}_t T = \kappa \Delta T, \quad -\infty < x < \infty, \quad 0 < t < \infty. $$

The analysis is fulfilled on assumption that a thermoelastic state of the solid is governed by the equilibrium equation

$$ \mu u_{i,jj} + (\lambda + \mu) u_{jj,i} = \beta_t K T_{,i}, \tag{4.29} $$

and the stress-strain-temperature relation

$$ \sigma_{ij} = 2\mu \varepsilon_{ij} + (\lambda\, \varepsilon_{ii} - \beta_t K T)\delta_{ij} \tag{4.30} $$

where u_i is the displacement vector, σ_{ij} is the stress tensor, ε_{ij} is the linear strain tensor, λ and μ are Lamé constants, $K = \lambda + 2\mu/3$, and β_t is the thermal coefficient of volumetric expansion.

Following classical theory, Povstenko represents stresses in terms of displacement potential Φ:

$$ \sigma_{ij} = 2\mu \left(\frac{\partial^2}{\partial x_i \partial x_j} - \delta_{ij} \frac{\partial^2}{\partial x_k^2} \right) \Phi \tag{4.31} $$

which is determined from the equation

$$ \Delta \Phi = mT, \quad m = \frac{1+\nu}{1-\nu} \frac{\beta_t}{3}, \tag{4.32} $$

where ν is the Poisson ratio.

In the Povstenko's article which is referred here, the problem is solved in two cases: one-dimensional and two-dimensional ones. In the first of them, the initial conditions are taken in the form

$$ T(x, 0) = p\delta(x), \quad 0 < \alpha < 1 $$

and

$$ T(x, 0) = p\delta(x), \quad \left. \frac{\partial T}{\partial t} \right|_{t=0} = 0, \quad 1 < \alpha < 2, $$

where p is a prescribed constant. Under these conditions, only the components

$$\sigma_{yy} = \sigma_{zz} = -2\mu m T$$

do not equal to zero. The solution $T(x,t)$ is written as

$$T(x,t) = \frac{p}{2\sqrt{\kappa}t^{\alpha/2}} M\left(\frac{|x|}{\sqrt{\kappa}t^{\alpha/2}}; \frac{\alpha}{2}\right).$$

Considering the axially symmetric two-dimensional case, Povstenko solves the equation

$${}_0^{\alpha}\mathsf{D}_t T = \kappa\left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r}\frac{\partial T}{\partial r}\right)$$

under initial conditions

$$T(r,0) = p\frac{\delta(r)}{2\pi r}, \quad 0 < \alpha < 2$$

and

$$\left.\frac{\partial T}{\partial t}\right|_{t=0} = 0, \quad 1 < \alpha < 2.$$

The nonzero components of the stress tensor are expressed in terms of the displacement potential as

$$\sigma_{zz} = \sigma_{rr} + \sigma_{\varphi\varphi} = -2\mu\Delta\Phi$$

and

$$\sigma_{rr} - \sigma_{\varphi\varphi} = 2\mu\left(\frac{\partial^2\Phi}{\partial r^2} - \frac{1}{r}\frac{\partial\Phi}{\partial r}\right).$$

The solution of the two-dimensional heat transfer equation, represented in the form

$$T(r,t) = \frac{p}{2\pi}\int_0^{\infty} F_{\alpha}(\xi,t) J_0(r\xi)\xi d\xi,$$

was specified for $\alpha = 1/2$ and $\alpha = 3/2$ by giving Laplace transforms of F-functions:

$$\widehat{F}_{1/2}(\xi,\lambda) = \frac{1}{\sqrt{\lambda}(\sqrt{\lambda} + \kappa\xi^2)}$$

and

$$\widehat{F}_{3/2}(\xi,\lambda) = \frac{\sqrt{\lambda}}{\lambda^{3/2} + \kappa\xi^2}.$$

Despite of many publications (see References to this Chapter) based on using time-fractional equations with only one time-derivative (like Eq. (4.26)), this approach

does not seem to be sufficiently justified. More appropriate for these aims Eq. (4.27) is used in the article [15] devoted to fractional generalization of the theory of magneto-thermoelasticity. These authors consider a thermoelastic medium of perfect conductivity permeated by an initial magnetic field **H**. This induces magnetic field **h** and electric field **E** satisfy (for slowly moving media) the linearized Maxwell equations

$$\operatorname{curl} \mathbf{h} = \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

$$\operatorname{curl} \mathbf{E} = -\mu_0 \frac{\partial \mathbf{h}}{\partial t},$$

$$\mathbf{E} = -\mu_0 \left[\frac{\partial \mathbf{u}}{\partial t} \times \mathbf{H} \right],$$

$$\operatorname{div} \mathbf{h} = 0.$$

Here the vectors **h** and **E** denote perturbations of the magnetic and electric fields, respectively, **J** is the electric current density vector, **H** is the initial magnetic field, and **u** is the displacement vector.

The second group of equations is the equations of motion,

$$\rho \frac{\partial^2 u_i}{\partial t^2} = \sigma_{ij,j} + \tau_{ij,j} + X_i,$$

where τ_{ij} is the Maxwell electromagnetic stress tensor,

$$T_{ij} = \mu_0 [H_i h_j + H_j h_i - \delta_{ij} (h_k H_k)],$$

$$\sigma_{ij} = 2\mu \varepsilon_{ij} + \lambda \varepsilon_{kk} \delta_{ij} - \gamma (T - T_0) \delta_{ij}, \quad \varepsilon_{ij} = (u_{i,j} + u_{j,i})/2,$$

and the heat conduction equation is written as

$$\left[\mathsf{D}_t + \tau_0^{\alpha+1} \mathsf{D}_t \right] (\rho C_E T + \gamma T_0 e) = \kappa T_{,ii}, \quad 0 < \alpha \leqslant 1.$$

The article ends with the conclusion, that the developed theory describes the behavior of the particles of an elastic body more realistically than the theory with integer-order derivatives, and that the result provides motivation to investigate perfectly conducting thermoelastic materials as a new class of applicable thermoelastic materials.

4.2.4 Some Irreversible Processes

4.2.4.1 Pair-Contact Process with Diffusion

The *pair-contact process with diffusion* (*PCPD*) is a classical reaction-diffusion process describing stochastically interacting particles of a single species which react spontaneously whenever two of them come into contact. In addition individual particles are allowed to diffuse. The interest in this model is first motivated by the fact that the PCPD exhibits a non-equilibrium phase transition caused by the competing character of fission and annihilation [24, 25, 36]. These stochastic processes describe the essential features of a vast number of growth phenomena of populations of infected individuals (in the language of a disease) near their extinction threshold and are relevant to a wide range of models in physics, chemistry, biology, and sociology.

In the standard version of the epidemic models the susceptible individuals can become contaminated by already infected neighboring individuals. At the same time infected individuals are subject to spontaneous healing or immunization processes. This process is described by the classical Langevin equation for the concentration $n(\mathbf{x}, t)$ of a percolating substance (infected individuals):

$$\frac{\partial n(\mathbf{x}, t)}{\partial t} = an(\mathbf{x}, t) - bn^2(\mathbf{x}, t) + K\,\Delta n(\mathbf{x}, t) + \zeta(\mathbf{x}, t).$$

The constant a controls the balance between offspring production and self-destruction, and plays the role of the deviation $p - p_c$ from the critical percolation probability. The infection of nearest neighbors (in epidemiological terms) is represented by the Laplacian, while the nonlinear term incorporates the exclusion principle on the lattice. The term $\zeta(\mathbf{x}, t)$ represents a multiplicative Gaussian noise field defined by the correlations

$$\langle \zeta(\mathbf{x}, t)\zeta(\mathbf{x}', t')\rangle = 2\Gamma n(\mathbf{x}, t)\delta(\mathbf{x} - \mathbf{x}')\delta(t - t').$$

The transition between survival and extinction of such a growing population is a nonequilibrium continuous phase transition phenomenon and is characterized by universal scaling laws.

In realistic situations the infection can be also long-ranged(lr). This is modeled by involving an integral term,

$$\left.\frac{\partial n(\mathbf{x}, t)}{\partial t}\right|_{\text{lr}} = \int d\mathbf{x}'\, p(\mathbf{x} - \mathbf{x}')n(\mathbf{x}')d\mathbf{x}'.$$

Following a suggestion of Mollison [56], Grassberger [23] introduced a variation of the epidemic processes in a d-dimensional space with infection probability distributions $p(r)$ which decays with the distance r according to a power-law,

$$p(r) \propto r^{-d-\sigma}, \quad r \to \infty.$$

Such a process can also be classified as an anomalous percolation taking place when a system is close to a phase transition. Complementing the short-range diffusion Laplacian operator by an integral operator representing long-range diffusion as a jumping motion [36] and passing to macroscopic scales yield an equation with the fractional Laplacian

$$\frac{\partial n(\mathbf{x},t)}{\partial t} = an(\mathbf{x},t) - bn^2(\mathbf{x},t) + K_2\Delta n(\mathbf{x},t) - K_\sigma(-\Delta)^{\sigma/2}n(\mathbf{x},t) + \zeta(\mathbf{x},t). \tag{4.33}$$

This process can be named as the reaction-superdiffusion. The major part of their article [36] is devoted to scaling analysis in the framework of the Wilson renormalization theory. In conclusion the authors stress that due to fractional Laplacian, new long-range determined universality classes come into play. These universality classes can be characterized by determining the critical exponents to first order in an ε-expansion around their upper critical dimension. There exist exact relationships between the exponents, and some critical exponents are given exactly as functions of spatial dimension and the exponent characterizing the long-range tail of the Lévy flight infection. Besides the renormalization group flow built in Hinrichsen [33] possesses a fixed point structure that allows to describe short-range and long-range infection in the same and to pass *continuously* from one behavior to another by varying the Lévy flight exponent. One can also add that introducing truncated distributions $p(r)$ with long tails of a power type allows to realize such transition only by changing scales, leaving the exponent untouched. The further development of this model can be traced by the reader in the works of the authors cited above. In particular, [33] considers also the time-fractional (that is, the *reaction-subdiffusion*) equation

$${}_0\mathsf{D}_t^\nu n(\mathbf{x},t) = an(\mathbf{x},t) - bn^2(\mathbf{x},t) + K_2\Delta n(\mathbf{x},t) + \zeta(\mathbf{x},t), \tag{4.34}$$

and even the combined case

$${}_0\mathsf{D}_t^\nu n(\mathbf{x},t) = an(\mathbf{x},t) - bn^2(\mathbf{x},t) + K_2\Delta n(\mathbf{x},t) - K_\sigma(-\Delta)^{\sigma/2}n(\mathbf{x},t) + \zeta(\mathbf{x},t). \tag{4.35}$$

4.2.4.2 Reaction-Superdiffusion Process

Because the equations considered above contain the random term $\zeta(\mathbf{x},t)$, their solutions are also random. Passing to the deterministic form of these equations is realized by substituting ζ for the deterministic reaction term $R(n)$. The latter is a non-linear function representing the reaction kinetics. Simple examples of such function used in analytical calculations are the Fisher-Kolmogorov term

$$R(n) = \gamma n(1-n)$$

and the Ginzburg-Landau term

$$R(n) = \gamma n(1 - n^2).$$

The nontrivial dynamics of these systems arises from the competition between the reaction kinetics and diffusion, which in classical theory are governed by the equation

$$\frac{\partial n}{\partial t} = K\Delta n(\mathbf{r}, t) + R(n(\mathbf{r}, t)).$$

In case of anomalous reaction-diffusion, the first term in right-hand side is modified by using fractional Laplacian or/and fractional time-derivative:

$$\frac{\partial n}{\partial t} = -K(-\Delta)^{\alpha/2} n(\mathbf{r}, t) + R(n(\mathbf{r}, t)),$$

$$\frac{\partial n}{\partial t} = K_0 \mathsf{D}_t^{1-\nu} \Delta n(\mathbf{r}, t) + R(n(\mathbf{r}, t)),$$

and

$$\frac{\partial n}{\partial t} = -K_0 \mathsf{D}_t^{1-\nu} (-\Delta)^{\alpha/2} n(\mathbf{r}, t) + R(n(\mathbf{r}, t))$$

respectively.

Observe that the all above equations describe isotropic diffusion. Del-Castillo-Negrete et al. [13] referring to transport in magnetized plasmas and in geophysical flows have focused on an asymmetric diffusion: Lévy flights in one ($x > 0$) direction, but Gaussian behaviour in the opposite ($x < 0$) direction. To solve this problem, the authors use the equation

$$\frac{\partial n}{\partial t} = K {}_0^{\alpha}\mathsf{D}_t n(x, t) + R(n), \quad 1 < \alpha < 2, \tag{4.36}$$

where $R(n) = \gamma n(x, t)[1 - n(x, t)]$. The physical motivation for using the left-fractional derivative rests on the fact that for $R = 0$, Equation (4.36) reduces to an asymmetric, space-fractional diffusion equation whose solution for a delta function initial condition,

$$n(x, 0) = \delta(x),$$

in the infinity domain $-\infty < x < \infty$ is

$$n(x, t) = (Kt)^{-1\alpha} g\left((Kt)^{-1\alpha} x; \alpha, \theta_\alpha\right).$$

Let us recall that $g(x; \alpha, \theta_\alpha)$ is the α-stable Lévy distribution with maximal skewness (because $\alpha > 1$, it is not a one-sided distribution, see Sect. 3.2 of Vol. I):

$$g(x;\alpha,\theta_\alpha)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\exp(ikx+(ik)^\alpha)dk.$$

In del-Castillo-Negrete et al. [13], calculations for a given initial and boundary conditions were performed numerically, but the numerical results were explained analytically using the *leading-edge approximation* [81]. In this approximation, in the limit $n \ll 1$, the reaction term $R(n)$ is linearized around the unstable phase:

$$\frac{\partial n}{\partial t}=K\,{}_{-\infty}\mathsf{D}_t^{\alpha}n(x,t)+\gamma n(x,t).$$

Substituting $n(x,t)=e^{\gamma t}\phi(x,t)$ leads to the following solution expressed in a quadrature:

$$n(x,t)=e^{\gamma t}\int_{-\infty}^{\infty}n\left(x-(Kt)^{1/\alpha}x',0\right)g(x';\alpha,\theta_\alpha)dx'.$$

In case $\alpha<2$, $n(x,t)$ falls as $x^{-\alpha}$ at a long time, but in case $\alpha=2$ one gets the asymptotic behavior $n(x,t)\propto e^{-\lambda(x-ct)}$, where λ and c are constants.

4.2.4.3 Reaction-Subdiffusion Process

One of the first works where the reaction-subdiffusion equation was derived is the article [26]. The authors followed in the usual way: starting from CTRF model with temporal memory of an inverse-power type, they performed the Laplace transform with respect to time and passed to the diffusion asymptotics with the use of Tauberian theorems As a result, they obtained:

$$\lambda\widehat{n}(\mathbf{r},\lambda)-n(\mathbf{r},0)=K\lambda^{1-\nu}\Delta\widehat{n}(\mathbf{r},\lambda)+\widehat{S}(\mathbf{r},\lambda).$$

Inverting the Laplace transform yields:

$$\frac{\partial n}{\partial t}=K\left({}_0\mathsf{D}_t^{1-\nu}\Delta n(\mathbf{r},t)+\mathcal{L}^{-1}\left({}_0\mathsf{D}_t^{-\nu}\Delta n(\mathbf{r},t)\right)_{t=0}\right)+R(n(\mathbf{r},t)).$$

A particular solution of this equation with the zero flux boundary conditions is the homogeneous steady-state solution $n(\mathbf{r},t)=\bar{n}$ defined by $R(\bar{n})=0$. In Henry and Wearne [26], the linear stability of this solution in the single species one-dimensional equation is investigated.

A similar equations, have been derived in Sokolov et al. [71]. These authors generalized reaction-diffusion schemes for a simple monomolecular conversion $A\to B$. The corresponding kinetic equations for local $A-$ and $B-$molecules concentrations are represented as

$$\frac{\partial n_A}{\partial t} = K\,{}_0\mathsf{T}_t(1-\alpha,k)\Delta n_A(\mathbf{x},t) - kn_a(\mathbf{x},t),$$

$$\frac{\partial n_B}{\partial t} = K\,{}_0\mathsf{D}_t^{1-\alpha}\Delta n_B(\mathbf{x},t) + kn_A(\mathbf{x},t) + K\left[{}_0\mathsf{D}_t^{1-\alpha} - {}_0\mathsf{T}_t(1-\alpha,k)\right]\Delta n_A(\mathbf{x},t),$$

where ${}_0\mathsf{T}_t(1-\alpha,k)$ stands for the fractional operator defined by

$${}_0\mathsf{T}_t(1-\alpha,k)f(t) = \left(\frac{d}{dt}+k\right)\int_0^t \frac{\exp[-k(t-t')]}{(t-t')^{1-\alpha}}f(t')dt',$$

turning to be a fractional derivative only for $k=0$. Discussion of these results shows that reaction-subdiffusion equations may not resemble the corresponding reaction-diffusion ones and are not obtained by a trivial change of the diffusion operator for a subdiffusion one.

Méndez and Ortega-Gejas [49] investigate front propagation problem by using the hyperbolic fractional reaction-diffusion equations. Starting from CTRF equation

$$n(x,t) = \int_0^t dt' q(t') \int_{-\infty}^{\infty} dx' p(x') n(x-x', t-t') + \int_0^t dt' Q(t') R(n(x,t-t')),$$

the authors apply the Laplace-Fourier transform and assuming

$$\widetilde{p}(k) = e^{-(\sigma k)^2}$$

and

$$\widehat{q}(\lambda) = e^{-(\tau\lambda)^\nu},$$

they get

$$\lambda[e^{(\tau\lambda)^\nu} - e^{-(\sigma k)^2}]\widehat{\widetilde{n}}(k,\lambda) = [e^{(\tau\lambda)^\nu} - 1]\widehat{\widetilde{R}}(k,\lambda).$$

Further, introducing

$$e^{-(\sigma k)^2} \simeq 1 - (\sigma k)^2$$

and

$$e^{(\tau\lambda)^\nu} \simeq 1 + (\tau\lambda)^\nu + (\tau\lambda)^{2\nu}/2$$

into this equation yields the hyperbolic fractional reaction-diffusion equation

$$\frac{\tau}{2}{}_0\mathsf{D}_t^{2\nu} n + {}_0\mathsf{D}_t^{\nu} n = K\frac{\partial^2 n}{\partial x^2} + \frac{\tau^\nu}{2}{}_0\mathsf{D}_t^{2\nu-1} R + {}_0\mathsf{D}_t^{\nu-1} R,$$

where $K = \sigma^2/\tau^\nu$. In order to check the existence of fronts as solutions of this equation on assumption that $R(n)$ is of the Fisher-Kolmogorov type, Méndez and

Ortega-Cejas have integrated it numerically by using a finite difference scheme based in the equivalence of the Grünwald-Letnikov representation.

Yuste and Lindenberg [86] applied fractional equations for modeling the coagulation dynamics $A + A \to A$ and the annihilation dynamics $A + A \to 0$ for particles moving subdiffusively in one dimension. For solving the first problem, the *empty interval method* has been developed. It is based on the probability $E(x, t)$ that an interval of length x is empty of particles at time t. As was shown in the work cited, this function obeys the equation

$$\frac{\partial E(x,t)}{\partial t} = K\, {}_0\mathsf{D}_t^{1-\alpha} \frac{\partial^2 E(x,t)}{\partial x^2}.$$

The second problem (annihilation) has been solved by using the probability $R(x, t)$ that an arbitrary interval of length x contains an even number of particles at time t. It is proved that this function satisfies the same equation as $E(x, t)$.

4.2.4.4 Dispersive Sedimentation

Normal diffusion problems involving external fields are often modeled in terms of a *Fokker-Planck equation* (*FPE*):

$$\frac{\partial p}{\partial t} = \left[-\frac{\partial}{\partial x} \frac{F(x)}{m\eta} + K \frac{\partial^2}{\partial x^2} \right] p(x,t).$$

Its fractional generalization (*FFPE*) for subdiffusion can be represented as

$$\frac{\partial p}{\partial t} = {}_0\mathsf{D}_t^{1-\alpha} \left[-\frac{\partial}{\partial x} \frac{F(x)}{m\eta_\alpha} + K_\alpha \frac{\partial^2}{\partial x^2} \right] p(x,t),$$

where m is the mass of the tracer particle, η_α is the generalized friction constant, and K_α denotes the generalized diffusion constant (see for details [50, 51]) and other works given in References). For investigation of the sedimentation process in the gravity field $F(x) = -mg$, FFPE becomes

$$\frac{\partial p}{\partial t} = {}_0\mathsf{D}_t^{1-\alpha} \left[-b \frac{\partial}{\partial x} + K_\alpha \frac{\partial^2}{\partial x^2} \right] p(x,t),$$

where $b = g/\eta_\alpha$. The Fourier-Laplace transform under initial condition $p(x, 0) = \delta(x)$ yields

$$\widehat{\widetilde{p}}(k,\lambda) = \frac{1}{\lambda + ikb\lambda^{1-\alpha} + k^2 K_\alpha \lambda^{1-\alpha}}.$$

The inverse Laplace transform expresses $\widetilde{p}(k,t)$ through the Mittag-Leffler function

$$\widetilde{p}(k,t) = E_\alpha\left[-(k^2K_\alpha + ikb)t^\alpha\right].$$

This formula together with an explicit expression for $E_\alpha(z)$ produces all spatial moments of $p(x,t)$. In particular, for the first moment we have

$$\langle X(t)\rangle = \frac{bt^\alpha}{\Gamma(1+\alpha)}.$$

An important measure for sedimentation processes is the speed with which the major part of the tracer is setting down. The effective setting velocity in the subdiffusion case can be defined by

$$V_\alpha = \frac{d\langle X(t)\rangle}{dt}$$

and reads as

$$V_\alpha = -\frac{gt^{\alpha-1}}{\eta_\alpha\Gamma(\alpha)}.$$

Observe that in the normal case $\alpha = 1$ we obtain the classical result

$$V_1 = -\frac{g}{\eta_1} = \text{const},$$

whereas in the subdiffusion case $\alpha < 1$ we see the decreasing speed.

4.3 Phase Transitions

4.3.1 Ornstein-Zernicke Equation

The Ornstein-Zernicke theory of critical phenomena is based on three theses [6]. The first one is the relation between temperature-dependent isothermal compressibility $\chi(t)$ and fluctuations of the number of molecules N in a large (in comparison with the correlation length) volume V of the system,

$$\chi(T) = -\frac{\langle N\rangle}{nV^2}\left(\frac{\partial V}{\partial P}\right)_T = \frac{\delta^2}{nk_\mathrm{B}T},$$

where $n = \langle N\rangle/V$ is the mean concentration of molecules, and $\delta^2 = (\langle N^2\rangle - \langle N\rangle^2)/\langle N\rangle$ is the ratio of the variance of number N to the Poisson variance with the same concentration. This formula means that the systems with high compressibility, e.g., diluted gases, are characterized by larger fluctuations of particle number than

solids. Expressing these fluctuations through the integral of the correlation function over the whole space

$$\delta^2 = 1 + n \int C(\mathbf{r}, T) d\mathbf{r},$$

we arrive at the formula

$$nk_{\mathrm{B}} T \chi(T) = 1 + n \int C(\mathbf{r}, T) d\mathbf{r}$$

connecting the correlation function with temperature.

It is known that for all stable systems $(\partial V/\partial P)_T < 0$, in a vicinity of a critical point of phase transition T_c the derivative of volume on pressure, $(\partial P/\partial V)_T$, turns into zero and $\chi(T) \to \infty, \quad T \to T_c$. This is an evidence of extremely high fluctuations of density in a critical region that generate the phenomena of critical opalescence, light scattering, plasma oscillations, and others. The arising divergence of the integral of the correlation function

$$\int C(\mathbf{r}, T_c) d\mathbf{r} = \infty$$

points to an infinite correlation radius caused by the presence of a long tail of the correlation function $C(\mathbf{r}, T_c)$. Theoretical analysis shows that this tail is of the power law type,

$$C(\mathbf{r}, T_c) \propto r^{-(d-2)}, \quad r \to \infty,$$

where d is a dimension of the space. At the same time, numerical simulations and some experiments show that

$$C(\mathbf{r}, T_c) \propto r^{-(d-2+\eta)},$$

where η is a positive number smaller than one.

The second thesis is reflected by the integral Ornstein-Zernicke equation allowing to pass on from one function $C(\mathbf{r}, T)$ to another function $c(\mathbf{r}, T)$ more convenient for analysing critical phenomena

$$C(\mathbf{r}_{12}, T) = c(\mathbf{r}_{12}, T) + n \int C(\mathbf{r}_{23}, T) c(\mathbf{r}_{13}, T) d\mathbf{r}_3.$$

This equation divides the total correlation function $C(\mathbf{r}, T)$ into two terms: the direct term $c(\mathbf{r}, T)$ describing interaction $1 \to 2$ with the nearest environment and therefore short-range, and the non-direct term expressed by an integral. The latter term is related to the interactions $1 \to 3 \to 2, \ 1 \to 3 \to 4 \to 2, \ \ldots, 1 \to 3 \to 4 \to \cdots \to n \to 2$ and therefore defines correlations at large distances. Both correlation functions are dimensionless. In a critical point, the role of short distances is small. Disregarding them and normalizing $c(\mathbf{r}, T)$ in a proper way, one can consider it as a

three-dimensional probability density function, and the equation may be interpreted as the equation of random walks.

The third thesis of the Ornstein-Zernicke theory is that the function $\tilde{c}(\mathbf{k}, T)$ is assumed to be an even analytical function of $k = |\mathbf{k}|$ in a vicinity of $k = 0$ for all temperatures, including T_c:

$$\tilde{c}(\mathbf{k}, T) - \tilde{c}(0, T) \propto k^2, \quad k \to 0.$$

This fact provides finiteness of the variance of a "path length" and a diffusion (Brownian) character of the correlation function asymptotic

$$C^{\text{as}}(\mathbf{r}, T_c) = Ar^{-1},$$

satisfying the stationary equation of normal diffusion,

$$-\Delta\, C^{\text{as}}(\mathbf{r}, T_c) = 4\pi A\delta(\mathbf{r}).$$

In the standard theory, the expansion up to the second member is used. It is presented in the form

$$\tilde{c}(\mathbf{k}, T) = \tilde{c}(0, T) - (R^2/n)k^2,$$

where $R(T)$ is the Debye length decay. In this case, the structural function takes the form

$$a_{\mathbf{k}} = \frac{1}{1 - nc(0, T) + R^2k^2} = \frac{R^{-2}}{\xi^{-2} + k^2},$$

where the quantity ξ depending on temperature and density is called the correlation length. When $T \to T_c$, it reduces to infinity. In this approximation, the inverse Fourier transformation gives the following result for the total correlation function:

$$C(\mathbf{r}, T) \propto \begin{cases} (R^2r)^{-1} \exp(-r/\xi), & T < T_c, \\ r^{-1}, & T = T_c. \end{cases}$$

One should note, however, that in case of a two-dimensional system, the approach under consideration leads to an absurd result: the correlation function grows logarithmically. Moreover, numerical simulations and a number of experiments with magnetic systems show that in a critical point the correlation function of d-dimensional system is better approximated by the formula proposed by Fisher,

$$C(\mathbf{r}, T_c) \propto r^{-(d-2+\eta)}, \tag{4.37}$$

where η is a positive number smaller than one. This formula practically weakens the third thesis of the theory, replacing it by the following requirement:

$$\tilde{c}(\mathbf{k}, T) - \tilde{c}(0, T) \propto k^{\alpha}, \; 0 < \alpha < 2 \quad k \to 0.$$

As one can see from our work [79], in this case, the asymptotical part of the solution is characterized by the Fourier transform $\widetilde{C}^{as}(\mathbf{k}, T_c)$ satisfying the equation

$$|\mathbf{k}|^{\alpha}\widetilde{C}^{\text{as}}(|\mathbf{k}|, T_c) = A(\alpha).$$

It presents the Fourier image of the equation with the Laplace operator of fractional order

$$(-\Delta)^{\alpha/2}C^{\text{as}}(\mathbf{r}, T_c) = A(\alpha)\delta(\mathbf{r}).$$

In framework of the fractional model of critical state of medium at $T \to T_c$, the analyticity of the direct correlation function transform is violated. As a consequence, the Poisson equation describing the asymptotical C^{as} reduces to the fractional Poisson equation. This is an evidence of the fractal structure of random excitations of medium at the phase transition [80].

4.3.2 Fractional Ginzburg-Landau Equation

Milovanov and Rasmussen [55] advocate an application of the fractional derivative formalism to the thermodynamics of second-type phase transitions in the presence of a coexisting nonlocal ordering which may influence the properties of the basic "symmetric" phase below the transition point. As an example, they mention highly correlated electron liquid states, such as electron liquids with fractionally charged excitations, as well as high-temperature superconducting fluid phases in copper-oxide compounds and their derivatives where the nonlocal ordering can be associated with the so-called stripy order.

The authors consider a system with two thermodynamically distinct phases, the "symmetric" (superconducting) phase below the transition temperature T_c, and asymmetric (normal) phase above the T_c. The deviation between the symmetric and normal phases is characterized by the order parameter, $\psi = \psi(x)$, which is assumed to be an analytical function of $x \in \mathbb{R}_1$.

Extremizing the conventional free energy expansion in the vicinity of the critical point $T \to T_c$,

$$F = F_n + \int_{-\infty}^{\infty}\left[\frac{\hbar^2}{4m}|\psi'| + a|\psi|^2 + \frac{b}{2}|\psi|^4\right]dx'$$

where F_n denotes the contribution from the normal phase, $\psi' = \partial\psi/\partial x$, the coefficient $a = \alpha(T - T_c)$ and coefficient $b > 0$ depends solely on the mass density of the substance yields the classical Ginzburg-Landau equation

$$-\frac{\hbar^2}{4m}\psi'' + a\psi + b|\psi|^2\psi = 0.$$

Assuming that blobs of the symmetric phase appear at random throughout the material as T approaches the critical range and *postulating* the free energy expansion for a long-range correlated system at criticality in the generalized form

$$F = F_n + \int_{-\infty}^{\infty} \left[\frac{\hbar^2}{4m} |\Delta_\mu * \psi'| + a|\psi|^2 + \frac{b}{2}|\psi|^4 \right] dx',$$

where

$$\Delta_\mu(x) = \frac{1}{2}\Phi_{1-\mu}(|x|) = \frac{|x|^{-\mu}}{2\Gamma(1-\nu)}$$

is the "smeared delta-function", and μ is expressed in terms of the Hausdorff fractal dimension D of the Cantor set:

$$\mu = 1 - D \in (0, 1).$$

Splitting the integration from $-\infty$ to ∞ into integrals over positive and negative semi-axis and taking into account parity of ψ ($\psi(-x) = \psi(x)$), Milovanov and Rasmussen obtain that

$$\Delta_\mu * \psi'(x) = \frac{1}{\Gamma(1-\mu)} \frac{\partial}{\partial x} \int_{-\infty}^{x} \frac{\psi(\xi) d\xi}{(x-\xi)^\mu} \equiv {}_{-\infty}\mathsf{D}_x^\mu \psi(x), \quad 0 \leqslant \mu \leqslant 1.$$

Thus, the free energy expansion becomes

$$F = F_n + \int_{-\infty}^{\infty} \left[A_\mu |{}_0\mathsf{D}_x^\mu \psi|^2 + a_\mu |\psi|^2 + \frac{1}{2} b_\mu |\psi|^4 \right] dx.$$

Varying it with respect to functions ψ and ψ^* considered as independent order parameters,

$$\delta F = \int_{-\infty}^{\infty} \left[A_\mu \, {}_0\mathsf{D}_x^\mu \psi \, {}_0\mathsf{D}_x^\mu \delta\psi^* + a_\mu \psi \delta\psi^* + b_\mu |\psi|^2 \psi \delta\psi^* \right] dx,$$

and performing integration by part, the authors come to a fractional generalization of the *Ginzburg-Landau equation*: The coefficient $A_\mu = \lambda^{2\mu-2}\hbar^2/4m$ where λ is the microscopic correlation length.

An important consequence of the fractional generalization is that the cross-over temperature T where a changes sign may not coincide with the transition temperature T_c when the effects of nonlocality play a role (i.e., when the index is smaller than 1) [55]. The authors believe as well that this fractional approach will be useful for modeling the unconventional superconductivity in materials with complex microscopic organization.

The interested reader is referred to Tarasov and Zaslavsky [75].

4.3.3 Classification of Phase Transitions

Based on fractional analysis, Hilfer [27–30] develops a new viewpoint on the theory of phase transitions that fractional derivatives play a more fundamental role than in the problems considered above.

The generalized Ehrenfest classification scheme has led to the hypothesis on existence of phase transitions of fractional (less than 1) order, called *non equilibrium transitions*.

According to the general thermodynamic postulates, the internal energy of a system $U = U(S, V, N)$ must be single-valued, convex, steadily increasing, differentiable (almost everywhere), and homogeneous (of the first order) function with respect to its arguments: entropy S, volume V, and number of particles N. In the classical model

$$0 < V < \infty, \quad 0 < N < \infty, \quad -\infty < S < \infty, \quad -\infty < U < \infty,$$

whereas in the quantum scheme, S and U must be bounded below. These conditions are necessary and sufficient to ensure the thermodynamic stability.

The classification of phase transitions is usually formulated in terms of free energy density or pressure, because other thermodynamic potentials are continuous for a wide class of interactions. The pressure $p = p(T, \mu)$ is the conjugate, convex with respect to the energy density $u(s, \rho)$ expressed through the entropy density $s = S/V$ and the density of the number of particles $\rho = N/V$:

$$p(T, \mu) = \sup_{s,\rho}[\mu\rho + Ts - u(s, \rho)],$$

where μ is a chemical potential and T is an absolute temperature.

Analysing phase transition involves the using of thermodynamic limits. Let us consider the thermodynamic process $\mathrm{C}: \mathbb{R} \to \mathbb{R}^2$, $\sigma \mapsto (T(\sigma), q(\sigma))$, such that $T(0) = T_c$ and $\mu(0) = \mu_c$ correspond to the critical point. Function $p = p(T(\sigma), \mu(\sigma))$ can be decomposed into regular and singular components

$$p = p_{\mathrm{reg}} + p_{\mathrm{sng}}.$$

In the Erenfest classification, a phase transition belongs to transitions of order n if and only if

$$J(C, n; \sigma) \equiv \mathsf{D}^n_\sigma p_{\mathrm{sng}}(T(\sigma), \mu(\sigma)) \sim A\ 1(\sigma) + B, \quad n = 1, 2, 3, \ldots, \tag{4.38}$$

for $\sigma \approx 0$, where $A,\ B \in \mathbb{R}$ and $1(\sigma)$ is the Heavyside step function. The condition (4.38) reflects the finite discontinuity of nth derivative of pressure at the point $\sigma = 0$. It can be presented in more general form,

$$\lim_{\sigma\to 0\pm}\frac{J(C,n;b\sigma)}{J(C,n;\sigma)}=1,\quad b>0. \tag{4.39}$$

This means that $J(C,n;\sigma)$ is a slowly varying function of the argument σ at $\sigma\to 0$. The condition (4.39) is assumed as a basis of a fractional differential classification of phase transitions, proposed in Hilfer [29]: a phase transition is called the phase transition of order ν, if and only if

$$\lim_{\sigma\to 0\pm}\frac{J(C,\nu;b\sigma)}{J(C,\nu;\sigma)}=1$$

for any positive b. Here

$$J(C,\nu;\sigma)={}_0\mathsf{D}^{\nu}_{\sigma}p_{\mathrm{sng}}(T(\sigma),\mu(\sigma)).$$

To understand whether phase transitions of fractional orders are allowed thermodynamically or not, Hilfer has considered model thermodynamical systems, for which

$$u(s)=as+b\sqrt{s^2+c^2}, \tag{4.40}$$

where $a,\ b,\ c>0$ and $a>b$. It is obvious that $T(s)=\partial u/\partial s>0$ and $\partial^2 u/\partial s^2>0$, so that $u(s)$ is a convex and steadily increasing function. It demonstrates the transitions of the order $\nu_u=1$ at the critical point $s_c=\pm\infty$. Note that the system (4.40) is characterized by a limited region of possible temperatures

$$a-b=T_{\min}<T<T_{\max}=a+b.$$

Pressure in the given system

$$p(T)=c\sqrt{b^2-(T-a)^2} \tag{4.41}$$

points to the existence of a phase transition of order

$$\nu_p=1/2,$$

when $T=T_{\min}$ and $T=T_{\max}$.

The simple example (4.40) shows that the thermodynamics allows two essentially different types of phase transitions: ordinary phase transitions of integer orders $\nu_p\geqslant 1$ and extraordinary (anomalous) transitions of fractional orders $\nu_p<1$ with limited range of permissible temperatures. The interest to this circumstance is conditioned by the fact that equilibrium thermodynamics formally admits the situation when the temperatures $T_{\min}$ and $T_{\max}$ bounding this domain can not be achieved in quasi-static thermodynamic processes. A quasi-static process is a consequence of state changes infinitely slower than the process of equilibrium establishing. It is reasonable to ask: is it correct to identify this temperature scale with the temperature scale of an

ideal gas when $\nu < 1$? In such systems, $T_{\min}$ plays a role of absolute zero, and $T_{\max}$ plays the role of $T = \infty$. In Hilfer [29], it was proposed to avoid this restriction by introducing many-valued thermodynamic potentials for interpretation of transitions of order $\nu < 1$. Later, Hilfer comes back to single-valued potentials [31].

The entropy density $s(T) = (\partial p/\partial T)_\mu$ following from (4.41) tends to $-\infty$ as $T \to T_{\min}$, but the third law of thermodynamics assumes the existence of a special temperature T_0 when the entropy density vanishes

$$s(T_0) = 0.$$

Because of the quantum-mechanical origin of the third law, the value T_0 can be considered as a minimum temperature of a quantum-mechanical system, while $T_{\min}$ keeps the meaning of the minimum temperature of a classical system ($T_0 > T_{\min}$).

4.4 Around Equilibrium

4.4.1 Relaxation to the Thermal Equilibrium

The central place among irreversible processes belongs to the relaxation process. Passing from integer orders to fractional orders of derivatives changes the relaxation law from exponential to algebraic form. This can be seen immediately from FFPE solution. Indeed, let us take the subdiffusion FFPE

$$\frac{\partial p}{\partial t} = {}_0\mathsf{D}_t^{1-\nu} \mathsf{L}_{\mathrm{FP}} p(x,t),$$

where

$$\mathsf{L}_{\mathrm{FP}} = \frac{\partial}{\partial x}\frac{V'(x)}{m\eta_\nu} + K_\nu \frac{\partial^2}{\partial x^2}.$$

Introducing the separation ansatz (as it is done in Metzler et al. [50])

$$p_n(x,t) = \phi_n(x) T_n(t)$$

yields two eigenequations

$$\frac{\partial T_n}{\partial t} = -l_{n,\nu}\, {}_0\mathsf{D}_t^{1-\nu} T_n,$$

$$\mathsf{L}_{\mathrm{FP}} \phi_n = -l_{n,\nu}\phi_n$$

for the eigenvalue $l_{n,\nu}$. The solution of the first equation is

$$T_n(t) = E_\nu(-l_{n,\nu} t^\nu).$$

Introducing the scaled potential $\Phi(x) = V(x)/k_{\mathrm{B}}T$ and the functions

$$\psi_n(x) = e^{\Phi(x)/2}\phi_n(x),$$

Metzler et al. represented the Green function of the equation under consideration as

$$p(x,t|x',0) = e^{[\Phi(x')-\Phi(x)]/2}\sum_n \psi_n(x)\psi_n(x')E_\nu(-l_{n,\nu}t^\nu).$$

The stationary solution is given by

$$p_{\mathrm{st}} = \lim_{t\to\infty} e^{[\Phi(x')-\Phi(x)]/2}\psi_0(x)\psi_0(x')$$

in accordance with the classical case ($\nu = 1$), and it is nothing but the Boltzmann distribution. However, the relaxation of other modes is not exponential but asymptotically algebraic, $\propto (l_{n,\nu}t)^{-\nu}$, $t \to \infty$. This investigations were continued by Sokolov [67], Sokolov et al. [69], and by other authors.

In some sense an alternative approach to this problem was developed in Feldman et al. [16]. The authors consider the problem in the framework of the fractionalised Liouville equation

$${}_0\mathsf{D}_t^\nu \rho(p_j,q_j,t) = -i\mathsf{L}_L\rho(p_j,q_j,t),$$

where L_L is the Liouvillian, expressed through the Poisson brackets as

$$\mathsf{L}_L f = -i\{\mathsf{H}, f\}.$$

Defining the *fractional equilibrium state* via equation

$${}_0\mathsf{D}_t^\nu \rho(p_j,q_j,t) = 0,$$

they obtain that $\rho \propto t^{-(1-\nu)}$. The same asymptotic behaviour are obtained and for averaged dynamical variables. However, these conclusion may be result of the chosen form of the fractional equation, which does not preserve the integral $\iint \rho dpdq$. It seems to me, the fractional equation

$$\frac{\partial\rho(p_j,q_j,t)}{\partial t} = -i\,{}_0\mathsf{D}_t^{1-\nu}\mathsf{L}_L\rho(p_j,q_j,t)$$

would be more appropriate for this aim.

Some fractional (in spatial variables) generalizations of basic equations of statistical mechanics are considered by Tarasov [73, 74].

4.4.2 Fractionalization of the Entropy

In recent years, a considerable amount of theoretical research activity has been stimulated by the conjecture that a certain class of dynamical systems might be governed by a peculiar thermal statistics, in which the usual Boltzmann-Gibbs exponential distribution

$$p_j = \frac{e^{-\epsilon_j}}{\sum\limits_j e^{-\epsilon_j}}, \quad \epsilon_j = E_j/kT$$

is replaced by a type of power-law distribution proposed by Tsallis in 1988 (see his book [77] and references therein)

$$p_j = \frac{e_q^{-\epsilon_j}}{\sum\limits_j e_q^{-\epsilon}}.$$

Tsallis' exponential function

$$e_q^t \equiv [1 + (1-q)t]^{1/(1-q)}, \; e_1^t = e^t$$

was introduced as a solution of the non-linear differential equation

$$\frac{dy}{dt} = y^q, \quad y(0) = 1, \quad q \in \mathbb{R}.$$

The inverse function (Tsallis' logarithm)

$$\ln_q(x) = \frac{x^{1-q} - 1}{1-q}, \quad x > 0, \quad \ln_1 x = \ln x$$

was used for constructing the Tsallis entropy:

$$S^{\mathrm{T}}[p_j] = -\sum_{j=1}^{n} p_j \ln_q p_j = \frac{1 - \sum\limits_j p_j^q}{q-1}, \quad p_j = p_j^{\mathrm{T}}.$$

The Tsallis logarithm possesses the following property:

$$\ln_q(y_A y_B) = \ln_q y_A + \ln_q y_B + (1-q)(\ln_q y_A)(\ln_q y_B).$$

This is why the thermodynamics constructed by Tsallis is often called the *nonextensive thermodynamics*. When $q \to 1$, the Tsallis entropy reduces to the Shennon entropy

$$S^{\mathrm{S}}[p_j] = -k \sum_{j=1}^{n} p_j \ln p_j .$$

The Tsallis entropy is not a unique nonextensive generalization of the classical one. The concept of entropy developed by L. Boltzmann for statistical description of the tendency of systems to the equilibrium state and later devised by C. Shennon to study problems with information transmission. Possibly, the first nonextensive entropy was generalization performed by Rénie. So, one can write the sequence: the Shannon entropy

$$S^{\mathrm{S}}[p_j] = -k \sum_{j=1}^{n} p_j \ln p_j ,$$

the Rényie entropy

$$S^{\mathrm{R}}[p_j] = \frac{1}{1-q} \ln \left(\sum_{j=1}^{n} p_j^q \right),$$

the Tsallis entropy

$$S^{\mathrm{T}}[p_j] = \frac{1 - \sum\limits_{j} p_j^q}{q-1},$$

the Ubriaco entropy

$$S^{\mathrm{U}}[p_j] = \sum_{j=1}^{n} p_j (-\ln p_j)^q .$$

The latter is based on observation that the Shannon entropy

$$S^{\mathrm{S}}[p_j] = -\sum_{j=1}^{n} p_j \ln p_j$$

can be represented as

$$S^{\mathrm{S}}[p_j] = \lim_{t \to -1} \frac{d}{dt} \sum_{j=1}^{n} e^{-t \ln p_j},$$

and Tsallis entropy is also representable by means of a derivative of a special kind (so-called the *Jackson q-derivative* $\mathsf{D}_t^{\mathrm{J}} f(t)$ [35]):

$$S^{\mathrm{S}}[p_j] = \lim_{t \to -1} \mathsf{D}_t^{\mathrm{J}} \sum_{j=1}^{n} e^{-t \ln p_j} \equiv \lim_{t \to -1} t^{-1} \frac{1 - q^{td/dt}}{1-q} \sum_{j=1}^{n} e^{-t \ln p_j} .$$

These facts inspired [78] to suggest a new generalization in the form

$$S^{\mathrm{U}}[p_j] = \lim_{t\to -1} \frac{d}{dt}\left({}_{-\infty}\mathsf{D}_t^{q-1} \sum_{j=1}^{n} e^{-t\ln p_j} \right) = \sum_{j=1}^{n} p_j (-\ln p_j)^q,$$

where $0 \leqslant q \leqslant 1$. He showed that this function is concave, positive definite, and non-additive. Maximizing the entropy subject to the usual constrains,

$$\frac{\partial}{\partial p_j}\left[S^{\mathrm{U}}[p_j] + \alpha\left(1 - \sum_{j=1}^{n} p_j\right) + \beta\left(\epsilon - \sum_{j=1}^{n} \epsilon_j p_j\right)\right] = 0$$

leads to an exponential probability distribution for $q = 1/2$,

$$p_j \propto e^{-q(\Omega^2 + \sqrt{\Omega^2+2}\Omega)}, \quad \Omega = \alpha + \beta\epsilon.$$

For an arbitrary value of q, p_j obey the equation

$$(-\ln p_j)^q - q(-\ln p_j)^{q-1} = \alpha + \beta\epsilon_j.$$

The question, where and why the generalized entropies should reflect a real physical situation, is in our opinion still open. The entropy is too abstract concept in order to get from it a simple clear answer. Another problem being not less important is to find such a form of thermodynamics which obeys the correspondence principle: demonstration of continuous transition from macroscopic to mesoscopic scales. Perhaps, a more obvious consideration of this problems will be performed in terms of probability distributions and stochastic process theory having a more clear physical sense (see also Compte and Jou [12]; Essex et al. [14]; Gaies and El-Akrmi [21])).

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Chapter 5
Electrodynamics

5.1 Electromagnetic Field

5.1.1 Maxwell Equations

5.1.1.1 Constitutive Equations

Fractionalization of the Maxwell system was started with the constitutive equations. Relying on hereditary interrelations between characteristics of the electromagnetic field in a matter (see Sect. 1.2.5 of Vol. I)

$$\mathbf{D}(t) = \int_{-\infty}^{t} \phi(t-\tau)\mathbf{E}(\tau)d\tau,$$

$$\mathbf{B}(t) = \int_{-\infty}^{t} \psi(t-\tau)\mathbf{H}(\tau)d\tau,$$

Westerlund [143] chosen the kernels in a power form,

$$\mathbf{D}(t) = \varepsilon \frac{1}{\Gamma(-\nu)} \int_{-\infty}^{t} (t-\tau)^{-\nu-1}\mathbf{E}(\tau)d\tau = \varepsilon {}_{-\infty}\mathsf{D}_t^{\nu}\mathbf{E}(t),$$

$$\mathbf{B}(t) = \mu \frac{1}{\Gamma(-\nu)} \int_{-\infty}^{t} (t-\tau)^{-\nu-1}\mathbf{H}(\tau)d\tau = \mu {}_{-\infty}\mathsf{D}_t^{\nu}\mathbf{H}(t),$$

with $\nu < 0$. One year earlier, Caputo [20] suggested the fractional-order version of **E**-**D** interrelation in the form[1]

[1] Because of an infinite lower limit, GC- and RL-derivatives are indistinguishable.

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$$\gamma \mathbf{D}^{(\nu)} + \alpha \mathbf{D} = \sigma \mathbf{E} + \varepsilon \mathbf{E}^{(\nu)}, \quad 0 < \nu < 1.$$

5.1.1.2 Intermediate Solutions

During 1992–2000, N. Engheta has been published a series of articles concerning application of fractional calculus to a wide class of problems in electromagnetic theory (see References). In [34], he introduced a fractional generalization of the curl operator considering solutions of the source-free Maxwell equations in vacuum. For the time-harmonic case with the frequency ω, the equations are written as

$$\frac{1}{ik_0}\mathsf{curl}(\eta_0 \mathbf{H}) = -\mathbf{E}, \tag{5.1}$$

$$\frac{1}{ik_0}\mathsf{curl}(\mathbf{E}) = \eta_0 \mathbf{H}, \tag{5.2}$$

$$\mathsf{div}(\eta_0 \mathbf{H}) = 0, \tag{5.3}$$

$$\mathsf{div}(\mathbf{E}) = 0, \tag{5.4}$$

where $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$, $k_0 = \omega\sqrt{\mu_0 \varepsilon_0}$, ε_0 is permittivity and μ_0 is permeability. Performing the spatial Fourier transform,

$$\frac{1}{ik_0} i\mathbf{k} \times (\eta_0 \widetilde{\mathbf{H}}) = -\widetilde{\mathbf{E}},$$

$$\frac{1}{ik_0} i\mathbf{k} \times \widetilde{\mathbf{E}} = \eta_0 \widetilde{\mathbf{H}},$$

$$i\mathbf{k} \cdot (\eta_0 \widetilde{\mathbf{H}}) = 0,$$

$$i\mathbf{k} \cdot \widetilde{\mathbf{E}} = 0,$$

Engheta used the fractional cross product operator $(ik_0)^{-\alpha}(i\mathbf{k}\times)^{\alpha}$ (see Sect. 4.7.3.1d of Vol. I), defined by the expression

$$(\mathbf{e}_z \times)^{\alpha} \widetilde{\mathbf{F}} = (i^{\alpha}/2)(\widetilde{F})_x + i\widetilde{F}_y)(\mathbf{e}_x - i\mathbf{e}_y) + [(-i)^{\alpha}/2](\widetilde{F})_x - i\widetilde{F}_y)(\mathbf{e}_x + i\mathbf{e}_y) + (0)^{\alpha}\widetilde{F}_z \mathbf{e}_z.$$

Taking $\mathbf{F} = \mathbf{e}_x$ and making a proper choice of branches for i^{α} and $(-i)^{\alpha}$, he showed that

$$(\mathbf{e}_z \times)^{\alpha} \mathbf{e}_x = \cos(\alpha\pi/2)\mathbf{e}_x + \sin(\alpha\pi/2)\mathbf{e}_y.$$

Evidently

$$(\mathbf{e}_z\times)^\alpha\mathbf{e}_x=\begin{cases}\mathbf{e}_x,\ \alpha=0;\\ \mathbf{e}_y,\ \alpha=1.\end{cases}$$

For other values of α between zero and unity, one gets the intermediate fractional cases of cross product operator.

Applying $(\mathbf{e}_z\times)^\alpha$ to both sides of the first two Maxwell equations yields

$$(ik_0)^{-\alpha-1}(i\mathbf{k}\times)^{\alpha+1}(\eta_0\widetilde{\mathbf{H}})=-(ik_0)^{-\alpha}(i\mathbf{k}\times)^\alpha\widetilde{\mathbf{E}},$$

$$(ik_0)^{-\alpha-1}(i\mathbf{k}\times)^{\alpha+1}\widetilde{\mathbf{E}}=(ik_0)^{-\alpha}(i\mathbf{k}\times)^\alpha(\eta_0\widetilde{\mathbf{H}}).$$

Using these formulas with appropriate mathematical choice of branches of eigenvalues of the fractional cross product operator, Engheta [34] has found a new set of solutions to the source-free Maxwell equations (5.1)–(5.4). In the $\mathbf{k}$-space, *fractional dual* fields have the form

$$\widetilde{\mathbf{E}}_{\rm fd}=(ik_0)^{-\alpha}(i\mathbf{k}\times)^\alpha\widetilde{\mathbf{E}},\quad \eta_0\widetilde{\mathbf{H}}_{\rm fd}=(ik_0)^{-\alpha}(i\mathbf{k}\times)^\alpha(\eta_0\widetilde{\mathbf{H}})$$

and becomes

$$\mathbf{E}_{\rm fd}=(ik_0)^{-\alpha}\mathsf{curl}^\alpha\mathbf{E},\quad \eta_0\mathbf{H}_{\rm fd}=(ik_0)^{-\alpha}\mathsf{curl}^\alpha(\eta_0\mathbf{H}) \tag{5.5}$$

in the coordinate space. For $\alpha=0$ we recognize here the original set of solutions satisfying the Maxwell equations (5.1)–(5.4),

$$\mathbf{E}_{\rm fd}=\mathbf{E},\quad \mathbf{H}_{\rm fd}=\mathbf{H},$$

when $\alpha=1$, these formulas represent the dual fields of the origin solutions to the Maxwell equations:

$$\mathbf{E}_{\rm fd}=(ik_0)^{-1}\mathsf{curl}\,\mathbf{E}=\eta_0\mathbf{H},\quad \eta_0\mathbf{H}_{\rm fd}=(ik_0)^{-1}\mathsf{curl}\,(\eta_0\mathbf{H})=-\mathbf{E}.$$

Recall that the electromagnetic duality principle states that if $(\mathbf{E},\,\eta\mathbf{H})$ is one set of field solutions to Maxwell equations then another set of their solutions is $(\eta\mathbf{H},\,-\mathbf{E})$. These two solution sets are termed as original and dual solutions to Maxwell equations respectively.

For all other values of $\alpha\in(0,1)$, Eq. (5.5) provide us with the intermediate solutions between the original fields and the dual fields (see also [83]).

5.1.1.3 Space-Fractional Maxwell's Equations

Referring to works [67, 99] which state that fractional integrals can be considered as approximations of integrals on fractals and ignoring the opposite point of view (see for example, [101][2], [102], or [115][3]).

Tarasov [116] extended the Maxwell equations to a fractal system with fractional dimension **D** bounded by fractal surface with dimension d as follows:

$$\oint_S \mathbf{E}\cdot d\mathbf{S}_2 = \frac{1}{\varepsilon_0}\int_V \rho dV_3 \quad \mapsto \quad \oint_S \mathbf{E}\cdot d\mathbf{S}_d = \frac{1}{\varepsilon_0}\int_V \rho dV_D,$$

$$\oint_L \mathbf{E}\cdot d\mathbf{I}_1 = -\frac{\partial}{\partial t}\int_S \mathbf{B}\cdot d\mathbf{S}_2 \quad \mapsto \quad \oint_L \mathbf{E}\cdot d\mathbf{I}_\gamma = -\frac{\partial}{\partial t}\int_S \mathbf{B}\cdot d\mathbf{S}_d,$$

$$\oint_S \mathbf{B}\cdot d\mathbf{S}_2 = 0, \quad \mapsto \quad \oint_S \mathbf{B}\cdot d\mathbf{S}_d = 0,$$

$$\oint_L \mathbf{B}\cdot d\mathbf{I}_1 = \int_S \mu_0\left(\mathbf{J}+\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}\right)\cdot d\mathbf{S}_2 \mapsto \oint_L \mathbf{B}\cdot d\mathbf{I}_\gamma = \int_S \mu_0\left(\mathbf{J}+\varepsilon_0\frac{\partial \mathbf{E}}{\partial t}\right)\cdot d\mathbf{S}_d.$$

Here

$$dV_D = c_3(D,\mathbf{r})dV_3, \quad c_3(D,\mathbf{r}) = \frac{2^{3-D}\Gamma(3/2)}{\Gamma(D/2)}r^{D-3}, \quad r=|\mathbf{r}|,$$

$$dS_d = c_2(d,\mathbf{r})dS_2, \quad c_2(d,\mathbf{r}) = \frac{2^{2-d}}{\Gamma(d/2)}r^{d-2},$$

$$d\mathbf{I}_\gamma = c_1(\gamma,\mathbf{r})d\mathbf{I}_\gamma, \quad c_1(\gamma,\mathbf{r}) = \frac{2^{1-\gamma}\Gamma(1/2)}{\Gamma(\gamma/2)}r^{\gamma-1}.$$

As a result, he obtained the following generalization of the differential Maxwell equations:

$$\mathrm{div}[c_2(d,\mathbf{r})\mathbf{E}] = \frac{1}{\varepsilon_0}c_3(D,\mathbf{r})\rho,$$

[2] "The aforesaid shows clearly, in the opinion of this writer, the erroneousness of the derivations in the paper and of the interpretation of the fractal Cantor set as a realization of the fractional integral. But what can one say about the underlying physical idea? May it be that this operation approximates, in a sense, the fractional integration?" (p. 1155). And then the author gives a negative answer.

[3] "The 'link' between the fractals and the fractional calculus is more artificial than natural. Although the Cantor set in the above-mentioned procedure forms self-similar properties, they are not fully adequate to those expected for the fractional integral/derivative." (p. 63).

$$\mathsf{curl}[c_1(\gamma, \mathbf{r})\mathbf{E}] = -c_2(d, \mathbf{r})\frac{\partial}{\partial t}\mathbf{B},$$

$$\mathsf{div}[c_2(d, \mathbf{r})B] = 0,$$

$$\mathsf{curl}[c_1(\gamma, \mathbf{r})\mathbf{B}] = \mu_0 c_2(d, \mathbf{r})\mathbf{J} + \varepsilon_0\mu_0 c_2(d, \mathbf{r})\frac{\partial \mathbf{E}}{\partial t}.$$

In connection with this approach I would like to mention another work [80]. Its authors obtained the Maxwell equations in space-time with a fractional dimension $D = D_s + D_t$ using the Euler-Lagrange density $\mathscr{L} = \mathscr{L}(\phi, \partial\phi_\mu)$ obeying the equation

$$\frac{\partial \mathscr{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathscr{L}}{\partial(\partial_\mu \phi)} = 0, \quad \mu = 0, 1, \ldots, N,$$

where $\phi = \phi(t, x^1, \ldots, x^N)$ and $\partial_\mu \phi = \left(\frac{\partial \phi}{\partial t}, \frac{\partial \phi}{\partial x^1}, \ldots, \frac{\partial \phi}{\partial x^N}\right)$, and the fractional volume element $d^D x$ and the fractional line element are given respectively as

$$d^D x = \prod_\mu d^{\alpha_\mu} x_\mu, \quad d^{\alpha_\mu} x_\mu = \frac{\pi^{\alpha_\mu/2}|x|^{\alpha_\mu - 1}}{\Gamma(\alpha_\mu/2)} dx_\mu,$$

and $\sum_{j=1}^N \alpha_j = D_s,\ \alpha_t = D_t$. The authors arrive at the conclusion that the Maxwell equations in a fractional space are local and moreover they have the same form as those obtained in an integer-dimensional space-time.

In 2008, Tarasov came back to the Maxwell equations in order to reformulate their non-local generalization in terms of fractional operators described in Sect. 4.7.3.1 of Vol. I:

$${}^{\alpha_1}\mathsf{div}_W \mathbf{E} = g_1 \rho,$$

$${}^{\alpha_2}\mathsf{curl}_W \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$

$${}^{\alpha_3}\mathsf{div}_W \mathbf{B} = 0,$$

$$g_2\, {}^{\alpha_4}\mathsf{curl}_W \mathbf{B} = \mathbf{j} + g_3^{-1}\frac{\partial \mathbf{E}}{\partial t},$$

where g_1, g_2, and g_3 are corresponding electromagnetic constants. Concluding his article, Tarasov noted that the nonlocal properties of electrodynamics can be considered as a result of dipole-dipole interactions with a fractional power-law screening that is connected with the integro-differentiation of a non-integer order.

5.1.1.4 Time-Fractional Maxwell Equations

Bogolyubov et al. [16] have introduced time-fractional version of Maxwell's system. Starting with fractionalization of the classical **j**-**v** interrelation,

$$\mathbf{j}(t) = qn\mathbf{v} \quad \mapsto \quad \mathbf{j}(t) = \frac{q}{\tau}\int_0^t g(t-t')n(t')\left(\frac{d\mathbf{r}(t')}{dt'}\right)dt',$$

$$g(t) = \frac{1}{\Gamma(1-\alpha)}\,\frac{1}{t^{\alpha}},$$

$$\mathbf{j}(t) = \frac{q}{\tau}\,{}_0\mathsf{D}_t^{\alpha-1}\left(n(t)\frac{d\mathbf{r}(t)}{dt}\right), \quad \mathbf{v}(t) = \frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha-1}\frac{d\mathbf{r}(t)}{dt} = \frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha}\mathbf{r}(t),$$

the authors represent the fractionalized Lorentz force as

$$\mathbf{F}(t) = \frac{q}{\tau}\mathbf{B} \times {}_0\mathsf{D}_t^{\alpha}\mathbf{r}(t).$$

The electric voltage arising during the motion of electrons along a closed smooth contour L is written as

$$\frac{1}{q}\oint_L \mathbf{F}\cdot d\mathbf{l} = \frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha}\oint_L [\mathbf{B}\times\mathbf{r}(t)]\cdot d\mathbf{l} = \frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha}\Phi,$$

where Φ is the magnetic flux through the field bounded by L. Continuing in the same spirit, the authors arrive at the following system:

$$\mathsf{div}\,\mathbf{D} = \rho,$$

$$\mathsf{curl}\,\mathbf{E} = -\frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha}\mathbf{B},$$

$$\mathsf{div}\,\mathbf{B} = 0,$$

$$\mathsf{curl}\,\mathbf{H} = \mathbf{j} + \frac{1}{\tau}\,{}_0\mathsf{D}_t^{\alpha}\mathbf{D}.$$

Kilicman et al. [62] solve a reduced system of the time-fractional Maxwell equations

$$\frac{\partial E_x}{\partial z} + \mu\ {}_0\mathsf{D}_t^{\alpha} H_y = 0,$$

$$\frac{\partial H_y}{\partial z} + \varepsilon\ {}_0\mathsf{D}_t^{\alpha} E_x + \sigma E_x = 0,$$

under the boundary conditions

$$E_x(z,0) = H_y(z,0) = 0,$$

$$\frac{\partial E_x(z,0)}{\partial z} = \frac{\partial H_y(z,0)}{\partial z} = 0,$$

$$\left.\frac{\partial E_x(z,t)}{\partial t}\right|_{t=0} = \left.\frac{\partial H_y(z,t)}{\partial t}\right|_{t=0} = 0,$$

and

$$E(0,t) = \begin{cases} 0, & t < 0; \\ f(t), & t \geq 0. \end{cases}$$

Solution of these equations obtained by the Sumudu transform is expressed as

$$E(z,t) = \int_0^{\infty} \frac{s}{\mu\varepsilon\sqrt{a^2 - 4b}} (\sin sz) ds \times \int_0^t \tau^{\alpha-1} [E_{\alpha,\alpha}(\lambda_1, \tau^{\alpha}) - E_{\alpha,\alpha}(\lambda_2 \tau^{\alpha})] f(\tau - t) d\tau,$$

and

$$H(z,t) = \int_0^{\infty} \frac{s^2}{\mu\varepsilon\sqrt{a^2 - 4b}} (\cos sz) ds \times \int_0^t \tau^{2\alpha-1} [E_{\alpha,2\alpha}(\lambda_1, \tau^{\alpha}) - E_{\alpha,2\alpha}(\lambda_2 \tau^{\alpha})] f(\tau - t) d\tau.$$

5.1.1.5 Bifractional Maxwell Equations

Using the fractional generalization of classical differential forms (Sect. 4.7.3.5 of Vol. I) Baleanu et al. [6] suggest fractionalized (with respect to spatial and temporal variables) Maxwell's equation as

$${}^{\nu}\mathsf{curl}\ \mathbf{E} = -\ {}_0^{\nu}\mathsf{D}_t \mathbf{B},$$

$${}^{\nu}\mathsf{div}\ \mathbf{B} = 0.$$

$$^{\nu}\mathrm{curl}\,\mathbf{H} = \mathbf{j} + {}^{\nu}_{0}\mathrm{D}_t\mathbf{D},$$

$$^{\nu}\mathrm{div}\,\mathbf{D} = \rho.$$

Based on these equations, the authors have explained fractional wave equations, fractional vector and scalar potentials, and fractional Poynting theorem.

5.1.2 Fractional Multipoles

One of the first steps in this direction was introducing a definition of fractional-order multipoles of electric-charge densities. He has showed that such multipoles effectively behave as "intermediate" sources bridging the gap between the cases of integer-order point multipoles such as point monopoles, point dipoles, point quadrupoles, etc.

When a point charge q located at the origin, its charge density is

$$\rho(\mathbf{r}) = q\delta(\mathbf{r}) = q\delta(x)\delta(y)\delta(z),$$

and the electrostatic scalar potential is expressed as

$$\Phi_1(\mathbf{r}) = \frac{q}{4\pi\varepsilon r} = \frac{q}{4\pi\varepsilon\sqrt{x^2+y^2+z^2}}.$$

In case of the point-dipole with the moment parallel to the z-axis $\rho_2(\mathbf{r}) = p\partial\delta(\mathbf{r})/\partial z$ and

$$\Phi(\mathbf{r}) = \Phi_2(\mathbf{r}) = -\frac{p\cos\theta}{4\pi\varepsilon(x^2+y^2+z^2)},$$

where θ is the angle defined by $\cos\theta = z/\sqrt{x^2+y^2+z^2}$ in the spherical coordinate system (r,θ,φ).

In order to find the potential $\Phi_{2^\alpha}(\mathbf{r})$ created by the fractional multipole

$$\rho_{2^\alpha,z}(\mathbf{r}) = ql^\alpha\,{}_{\infty}\mathrm{D}^{\alpha}_{z}\delta(\mathbf{r}) = \delta(x)\delta(y)\frac{ql^\alpha}{\Gamma(-\alpha)}z_+^{-\alpha-1}$$

(an arbitrary constant of dimension length l is introduced here for dimensionality)[4] Engheta [30] applies to the Poisson equation

$$\Phi(\mathbf{r}) = -\frac{q}{\varepsilon}\delta(\mathbf{r})$$

[4] Observe that for $\alpha = 0$ we get the original point monopole $\rho_1(\mathbf{r})$ and for $\alpha = 1$ we obtain a charge distribution of dipole $\rho_2(\mathbf{r})$.

the operator $l^{\alpha}\ {}_{-\infty}\mathsf{D}_z^{\alpha}$ and obtains the equation

$$\Phi_{2^{\alpha}}(\mathbf{r}) = -\frac{ql^{\alpha}}{\varepsilon}\ {}_{-\infty}\mathsf{D}_z^{\alpha}\delta(\mathbf{r}),$$

where

$$\Phi_{2^{\alpha},z}(\mathbf{r}) = \frac{ql^{\alpha}}{4\pi\varepsilon}\ {}_{-\infty}\mathsf{D}_z^{\alpha}\left[\frac{1}{\sqrt{x^2+y^2+z^2}}\right].$$

For $\alpha \in (0, 1)$, this equation comes down to

$$\Phi_{2^{\alpha},z}(\mathbf{r}) = \frac{ql^{\alpha}}{4\pi\varepsilon\Gamma(1-\alpha)}\frac{\partial}{\partial z}\int_{-\infty}^{z}\frac{(z-\zeta)^{-\alpha}d\zeta}{\sqrt{x^2+y^2+\zeta^2}}$$

and after some algebra takes the form

$$\Phi_{2^{\alpha},z}(\mathbf{r}) = \frac{ql^{\alpha}r^{-\alpha-1}\Gamma(\alpha+1)}{4\pi\varepsilon r^{1+\alpha}}P_{\alpha}(-\cos\theta),$$

where $P_{\alpha}(-\cos\theta)$ is the Legendre polynomial of the first order and the noninteger degree α.

The obtained expression with $\alpha \in (0, 1)$ represents the set of intermediate cases between potentials of the point monopole and point dipole. The θ-angular variation of $\Phi_{2^{\alpha},z}$ do not have periodicity and must have discontinuity in its derivative at some angle, say at $\theta = 0$. It is also worth noting that $r^{-1-\alpha}$ dependence does not violate Gauss' law, since the source is not localized and is extended to $z = \infty$: the total flux over the surface of the closed sphere of radius R is expressed as

$$\oint_{r=R}\mathbf{E}\cdot d\mathbf{S} = \frac{ql^{\alpha}}{\Gamma(1-\alpha)\varepsilon R^{\alpha}} = \int_{r<R}\rho_{2^{\alpha}}(\mathbf{r})dV.$$

The numerical results can be find in Engheta [30–32]. Rousan et al. [100] apply the fractional calculus techniques to consideration of continuous deformation of a point mass distribution into a uniform semi-infinite line along one of the coordinate axis. The continuous deformation is dependent on a parameter $\alpha \in [-1, 0]$ corresponding to the fractional order of differentiation.

Using fractional calculus and the knowledge of electrostatic potentials of fractional multipoles, N. Engheta has shown that a theory for electrostatic "fractional" image methods can be developed for perfectly conducting wedges and cones. Another interesting application of fractional calculus in electrostatic-image theory is the mathematical link between the electrostatic images for a dielectric sphere and a perfectly conducting sphere generalizing the known Lord Kelvin result.

Further development of these ideas can be found, in particular, in publications of Machado with coworkers (see References).

5.1.3 A Link Between Two Electrostatic Images

Consider a special application of the fractional method to electrostatic image theory discussed at point 4 of the Engheta work cited, namely, the mathematical link between the electrostatic images for a dielectric sphere and a perfectly conducting sphere. Let a point charge Q be placed in front of a perfectly conducting ungrounded sphere of radius a, at a distance d from its centers. The image charges consists of an image point source $-aQ/d$, located at distance a^2/d away from the sphere's center on the line connecting the real point charge Q and the center, and another image point source aQ/d, located at the center of the sphere. Its volume charge density is written as

$$\rho_c(z) = -\frac{aQ}{d}\delta_-\left(z - \frac{a^2}{d}\right) + \frac{cQ}{d}\delta_+(z),$$

where z is the coordinate along the line connecting the center to the real point charge, and $\delta_+(z) \equiv \delta(z-0)$ and $\delta_-(z) \equiv \delta(z+0)$ are right-handed and left-handed delta-functions. When the sphere is made of a dielectrics with permittivity $\varepsilon = \varepsilon_r\varepsilon_0$, where ε_0 is a permittivity of a free space, the density becomes [68]

$$\rho_d(z) = \rho_d^{\mathrm{I}}(z) + \rho_d^{\mathrm{II}}(z),$$

where

$$\rho_d^{\mathrm{I}}(z) = -\frac{\varepsilon_r - 1}{\varepsilon_r + 1}\frac{aQ}{d}\delta_-\left(z - \frac{a^2}{d}\right)$$

and

$$\rho_d^{\mathrm{II}}(z) = \frac{\varepsilon_r - 1}{(\varepsilon_r + 1)^2}\frac{Q}{a}\left(\frac{zd}{a^2}\right)^{-\varepsilon/(\varepsilon+1)}.$$

The first term remains the same (up to a multiplier), but the second becomes quite different: it represents now the image source distributed as $z^{-\varepsilon_r/(\varepsilon_r+1)}$ along the connecting line with a singularity at the origin. As noted in Engheta [29], the latter can be expressed through the fractional derivative

$$\rho_d^{\mathrm{II}}(z) = \frac{\varepsilon_r - 1}{(\varepsilon_r + 1)}\Gamma\left(\frac{\varepsilon_r + 2}{\varepsilon_r + 1}\right)\frac{aQ}{d}\,{}_{-\infty}\mathrm{D}_\zeta^{-1/(\varepsilon_r+1)}\delta_+(z), \quad \zeta = \frac{zd}{a^2}.$$

5.1.4 "Intermediate" Waves

As well known, the canonical solutions for the scalar Helmholtz equation in one-, two-, and three-dimensional cases are identified as plane, cylindrical, and spherical waves, respectively. The corresponding sources can be one-, two-, and three-dimensional delta-functions,

$$\left[+k^2\right] G_d(\mathbf{r};k) = -\delta_d(\mathbf{r}), \quad d = 1, 2, 3,$$

$$\delta_1(\mathbf{r}) \equiv \delta(x) = \int_{-\infty}^{\infty} \delta_2(x, y)dy, \quad \delta_2(x, y) = \int_{-\infty}^{\infty} \delta_3(x, y, z)dz.$$

The solutions for $d = 1$ and 2 are

$$G_1(\mathbf{r};k) = \frac{i}{2k} e^{ik|x|},$$

$$G_2(\mathbf{r};k) = \frac{i}{4} H_0^{(1)}\left(k\sqrt{x^2+y^2}\right),$$

where $H_0^{(1)}$ is the zeroth-order Hankel function of the first kind. As shown in Engheta [33], the symmetric source of the intermediate dimension D, $1 < D < 2$ is

$$\delta_D(x, y) = \frac{1}{2}\left[{}_{-\infty}\mathsf{D}_y^{D-2} + {}_{-\infty}\mathsf{D}_{-y}^{D-2}\right]\delta(x)\delta(y) = \frac{\delta(x)|y|^{1-D}}{2\Gamma(2-D)}.$$

With this source, the solution of the Helmholtz equation, for the observation points on the symmetric $(x - z)$plane and for $x > 0$, is represented through the Hankel function of order $(D - 2)/2$ [33]:

$$G_D(x, y = 0; k) = -\frac{i\Gamma((D-1)/2)\cos(D\pi/2)}{4\sqrt{\pi}} \left(\frac{x}{2k}\right)^{(2-D)/2} H_{(D-2)/2}^{(1)}(kx),$$

$$1 < D < 2, \quad x > 0.$$

In the fare zone $k\rho \gg 1$ $(\rho = \sqrt{x^2+y^2},\ \varphi = \tan^{-1}(y/x))$,

$$\begin{aligned} G_D(x, y; k) \cong & (2i)^{-1}(2\pi k\rho)^{-1/2}\cos(D\pi/2)(k\sin|\varphi|)^{D-2}e^{ik\rho - i\pi/4} \\ & + i[4k^{2-D}\Gamma(2-D)(k|y|^{D-1}]^{-1}e^{ik|x|}, \quad 1 < D < 2. \end{aligned}$$

We see that far-zone solution for intermediate values of D consists of two parts: a cylindrical wave dropping as $\rho^{-1/2}$ and a nonuniform plane wave, which propagates in the x-direction with amplitude dropping as $|y|^{1-D}$. For $D = 1$, the latter disappears and only plane wave remains. When $D = 2$, the second term vanishes due to the $\Gamma(2 - D)$ and the first term becomes

$$G_{D\to 2}(x, y; k) \cong \frac{i}{4}\sqrt{\frac{2}{\pi k\rho}} e^{ik\rho - i\pi/4} \sim \frac{i}{\pi} H_0^{(1)}(k\rho), \quad k\rho \to \infty.$$

5.2 Optics

5.2.1 *Fractional Differentiation Method*

The aim of astrophotography is to obtain as much detail as possible on a targeted cosmic image. Long exposure astrophotography required for achieving this is vulnerable to various types of atmospheric noise. Removing this noise depletes the image detail and in consequence objects appear fainter and weaker in contrast. One of the most important problem is the edge detection of an image. Ordinary methods are based on using expressions containing the first-order derivatives. The second-order derivatives applications lead to the Laplacian calculations. There exists some experience in using dipole and quadrupole moments of the image maps. Fractional Fourier transformations are used for this aim as well.

As proposed in You et al. [145], fractional differentiation can help scan and examine an image to detect the faint, weaker edges, and details to enhance and display them. In outline, the algorithm looks as follows. Let $f(x, y)$ be a two-dimensional map requiring such processing. The partial fractional derivatives ${}_0\mathsf{D}_x^\nu f(x, y)$ and ${}_0\mathsf{D}_y^\nu f(x, y)$ are numerically calculated and the fractional gradient is defined as

$$\mathbf{G}^\nu \equiv \nabla^\nu f(x, y) = \mathbf{e}_x \, {}_0\mathsf{D}_x^\nu f(x, y) + \mathbf{e}_y \, {}_0\mathsf{D}_y^\nu f(x, y).$$

In image processing, the magnitude of the gradient

$$G^\nu(x, y, c) = \sqrt{[G_x^\nu(x, y, c)]^2 + [G_y^\nu(x, y, c)]^2}$$

is evaluated on the function $f(x, y)$ given by the image map $b(x, y, c)$ for each color tone c. For each color, the maximum value on the image map is defined and then the output map is determined as

$$b_G(x, y, c) = N \left(\frac{G^\nu(x, y, c)}{G_{\max}^\nu(c)} \right)^\alpha,$$

where $N(= 255)$ is the total number of color tones and α is a parameter suitable to adjust the image visibility.

Sparavigna and Milligan [114] explain the role of fractional differentiation in the following way. "We observe a galaxy and stars in the background and in front of it. We evaluated the fractional gradient and obtained images with different values of the fractional parameter ν. The value of α is fixed at 0.5. The map, obtained for $\nu = 1$, which is working as the usual edge detector based on the ordinary gradient, almost removes the galaxy and puts in evidence the stars. The galaxy regains its visibility, when parameter ν approaches the zero value. In this case we have the original image with a resulting appearance according to α parameter."

The pictures shown in Mathieu et al. [75], Sparavigna [113], Sparavigna and Milligan [114] look very convincingly and the method itself seems handy enough.

Nevertheless, I would like to bring here another approach which is more interesting from (my) theoretical point of view. I mean the works of Blackledge [13–15], where the process of image forming is considered in terms of wave scattering theory.

5.2.2 Wave-Diffusion Model of Image Transfer

Let us recall the transition from the wave description of light propagation to the diffusion model in framework of which the process can be considered as a chaotic walk of mutually independent quants of light. Representing the solution $u = u(\mathbf{r}, t)$ of the three-dimensional wave equation

$$\Delta u - \frac{1}{c_0^2}\frac{\partial^2 u}{\partial t^2} = 0$$

as the product of a function ϕ slowly varying with time and the fast varying harmonic multiplier $\exp(i\omega t)$, so that

$$\left|\frac{\partial^2 \phi}{\partial t^2}\right| \ll 2\omega \left|\frac{\partial \phi}{\partial t}\right|,$$

we obtain

$$\frac{\partial^2 u}{\partial t^2} \cong \exp(i\omega t)\left(2i\omega\frac{\partial \phi}{\partial t} - \omega^2\phi\right)$$

and the initial wave equation reduces to

$$(\Delta + k^2)\phi = \frac{2ik}{c_0}\frac{\partial \phi}{\partial t}$$

with $k = \omega/c_0$ (c_0 is the light speed). Evidently, the complex conjugate function u^* contains multiplier ϕ^* obeying the conjugate equation

$$(\Delta + k^2)\phi^* = -\frac{2ik}{c_0}\frac{\partial \phi^*}{\partial t}.$$

This couple equations can be combined into the expression

$$\phi^*\Delta\phi - \phi\Delta\phi^* = \frac{2ik}{c_0}\left(\phi^*\frac{\partial \phi}{\partial t} + \phi\frac{\partial \phi^*}{\partial t}\right)$$

which can be rewritten in the form

$$\Delta I - 2\nabla\cdot(\phi\nabla\phi^*) = \frac{2ik}{c_0}\frac{\partial I}{\partial t},$$

where $I = \phi^* \phi = |\phi|^2$. Assuming $k = k_0 - ik_1$ and equating the real and imaginary parts yield

$$D\Delta I + 2\mathrm{Re}[\nabla \cdot (\phi \nabla \phi^*)] = \frac{\partial I}{\partial t}, \quad D = \frac{c_0}{2k_1},$$

and

$$\mathrm{Im}[\nabla \cdot (\phi \nabla \phi^*)] = -\frac{k_0}{c_0} \frac{\partial I}{\partial t}$$

respectively. As a result, we arrive at the diffusion equation for the intensity of light

$$\frac{\partial I}{\partial t} = D\Delta I(\mathbf{r}, t).$$

Talking about an image in the fixed plain $z = z_0$, we reduce the diffusion equation to two-dimensional version

$$\frac{\partial I(x, y; t)}{\partial t} = D\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right] I(x, y; t)$$

subjected to the initial condition $I(x, y; 0) = I_0(x, y)$. The resulting image at time t is represented by the two-dimensional convolution of

$$I(x, y; t) = \frac{1}{4\pi Dt} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\left(\frac{(x - x')^2 + (y - y')^2}{4Dt}\right)\right] I_0(x', y') dx' dy'$$

[13, 15].

Comparing two types of equations, the wave equation

$$\sigma^2 \frac{\partial^2 G^{(2)}}{\partial t^2} - \Delta G^{(2)}(\mathbf{r}, \mathbf{r}_0; t) = \delta(\mathbf{r} - \mathbf{r}_0)\delta(t)$$

and the diffusion equation

$$\sigma^1 \frac{\partial G^{(1)}}{\partial t} - \Delta G^{(1)}(\mathbf{r}, \mathbf{r}_0; t) = \delta(\mathbf{r} - \mathbf{r}_0)\delta(t),$$

Blackledge [14] has suggested to consider the time-fractional equation

$$\sigma^\nu \frac{\partial^\nu G^{(\nu)}}{\partial t^\nu} - \Delta G^{(\nu)}(\mathbf{r}, \mathbf{r}_0; t) = \delta(\mathbf{r} - \mathbf{r}_0)\delta(t) \tag{5.6}$$

with $\sigma = 1/D$ as an intermediate case between the wave and diffusion mechanisms of image blurring with $1 < \nu < 2$. Involving the fractional calculus is justified in terms of the generalization of a random walk model. He gives an example of application including image enhancement of star fields and other cosmic bodies imaged through

interstellar dust clouds. Giving the physical validation to this approach, Blackledge [14] writes:

"Fractional diffusion models apply to scattering processes that occur in a tenuous and extremely rarefied medium. In applied optics, one of the most common examples of this phenomenon occurs in astronomy and the processes associated with light scattering from cosmic dust which is composed of particles which are a few molecules to the order of 10^{-4} m in size."

After the Fourier transform with respect to time,

$$g(\mathbf{r}, \mathbf{r}_0; \omega) = \int_{-\infty}^{\infty} G(\mathbf{r}, \mathbf{r}_0; t) e^{i\omega t} dt,$$

this equation becomes

$$(\Delta + \Omega_\nu^2) g(\mathbf{r}, \mathbf{r}_0; \omega) = -\delta(\mathbf{r} - \mathbf{r}_0)$$

with

$$\Omega_\nu = \pm i (i\omega\sigma)^{\nu/2}.$$

Generalizing the classical results for Green functions in one-, two-, and three-dimensions, Blackledge [13] finds:

$$\begin{aligned} G_1^{(\nu)}(x, x_0; t) &= \frac{i}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2\Omega_\nu} e^{i(\Omega_\nu |x - x_0| + \omega t)} d\omega \\ &\sim \frac{1}{2\sigma^{\nu/2} t^{1-\nu/2}} - \frac{1}{2} |x - x_0| \delta(t), \quad \sigma \to 0, \\ G_2^{(\nu)}(\mathbf{r}, \mathbf{r}_0; t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\exp\{i(\omega t + \pi/4) - (i\omega\sigma)^{\nu/2} |\mathbf{r} - \mathbf{r}_0|\}}{\sqrt{8\pi i |\mathbf{r} - \mathbf{r}_0|} (i\omega\sigma)^{\nu/4}} d\omega \\ &\sim \frac{1}{\sqrt{8\pi |\mathbf{r} - \mathbf{r}_0|} \sigma^{\nu/4} t^{1-\nu/4}}, \quad \sigma \to 0, \end{aligned}$$

and

$$\begin{aligned} G_3^{(\nu)}(\mathbf{r}, \mathbf{r}_0; t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\exp\{i\omega t - (i\omega\sigma)^{\nu/2} |\mathbf{r} - \mathbf{r}_0|\}}{4\pi |\mathbf{r} - \mathbf{r}_0|} d\omega \\ &\sim \frac{\delta(t)}{4\pi |\mathbf{r} - \mathbf{r}_0|}, \quad \sigma \to 0. \end{aligned}$$

Thus, the transformation of an image in this case is given by

$$I(x, y; t) = \frac{1}{2\sqrt{2\pi} (Dt)^{1-\nu/4}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{I_0(x', y') dx' dy'}{[(x - x')^2 + (y - y')^2]^{1/4}}.$$

Compared with the Gaussian, the function $R^{-1/2}$ decays more rapidly and hence the image will be less blurred in the fractional diffusion model. This fact is used in a

range of applications of optical diffusers technology including the de-pixelation of liquid crystal displays. In this case, one has to solve the inverse fractional diffusion problem called *fractional de-diffusion*: finding the initial image $I(\mathbf{r}, 0)$ from the observed data $I(\mathbf{r}, t)$ [13].

Another version of fractional approach based on the Lévy motion equation was proposed in Blackledge and Blackledge [15], where a special consideration to the reduction of noise in *Magnetic Resonance Imaging* (*MRI*) is given. "*Diffusion Weighted MRI* (*DWI*) is now starting to be used routinely in the clinic, as it provides excellent morphological contrast and also quantitative information in the form of an Apparent Diffusion Coefficient", the authors write.

We shall discuss this model below in connection with image transfer through turbulent media.

5.2.3 Superdiffusion Transfer

Propagation of light through a turbulent optically transparent medium is often described in terms of multiple scattering theory. The differential scattering cross-section $w(\theta)$ per unit length and unit solid angle is connected to the spectral density (Fourier transform) of the dielectric correlations function

$$\Phi_\varepsilon(\mathbf{q}) = \frac{1}{8\pi^3}\int_{\mathbb{R}^3} e^{-i\mathbf{qr}}\langle[\varepsilon(\mathbf{r}_1) - \overline{\varepsilon}][\varepsilon(\mathbf{r}_1 + \mathbf{r}) - \overline{\varepsilon}]\rangle d\mathbf{r}.$$

In a statistically homogeneous and isotropic turbulent medium, the spectral density depends only on the absolute value of the vector argument, $q = |\mathbf{q}|$, and this link is expressed as

$$w(\theta) = (1/2)\pi k_0^4 \Phi_\varepsilon(q),$$

where $q = 2k_0 \sin(\theta/2)$, and k_0 is a wave number in the homogeneous medium with the dielectric constant $\overline{\varepsilon}$. According to Kolmogorov's two-thirds law,

$$\Phi_\varepsilon(q) = Cq^{-11/3}, \quad C = \text{ const},$$

in the inertial interval of wave numbers (see Eq. (26.31) in the book [103]). This validates the small angle approximation with

$$w(\theta) \approx A\theta^{-\alpha-2}, \ A = \text{const}, \ \alpha = 5/3. \tag{5.7}$$

Let us assume that a one-directional point source is placed at origin and emits a photon along z-axis. In the small-angle approximation, z-coordinate of the photon is equated to its travel. For characterizing the deviation of the photon from the initial direction, the two-dimensional vector $\mathbf{u} = \{\Omega_x, \Omega_y\}$ is used. The kinetic equation for the angular distribution density $f(\mathbf{u}, z)$ reads

$$\frac{\partial f(\mathbf{u},z)}{\partial z}=\int_{\mathbb{R}^2}[f(\mathbf{u}-\mathbf{u}',z)-f(\mathbf{u},z)]w(|\mathbf{u}'|)d\mathbf{u}',\ \ f(\mathbf{u},0)=\delta(\mathbf{u}).$$

Developing as series in $\mathbf{u}'$ and making integration (i.e., averaging over the scattering angle), one arrives at the diffusion equation for the main asymptotical part of the solution $f^{\mathrm{as}}(\mathbf{u},z)$:

$$\frac{\partial f^{\mathrm{as}}(\mathbf{u},z)}{\partial z}=\frac{\langle u^2\rangle}{2}\Delta_2 f^{\mathrm{as}}(\mathbf{u},z),$$

where $\Delta_2=\partial^2/\partial u_x^2+\partial^2/\partial u_y^2$ and $\langle u^2\rangle=\int_{\mathbb{R}^2}w(|\mathbf{u}|)u^2d\mathbf{u}$ is the mean square scattering angle per unit length.

In the problem under consideration, this integral diverges instead of expanding into series, we shall use anzatz (5.7):

$$\frac{\partial f^{\mathrm{as}}(\mathbf{u},z)}{\partial z}=AJ(\mathbf{u}),\quad J(\mathbf{u})=\int_{\mathbb{R}^2}[f^{\mathrm{as}}(\mathbf{u}-\mathbf{u}',z)-f^{\mathrm{as}}(\mathbf{u},z)]|\mathbf{u}'|^{-\alpha-2}d\mathbf{u}'.$$

When $\alpha\geq 1$, the collision integral diverges and we have to use the Adamar regularization procedure, replacing the divergent integral $J(\mathbf{u})$ by its finite (in Hadamar's sense) part:

$$\begin{aligned}J(\mathbf{u})\to\text{p.f. }J(\mathbf{u})&\equiv\int_{\mathbb{R}^2}[f^{\mathrm{as}}(\mathbf{u}-2\mathbf{u}',z)-2f^{\mathrm{as}}(\mathbf{u}-\mathbf{u}',z)+f^{\mathrm{as}}(\mathbf{u},z)]|\mathbf{u}'|^{-\alpha-2}d\mathbf{u}'\\&=-c(\alpha)(-\Delta_2)^{\alpha/2},\end{aligned}$$

where

$$c(\alpha)=\frac{\pi^2(1-2^{1-\alpha})}{[\Gamma(1-\alpha/2)]^2\sin(\alpha\pi/2)}.$$

Consequently, the asymptotical behavior of the angle distribution $f^{\mathrm{as}}(\mathbf{u},z)$ of photons traversed path z in a turbulent medium is described by the equation

$$\frac{\partial f^{\mathrm{as}}(\mathbf{u},z)}{\partial z}=-c(\alpha)A(-\Delta_2)^{\alpha/2}f^{\mathrm{as}}(\mathbf{u},z),\quad f(\mathbf{u},0)=\delta(\mathbf{u}).$$

The solution to this equation is expressed through the two-dimensional isotropic Lévy-Felgheim stable density with the characteristic exponent $\alpha=5/3$,

$$g_2(r;\alpha)=\frac{1}{2\pi}\int_0^\infty e^{-k^\alpha}J_0(kr)kdk$$

by the relation

$$f^{\mathrm{as}}(\mathbf{u},z)=[c(\alpha)Az]^{-2/\alpha}g_2([c(\alpha)Az]^{-1/\alpha}|\mathbf{u}|;\alpha).$$

Let us note three characteristic differences of scattering in a turbulent medium from analogous process in a medium with small-scale fluctuations of the refraction coefficient. The angle distribution width of the scattered photons grows with depth

proportionally to $z^{3/5}$, but not to $z^{1/2}$, distribution tails have the power-law form but not the Gaussian one, and x- and y-projections of the vector $\mathbf{u}$ are not statistically independent anymore.

Note that in frame-work of the small-angle approximation a simple interrelation takes place, which links angular and radial particle distributions [64].

5.2.4 Subdiffusion and Combined (Bifractional) Diffusion Transfer Models

Considering above discussed models, we assumed that the medium is homogeneous, at least at large scales. Even when we talked about turbulence, we kept in mind a uniformly turbulent medium, with no domains being free from signs of turbulence[5]. This assumption is expressed by the first order of the time-derivative beginning the diffusion equations. If we observe a space intermittency when the turbulent domains alternate with calm regions, this term must be replaced by an integral one (see Sect. 3.6.3 of Vol. I). If the mean distance between inhomogeneities is finite, the equation for an asymptotic term of its solution contains the first-order time derivative as before. However, if it is infinite, and we have some reasons to consider that the distance probability distribution is of the power type with the exponent $\nu \in (0, 1]$, we arrive at the subdiffusion model of image transfer. It is described by the equation,

$$\left[\sigma^{\nu}\, {}^{\nu}_{0}\mathsf{D}_t - \Delta_2\right] f(x, y; t) = f_0(x, y)\delta(t). \quad 0 < \nu \leq 1,$$

which is similar to Eq. (5.6) except the range of admissible values of ν. If the inhomogeneities have the turbulent character, the equation becomes of time-space-fractional type:

$$\left[\sigma^{\nu}\, {}^{\nu}_{0}\mathsf{D}_t + (-\Delta_2)^{\alpha/2}\right] f(x, y; t) = f_0(x, y)\delta(t), \quad 0 < \nu \leq 1, \quad 0 < \alpha \leq 2.$$

The last equation represents a more universal transfer model, combining properties of the subdiffusion model (sharp peak at the center of an image) and of the superdiffusion one (long tails at the periphery of an image). The presence of two independent parameters (α and ν) gives a more freedom in tuning a reconstruction algorithm.

[5] Recall that in the image transfer problem the variable t is usually associated with the penetration distance.

5.3 Laser Optics

5.3.1 Laser Beam Equation

Usually, the interaction of nearly resonant strong laser light with matter is studied on assumption that the laser light is monochromatic. However, a real quantum system will unavoidably interact with their surrounding environments, and this effect should often be taken into account. The influence of noise (jump-type) on the atom-field interactions was first introduced by Burshtein [19] in quantum optics. A few models were developed to describe adequately the influence of phase fluctuations: the diffusion or Brownian-motion-type process, the telegraph model which assumed phase jumping between two values at random times, both amplitude and phase fluctuations model assuming a complex Gaussian field.

We dwell here at the work [63], where a non-Markovian jump model of a phase-modulated stochastic field has been suggested. The field is written as

$$E(t) = \varepsilon \exp[-i\omega_L t - i\phi] + \text{c.c.},$$

where the phase $\phi(t)$ is assumed to undergo sudden jumps at random times and the phase jumps β are described by the same Gaussian distribution $f(\beta)$ with $\langle\beta\rangle = 0$. Each phase jump β depends on the preceding jump β' according to the conditional probability density

$$p(\beta|\beta') = \frac{1}{b[2\pi(1-\gamma^2)]^{1/2}} \exp\left[-\frac{(\beta-\gamma\beta')^2}{2b^2(1-\gamma^2)}\right].$$

Here, $b = \sqrt{\langle\beta^2\rangle}$, γ is the correlation coefficient for successive phase jumps. For $\gamma = 0$, the jumps are uncorrelated and consequently independent, so that the model becomes Markovian. For $\gamma = 1$, the jumps are fully correlated (they have the same size, $\beta = \beta'$). For $\gamma = -1$, the jumps are fully anticorrelated ($\beta = -\beta'$), and the system jumps between two states. This is a telegraph process, which had been studied by Eberly et al. [27]. A very interesting feature of the model offered in Kofman et al. [63] is the dependence of the resonance fluorescence spectral components on the field intensity. In the small-jump limit, when $b^2 \ll 1$ and $|\gamma|$ is not too close to 1 the non-Markovian equation

$$\frac{dk}{dt} = -\nu_0 k - \frac{2\gamma\nu_0}{\tau_0}\int_0^t e^{-\nu_\beta(t-t')}k(t')dt',$$

for correlation function $k(t)$ of the input field. Here τ_0 is the average time between jumps, $\nu_\beta = (1-\gamma)/\tau_0$ and $\nu_0 = b^2/2\tau_0$. For $\gamma = 0$, the second term vanishes, and the equation is the Markovian equation leading to Lorentzian line shape, while for $\gamma \neq 0$ it is clearly non-Markovian. Its solution is

$$k(t) = \left[1 + \frac{\gamma b^2}{(1-\gamma)^2}\right] e^{-\nu t} - \frac{\gamma b^2}{(1-\gamma)^2} e^{-\nu_\beta t},$$

where $\nu = \nu_0(1+\gamma)/(1-\gamma)$.

5.3.2 *Propagation of Laser Beam Through Fractal Medium*

Nakhushev [81] (see also [82]) considers propagation of the laser radiation through a layer $(z_0 \le z \le z_0 + l)$ of an absorbing medium of the fractal type. Replacing the standard continuity equation for molecular concentration $n(z,t)$ and its current density $q(z,t)$,

$$\frac{\partial n}{\partial t} = -\frac{\partial q}{\partial z}$$

by the fractional counterpart of the equation,

$${}_\tau \mathsf{D}_t^\alpha n = -\frac{\partial q}{\partial z}, \quad 0 < \alpha < 1,$$

the author adds it by the second Fourier law with a diffusion coefficient which is a linear function of the concentration itself,

$$q = -\frac{\partial[(an+b)n]}{\partial z}, \; a, b = \text{const} > 0.$$

Nakhushev interprets a as a quantity characterizing the synergetic character of interaction of the laser beam with the absorbing medium.[6] Combining both this equations yields

$${}_\tau \mathsf{D}_t^\alpha n = \frac{\partial^2[(an+b)n]}{\partial z^2}.$$

After differentiation of the equation with respect to t and introducing

$$\bar{n}(z,t) = \frac{1}{l}\int_{z_0}^{z_0+l} n(z,t)dz,$$

it can be approximated by

$${}_\tau \mathsf{D}_t^{\alpha+1} n = 2a\frac{\partial \bar{n}}{\partial t}\frac{\partial^2 n}{\partial z^2}.$$

Under some special non-local condition called the *Samarsky condition*,

[6] It seems to be a somewhat bold statement.

$$\frac{\partial}{\partial t}\int_{z_0}^{z_0+l} n(z,t)dz = \frac{l}{2a}|t-\tau|^m \text{sign}(t-\tau),\ m>0,$$

passing to new variables $x = z - z_0,\ y = t - t_1,\ u(x,y) = n(x+z_0, y+t_1)$ yields

$${}_0\mathrm{D}_y^{\alpha+1}u(x,y) + \text{sign } y\ |y|^m \frac{\partial^2 u}{\partial x^2} = 0.$$

This is a fractional generalization of the Hellerstadt equation [82].

5.3.3 Free Electron Lasers

The *free electron laser* (*FEL*) was developed at the Stanford University (USA) in the late seventies of the last century. As opposed to ordinary lasers, the active medium in an FEL is not made of atoms and molecules, it consists of a beam of free electrons, propagating at relativistic velocities in a spatially periodic magnet (undulator), where the electrons execute transverse oscillations and emit synchrotron radiation under action of the Lorentz force (see for details [24]). The dimensionless complex-field amplitude a is described by the first-order integro-differential equation of Volterra type derived from the complete set of coupled Lorentz-Maxwell equations,

$$\frac{da(t)}{dt} = \mu\int_0^t \tau e^{i\mu_0\tau}a(t-\tau)d\tau, \quad a(0)=1,$$

where t is the dimensionless interaction time ranging from 0 to 1, $\mu = -i\pi g_0$, g_0 is the small-signal gain coefficient, and ν is the detuning parameter. The exact closed form solution of this equations suitable for numerical computation is given in Dattoli et al. [23]:

$$a(t) = \sum_{n=0}^{\infty}\mu^n t^{3n}\Phi^*(2n, 3n+1; i\nu t),$$

where

$$\Phi^*(a,c;z) = \frac{1}{\Gamma(a)\gamma(c-a)}\int_0^1 u^{a-1}(1-u)^{c-a-1}e^{zu}du, \quad \text{Re}\, a>0, \quad \text{Re}\, c>0$$

is the modified Kummer function, obeying the integral summation formula

$$\int_0^t \Phi^*(a,c;k\tau)\Phi^*(a',c';k(t-\tau))\tau^{c-1}(t-\tau)^{c'-1}d\tau = \Phi^*(a+a', c+c'; kt).$$

The fractionalized FEL equation

$$ {}_0\mathsf{D}_t^{\alpha} a(t) = \mu \int_0^t \tau e^{i\nu\tau} a(t-\tau) d\tau $$

with initial conditions

$$ {}_0\mathsf{D}_t^{\alpha-k} a(t)\Big|_{t=0} = b_k, \quad k = 1, 2, \ldots, n, \quad (n-1 < \alpha \le n). $$

was considered in Boyadjiev and Dobner [17]. The authors have shown that if $a(t)$ has a summable fractional derivative, this Cauchy type problem has a unique solution of the form

$$ a(t) = a_0(t) + \sum_{j=1}^{\infty} \mu^j a_j(\alpha, \nu; t), $$

where

$$ a_0(t) = \sum_{k=0}^{n} \frac{b_k}{\Gamma(\alpha - k + 1)} t^{\alpha-k}. $$

To prove this statement, they bring the initial equation to the form

$$ a(t) = a_0(t) + \mu \, {}_0\mathsf{I}_t^{\alpha} \left[\int_0^t (t-\theta) a(\theta) d\theta \right]. $$

Using the Dirichlet rule

$$ \int_0^t d\tau \int_0^{\tau} d\theta \cdots = \int_0^t d\theta \int_{\theta}^t d\tau \cdots $$

yields the equivalent integral form of the equation

$$ a(t) = a_0(t) + \mu \int_0^t (t-\theta)^{\alpha+1} \Phi^*(2, \alpha+2; i\nu(t-\theta)) a(\theta) d\theta. $$

Since this is a Volterra integral equation with continuous kernel, its solution can be computed as the limit of the sequence

$$ a_m(t) = a_0(t) + \mu \int_0^t (t-\theta)^{\alpha+1} \Phi^*(2, \alpha; i\nu(t-\theta)) a_{m-1}(\theta) d\theta. $$

Thus, the solution of the problem under consideration is represented as

$$ a(t) = a_0(t) + \int_0^t \left(\sum_{j=1}^{\infty} k(j, t, \theta) \right) a_0(\theta) d\theta, $$

where

$$k(j,t,\theta) = \mu^j (t-\theta)^{j(\alpha+2)-1} \Phi^*(2j,\, j(\alpha+2);\, i\nu(t-\theta)).$$

Applying the integral summation formula for Φ^* completes the proof.

5.4 Dielectrics

5.4.1 Phenomenology of Relaxation

Let us come back to Sect. 2.5.1 of Vol. I, the polarization $P(t)$ of a dielectric, placed in an alternative electric field, consist of two parts: an instantaneous $P_1(t)$ being proportional to the magnitude of the field intensity at the same instant $E(t)$,

$$P_1(t) = \chi_1 E(t),$$

and a retarded one

$$P_2(t) = \int_{-\infty}^{t} K(t-t')E(t')dt'.$$

Denote via $\chi_2 E$ the extreme value of P_2 with constant E and $t \to \infty$. In classical relaxation theory it is supposed that the rate of change of the retarded component is proportional to the difference between its extreme and current values:

$$\frac{dP_2}{dt} = \frac{1}{\tau}(\chi_2 E - P_2),$$

where τ is the relaxation time. If initially the electric field and the polarization equal zero, and the constant field E_0 is applied instantaneously at $t = 0$, then

$$\frac{dP_2}{dt} + \frac{1}{\tau}P_2 = \frac{1}{\tau}\chi_2 E_0.$$

The solution of the equation is of the form:

$$P(t) = P_1(t) + P_2(t) = [\chi_1 + \chi_2(1 - e^{-t/\tau})]E_0.$$

In the case of steady state in alternative harmonic field

$$E(t) = E_0 e^{i\omega t},$$

the following relation holds

$$\mathsf{D}_t \;\Leftrightarrow\; i\omega, \tag{5.8}$$

yielding to the formula

$$P(t) = P_1(t) + P_2(t) = \left[\chi_1 + \frac{\chi_2}{1 + i\omega\tau}\right] E(t).$$

Passing to complex dielectric permittivity $\varepsilon(\omega)$ and denoting

$$\varepsilon_0 = \varepsilon(0) = 1 + 4\pi(\chi_1 + \chi_2)$$

and

$$\varepsilon_\infty = \varepsilon(\infty) = 1 + 4\pi\chi_1,$$

we get

$$\frac{\varepsilon(\omega) - \varepsilon_\infty}{\varepsilon_0 - \varepsilon_\infty} = \frac{1}{1 + i\omega\tau}.$$

Treating this function as the Laplace transform of some function $f(t)$ with argument $\lambda = i\omega$

$$\widehat{f}(i\omega) = \int_0^\infty e^{-i\omega t} f(t)dt,$$

we rewrite this relation in the form

$$i\omega\tau\,\widehat{f}(i\omega) + \widehat{f}(i\omega) = 1.$$

Coming back to the time variable, we obtain the differential equation of the Debye relaxation

$$\tau\frac{df}{dt} + f(t) = \delta(t),$$

which has the exponential solution

$$f(t) = \frac{1}{\tau}e^{-t/\tau}.$$

Observe that two functions are often used for the description of the relaxation, $f(t)$ and $\phi(t) = \int_t^\infty f(t')dt'$, having the sense of a probability density and a cumulative probability function respectively. In the Debye case, they obey the equations

$$\frac{df(t)}{dt} + \frac{1}{\tau}f(t) = \frac{1}{\tau}\delta(t),$$

and

$$\frac{d\phi(t)}{dt} + \frac{1}{\tau}\phi(t) = \delta(t).$$

Generalizing the operational relation (5.8) to the case of a fractional power

Table 5.1 Three popular relaxation processes

Process	Fourier transform	Equation
Cole-Cole	$\widehat{f}(i\omega) = [1 + (i\omega\tau)^{\alpha}]^{-1}$	$\left[\tau^{\alpha}\ {}_{-\infty}\mathsf{D}_t^{\alpha} + 1\right] f(t) = \delta(t)$
Cole-Davidson	$\widehat{f}(i\omega) = [1 + i\omega\tau]^{-\beta}$	$\left[\tau \mathsf{D}_t + 1\right]^{\beta} f(t) = \delta(t)$
Havriliak-Negami	$\widehat{f}(i\omega) = [1 + (i\omega\tau)^{\alpha}]^{-\beta}$	$\left[\tau^{\alpha}\ {}_{-\infty}\mathsf{D}_t^{\alpha} + 1\right]^{\beta} f(t) = \delta(t)$

$$ {}_{-\infty}\mathsf{D}_t^{\alpha} \ \Leftrightarrow\ (i\omega)^{\alpha},\ \alpha \in (0, 1), \tag{5.9} $$

one can expect validity of the relations, given in Table 5.1.

We now concentrate on the less formal derivation of the above equations.

5.4.2 *Cole-Cole Process*: *Macroscopic View*

The simplest macroscopic way to embed a fractional derivative in the relaxation theory is the fractional generalization of the classical relation between current $i(t)$ and voltage $u(t)$ for an ideal capacitor—the *fractional analogue of Ohm's law*:

$$ i(t) = K_{\alpha}\ {}_a\mathsf{D}_t^{\alpha} u(t), \quad t > 0, \quad 0 < \alpha < 1. \tag{5.10} $$

This relation describes a chain element being in a certain case intermediate between the ideal dielectric with the capacity $C = K_1$ (when $\alpha = 1$) and the ordinary conductor with the resistance $R = 1/K_0$ (when $\alpha = 0$).

The *fractional Ohm law* (5.10) is grounded on the above Curie-von Schweidler law [22, 138], stating that the current appearing in a capacitor when stepwise voltage $\Delta u(t')$ is applied, changes according to the law

$$ i(t) = A\Delta u(t')(t - t')^{-\alpha}, \quad t > t'. \tag{5.11} $$

Rewriting Eq. (5.11) for arbitrary $u(t)$ and performing simple transformations, we find that (for $t > t_j$)

$$ i(t) = \frac{K_{\alpha}\Delta u(t_j)}{\Gamma(1-\alpha)} \frac{d}{dt} \int_{t_j}^{t} \frac{dt'}{(t-t')^{\alpha}} = A\Delta u(t_j)(t - t_j)^{-\alpha}, \quad A = K_{\alpha}/\Gamma(1-\alpha). $$

If $u(t')$ varies continuously and quite rapidly approaches the zero when $t' \to -\infty$, then Eq. (5.11) takes the form

$$ i(t) = A \int_{-\infty}^{t} \frac{du(t')}{(t-t')^{\alpha}} = K_{\alpha}\ {}_{-\infty}^{\alpha}\mathsf{D}_t u(t) = K_{\alpha}\ {}_{-\infty}\mathsf{D}_t^{\alpha} u(t). \tag{5.12} $$

(R-L and G-C derivatives coincides when the lower limit is $-\infty$). When the active resistance R and the electromotive force (*EMF*) $\mathscr{E}(t)$ are embedded in the contour under consideration, the Kirchhoff equation

$$i(t)R + u(t) = \mathscr{E}(t)$$

governs the process. Inserting here the current (5.12), we obtain

$$\left[\tau^{\alpha}\, {}_{-\infty}\mathsf{D}_t^{\alpha} + 1\right] u(t) = \mathscr{E}(t),$$

where

$$\tau^{\alpha} = K_{\alpha} R.$$

Assuming $\mathscr{E}(t) = e^{i\omega t}$, we obtain the solution of the above equation in the form of Cole-Cole response function

$$\left[(i\omega\tau)^{\alpha} + 1\right] \widehat{f}_{\alpha}(i\omega) = 1, \quad 0 < \alpha < 1, \tag{5.13}$$

5.4.3 Microscopic View

Gaseous or liquid dielectric can be imagined as a set of polar molecules freely moving and rotating in a boiling broth simulating an action of heat motion of surrounding molecules. Every molecule possesses its own dipole moment which is subjected to both an external field and internal viscous forces as well as inertial ones. The most important property of the model is the independence of dipoles: they don't "see" each other. In the absence of the external field their dipole moments are distributed uniformly over all directions. Switching up the field causes all the dipoles to turn a bit, inducing the total dipole moment. As the field is switched off the polarization starts to disperse, representing the relaxation process.

Let us mentally place all the dipole vectors $\mathbf{P}_j$ at the center of the sphere of radius $|\mathbf{P}_j|$. Now each of the dipoles is represented by a point on the spherical surface (Fig. 5.1), and all of them move chaotically over this surface. If initially the points form a small packet near the North pole, some time later they will be distributed over the surface almost uniformly. The classical theory describes this process as a Brownian motion of mutually independent points over the sphere. The corresponding equation written in terms of angular variables ϑ and φ reads

$$\frac{\partial p(\vartheta, \varphi, t)}{\partial t} = K \Delta_{\vartheta,\varphi} p(\vartheta, \varphi, t) + \frac{\delta(\cos\vartheta - 1)}{2\pi}\delta(t),$$

where $p(\vartheta, \varphi, t)$ is the probability density of a point on a sphere at time t, K is the diffusion coefficient in the direction space, and $\Delta_{\vartheta,\varphi}$ stands for angular part of the Laplacian. Multiplying both parts of the equation by $\varphi_0 \cos\vartheta \sin\vartheta d\vartheta d\varphi$ and integrating over the sphere, we get

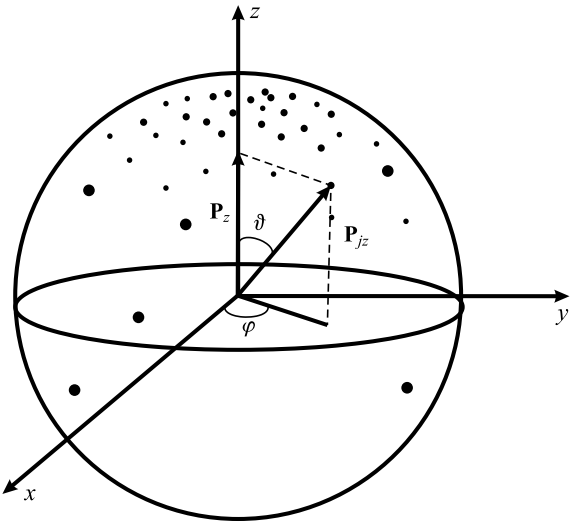


Fig. 5.1 The random dipole distribution

$$\frac{d\phi(t)}{dt} = -2K\phi(t) + \phi_0\delta(t). \tag{5.14}$$

This formula leads to the Debye law of relaxation. How should be changed the model to obtain a qualitatively different behavior of the process.

Discussing the reasons why relaxation in solids differs from that in gases, Jonscher notes equitably that dipole moments carriers in solids can not be considered as free and independent of each other. It is natural to assume that in disordered solid they detained each other, and have to wait for an empty place to become vacant due to the heat motion so that the dipole can make a rotation decreasing its potential energy. Dipole looks as if it is trapped (in direction space) and stays there for some random time T, after which it makes a jump in another direction close to a previous one. The probability initially concentrated at the pole of the sphere, eventually distributes over the sphere according to the subdiffusion law

$${}_0\mathsf{D}_t^{\alpha} p(\vartheta, \varphi, t) = K\Delta_{\vartheta,\varphi} p(\vartheta, \varphi, t) + \frac{\delta(\cos\vartheta - 1)}{2\pi}\frac{t^{-\alpha}}{\Gamma(1-\alpha)},$$

and as $t \to \infty$ it uniformly covers the whole sphere. The mean value of z-projection of a dipole moment satisfies the equation

$${}_0\mathsf{D}_t^{\alpha}\phi(t) = -2\,K\phi(t) + \phi_0\frac{t^{-\alpha}}{\Gamma(1-\alpha)},$$

generalizing (5.14) to the subdiffusion process.

5.4.4 Memory Phenomenon

Recall that for $\alpha = 1$, $f_\alpha(t) = f_1(t)$ decreases exponentially depending only on the initial state of the process, $f_1(0)$, but not of its prehistory $f_1(t)$, $t < 0$, while for $\alpha < 1$ the behavior of $f_\alpha(t)$ at $t > 0$ is determined not only by $f_\alpha(0)$ but by the prehistory $f_\alpha(t)$, $t < 0$. In other words, we observe here some type of memory. The memory effect was noticed by several authors of experimental and theoretic works. One of them named his article with an expressive title "Dead matter has memory!" [142]. The following experiment is described in it.

Three identical capacitors had been short-circuited for at least 19 min before a step voltage $U_0 = 100$ V was applied and the current was measured. However, each capacitor had its own prehistory. Capacitor 1 had been charged to +500 V for 16.7 h and then shorted for 15 min before the measurement. Capacitor 2 had been charged to −500 V for 16.7 h and then shorted for 10 min before the measurement. Capacitor 3 was not subjected to any voltage. After about 10 min the measured currents became notably different from each other and to the 20th second they were 2–3 times different from the third (test) capacitor curve. The author reports that such behavior is in agreement with fractional differential equation.

In our works [131, 132] we have investigated the solution behavior at small deviations of the derivative order from the unit, with a view to follow up the transition from Markovian ($\alpha = 1$) to non-Markovian ($\alpha \neq 1$) process, in other words to observe how memory appears.

Consider a capacitor charging-discharging process, following the next scheme. The charging EMF is turned on at $t = -\theta < 0$ staying constant up to $t = 0$, and turned off at $t = 0$, when relaxation is activated:

$$\mathscr{E}(t) = \begin{cases} 0, & t < -\theta; \\ \mathscr{E}_0, & -\theta \leqslant t < 0; \\ 0, & t \geq 0. \end{cases}$$

Writing the solution of the equation ${}_0\mathsf{D}_t^\alpha u(t) + \tau^{-\alpha} u(t) = \mathscr{E}(t)$ via the Green function in the form of Mittag-Leffler, and representing the latter in the power series form, we get

$$u(t) = \mathscr{E}_0 \int_{-\theta}^{0} \sum_{j=0}^{\infty} \frac{(t-t')^{\alpha-1}}{\Gamma(\alpha j + \alpha)} \left[-\left(\frac{t-t'}{\tau} \right)^\alpha \right]^j dt', \quad t \geq 0.$$

Changing the order of integration and addition, and using the gamma function property, we have

$$u(t) = \tau^\alpha \mathscr{E}_0 \sum_{j=0}^{\infty} \frac{(-1)^j}{\Gamma(\alpha j + \alpha + 1)} \left[\left(\frac{t+\theta}{\tau} \right)^{\alpha j + \alpha} - \left(\frac{t}{\tau} \right)^{\alpha j + \alpha} \right]. \tag{5.15}$$

Passing to addition over $k = j + 1$ and taking into account the relation

$$-\sum_{k=1}^{\infty} \frac{(-1)^k (t/\tau)^{\alpha k}}{\Gamma(\alpha k + 1)} = 1 - E_\alpha(-(t/\tau)^\alpha),$$

we come to the following equation for the ratio $u(t)/u(0),\ t > 0$:

$$u(t)/u(0) = \frac{E_\alpha(-t^\alpha/\tau^\alpha) - E_\alpha[-(t+\theta)^\alpha/\tau^\alpha]}{1 - E_\alpha[-\theta^\alpha/\tau^\alpha]}. \tag{5.16}$$

Each value of α produces a family of curves with relation to different charging times θ, and only one value α ($\alpha = 1$) produces a single curve independent on θ. Indeed,

$$\begin{aligned} |u(t)/u(0)|_{\alpha=1} &= \frac{E_1(-t/\theta) - E_1[-(t+\theta)/\tau]}{1 - E_1(-\theta/\tau)} \\ &= \frac{1 - \exp(-\theta/\tau)}{1 - \exp(-\theta/\tau)} \exp(-t/\tau) = \exp(-t/\tau), \quad t \geq 0. \end{aligned}$$

At large times $t \gg \theta$ the difference and the charging time θ in the sum (5.16) can be interpreted as differential. Then it follows from (5.15–5.16) that

$$u(t) \sim \theta \mathscr{E}_0 t^{\alpha-1} \sum_{j=0}^{\infty} \frac{(-1)^j}{\Gamma(\alpha j + \alpha)} \left(\frac{t}{\tau}\right)^{\alpha j} = \theta \mathscr{E}_0 t^{\alpha-1} E_{\alpha,\alpha}[-(t/\tau)^\alpha], \quad t \gg 0.$$

This function has a power-tail: $u(t) \propto t^{-\alpha-1}$, $t \to \infty$. When $t \ll \theta$, the second term in (5.15) becomes neglegibly small for $t \gg \tau$: $u(t) \propto t^{-\alpha} (\tau \ll t \ll \theta)$.

Charging-discharging curves constructed with the use of Eq. (5.15) for $\alpha = 0.998$ are shown at Fig. 5.2. Charging times are $\theta = 10$ s (upper line), 7.5, 5.0 and 2.5 s (lower line). The voltage falls down accordingly to the Debye exponential law for a long time, but eventually we observe a split of the curves with different values of θ and transition to the non-Debye power law. It looks like the memory returns into the system after some interval of time. Such behavior was named *a memory regeneration effect* [131]. For $\alpha = 1$ relaxation follows the Debye law independently on θ: the memory is absent.

Experimental measurements were carried out in the following way [132]. Firstly, a capacitor was shunted with a resistor and amperemeter. Then the necessary voltage was applied by a power supply and the capacitor was charged during the time θ. Then it was shunted again. The measurements were performed on a capacitor, consisting of technical paper and oil. Its capacitance was equal to 2×10^{-6} F, the resistor had the resistance 2×10^5 Ohm and the power supply voltage was equal to 200 V. The results shown at the Fig. 5.2 confirms theoretical predictions (Fig. 5.3). Relaxation doesn't depend on charging for some interval of time, but then the process continues accordingly to the prehistory.

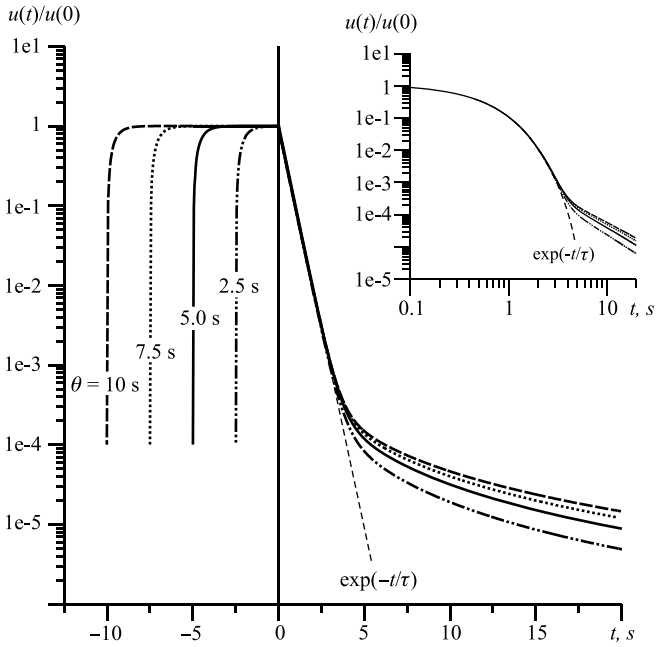


Fig. 5.2 Charging-discharging curves calculated for different values of θ

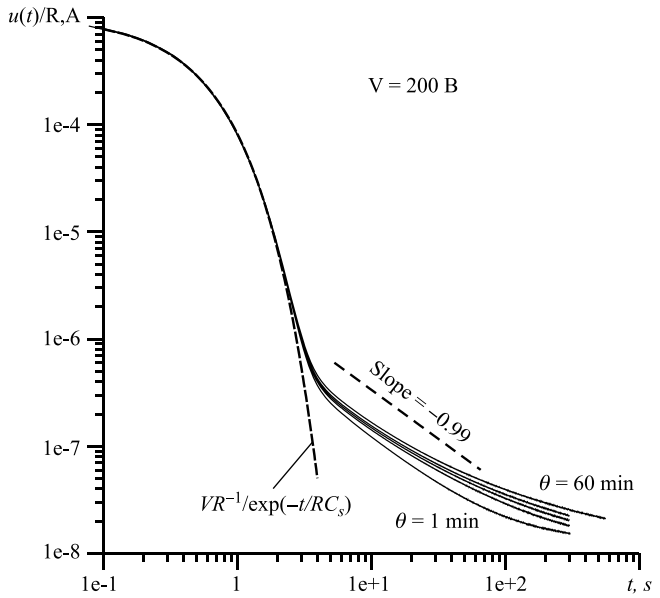


Fig. 5.3 Experimental discharging curves of a paper capacitor

In Fig. 5.4, the calculated discharging curves (points) are compared with experimental data (solid lines) with the use of a single fitting parameter α. Note a good agreement of the theoretical solution with the experimental data, though a small difference is observed at large times. At large θ, we observe the transition of the process from one decay law ($t^{-\alpha}$) to another ($t^{-\alpha-1}$) with accordance with theoretical predictions. Such variation of power law exponent was practically observed in the experiments with electrolitic capacitor (Fig. 5.5).

In conclusion, it is necessary to say a few words concerning the results. The term "regeneration" shouldn't mislead the reader: there is not any regeneration mechanism, we deal with the *relative contributions of two components*

$$u(t)/u(0) = e^{-t/\tau} + \eta_\alpha(t).$$

The first of them doesn't depend on its prehistory, while the second one does. They both descend with time, but the first (exponential) term falls down much faster at the beginning. Consequently, the second term, being neglegible compared with the first one, becomes dominant at large times, where curve splitting is getting clear.

5.4.5 Cole-Davidson Process

Following the work [86], consider another type of relaxation, namely that of Cole-Davidson. In order to describe this type of relaxation, Nigmatullin writes an ordinary exponential relaxation equation in the following form:

$$\exp(-\Omega_0 t)\mathsf{D}_t \exp(\Omega_0 t)\phi(t) = 0, \tag{5.17}$$

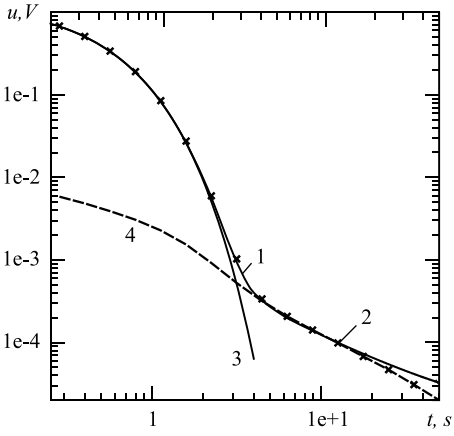


Fig. 5.4 Comparing of numerical results (2) with experimental data (1) for a paper capacitor $\theta = 300$, $\tau \approx 0.4\ \mathrm{c}^{-1}$. α was chosen equal to 0.998. (3) is an exponential function; (4) is the difference between theoretical curve and exponential function

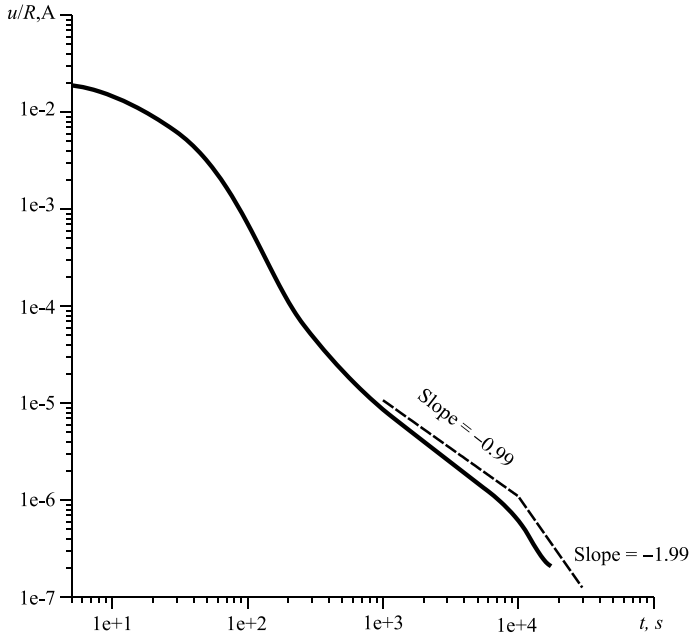


Fig. 5.5 Experimental discharging curve for an electrolitic capacitor ($\theta = 3600$)

where Ω_0 is a constant, describing the interaction of a macroscopic dipole with the thermostat [41]. In particular, the Eq. (5.17) holds true for those systems, whose relaxation at a microlevel can be described by the model of a deep potential hole with two equilibrium locations (Fröhlich relaxer) [41, 44, 53, 54]. The relaxation process in this model starts when an external field produces some difference in energy for both locations. Frelich relaxator model is well known to hold for a wide class of dielectrics. Some authors made an attempt to modify the model, using the models of hopping-charge transfer and ion conductivity [53, 54], in order to use it for describing non-exponential relaxation. However, all these attempts were based on the concept of *relaxation time distribution* (*RTD*), representing the non-Debye relaxation law $f(t)$ as a superposition of Debye ones with the time distribution density $p(\tau)$:

$$f(t) = \int_0^\infty e^{-t/\tau} p(\tau) d\tau. \tag{5.18}$$

In the work [86], the Fröhlich relaxer model is modified in different way. The function $G(t) = \exp(\Omega_0 t) f(t)$ is introduced. If $f(t)$ is a solution of Eq. (5.17), then $G(t) = f(0)$ is constant and

$$\mathsf{D}_t G(t) = 0. \tag{5.19}$$

Suppose that at some instants the system stays in an equilibrium state. In other words, at these instants $G(t) = 0$ but not $f(0)$ and we observe an equilibrium state instead of a non-equilibrium one. This way is caused, for example by heat fluctuations of local fields in dielectric, screening an external field. Besides that, suppose the instants when $G(t) = f(0)$ to be distributed over a self-similar (fractal) set. In other words, at the instants, coinciding with the points of some self-similar set, $G(t) = f(0)$, while at the ones, coinciding with the voids, $G(t) = 0$. Then, by integrating function $G(t)$ and averaging over different realizations of the self-similar set, one can obtain [86]

$$ {}_0\mathsf{D}_t^{-1} G(t) = A\ {}_0\mathsf{D}_t^{-\beta}[f(0)]. \tag{5.20} $$

where A is a constant, depending on the fractal set structure, where the $G(t)$ is distributed on and $\beta \in (0, 1]$ is its fractal dimension.

Using fractional integration and differentiation operators properties, the expression (5.17) can be written in the form

$$ {}_0\mathsf{D}_t^{\beta}[G(t)] = 0. \tag{5.21} $$

Stated differently, if during a relaxation process, the system interaction with an external field has an intermittent self-similar character, then Eq. (5.19) for the function $G(t)$ is substituted by Eq. (5.21), the equation for relaxation function $f(t)$ can be written in the form

$$ \exp(-\Omega_0 t)\ {}_0\mathsf{D}_t^{\beta}[\exp(\Omega_0 t) f(t)] = 0 \tag{5.22} $$

or using the operational relation [87]

$$ \exp(-\Omega u\ {}_0\mathsf{D}_t^{1-\varepsilon})\ {}_0\mathsf{D}_t^{\alpha} \exp(\Omega u\ {}_0\mathsf{D}_t^{1-\varepsilon}) = ({}_0\mathsf{D}_t^{\varepsilon} + \Omega\varepsilon)^{\alpha/\varepsilon}, $$

$$ 0 < \varepsilon \leqslant 1, \quad \alpha \leqslant \varepsilon. $$

This gives the formula of Cole-Davidson better visibility:

$$ {}_0(\mathsf{D}_t + \Omega_0)^{\beta} f(t) = 0. \tag{5.23} $$

The following conclusions are stated in the cited work.

(1) The Cole-Davidson process can occur in those systems, which both the Frölich relaxator model and the non-Debye relaxation can be applied to. Possibly, such conditions can be held in segnetoelectrics, ion conductive crystals, and also in liquid and plastic crystals.
(2) Relaxation function of such systems obeys the equation

$$ \left[\mathsf{D}_t + \Omega_0\right]^{\beta} f(t) = 0, $$

and has the form

$$f(t) = C\frac{t^{\beta-1}}{\Gamma(\beta)}e^{-\Omega_0 t}.$$

(3) The parameter β "has a sense of fractal dimension of the set, which the instants liable for the interaction between the elementary components of the physical system and external field are statistically distributed on".

5.4.6 Havriliak-Negami Process

A more generalized approximation, including all the three previous laws, is given by the two-parametric formula of Havriliak-Negami (HN)

$$\tilde{f}_{\alpha}^{\beta}(i\omega) = \frac{1}{[1+(i\omega\tau)^{\alpha}]^{\beta}}, \quad 0<\alpha<1, \quad 0<\beta<1. \tag{5.24}$$

The solution of the corresponding fractional-differential equation

$$\left[1+(\tau\ {}_0\mathsf{D}_t)^{\alpha}\right]^{\beta} f(t) = \delta(t),$$

based on the expansion of fractional power of the operational sum in the infinite Newtonian series

$$\left[1+(\tau\ {}_0\mathsf{D}_t)^{\alpha}\right]^{\beta} = \sum_{n=0}^{\infty}\binom{\beta}{n}(\tau_0\mathsf{D}_t)^{\alpha(\beta-n)},$$

is obtained in the work [88] and has the form

$$f(t) = -\frac{1}{\Gamma(\beta)}\sum_{n=0}^{\infty}\frac{(-1)^n\Gamma(n+\beta)}{n!\Gamma(\alpha(n+\beta))}\left(\frac{t}{\tau}\right)^{\alpha(n+\beta)}.$$

Since parameter α, as we have seen in the Cole-Cole model, is connected with the power distribution of trapping time, and β characterizes fractal distribution of relaxation instances in the Cole-Davidson model, it is then natural to suppose that the Havriliak-Negami process is attributed to both above mentioned properties.

Finally, in the conclusion of the chapter we put the Havriliak-Negami relaxation in the form of distributed relaxation rates [119]

$$f_{\alpha}^{\beta}(t) = \int_0^{\infty} e^{-t/\tau}\, p_{\alpha}^{\beta}(\tau)d\tau,$$

$$p_{\alpha}^{\beta}(\tau) = \frac{1}{\pi}\left|\frac{10^{s\alpha\beta}\sin(\beta\theta)}{\left[10^{2s\alpha}+2\times 10^{s\alpha}\cos(\alpha\pi)+1\right]^{\beta/2}}\right|,$$

where $s = \log(\tau/\bar{\tau})$, $\bar{\tau}$ is the most probable value of τ, and

$$\theta = \operatorname{arctg}\left[\frac{\sin(\alpha\pi)}{10^{s\alpha} + \cos(\alpha\pi)}\right].$$

Comparison of these formulas with (5.24) gives, to my mind, a convincing argument towards time approach of the physical interpretation of non-Debye relaxation search.

Weron et al. [139] have derived the following Monte-Carlo algorithm for simulation the random waiting time T obeying the Havriliak-Negami distribution:

$$T = \tau G_{\beta}^{1/\alpha} S_{+}(\alpha)$$

where the random variable G_β is distributed according to the gamma distribution with the scale parameter 1 and shape parameter β:

$$\mathsf{P}(G_\beta < t) = \frac{1}{\Gamma(\beta)} \int_0^t \tau^{\beta-1} e^{-\tau} d\tau.$$

Uchaikin [121] has found that the HN-function

$$\widehat{f}_{\alpha}^{\beta}(z) = \frac{1}{[1 + z^{\alpha}]^{\beta/\alpha}} \tag{5.25}$$

can satisfactorily be approximated by the function

$$\widehat{f}_{\gamma\delta\varepsilon}(z) = \frac{1}{c_0 + cz^{\gamma} + dz^{\delta} + ez^{\varepsilon}}, \quad \gamma < \delta < \varepsilon \tag{5.26}$$

in the case $\alpha < \beta$ (Fig. 5.6). The constants c_0, c, d, e, γ, δ, and ε may be determined from the conditions

$$\lim_{z\to\infty} \left[\widehat{f}_{\gamma\delta\varepsilon}(z)/\widehat{f}_{\alpha}^{\beta}(z)\right] = 1,$$

$$\lim_{z\to 0} \left[1 - \widehat{f}_{\gamma\delta\varepsilon}(z)\right] / \left[1 - \widehat{f}_{\alpha}^{\beta}(z)\right] = 1,$$

$$\widehat{f}_{\gamma\delta\varepsilon}(1) = \widehat{f}_{\alpha}^{\beta}(1),$$

and

$$\widehat{f}'_{\gamma\delta\varepsilon}(1) = \widehat{f}'^{\beta}_{\alpha}(1).$$

The two first conditions yield $c_0 = e = 1$, $c = \beta/\alpha$, and $\gamma = \alpha$. From the third we have

$$d = 2^{\beta/\alpha} - 2 - \beta/\alpha,$$

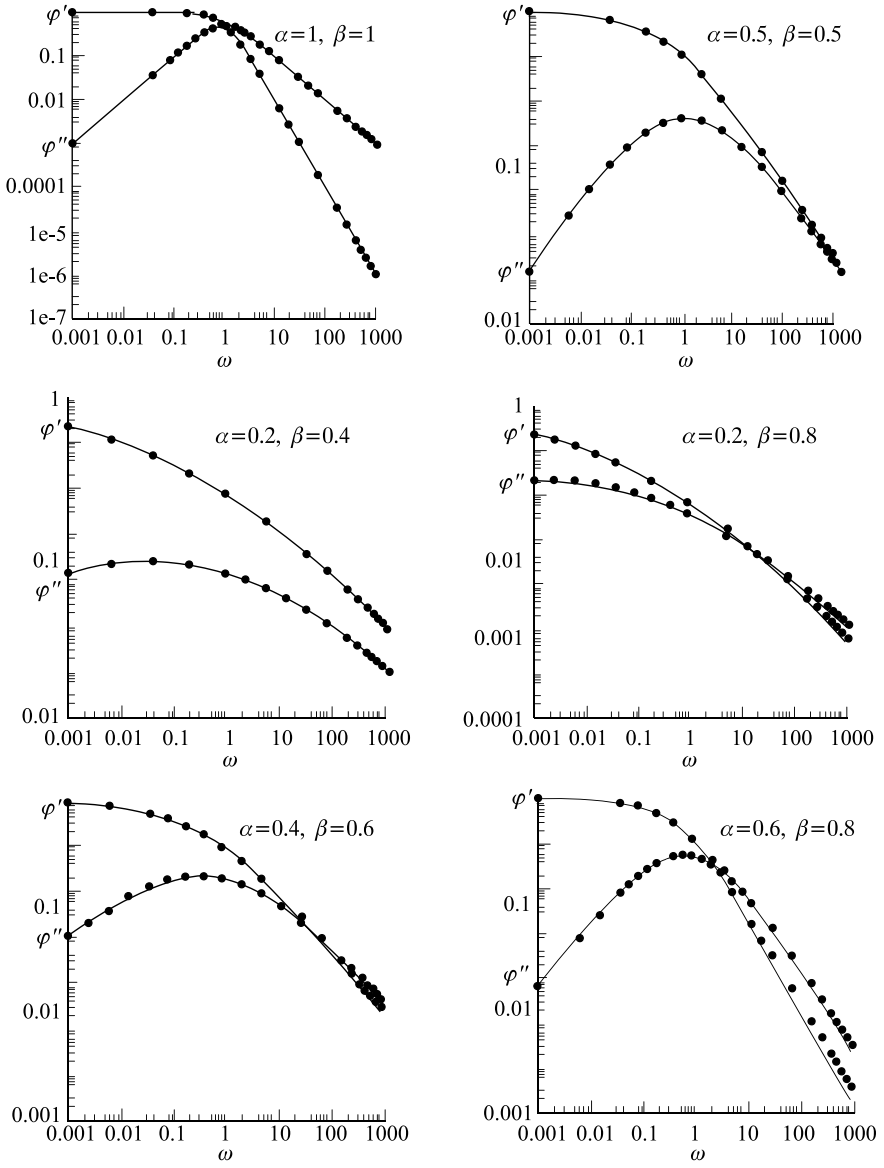


Fig. 5.6 Comparison approximation (5.26) of the Havriliak-Negami law (circles) with exact expression (5.28) (lines)

and from the fourth

$$\delta = \beta \left[2^{(\beta-\alpha)} - 2/d \right].$$

The case $\beta = 2\alpha$ obtained by the limit transition leads to the exact result

$$\widehat{f}_{\gamma\delta\varepsilon}(z) = \frac{1}{(1+z^{\alpha})^2}.$$

Results of comparative calculations for real and imaginary components of function $\tilde{f}(i\omega)$ plotted in Fig. 5.6 show that Eq. (5.26) fits HN-function with an acceptable accuracy.

Equation (5.26) represents the differential equation with fractional derivatives

$$\left[c_0 + c\ {}_0\mathsf{D}_t^{\gamma} + d\ {}_0\mathsf{D}_t^{\delta} + e\ {}_0\mathsf{D}_t^{\varepsilon}\right] f_{\gamma\delta\varepsilon}(t) = \delta(t)$$

in Fourier-space. Taking into account that $\gamma < \delta < \varepsilon$, we can write $\widehat{f}_{\gamma\delta\varepsilon}(z)$ in the form [93]

$$\begin{aligned}
\widehat{f}_{\gamma\delta\varepsilon}(z) &= \frac{1}{ez^{\varepsilon} + dz^{\delta}} \frac{1}{1 + \frac{cz^{\gamma}+c_0}{ez^{\varepsilon}+dz^{\delta}}} = \frac{e^{-1}z^{-\delta}}{z^{\varepsilon-\delta} + e^{-1}d} \frac{1}{1 + \frac{e^{-1}cz^{\gamma-\delta}+e^{-1}c_0z^{-\delta}}{z^{\varepsilon-\delta}+e^{-1}d}} \\
&= \sum_{m=0}^{\infty}(-1)^m \frac{e^{-1}z^{-\delta}}{(z^{\varepsilon-\delta} + e^{-1}d)^{m+1}} \left(\frac{c}{e}z^{\gamma-\delta} + \frac{c_0}{e}z^{-\delta}\right)^m \\
&= \sum_{m=0}^{\infty}(-1)^m \frac{e^{-1}z^{-\delta}}{(z^{\varepsilon-\delta} + e^{-1}d)^{m+1}} \sum_{k=0}^{m}\binom{m}{k} \frac{c^k c_0^{m-k}}{e^m} z^{\gamma k-\delta m} \\
&= \frac{1}{e}\sum_{m=0}^{\infty}(-1)^m \left(\frac{c_0}{e}\right)^m \sum_{k=0}^{m}\binom{m}{k}\left(\frac{c}{c_0}\right)^k \frac{z^{\gamma k-\delta m-\delta}}{(z^{\varepsilon-\delta} + e^{-1}d)^{m+1}}.
\end{aligned}$$

The term-by-term inversion based on the general expansion theorem for the Laplace transform yields the final expression for the version of HN function under consideration

$$\begin{aligned}
f_{\gamma\delta t}(t) = \frac{1}{e}\sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{-c_0}{e}\right)^m \\
\times \sum_{k=0}^{m}\binom{m}{k}\left(\frac{c}{c_0}\right)^k t^{\varepsilon(m+1)-\gamma k-1} E^{(m)}_{\varepsilon-\delta,\varepsilon+\delta m-\gamma k}\left(-\frac{d}{e}t^{\varepsilon-\delta}\right).
\end{aligned}$$

This is a general solution for all relaxation processes belonging to the region $L(\alpha < \beta)$.

The equation with three fractional derivatives of different orders γ, δ, and ε allows us to propose that three different mechanisms are involved in the relaxation process. All of them manifest themselves due to jumps of the observed physical characteristics as quantum mechanics dictates, and the response function is an average function. One can suppose that one of them is the change in orientation of macrodipoles in solids, and the second of these mechanisms is the decay of macrodipole moments in magnitude (this process is absent in the Debye relaxation).

5.5 Semiconductors

5.5.1 Diffusion in Semiconductors

The current density $j(x,t)$ flowing through a p-n-junction at $x=0$ is expressed through the hole concentration $\rho(x,t)$ by the relationship

$$j(t) = \left[eK\mathsf{D}_x - e\mu E(0)\right]\rho(x,t)\Big|_{x=0},$$

where e is the electron charge, K, μ, and τ are the diffusion coefficient, mobility, and mean life time of holes, respectively, E is the electric field of intensity of uncompensated charges. The charges concentration satisfies the equation

$$\left[\mathsf{D}_t - K\mathsf{D}_x^2 + \mu E(x)\mathsf{D}_x + 1/\tau\right]\rho(x,t) = 0, \quad x \geq 0, \quad t > 0 \tag{5.27}$$

with the initial-boundary condition

$$\rho(x,0) = \rho(\infty,t) = 0.$$

The current density $j(t)$ is an easily measured quantity and if the boundary concentration is wanted to be found, it is convenient to split Eq. (5.27) following the Babenko [5] approach. Assuming for simplicity that $E = \text{const}$, $K = 1$ and introducing notation $\beta = \mu E$, $\gamma = 1/\tau$, $\delta = \beta^2/4 + \gamma$, we obtain the following relation:

$$j(t) = e^{-\delta t}({}_0\mathsf{D}_t^{1/2} + \delta/2)e^{\delta t}\rho(0,t).$$

The solution of this equation has the form

$$\rho(0,t) = \frac{1}{e}\int_0^t \left[\frac{e^{-\delta(t-\theta)}}{\sqrt{\pi(t-\theta)}} - \frac{\beta}{2}e^{-\gamma(t-\theta)}\operatorname{erfc}\left(\frac{\beta\sqrt{t-\theta}}{2}\right)\right] j(\theta)d\theta. \tag{5.28}$$

As a matter of fact, the electric field E and consequently the coefficient β depends on x. In particular, for a diode microwave frequency multiplier diode,

$$\beta(x) = -\frac{2}{\sqrt{\pi}l}\frac{\exp(-x^2/l^2)}{1+\operatorname{erfc}(x/l)}, \quad l = \text{const}.$$

In this case, the hole concentration is written as

$$\begin{aligned}\rho(0,t) = (1/e)\exp[-(l^2/\pi+\gamma)t]({}_0\mathsf{D}_t^{-1/2} + \pi^{-1/2}l^{-1}\,{}_0\mathsf{D}_t^{-1} \\ +2\pi^{-1}l^{-2}\,{}_0\mathsf{D}_t^{-3/2} + \cdots)j(t).\end{aligned}$$

Further information about this solution can be found in Babenko [5]. We just note that fractional derivatives (semi-integrals) appear here as mathematical tools and they are not connected with physics of the process. In the following part of this chapter, the case of fractional derivatives lies in physical features of transport processes in disordered semiconductors.

5.5.2 *Dispersive Transport: Transient Current Curves*

The measurements in the time-of-flight experiments in semiconductors are carried out so that the transient photocurrent density in the sample of the thickness L is determined as thickness-averaged conduction current density:

$$I(t) = \frac{1}{L}\int_0^L j(x,t)dx. \tag{5.29}$$

At the initial stage $t \ll t_{\mathrm{T}}$ the integral in the last equation is independent of L. According to Eq. (2.16) of Vol. I., $t_{\mathrm{T}} \propto L^{1/\alpha}$. Hence,

$$A \propto L^{-1}, \qquad B \propto A\, t_{\mathrm{T}}^{2\alpha} \propto L, \qquad I_{\mathrm{T}} \propto L^{-1/\alpha}.$$

Thus, the universality of the transient current curves is the self-similarity property at time scales

$$I(t; L_2) \approx (L_2/L_1)^{-1/\alpha} I\left(t(L_2/L_1)^{-1/\alpha}; L_1\right), \tag{5.30}$$

where $I(t; L_1)$ and $I(t; L_2)$ are the time dependences of the transient current in the samples with the thicknesses L_1 and L_2, respectively.

The conduction current density $j(x,t)$ is proportional to the density distribution $p(t|x)$ of the first reaching time:

$$j(x,t) = eNp(t|x),$$

where e is the elementary charge and N is the number of photoinjected carriers per unit area of the illuminated electrode. The product $p(t|x)dt$ is the probability that a walking particle (a charge carrier in our case) reaches a coordinate of x during the interval $[t, t+dt)$. The coordinate distribution density $p(x|t)$ and the probability flux $p(t|x)$ are related by the probability conservation equation

$$\frac{\partial p(x|t)}{\partial t} + \frac{\partial p(t|x)}{\partial x} = \delta(x)\delta(t). \tag{5.31}$$

According to Eq. (2.15) of Vol. I, the asymptotic behavior for the transient current and the distribution of the first reaching times at long times is a power law with the

exponent α. According to Eqs. (5.30) and (5.31), the function $p(t|x)$ is self-similar in time:

$$p(t|x_2) = \left(\frac{x_2}{x_1}\right)^{-1/\alpha} p\left(t\left(\frac{x_2}{x_1}\right)^{-1/\alpha}\middle| x_1\right). \tag{5.32}$$

5.5.3 Stability as a Consequence of Self-similarity

The time of reaching the coordinate $2x$ is a random variable and is the sum of two terms, which are independent random times during which a carrier passes the $(0, x)$ and $(x, 2x)$ layers. The corresponding distribution is expressed as the convolution of the distributions of the times of passing two layers:

$$p(t|2x) = 2^{-1/\alpha}\, p(2^{-1/\alpha}t|x) = \int_0^t dt\; p(t-t'|x)\; p(t'|x). \tag{5.33}$$

The first equality in this equation follows from the self-similarity property given by Eq. (5.30). The solution of Eq. (5.33) is the stable density with the characteristic exponent α:

$$p(t|x) = \left(\frac{x}{K}\right)^{-1/\alpha} g_+\left(t\left(\frac{x}{K}\right)^{-1/\alpha}; \alpha\right). \tag{5.34}$$

Thus, the distribution density of the first reaching times is the one-side stable density with the characteristic exponent equal to the dispersion parameter.

Substituting of the conduction current density $j(x, t) = eNp(t|x)$ into Eq. (5.29) provides the following expression for transient current:

$$I(t) = \frac{eKN\alpha}{L} t^{\alpha-1} \int_{\zeta_0}^{\infty} \zeta^{-\alpha} g_+(\zeta; \alpha)\, d\zeta, \tag{5.35}$$

where $\zeta_0 = t\,(L/K)^{-1/\alpha}$. Figure 5.7 shows the transient current curves calculated by Eq. (5.35) for various dispersion parameters α.

Photocurrent calculated by Eq. (5.35) is shown in Fig. 5.8 in comparison with the experimental data and results obtained from the Arkhipov-Rudenko τ-equation [3]. According to this figure, the fractional differential equation provides better agreement with the experimental data. Arkhipov et al. [4] derived their equation on the assumption that most carriers are captured in the traps so deep (below the demarcation level) that the release from them t is slow. This assumption is invalid for α values close to one. The equation allows one to express all the results in terms of elementary functions for the case of the exponential density of localized states.

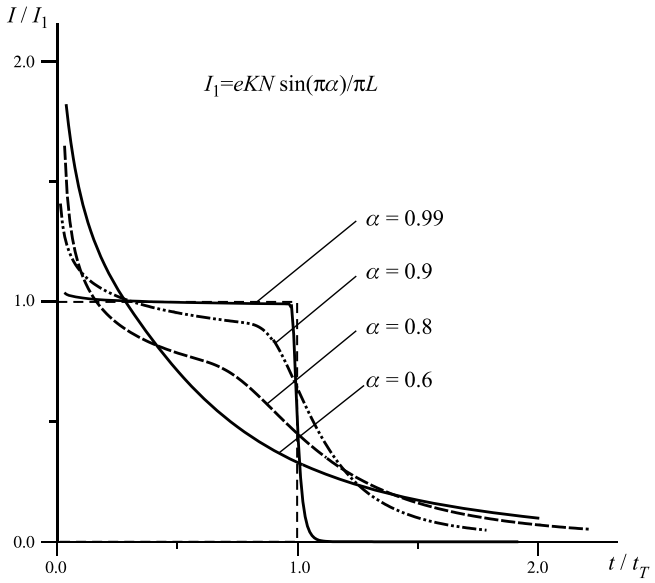


Fig. 5.7 Theoretical curves of transient current (5.35) for various dispersion parameters

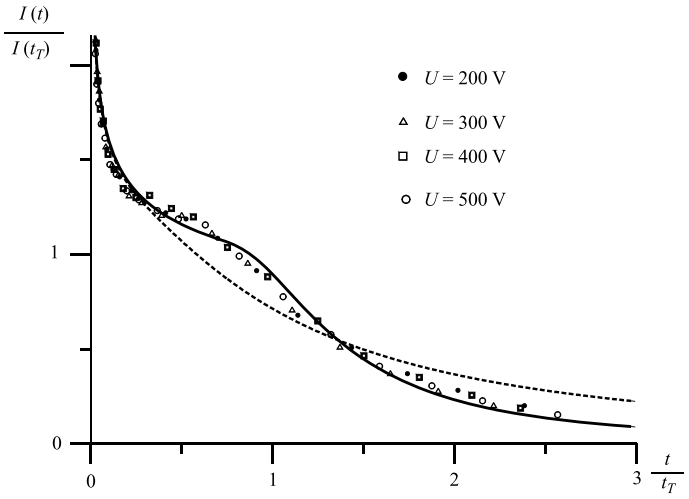


Fig. 5.8 Transient current curves and experimental data (digitalized from the figure from Scher and Montroll [108]) for trinitrofluorene-polyviynilcarbazole (TNF-PVC) organic complex, which are obtained by Gill. The dotted line is obtained from the Arkhipov-Rudenko τ-equation [3]. The solid line is the transient current given by Eq. (5.35). The dispersion parameter is $\alpha = 0.8$

According to Eqs. (5.31) and (5.34) the distribution density of the photoinjected carriers in the coordinate parallel to the field is given by the expression

$$f(x,t) = p(x|t) = -\frac{\partial}{\partial x}\int_0^t p(t|x)dt = \frac{t}{\alpha K}\left(\frac{x}{K}\right)^{-1/\alpha-1} g_+\left(t\left(\frac{x}{K}\right)^{-1/\alpha};\alpha\right). \tag{5.36}$$

Figure 5.9 shows the plots of this density.

5.5.4 *Fractional Equations as a Consequence of Stability*

Functions (5.34) and (5.36) are the solutions of the equations with fractional derivatives:

$$K^{-1}{}_0\mathsf{D}_t^{\alpha} p(t|x) + \frac{\partial}{\partial x} p(t|x) = \delta(x)\delta(t), \tag{5.37}$$

$${}_0\mathsf{D}_t^{\alpha} f(x,t) + K\frac{\partial}{\partial x} f(x,t) = \delta_{\alpha}(t)\delta(x),$$

where $\delta_{\alpha}(t) = t^{-\alpha}/\Gamma(1-\alpha)$. Here, a one-directional motion of the carriers is considered. When carriers move in both directions the second equation becomes of diffusion type. In the presence of a flux in the negative direction and/or the scattering of the carriers, the diffusion term appears in the last equation:

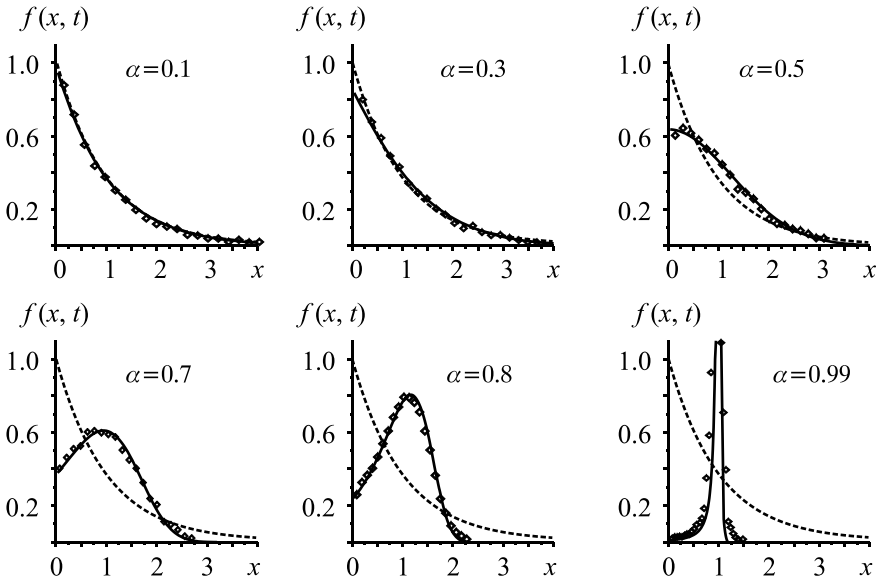


Fig. 5.9 Total carrier density normalized to one. The points are the Monte Carlo results obtained in the Scher-Montroll random walk scheme, dashed lines are plotted according to the Arkhipov-Rudenko τ-equation [4], and the solid lines are obtained from Eq. (5.36)

$$ {}_0\mathsf{D}_t^{\alpha} f(x,t) + K\frac{\partial f(x,t)}{\partial x} - D\frac{\partial^2 f(x,t)}{\partial x^2} = \delta_{\alpha}(t)\delta(x). \tag{5.38} $$

The fundamental solutions of Eq. (5.38) in the limit $\alpha \to 1$ are transferred to the Gaussian density and the equation is modified to the classical one-dimensional Fokker-Planck equation.

Thus, the conclusions of this study are as follows.

- According to the experimental facts (the universality property of transient current curves and the power-law dependence of the time-of-flight on the sample thickness), the nonequilibrium carrier densities in dispersive transport are necessarily expressed in terms of the stable densities and satisfy the fractional differential equations.
- In contrast to the solutions of the Arkhipov-Rudenko τ-equation, the solutions of the fractional differential equations satisfy the correspondence principle; i.e., these solutions in the limit $\alpha \to 1$ are transferred to the solutions for normal transport and the equations are modified to the classical one-dimensional Fokker-Planck equation.
- The stable probability flux density in the dispersion diffusion is the evidence of the Scher-Montroll hypothesis about the power-law distribution of the waiting times.
- The fractional differential model is consistent with the Scher-Montroll theory and multiple trapping model, but the fractional differential model allows the description of normal and dispersive transports in the framework of a unified formalism.
- A fractional differential equation can be a phenomenological basis for the description of dispersive transport in disordered semiconductors.

5.6 Conductors

5.6.1 Skin-Effect in a Good Conductor

Harsoyo [46] arrives at a fractional derivative when considering the skin-effect in a good conductor in the framework of elementary classical electrodynamics. A homogeneous conductor medium fills the half-space $x > 0$, its conductivity σ is assumed real in the range of working frequencies. Combining the continuity equation

$$ \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0, $$

the constitutive equation

$$ \mathbf{j} = \sigma \mathbf{E}, $$

and the Gauss equation

$$ \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, $$

Harsoyo derives the following differential equation for the charge density in the conductor,

$$\frac{\partial \rho}{\partial t} + \frac{\rho}{\tau} = 0,$$

where $\tau = \varepsilon_0/\sigma$. Following the standard way, that is taking into account that the conduction current is much greater than the displacement current, the author uses the reduced Maxwell equation

$$\mu_0 \sigma \frac{\partial \mathbf{E}}{\partial t} - \Delta \mathbf{E} = 0.$$

Inserting

$$\mathbf{E} = E_1(x) e^{i\omega t} \mathbf{e}_z$$

leads to the following expressions for the field and current respectively:

$$\mathbf{E} = \mathbf{E}_0 e^{-x/\delta} e^{i(\omega t - x/\delta)},$$

$$\mathbf{j} = \mathbf{j}_0 e^{-x/\delta} e^{i(\omega t - x/\delta)},$$

$$\delta = \sqrt{\frac{2}{\mu_o \sigma \omega}}.$$

If the conductivity σ is very high, the skin-depth δ is very small, and one can say that the current flows only through a thin surface layer. Finally, the surface current density take the form

$$\mathbf{J} = \int_0^\infty \mathbf{j}(x,t)dx = i\mathbf{E}_0 \sqrt{\frac{\sigma}{\mu_0 \omega}} e^{i(\omega t + \pi/4)}.$$

Observing that action of the Weyl fractional operator

$${}_t\mathsf{D}^\alpha_\infty f(t) = \frac{1}{\Gamma(\alpha)} \int_t^\infty (t' - t)^{\alpha - 1} f(t')dt', \quad \alpha < 0$$

on sinusoidal functions is determined by expressions

$$\begin{aligned} {}_t\mathsf{D}^\alpha_\infty \cos(\omega t) &= \omega^\alpha \cos(\omega t - \alpha\pi/2), \\ {}_t\mathsf{D}^\alpha_\infty \sin(\omega t) &= \omega^\alpha \cos(\omega t - \alpha\pi/2), \\ {}_t\mathsf{D}^\alpha_\infty e^{i\omega t} &= \omega^\alpha e^{i(\omega t - \alpha\pi/2)}. \end{aligned}$$

Harsoyo [46] represents the surface current as

$$\mathbf{J}(t) = i\mathbf{E}_0\sqrt{\frac{\sigma}{\mu_0}}\,{}_t\mathsf{D}_\infty^{-1/2}e^{i\omega t}.$$

If an incident wave has the amplitude $\mathbf{E}_0 f(t)$ then the surface current is expressed as

$$\mathbf{J}(t) = i\mathbf{E}_0\sqrt{\frac{\sigma}{\mu_0}}\,{}_t\mathsf{D}_\infty^{-1/2}f(t).$$

5.6.2 *Electrochemistry*

In the seventies, the fractional calculus gives rise a New approach to analysis in electrochemistry—the half-integral and half-differential methods [89, 90]. Based on the main problem in the electrochemical analysis is to define the concentration $\rho(x,t)$ of electroactive elements on an electrode's surface ($x = 0$). In contradistinction to direct measurements of $\rho(0,t)$, the surface current density

$$j_x(0,t) = -K\frac{\partial\rho(x,t)}{\partial x}\Big|_{x=0}$$

is much easier to measure. Using this formula, however, required the solution of diffusion equation in the right half-space (inside the electrode)

$$\frac{\partial\rho(x,t)}{\partial t} = K\frac{\partial^2\rho(x,t)}{\partial x^2}.$$

In the work [43], it was proposed to find a square root of the operators in both parts of the equation and substitute $x = 0$ into the result equation:

$${}_0\mathsf{D}_t^{1/2}\rho(0,t) = K^{1/2}\frac{\partial\rho(x,t)}{\partial x}\Big|_{x=0} = -K^{-1/2}j_x(0,t).$$

Solving the above equation

$$\rho(0,t) = -K^{-1/2}\,{}_0\mathsf{D}_t^{-1/2}j_x(0,t) = -K^{-1/2}\,{}_0\mathsf{I}_t^{1/2}j(0,t) \equiv \frac{1}{\sqrt{\pi}}\int_0^t\frac{j_x(0,\tau)d\tau}{\sqrt{t-\tau}}$$

formed the basis of a new method in electrochemistry. All physical characteristics are taken on the surface $x = 0$ and no more redundant information is used. This is exactly the same approach developed by Babenko in thermodynamics (Sect. 4.1.1).

5.6.3 Rough Surface Impedance

Characteristics of electrochemical devices and processes are related to properties of metallic electrode surfaces contacting with liquid or solid electrolytes. In classical theory the influence of a boundary on the alternating current flowing through a system is described by a boundary capacitance inserted in parallel with the ohmic resistance of the electrolyte. At the same time the real part of the impedance doesn't depend on frequency and the imaginary part is inversely proportional to it. Nevertheless, many experiments (starting with [144]) show the different behavior in most cases: at least in restricted range the frequency dependence on impedance is characterized by the additional power type term $A(i\omega)^{-\eta}$ with η between zero and one, $0 < \eta < 1$. This term is called the *constant phase element* (CPE). The presence of CPE is associated with the surface roughness: the smoother the surface is, the closer η to the unity is. It is shown in the work [25], that the imitation of a rough surface via pores leads to the power dependence, but with a definite exponent $\eta = 1/2$, only if one doesn't assume a particular space structure of resistance and capacitance distribution.

Some ground for modeling such structures is given by investigations of surfaces using electronic microscopes. The absence of natural scale of length causing different zoomings to show almost similar pictures of inhomogeneities distributions, makes it possible to apply the concept of fractals. In the works [59, 70], the Cantor set was used as a model of the fractal. (See Sect. 2.1.4 of Vol. I) At each step of constructing, the middle part of every section is removed so that only two sections remains with the same length equal $1/a$ ($a < 2$) of the former length.

Figure 5.10 shows an equivalent scheme of a surface obtained as described above. The input impedance of this chain can be written in the form of an infinite chain fraction.

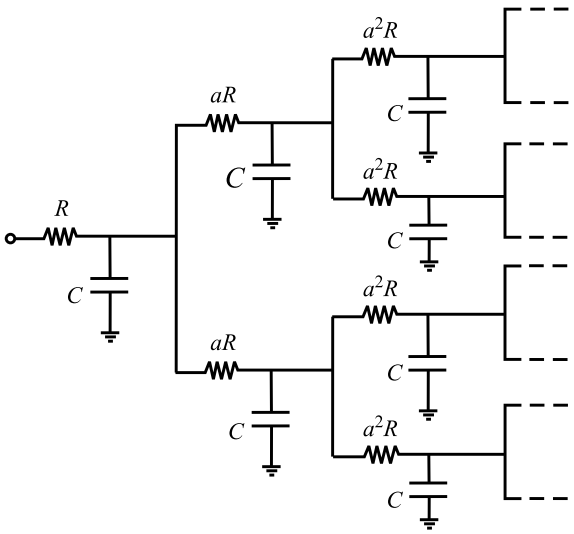


Fig. 5.10 Equivalent scheme of rough electrode's surface

$$Z(i\omega) = R + \cfrac{1}{i\omega C + \cfrac{2}{aR + \cfrac{1}{i\omega C + \cfrac{2}{a^2 R + \cdots}}}}.$$

Practical calculations are performed for finite chain fraction with the recursion number n and the solution behavior is investigated against the increase of n. At low frequences the real part of impedance gets a plateau whose height rises $a/2$ times with every recursion. At high frequences the impedance takes its extremal value R. At intermediate frequences the system possess the property of CPE with $\eta = 1 - D$, where $D = \ln 2/\ln a$ is the fractal dimension of the Cantor set. The imaginary part $Z(i\omega)$ is inversely proportional to frequency at high and low frequences and corresponds to CPE in the intermediate region. Thus the frequency representation of generalized Ohm's law for rough surface has the form

$$\widehat{I}(\omega) = [Z(i\omega)]^{-1}\widehat{U}(i\omega) = A^{-1}(i\omega)^{\eta}\widehat{U}(i\omega),$$

and in time representation it contains a fractional derivative:

$$I(t) = A^{-1}\,{}_0\mathsf{D}_t^{\eta} U(t).$$

As it is stated in Liu et al. [70], the power dependence of impedance on frequency and therefore the fractional order of the derivative are connected with the competition of resistive and capacitive currents. At low frequency signal propagates farther through a chain, before it can escape through surface capacity, therefore the low frequency impedance is larger. Of course, real surfaces are self-similar only in a finite frequency range, which ultimately defines the frequency interval, where phase constancy (loss angle) takes place.

5.6.4 Electrical Line

Consider a homogeneous electrical transmission line of length l [2]. Let L, C, and R be self-induction, capacitance, and resistance per unit length of the line. One end of the line (at $x = 0$) is subjected to an electromotive force $E_0(t)$, and another end is connected to an impedance Z (Fig. 5.11).

Let $U(x,t)$ be the potential difference between a point x of the line and ground (voltage), and $I(x,t)$ be the current at the point instant t. Voltage decrease per unit of length consists of two terms caused by self-induction and by resistance:

$$\frac{\partial U}{\partial x} = -L\frac{\partial I}{\partial t} - RI. \tag{5.39}$$

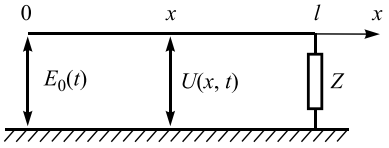


Fig. 5.11 On a derivation of the telegraph equation

The current difference per section equals the sum of current leakage through the capacitance ($C\partial U/\partial t$) and through isolation (GU):

Excluding current, we come to equation

$$LC\frac{\partial^2 U}{\partial t^2} + (RC + LG)\frac{\partial U}{\partial t} + RGU = \frac{\partial^2 U}{\partial x^2}, \tag{5.40}$$

which is known as the *telegraph equation*.

Return to the system of Eqs. (5.39)–(5.40) and perform the Laplace transform over time:

$$\begin{aligned} \frac{d\widehat{U}}{dx} &= -L(L\lambda + R)\widehat{I}(x, \lambda), \\ \frac{d\widehat{I}}{dx} &= -(C\lambda + G)\widehat{U}(x, \lambda). \end{aligned} \tag{5.41}$$

This system of equations of the first order is reduced to a single equation of the second order which can be written both for voltage

$$\frac{d^2\widehat{U}}{dt^2} - \mu^2\widehat{U}(x, \lambda) = 0, \tag{5.42}$$

or for current

$$\frac{d^2\widehat{I}}{dt^2} - \mu^2\widehat{I}(x, \lambda) = 0,$$

where

$$\mu^2 = (C\lambda + G)(L\lambda + R).$$

The general solution of Eq. (5.42) is

$$\widehat{U}(x, \lambda) = Ae^{\mu x} + Be^{-\mu x}. \tag{5.43}$$

From Eq. (5.41) it follows

$$\widehat{I}(x, \lambda) = \sqrt{(C\lambda + G)/(L\lambda + R)}(Be^{-\mu x} - Ae^{\mu x}).$$

Constants A and B are defined by boundary conditions:

$$\widehat{U}(0, \lambda) = A + B, \tag{5.44}$$

$$\widehat{U}(l,\lambda) = Ae^{\mu l} + Be^{-\mu l}. \tag{5.45}$$

It is clear that

$$\widehat{U}(0,\lambda) = \widehat{E}_0(\lambda).$$

Introducing the impedance $Z(\lambda)$ via the relation

$$\widehat{U}(l,\lambda) = \widehat{I}(l,\lambda)Z(\lambda),$$

yields the second condition in the form

$$Ae^{\mu l} + Be^{-\mu l} = Z(\lambda)\sqrt{(C\lambda+G)/(L\lambda+R)}\left(Be^{-\mu l} - Ae^{\mu l}\right).$$

Assuming

$$\gamma = Z(\lambda)\sqrt{(C\lambda+G)/(L\lambda+R)},$$

we get two equations to define the constants:

$$A + B = \widehat{E}_0(\lambda),$$

$$(\gamma+1)Ae^{\mu l} = (\gamma-1)Be^{-\mu l}.$$

Let us close the end of the line on the impedance equal to *characteristic resistance of the line* $Z_c = \sqrt{(L\lambda+R)/(C\lambda+G)}$, then $\gamma = 1$ and from the Eqs. (5.44), (5.45) we have $A = 0$, $B = E_0(\lambda)$, and

$$\widehat{I}(\lambda) = \frac{\widehat{U}(\lambda)}{Z_c(\lambda)},$$

where

$$\widehat{U}(\lambda) = \widehat{E}_0(\lambda)\exp\left[-\sqrt{(L\lambda+R)(C\lambda+G)}\,x\right].$$

This case corresponds to a half-infinite line: the latter doesn't have a reflected wave. It is seen from Eq. (5.43). To keep a physical sense when $x \to \infty$, particularly staying restricted by magnitude, the constant A must equal zero.

In the case when self-induction is small against capacitance (in the sense of influence on the process), and isolation is very good (say an underground cable)

$$Z_c = \sqrt{\frac{L\lambda+R}{C\lambda+G}} \simeq \sqrt{\frac{R}{C}}\lambda^{-1/2},$$

and we arrive at the relation

$$\widehat{I}(\lambda) = \sqrt{\frac{C}{R}}\lambda^{1/2}\widehat{U}(\lambda),$$

equivalent to an equation with a fractional derivative

$$I(t) = \sqrt{\frac{C}{R}}\, {}_0\mathsf{D}_t^{1/2} U(t).$$

5.6.5 *Josephson Effect*

The Josephson effect is a phenomenon of supercurrent across two superconductors coupled by a weak link. The weak link can consist of a thin insulating barrier (superconductor-insulator-superconductor junction), a short section of non-superconducting metal, or a physical constriction that weakens the superconductivity at the point of contact. Such a device is known as a *Josephson junction*.

For a long time, most of researchers have been used for describing this effect the sine Gordon equation. However, this approach fails to solve this problem in case of small scales being essentially less than the London length. An adequate description of the effect at small scales can be reached by taking the natural nonlocality of the process into account [51].

Speaking more exactly, the nonlocal generalization of the sine Gordon equation has been *derived* from the Maxwell equations. Starting point of the derivation is the nonlocal interrelation between magnetic strength and magnetic induction

$$\mathbf{H}(\mathbf{r}, t) = \int_{-\infty}^{t} dt' \int d\mathbf{r}' \mu^{-1}(\mathbf{r} - \mathbf{r}', t - t') \mathbf{B}(\mathbf{r}', t').$$

The resulting equation for the dynamics of the phase shift $\varphi(x, t)$ in Josephson junction reads (see, for examples, Lomtev [69])

$$\left[\beta \frac{\partial}{\partial t} + \frac{\partial^2}{\partial x^2}\right] \varphi(x, t) - \frac{\lambda_J^2}{\lambda \omega_J^2} \int_{-\infty}^{\infty} dx' \int_{-\infty}^{\infty} dt' K\left(\frac{x - x'}{2\lambda_{\text{eff}}}, t - t'\right) \frac{\partial^2 \varphi(x', t')}{\partial x'^2}$$
$$= -\omega_J^2 \sin \varphi(x, t),$$

where β is a dissipation parameter, ω_J Josephson frequency, λ_J Josephson penetration depth, $\lambda_{\text{eff}} = \lambda^2/d$, and

$$K\left(\frac{x - x'}{2\lambda_{\text{eff}}}, t - t'\right) = \int_0^{\infty} \frac{dq}{\pi} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{2\lambda_{\text{eff}} J_0[q(x - x')] \exp[-i\omega(t - t')]}{\mu(\mathbf{q}, \omega) + 2q\lambda_{\text{eff}}}.$$

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Chapter 6
Quantum Mechanics

6.1 Atom Optics

6.1.1 Atoms in an Optical Lattice

The rapid progress in atom optics observed during last two decades stimulates development of new effective technologies (see Bardou et al. [6] and references therein). In the process of laser cooling distribution centered around zero velocity have been created by velocity selective population trapping. In Kondrashin et al. [27] these considerations are extended to engineer an atomic beam with a sharply defined nonzero velocity distribution that is an atomic pencil. This mechanism for momentum engineering heavily relies on anomalous kinetics of atoms represented by Lévy flights [7].

The behaviour of atoms with a general type Λ-configuration of internal levels in a field of two counterpropagating laser beams follows from the quantum equations for the atomic density matrix taking into account the recoil effect due to induce and spontaneous transitions. In the weak saturation regime, $g = dE/\hbar\Gamma \ll 1$ and provided $t \gg 1/\Gamma$ (Γ is the total width of the two levels, $\Gamma = \gamma_0 + \gamma$), the equations for the ground state density matrix elements $n(p,t) = \rho_{11}(p,t) + \rho_{22}(p,t)$ and $\mu(p,t) + i\nu(p,t)$ read

$$\begin{aligned}
\frac{\partial n}{\partial t} &= g^2\Gamma\left[-F(p,t) + \frac{\gamma}{2\Gamma}\int_{-1}^{1} F(p + 2\hbar k x, t)dx\right], \\
\frac{\partial \mu}{\partial t} &= -g^2\Gamma F(p,t) + \frac{2k(p-p_0)}{m}\nu(p,t), \\
\frac{\partial \nu}{\partial t} &= -g^2\Gamma G(p,t) - \frac{2k(p-p_0)}{m}\mu(p,t),
\end{aligned} \tag{6.1}$$

where the notation

$$\frac{n + \mu + i\nu}{1 + i(2k/m\Gamma)(p-p_0)} = F + iG$$

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is used. One should note (see for details [27]) that $(n+\mu)/2 \equiv n_{\rm C}$ and $(n-\mu)/2 \equiv n_{\rm NC}$ are the populations of the coupled state $|\psi_{\rm C}\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$ and the non-coupled state $|\psi_{\rm NC}\rangle = (|1\rangle - |2\rangle)/\sqrt{2}$, and μ is their difference. The trapping process in momentum space occurs in the vicinity of the dark state $|\psi_{\rm NC}(p_0)\rangle$ with the non-zero momentum p_0.

Integrating the above equation for $n(p,t)$ gives the distribution formula

$$n(p,t) = F(p,t) + g^2\Gamma \int_0^t K(p,t-t')F(p,t')dt',$$

relating it to F in a local and a non-local way in time. The latter dominates at large times where it can be approximated by a scaled δ-function

$$K(p,t) \cong \frac{\hbar k}{4\omega_{\rm R}}\sqrt{\frac{\pi g^2\Gamma}{t}}\delta(p-p_0),$$

where $\omega_{\rm R} = \hbar k^2/2m$ denotes the recoil frequency. Thus, we obtain

$$n(p,t) = F(p,t) + \delta(p-p_0)N_{\rm tr}(t), \tag{6.2}$$

where the total number of trapped atoms is given by expression

$$N_{\rm tr}(t) = A\int_0^t \frac{F(p_0,t')dt'}{\sqrt{\pi(t-t')}},$$

with the constant

$$A = \frac{\pi g^3\Gamma^{3/2}\hbar k}{4\omega_{\rm R}}.$$

On substituting Eq. (6.2) into Eq. (6.1) one arrives at the equation

$$\frac{\partial F(p,t)}{\partial t} + \delta(p-p_0)A\frac{\partial}{\partial t}\int_0^t \frac{F(p_0,t')dt'}{\sqrt{\pi(t-t')}}$$

$$= g^2\Gamma\left[-F(p,t) + \frac{\gamma}{2\Gamma}\int_{-1}^1 F(p+2\hbar kx,t)dx\right],$$

the left-hand side of which contains the semi-derivative with respect to time.

In order to solve the fractional kinetic equation and find the time evolution within the whole region $1/(g^2\Gamma) \ll t \ll \gamma/(g\omega_{\rm R})^2$, Kondrashin et al. [27] pass to the dimensionless variables $\tau = 2\gamma g^2 t/3, \quad u = p/\hbar k$, expand the integrand on the right-hand side of the equation to second order and perform the Laplace transformation in time. As a result, they obtain

$$\left(\frac{\partial^2}{\partial u^2} - \lambda - \lambda_0\right)\widehat{F}(u,\lambda) = -\delta(u) + \sigma_0\delta(u-u_0)\sqrt{\lambda}\widehat{F}(u_0,\lambda).$$

Here $\lambda_0 = 3\gamma_0/2\gamma$, $\sigma_0 = (\pi/4)\sqrt{3/2}(g^2\Gamma/\omega_{\mathrm{R}})$, and $u_0 = p_0/\hbar k$. Solution of such equation has been considered in Schaufler and Yakovlev [50]. Kondrashin et al. [27] uses the result relating to the special value $u = u_0$ assumed to be positive. It is given by the expression

$$\widehat{F}(u_0,\lambda) = \frac{\exp(-\sqrt{\lambda+\lambda_0}u_0)}{2\sqrt{\lambda+\lambda_0}} + \sigma_0\sqrt{\lambda},$$

and the total number of atoms obtained by inverse Laplace transformation decreases asymptotically as

$$N(t) \approx N_{\mathrm{tr}}(t) \sim \frac{\tau_0}{\pi t}\exp\left(-\frac{u_0}{\sqrt{\tau_0}}\right),$$

where $\tau_0 = 1/\lambda_0$. This means that the waiting time distribution is characterized by a long power tail, $\mathsf{P}(T > t) \propto t^{-1/2}$.

6.1.2 *Laser Cooling of Atoms*

Laser cooling and trapping of atoms is a new technological process for obtaining atom ensembles with temperatures in the range of nanokelvin, i.e., with velocities of order mm/s. This process enriched sufficiently wide field of nanophysical studies from spectroscopy of ultra-high resolution and atomic clock to atomic interferometry and Bose-Einstein condensation [6, 44]. Due to exchange of momentum with photons of laser radiation and due to an acquisition of momentum after spontaneous radiation, the process of atomic momentum evolution looks like random walk in momentum space. The idea of new cooling method consists in creation of a trap in a vicinity of zero ($\mathbf{p} = 0$) in the momentum space. Atoms can be trapped in it after random walk in the space and they can stay there for a long time. In mathematical sense, cooling characterizing in the momentum space by increase of concentration of atoms in a vicinity of $\mathbf{p} = 0$ and decrease of it out of the trap is conditioned by competition of two processes, momentum diffusion and going to zero of escape probability from the point $\mathbf{p} = 0$. History of each atom is characterized by consequence of mutually independent random variables $T_1, \Theta_1, T_2, \Theta_2, T_3, \Theta_3, \ldots$, where Θ_j is a random duration of j th trapping, T_j is a random duration of diffusion between j th and $j+1$ th trapping that is called *recirculation time*. Distribution and densities of these random variables we denote $q_T(t)$ and $q_\Theta(t)$.

If the frequency of jumps in a vicinity of $\mathbf{p} = 0$ obeys the following law

$$R(\mathbf{p}) \simeq \frac{1}{\tau_0}\left(\frac{p}{p_0}\right)^{\nu}, \quad p < p_0,$$

then

$$\mathsf{P}(\Theta_j > \theta) \sim A\theta^{-\alpha}, \quad \theta \to \infty,$$

where the exponent α is connected with ν and the dimension of momentum space D through the relation $\alpha = D/\nu$.

Concentration of cooled atoms $N_0(t)$ is proportional to the probability of detecting of atom in the origin $f_0(t)$. The last function is expressed with the use of the integral

$$f_0(t) = \int_0^t [F(t') - F'(t')]dt'$$

through the difference of transition rates $F(t)$ and $F'(t)$ satisfying the system of integral equations (at the condition, that the motion starts from release from a trap):

$$F(t) = q_T(t) + \int_0^t F'(t-t')q_T(t')dt',$$
$$F'(t) = \int_0^t F(t-t')\, q_\Theta(t')dt'.$$

Applying the Laplace transformation and solving the algebraic equation system, we obtain

$$\widehat{f_0}(\lambda) = \frac{\widehat{q_T}(\lambda)}{\lambda} \cdot \frac{1 - \widehat{q}_\Theta(\lambda)}{1 - \widehat{q_T}(\lambda)\widehat{q}_\Theta(\lambda)}.$$

As follows from physical argumentations

$$1 - \widehat{q}_\Theta(\lambda) \sim A\lambda^\alpha, \quad 1 - \widehat{q_T}(\lambda) \sim B\lambda^\beta,$$

where $0 < \alpha, \beta < 1$. In the domain of small values of λ the function $\widehat{f_0}(\lambda)$ satisfies the equation

$$A\lambda^\alpha \widehat{f_0}(\lambda) + B\lambda^\beta \widehat{f_0}(\lambda) = A\lambda^{\alpha-1}.$$

Applying inverse Laplace transformation, we obtain

$$[{}_0\mathsf{D}_t^\alpha + b_0 \mathsf{D}_t^\beta] f_0(t) = \frac{1}{\Gamma(1-\alpha)} t^{-\alpha}.$$

Solution obtained with the use of Green's function method has the form

$$f_0(t) = t^{\alpha-\beta} E_{\alpha-\beta,\alpha+1-\beta}(-bt^{\alpha-\beta}).$$

This solution is in accordance with the results of Monte Carlo simulation based on the integral equations [50].

6.1.3 Atomic Force Microscopy

Atomic force microscopy (*AFM*) is a microscopic technique imaging a surface topography by using attractive and repulsive interaction forces between a few atoms attached at a tip on a cantilever and a sample. It plays an important role in nano-scale science and technology. By using AFM, not only the topographic imaging, but also the force between the tip and the sample can be measured as a function of distance. In the case of attractive forces, there are three main contributions causing AFM: short-range chemical force, van der Waals force, and electrostatic force.

The AFM consists of a *cantilever* with a sharp tip at its end that is used to scan the specimen surface. When the tip is brought into proximity of a sample surface, forces between the tip and the sample lead to a deflection of the cantilever. The simplified equation of the cantilever motion can be written as

$$m\ddot{q} = -kq + F_{\text{ts}}, \tag{6.3}$$

where $F_{\text{ts}}(z)$ is the force between the tip and the sample and its solution is assumed to be harmonic and given by the Fourier series

$$q(t) = \sum_{n=1}^{\infty} a_n \cos(n\omega t). \tag{6.4}$$

By inserting Eq. (6.4) into Eq. (6.3), multiplying by $\cos(n'\omega t)dt$ and integrating over a period T, the coefficients a_n are computed and the first-order term in the frequency shift $\Delta f = f - f_0,\ f_0 = (1/2\pi)\sqrt{k/m}$, we get

$$\Delta f = -\frac{f_0}{kA}\int_0^T F_{\text{ts}}(z+q)\cos(\omega_0 t)dt,$$

where A is the amplitude in $q(t) \cong A\cos(\omega t)$:

$$\frac{\omega_0^2 - \omega^2}{\omega_0^2} \cong -2\frac{\Delta\omega}{\omega_0} = -2\frac{\Delta f}{f_0}.$$

In Eq. (6.4) all Fourier coefficients a_n with $n \neq 1$ are small compared the amplitude of the harmonic term. In $n = 1$ approximation, the frequency shift as a function of the minimum distance between the sample and the tip d (Fig. 6.1) is

$$\Delta f(d) = \frac{f_0}{\pi k a_1}\int_{-1}^{1} F_{\text{ts}}[d + a_1(1+u)]K(u)du,$$

where the kernel

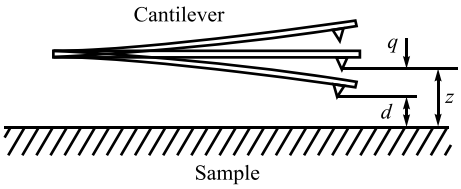


Fig. 6.1 Tip and sample in dynamic AFM

$$K(u) = \frac{u}{\sqrt{1-u^2}}$$

weights the tip-sample interacting force in the integration over a period. On approximating the latter by the leading divergent form in the vicinity of $u = -1$,

$$K(u) = \frac{u}{\sqrt{(1+u)(1-u)}} \simeq \frac{1}{\sqrt{2(1+u)}},$$

the effective force gradient $k_{\rm ts}^{\rm eff} \equiv 2k\Delta f/f_0$ can be represented as

$$k_{\rm ts}^{\rm eff}(d) = \frac{\sqrt{2}}{\pi A^{3/2}} \int_d^\infty \frac{F_{\rm ts}(s)ds}{\sqrt{s-d}} \equiv \frac{\sqrt{2/\pi}}{A^{3/2}}\, {}_d\mathsf{D}_\infty^{-1/2} F_{\rm ts}(d).$$

The inverse operation gives

$$F_{\rm ts}(s) = \frac{A^{3/2}}{\sqrt{2/\pi}}\, {}_s\mathsf{D}_\infty^{1/2} k_{\rm ts}^{\rm eff}(s) = -\frac{A^{3/2}}{\sqrt{2}} \int_s^\infty \frac{dk_{\rm ts}^{\rm eff}(s')/ds'}{\sqrt{s'-s}} ds'.$$

By using this equation, the tip-sample interacting force can be calculated from experimentally measured frequency shift data.

6.2 Quantum Particles

6.2.1 Kinetic-Fractional Schödinger equation

Intrusion of fractional calculus into quantum mechanics began with works of Laskin [29–32] by simple replacement of the exponent in the kinetic terms of Hamilton's operator:

$$\mathsf{K} = \frac{\mathsf{p}^2}{2m} = -\frac{\hbar^2\Delta}{2m} \quad \mapsto \quad \mathsf{K}_\alpha = K_\alpha(-\hbar^2\Delta)^{\alpha/2} \equiv -K_\alpha(\hbar\nabla)^\alpha, \quad 1 < \alpha \leqslant 2,$$

where K_α is a coefficient the physical dimension of which is $\mathrm{erg}^{1-\alpha}\cdot\mathrm{cm}^\alpha\cdot\mathrm{sec}^{-\alpha}$. As a result, the generalized (fractional) Schrödinger equation takes the form

$$i\hbar\frac{\partial\Psi(\mathbf{r},t)}{\partial t}=\left[K_\alpha(-\hbar^2\Delta)^{\alpha/2}+V(\mathbf{r},t)\right]\Psi(\mathbf{r},t). \tag{6.5}$$

The motivation of this action is based on the fractal concept applied to the Feynman path integrals over the measure generated by Brownian motion. The natural generalization of the Brownian motion is the Lévy motion. "If the path integral over Brownian trajectories leads to the Schrödinger equation, then the path integral over Lévy trajectories leads to the fractional Schrödinger equation. This is the main point for the term *fractional Quantum Mechanics*" [30].

The fractional Hamilton operator

$$\mathsf{H}_\alpha=\mathsf{K}_\alpha+V(\mathbf{r},t)$$

is an Hermitian operator in the space with scalar product

$$(\phi,\chi)=\int\phi^*(\mathbf{r},t)\chi(\mathbf{r},t)d\mathbf{r},$$

that is

$$(\phi,\mathsf{H}_\alpha\chi)=(\mathsf{H}_\alpha\phi,\chi),$$

and

$$E=(\Psi,\mathsf{H}_\alpha\Psi)$$

is a real value.

Laskin derived also the fractional Schrödinger equation from the Chapmen-Kolmogorov equation for the wave function as for "probability density" function

$$\Psi(\mathbf{r},t+\tau)=\int K(\mathbf{r},t+\tau|\mathbf{r}',t)\Psi(\mathbf{r}',t)d\mathbf{r}'$$

with the Feynman approximation of the kernel for $\tau\to 0$ leading to the infinitesimal relation

$$\Psi(\mathbf{r},t+\tau)\sim$$

$$\frac{1}{(2\pi\hbar)^3}\int d\mathbf{r}'\int d\mathbf{p}\,\exp\left\{(i/\hbar)[\mathbf{p}(\mathbf{r}'-\mathbf{r})-\epsilon K_\alpha|\mathbf{p}|^\alpha-\epsilon V((\mathbf{r}+\mathbf{r}')/2,t)]\right\}\Psi(\mathbf{r}',t).$$

Performing series expansion in the left-hand side, using the relation

$$\frac{1}{(2\pi\hbar)^3}\int d\mathbf{p}\,e^{(i/\hbar)\mathbf{pr}}|\mathbf{p}|^\alpha\int d\mathbf{r}'e^{-(i/\hbar)\mathbf{pr}'}\Psi(\mathbf{r}',t)=(-\hbar^2\Delta)^{\alpha/2}\Psi(\mathbf{r},t)$$

in the right-hand side of the equation, and taking the limit as $\tau\to 0$, he arrived at

the fractional Schrödinger equation (6.5)

$$i\hbar\frac{\partial\Psi(\mathbf{r},t)}{\partial t} = K_\alpha(-\hbar^2\Delta)^{\alpha/2}\Psi(\mathbf{r},t) + V(\mathbf{r},t)\Psi(\mathbf{r},t).$$

The solution of the fractional Schrödinger equation for a free particle moving along the x-axis with $p_x = p$

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = K_\alpha\left(-\hbar^2\frac{\partial^2}{\partial x^2}\right)^{\alpha/2}\Psi(x,t) + V(x,t)\Psi(x,t).$$

is

$$\Psi(x,t) = C\exp\left\{(i/\hbar)[px - K_\alpha|p|^\alpha t]\right\},$$

where C is a normalization constant. Localized states are described by superpositions of this plane waves:

$$\Psi(x,t) = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty}\exp\left\{(i/\hbar)[px - K_\alpha|p|^\alpha t]\right\}\varphi(p)dp.$$

Taking the weight function $\varphi(p)$ in the form

$$\varphi(p) = A_\gamma\exp\left\{-\frac{|p-p_0|^\gamma l^\gamma}{2\hbar^\gamma}\right\}, \quad p_0 > 0,$$

where $A_\gamma = \sqrt{\pi\gamma l/\Gamma(1/\gamma)}$, l is a space scale, $\alpha \in (1,2]$ and $\gamma < \alpha$, Laskin (2000) has found probability distribution functions for coordinate x and momentum p of the particle in the wave packet, their expectation values

$$\langle x(t)\rangle = \alpha K_\alpha p_0^\alpha t, \quad \langle p(t)\rangle = p_0,$$

and has proved the generalization Heisenberg uncertainty relation,

$$\langle|\Delta x|^\nu\rangle^{1/\nu}\langle|\Delta p|^\nu\rangle^{1/\nu} \geqslant \hbar(2\alpha)^{-1/\nu}, \quad \nu < \alpha, \quad 1 < \alpha \leqslant 2.$$

Acting in a usual way leads to the continuity equation

$$\frac{\partial\rho(\mathbf{r},t)}{\partial t} + \operatorname{div}\mathbf{j}(\mathbf{r},t) = 0,$$

where

$$\rho(\mathbf{r},t) = \Psi^*(\mathbf{r},t)\Psi(\mathbf{r},t)$$

is the probability density function and

$$\mathbf{j}(\mathbf{r},t) = -i\hbar\left[\Psi^*(\mathbf{r},t)(-\hbar^2\Delta)^{\alpha/2-1}\nabla\Psi(\mathbf{r},t) - \Psi(\mathbf{r},t)(-\hbar^2\Delta)^{\alpha/2-1}\nabla\Psi^*(\mathbf{r},t)\right] \tag{6.6}$$

is the term named "probability current density vector".[1]

The next standard problem considered in frame of the fractional approach relates to a particle in a one-dimensional square potential well with an infinite depth:

$$V(x) = \begin{cases} 0, & |x| \leqslant a; \\ \infty, & |x| > a. \end{cases}$$

The eigenfunctions and eigenvalues of the Hamiltonian have been found in the form

$$\phi_n(x) = \begin{cases} a^{-1/2}\cos[(n+1/2)\pi x/a], & \text{even states;} \\ a^{-1/2}\sin(n\pi x/a), & \text{odd states,} \end{cases}$$

$$E_n = K_\alpha\left(\frac{\pi\hbar}{a}\right)^\alpha n^\alpha,$$

where $n = 0, 1, 2, \ldots$. However, these results were subject to criticism in Jeng et al. [22] for the reason indicated in Sect. 5.2.20 of Vol. I.

The finite square well with potential

$$V(x) = \begin{cases} -V_0, & |x| < a; \\ 0, & |x| \geqslant a, \end{cases}$$

where V_0 is a positive constant, has been considered by Dong et al. [11]. The authors write that in the region $|x| < a$ the fractional equation reads

$$K_\alpha\left(-\hbar^2\frac{\partial^2}{\partial x^2}\right)^{\alpha/2}\phi(x) = (E+V_0)\phi(x),$$

and its solutions are $\cos(k_1 x)$ and $\sin(k_1 x)$ with $k_1 = [(E+V_0)/K_\alpha]^{1/\alpha}/\hbar$. Meanwhile, in the outer region, where the equation reads

$$K_\alpha\left(-\hbar^2\frac{\partial^2}{\partial x^2}\right)^{\alpha/2}\phi(x) = E\phi(x),$$

and its solutions are claimed as

[1] The physical sense of the word "current" connected to the image of continuously flowing fluid, does not correspond to the case under consideration (if only α is not equal to 2), because the fractional Laplacian is nonlocal and the result (6.6) obtained with its application can not be interpreted as a flow.

$$\phi_1(x) = \cos(k_2 x \cos(\pi/\alpha))e^{-k_2|x|\sin(\pi/\alpha)}$$

and

$$\phi_2(x) = \sin(k_2 x \cos(\pi/\alpha))e^{-k_2|x|\sin(\pi/\alpha)}$$

with $k_2 = (-E/K_\alpha)^{1/\alpha}/\hbar$ (the energy E is assumed to be negative, $-V_0 < E < 0$). Then, the solutions for different regions are joined at boundary points by continuity. However, this procedure is valid only if α is integer. If it is not the case, the fractional Laplacian is nonlocal,

$$\left(-\hbar^2 \frac{\partial^2}{\partial x^2}\right)^{\alpha/2} = (-\hbar^2\Delta)^{\alpha/2} \equiv {}_{-\infty}[(-\hbar^2\Delta)^{\alpha/2}]_{\infty}$$

and independent solving the equation in different regions is impossible.

The next problem considered by Laskin two years later relates to the Bohr atom and can be treated as a fractional eigenvalue problem:

$$K_\alpha(-\hbar^2\Delta)^{\alpha/2}\phi(\mathbf{r}) - \frac{Ze^2}{|\mathbf{r}|}\phi(\mathbf{r}) = E\phi(\mathbf{r}).$$

The energy levels of the fractional hydrogen-like atom are obtained as[2]

$$E_n = -\frac{(\alpha-1)E_0}{n^{\alpha/(\alpha-1)}}$$

with

$$E_0 = \left(\frac{(Ze^2)^\alpha}{\alpha^2 K_\alpha \hbar^\alpha}\right)^{1/(\alpha-1)}.$$

The frequency of radiation associated with the transition $k \to n$ is

$$\omega_{k\to n} = \frac{(\alpha-1)E_0}{\hbar}\left[\frac{1}{n^{\alpha/(\alpha-1)}} - \frac{1}{k^{\alpha(\alpha-1)}}\right].$$

Observe that when $\alpha \to 2$, all these results convert to the classic Bohr model.

Further, Laskin considers the one-dimensional oscillator with the Hamiltonian

$$\mathsf{H}_{\alpha,\beta} = K_\alpha\left(-\hbar^2\frac{\partial^2}{\partial x^2}\right)^{\alpha/2} + b^2|x|^\beta$$

where b is a constant with physical dimension $[b] = \mathrm{erg}^{1/2}\cdot\mathrm{cm}^{-\beta/2}$. A routine use of the Bohr-Sommerfeld quantization rule yields

[2] Recall $\alpha > 1$.

$$\oint p dx = \frac{4}{K_\alpha^{1/\alpha}} \int_0^{x_m} \left(E - b^2|x|^\beta\right)^{1/\alpha} dx = 2\pi\hbar(n + 1/2),$$

which gives the following spectrum

$$E_n = \left(\frac{\pi\hbar\beta K_\alpha^{1/\alpha} b^{2/\beta}}{2\mathrm{B}(1/\beta, 1/\alpha + 1)}\right)^{\alpha\beta/(\alpha+\beta)} \left(n + \frac{1}{2}\right)^{\alpha\beta/(\alpha+\beta)}.$$

6.2.2 Potential-Fractional Schrödinger Equation

When a particle is placed in an infinitely deep one-dimensional box of width $2a$,

$$V(x) = \begin{cases} 0, & -a < x < a; \\ \infty, & \text{otherwise.} \end{cases}$$

Inside the box, the Schrödinger equation

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2}$$

has the solution

$$\Psi(x,t) = e^{-i(E_n/\hbar)t}\psi_n(x), \quad \psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad E_n = \frac{(\pi\hbar n)^2}{2ma^2}.$$

Ajlouni et al. [2] have considered the problem in case when the box contains a dissipative medium described by a generalized potential

$$U(x) = \frac{i\gamma}{2}\left[x^{(1/2)}\right]^2,$$

so the Lagrangian has the form

$$L = \frac{1}{2}m\dot{x}^2 - \frac{i\gamma}{2}\left[x^{(1/2)}\right]^2$$

and obeys the Euler-Lagrange equation

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = (-1)^{1/2}\, {}_0\mathsf{D}_t^{1/2}\left(\frac{\partial L}{\partial x^{(1/2)}}\right) + \frac{\partial L}{\partial x}.$$

Passing to the Hamilton operator

$$\mathsf{H} = \frac{[\mathsf{p}^{(1/2)}]^2}{2m} + x^{(1/2)}\mathsf{p} + \frac{\gamma}{2i}[x^{(1/2)}]^2$$

and taking into account the relation

$$\mathsf{p}^{(\alpha)} = \frac{h}{i}\frac{\partial}{\partial x^{(\alpha)}},$$

we arrive at the fractional Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial[x^{(1/2)}]^2} + \frac{h}{i}x^{(1/2)}\frac{\partial}{\partial x} + \frac{\gamma}{2i}[x^{(1/2)}]^2\right]\Psi(x,t). \tag{6.7}$$

Using the method of separation of variables, Ailouni et al. [2] represent the solution as

$$\Psi = F(x^{(1/2)}, x)T(t)$$

and find the following equations for the multipliers: the time-dependent equation

$$i\hbar\frac{dT}{dt} = E_0 T$$

with the solution

$$T(t) = e^{-i(E_0/\hbar)t},$$

and the other time-independent equation

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2} + \frac{\gamma}{2i}y^2 + \frac{\hbar}{i}y\frac{\partial}{\partial x}\right]F(x,y) = E_0 F(x,y),$$

where y stands for $x^{(1/2)}$. Substituting $x = uy$ into this equation and separating variables y and u yield

$$F(uy, y) = U(u)Y(y),$$

$$-i\hbar\frac{d}{du}U(u) = E_x U(u),$$

$$\left[\frac{d^2}{dx^2} - \left(\frac{m\gamma}{i\hbar^2}\right)y^2\right]Y(y) = -2E_y\left(\frac{m}{\hbar^2}\right)Y(y).$$

Under the boundary conditions $\Psi(-a,t) = \Psi(a,t) = 0$, the first of the equations has the solution

$$U(u) = U_0\cos[(E_x/\hbar)u],$$

meanwhile the solution of the second of them reads as

$$Y_n(y) = Y_0 H_n\left[\left(\frac{m\gamma}{i\hbar^2}\right)^{1/4} y\right] \exp\left[-\left(\frac{m\gamma}{i\hbar^2}\right)^{1/2} \frac{y^2}{2}\right]$$

where $H_n(z)$ are Hermit polynomials (see Arfken [5]). Finally, we get the family of wave functions

$$\Psi_n = AH_n\left[\left(\frac{m\gamma}{i\hbar^2}\right)^{1/4} x^{(1/2)}\right] \exp\left[-\left(\frac{m\gamma}{i\hbar^2}\right)^{1/2} \frac{[x^{(1/2)}]^2}{2}\right] \cos\left(\frac{E_n x}{\hbar x^{1/2}}\right) e^{-i(E_0/\hbar)t} \tag{6.8}$$

with

$$E_n = \frac{n\pi\hbar}{2a} + \frac{(2n+1)\hbar}{2}\left(\frac{i\gamma}{m}\right), \quad n = 1, 2, \ldots.$$

6.2.3 Time-Fractional Schrödinger Equation

To avoid the problems with physical dimensions when embedding the fractional time derivative, we will express lengths, time, masses, and energies in the Planck units, so the variables

$$\xi = x/\sqrt{G\hbar/c^3}, \quad \tau = t/\sqrt{G\hbar/c^3}, \quad \mu = m/\sqrt{\hbar c/G}, \quad \epsilon = E/\sqrt{\hbar c^5/G},$$

and

$$U(\xi, \tau) = V(x, t)/\sqrt{\hbar c^5/G}.$$

The Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V(x, t)\Psi(x, t)$$

becomes

$$i\frac{\partial\psi}{\partial\tau} = -\frac{1}{2\mu}\frac{\partial^2\psi}{\partial\xi^2} + U(\xi, \tau)\psi(\xi, \tau).$$

The time-fractional counterpart of this equation introduced by Naber [40] looks as

$$i^\nu\, {}_0\mathsf{D}_\tau^\nu\psi = -\frac{1}{2\mu}\frac{\partial^2\psi}{\partial\xi^2} + U(\xi, \tau)\psi(\xi, \tau), \quad \nu \in (0, 1]. \tag{6.9}$$

Explaining why he has preferred it to another possible form,

$$i\,{}_0^{\nu}\mathsf{D}_{\tau}\psi = -\frac{1}{2\mu}\frac{\partial^2\psi}{\partial\xi^2} + U(\xi,\tau)\psi(\xi,\tau), \tag{6.10}$$

Naber submits two reasons. The first reason: when performing a Wick rotation the imaginary unit is raised to the same power as the time coordinate. The second involves the temporal behavior of the solution. "For Eq. (6.9), changing the order of the derivative moves the pole (from the inverse Laplace transform) up or down the negative imaginary axis. Hence, the temporal behavior of the solution will not change. For Eq. (6.10), changing the order of the derivative moves the pole to almost any desired location in the complex plane. Physically, this would mean that a small change in the order of the time derivative, in Eq. (6.10) could change the temporal behavior from sinusoidal to growth or to decay. Due to the simpler physical behavior of Eq. (6.9) and the role of 'i' in a Wick rotation, Eq. (6.9) is the best candidate for a time-fractional Schrödinger equation" [40].

By using the known identity

$${}_0^{1-\nu}\mathsf{D}_{\tau}\,{}_0^{\nu}\mathsf{D}_{\tau}\psi(\tau) = \frac{d\psi}{d\tau} - \frac{[{}_0^{\nu}\mathsf{D}_{\tau}\psi(\tau)]_{\tau=0}}{\tau^{1-\nu}\Gamma(\nu)},$$

the fractional equation can be represented in an ordinary evolution form

$$\frac{\partial\psi}{\partial t} = -\frac{1}{i^{\nu}}\left[\frac{\partial^2({}_0^{1-\nu}\mathsf{D}_{\tau}\psi)}{2\mu\partial\xi^2} - {}_0^{1-\nu}\mathsf{D}_{\tau}(\mathrm{V}\psi)\right] + \frac{[{}_0^{\nu}\mathsf{D}_{\tau}\psi(\xi,\tau)]_{\tau=0}}{\tau^{1-\nu}\Gamma(\nu)}$$

with a time-dependent Hamiltonian. Thus, one can not expect probability to be conserved. Moreover, the nonlocal Hamiltonian character means that the solution is not invariant under time reversal. The third term vanishes as $\tau \to \infty$.

For a free particle, the probability current equation can be constructed just as for the ordinary integer-order Schrödinger equation,

$$\begin{aligned}\frac{\partial(\psi^*\psi)}{\partial\tau} &= \psi^*\left(-\frac{1}{2\mu i^{\nu}}\frac{\partial^2\grave{\psi}}{\partial\xi^2} + \frac{[{}_0^{\nu}\mathsf{D}_{\tau}\psi]_{\tau=0}}{\tau^{1-\nu}\Gamma(\nu)}\right)\\ &\quad + \left(-\frac{1}{2\mu(-i)^{\nu}}\frac{\partial^2\grave{\psi}^*}{\partial\xi^2} + \frac{[{}_0^{\nu}\mathsf{D}_{\tau}\psi^*]_{\tau=0}}{\tau^{1-\nu}\Gamma(\nu)}\right)\psi,\end{aligned}$$

where $\grave{\psi} = {}_0^{1-\nu}\mathsf{D}_{\tau}\psi$. Simple transformations lead to the balance equation

$$\frac{\partial P}{\partial\tau} + \frac{\partial J_{\xi}}{\partial\xi} = S(\xi,\tau),$$

where

$$P = P(\xi,\tau)$$

is the probability density function,

$$J_\xi = \frac{1}{2\mu}\left(\frac{\partial\psi^*\overset{\backprime}{\psi}}{i^\nu\partial\xi} + \frac{\partial\overset{\backprime}{\psi}^*\psi}{(-i)^\nu\partial\xi}\right)$$

is the probability current projection, and

$$S(\xi,\tau) = \frac{1}{2\mu}\left(\frac{\partial\psi^*}{i^\nu\partial\xi}\frac{\partial\overset{\backprime}{\psi}}{\partial\xi} + \frac{\partial\overset{\backprime}{\psi}^*}{(-i)^\nu\partial\xi}\frac{\partial\psi}{\partial\xi}\right) + \frac{\psi^*[{}_0^\nu\mathsf{D}_\tau\psi]_{\tau=0} + \psi[{}_0^\nu\mathsf{D}_\tau\psi^*]_{\tau=0}}{\tau^{1-\nu}\Gamma(\nu)}$$

is the source-sink term. Observe that $\nu \to 1$ leads to ordinary quantum mechanics relations.

Applying the Fourier transform

$$\psi(\xi,\tau) \mapsto \widetilde{\psi}(k,\tau) = \int_{-\infty}^{\infty} e^{ik\xi}\psi(\xi,\tau)d\xi$$

to the partial fractional equation for a free particle

$$i^\nu\,{}_0^\nu\mathsf{D}_\tau\psi(\xi,\tau) = -\frac{1}{2\mu}\frac{\partial^2\psi(\xi,\tau)}{\partial\xi^2}$$

yields the ordinary fractional equation

$${}_0^\nu\mathsf{D}_\tau\widetilde{\psi}(k,\tau) = \frac{k^2}{2\mu}\widetilde{\psi}(k,\tau),$$

which has the solution

$$\widetilde{\psi}(k,\tau) = AE_\nu[\omega(-it)^\nu], \quad \omega = k^2/2\mu.$$

Inverse Fourier transforming gives $\psi(\xi,\tau)$ as a sum of two terms: a Schrödinger like

$$\psi_{\mathrm{S}}(\xi,\tau) = \frac{1}{2\pi\nu}\int_{-\infty}^{\infty} e^{-ik\xi}Ae^{-i\omega^{1/\nu}t}dk$$

and a decay term $\psi_{\mathrm{D}}(\xi,t)$ vanishing as time goes to infinity.

In case of a particle in a potential well

$$U(\xi) = \begin{cases} 0, & 0 < \xi < L; \\ \infty, & \text{elsewhere,} \end{cases}$$

the solution is sought by the separation variables method which gives $\psi(\xi,\tau)$ again as a sum of two terms, nonvanishing of them is represented as

$$\psi_n(\xi,\tau)=\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi\xi}{L}\right)E_\nu(\omega_n(-it)^\nu)\sim\frac{1}{\nu}\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi\xi}{L}\right)e^{-i\omega^{1/\nu}\tau},\ \tau\to\infty,$$

$$n=1,2,3,\ldots,$$

where

$$\omega_n=\frac{1}{2\mu}\left(\frac{n\pi}{L}\right)^2.$$

Since Hamiltonian depends on time, so do energy levels. In $\tau\to\infty$ limit they become

$$\epsilon_n(\infty)=\frac{(n\pi)^{2/\nu}}{\nu^2(2\mu L^2)^{1/\nu}}.$$

It is easy to see, that for $\nu=1$ we arrive at classical results of quantum mechanics (recall, that in our unites $\hbar=1$).

6.2.4 *Fractional Heisenberg Equation*

Tarasov [58] considers fractional generalization of the Heisenberg equation for a Hamiltonian system

$$\frac{d\mathsf{A}_t}{dt}=-\mathsf{L}\mathsf{A}_t\equiv-\frac{1}{i\hbar}(\mathsf{H}\mathsf{A}_t-\mathsf{A}_t\mathsf{H})$$

having the form

$$\frac{d\mathsf{A}_t}{dt}=-\mathsf{L}^\alpha\mathsf{A}_t.$$

The solution of the equation can be presented as

$$\mathsf{A}_t(\alpha)=\mathsf{G}_t^\alpha\mathsf{A}_0,$$

where the superoperator G_t^α according to the Bochner-Phillips formula can be expressed through the subordinator

$$f_\alpha(t,s)=t^{-1/\alpha}g_+(st^{-1/\alpha}).$$

In particular, if $\alpha=1/2$,

$$f_{1/2}(t,s)=\frac{t}{2s\sqrt{\pi s}}e^{-t^2/4s}.$$

For a free particle, $\mathsf{H}=p^2/2m$. Taking firstly $\mathsf{A}_t=\mathsf{x}_t$ and then $\mathsf{A}_t=\mathsf{p}_t$ yields

$$\frac{d\mathsf{x}_t}{dt} = -\frac{1}{m^\alpha}\mathsf{L}^\alpha \mathsf{x}_t$$

and

$$\frac{d\mathsf{p}_t}{dt} = 0$$

respectively. The solutions of them are

$$\mathsf{x}_t = \mathsf{x}_0 + \frac{1}{m}F_\alpha(t)\mathsf{p}_0$$

and

$$\mathsf{p}_t = \mathsf{p}_0$$

where

$$F_\alpha(t) = \int_0^\infty f_\alpha(t, s)sds.$$

In integer-order quantum mechanics, $f_1(t, s) = t^{-1}\delta(s/t - 1)$, and therefore

$$\mathsf{x}_t = \mathsf{x}_0 + \frac{1}{m}\mathsf{p}_0 t,$$

but the $1/2$-order mechanics gives

$$\mathsf{x}_t = \mathsf{x}_0 + \frac{1}{2m}\mathsf{p}_0 t^2.$$

Tarasov has considered the harmonic oscillator as well, described by the Hamiltonian

$$\mathsf{H}(\mathsf{x}, \mathsf{p}) = \frac{1}{2m}\mathsf{p}^2 + \frac{m\omega^2}{2}\mathsf{q}^2,$$

where t and p are dimensionless variables. The solutions of fractional equations of motion have been found in the forms

$$\mathsf{q}_t = C_\alpha(t)\mathsf{q}_0 + \frac{1}{m\omega}S_\alpha(t)\mathsf{p}_0,$$

$$\mathsf{p}_t = C_\alpha(t)\mathsf{p}_0 - m\omega S_\alpha(t)\mathsf{q}_0,$$

where

$$C_\alpha(t) = \int_0^\infty f_\alpha(t, s)\cos(\omega s)ds,$$

and

$$S_\alpha(t) = \int_0^\infty f_\alpha(t, s) \sin(\omega s) ds.$$

The solutions are characterized by the fractional damping of oscillations and powerlike tails.

6.2.5 The Fine Structure Constant

It is known that the asymptotic limit of quantum field dynamics, dominated by large and persistent perturbations, may be described as an anomalous diffusion process. Goldfain [17] uses fractional calculus as an appropriate tool to handle this highly non-trivial regime and shows that the fine structure constant can be recovered from the fractional evolution equation of the density matrix under standard normalization conditions.

To properly describe the short-distance behavior of the quantum electrodynamics, a vacuum polarization correction known as the Uehling term needs to be added to the classical Coulomb potential. In non-dimensional form, the result reads

$$H_d^0 = -\frac{e_0^2}{4\pi r}[(1 - b \ln r^0) - O(e_0^4)]$$

where

$$b = \frac{e_0^2}{6\pi^2}$$

and e_0^2 represents the electric charge at the reference momentum scale. Thus, the corresponding density matrix obeys the equation (in dimensionless variables)

$$\frac{\partial \rho(r, t)}{\partial t} = \left[\frac{e_0^2}{4\pi r}(1 - b \ln r)\right] \rho(r, t).$$

Goldfain [17] suggests the fractional generalization

$${}_0\mathsf{D}_t^\nu \rho(r, t) = \left[\frac{e_0^2}{4\pi r}(1 - b \ln r)\right] \rho(r, t),$$

whose solution is

$$\rho(r, t) = C E_{\nu,1}\left[\frac{e_0^2}{4\pi r}(1 - b \ln r) t^\nu\right].$$

For finite time intervals satisfying the constraint

$$t \ll \frac{4\pi r}{e_0^2(1 - b \ln r)},$$

the solution may be presented in a linearized form, that is

$$\rho(r,t) = C\left[1 + \frac{e_0^2}{4\pi r}\frac{(1 - b\ln r)}{\Gamma(1+\nu)}t^{\nu}\right].$$

Let k be the sliding momentum scale, k_{ref} the reference momentum scale and $e_0^2 = e^2(k_{\text{ref}})$. Denoting by r_c the distance cutoff assigned to the high energy region, and taking into account that in this region $t \sim r_c$, Goldfain formulates the main result of his paper in the form of equality

$$(1 - r_c) - \frac{r_c^{\nu}\alpha}{\Gamma(1+\nu)}\ln r_c\left[1 - \frac{b\ln r_c}{2}\right] = 1.$$

Based on universality of the approach to criticality in statistical physics the author takes for ν the critical exponent 0.626 found for the traditional Ising model [24]. Making reference to studies performed at TEVATRON on Drell-Yan processes, Goldfain uses the value

$$r_c = \frac{5.69 \times 10^{-19}}{10^{-15}} = 5.69 \times 10^{-4}$$

for finding the fine structure constant

$$\alpha = \frac{3\pi}{2\ln r_c}\left[1 - \sqrt{1 + \frac{4r_c^{1-\nu}\Gamma(1+\nu)}{3\pi}}\right] = \frac{1}{137},$$

which turns out in close agreement with the experimental value [24]

$$\alpha_{\text{exp}} = \frac{1}{137.036}.$$

6.3 Fractons

6.3.1 Localized Vibrational States (fractons)

The inelastic scattering of extended-electronic states off running-wave (*phonon*) states has been investigated long ago [71], but the analogous process for *localized* (in Anderson's definition) vibrational states attracted the researchers' attention only in the early eighties and then this phenomena is intensively investigated (see Ben Avraham and Havlin [10] and references therein). Examples of such vibrations are collective sinusoidal oscillations of clots of turbulent magnetic field (magnetic domains) relatively to a common equilibrium position. Localization can occur by

virtue of the geometrical connectivity properties of the atomic network, in which case one refers to the vibrational excitations as *fractons*. Fractons play an important role in superconductivity mechanisms such as Cooper pairing in organic polymers and ceramic compounds, and also in other physical systems and processes [70].

Fractons are characterized by a fracton dimension $d_f < 2$. Only the vibrations of random fractal networks were considered for which localization of the vibrational states occurred if the fracton dimensionality, d_f, was less than 2. In this model, the density of states is governed by d_f, and the vibrational localization length scales with a negative power of frequency. One expects $1 < d_f < 2$ for a mechanical model where scalar elasticity is dominant. In the case of a purely mechanical rotationally invariant model, d_f can be less than unity, implying a weak divergence of the vibrational density of states with decreasing frequency. Impurities in an otherwise translationally invariant atomic network can also result in vibrational localization. One refers to these excitations as localized phonons.

Throughout [13], the results are expressed in terms which are equally applicable to a vibrational structure with only "scalar" forces between vibrational atoms and for the structures for which purely mechanical central and "bending" forces are appropriate. Both assumptions lead to identical scalable forms for the dispersion law and density of vibrational states, with only the constant θ, involved in the range dependence of the force constant, changing its value. In particular, the range dependence of the scalar force constant, K_s, depends on θ as

$$K_s \propto r^{-D-2-\theta},$$

where D is the fractal dimension. The force constant, $K_{c,b}$, for central and bending forces varies as

$$K_{c,b} \propto r^{-\tilde{\zeta}_E}.$$

Hence, if one identifies $\theta_{c,b} = \tilde{\zeta}_E + D - 2$, all the results for scalar forces go over directly to those one would have found for central and bending forces.

In particular, the spatial Fourier transform of the overlap of two wave functions $\phi_a(\mathbf{r})$ of the same mode index a was evaluated:

$$I_a(\mathbf{k}) = \int d\mathbf{r} e^{i\mathbf{kr}} I_a(\mathbf{r}), \tag{6.11}$$

$$I_a(\mathbf{r}) = \int d\mathbf{r}' \phi_a(\mathbf{r} + \mathbf{r}') \phi_a^*(\mathbf{r}'). \tag{6.12}$$

In order to evaluate $I_a(\mathbf{k})$ within the fracton regime, they write the fracton wavefunction assuming simple exponential localization:

$$\phi_a(\mathbf{r}) = A_a \exp\left\{-(1/2)(r/\lambda_a)^{d_\phi}\right\}, \tag{6.13}$$

where A_a is the normalization coefficient, and d_ϕ characterizes the localization of the vibrational wave function in a fractal network. From Eqs. (6.11) and (6.12), it follows that

$$I_a(\mathbf{k}) = |\phi_a(\mathbf{k})|^2,$$

where $\phi_a(\mathbf{k})$ is the Fourier transform of the wave function in Eq. (6.13):

$$\phi_a(\mathbf{k}) = 2A_a\pi^{D/2}(2/k)^{D/2-1}\int_0^\infty \exp\left\{-(1/2)(r/\lambda_a)^{d_\phi}\right\} J_{D/2-1}(qr)r^{D/2}dr. \tag{6.14}$$

Equations (6.13) and (6.14) labelled in Entin-Wohlman et al. [13] by (6.11) and (6.13), respectively, are nothing but the characteristic function and the distribution density of multivariate spherically symmetric stable law expanded to the fractional dimension D.

In the mechanical model with the predominance of scalar elasticity $1 < d_f < 2$, in the isotropic that is invariant with respect to rotation model $d_f < 1$. The frequency dependence of density of fracton states has a power type [1]

$$\rho(\omega) \propto \omega^{d_s-1},$$

where $d_s = d_f/\sigma$ is a spectral dimension of the ensemble, and the constant σ is determined by a connectedness index θ of the fractal structure [38]:

$$\sigma = 1 + \theta/2.$$

This index equals 0 for any "smooth" sets and is positive for fractal sets containing internal cavities and voids. For fractals embedded in the two-dimensional space, $0.75 \lesssim \sigma \lesssim 1.42$, meanwhile for fractals in the three-dimensional space, $0.75 \lesssim \sigma \lesssim 2.23$.

Fractal excitations are characterized by a non-linear dispersion relation of a power kind [41]

$$\omega = \Lambda|k|^\sigma = \Lambda(k^2)^{\sigma/2}. \tag{6.15}$$

A more detailed presentation of these ideas can be found in the remarkable book [10] and the review [70].

6.3.2 Weak Fracton Excitations

Following standard recipes in quantum mechanics, let us multiply Eq. (6.15) by the wave function $\psi(x,t)$ and perform the substitutions

$$\omega \mapsto i\partial/\partial t, \quad k \mapsto -i\partial/\partial x, \quad k^2 \mapsto -\partial^2/\partial x^2$$

(observe that we use the units in which $\hbar = 1$). As a result, we obtain the linear Shrödinger equation of fractional order

$$i\frac{\partial \Psi}{\partial t} = -i^{-\sigma}\Lambda\left(-\frac{\partial^2}{\partial x^2}\right)^{\sigma/2}\Psi(x,t). \tag{6.16}$$

The spatial multiplier $\psi(x)$ of the wave function

$$\Psi(x,t) = \psi(x)e^{i\omega t}$$

satisfies the fractional analogue of the steady-state Shrödinger equation

$$-(-\Delta)^{\sigma/2}\psi(x) = \lambda\psi(x), \quad \lambda \equiv i^{\sigma}\omega/\Lambda.$$

Zeleny and Milovanov [70] approximated its solution by

$$\psi(x) \sim \exp\left(-\frac{\omega(i|x|)^{\sigma}}{\Lambda\Gamma(1+\sigma)}\right)$$

in the central part ($\omega|x|^{\sigma} \lesssim \Lambda$) and by

$$\psi(x) \sim \left(\frac{\Lambda}{\omega\Gamma(-\sigma)}\right)^{1/2} x^{-(\sigma+1)/2}, \quad x \to \infty$$

in the asymptotical region. The slow decay of the fracton wave function at $x \to \infty$ points to a larger probability of finding fracton beyond the central part.

6.3.3 Non-linear Fractional Shrödinger Equation

The linear fractional differential equation (6.16) derived from the relation (6.15) is valid only when the dependence of frequency upon ω on the fracton wave intensity can be ignored.

With increasing the wave amplitude $|\Psi(x,t)|$, one has to involve corresponding corrections. In the first approximation, Eq. (6.15) is replaced by the relation

$$\omega = \Lambda|k|^{\sigma} - \zeta|\Psi(x,t)|^2, \quad \zeta > 0,$$

pointing that the oscillation frequency decreases with increasing the amplitude, and the fractional equation becomes non-linear:

$$i\frac{\partial \Psi(x,t)}{\partial t} = -i^{-\sigma}\Lambda\left(-\frac{\partial^2}{\partial x^2}\right)^{\sigma/2}\Psi(x,t) - \zeta|\Psi(x,t)|^2\Psi(x,t). \tag{6.17}$$

According to the interpretation given Zeleny and Milovanov [70], this equation reflects the phenomenon of *self-compression* (*self-focusing*) of fracton excitation, when oscillations in the center (the "kernel") become stronger, but in the tails weaker.

In competition with this process, another process, diffraction causes "kernel spreading". Involving anomalous turbulent diffusion into fracton excitation dynamics supplements the Eq. (6.17) with another fractional Laplacian responsible for this process:

$$i\frac{\partial\Psi(x,t)}{\partial t} = -i^{-\sigma}\Lambda\left(-\frac{\partial^2}{\partial x^2}\right)^{\sigma/2}\Psi(x,t) \\ -K_\beta\left(-\frac{\partial^2}{\partial x^2}\right)^{\beta}\Psi(x,t) - \zeta|\Psi(x,t)|^2\Psi(x,t), \quad 0 < \beta \leqslant 1.$$

6.3.4 *Fractional Ginzburg-Landau Equation*

Basing on the developed propositions of fracton dynamics, Zeleny and Milovanov [70] predict an interesting phenomenon named *fracton self-delocalization*. They explain the essence of the phenomenon as follows. "The non-linear dependence of fracton frequency on amplitude produces a difference between phases of oscillations in the kernel and those in the tail of excitation. With the course of time, initial phase heterogeneities grow up to a certain time and then break up into oscillators with almost independent phases. The phase mixing has an effect on a mutual arrangement of structure elements of the fractal participating in oscillations. This process violates a relation between nearest neighbors which can become distant and their places can be occupied by some other fractons. Because of strong mixing, the index θ vanishes, the conditions requiring localization of excitations disappear and it (the excitation, in other words, the fracton) wins a long-expected freedom. The wave makes a kind of corridor with the Euclidian connectivity along which it freely runs through the medium. This phenomenon can be considered as a *strange soliton* for which the fractal distribution is transparent."

For wave fields that are stable relative to self-compression, Zeleny and Milovanov [70] derive the equation

$$K_\beta\left(-\frac{\partial^2}{\partial x^2}\right)^{\beta}\psi(x) + \omega\psi(x) + \zeta|\psi(x)|^2\psi(x) = 0, \tag{6.18}$$

which is a fractional analogue of the Ginzburg-Landau equation,

$$-K\left(\frac{\partial^2}{\partial x^2}\right)\psi(x) + \omega\psi(x) + \zeta|\psi(x)|^2\psi(x) = 0.$$

The latter associates the fractional character of operators in Eq. (6.18) with coherent forming a non-linear wave field—for example, a condensate wave function below the point of transition into the superconducting state—under influence of long-range correlations introducing an essential non-locality into the behavior of the system as a whole.

6.4 Quantum Dots

6.4.1 Fluorescence of Nanocrystals

Such chromophores as colloidal semiconductor nano-crystals, also known as *quantum dots* (QDs), have attracted the attention of researchers as having a number of advantages in comparison to organic dye molecules. The higher photostability, wide band-edge absorption, and narrow emission tunable across the complete visible spectrum make them very attractive. However, their performance is limited by a severe intermittency in emission. The main of them is intermittency: single QDs emit measurable fluorescence during limited time intervals (t_{on}) separated by dark periods (t_{off}) when no photons except experimental background are detected. This blinking character of fluorescence limits the brightness of QDs and their applicability in spectroscopy of single-biological molecules [25, 65] and quantum information processing using single-photon sources [35]. The mechanism of QDs blinking is not quite understood yet and represents a question of relevance not only from a fundamental point of view of confined semiconductor physics but also from a technological standpoint because it might allow us to develop strategies to obtain QDs with the highest performance. All this motivates theoretical investigations and stipulates for developing numerical modeling of the process with the use of experimental data. We consider below one of such models.

6.4.2 Binary Model

As it is noted in 2.6.3 of Vol. I, experiments show that the on-time $T_1 \equiv T_{\text{on}}$ and the off-time $T_0 \equiv T_{\text{off}}$ are random variables which in the first approximation can be considered as mutually independent and distributed according to power laws

$$\mathsf{P}(T_1 > t) \equiv \Psi_1(t) \propto t^{-\alpha}, \quad \mathsf{P}(T_0 > t) \equiv \Psi_0(t) \propto t^{-\beta}, \quad 0 < \alpha, \beta < 1.$$

From probabilistic point of view, blinking fluorescence can be considered as a random process of instantaneous transitions of the system between two states at random time moments. An example of such trajectory is presented in Fig. 6.2a. Within the framework of this model, the correlation function and the Mandel parameter were

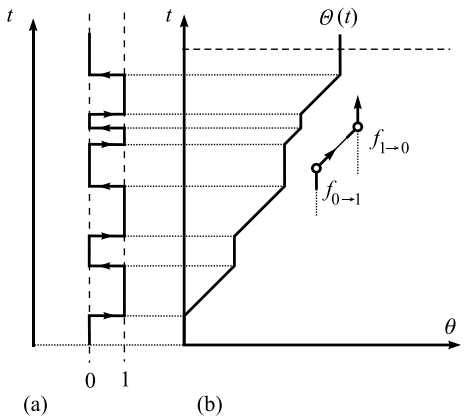


Fig. 6.2 Stochastic model of blinking fluorescence. **a** Random trajectory and **b** its representation in the coordinates (θ, t)

found [23, 36]. In reality, the situation is certainly more complicated, nevertheless the two states model (*binary model*) can be useful for making some statistical inferences.

Let $\Theta(t),\ t > 0$ be a total duration of radiation during the observation interval $(0, t)$ beginning with one of transitions (Fig. 6.2b). We will use the following notations:

1 and 0 are the on- and off-states respectively;

ε_1 is the probability that the initial transition is $0 \to 1$;

ε_2 is the probability that the initial transition is $1 \to 0$;

$p_1(\theta|t)d\theta = \mathsf{P}(\Theta(t) \in d\theta)$ is the probability that the total time of radiation during the observation time t belongs to the interval $(\theta, \theta + d\theta)$;

$f_{1\to 0}(\theta, t)d\theta dt$ is the mean number of transitions $1 \to 0$ from $(\theta, \theta + d\theta)$ during the time $(t, t + dt)$;

$f_{0\to 1}(\theta, t)d\theta dt$ is the mean number of transitions $0 \to 1$ into the interval $(\theta, \theta + d\theta)$ during the time $(t, t + dt)$,

$$\Psi_n(t) = \int_t^\infty \psi_n(t')dt' = \mathsf{P}(T_n > t), \quad n = 0, 1.$$

The integral equations connecting the densities introduced above have the form:

$$p_1(\theta|t) = \int_0^\theta f_{0\to 1}(\theta - t', t - t')\Psi_1(t')dt' + \int_0^t f_{1\to 0}(\theta, t - t')\Psi_0(t')dt',$$

$$f_{1\to 0}(\theta, t) = \int_0^\theta f_{0\to 1}(\theta - t', t - t')\psi_1(t')dt' + \varepsilon_0\delta(\theta)\delta(t),$$

$$f_{0\to 1}(\theta, t) = \int_0^t f_{1\to 0}(\theta, t - t')\psi_0(t')\,dt' + \varepsilon_1\delta(\theta)\delta(t).$$

Applying the double Laplace transformation

$$\hat{p}_1(\lambda_1|\lambda) = \int_0^\infty dt \int_0^t d\theta\, e^{-\lambda t - \lambda_1 \theta} p_1(\theta|t),$$

we obtain

$$\begin{aligned}
\widehat{p}_1(\lambda_1|\lambda) &= \widehat{f}_{0\to1}(\lambda_1,\lambda)\,\widehat{\Psi}_1(\lambda+\lambda_1) + \widehat{f}_{1\to0}(\lambda_1,\lambda)\,\widehat{\Psi}_0(s),\\
\widehat{f}_{1\to0}(\lambda_1,\lambda) &= \widehat{f}_{0\to1}(\lambda_1,\lambda)\,\widehat{\psi}_1(\lambda+\lambda_1) + \varepsilon_0,\\
\widehat{f}_{0\to1}(\lambda_1,\lambda) &= \widehat{f}_{1\to0}(\lambda_1,\lambda)\,\widehat{\psi}_0(s) + \varepsilon_1.
\end{aligned}$$

Solving this system, we arrive at the transforms of jump rates

$$\begin{aligned}
\widehat{f}_{1\to0}(\lambda_1,\lambda) &= \frac{\varepsilon_1\widehat{\psi}_1(\lambda_1+\lambda)+\varepsilon_0}{1-\widehat{\psi}_1(\lambda+\lambda_1)\widehat{\psi}_0(\lambda)},\\
\widehat{f}_{0\to1}(\lambda_1,\lambda) &= \frac{\varepsilon_0\widehat{\psi}_0(\lambda)+\varepsilon_1}{1-\widehat{\psi}_1(\lambda+\lambda_1)\widehat{\psi}_0(\lambda)},
\end{aligned}$$

and the expression for the image of unknown density:

$$\begin{aligned}
\widehat{p}_1(\lambda_1|\lambda) = &\frac{\varepsilon_0\widehat{\psi}_0(\lambda)+\varepsilon_1}{1-\widehat{\psi}_1(\lambda+\lambda_1)\widehat{\psi}_0(\lambda)}\,\frac{1-\widehat{\psi}_1(\lambda+\lambda_1)}{\lambda+\lambda_1}\\
&+\frac{\varepsilon_1\widehat{\psi}_1(\lambda+\lambda_1)+\varepsilon_0}{1-\widehat{\psi}_1(\lambda+\lambda_1)\widehat{\psi}_0(\lambda)}\,\frac{1-\widehat{\psi}_0(\lambda)}{\lambda}.
\end{aligned} \tag{6.19}$$

The latter expression coincides accurately within notions with the result obtained in Jung et al. [23] for the characteristic function of PDF of fluorescence intensity. This transform was used by the authors for asymptotical analysis of the Mandel parameter. We shall obtain the explicit expression for asymptotical distribution density $p_1(\theta|t)$ and show that this density satisfies the fractional differential equation.

6.4.3 *Fractional Transport Equations*

Tang and Marcus [57] proposed a mechanism of blinking based on a diffusion-controlled electron-transfer model, where a light-induced one-dimensional diffusion process in energy space is considered. This mechanism leads to the distribution densities of on- and off-intervals in the form of "*fractional exponential function*":

$$\psi_1(t) = c_1^\alpha t^{\alpha-1}\,E_{\alpha,\alpha}(-c_1^\alpha t^\alpha), \qquad \psi_0(t) = c_0^\beta t^{\beta-1}\,E_{\beta,\beta}(-c_0^\beta t^\beta).$$

When $\alpha < 1$ and $\beta < 1$, these densities have "heavy" power-law tails. If $\alpha \to 1$ and $\beta \to 1$, "fractional exponents" reduce to ordinary exponents. Thus, a change of only one parameter in each density $\psi_1(t)$ and $\psi_0(t)$ allows to model power-law on- and off-distributions as well as exponential ones.

Substituting the Laplace transforms of these functions:

$$\widehat{\psi}_1(\lambda) = \frac{1}{1 + (\lambda/c_1)^{\alpha}}, \qquad \widehat{\psi}_0(\lambda) = \frac{1}{1 + (\lambda/c_0)^{\beta}}$$

into the expression (6.19), we obtain

$$\widehat{p}_1(\lambda_1|\lambda) = \frac{c_1^{-\alpha}(\lambda+\lambda_1)^{\alpha-1} + c_0^{-\beta}\lambda^{\beta-1} + c_1^{-\alpha}c_0^{-\beta}\lambda^{\beta}(\lambda+\lambda_1)^{\alpha}\left[\varepsilon_1(\lambda+\lambda_1)^{-1} + \varepsilon_0\lambda^{-1}\right]}{c_1^{-\alpha}(\lambda+\lambda_1)^{\alpha} + c_0^{-\beta}\lambda^{\beta} + c_1^{-\alpha}c_0^{-\beta}(\lambda+\lambda_1)^{\alpha}\lambda^{\beta}}. \tag{6.20}$$

Let us rewrite the last relation in the form:

$$\begin{aligned}&\lambda^{\beta}\,\widehat{p}_1(\lambda_1|\lambda) + C(\lambda+\lambda_1)^{\alpha}\,\widehat{p}_1(\lambda_1|\lambda) + c_1^{-\alpha}(\lambda+\lambda_1)^{\alpha}\lambda^{\beta}\,\widehat{p}_1(\lambda_1|\lambda)\\ &= C(\lambda+\lambda_1)^{\alpha-1} + \lambda^{\beta-1} + c_1^{-\alpha}\lambda^{\beta}(\lambda+\lambda_1)^{\alpha}\left[\varepsilon_1(\lambda+\lambda_1)^{-1} + \varepsilon_0\lambda^{-1}\right].\end{aligned}$$

Here $C = c_0^{\beta}/c_1^{\alpha}$. Performing the double inverse Laplace transformation and taking into account that

$$\begin{aligned}&\frac{1}{(2\pi i)^2}\int_{\Gamma} d\lambda \int_{\Gamma'} d\lambda_1\ \exp(\lambda t + \lambda_1\theta)(\lambda+\lambda_1)^{\alpha}\widehat{p}_1(\lambda_1|\lambda)\\ &= \frac{1}{2\pi i}\int_{\Gamma} d\lambda\ \exp(\lambda t)\exp(-\lambda\theta)\ {}_0\mathsf{D}_{\theta}^{\alpha}\exp(\lambda\theta)\widehat{p}_1(\theta|\lambda) = \mathsf{E}_t^{-\theta}\,{}_0\mathsf{D}_{\theta}^{\alpha}\mathsf{E}_t^{\theta}\,p_1(\theta|t),\end{aligned}$$

where E_t^a is the time-shift operator:

$$\mathsf{E}_t^a\,p_1(\theta|t) = p_1(\theta|t+a),$$

we arrive at the fractional equation for the PDF of total on-time duration:

$$\begin{aligned}&{}_0\mathsf{D}_t^{\beta}p_1(\theta|t) + C\ \mathsf{E}_t^{-\theta}\,{}_0\mathsf{D}_{\theta}^{\alpha}\mathsf{E}_t^{\theta}\ p_1(\theta|t) + c_1^{-\alpha}\ {}_0\mathsf{D}_t^{\beta}\ \mathsf{E}_t^{-\theta}\,{}_0\mathsf{D}_{\theta}^{\alpha}\mathsf{E}_t^{\theta}\ p_1(\theta|t)\\ &= C\ \delta(t-\theta)\frac{t^{-\alpha}}{\Gamma(1-\alpha)} + \delta(\theta)\frac{t^{-\beta}}{\Gamma(1-\beta)} - \frac{c_1^{-\alpha}\theta^{-\alpha}(t-\theta)^{-\beta}}{\Gamma(1-\alpha)\Gamma(1-\beta)}\left(\frac{\alpha\varepsilon_0}{\theta} + \frac{\beta\varepsilon_1}{t-\theta}\right).\end{aligned}$$

Let the blinking process be studied since the moment of transition off $\to$ on. Then $\varepsilon_1 = 1$ and $\varepsilon_0 = 0$. According to the Tauberian theorem, the long-time ($t \gg c_1^{-1},\ c_0^{-1}$) asymptotics of the expression (6.20) has the form:

$$\widehat{p}_1(\lambda_1|\lambda) \sim \frac{C(\lambda+\lambda_1)^{\alpha-1} + \lambda^{\beta-1}}{C(\lambda+\lambda_1)^{\alpha} + \lambda^{\beta}}, \qquad C = c_0^{\beta}/c_1^{\alpha}. \tag{6.21}$$

Rewriting the last expression in the form

$$\left[C(\lambda+\lambda_1)^{\alpha}+\lambda^{\beta}\right]\widehat{p}_1(\lambda_1|\lambda)=C(\lambda+\lambda_1)^{\alpha-1}+\lambda^{\beta-1}.$$

and performing the double inverse Laplace transformation leads us to the fractional equation for the density of total on-time duration:

$${}_0\mathsf{D}_t^{\beta}\,p_1(\theta|t)+C\;\mathsf{E}_t^{-\theta}\,{}_0\mathsf{D}_{\theta}^{\alpha}\mathsf{E}_t^{+\theta}\;p_1(\theta|t)=C\;\delta(t-\theta)\frac{t^{-\alpha}}{\Gamma(1-\alpha)}+\delta(\theta)\frac{t^{-\beta}}{\Gamma(1-\beta)}.$$

It is known (see, for example Uchaikin [60]) that the expression

$$\frac{\lambda^{\beta-1}}{C\lambda_1^{\alpha}+\lambda^{\beta}}$$

represents the double Laplace transform of the function:

$$(Ct^{\beta})^{-1/\alpha}\;q_+\left(\theta(Ct^{\beta})^{-1/\alpha};\,\alpha,\,\beta\right),$$

where $q_+(t;\,\alpha,\,\beta)$ is the one-sided fractionally stable density

$$q_+(t;\,\alpha,\,\beta)=\int_0^{\infty}d\tau\;g_+(\tau;\,\beta)g_+(t\tau^{-\beta/\alpha};\,\alpha)\;\tau^{\beta/\alpha}.$$

Finding an inverse transform for Eq. (6.21)

$$\begin{aligned}p(\theta|t)=&\,[C(t-\theta)^{\beta}]^{-1/\alpha}\;q_+\left(\theta[C(t-\theta)^{\beta}]^{-1/\alpha};\,\alpha,\,\beta\right)\\&+(C^{-1}\theta^{\alpha})^{-1/\beta}\;q_+\left((t-\theta)(C^{-1}\theta^{\alpha})^{-1/\beta};\,\beta,\,\alpha\right).\end{aligned}\tag{6.22}$$

and taking into account the relation for fractional stable functions

$$Q_+\left(\xi(C\tau^{\beta})^{-1/\alpha};\,\alpha,\,\beta\right)=1-Q_+\left(\tau(C^{-1}\xi^{\alpha})^{-1/\beta};\,\beta,\,\alpha\right),$$

we rewrite the Eq. (6.22) in the form:

$$p_1(\theta|t)=\left[1+\frac{\alpha}{\beta}\frac{t-\theta}{\theta}\right](C\theta^{\alpha})^{-1/\beta}\;q_+\left((t-\theta)(C\theta^{\alpha})^{-1/\beta};\,\beta,\,\alpha\right).\tag{6.23}$$

The comparison of solution (6.23) with results of Monte Carlo simulation presented in Fig. 6.3 shows good accordance.

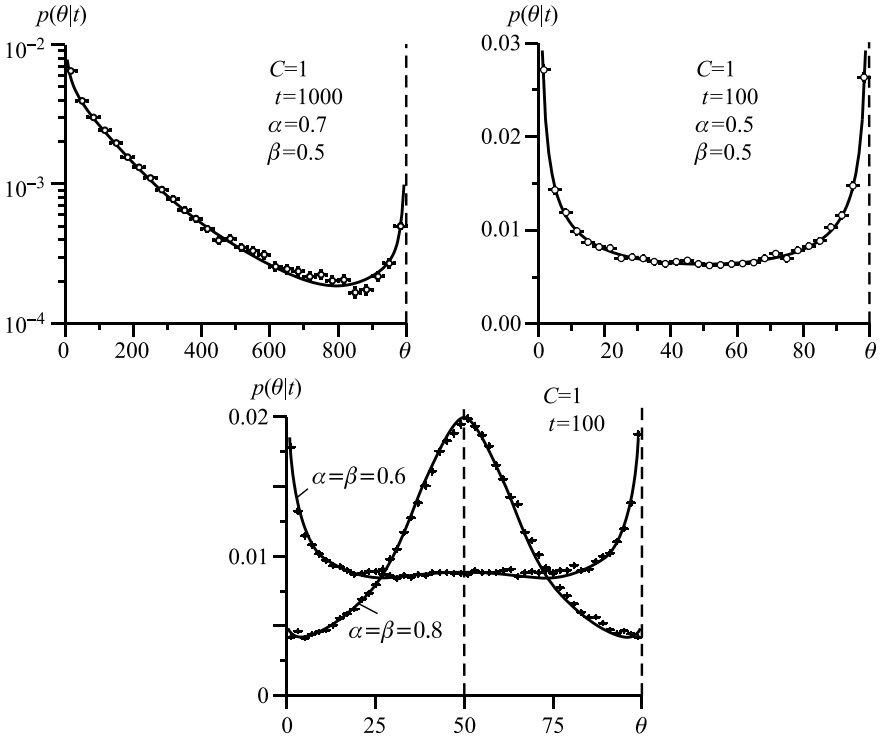


Fig. 6.3 Comparison of the analytical solutions (lines) with the results of Monte Carlo simulation (dots)

6.4.4 Quantum Wires

In transfer matrix theory, the electronic transport is seen as a scattering problem. The zero-temperature conductance g is related to the quantum-mechanical transmission matrix through the Landauer formula. Dorokhov [12] and Mello et al. [37] have derived an equation known as the DMPK-equation for the joint probability distribution of the transmission eigenvalues in the weak localization regime. The generalized DMPK-equation has been proposed in Muttalib and Klauder [39] to describe the electron transport in strongly disordered systems. These equations of the Fokker-Planck type in both cases (weak and strong localization) have been derived on the assumption of regular space distribution of scatterers. However, experimental works of Kohno and Yoshida [26], Barthelemy et al. [8], and Hegger et al. [20], and numerical simulations of Leadbeater et al. [33] have shown that disorder in mesoscopic systems can be of fractal (self-similar) type. Conductance of such systems can not be described in frames of the quasi-one-dimensional DMPK approach and its multidimensional generalizations. Sibatov [54] has derived a fractional analogue of the DMPK-equation for the joint probability distribution of the transmission eigenvalues

$$l^{\alpha}\,{}_0\mathsf{D}_L^{\alpha} P(\{\lambda\}, L) = K \frac{2}{\beta N + 2 - \beta} \times \sum_{j=1}^{N} \frac{\partial}{\partial \lambda_j} \left[\lambda_j (1 + \lambda_j) J(\lambda) \frac{\partial}{\partial \lambda_j} \frac{P(\{\lambda\}, L)}{J(\lambda)} \right] + \frac{(L/l)^{-\alpha}}{\Gamma(1-\alpha)}\, \delta(\lambda)$$

in case of fractal disorder characterized by a heavy-tailed distribution of distances between scatterers

$$\mathsf{P}(d > x) \sim \frac{(x/l)^{-\alpha}}{\Gamma(1-\alpha)}, \quad l > 0, \quad x \to \infty.$$

Here, λ_j are eigenvalues of quantum-mechanical transfer matrix t, N number of parallel channels, $P(\{\lambda\}, L)$ the N-dimensional probability distribution function of random vector $(\lambda_1, \lambda_2, \ldots, \lambda_N)$. The integer parameter β equals 1 in a zero magnetic field and 2 in a time-reversal-symmetry breaking magnetic field. In the case of zero field and strong spin-orbit scattering $\beta = 4$. The Jacobian $J = \prod_{i<j} |\lambda_j - \lambda_i|^{\beta}$ corresponds to the transformation from the transfer matrix space to the eigenvalue space. Solutions of this equation lead to a new class of universal conductance distributions related to the Lévy stable statistics. Analytical results agree with Monte-Carlo simulations of disordered wires and with particular results obtained in Beenakker et al. [9] and Falceto and Gopar [14].

6.5 Quantum Decay Theory

6.5.1 Krylov-Fock Theorem

The problem considered above is closely related to the general quantum decay theory.

In quantum mechanics, the state of an unstable physical system is described by the so-called state vector $|\psi(t)\rangle$, which is a solution of the time-dependent Cauchy problem for the Schrödinger equation

$$i \frac{\partial |\psi(t)\rangle}{\partial t} = \mathsf{H} |\psi(t)\rangle, \tag{6.24}$$

where $|\psi(0)\rangle$ is a given initial state vector and again the units are chosen so that the Planck constant $\hbar = 1$. Let $\{|\phi_E\rangle, |\phi_k\rangle\}$ be the complete system of eigenvectors of the operator H ($|\phi_E\rangle$ corresponds to the absolutely continuous component of its spectrum, and $|\phi_k\rangle$ corresponds to the discrete component), i.e.,

$$\mathsf{H}|\phi_E\rangle = E|\phi_E\rangle, \qquad \langle \phi_{E'} | \phi_E \rangle = \delta(E' - E),$$

$$H|\phi_k\rangle = E_k|\phi_k\rangle, \qquad \langle\phi_k|\phi_l\rangle = \delta_{kl},$$

where $\delta(E' - E)$ is the Dirac delta function, and δ_{kl} is the Kronecker symbol.

We are interested in the probability $P(t)$ that at a time t the system is in the initial state $|\psi_0\rangle$. According to the laws of quantum mechanics,

$$P(t) = |\langle\psi(0)|\psi(t)\rangle|^2.$$

Solving the Cauchy problem Eq. (6.24) for the Schrödinger equation, we assume that $\langle\psi(0)|\psi(0)\rangle = 1$. In this case, the Krylov-Fock [28] theorem yields

$$f(t) = \langle\psi(0)|\psi(t)\rangle = \sum_k |c_k|^2 \exp(-iE_k t) + \int_0^\infty |c(E)|^2 \exp(-iEt)\, dt,$$

where c_k and $c(E)$ are the Fourier coefficients in the expansion of the vector $|\psi(0)\rangle$ in the complete system $\{|\phi_E\rangle, |\phi_k\rangle\}$ of eigenvectors

$$|\psi(0)\rangle = \sum_k c_k|\phi_k\rangle + \int_0^\infty c(E)|\phi_E\rangle dE.$$

Thus, $f(t)$ can be interpreted as the characteristic function of some distribution having discrete components (probabilities of isolated values) $|c_k|^2$ and absolutely continuous component (i.e., density) $|c(E)|^2$. Instability of the system means that the probability $P(t) = |f(t)|^2$ of the system returning to the original state at time t tends to zero as $t \to \infty$.

Since $f(t)$ is a characteristic function, $|f(t)| \to 0$ only if the discrete components of the spectrum of H are missing, i.e., $c_k = 0$. In this case

$$f(t) = \int_0^\infty \mathrm{w}(E) \exp(-iEt)\, dE, \tag{6.25}$$

where $\mathrm{w}(E) = |c(E)|^2$ denotes the density of the energy distribution of the decaying physical system described by Eq. (6.24).

It turns out that for a very broad class of unstable physical systems the densities $\mathrm{w}(E)$ are meromorphic functions (see Krylov and Fock [28]). For a number of reasons, the case of a function $\mathrm{w}(E)$ having only two simple poles (they are complex conjugated in view of the condition $\mathrm{w}(E) \geqslant 0$) is of great interest. In this case it is obvious that

$$\mathrm{w}(E) = A[(E - E_0)^2 + \Gamma^2]^{-1}, \qquad E \geqslant 0,$$

where A is a normalizing constant, and E_0 and Γ are the most probable value and the measure of dispersion (with respect to E_0) of the system's energy. For actual unstable systems the ratio Γ/E_0 is very small, as a rule (10^{-15}, or even smaller). Therefore, to compute $P(t)$ we can, without adverse effects, replace the lower limit 0 in integral

Eq. (6.25) by $-\infty$, after which the density function $\mathrm{w}(E)$ and the probability $P(t)$ connected with it take the approximate expressions

$$\mathrm{w}(E) \approx \frac{\Gamma}{\pi}[(E - E_0)^2 + \Gamma^2]^{-1}, \quad P(t) = |f(t)|^2 \approx \exp(-2\Gamma t).$$

It is clear from the first relation (the Lorentz distribution of the energy of the unstable system) that we are dealing with the Cauchy distribution, and it is clear from the second relation that the lifetime for unstable systems of the type under consideration behaves according to the exponential law.

6.5.2 *Weron-Weron Theorem*

Thus, the Cauchy law appears here only as a more or less good approximation of the real energy distribution for unstable systems. And there are situations where the replacement of 0 by $-\infty$ in Eq. (6.25) is unacceptable, because the corresponding law $P(t)$ of decay of the system differs essentially from the exponential law.

We give here a result of Hack [19]. Imposing the constraint normally applied in quantum theory that the self-adjoint Hamiltonian H is lower semi-bounded, i.e., that the energy spectrum is bounded below, Hack established that $P(t)$ cannot decay exponentially fast as $t \to \infty$, i.e.,

$$P(t) > Ce^{-at}$$

for $t > T$, where C, a, and T are positive constants.

The following theorem proved in Weron and Weron [66] leads directly to the stretched exponent law.

Theorem 6.1 *The non-decay probability function for many-body weakly interacting quantum system is of the form*

$$P(t) = \exp\{-at^{\alpha}\}, \quad a > 0, \quad 0 < \alpha < 1.$$

Representing the amplitude $f(t)$ as

$$f(t) = \langle\psi|\exp(-\mathsf{L}t)|\psi\rangle = \int_0^{\infty} \exp(-Et)\mathrm{w}(E)\,dE$$

where L is the development operator governing the dynamic evolution of the quantum system under investigation and $\mathrm{w}(E)$ is the probability density of the state $|\psi\rangle$ associated with the continuous spectrum of the development operator L, the authors

conclude that we observe an arbitrariness in the specification of ψ and $\mathrm{w}(E)$. In general, one considers ψ to represent a decaying state for a many-body system, and therefore the number of components in the system should not influence the decay. In other words, the same decaying law should be obtained for one portion or several portions of the system. Consequently, in a weakly interacting quantum system, microscopic energies can be considered as independent identically distributed energy random variables. The microscopic energy distribution $\mathrm{w}(E)\,dE$ associated with the decaying system is identified to be the limit distribution of normalized sums of the microscopic energy random variables. According to the generalized limit theorem, the limit $\mathrm{w}(E)\,dE$ has α-stable distribution $0 < \alpha \leqslant 2$. Since $\mathrm{w}(E)$ is associated from the above construction with the development operator L, it has to have positive support. This holds only when $\mathrm{w}(E)dE$ has a completely asymmetric ($\beta = 1$, $0 < \alpha < 1$) stable distribution [66].

6.5.3 Nakhushev Fractional Equation

Nakhushev [42] has derived a fractional equation for the stretched exponent

$$u(t) = u(0)\exp\left[-\left(\frac{t}{\tau}\right)^{\alpha}\right], \quad \tau = \text{const}, \quad 0 < \alpha < 1.$$

Passing to the dimensionless variables

$$x = \frac{t}{\tau}, \quad y = \ln[u(\tau x)/p]$$

reduces to a simple algebraic form

$$y = -x^{\alpha}.$$

This function satisfies the fractional equation

$${}_0\mathsf{D}_x^{\alpha+1} y(x) = 0. \tag{6.26}$$

Nakhushev [42] shows that any solution of Eq. (6.26) is representable in the form

$$y(x) = (ax + b)x^{\alpha-1}, \tag{6.27}$$

where a and b are arbitrary constants. Indeed, let $y(x)$ be a solution of Eq. (6.26) from the class $L[0, l]$, then the equation

$$\frac{d^2}{dx^2}\,{}_0\mathsf{D}_x^{\alpha-1} y(x) = 0$$

implies

$$ {}_0\mathsf{D}_x^{\alpha-1} y(x) = Ax + B, $$

where A and B are constant values. For any function $f(x) \in L(0, l)$ and almost all $x \in [0, l]$, the following equality takes place:

$$ {}_0\mathsf{D}_x^{\nu}\, {}_0\mathsf{D}_x^{-\nu} f(x) = f(x), \quad \forall \nu > 0. $$

Inserting here $y(x)$ and $\nu = 1 - \alpha$, we get

$$ y(x) = {}_0\mathsf{D}_x^{1-\alpha}(Ax + B). $$

For any ν and $\mu > -1$,

$$ {}_0\mathsf{D}_x^{\nu} x^{\mu} = \frac{\Gamma(1+\mu)}{\Gamma(1+\mu-\nu)} x^{\mu-\nu}, $$

therefore

$$ {}_0\mathsf{D}_x^{1-\alpha} x = \frac{1}{\Gamma(1+\alpha)} x^{\alpha}, $$

and

$$ {}_0\mathsf{D}_x^{1-\alpha} 1 = \frac{1}{\Gamma(\alpha)} x^{\alpha-1}. $$

As a result, we have

$$ y(x) = \frac{A}{\Gamma(1+\alpha)} x^{\alpha} + \frac{B}{\Gamma(\alpha)} x^{\alpha-1} = (ax + b) x^{\alpha-1} $$

with $\alpha = A/\Gamma(1+\alpha)$ and $b = B/\Gamma(\alpha)$. On the other hand, any function representable in the form of Eq. (6.27) is a solution of Eq. (6.26). Really,

$$ \begin{aligned} {}_0\mathsf{D}_x^{\alpha+1}\left[(ax+b)x^{\alpha-1}\right] &= a\, {}_0\mathsf{D}_x^{\alpha+1} x^{\alpha} + b\, {}_0\mathsf{D}_x^{\alpha+1} x^{\alpha-1} \\ &= b\, {}_0\mathsf{D}_x^{\alpha+1} x^{\alpha-1} = b \frac{\partial^2 \Gamma(\alpha)}{\partial x^2} = 0. \end{aligned} $$

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Chapter 7
Plasma Dynamics



7.1 Resonance Radiation Transport

7.1.1 A Role of the Dispersion Profile

Two main characteristics govern the radiation transport of excitations in plasma: the absorption coefficient $k(\nu)$ and the frequency spectrum $\varepsilon(\nu)$ of the radiation emitted from a given volume element. If the system was in temperature equilibrium, the principles of thermodynamics would provide us with the simple proportionality between these coefficients,

$$\varepsilon(\nu) \propto k(\nu),$$

which is a form of Kirchhoff's law. Strictly speaking, the radiation under consideration is certainly not in temperature equilibrium with the gas. Nevertheless, there is accepted and used the hypothesis that the system of photons and atomic excitations is in a state of *local thermodynamic equilibrium*, with negligible interaction between it and the mass motion of the atoms.

There exist a few mechanisms having an influence on forming the dispersion profile $k(\nu)$. In each case, this is a unimodal function with maximum at some frequency ν_0. In what follows, we introduce notation $z = \nu - \nu_0$ normalized to unity *dispersion distribution function* $f(z)$,

$$\int_{-\infty}^{\infty} f(z)dz = 1.$$

Writing the above-mentioned proportionality between two coefficients as

$$k(z) = Cf(z)$$

and considering ν as a random variable, one can express the interrelation between

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conditional and unconditional free path probability densities as

$$\mathsf{P}(R > r) = \int_{-\infty}^{\infty} \mathsf{P}(R > r|z) f(z) dz = \int_{-\infty}^{\infty} e^{-k(z)r} f(z) dz = \int_{-\infty}^{\infty} e^{-Cf(z)r} f(z) dz.$$

It is necessary to stress that this formula is valid only for photons, which frequency lies in the vicinity of a spectral line. This is a result of the total redistribution frequencies model, assuming the mutual independence of absorbed and radiated photons frequencies, and this assumption in its turn follows from the local equilibrium hypothesis. The only characteristics involved in this distribution is the spectral density $f(\omega)$ called the *dispersion profile*. The dispersion profile $f(\omega)$ is a high narrow peak around the central frequency of the line, consequently, the probability

$$P(dz, r) = e^{-Cf(z)r} f(z) dz$$

is very small for frequencies in the vicinity of $z = 0$, but becomes significant out of this region. In other words, free paths of photons are short in the region of high probability density $f(z)$ and long out of this region. Because each absorption of the photon by an atom is attended by confinement during random trapping time, this form yields the original behaviour of resonance photons called the *imprisonment phenomenon*: neighboring atoms exchange by the photon many times before the frequency appears out of the opaque domain and photon can escape from its prison.

The unconditional free path distribution essentially depends on the dispersion profile, formed under action of various causes. Let us consider the most significants of them.

Describing absorption-radiation processes in a low-temperature low-density plasma, one can consider it as a system of non-interacting atoms at rest. In this case, the main mechanism responsible for the dispersion profile is the radiation process itself, generating

$$f_0(z) = \frac{\Delta}{\pi[\Delta^2 + z^2]}. \tag{7.1}$$

This distribution, known as *Lorentz dispersion profile*, is nothing but the Cauchy distribution. The constant Δ is given by classical electrodynamics as

$$\Delta = 1/\tau_{\text{rad}},$$

where τ_{rad} is the lifetime of the excited state (see [31, 37]).

With increasing the density of the plasma, two new factors become essential: collisions and Stark effect produced by surrounding ions. The collisional broadening leads to the same Lorentz profile with the parameter Δ equals to the collision frequency,

$$\Delta = 1/\tau_{\text{coll}}.$$

Let us consider the second source of the profile broadening.

Each atom (ion) in plasma is surrounded by other ions creating electric fields which act on the atom under consideration. The main result of this action is an energy levels shift involving the corresponding shift in radiation frequency called the *Stark effect*. For hydrogen-like atoms, the frequency shift $S(E)$ induced by the electrical field $\mathbf{E}$ is a linear function of E (the *linear Stark effect*). For other atoms, the linear dependence is observed only in strong fields; in the weak fields, $S(E)$ is proportional to E^2 (the *quadratic Stark effect*). Different radiating atoms are in various microscopic environments. As a simple model of this situation, they usually consider a single atom placed in a random electrical field $\mathbf{E}$ with probability distribution density $g(\mathbf{E})$. In the absence of reasons creating anisotropy, this is isotropic and its density depends only on the absolute value of the strength

$$w(E) = 4\pi g(\mathbf{E})E^2. \tag{7.2}$$

Let the frequency shift be a monotonically growing function of E,

$$\nu = \nu_0 + S(E);$$

then distribution (7.2) induces the profile of the spectral line

$$f(z) = w(S^{-1}(z))dS^{-1}(z)/dz, \tag{7.3}$$

where $S^{-1}(z)$ is the inverse to $S(E)$ function. In a linear case,

$$S(E) = s_1 E,$$

and Eq. (7.3) takes the form

$$f_1(z) = w(z/s_1)/s_1, \tag{7.4}$$

and in a square-law case,

$$S(E) = s_2 E^2;$$

hence

$$f_2(z) = \frac{1}{2} w\left(\sqrt{z/s_2}\right)/\sqrt{zs_2}. \tag{7.5}$$

Thus, the problem is reduced to evaluation of function (7.2).

The distribution (7.2) was calculated in Holtsmark [30] under the assumption that a given number of point sources (point charges, dipoles, or quadrupoles) is distributed within a spherical volume centered at the atom, uniformly and independently of each other. In case of point charges (ions), the vector $\mathbf{E}$ is distributed according to the three-dimensional stable law with the characteristic exponent 3/2,

$$g_{\mathrm{ch}}(\mathbf{E}) = g_3(\mathbf{E}; 3/2)$$

(see also [62, 75]). The absolute value of this vector is distributed with the density

$$w_{\rm ch}(E) = (a\rho)^{-2/3} H((a\rho)^{-2/3} E); \tag{7.6}$$

where $a = 2\pi(4/15)^{2/3}\langle|e|^{3/2}\rangle^{2/3}$, ρ is the concentration of field sources, and

$$H(x) = \frac{2}{\pi x}\int_0^\infty e^{-(y/x)^{3/2}} y \sin y \, dy \tag{7.7}$$

is the *Holtsmark distribution formula*. This is nothing but the probability density function for magnitude of vector distributed according to three-dimensional stable law with the characteristic exponent 3/2. Equation (7.6) has the asymptotics

$$w_{\rm ch}(E) \sim W_{\rm ch} E^{-5/2}, \quad x \to \infty.$$

For dipoles and quadrupoles Holtsmark found that

$$g_{\rm dip}(\mathbf{E}) = c_{\rm d}^{-3} g_3(c_{\rm d}^{-1}\mathbf{E}; 1)$$

and

$$g_{\rm quad}(\mathbf{E}) = c_{\rm q}^{-3} g_3(c_{\rm q}^{-1}\mathbf{E}; 3/4)$$

respectively with scale constants $c_{\rm d}$ and $c_{\rm q}$. Correspondingly,

$$w_{\rm dip}(E) \sim W_{\rm dip} E^{-2}, \quad E \to \infty,$$

and

$$w_{\rm quad}(E) \sim W_{\rm quad} E^{-7/4}, \quad E \to \infty.$$

Thus, the distribution derived by Holtsmark is related to the observable structure of spectral lines, though it is necessary to say that experiments reveal the combined impact of all reasons on the line width. As numerical calculations show [32], the center of the profile is formed by the Doppler mechanism of widening, giving the normal distribution, and the far wings coincide with the Lorentz distribution.

7.1.2 *Fractional Biberman-Holstein Equation*

Let us recall the way of derivation of the balance equation for the density of excited atoms $N(\mathbf{r}, t)$ following [29]. During the infinitesimal time interval $(t, t + dt)$, the mean number of excited atoms in a volume element $d\mathbf{r}$, $N(\mathbf{r}, t)d\mathbf{r}$ changes because of photon absorbtion and emission processes according to equality

$$N(\mathbf{r}, t)d\mathbf{r} \to N(\mathbf{r}, t + dt)d\mathbf{r} = N(\mathbf{r}, t)d\mathbf{r} + Q_+(d\mathbf{r}, dt) - Q_-(d\mathbf{r}, dt). \tag{7.8}$$

The decrease in the number of excited atoms in $d\mathbf{r}$ is proportional to their concentration at the point $\mathbf{r}$,

$$Q_-(d\mathbf{r}, dt) = \gamma N(\mathbf{r}, t) d\mathbf{r} dt, \tag{7.9}$$

meanwhile, to obtain the increase in this quantity, we must sum contributions of all other volume elements. The contribution of such an element $d\mathbf{r}'$ is given by the number of photons emitted therein in time dt, $\gamma N(\mathbf{r}', t) d\mathbf{r}' dt$, multiplied by the probability of one of these photons being trapped in $d\mathbf{r}$, $K(|\mathbf{r} - \mathbf{r}'|)d\mathbf{r}$. Integrating over the whole volume V occupied by the plasma[1], we obtain

$$Q_+(d\mathbf{r}, dt) = \gamma dt d\mathbf{r} \int_V d\mathbf{r}' K(|\mathbf{r} - \mathbf{r}'|) N(\mathbf{r}', t). \tag{7.10}$$

Writing this formula, we have implicitly assumed that the time of flight of photon is negligible small compared to the excitation lifetime. Inserting Eqs. (7.9) and (7.10) in (7.8), we arrive at the integro-differential equation for excitation transport in plasma,

$$\frac{\partial N(\mathbf{r}, t)}{\partial t} = \gamma \int d\mathbf{r}' K(|\mathbf{r} - \mathbf{r}'|) \left[N(\mathbf{r}', t) - N(\mathbf{r}, t) \right], \tag{7.11}$$

called the *Biberman-Holstein equation* [9, 29].

The kernel $K(r)$ is the photon transfer kernel averaged over the random frequency:

$$K(r) = \left\langle k(\omega) \frac{e^{-k(\omega)r}}{4\pi r^2} \right\rangle = -\frac{d\mathsf{P}(R > r)}{4\pi r^2 dr}.$$

As we saw above, a few practically interesting cases are described by dispersion distributions with power type of wings. Let us introduce a one-sided probability density function

$$\phi(z) = f(z) + f(-z), \quad z > 0.$$

According to aforesaid, $k(z) = C\phi(z)$ and $\phi(z) \sim Bz^{-\beta-1}$, so

$$\mathsf{P}(R > r) = \int_0^\infty e^{-C\phi(z)r} \phi(z) dz = -C^{-1} \frac{d}{dr} \int_0^\infty e^{-C\phi(z)r} dz.$$

Replacing $\phi(z)$ by its asymptotical expression and performing the integration yields

$$\int_0^\infty e^{-C\phi(z)r} dz \sim \int_0^\infty e^{-BCrz^{-\beta-1}} dz = B^{1/(\beta+1)} \Gamma\left(\frac{\beta}{\beta+1}\right) r^{1/(\beta+1)},$$

and we have

[1] The walls of this volume are assumed to be non-concave and non-reflecting.

$$\mathsf{P}(R > r) \sim -\frac{1}{C}\frac{d}{dr}\left[B^{1/(\beta+1)}\Gamma\left(\frac{\beta}{\beta+1}\right)r^{1/(\beta+1)}\right]$$
$$= \frac{1}{\beta}B^{1/(\beta+1)}\Gamma\left(\frac{2\beta+1}{\beta+1}\right)r^{-\beta/(\beta+1)}.$$

The passage to the fractional BH equation is realized by replacing the transport kernel $K(r)$ by its asymptotical expression

$$K(r) = \frac{d\mathsf{P}(R > r)}{4\pi r^2 dr} \sim \frac{A}{r^{\alpha+3}},$$

$$A = \frac{A^{1/(\beta+1)}}{4\pi(\beta+1)}\Gamma\left(\frac{2\beta+1}{\beta+1}\right), \quad \alpha = \frac{\beta}{\beta+1},$$

and the Biberman-Holstein equation takes the form of a fractional equation [63]:

$$\frac{\partial N(\mathbf{r},t)}{\partial t} = \gamma A\int_V \frac{N(\mathbf{r}',t) - N(\mathbf{r},t)}{|\mathbf{r}'-\mathbf{r}|^{\alpha+3}}d\mathbf{r}' \equiv -\gamma A C_\alpha(-\Delta)_V^{\alpha/2}N(\mathbf{r},t).$$

Knowing the time-space distribution of excited atoms in plasma, one can easily compute various characteristics of light field necessary for constructing modern lighting devices (see for details [3, 4, 38, 58, 64]).

7.1.3 Fractional Boltzmann Equation

For the sake of convenience, they usually identify radiated and absorbed photons, in other words, they say that an atom absorbing a photon radiates "the same photon" with frequency and direction changed after an interaction with the atom, or shorter, the photon is scattered on the atom. This process is described by the linear Boltzmann equation

$$\left(\frac{\partial}{\partial t} + c\Omega\nabla\right)\Phi(\mathbf{r},\Omega,\omega,t) + c\sigma(\omega)\Phi(\mathbf{r},\Omega,\omega,t)$$
$$= \int\int d\Omega' d\omega' w(\omega \leftarrow \omega')W(\Omega \leftarrow \Omega')c\sigma(\omega')\Phi(\mathbf{r},\Omega',\omega',t)$$
$$+ \Phi_0(\mathbf{r},\Omega,\omega)\,\delta(t),$$

which can be transformed to a pure integral form:

$$\Phi(\mathbf{r}, \Omega, \omega, t) = \Phi_0(\mathbf{r}, \Omega, \omega, t) + \int_0^\infty d\xi\ e^{-\sigma(\omega)\xi} \int\int d\Omega' d\omega'\ w(\omega \leftarrow \omega') W(\Omega \leftarrow \Omega') \sigma(\omega') \times \Phi(\mathbf{r} - \Omega\ \xi, \Omega', \omega', t - \xi/c).$$

Here $\Phi(\mathbf{r}, \Omega, \omega, t) d\mathbf{r} d\Omega d\omega$ is the mean number of photons at the moment t in volume $d\mathbf{r} = dxdydz$ flying into solid angle $d\Omega$ with frequencies in $[\omega, \omega + d\omega)$. Further, we use the standard assumptions [38].

- *Assumption* 1. The complete frequency redistribution takes place:

$$w(\omega \leftarrow \omega') = w(\omega).$$

- *Assumption* 2. The local thermodynamic equilibrium is valid:

$$\sigma(\omega) = C w(\omega).$$

Passing to the photon collision density

$$F(\mathbf{r}, \Omega, \omega, t) \equiv c\sigma(\omega)\Phi(\mathbf{r}, \Omega, \omega, t),$$

we arrive at

$$F(\mathbf{r}, \Omega, \omega, t) = F_0(\mathbf{r}, \Omega, \omega, t) + \int_0^\infty d\xi\ w(\omega)\sigma(\omega) e^{-\sigma(\omega)\xi} \int d\Omega' \int d\omega'\ W(\Omega \leftarrow \Omega') \times F(\mathbf{r} - \Omega\ \xi, \Omega', \omega', t - \xi/c).$$

Integrating over frequencies

$$F(\mathbf{r}, \Omega, t) = \int_0^\infty F(\mathbf{r}, \Omega, \omega, t) d\omega,$$

we obtain the following integral equation

$$F(\mathbf{r}, \Omega, t) - F_0(\mathbf{r}, \Omega, t) = \int_0^\infty d\xi\ p(\xi) \int d\Omega' W(\Omega \leftarrow \Omega')\ F(\mathbf{r} - \Omega\ \xi, \Omega', t - \xi/c),$$

where the kernel is given by

$$p(\xi) = \left\langle \sigma(\omega) e^{-\sigma(\omega)\xi} \right\rangle = \int_0^\infty w(\omega)\sigma(\omega) e^{-\sigma(\omega)\xi} d\omega.$$

Characteristic feature of resonance radiation propagation is the fact that absorption spectrum (as an approximation of thermodynamic equilibrium and total redistribution of frequency) is proportional to the spectrum of radiation

$$\sigma(\omega) = Cw(\omega).$$

Calculations lead us to the following asymptotic properties of the path length distribution.

$$P(\xi) = \left\langle e^{-\sigma(\omega)\xi} \right\rangle \sim \frac{\xi^{-\nu}}{\gamma\Gamma(1-\nu)},$$

$$p(\xi) = -P'(\xi) = \left\langle \sigma e^{-\sigma(\omega)\xi} \right\rangle \sim \frac{\upsilon\xi^{-\nu-1}}{\gamma\Gamma(1-\nu)}.$$

To satisfy this condition, we choose the distribution in the form of "fractional exponents":

$$p(\xi) = \gamma\xi^{\nu-1}E_{\nu,\nu}(-\gamma\xi^{\nu}), \tag{7.12}$$

where $E_{\alpha,\beta}(x)$ is the two-parameter Mittag-Leffler function (see for example (Appendix A.2.1)).

Generalized Boltzmann equation for the case of path length distribution (7.12) is obtained in the form

$$\left(\frac{\partial}{\partial t} + c\Omega\nabla\right)^{\nu} \Phi(\mathbf{r}, \Omega, t) + \mu\Phi(\mathbf{r}, \Omega, t)$$
$$= \mu \int W(\Omega \leftarrow \Omega')\Phi(\mathbf{r}, \Omega', t)d\Omega' + \left(\frac{\partial}{\partial t} + c\Omega\nabla\right)^{\nu-1} \left[\Phi_0(\mathbf{r}, \Omega)\,\delta(t)\right], \tag{7.13}$$

where $\mu = c\gamma$ and the operator

$$\left(\frac{\partial}{\partial t} + c\Omega\nabla\right)^{\nu} \Phi(\mathbf{r}, \Omega, t)$$
$$= \frac{1}{\Gamma(1-\nu)} \left(\frac{\partial}{\partial t} + c\Omega\nabla\right) \int_0^t \frac{\Phi(\mathbf{r} - c\Omega(t-\tau), \Omega, \tau)}{(t-\tau)^{\nu}} d\tau$$

is the fractional generalization of the material derivative.

Ordinary material derivative is determined as

$$\left(\frac{\partial}{\partial t} + \mathbf{c}\frac{\partial}{\partial \mathbf{r}}\right) f(\mathbf{r}, t) = \lim_{h\downarrow 0} \frac{f(\mathbf{r}, t) - f(\mathbf{r} - \mathbf{c}h, t - h)}{h}.$$

The operator

$$\mathsf{A} \equiv -\left(\frac{\partial}{\partial t} + \mathbf{c}\frac{\partial}{\partial \mathbf{r}}\right)$$

is the infinitesimal operator generating the semigroup of operators

$$\mathsf{T}^{-h} f(\mathbf{r}, t) = f(\mathbf{r} - \mathbf{c}h, t - h).$$

Consequently, we have for $(-\mathsf{A})^{\alpha}$

$$\begin{aligned}(-\mathsf{A})^{\alpha} f(\mathbf{r}, t) &= \left(\frac{\partial}{\partial t} + \mathbf{c}\frac{\partial}{\partial \mathbf{r}}\right)^{\alpha} f(\mathbf{r}, t) \\ &= \lim_{h \downarrow 0} \frac{\alpha}{\Gamma(1-\alpha)} \int_h^{\infty} s^{-\alpha-1} [f(\mathbf{r}, t) - f(\mathbf{r} - \mathbf{c}s, t - s)]\, ds.\end{aligned}$$

By integrating by parts, the last operator can be reduced to the form

$$\begin{aligned}&\left(\frac{\partial}{\partial t} + \mathbf{c}\frac{\partial}{\partial \mathbf{r}}\right)^{\alpha} f(\mathbf{r}, t) \\ &= \frac{1}{\Gamma(1-\alpha)} \left(\frac{\partial}{\partial t} + \mathbf{c}\frac{\partial}{\partial \mathbf{r}}\right) \int_{-\infty}^{t} \frac{f(\mathbf{r} - \mathbf{c}(t-\tau), \tau)}{(t-\tau)^{\alpha}} d\tau.\end{aligned}$$

If we consider the case of a space-homogeneous distribution of particles $\Phi(\mathbf{r}, \Omega, t) \equiv \Phi(\Omega, t)$ this operator becomes the fractional Riemann-Liouville derivative of order ν

$$\left(\frac{\partial}{\partial t} + c\Omega\nabla\right)^{\nu} \Phi(\mathbf{r}, \Omega, t) \mapsto {}_0\mathsf{D}_t^{\nu} \Phi(\Omega, t).$$

In a stationary problem, when the function $\Phi(\mathbf{r}, \Omega)$ does not depend on time,

$$\left(\frac{\partial}{\partial t} + c\Omega\nabla\right)^{\nu} \Phi(\mathbf{r}, \Omega) \mapsto c(\Omega\nabla)^{\nu} \Phi(\mathbf{r}, \Omega, t),$$

this operator represents fractional generalization of the directional derivative.

In one-dimensional case, when photons can fly only along of x-axis, the fractional generalization of the Boltzmann equation is of the form

$$\left(\frac{\partial}{\partial t} + c\Omega_x \frac{\partial}{\partial x}\right)^{\nu} \Phi(x, \Omega_x, t) + \mu\Phi(x, \Omega_x, t)$$

$$= \mu J(x, \Omega_x, t) + \left(\frac{\partial}{\partial t} + c\Omega_x \frac{\partial}{\partial x}\right)^{\nu-1} S(x, \Omega_x, t).$$

Using expressions for the collision integral

$$J(x, \Omega_x, t) = \frac{1}{2}\left[\Phi_+(x, t) + \Phi_-(x, t)\right] [\delta(\Omega_x - 1) + \delta(\Omega_x + 1)]$$

and for the one-dimensionally "isotropic" instantaneous source

$$S(x, \Omega_x, t) = \frac{1}{2}\,[\delta(\Omega_x - 1) + \delta(\Omega_x + 1)]\delta(x)\delta(t),$$

we obtain the following equations:

$$\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)^{\nu} \Phi_+(x,t) = \frac{\mu}{2}[\Phi_-(x,t) - \Phi_+(x,t)] + \frac{t^{-\nu}}{2\Gamma(1-\nu)}\,\delta(x - ct),$$

$$\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)^{\nu} \Phi_-(x,t) = \frac{\mu}{2}[\Phi_+(x,t) - \Phi_-(x,t)] + \frac{t^{-\nu}}{2\Gamma(1-\nu)}\,\delta(x + ct),$$

$$\Phi(x,t) = \Phi_+(x,t) + \Phi_-(x,t).$$

From these equations, one follows

$$\left[\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)^{\nu} + \left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)^{\nu} + \frac{2}{\mu}\left(\frac{\partial^2}{\partial t^2} - c^2\frac{\partial^2}{\partial x^2}\right)^{\nu}\right]\Phi(x,t)$$

$$= \frac{t^{-\nu}}{2\Gamma(1-\nu)}\,[\delta(x - ct) + \delta(x + ct)] - \frac{\nu\, t}{\mu\,[\Gamma(1-\nu)]^2}\left(\frac{c^2t^2 - x^2}{4c^2}\right)^{-1-\nu} \tag{7.14}$$

In the long-time asymptotics, Eq. (7.14) passes into the following one

$$\left[\left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)^{\nu} + \left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right)^{\nu}\right]\Phi(x,t) = \frac{t^{-\nu}}{2\Gamma(1-\nu)}\,[\delta(x - ct) + \delta(x + ct)],$$

solution of which is expressed in terms of elementary functions (see [61])

$$\Phi(x,t) = \frac{2\sin\pi\nu}{\pi}\,\frac{\left(1 - x^2/c^2t^2\right)^{\nu-1}}{(1 - x/ct)^{2\nu} + (1 + x/ct)^{2\nu} + 2\left(1 - x^2/c^2t^2\right)^{\nu}\cos\pi\nu}. \tag{7.15}$$

The quantitative analysis of the distribution can be performed using the method of moments. Due to finiteness of photon velocity, its position is a bounded random variable and according to Cramer's criterion, is unambiguously determined by its moments the probability density function. Moments can be found from a characteristic function. For the one-dimensional case, the Fourier-Laplace transformation of Eq. (7.14) leads to the following expression:

$$\widetilde{\Phi}(k,\lambda) = \frac{(\lambda - ick)^{\nu-1} + (\lambda + ick)^{\nu-1} + \dfrac{2\lambda}{\mu}\left(\lambda^2 + c^2k^2\right)^{\nu}}{(\lambda - ick)^{\nu} + (\lambda + ick)^{\nu} + \dfrac{2\lambda}{\mu}\left(\lambda^2 + c^2k^2\right)^{\nu}}.$$

In the asymptotics of small k and λ corresponding to the diffusion limit, we obtain

$$\widetilde{\Phi}(k,\lambda) \sim \frac{(\lambda - ick)^{\nu-1} + (\lambda + ick)^{\nu-1}}{(\lambda - ick)^{\nu} + (\lambda + ick)^{\nu}}.$$

Moments are determined by the relation

$$\tilde{m}_n(\lambda) = (-i)^n \frac{\partial^n \Phi(k,\lambda)}{\partial k^n}.$$

In isotropic case, Fourier-Laplace transformation of Eq. (7.13) leads to the following asymptotical form for $\widetilde{\Phi}(\mathbf{k}, t)$

$$\widetilde{\Phi}(\mathbf{k},t) \sim \frac{\int (\lambda - ic\Omega\,\mathbf{k})^{\nu-1}\, d\Omega}{\int (\lambda - ic\Omega\,\mathbf{k})^{\nu}\, d\Omega}.$$

Let us sketch the computational algorithm for the process, realized in the Monte Carlo code and used for computing the resonance radiation and excitation distribution in plasma. An important advantage of this method is simplicity of taking into account boundary conditions and inhomogeneity of the medium.

This method can be considered as statistical simulation of transport process characterized by known distributions of independent random parameters determining a random trajectory. A large set of independent trajectories allows us to estimate statistically a process under investigation.

The trajectory is simulated as follows.

1. A position of excited atom at the initial time is sampled. PDF of its radius-vector $\mathbf{R}_0$ is determined by given initial distribution of excitations $N(\mathbf{r}, 0)$ through the relation $p_{\mathbf{R}} = N(\mathbf{r}, 0)/\int N(\mathbf{r}, 0) d\mathbf{r}$.
2. The random time till V (under the integral symbol) photon emission is generated according to the exponential distribution $p_T(t) = \gamma e^{-\gamma t}$.
3. The frequency of emitted photon is sampled according to PDF $w(\omega)$.
4. A direction Ω of photon emission is generated. This direction is characterized by random variables $\cos\Theta$ and Ψ, which are uniformly distributed (in isotropic case) in $[-1, 1]$ and $[0, 2\pi)$, respectively.
5. A photon path length R is simulated in infinite homogeneous medium, $p_R(\xi) = \sigma(\omega) e^{-\sigma(\omega)\xi}$. If the whole segment $[\mathbf{R}_0, \mathbf{R}_0 + \Omega R]$ keeps within the volume occupied by plasma, its second end is taken as the position of another excited atom. This atom absorbs the photon. If this segment traverses absorbing or transparent outer boundary, the trajectory comes to the end.
6. For new excited atom, the waiting time till the moment of photon emission is simulated and all operations are repeated before the end of trajectory and so on.

From physical point of view, the fractional operator in the Boltzmann equation means that anomalously long free paths arising from time to time tear the trajectory into more or less localized clusters which look separated at any scale.

Solutions obtained by Monte Carlo simulations are presented in Figs. 7.1 and 7.2 (points). One-dimensional solutions listed in the previous subsection agree with the numerical results.

Numerical results obtained in frames of the Lévy walk model for the transparency of the layer are compared with the data of the classical experiments described in [9]. In the experimental study [9], the layer of mercury vapor was illuminated by resonance radiation 253.7 nm, and a layer transparency χ was measured for various concentrations of absorbing atoms, i.e., at different transmission densities. This transparency is the ratio of the transmitted radiation intensity to the incident intensity. The comparison shows a good agreement of numerical results with experimental data (Fig. 7.3).

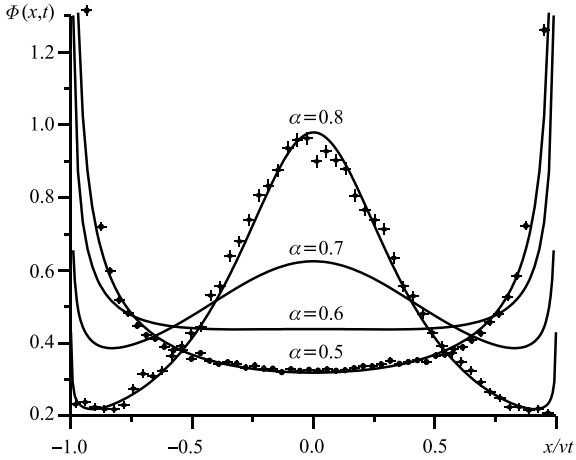


Fig. 7.1 Comparison of analytical solutions Eq. (7.15) (solid lines) with the results of Monte Carlo simulation (points) for isotropically one-dimensional case as functions of dimensionless variable x/ct

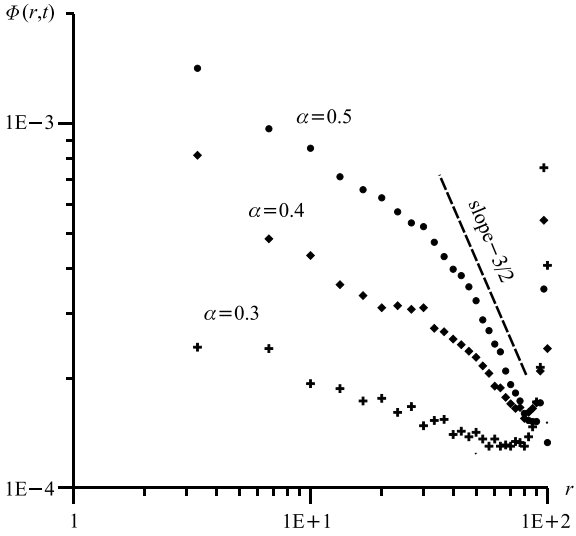


Fig. 7.2 Results of the Monte Carlo simulation for the spatial PDF $\Phi(r, t)$ in the isotropic 3D case

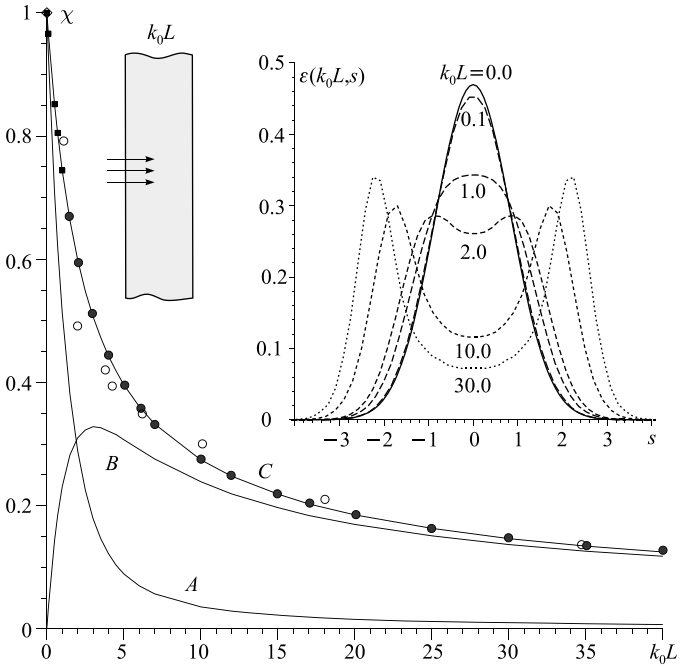


Fig. 7.3 The total current of the resonance radiation (C), its direct (A), and scattered (B) components behinds the layer of mercury steam of dimensionless thickness (k_0L). Open circles show experimental data, filled circles represent our numerical results. The inner part of the figure shows frequency spectra of scattered radiation

The numerical simulation was made for the Doppler spectral line shape. The total current passing through the layer consists of two parts: direct current of non-interacting photons (A), and indirect (scattered) current of multiple interacting photons (B). For large thickness the second part dominates. The inner figure shows the spectrum of outgoing radiation as a function of the rescaled frequency $s = 2(\nu - \nu_0)/\beta$. One can see that with growing thickness the spectrum changes from unimodal Gaussian spectrum to the spectrum of bimodal saddle shape. The saddle point falls while the depth grows. This means that mainly such photons leave the layer free paths of which are close to the thickness of the layer.

7.2 Turbulent Dynamics of Plasma

7.2.1 Diffusion in Plasma Turbulence

A decisive factor exerting influence on diffusion processes in plasma is turbulence. For this reason, the turbulence diffusion essentially varies from the ordinary molecular diffusion. The confinement time scaling in low confinement mode plasmas [28, 70], fast propagation and nonlocal transport phenomena [16, 26], and the non-Gaussianity and long-range correlations of fluctuations [12, 72] demonstrate anomalous ("non-diffusive", as they say sometimes) transport in plasmas. Anomalous diffusion character is observed in laboratory plasma devices [11, 12, 72], magnetized turbulent flows [39, 40], cosmic electrodynamics [15, 35, 36], cosmic rays propagation in Galaxy and so on.

The specific feature of particles motion in plasma is caused by inhomogeneous and continuously varying magnetic field. The motion of charges across the force lines is accompanied with the intensive rotation around these lines slowing down their travelling in space (the trap effects). Leaving such a trap, the particle can find itself in a more calm region with the low magnetic field and perform a distant fast flight through this region. The standard diffusion paradigm fails to describe such particularities of the process which make it quite different from the Brownian motion.

To reach an agreement with the observed features, del-Castillo-Negrete et al. [20] have proposed a model that incorporates in a unified way the space-time nonlocality, non-Gaussianity, and anomalous diffusive scaling believed to be present in fusion plasmas. This model is based on the CTRF asymptotic behavior and expressed in terms of fractional calculus.

Concluding the description of the Balescu hybrid kinetics equation for plasma (Eq. (1.21), Sect. 1.3.5 of Vol. 1) we mentioned that in the weak turbulence limit hereditarity dissapears and the process becomes Markovian. Whereas analysing strong turbulence ($\mathsf{Ku} \gg 1$) requires to account the retardation function $\Lambda(\tau)$ form. Passing to the Laplace images:

$$\lambda \widehat{n}(\mathbf{x}, \lambda) = \widehat{\Lambda}(\lambda) \Delta_2 \widehat{n}(\mathbf{x}, \lambda) + \widehat{S}(\mathbf{x}, \lambda) \tag{7.16}$$

and following the self-similarity principle, confirmable by numerical modeling, we choose the kernel transformant in the power function form

$$\widehat{\Lambda}(\lambda) = K \lambda^{1-\nu}, \quad 0 < \nu \leqslant 1.$$

The inverse Laplace transform of Eq. (7.16) leads to subdiffusion equation

$${}_0\mathsf{D}_t^{\nu} n(\mathbf{x}, t) = K \Delta_2 n(\mathbf{x}, t) + Q(\mathbf{x}, t).$$

Thus, memory influence slows down a diffusion process. The diffusion packet size increases as $t^{\nu/2}$. Within Balescu's theory the limit case $\nu \to 1$ corresponds to the

normal diffusion, which to some extent contradicts Monin's model, because the latter was derived on the additional assumption about the Markovian property of the process under consideration. An evident generalization of both approaches is the equation

$$ {}_0\mathsf{D}_t^{\nu} n(\mathbf{x}, t) = -K(-\Delta_2)^{\alpha/2} n(\mathbf{x}, t) + Q(\mathbf{x}, t), \tag{7.17} $$

whose fundamental solutions (i.e., when $Q(\mathbf{x}, t) = \delta(\mathbf{x})\delta(t)$) are expressed via fractional stable distributions and comprehend both subdiffusive ($\nu < \alpha/2$) and superdiffusive ($\nu > \alpha/2$) regimes.

The one-dimensional analog[2] of Eq. (7.17) was derived in Saichev and Zaslavsky [57] and rederived on the same assumption in del-Castillo-Negrete et al. [20].

To validate the fractional diffusion description of the transport in the plasma turbulence, del-Castillo-Negrete et al. [20] have carried out numerical calculations in frame of a model of three-dimensional, resistive, pressure-gradient-driven plasma turbulence. The model is based on the electrostatic approximation of the reduced resistive magnetohydrodynamic equations, and it describes the evolution of the [**E**, **B**] velocity streamfunction Φ and the pressure p,

$$ \left(\frac{\partial}{\partial t} + \mathbf{V} \cdot \nabla + \langle V_\theta \rangle \frac{1}{r} \frac{\partial}{\partial \theta} \right) \Delta_\perp \Phi = -\frac{1}{\eta m_i n_0 R_0} \Delta_\parallel \Phi + \frac{B_0}{m_i n_0 r_c} \frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \Delta_\perp^2 \Phi, $$

$$ \left(\frac{\partial}{\partial t} + \mathbf{V} \cdot \nabla + \langle V_\theta \rangle \frac{1}{r} \frac{\partial}{\partial \theta} \right) p = \frac{\partial \langle p \rangle}{\partial r} \frac{1}{r} \frac{\partial \Phi}{\partial \theta} + \chi_\perp \Delta_\perp p + \chi_\parallel \Delta_\parallel p. $$

Here η denotes the resistivity, μ the viscosity, $\chi_\perp$ and $\chi_\parallel$ the perpendicular and parallel thermal conductivities.

The authors used $25 \cdot 10^3$ tracer particles with initial conditions randomly distributed on θ and z on the surface a cylinder of radius $r = a/2$. In they evolution, the particles are advected by the [**E**, **B**] flow velocity **V** and either get trapped in eddies for long times, or jump over several eddies independently of each other. Comparison of fractional model ($\alpha = 3/4,\ \nu = 1/2$) results with turbulent transport calculations has demonstrated satisfactory fit.

7.2.2 *Stationary States and Fractional Dynamics*

Plasmas belong to long range systems every constituent of which is simultaneously undergone by action of microscopic "actors", resulting in a complex dynamical picture. Such systems often display a slow relaxation to equilibrium. They are in fact trapped in long-lasting out of equilibrium regimes, termed Quasi Stationary States. These regimes are considered in the framework of so called *Hamiltonian Mean Field* (HMF) model, which describes the coupled evolution of N rotators, populating the

[2] The three-dimensional version of Eq. (7.17) was derived and solved in Uchaikin and Zolotarev [62].

unitary circle and interacting via a cosines like potential. In the limit of infinite system size the discrete HMF model is described by the Vlasov equation for the evolution of the single particle distribution function.

To make this description more adequate to systems with long-range interactions, the so-called α-HMF model has been offered [18]. We touch here with its fractional modification which is developed in [65]. Introducing the governing Hamiltonian

$$H = \sum_{i=1}^{N} \left[\frac{p_i^2}{2} + \frac{1}{2\tilde{N}} \sum_{j \neq i}^{N} \frac{1 - \cos(q_i - q_j)}{||i - j||^{\alpha}} \right],$$

where q_i stands for the orientation of the rotor occupying the lattice position i, while p_i labels the conjugate momentum. The quantity $||i - j||$ denotes the shortest distance of the circle of circumference N and $\alpha \in [0, 1]$ (the case $\alpha = 0$ corresponds to the HMF). For N even

$$\tilde{N} = \left(\frac{2}{N} \right)^{\alpha} + 2 \sum_{i=1}^{N/2-1} \frac{1}{i^{\alpha}},$$

which guarantees extensivity of the system. The equations of motion derived from the above Hamiltonian are

$$\frac{dq_i}{dt} = p_i(t),$$

$$\frac{dp_i}{dt} = -C_i \sin q_i + S_i \cos q_i,$$

where

$$C_i = \frac{1}{\tilde{N}} \sum_{j \neq i} \frac{\cos q_j}{||i - j||^{\alpha}},$$

$$S_i = \frac{1}{\tilde{N}} \sum_{j \neq i} \frac{\sin q_j}{||i - j||^{\alpha}}.$$

For large N

$$\tilde{N} \sim \frac{2}{1 - \alpha} \left(\frac{N}{2} \right)^{1-\alpha}.$$

Passing from the discrete variables i, j to continuous ones, $x = i/N$ and $y = j/N$, we arrive at the quantities expressed in terms of fractional potentials

$$C(x) = \frac{1 - \alpha}{2^{\alpha}} \int_{-1/2}^{1/2} \frac{\cos q(y)}{|x - y|^{\alpha}} dy = 2^{-\alpha} \Gamma(2 - \alpha)\, {}_{-1/2}\mathbb{I}_{1/2}^{1-\alpha}[\cos q(x)].$$

A similar relation holds for the $S(x)$. Thus, the α-HMF dynamics characterizes the evolution of the scalar fields $q(x,t)$ and $p(x,t)$ via space-fractional integro-differential equations

$$\frac{\partial q}{\partial t} = p(x,t),$$

$$\frac{\partial p}{\partial t} = 2^{-\alpha}\Gamma(2-\alpha)\left[-\sin q(x)\,{}_{-1/2}\mathbb{I}_{1/2}^{1-\alpha}\cos q(x) + \cos q(x)\,{}_{-1/2}\mathbb{I}_{1/2}^{1-\alpha}\sin q(x)\right].$$

As the authors of the cited work notice, the microscopic dynamics of the particles in such states can be mapped into a pendulum motion and is hence integrable. Interestingly, the spatial organization is locally scale free. The functions $q(x,t)$ and $p(x,t)$ are thus very complicated along the spatial direction, while displaying a regular time evolution and no chaos.

Chechkin and Gonchar [13, 14] used the fractional Fokker-Planck equation to describe 3d-motion of a charge in magnetic field $\mathbf{B} = B\mathbf{e}_z$ taking into account a friction force $-\eta m\mathbf{v}$ and a random electric field $\mathbf{E}$. The latter was represented by homogeneous isotropic stationary Lévy white noise with intensity K and exponent α. The Lévy noise is a sequence of independent stationary increments of the Lévy motion just like the white Gaussian noise in a sequence of Brownian motion increments. Increment characteristic function is

$$\widetilde{p}(\mathbf{k}, \Delta t) = e^{-K|\mathbf{k}|^\alpha \Delta t}, \quad 0 < \alpha \leqslant 2.$$

When $\alpha \to 2$ the Lévy noise becomes the Gaussian one.

The corresponding equation for velocity distribution density $f(\mathbf{x}, \mathbf{v}, t)$ of a charged particle has the form:

$$\frac{\partial f}{\partial t} + \Omega[\mathbf{v}, \mathbf{e}_z]\nabla_{\mathbf{v}} f = \eta\nabla_{\mathbf{v}}(\mathbf{v}f) - K(-\Delta_{\mathbf{v}})^{\alpha/2} f.$$

With the Fourier transform

$$\widetilde{f}(\mathbf{k}, t) = \int d\mathbf{v} e^{i\mathbf{k}\cdot\mathbf{v}} f(\mathbf{v}, t)$$

it can be reduced to the form

$$\frac{\partial \widetilde{f}}{\partial t} + (\Omega[\mathbf{k}, \mathbf{B}] + \eta\mathbf{k})\nabla_{\mathbf{k}}\widetilde{f} = -K|\mathbf{k}|^\alpha \widetilde{f}.$$

The solution of the latter equation (with the initial condition $\mathbf{v}(0) = 0$) is given by the formula

$$\widetilde{f}(\mathbf{k}, t) = \exp\left\{-(K/\alpha\eta)\left(1 - e^{-\alpha\eta t}\right)|\mathbf{k}|^\alpha\right\},$$

so the velocity distribution is expressed via the isotropic Lévy-Feldheim density:

$$f(\mathbf{v},t)=\left[(K/\alpha\eta)\left(1-e^{-\alpha\eta t}\right)\right]^{-3/\alpha}g_3(\left[(K/\alpha\eta)\left(1-e^{-\alpha\eta t}\right)\right]^{-1/\alpha}\mathbf{v};\alpha).$$

Within short times

$$f(\mathbf{v},t)\sim(Kt)^{-3/\alpha}g_3((Kt)^{-1/\alpha}\mathbf{v};\alpha),$$

and we deal with the Lévy motion, but in the long times limit we come to stationary velocity distribution

$$f(\mathbf{v},\infty)=(K/\alpha\eta)^{-3/\alpha}g_3((K/\alpha\eta)^{-1/\alpha}\mathbf{v};\alpha).$$

When $\alpha=2$ it coincides with the equilibrium Maxwell distribution, in other cases it essentially differs from it by power type asymptotic tails at large velocities:

$$f(\mathbf{v},\infty)\propto|\mathbf{v}|^{-\alpha-d},\quad|\mathbf{v}|\to\infty.$$

7.2.3 *Perturbative Transport*

Perturbative experiments in magnetically confined fusion plasmas have shown that edge cold pulses travel to the center of the device on a time scale of a few microseconds, i.e., much faster than expected on the basis of ordinary local diffusive transport. To estimate a role of non-locality as compared to nonlinearity, del-Castillo-Negrete et al. [21] have used a transport model in frame of which the flux-temperature interrelation has the non-local character,

$$q(x,t)=-\chi n(x,t)\frac{\partial}{\partial x}\int_a^b K(x-x')T(x',t)dx'$$

and its kernel is an algebraic decaying function, so that the interrelation can be represented as

$$q(x,t)=-\frac{\chi n(x,t)}{\Gamma(2-\alpha)}\frac{\partial}{\partial x}\left[l\int_a^x\frac{T(x',t)dx'}{(x-x')^{\alpha-1}}+r\int_x^b\frac{T(x',t)dx'}{(x'-x)^{\alpha-1}}\right],$$

where $1<\alpha<2$, and

$$l=-\frac{1-\theta}{2\cos(\alpha\pi/2)},\quad r=-\frac{1+\theta}{2\cos(\alpha\pi/2)},\quad-1<\theta<1.$$

Using the regularization procedure,

$$ {}_a\mathsf{D}_x^{\alpha-1} T(x) \mapsto {}_a\mathsf{D}_x^{\alpha-1}[T(x) - T(a) - T'(a)(x-a)] \equiv {}_a^{\alpha}\mathscr{D}_x T(x) $$

and

$$ {}_x\mathsf{D}_b^{\alpha-1} T(x) \mapsto {}_x\mathsf{D}_b^{\alpha-1}[T(x) - T(b) - T'(b)(b-x)] \equiv {}_x^{\alpha}\mathscr{D}_b T(x) $$

allows to define the finite-size domain non-local flux as

$$ q(x,t) = -\chi n(x,t)[l\, {}_a^{\alpha}\mathscr{D}_x - r\, {}_x^{\alpha}\mathscr{D}_b T(x)]T(x,t). $$

As authors conclude at the end of the cited article, this relatively simple linear model has allowed them to make evident the crucial and previously overlooked role played by non-locality, independent of further potential complications due to non-linearity. However, to make further progress it would be of interest to solve the fractional diffusion model incorporating a critical gradient condition.

For numerical investigation of the interplay of normal and fractional diffusion on the whole x-axis, del-Castillo-Negrete [19] has involved the equation

$$ {}_0^{\nu}\mathsf{D}_t f(x,t) = \chi_\alpha[l\, {}_{-\infty}\mathsf{D}_x^{\alpha} + r\, {}_{\infty}\mathsf{D}_x^{\alpha}] f(x,t) + \chi_2 \frac{\partial^2 f(x,t)}{\partial x^2}. $$

Its Green's function, that is the solution $f(x,t)$ under initial condition $f(x,0) = \delta(x)$, has been found in the form

$$ G(x,t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} E_\nu(\widehat{\Lambda}_\alpha(k) t^\nu) dk, $$

where

$$ \widehat{\Lambda}_\alpha(k) = -\chi_\alpha |k|^\alpha \left[1 + 2i\theta \tan(\alpha\pi)\mathrm{sign}(k)\right] - \chi_2 k^2. $$

7.2.4 *Electron-Acoustic Waves*

When the plasma consists of two species of electrons with different temperatures, electron-acoustic waves are observed. Their propagation plays an important role both in laboratory and in space plasma. El-Wakil et al. [22–24] have theoretically investigated a homogeneous unmagnetized collisionless plasma consisting of a cold electron fluid and isothermal ions with two different temperatures obeying Boltzmann type distributions. As these authors state, the reason for using the fractional differential equation lies in presence of non-conservative forces.

The starting point of this article is the classical system of the normalized one-dimensional equations

$$ \frac{\partial n_e}{\partial t} + \frac{\partial (u_e n_e)}{\partial x} = 0, $$

$$\frac{\partial u_e}{\partial t} + u_e \frac{\partial u_e}{\partial x} - \frac{\partial \phi}{\partial x} = 0,$$
$$\frac{\partial^2 \phi}{\partial x^2} - n_e + n_1 + n_h = 0,$$
$$n_1 = \mu \exp[-\phi/(\mu + \nu\beta)],$$

and

$$n_h = \nu \exp[-\phi/(\mu + \nu\beta)].$$

In these equations, $n_e \equiv n_e(x,t)$ is the cold electron density normalized by equilibrium value n_{e0}, $u_e \equiv u_e(x,t)$ is their fluid velocity normalized by $C_{\text{eff}} = (T_{\text{eff}}/m_e)^{1/2}$, n_1 and n_h are the densities with the low (T_1) and high (T_h) temperatures respectively, $\beta = T_1/T_h$, $\phi \equiv \phi(x,t)$ is the electric potential normalized by T_{eff}/e, x is the space coordinate normalized to the effective Debye length, and t is the time variable normalized to the inverse of the cold electron plasma frequency. The neutrality condition reads $\mu + \nu = 1$.

According to the general idea of the perturbation theory, a small dimensionless parameter ε is introduced and used for the power series expansion of the physical quantities n_e, u_e, and ϕ about their equilibrium variables:

$$n_e(\xi,\tau) = 1 + \varepsilon n_1(\xi,\tau) + \varepsilon^2 n_2(\xi,\tau) + \varepsilon^3 n_3(\xi,\tau) + \cdots,$$
$$u_e(\xi,\tau) = \varepsilon u_1(\xi,\tau) + \varepsilon^2 u_2(\xi,\tau) + \varepsilon^3 u_3(\xi,\tau) + \cdots,$$

and

$$\phi(\xi,\tau) = \varepsilon\phi_1(\xi,\tau) + \varepsilon^2\phi_2(\xi,\tau) + \varepsilon^3\phi_3(\xi,\tau) + \cdots,$$

with $\tau = \varepsilon^{3/2}t$, $\xi = \varepsilon(x - \lambda t)$, λ being the wave speed normalized by C_{eff} and the boundary conditions $n_e = 1, \quad u_e = 0, \quad \phi = 0$ as $|\xi \to \infty|$. In the first-order approximation, substituting these series into the above equations system yields the Korteweg-de Vries equation for $\Phi \equiv \phi_1(\xi,\tau)$:

$$\frac{\partial \Phi}{\partial \tau} + A\Phi\frac{\partial \Phi}{\partial \xi} + B\frac{\partial^3 \Phi}{\partial \xi^3} = 0, \tag{7.18}$$

where

$$A = \frac{\lambda^3}{2}\left[\frac{\mu + \nu\beta^2}{(\mu + \nu\beta)^2} - \frac{3}{\lambda^4}\right], \quad B = \frac{\lambda^3}{2}, \quad \lambda = \pm 1.$$

Involving a potential function $V(\xi,\tau)$, such that

$$V_\xi(\xi,\tau) \equiv \frac{\partial V(\xi,\tau)}{\partial \xi} = \Phi(\xi,\tau),$$

one can show that the regular KdV Eq. (7.18) is generated by the Lagrangian

$$L(V_\tau, V_\xi, V_{\xi\xi}) = -\frac{1}{2}V_\xi V_\tau - \frac{1}{6}AV_\xi^3 + \frac{1}{2}V_{\xi\xi}^2.$$

El-Wakil et al. [25] generalize it to the fractional form as

$$L({}_0V_\tau^{(\alpha)}, V_\xi, V_{\xi\xi}) = -\frac{1}{2}{}_0V_\tau^{(\alpha)}V_\tau - \frac{1}{6}AV_\xi^3 + \frac{1}{2}V_{\xi\xi}^2, \quad 0 \leqslant \alpha < 1.$$

According to Agraval's [1, 2] method, the variational principle

$$\delta\int_{\mathbb{R}} d\xi \int_0^b d\tau \left[{}_\tau\mathsf{D}_b\left(\frac{\partial L}{\partial\, {}_0V_\tau^{(\alpha)}}\right) - \frac{\partial}{\partial\xi}\left(\frac{\partial L}{\partial V_\xi}\right) + \frac{\partial^2}{\partial\xi^2}\left(\frac{\partial L}{\partial V_{\xi\xi}}\right)\right]\delta V = 0$$

leads to the fractional equation

$${}_0\mathbb{D}_b^\alpha\Phi + A\Phi\frac{\partial\Phi}{\partial\xi} + B\frac{\partial^3\Phi}{\partial\xi^3} = 0,$$

where ${}_0\mathbb{D}_b^\alpha = (1/2)\,{}_0\mathsf{D}_\tau^\alpha + (-1)^k\,{}_\tau\mathsf{D}_b^\alpha$, $k-1 < \alpha \leqslant k$ is the Riesz fractional operator.

For solving this non-linear fractional equation, El-Wakil et al. [24] used variational-iteration method. The estimated electric field of the electrostatic structures turned out to be in good agreement with the observed solitary wave structures in the Earth?s plasma sheet boundary layer. Authors conclude that this model could be useful to interpret the compressive and rarefactive bipolar pulses of the Broadband Electrostatic Noise type emissions in the regions where the electron beams are not present.

7.3 Wandering of Magnetic Field Lines

7.3.1 Normal Diffusion Model

A magnetic field line $\mathbf{H}(\mathbf{r})$ is a curve that is tangential to $\mathbf{H}(\mathbf{r})$ at every point $\mathbf{r}$ along the line. Mathematical description of such a line may be given by the parametric equations

$$\frac{dx}{H_x} = \frac{dy}{H_y} = \frac{dz}{H_z}. \tag{7.19}$$

In the presence of magnetic fluctuations in plasma, the direction of this line wanders due to the magnetic field, $\delta\mathbf{H}$, associated with the fluctuations. Thus, if two test particles begin their motion along the lines at the same point but at two different instants of time, each of them meets a different detailed spectrum of fluctuations, and so its guiding center follows a different path. A large number of such particles

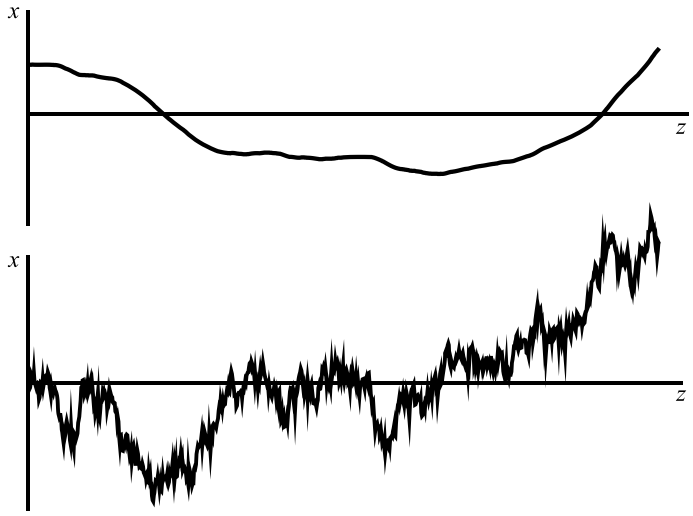


Fig. 7.4 A wandering magnetic line (upper panel) and its Brownian counterpart (lower panel)

starting from the same point at different times spread out from the mean direction of the magnetic field, as illustrated in Fig. 7.4. This phenomenon is referred to as *wandering of magnetic field lines*.

Let the average field be $\mathbf{H}_0$,

$$\langle \mathbf{H} \rangle = \mathbf{H}_0 = H_0 \mathbf{e}_z,$$

then $\langle \delta \mathbf{H} \rangle = 0$ and in case of homogeneous turbulence

$$\langle H_i(z_2) H_j^*(z_1) \rangle = \langle \delta H_i(z_2) \delta H_j^*(z_1) \rangle = C_{ij}(\zeta), \quad \zeta = |z_2 - z_1|,$$

where $C_{ij}(z_2, z_1)$ is the correlation function of the line fluctuations. In the absence of detailed information, the fluctuations are often assumed to be gaussian,

$$C_{ij}(\zeta) \propto \exp(-\zeta^2 / 2\zeta_0^2),$$

where ζ_0 is a *correlation length*. Such representation of C_{ij} provokes to interpret the random lines as trajectories of a normal diffusion in the xy-plane process. Let $N(\mathbf{r}_\perp, z) dx dy$ be the mean number of field lines intersecting the plane $z = \text{const}$ in the rectangle $(x, x + dx) \times (y, y + dy)$. According to this model, the density of the lines obeys the diffusion equation

$$\frac{\partial N}{\partial z} = K_L \Delta_\perp N(\mathbf{r}_\perp, z) \tag{7.20}$$

with

$$K_L = \frac{1}{4}\lim_{\Delta z\to 0}\frac{\langle[\Delta \mathbf{R}_\perp]^2\rangle}{\Delta z} = \frac{1}{2}\lim_{\Delta z\to 0}\frac{\langle[\Delta X]^2\rangle}{\Delta z} = \frac{1}{2}\lim_{\Delta z\to 0}\frac{\langle[\Delta Y]^2\rangle}{\Delta z}.$$

Evidently, Eq. (7.20) describing normal diffusion takes place only under condition that

$$\left\langle[\Delta X(z)]^2\right\rangle = \frac{1}{H_0^2}\int_0^z dz_1\int_0^z dz_2\langle\delta H_x(z_1)\delta H_x(z_2)\rangle \sim z, \quad z\to 0. \tag{7.21}$$

However, as will be stated below, this condition is not necessary realized.

7.3.2 Shalchi-Kourakis Equations

Assuming homogeneous turbulence, one can reduce the field-line mean square displacement (MSD) as follows:

$$\langle[\Delta X(z)]^2\rangle = \frac{1}{H_0^2}\int_0^z dz_1\int_0^z dz_2 C_{xx}(\zeta) = \frac{2}{H_0^2}\int_0^z (z-\zeta)C_{xx}(\zeta)d\zeta.$$

In order to continue evaluation of the MSD, we pass to the Fourier representation of the field fluctuations,

$$\delta H_x(\mathbf{r}) = \frac{1}{(2\pi)^3}\int d\mathbf{k}\widetilde{\delta H}_x(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}$$

and

$$C_{xx}(z) = \frac{1}{(2\pi)^6}\int d\mathbf{k}\int d\mathbf{k}'\left\langle\widetilde{\delta H}_x(\mathbf{k})\widetilde{\delta H}_x^*(\mathbf{k}')e^{-i[\mathbf{k}\cdot\mathbf{R}(z)-\mathbf{k}'\cdot\mathbf{R}(0)]}\right\rangle.$$

Applying Corrsin's independence hypothesis [17] and assuming that the magnetic fields for different wave vectors $\mathbf{k}$ and $\mathbf{k}'$ are uncorrelated, lead to

$$C_{xx}(z) = \int\left\langle e^{-i\mathbf{k}\cdot\Delta\mathbf{R}(z)}\right\rangle P_{xx}(\mathbf{k})d\mathbf{k}$$

with

$$P_{xx}(\mathbf{k}) = (2\pi)^{-6}\langle\widetilde{\delta H}_x(\mathbf{k})\widetilde{\delta H}_x^*(\mathbf{k})\rangle.$$

Following [51], we shall use Gaussian approximation for the characteristic function,

$$\left\langle e^{-i\mathbf{k}\cdot\Delta\mathbf{R}(z)}\right\rangle = \exp\left\{-(1/2)\langle[\Delta X(z)]^2\rangle k_x^2 - (1/2)\langle[\Delta Y(z)]^2\rangle k_y^2 - ik_\parallel z\right\}.$$

For axisymmetric turbulence

$$\left\langle [\Delta X(z)]^2 \right\rangle = \left\langle [\Delta Y(z)]^2 \right\rangle,$$

$$\left\langle [\Delta X(z)]^2 \right\rangle = \frac{2}{H_0^2} \int d\mathbf{k} P_{xx}(\mathbf{k}) \int_0^z (z-\zeta) \cos(k_\parallel \zeta) e^{-(1/2)\langle [\Delta X(\zeta)]^2 \rangle k_\perp^2} d\zeta. \quad (7.22)$$

Consecutive differentiation yields

$$\frac{d}{dz} \left\langle [\Delta X(z)]^2 \right\rangle = \frac{2}{H_0^2} \int d\mathbf{k} P_{xx}(\mathbf{k}) \int_0^z \cos(k_\parallel \zeta) e^{-(1/2)\langle [\Delta X(\zeta)]^2 \rangle k_\perp^2} d\zeta \quad (7.23)$$

and

$$\frac{d^2}{dz^2} \left\langle [\Delta X(z)]^2 \right\rangle = \frac{2}{H_0^2} \int d\mathbf{k} P_{xx}(\mathbf{k}) \cos(k_\parallel z) e^{-(1/2)\langle [\Delta X(z)]^2 \rangle k_\perp^2}. \quad (7.24)$$

The Shalchi-Kourakis equations (7.22)–(7.24) provide a general basis for a quantitative description of stochastic magnetic field in plasma in terms of the field-line wandering representation.

7.3.3 Theoretical Evidence of Superdiffusion Wandering

Bieber et al. [10] proposed the *hybrid slab 2D* model within which the turbulent fields are described as a superposition of a one-dimensional *slab model* ($\mathbf{k} \parallel \mathbf{H}_0$) and a two-dimensional (2D) model ($\mathbf{k}_\perp \mathbf{H}$). The correlation function has the form

$$P_{xx}(\mathbf{k}) = P_{xx}^{\mathrm{sl}}(\mathbf{k}) + P_{xx}^{\mathrm{2D}}(\mathbf{k})$$

with

$$P_{xx}^{\mathrm{sl}}(\mathbf{k}) = g^{\mathrm{sl}}(k_\parallel) \frac{\delta(k_\perp)}{k_\perp}$$

and

$$P_{xx}^{\mathrm{2D}}(\mathbf{k}) = g^{\mathrm{2D}}(k_\perp) \frac{\delta(k_\parallel)}{k_\perp} \left(1 - \frac{k_x^2}{k^2}\right).$$

Here

$$g^{\mathrm{sl}}(k_\parallel) = \frac{C(\nu)}{2\pi} \frac{l_{\mathrm{sl}} \delta H_{\mathrm{sl}}^2}{(1 + k_\parallel^2 l_\parallel^2)^\nu}$$

is the slab wave spectrum (l_{sl} is the slab-bendover-scale, 2ν is the inertial-range spectral index, δH_{sl}^2 is the strength of slab turbulence, and $C(\nu) = \frac{1}{2\sqrt{\pi}} \frac{\Gamma(\nu)}{\Gamma(1-\nu/2)}$). The 2D wave spectrum is given by a similar expression:

$$g^{2D}(k_\perp) = \frac{2C(\nu)}{\pi} \frac{l_{2D}\delta H_{2D}^2}{(1+k_\perp^2 l_\perp^2)^\nu}$$

(see for details [51]). Observe that for pure slab turbulence, Eq. (7.23) immediately reveals the normal diffusion behavior:

$$\langle [\Delta X(z)]^2 \rangle = 2K^{sl} z, \quad K_{sl} = \pi C(\nu) l_{sl} \frac{\delta H_{sl}^2}{H_0^2}, \quad z > 0.$$

Combining both components into the hybrid model reduces Eq. (7.24) to the form

$$\frac{d^2}{dz^2} \langle [\Delta X(z)]^2 \rangle = \frac{2}{H_0^2} C_{xx}^{sl}(z) + \frac{2\pi}{H_0^2} \int_0^\infty dk_\perp g^{2D}(k_\perp) e^{-(1/2)\langle [\Delta X(z)]^2 \rangle k_\perp^2}. \quad (7.25)$$

One can estimate that the fist term in this sum has the asymptotic behaviour $\propto z^{\nu-1} e^{-z/l_{sl}}$, $z \to \infty$. Neglecting the 2D spectrum change in the integrand, we reduce Eq. (7.25) to

$$\frac{d^2}{dz^2} \langle [\Delta X(z)]^2 \rangle = \frac{2\pi g^{2D}(0)}{H_0^2} \int_0^\infty dk_\perp e^{-(1/2)\langle [\Delta X(z)]^2 \rangle k_\perp^2} = \frac{2\pi}{H_0^2} \sqrt{\frac{\pi}{2}} \frac{g^{2D}(0)}{\sqrt{\langle [\Delta X(z)]^2 \rangle}}.$$

The last one has the solution

$$\langle [\Delta X(t)]^2 \rangle = \left[\sqrt{\frac{\pi}{2}} \frac{9\pi g_{2D}(0)}{2H_0^2} \right]^{2/3} z^{4/3}.$$

Of course, this result is valid only for long distances z. Consequently, one may conclude (as it is done in [51]) that the wandering of magnetic field lines in the hybrid slab/2D model of turbulence reveal superdiffusion behaviour.

7.3.4 *Fractional Brownian Motion for Simulating Magnetic Lines*

Let us compare the shape of magnetic field line depicted in the upper panel of Fig. 7.4 with the Brownian form, corresponding to the standard diffusion equation (the lower panel). The curves are quite different. It is impossible to imagine the motion of a charged particle along the Brownian a line. The smooth line looks more natural, but it doesn't belong to the Brownian ensemble. How should the ensemble be modified in order to become more appropriate as a magnetic field lines ensemble?

Recall that Brownian motion $B(t)$ can be defined as Gaussian process with mean value $\langle B(t) \rangle = B(0) = 0$ correlation function

$$\langle B(t)B(s)\rangle = (1/2)\left(|t| + |s| - |t-s|\right).$$

The fractional Brownian motion (fBm) $B^H(t)$, $H \in (0,1)$ differs from Bm by correlation function having form

$$\langle B(t)B(s)\rangle = (1/2)\left(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\right).$$

The *Hurst number* H determines the self-similarity order of the process

$$B^H(at) \stackrel{d}{=} a^H(t)$$

and a type of memory of the fractional motion: short memory of antipersistent (subdiffusion) with $H < 1/2$, absence of memory (ordinary Bm) in case $H = 1/2$, and long memory of persistent (superdiffusion type) when $H > 1/2$.

As shown in [66], the PDF $f^H(x,t)$ of the fBm process obeys the diffusion equation with time-dependent diffusivity KT^{2H-1}:

$$\frac{\partial f^H(x,t)}{\partial t} = 2HKt^{2H-1}\frac{\partial^2 f^H(x,t)}{\partial x^2}.$$

Under initial condition

$$f^H(x,0) = \delta(x)$$

it has the solution

$$f^H(x,t) = \frac{1}{\sqrt{4\pi Kt^{2H}}}\exp\left(-\frac{x^2}{4Kt^{2H}}\right),$$

$$\sigma = \sqrt{2K}t^H.$$

Typical realizations of the fractional Brownian motion are shown in Fig. 3.6 in Vol. I. One can clearly see the smoothing influence of integrating Brownian curve.

7.3.5 Compound Model

One of important application of the diffusion theory under generalization is the problem propagation of charged *cosmic rays* (CR) through interstellar magnetic field. As a first approximation, the model of isotropic ordinary diffusion has been used for a long time (see the next chapter). Generally speaking, this is a not so bad approximation, especially if the diffusivity is taken as an increasing function of the particle energy, but the nature of this phenomenon is not quite clear yet. Why the angular flux of CR observed near the Earth is so isotropic, although their sources

are belived to be local? There are suggested a few mechanisms of isotropization of cosmic rays giving various relaxation times. Two following hypothesis can be considered as extremal in some sense: (i) scattering charged particles on small-scale fluctuations of weak-turbulent magneto-active interstellar plasma and (ii) random walk of particles between large-scale magnetic clouds. In the second case the most part of time is taken by the particle for motion along magnetic field lines which seem to be smooth enough but having some points of brakes, sharp bends, intersections, magnetic clotting, plugs, and traps. The particle can slip through this irregularity and continue its motion in the direction being close to the previous one, or be reflected from this region and start motion in the opposite direction. The lines can in their turn perform the "diffusion dance".

Such patterns of the process inspire the idea of separate consideration of these two stochastic motions: the particle motion along the line and the line deflection with regard to the initial straight line. This model called the *compound diffusion model* was firstly suggested by Russian astrophysicist Getmantsev [27] and later investigated in details in [67–69, 71]. We will call the *standard compound model* the case when longitudinal and lateral processes are described in frame of standard diffusion theory. In this case, the typical path of the particle along the line during time t is $s(t) \simeq \sqrt{D_\parallel t}$, and the lateral displacement of the line at this point is $r(s) \simeq \sqrt{D_L s}$, where $D_\parallel$ is the longitudinal diffusivity of the particle, while D_L denotes the lateral diffusivity of the line. As a result, we arrive at the equality

$$r(s(t)) \simeq \sqrt{L\sqrt{\kappa_\parallel t}} \simeq L^{1/2}\kappa_\parallel^{1/4} t^{1/4},$$

obtained by Getmantsev [27]. He was first who offered the separated description of longitudinal and lateral diffusion processes.

Let us denote by $p_\parallel(z, t \leftarrow z_0, t_0)$ the longitudinal propagator, that is the pdf of longitudinal coordinate counting along magnetic line at the moment t under the initial condition $(z(t_0)) = z_0$, and $q(\mathbf{r} - \mathbf{r}_0|z - z_0)$ is the PDF of the lateral displacement of magnetic line at the point z ($\mathbf{r}$ is a two-dimensional vector). The resulting lateral PDF is given by the lateral propagator:

$$p_\perp(\mathbf{r}, t \leftarrow \mathbf{r}_0, t_0) = \int_{-\infty}^{\infty} q(\mathbf{r} - \mathbf{r}_0|z - z_0) p_\parallel(z, t \leftarrow z_0, t_0) dz. \tag{7.26}$$

In frame of standard compound model, the longitudinal propagator and PDF of lateral displacements of magnetic field line obey the ordinary one-dimensional and two-dimensional diffusion equations respectively:

$$\frac{\partial p_\parallel(z, t \leftarrow z_0, t_0)}{\partial t} = D_\parallel \frac{\partial^2 p_\parallel(z, t \leftarrow z_0, t_0)}{\partial z^2} + \delta(z - z_0)\delta(t - t_0), \tag{7.27}$$

$$\frac{\partial q(\mathbf{r} - \mathbf{r}_0, z - z_0)}{\partial z} = D_L \Delta_\mathbf{r} q(\mathbf{r} - \mathbf{r}_0, z - z_0) + \delta(\mathbf{r} - \mathbf{r}_0)\delta(z - z_0). \tag{7.28}$$

The combined solution for the particle concentration in 3d-space is

$$n(\mathbf{r},t) = n_0 \frac{\exp\left(-\frac{(z-z_0)^2}{4D_\parallel t}\right)}{\sqrt{4\pi D_\parallel t}} \frac{\exp\left(-\frac{(\mathbf{r}-\mathbf{r}_0)^2}{4D_L(z-z_0)}\right)}{4\pi D_L(z-z_0)}.$$

Zaburdaev [71] has shown that the random-walk process just considered, namely, the one whose argument is also a random-walk process, obeys a subdiffusion scaling, as follows. Considering an initial particle distribution as uniform along the z-axis[3] and keeping in mind, that in this situation, particles with different coordinates z_0 can occur at the same magnetic field line, he writes down the total concentration N as

$$N(\mathbf{r},t) = 2n_0 \int_{-\infty}^{z} \frac{\exp\left(-\frac{(z-z_0)^2}{4D_\parallel t}\right)}{\sqrt{4\pi D_\parallel t}} \frac{\exp\left(-\frac{(\mathbf{r}-\mathbf{r}_0)^2}{4D_L|z-z_0|}\right)}{4\pi D_L|z-z_0|} dz_0.$$

Taking the Fourier-Laplace transform with respect to space-time variables yields

$$\widehat{\widetilde{N}}(\mathbf{k},t) = \frac{1}{\sqrt{\lambda D_\parallel}} \frac{n_0}{\sqrt{\lambda/D_\parallel} + D_L k^2}.$$

In terms of space-time variables, this expression relates to the fractional differential equation

$${}_0\mathsf{D}_t^{1/2} N(\mathbf{r},t) = D_L\sqrt{\pi D_\parallel}\,\Delta_\perp N(\mathbf{r},t) + \frac{n_0}{\sqrt{t}}\delta(\mathbf{r}).$$

The author concludes his article by noting that "the problem considered above constitutes one of the few examples of the rigorous derivation of an equation with fractional derivatives and thereby shows the naturalness and importance of this approach to describing stochastic processes in which the subdiffusive behavior of the particles is an inherent feature of the physical phenomenon" [71].

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Chapter 8
Cosmic Rays



8.1 Unbounded Anomalous Diffusion

8.1.1 Space-Fractional Equation for Cosmic Rays Diffusion

The origin of cosmic rays (CR) has been intriguing scientists since 1912 when Victor Hess carried out his famous balloon flight to measure the ionization rate in the upper atmosphere. The total energy density of CR (nuclei, protons, pions, electrons, and gamma-rays) turns out to be comparable to that of the interstellar radiation and magnetic fields, and turbulent motions of the interstellar gas. This makes CR one of the essential factors determining the dynamics and processes in the interstellar medium (ISM). Major cosmic sources, or more strictly, accelerators are supernova remnants, and to the less degree are pulsars, compact objects in close binary systems, and stellar winds. CR are stored in the Galaxy for tens of millions years before escaping into the intergalactic space. In order to explain such a long time one has to assume that trajectory of CR has the form of very tortuous lines, something like Brownian paths. Thus, it was very natural to start modeling this process with the use of the diffusion model described by the standard diffusion equation [27]. In framework of this model, only a few numerical parameters determinate the transport of CR, and it seemed sufficient for the first time, the more so because we don't know a real picture of interstellar magnetic field with the exception of its turbulent character and rough estimations of its values. Nevertheless, this model forms the theoretical basis for computing isotopic abundances, energy spectra, and angular distribution, and pursuit to reach an agreement between theoretical and observation data motivates the further development of the model.

The standard diffusion equation describes the Brownian motion, which is a self-similar stochastic process equally looking at different scales. Any part of its sample trajectory consists of infinite number of turning-points: it is a continuous but nowhere differentiable curve. Followers of the diffusion approach ignore this peculiarity of

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Nonlinear Physical Science, https://doi.org/10.1007/978-981-96-0582-8_8

the Brownian motion and take the model only because of its random (or chaotic) character as if it would be the only random process. Of course, it is not the case.

As can be concluded from modern astronomical observations [6], magnetic field lines have long smooth parts up to a few tenths or even hundreds parsecs [31] which may produce long parts of cosmic rays trajectories. These parts interpreted as free paths have random length R, and there are no any reason to consider them as exponentially distributed random variables. Quite the contrary, it is more logical to take for them algebraic distribution, $\mathsf{P}(R > r) \propto r^{-\alpha}, \quad \alpha \in (0, 1)$. Limiting α by upper value 1 becomes clear if we recall that in case $\alpha > 1$ the equation becomes again of classical diffusion form at long-time asymptotics, whereas the case $\alpha < 1$ reveals self-similarity, in other words, fractality. The choice of the fractional diffusion model for CR modeling was supported by information that interstellar magnetic field heterogeneities were testified to take large-scaled (fractal) character [35]. The supernova remains analysis shows the presence in this region of gas components with different physical parameters ($T_e \sim 5 \div 10^6$ K, $n_e \sim 0.1 \div 10^3$ m^{-3}), which can be the sequence of extreme heterogeneity of interstellar medium. These facts and other data concerning the heterogeneities of matter density ρ and magnetic field intensity $H \propto \rho^q$, $q \sim 1/3 \div 1/2$ [58] within the length range 100–150 pc, giving rise to uncertainties of diffusion model to be applicable to cosmic rays transfer description, stimulated the use of the superdiffusion model, based on the fractional Laplacian operator. Without account of energy losses and nuclear interactions, the corresponding equation is of the form

$$\left[\frac{\partial}{\partial t} + D(E)(-\Delta)^{\alpha/2}\right] N(\mathbf{r}, t, E) = S(\mathbf{r}, t, E). \tag{8.1}$$

Here $N(\mathbf{r}, t, E)$ is the concentration of the CR particles (protons) with energy E, $D(E)$ their time-dependent diffusivity, and $S(\mathbf{r}, t; E)$ their source density. I'd like to stress again, that using this equation, we neglect effects of slowdown, acceleration, and collision with interstellar matter, this is why the energy E does not change its value in process of CR motion and serves as a parameter.

The common CR source is represented as a set of points sources related to supernova bursts producing CR during some time T. The CR concentration at the observation point $\mathbf{r}$ is expressed through the Green function as

$$N(\mathbf{r}, t; E) = \int_{\infty}^{t} dt' \int d\mathbf{r}' G(\mathbf{r} - \mathbf{r}', t - t'; E) S(\mathbf{r}', t', E)$$

8.1.2 The "Knee"-Problem

Our first application of this equation to CR physics was connected with the energy spectrum problem: explanation of the observed steepening of the all-particle spectrum at $3 \cdot 10^{15}$ eV (the "knee") and the flattering of the spectrum around 10^{19} eV (the "ankle"). This problem was animatedly discussed at that time (see, for example, the reviews by [6, 15, 20, 30, 55, 56] and others). Our model turned out to be able to give a variant of such explanation based on the hypothesis of the fractal structure of the galaxy. Increasing of the exponent η in power representation of the $E^{-\eta}$ spectrum while passing from the region $\sim 10^2$ Gev/nucleon into that of $\sim 10^5$ Gev/nucleon was interpreted as a consequence of the anomalously large free paths of particles (Lévy "flights"). The absence of the definite scales bounded these regions opens the way of using the self-similarity hypothesis, which is natural for the turbulent type medium and leads to the power character of paths distribution appearing the Lévy flights.

Let us consider the contribution of one point source with the power type spectrum $S(E) = S_0 E^{-p}$ placed at point $\mathbf{r}$ with respect to the observation point and emitting CR with a constant rate during time-interval $[0, T]$. The observed concentration of CR at time t is

$$N(\mathbf{r}, t; E) = S_0 E^{-p} \int_{\max[0,t-T]}^{t} G(\mathbf{r}, \tau; E) d\tau$$

Observe that we consider the random motion of CR in space as a diffusion of particles against some background, but the background itself is supposed to be time-independent, isotropic, and homogeneous because all effects of inhomogeneities are included into the diffusion coefficient $D(E)$; for this reason G is a space-time translation invariant function $G(r, t; E)$ satisfying the equation

$$\frac{\partial G}{\partial t} = -D(E)(-\Delta)^{\alpha/2} G(r, t; E) + \delta(\mathbf{r})\delta(t)$$

and zero boundary conditions at infinity. It is expressed in terms of 3d isotropic Lévy-Felgheim distribution as

$$G(r, t; E) = [D(E)t]^{-3/\alpha} g_3(r[D(E)t]^{-1/\alpha}; \alpha) 1(t), \tag{8.2}$$

$$g_3(r; \alpha) = \begin{cases} \dfrac{1}{2\pi^2 \alpha} \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \Gamma\left(\frac{2n+3}{\alpha}\right) r^{2n}, & r \to 0; \\ \dfrac{1}{2\pi^2 r} \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n!} \Gamma(n\alpha + 2) \sin\left(\frac{\pi \alpha n}{2}\right) r^{-n\alpha-2}, & r \to \infty. \end{cases}$$

Graphs of the stable densities are shown in Fig. 8.1.

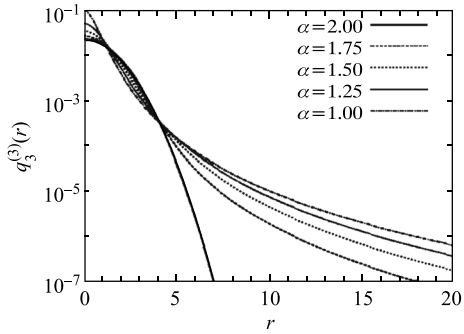


Fig. 8.1 Three-dimensional stable densities $q_3^{(\alpha)}(r)$ for different values of α

Using this Greens function, we can represent a contribution of one single point source as

$$N(r,t;E) = S_0 E^{-p}[D(E)]^{-3/\alpha} \int_{\max[0,t-T]}^{t} \tau^{-3/\alpha} g_3(r[D(E)\tau]^{-1/\alpha};\alpha)d\tau. \quad (8.3)$$

Taking into account the energy dependence diffusivity $D(E) = D_0 E^{\delta}$, representing the sought-for spectrum as $N = N_0 E^{-\eta}$ and using the interrelations

$$\frac{dg_m(r;\alpha)}{dr} = -2\pi r g_{m+2}(r;\alpha)$$

between isotropic stable densities, one obtains the spectral exponent for observed particles:

$$\eta = p + \frac{\delta}{\alpha}\Xi, \quad (8.4)$$

where

$$\Xi = 3 - \frac{2\pi r^2}{[D(E)]^{2/\alpha}} \frac{\int_{\max[0,t-T]}^{t} \tau^{-5/\alpha} g_5\left(r[D(E)\tau]^{-1/\alpha};\alpha\right) d\tau}{\int_{\max[0,t-T]}^{t} \tau^{-3/\alpha} g_3\left(r[D(E)\tau]^{-1/\alpha};\alpha\right) d\tau}. \quad (8.5)$$

Let E_0 be a solution of the equation $\Xi(E) = 0$. One can see from Eqs. (8.4) and (8.5) that at $E = E_0$ the spectral exponent for observed particles η is equal to the spectral exponent p for particles produced by the source. At $E \ll E_0$ or $E \gg E_0$, we have $\eta_{E\ll E_0} \approx p - \delta$, and $\eta_{E\gg E_0} \approx p + \delta$ correspondingly. In other words, the spectral exponent of observed particles increases with energy on 2δ. Let us also note that this property of the energy spectrum is missing in the normal diffusion case.

From experimental values at $E = E_0$, $\eta_{E\ll E_0}$, and $\eta_{E\gg E_0}$ one can find the main parameters of the model: D_0, δ, and p. For example, taking $E_0 = 3 \cdot 10^4$ GeV, $\eta_{E\ll E_0} \approx 2.65$, and $\eta_{E\ll E_0} \approx 3.15$ [6, 91], we obtain $p \approx 2.9$ and $\delta \approx 0.25$. Our

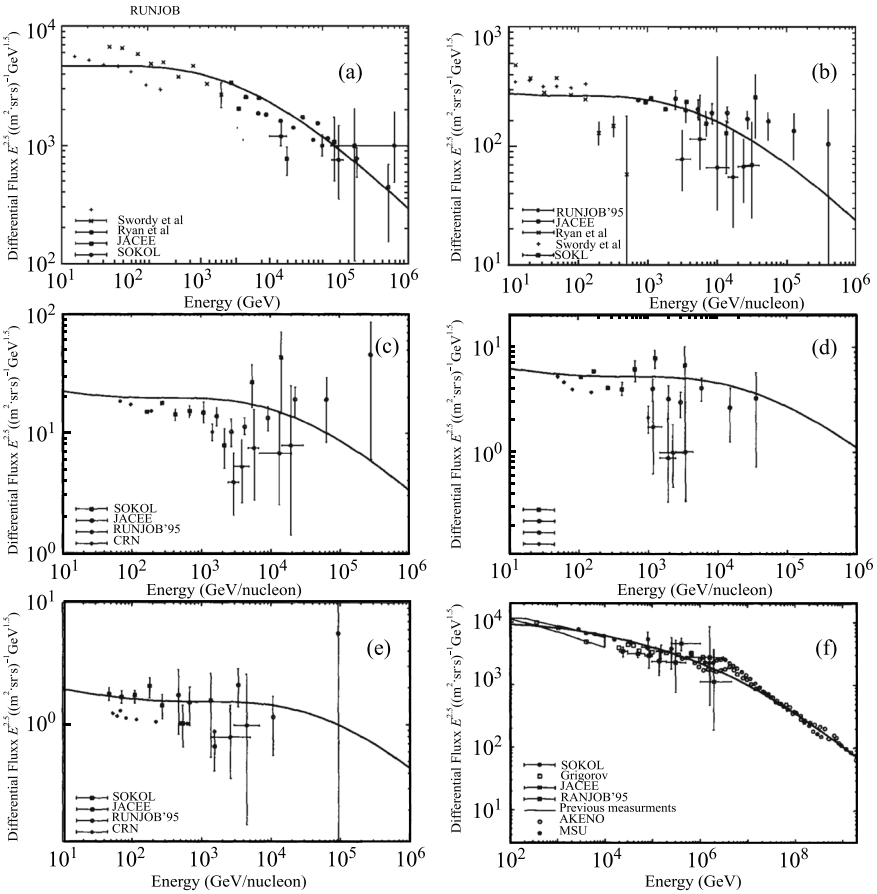


Fig. 8.2 Comparison of our calculation of spectra (solid curve) with experimental data from direct measurements (review (Shibata 1999), the MSU [25], and AKENO [54, 91] arrays). We have: **a** proton spectrum, **b** helium nuclei spectrum, **c** CNO nuclei spectrum, **d** Ne-Si nuclei spectrum, **e** Fe group nuclei spectrum, and **f** all particle spectrum

calculations show also that the best fit of experimental data of direct measurements [2, 28, 29, 59] (Shibata 1999) of nuclei H, He, CNO, Ne-Si, and Fe-group (found in review (Shibata 1999)) and all particles [25, 54, 91] (Shibata 1999) are reached at $\alpha \approx 5/3$, $\delta \approx 0.25$, and $p \approx 2.9$ ($r \sim 200$ pc, $t \sim 10^5$ years, $T \sim t$) (see Fig. 8.2).

Thus, the following conclusion was derived from these calculations in our article [37, 38].

1. The "knee" in primary cosmic ray spectrum at $E \sim 3 \cdot 10^6$ GeV and the observed distinction in spectral exponent of protons and other nuclei can be explained by superdiffusion propagation of cosmic rays in fractal interstellar medium.

2. The source spectral exponent found from experimental data in this approach is $p \approx 2.9$.
3. The energy dependence of particle spectra does not contradict the hypothesis that the sources of particles with $E \lesssim 10^{18}$eV are the bursts of nearby supernovae in the last $\sim 10^5$ years.

8.1.3 Trapping CR by Stochastic Magnetic Field

The anomality of the CR diffusion process considered above is produced by torn and wrinkled (fractal) patterns of interstellar magnetic field. By virtue of the fractal structure, the particles have a chance to make a long path up to the next crucial changing its direction (Lévy-flight). However, stochastic character of ISMF is responsible to another important effect: trapping effect [6, 17]. This effect produces the subdiffusional (i.e., time-fractional) term in the diffusion equation.

This point of view arouses in connection to the problem of perpendicular diffusion due to the effect of the large scale random magnetic field. The idea that the CR diffusion in the Galaxy can be considered as a compound process when charged particles perform diffusion along random magnetic force lines which by turn execute a random walk in space was suggested as far back as half a century [26]. It was shown that this assumption leads to the subdiffusion regime the diffusion packet width growing with time as

$$\Delta(t) \propto t^{1/4}.$$

The relative role of these processes depends on plasma parameters. It is necessary to say that the actions of random walk of the magnetic field lines and random particles motions are not independent. So, strictly speaking, these processes should be studied together. Such consideration first carried out by Chuvilgin and Ptuskin [13] has led to the fractional subdiffusion equation. This work is of particular interest for us, because its authors started the derivation with the classical kinetic

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f + \mathbf{F} \cdot \frac{\partial f}{\partial \mathbf{p}} = 0, \quad \mathbf{F} = q\mathbf{E} + \frac{q}{c}[\mathbf{v}, \mathbf{H}],$$

for the distribution density function $f(\mathbf{r}, \mathbf{p}, t)$. Let us sketch in the way of derivation rather more detailed.

Considering $\mathbf{H}$, $\mathbf{E}$, $\mathbf{F}$, and f as random fields with the averaged (over small scale fluctuations) values labeled by $\langle \cdots \rangle$, and marking the variations by subscript 1,

$$\mathbf{H} = \langle \mathbf{H} \rangle + \mathbf{H}_1, \quad \langle \mathbf{H}_1 \rangle = 0,$$

and so on, one can derive the equation for fluctuating part of the particle distribution. In the framework of quasilinear approximation of a small-scale weak-turbulent plasma [64]

$$f_1 \ll \langle f \rangle,$$

one finds

$$\frac{\partial \langle f \rangle}{\partial t} + \mathbf{v}\nabla\langle f \rangle + \langle \mathbf{F} \rangle \frac{\partial \langle f \rangle}{\partial \mathbf{p}} = R.$$

The right-hand side of the equation is a collision term describing scattering of particles by small-scale field fluctuations. It is expressed through the convolution of $\mathbf{F}_1 \cdot \partial\langle f \rangle/\partial\mathbf{p}$ with corresponding Green's function of the linear equation (see for details [18]). Approximating this term by $-(f - \bar{f})/\tau$ with

$$\bar{f} = \bar{f}(\mathbf{r}, p, t) = \frac{1}{4\pi} \int_{4\pi} f(\mathbf{r}, p\Omega, t) d\Omega$$

yields

$$\frac{\partial \langle f \rangle}{\partial t} + \mathbf{v} \cdot \nabla\langle f \rangle + \frac{q}{c}[\mathbf{v}, \langle \mathbf{H} \rangle] \cdot \frac{\partial \langle f \rangle}{\partial \mathbf{p}} = -\frac{\langle f \rangle - \bar{f}}{\tau}, \tag{8.6}$$

where τ is the characteristic scattering time on small-scale inhomogeneities and

$$n(\mathbf{r}, p, t) \equiv \langle \bar{f} \rangle$$

is an isotropic part of the distribution density f averaged over small-scale fluctuations (angle bracket) and over all directions (bar accent). Let us denote another (anisotropic) part by $n_1(\mathbf{r}, \mathbf{p}, t)$, so that

$$\langle f \rangle = n + n_1, \quad \overline{n_1} = 0.$$

One can make certain that for large distances ($R \gg |\mathbf{v}|\tau$) and long times ($t \gg R/|\mathbf{v}|$) being of interested in cosmic ray problems, the distribution function is nearly isotropic ($\delta n \ll n$). In frame of these assumptions, averaging Eq.(8.6) over angles on assumption leads to the equation

$$\frac{\partial n}{\partial t} + \nabla(\overline{\mathbf{v} n_1}) = 0,$$

which can be reduced to the ordinary diffusion equation

$$\frac{\partial n}{\partial t} - \nabla_i \kappa_{ij} \nabla_j \mathbf{n} = 0$$

with the diffusion tensor κ_{ij} containing a longitudinal component

$$\kappa_{\parallel} = \frac{\mathbf{v}^2 \tau}{3},$$

perpendicular component

$$\kappa_\perp = \frac{\kappa_\parallel}{1 + (\omega_H \tau)^2},$$

and asymmetric component

$$\kappa_A = -\frac{\kappa_\parallel \omega_H \tau}{1 + (\omega_H \tau)^2}.$$

One can see from here that at $\omega_H \tau \gg 1$ (the case of strong magnetization), the ratios $\kappa_\perp/\kappa_\parallel \simeq (\omega_h \tau)^{-2} \ll 1$, $\kappa_A/\kappa_\parallel \simeq (\omega_H \tau)^{-1} \ll 1$, and particle displacement across the average magnetic field is small.

Now, we assume that magnetic field is time-independent ("frozen") and its lines deviate only slightly from rectilinear lines representing the average field $\mathbf{H}_0 = \text{const}$. For these reason, the length parameter along the field line can be approximated by the value of displacement along $\mathbf{H}_0$. Magnetic field large scale fluctuations $\Delta\mathbf{H} = \mathbf{H} - \mathbf{H}_0$ give rise to the corresponding fluctuations in κ and n. Denoting large-scale averaging by $\langle\langle \cdots \rangle\rangle$ and deviations from these averages by $\Delta\kappa$ and Δn, one can write

$$\frac{\partial N}{\partial t} - \nabla_i \langle\langle \kappa_{ij} \rangle\rangle \nabla_j N - \nabla_i \langle\langle \Delta\kappa_{ij} \nabla_j \Delta n \rangle\rangle = 0.$$

The fluctuations Δn contain rapid random spatial changes and undergo slow changes as well. Performing the Fourier transformation over rapid space variables and keeping only slow space variables, Chuvilgin and Ptuskin [13] obtained a closed kinetic equation for perpendicular diffusion of charge particles in the random magnetic field $\Delta\mathbf{H}(\mathbf{r}) = \mathbf{A}(\mathbf{r}) H_0,\ A \ll H_0,\ \mathbf{r} = (\mathbf{r}_\perp, z)$ (see Eq. (B11) in their article)

$$\frac{\partial N(\mathbf{r},t)}{\partial t} - (1 - \langle\langle A^2 \rangle\rangle)\kappa_\parallel \frac{\partial^2 N(\mathbf{r},t)}{\partial z^2}$$
$$-\langle\langle A^2 \rangle\rangle \frac{\kappa_\parallel}{2} \Delta_\perp \left[N(\mathbf{r},t) - \frac{1}{2} \int_1^\infty N(\mathbf{r}, t - yL^2/\kappa_\parallel) y^{-3/2} dy \right] = 0,$$

where L is the correlation length of field fluctuations. Using the integral

$$\int_1^\infty y^{-3/2} dy = 2,$$

one can transform the expression in square brackets as follows:

$$\left[N(\mathbf{r}_\perp, t) - \frac{1}{2} \int_1^\infty N(\mathbf{r}_\perp, t - yL^2/\kappa_\parallel) y^{-3/2} dy \right]$$
$$= \frac{1}{2} \int_1^\infty \left[N(\mathbf{r}_\perp, t) - N(\mathbf{r}_\perp, t - yL^2/\kappa_\parallel) \right] y^{-3/2} dy$$
$$= L\sqrt{\frac{\pi}{\kappa_\parallel}} \left\{ \frac{1}{\Gamma(-1/2)} \int_{L^2/\kappa_\parallel}^\infty \left[N(\mathbf{r}_\perp, t - \tau) - N(\mathbf{r}_\perp, t) \right] \tau^{-3/2} d\tau \right\}.$$

As noted by Chuvilgin and Ptuskin, the compound-diffusion equation is valid on time intervals $t \gg t_d = L^2/\kappa_\parallel$. In this case, the lower limit of integration can be

replaced by 0 and contents of the curly brackets become the semi-derivative (in the Marchaud form) with respect to time. By noting

$$\frac{\langle\langle A^2\rangle\rangle\kappa_\parallel\sqrt{t_d}}{2}=\left(\frac{\Delta H}{H_0}\right)^2\frac{L\sqrt{\kappa_\parallel}}{2}=\sqrt{\frac{\kappa_\parallel}{\pi}}K_L=\frac{1}{\sqrt{\pi}}K_\perp$$

and

$$(1-\langle\langle A^2\rangle\rangle)\kappa_\parallel=K_\parallel$$

with $K_L=\langle\Delta\mathbf{r}_\perp^2\rangle/(2\Delta z)$, $K_\perp=\sqrt{\kappa_\parallel}K_L$ and $K_\parallel$ being effective coefficients for the magnetic line diffusion, cross-field compound diffusion, and parallel to the mean magnetic field diffusion of particles respectively, we arrive at the time-fractional equation coinciding with Eq. (2.51) from (Webb et al. [90]):

$$\frac{\partial N}{\partial t}=K_\parallel\frac{\partial^2 N}{\partial z^2}+K_{\perp 0}\mathsf{D}_t^{1/2}\Delta_\perp N(\mathbf{r}_\perp,z,t).$$

Inserting $\delta(\mathbf{r}_\perp)\delta(z)\delta(t)\equiv\delta(\mathbf{r})\delta(t)$ into its left-hand side, we obtain the equation for its Greens function $G(\mathbf{r},t)$. Performing the Fourier-Laplace transformation with respect to spatial and time variables,

$$G(\mathbf{r}_\perp,z,t)\mapsto\widetilde{G}(\mathbf{k}_\perp)=\int_{\mathbb{R}^2}d\mathbf{r}_\perp\int_{-\infty}^{\infty}dz\int_0^\infty dt e^{i(\mathbf{k}_\perp\mathbf{r}_\perp+k_z z)-\lambda t}G(\mathbf{r}_\perp,z,t),$$

and solving the obtained algebraic equation

$$[\lambda+K_\parallel k_z^2+\lambda^{1/2}K_\perp k_\perp^2]\widetilde{G}(\mathbf{k}_\perp,k_z,\lambda)=1,$$

we obtain after inverting

$$G(\mathbf{r}_\perp,z,t)=\frac{1}{2\pi i}\int_C\frac{\exp\left\{-(\lambda/K_\parallel)^{1/2}[z^2+r_\perp^2K_\parallel/(K_\perp\lambda^{1/2})]^{1/2}\right\}}{4\pi K_\perp\lambda^{1/2}[z^2+r_\perp^2K_\parallel/(K_\perp\lambda^{1/2})]^{1/2}}e^{\lambda t}d\lambda.$$

If we are interested in the perpendicular diffusion only, the diffusion equation is reduced to

$$\frac{\partial N_\perp}{\partial t}=K_{\perp 0}\mathsf{D}_t^{1/2}\Delta_\perp N_\perp(\mathbf{r}_\perp,t),$$

or

$${}_0\mathsf{D}_t^{1/2}N_\perp(\mathbf{r}_\perp,t)=K_\perp\Delta_\perp N_\perp(\mathbf{r}_\perp,t)+S(\mathbf{r}_\perp,0)\delta_{1/2}(t),$$

and the Green function becomes

$$G_\perp(\mathbf{r}_\perp,t)=\int_{-\infty}^{\infty}G(\mathbf{r}_\perp,z,t)dz=\frac{1}{(2\pi)^2 i\,K_\perp\sqrt{t}}\int_{\mathbb{C}}K_0\left(\frac{r_\perp s^{1/4}}{\sqrt{K_\perp\sqrt{t}}}\right)e^s\frac{ds}{\sqrt{s}},$$

where K_0 is a modified Bessel function of the second kind of order zero. These representations for both Green's functions are given in Webb et al. [90]; Eqs. (3.7) and (3.23) in Vol. I.

Alternative representations can be obtained by direct applying the explicit expressions for one-dimensional Green's functions (Sect. 5.1.9 in Vol. I) to the Fourier transformed diffusion equations. So, for three-dimensional diffusion we have

$$\frac{d\widetilde{G}}{dt} + K_\perp k_{\perp 0}^2 \mathsf{D}_t^{1/2} \widetilde{G}(\mathbf{k}_\perp, k_z, t) + K_\parallel k_z^2 \widetilde{G}(\mathbf{k}_\perp, k_z, t) = \delta(t).$$

Observe that at $t \to \infty$ this equation is reduced to

$$K_\perp k_{\perp 0}^2 \mathsf{D}_t^{1/2} \widetilde{G}(\mathbf{k}_\perp, k_z, t) + K_\parallel k_z^2 \widetilde{G}(\mathbf{k}_\perp, k_z, t) = \delta_{1/2}(t).$$

Inserting $\alpha = 1,\ \beta = 1/2, a = 1,\ b = K_\perp k_\perp^2$, and $c = K_\parallel k_z^2$ into Eq. (5.5) of vol. 1 yields

$$\widetilde{G}(\mathbf{k}_\perp, k_z, t) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (K_\parallel k_z^2)^n t^n E_{1/2,1+n/2}^{(1/2)}(-K_\perp k_\perp^2 \sqrt{t}),$$

and

$$G(\mathbf{r}_\perp, z, t) = \frac{1}{(2\pi)^3} \int \int e^{-i(\mathbf{k}_\perp \mathbf{r}_\perp + k_z)} \widetilde{G}(\mathbf{k}_\perp, k_z, t) d\mathbf{k}_\perp dk_z.$$

Hence, the second-order spatial moments follow:

$$\langle z^2 \rangle = -\left(\frac{\partial^2 \widetilde{G}}{\partial k_z^2} \right)_{\mathbf{k}=0} = 2K_\parallel t,$$

$$\langle r^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = -\left(\frac{\partial^2 \widetilde{G}}{\partial k_x^2} + \frac{\partial^2 \widetilde{G}}{\partial k_y^2} \right)_{\mathbf{k}=0} = 8(K_\perp/\sqrt{\pi}) t^{1/2}.$$

These equations demonstrate us ordinary diffusion of cosmic ray particles along the magnetic field but subdiffusion in perpendicular directions.

Observe that the main conclusion we came up is the presence of time-fractional derivatives, indicating to existence of a trapping mechanism.

8.1.4 Bifractional Anomalous CR Diffusion

Let us come back to isotropic diffusion. Including the trapping mechanism into Eq. (8.1) by substitution of fractional derivative operator for the integer-order one yields bifractional diffusion equation

$$\left[b_0 \mathsf{D}_t^{\beta} + a(-\Delta)^{\alpha/2}\right] N(\mathbf{r}, t) = \delta(\mathbf{r})\delta_{\nu}(t)$$

(we introduce here the additional coefficient b compensating the time-dimension and, for the brevity, omit the constant parameter E). The reader who reads sequentially this book does not need any additional explanation. But for the reader who wants to understand the physical sense immediately, without reference to the previous chapters, the following arguments may be useful.

For adequate understanding of this equation, we pass to Fourier-Laplace variables,

$$N(\mathbf{r}, t) \mapsto \tilde{p}(\mathbf{k}, \lambda) = \int_0^{\infty} dt\ e^{-\lambda t} \int d\mathbf{r}\ e^{i\mathbf{k}\mathbf{r}} p(\mathbf{r}, t).$$

The equation becomes

$$[b\lambda^{\beta}] + a|\mathbf{k}|^{\alpha}\ \tilde{N}(\mathbf{k}, \lambda) = b\lambda^{\beta-1}.$$

Let us rewrite it in the following asymptotically ($\lambda \to 0$, $|\mathbf{k}| \to 0$) equivalent form:

$$[1 - (1 - b\lambda^{\beta})(1 - a|\mathbf{k}|^{\alpha})]\ \tilde{N}(\mathbf{k}, \lambda) = \frac{1 - (1 - b\lambda^{\beta})}{\lambda}. \tag{8.7}$$

The terms $1 - b\lambda^{\beta}$ and $1 - a|\mathbf{k}|^{\alpha}$ can be considered as main asymptotical terms in integral transforms of some time and space probability densities $q(t)$ and $p(\mathbf{r})$

$$\tilde{q}(\lambda) = \int_0^{\infty} e^{-\lambda t} q(t) dt \sim 1 - b\lambda^{\beta}, \quad \lambda \to 0 \tag{8.8}$$

and

$$\tilde{p}(\mathbf{k}) = \int_{\mathbb{R}^d} e^{-i\mathbf{k}\mathbf{r}} p(\mathbf{r}) d\mathbf{r} \sim 1 - a|\mathbf{k}|^{\alpha}, \quad |\mathbf{k}| \to 0. \tag{8.9}$$

Thus, Eq. (8.9) can be considered as the asymptotical representation of some "exact" equation

$$[1 - \tilde{q}(\lambda)\tilde{p}(\mathbf{k})]\ \tilde{N}(\mathbf{k}, \lambda) = \frac{1 - \tilde{q}(\lambda)}{\lambda},$$

looking in time-space variables as

$$N(\mathbf{r}, t) = \int_0^t \int d\mathbf{r}'\ q(t') p(\mathbf{r}') N(\mathbf{r} - \mathbf{r}', t - t') + \delta(\mathbf{r}) Q(t), \tag{8.10}$$

where

$$Q(t) = \int_t^{\infty} q(t') dt'.$$

Equation (8.10) describes the following jump process. A particle placed at origin $\mathbf{r} = 0$ at time $t = 0$ stays there during a random time T, which is distributed with the density $q(t)$. Evidently, PDF for random position of the particle at observation time t conditioned on $T > t$ is $\delta(\mathbf{r})$, but conditioned on $T \in (t', t' + dt') < t$ is $\int p(\mathbf{r}')N(\mathbf{r} - \mathbf{r}', t - t')d\mathbf{r}'\, q(t')dt'$. Dividing the interval $(0, t)$ into elements Δt; and using the total probability rule, we can write

$$N(\mathbf{r}, t) = \sum_i N(\mathbf{r}, t|T \in \Delta t_i)\mathsf{P}(T \in \Delta t_i) + N(\mathbf{r}, t|T > t)\mathsf{P}(T > t).$$

Taking into account that

$$N(\mathbf{r}, t|T \in \Delta t_i) = \int d\mathbf{r}' p(\mathbf{r}')N(\mathbf{r} - \mathbf{r}', t - t_i),$$

$$\mathsf{P}(T \in \Delta t_i) \cong q(t_i)\Delta t_i,$$

$$N(\mathbf{r}, t|T > t) = \delta(\mathbf{r}),$$

$$\mathsf{P}(T > t) = Q(t) = \int_0^t q(t)dt$$

and passing from summing to integrating in time, we confirm the sense of Eq. (8.9).

Thus, one should conclude that the bifractional equation describes a CTRF process, posessing following properties: (1) waiting time and jump vector are independent of each other, and (2) the jumps are instantaneous, the particle disappears at point $\mathbf{r}$ at time t and appears at point $\mathbf{r} + \mathbf{R}'$ at the same time t independently of the displacement distance $\mathbf{r}'$. When we are dealing with a normal diffusion ($\alpha = 2$, $\beta = 1$), random variables T and $\mathbf{R}$ become negligibly small in the asymptotical process but in case of anomalous diffusion ($\alpha < 2$ or/and $\beta < 1$) conditions (8.7)–(8.8) generate long tail distributions of inverse power type, and some of random summands T_j, $\mathbf{R}_j$ are observable at any scales.

Recalculating the energy spectra on the base of the bifractional equation [39, 42, 43] we obtain for the Green function

$$G(\mathbf{r}, t) = (Dt^\beta)^{-3/\alpha}\Psi_3^{(\alpha,\beta)}\left(r(Dt^\beta)^{-1/\alpha}\right), \tag{8.11}$$

where

$$\Psi_3^{(\alpha,\beta)}(r) = \int_0^\infty g_3^{(\alpha)}(r\tau^\beta)g_+(\tau; \nu)\tau^{3\beta/\alpha}d\tau$$

is the fractional stable density (see Fig. 8.3); the energy spectrum from an instantaneous source

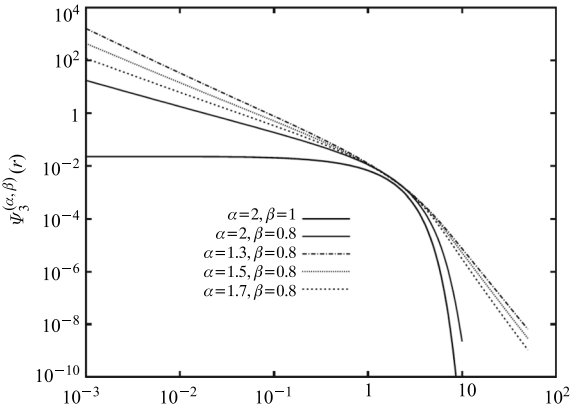


Fig. 8.3 Graphs of $\Psi_3^{(\alpha,\beta)}(r)$ normalized to a common point

$$N(r,t,E) = S_0 E^{-p}[D(E)t^{\beta}]^{-3/\alpha}\Psi_3^{(\alpha,\beta)}(r[D(E)t^{\beta}]^{-1/\alpha}) \propto \begin{cases} E^{-p+\delta}, & E \to 0; \\ E^{-p-\delta}, & E \to \infty, \end{cases}$$

the energy spectrum from a finite duration source

$$N(r,t,E) = \frac{S_0 E^{-p}}{D(E)^{3/\alpha}} \int_{\max[0,t-T]}^{t} \tau^{-3\beta/\alpha}\Psi_3^{(\alpha,\beta)}(|r|[D(E)\tau^{\beta}]^{-1/\alpha})$$

$$\propto \begin{cases} E^{-p+\delta}, & E \to 0; \\ E^{-p-\delta/\beta}, & E \to \infty. \end{cases}$$

Numerical calculations [36] have shown that under condition

$$\alpha = 1.7,\ \beta = 0.8,\ p = 2.9,\ \delta = 0.27,\ D_0 = (1 \div 4) \cdot 10^3\ \mathrm{pc}^{1.7}\mathrm{y}^{0.8},$$

this model is in a position to explain the observed change in spectral exponents of proton and other nuclei and mass composition at $E = 10^2 \div 10^5$ GeV/nucleon. The contribution of the nearby ($r \leq 1$ kpc) relatively young ($t < 10^5$ y) defines the spectrum in the high-energy region and provides the "knee". The only one problem remained unsolved: the level of anisotropy given by this model turned out much higher than observations showed. To overcome this trouble, the experiments with various parameters of the model were continued. In the course of those investigations, Lagutin and Tyumentsev [40] changed the parameters from values $\alpha = 1.7,\ \beta = 1 \div 0.8$ used in our first works in 2001 to $\alpha = 0.3$ and $\beta = 0.8$ in 2004; thus, the ratio β/α increased from 0.47 to 2.67. The former values were not unreasonable, whereas the latter values seem to be absurd. Indeed, in this case, the cloud of CRs instantaneously emitted by a point source spreads according to the law

$$R_c \propto t^{\beta/\alpha} = t^{2.67},$$

whereas the limit speed of the particles is the light speed c and the diffusion packet cannot spread faster then $R_c = ct$. This modification of the model is criticized in our recent works [71, 74, 79, 80]. Let us consider the arguments more detailed.

8.2 Bounded Anomalous Diffusion

8.2.1 Fractal Structures and Finite Speed

Since the most important property of space plasma is turbulence, manifesting self-similar structures of a power-law (fractal) type, the application of the fractional derivative technique to diffusion in this medium seems to be appropriate in this case.

Application of the fractional differential equation to anomalous diffusion is based on the continuous time random walk model introduced by Montroll and Weiss [53]. In this model, the walk of a particle is represented by a sequence of instantaneous jumps with random lengths at random times between which the particle is at rest. The lengths of the jumps and intervals during which the particle is at rest (in "a trap") are independent of each other. To illustrate the relation of the walk scheme with the real transport of CRs in the galactic magnetic field, let us divide space into cubic cells and specify each particle inside the i th cell by the radius vector $\mathbf{r}_i$ of the center of this cell (Fig. 8.4). At a random time T after entering this cell, the particle passes to one of the six neighboring cells and the corresponding vector jumps to the center of this new cell at the time of the intersection of the interface between the old and new cells. After a random time T', the particle passes to the next neighboring cell and the vector again instantaneously jumps, etc. If these cells were identical in the properties, the walk of the specifying vector would be discretized Brownian motion, which is transferred to ordinary diffusion with an increase in the scale. In case of a regular medium all cells are identical and free motion regime is absent.

However, the strongly turbulent high magnetic field is not inherent in each cell. According to the current representation, which was formed five decades ago, most of

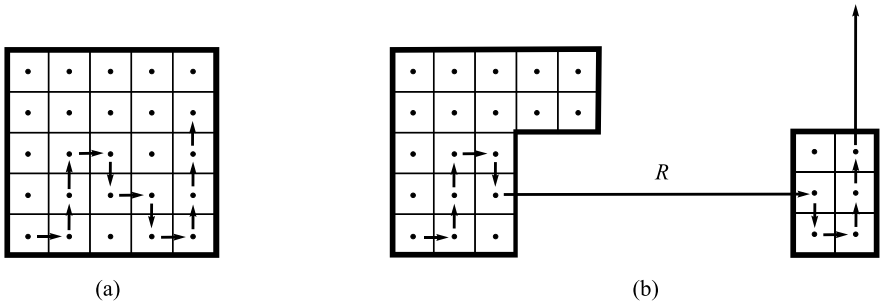


Fig. 8.4 The coarse-grained trajectory in regular (**a**) and fractal (**b**) medium

the space between magnetic clouds is filled with lower quieter fields whose smooth field lines can hold individuality at a long distance. The charged particles of CRs move along these field lines in helical paths and from time to time enter trap clouds, where they can stay for a long time and "forget" their initial direction of motion. In this case, a jump from one trap to another is not instantaneous as in the above case: the particle rather intersects "almost empty" cells, covering large distances R. The distribution of R in a fractal medium involves a long power-law tail. These transitions require the consideration of a finite velocity of the particle, more precisely, the leading center of the particle.

8.2.2 Equations of the Bounded Anomalous Diffusion Model

The most important consequence of the finiteness of the velocity of free particles is the finiteness of the spatial distribution: the probability density beyond a sphere with the radius vt and the center at the instantaneous point source is zero in this case. Let us refer to this process as bounded anomalous diffusion in order to distinguish it from unbounded anomalous diffusion in which the motion of a particle is represented as a sequence of instantaneous jumps from one point of space to another: the particle arrives at the latter point at the same time at which it leaves the former point at any distance between these two points. The delay time (in a trap) is not related to this distance and to the motion as a whole. If the traps are removed from this model, it becomes senseless, because the particle instantaneously flies to infinity and leaves the system under consideration. The bounded anomalous diffusion model involves not only the time spent in the traps, but also the time taken for the motion of the particle and, for this reason, is meaningful even in the absence of traps. Finally, since the bounded anomalous diffusion propagator vanishes beyond the sphere with the radius vt, all of its moments are finite.

The effect of the finiteness of the velocity of free motion on the walk process described above, which was investigated in [66–70, 81, 95], significantly changes the continuous time random walk model. In this case, the particle at the observation time can be in one of two states, rest and motion. The corresponding components of the probability density are denoted as $\psi_0(\mathbf{r}, t)$ with the speed V and $\psi_1(\mathbf{r}, t)$ so that the total density is

$$\psi(\mathbf{r}, t) = \psi_0(\mathbf{r}, t) + \psi_1(\mathbf{r}, t).$$

The rates of the $1 \to 0$ and $0 \to 1$ transitions per unit volume near the point $\mathbf{r}$ are denoted as $F_{1\to 0}(\mathbf{r}, t)$ and $F_{0\to 1}(\mathbf{r}, t)$, respectively. It is obvious that the particle passing to the state of rest at the point $\mathbf{r}$ at the time $t - t'$ remains in this state at the observation time $\mathbf{t}$ with the probability $Q(t') = \int_{t'}^{\infty} q(t)dt$, and the particle leaving the trap at the point $\mathbf{r} - \mathbf{r}'$ intersects a unit area at the point $\mathbf{r}$ without an interaction with the probability $P(\mathbf{r}') = \int_0^{\infty} p(\mathbf{r}' + \xi\Omega)d\xi, \quad \Omega = \mathbf{r}'/r'$. Since this transition takes r'/V seconds,

$$\psi(\mathbf{r}, t) = \int_0^{\infty} dt' Q(t') F_{1\to 0}(\mathbf{r}, t - t') \tag{8.12}$$

$$+\ V^{-1}\int d\mathbf{r}'P(\mathbf{r}')F_{0\to1}(\mathbf{r}-\mathbf{r}',t-r'/V),$$

The factor V^{-1} appears in front of the integral because the integral gives the flux of particles. The transition rates (if the particle begins its evolution in a trap at the origin of the coordinates at the initial time) are related as

$$F_{1\to0}(\mathbf{r},t)=\int d\mathbf{r}'p(\mathbf{r}')F_{0\to1}(\mathbf{r}-\mathbf{r}',t-r'/V)+\delta(\mathbf{r})\delta(t), \tag{8.13}$$

$$F_{0\to1}(\mathbf{r},t)=\int_0^\infty d\tau q(\tau)F_{1\to0}(\mathbf{r},t-\tau). \tag{8.14}$$

The Fourier-Laplace transformation

$$\psi(\mathbf{r},t)\mapsto\psi(\mathbf{k},\lambda)=\int d\mathbf{r}\int_0^\infty e^{i\mathbf{kr}-\lambda t}\psi(\mathbf{r},t)dt$$

reduces the system of Eqs. (8.12)–(8.14) to the form

$$L_V(\lambda,\mathbf{k})\psi(\mathbf{k},\lambda)\equiv[1-p(\mathbf{k},\lambda/V)q(\lambda)]\psi(\mathbf{k},\lambda)=S_V(\lambda,\mathbf{k}),$$

where

$$S_V(\lambda,\mathbf{k})=Q(\lambda)+(1/V)P(\mathbf{k},\lambda/V)q(\lambda)$$

and

$$p(\mathbf{k},\lambda/V)=\int p(\mathbf{r})e^{-(\lambda/V)r}e^{i\mathbf{kr}}d\mathbf{r}.$$

The quantity $P(\mathbf{k},\lambda/V)$ is determined similarly. If the tails of the distributions of R and T are of a power-law character with exponents $\alpha\in(0,2)$ and $\beta\in(0,1]$, respectively,

$$\mathsf{P}(R>r)\sim\frac{A}{\Gamma(1-\alpha)}r^{-\alpha},\ r\to\infty,$$

$$\mathsf{P}(T>t)-Q(t)\sim\frac{B}{\Gamma(1-\beta)}t^{-\beta},\ t\to\infty.$$

Then, using Tauberian theorems, one can show that for $\mathbf{k}\to0,\ \lambda\to0$

$$L_V(\lambda,\mathbf{k})\sim B\lambda^\beta+\begin{cases}\langle R\rangle\langle\lambda/V-i\mathbf{k}\Omega\rangle+A\langle(\lambda/V-i\mathbf{k}\Omega)^\alpha\rangle, & \alpha>1,\\ A\langle(\lambda/V-i\mathbf{k}\Omega)^\alpha\rangle, & \alpha<1,\end{cases}$$

where Ω is a random direction of the motion, which is assumed to be isotropically distributed. The comparison of different terms at $\lambda \to 0$ provides the following conclusions.

The asymptotic expressions at $V = \infty$ for both cases have the same form

$$L_V(\lambda, \mathbf{k}) \sim B[\lambda^\beta + C_\infty |\mathbf{k}|^\alpha],$$
$$C_\infty = A|\cos(\alpha\pi/2)|/[(\alpha+1)B],$$

This expression provides fractional differential equation of unbounded anomalous diffusion. At $V < \infty$, $\alpha = 2$, $\beta = 1$ (in view of the mentioned isotropy, $\langle \Omega \rangle = 0$), the transport operator has the form

$$L_V(\lambda, \mathbf{k}) \sim B\lambda^\beta + \langle R \rangle \langle \lambda/V - i\mathbf{k}\Omega \rangle + A\langle (\lambda/V - i\mathbf{k}\Omega)^2 \rangle$$
$$= (A/V^2)\lambda^2 + (B + \langle R \rangle / V)\lambda - (A/3)k^2,$$

corresponding to the telegraph equation

$$\left[\frac{A}{V^2} \frac{\partial^2}{\partial t^2} + \left(B + \frac{\langle R \rangle}{V} \right) \frac{\partial}{\partial t} - \frac{A}{3} \Delta \right] \psi(\mathbf{r}, t) = S_V(\mathbf{r}, t),$$

which describes the bounded normal diffusion. Under the same conditions at $\beta < 1$, the following fractional variant of the subdiffusion telegraph equation is obtained:

$$\left[\frac{A}{V^2} \frac{\partial^2}{\partial t^2} + \frac{\langle R \rangle}{V} \frac{\partial}{\partial t} + B_0 \mathsf{D}_t^\beta - \frac{A}{3} \Delta \right] \psi(\mathbf{r}, t) = S_V(\mathbf{r}, t).$$

In both bases, the form of the "deep" time asymptotic expression for $\psi(\mathbf{r}, t)$ is independent of the velocity and leads to the equations of bounded diffusion and subdiffusion, respectively. The same property is observed at $\alpha < 1$ when $\beta < \alpha$. If $\alpha < \beta \leq 1$ (this relation between the parameters is used by Lagutin and Tyumentsev), the asymptotic equation for $\psi(\mathbf{r}, t)$ has the form unusual for a diffusion process with a pseudodifferential operator $(\mathsf{D}_t + V\Omega\nabla)^\alpha$ averaged over the directions

$$(A/V^\alpha)\langle (\mathsf{D}_t + V\Omega\nabla)^\alpha \rangle \psi(\mathbf{r}, t) = S_V(\mathbf{r}, t).$$

Here, the operator

$$(\mathsf{D}_t + V\Omega\nabla)^\nu \psi(\mathbf{r}, t)$$
$$= \frac{1}{\Gamma(1-V)} (\mathsf{D}_t + V\Omega\nabla) \int_0^t \frac{\psi(\mathbf{r} - V\Omega(t-\tau), \tau)}{(t-\tau)^V} d\tau$$

is the fractional generalization of the material derivative that is agree with results obtained by Sokolov and Metzler [62] for one-dimensional Lévy walks.

If we consider the case of homogeneous distribution of particles $\psi(\mathbf{r}, t) \equiv \psi(\mathbf{t})$ this operator becomes the fractional Riemann-Liouville derivative with respect to

time:

$$(\mathsf{D}_t + V\Omega\nabla)^\alpha \psi(\mathbf{r}, t) \mapsto {}_0\mathsf{D}_t^\alpha \psi(\mathbf{t}).$$

In the stationary problem, when the function $\psi(\mathbf{r}, t) = \psi(\mathbf{r})$ does not depend on time,

$$(\mathsf{D}_t + V\Omega\nabla)^\alpha \psi(\mathbf{r}) \mapsto (V\Omega\nabla)^\alpha \psi(\mathbf{r}),$$

this operator represents fractional generalization of directional derivative.

The dependence on the velocity remains at any time, but the dependence on disappears. The distribution $\psi(\mathbf{r}, t)$ corresponding to this case has a specific U-shape in the region bounded by the radius vt beyond which it vanishes [81, 95].

8.2.3 The Bounded Anomalous Diffusion Propagator

In one-dimensional case, for $\alpha < 1$, the equation of bounded anomalous diffusion without traps takes the form:

$$\left[\left(\tfrac{\partial}{\partial t} - V\tfrac{\partial}{\partial x}\right)^\alpha + \left(\tfrac{\partial}{\partial t} + V\tfrac{\partial}{\partial x}\right)^\alpha\right]\Phi(x, t) = \tfrac{t^{-\alpha}}{2\Gamma(1-V)}\left[\delta(x - Vt) + \delta(x + Vt)\right].$$

Solutions of this equation can be expressed through elementary functions [78]

$$\Phi(x, t) = \frac{2\sin\pi\alpha}{\pi} \frac{\left(1 - x^2/V^2t^2\right)^{\alpha-1}}{(1 - x/Vt)^{2\alpha} + (1 + x/Vt)^{2\alpha} + 2\left(1 - x^2/V^2t^2\right)^\alpha \cos\pi\alpha}.$$

Solutions can also be written in terms of fractional stable densities [78], which are useful for probabilistic interpretation of these distributions.

In Fig. 8.5a, the analytical solutions (lines) are compared with the results of Monte Carlo simulated random walks with finite velocity of motion (points). In Fig. 8.5b, the influence of ballistic restriction on the dynamics of diffusion packet spreading is demonstrated schematically.

To make clear the role of correlations between path lengths and waiting times in the model under consideration, we compare propagators of bounded and unbounded anomalous diffusion. In the latter model we take identical distributions for waiting times and path lengths ($\alpha = \beta$). In the model of unbounded anomalous diffusion path lengths and waiting times in traps are independent. We take the model of bounded anomalous diffusion without traps, time delay is provided by finiteness of rays propagation velocity. The results for the one-dimensional case are presented in Fig. 8.6a, c, and for the 3D-model in Fig. 8.6b, d. From the graphs, we can see that for $\alpha = \beta < 1$, distinctions are very strong, shapes of packets, spreading laws, behaviors near the ballistic boundaries and near the source are essentially different. When $1 < \alpha = \beta < 2$, distinctions are also sufficient despite the fact that mean path length is finite. In one

case distributions are bounded, in another case they are unbounded. In the model of bounded anomalous diffusion, the front near $|\mathbf{r}| = vt$ appears. The densities differ quantitatively near the source as well. All these facts say that the results obtained on the base of unbounded anomalous diffusion and presented in the works [38, 40–43] need to be re-examined.

8.3 Acceleration of Cosmic Rays

8.3.1 CR Reacceleration

In this section, the CTRW approach is considered in application to the description of the acceleration (more precisely, reacceleration) of cosmic rays. In contrast to the preceding problem, this concerns the behavior of particles in the momentum space. Successive interactions (collisions) of a charged particle with more or less localized inhomogeneities of the magnetic field from slowly moving magnetic clouds mentioned by Fermi in his pioneering works [23, 24] to strong shock waves in the supernovae remnants mentioned by Berezhko and Krymskii in review [5] can be considered as instantaneous jumps from one point of the momentum space to another one. The momenta $\Delta\mathbf{p}_i$ acquired by the particle in these collisions are random and, even for their isotropic distribution, the point

$$\mathbf{p} = \mathbf{p}_0 + \Delta\mathbf{p}_1 + \Delta\mathbf{p}_2 + \Delta\mathbf{p}_3 + \cdots$$

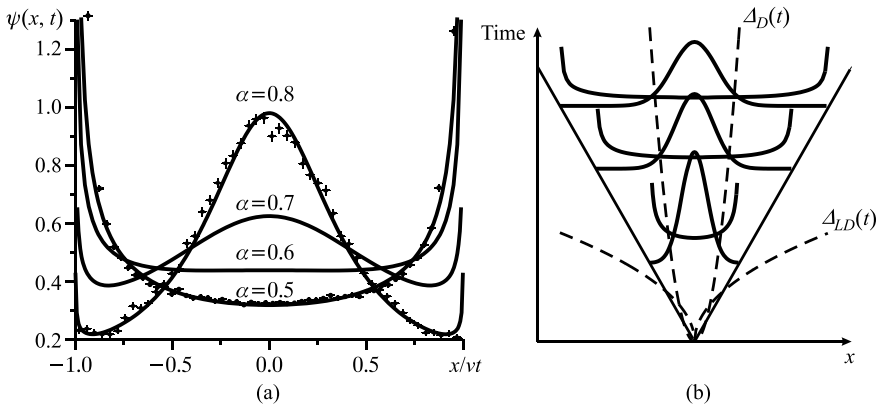


Fig. 8.5 Comparison of analytical solutions (lines) with results of numerical Monte Carlo simulation (points) in one-dimensional case (**a**). Evolution of the Gaussian diffusion packet and the propagator of the bounded anomalous diffusion model, $\alpha = 0.5$ (**b**). Dashed lines represent the time dependencies of the diffusion packet width in both cases

representing the particle in the momentum space moves away from the point (momentum) of the acceleration injection $\mathbf{p}_0$ similar to a Brownian particle; this behavior means the further acceleration (reacceleration) of the particle. However, only a certain fraction of the particles moving away from the center are accelerated. This fluctuation component of the mechanism of the acceleration of cosmic rays is analyzed in this work.

From the statistical point of view, the main consequence of the Fermi conclusion is that the exponential increase in the energy of the accelerated particle with time, $E = E_0 e^{at}$, and the exponential distribution of the age of the detected particles, $d\mathsf{P} = \exp(-t/\tau)dt/\tau$, is sufficient for the formation of the power-law energy spectrum $N(E)$. That is all. There are no other sources of fluctuations taken into account in the Fermi model. This result is evident: What fluctuations can else exist if an

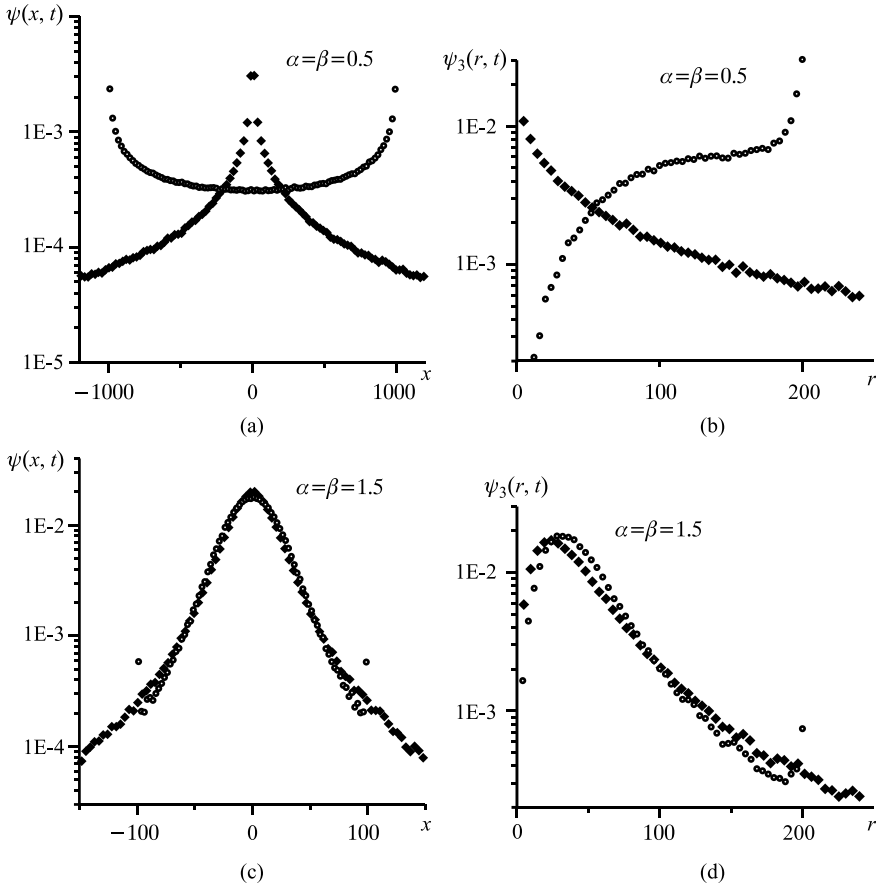


Fig. 8.6 Comparison of propagators of the bounded anomalous diffusion (circles) and unbounded diffusion (rhombs) for $\alpha = 0.5$ and 1.5 in the 1D case (**a** and **c**) and 3D case (**b** and **d**)

increase in the energy by a factor of e requires 10^8 collisions according to the Fermi estimation? A more significant effect could be produced by more rare acts with more large acceleration in each of them. Among these processes are the aforementioned interactions with strong shock waves, when even a single interaction can increase the energy of the particle by a factor of $7 \div 13$ [6]. To this end, it is appropriate to pass from the degenerate spectral function $\delta(E - E_0 e^{at})$ characterizing the determinate process of the Fermi acceleration to the continuous function $n(E, t)$ related to the momentum distribution $f(\mathbf{p}, t)$ as

$$n(E, t) = \int \delta(E - E(\mathbf{p}))f(\mathbf{p}, t)d\mathbf{p}.$$

Under the Fermi assumption that the parameter τ is independent of the energy, the desired spectrum can be represented in the form

$$N(E) \equiv N(E; \tau) = \tau^{-1}\left[\int_0^\infty n(E, t)e^{-t/\tau}dt\right] = \tau^{-1}\widehat{n}(E, \tau^{-1}), \tag{8.15}$$

where

$$\widehat{n}(E, \lambda) \equiv \int_0^\infty e^{-\lambda t}n(E, t)dt$$

is the Laplace transform of the spectral function in the time variable. The effect of the kinetic fluctuations is now taken into account at the stage of the construction of equations for the distributions $f(\mathbf{p}, t)$ or $n(E, t)$ by the inclusion of additional terms containing differential and integral operators. The theory of the acceleration of cosmic rays is under development and this work is focused on the role of fractional derivatives in this development.

8.3.2 Fractional Kinetic Equations

In the momentum space, a motionless point represents the particle moving with a constant momentum (velocity, energy). The exponent ν characterizes now the tail part of the distribution of the random duration ΔT between the successive collisions of the moving particle $Q(t) = \mathsf{P}(\Delta T > t),\ t \to \infty$. For an ultrarelativistic particle ($v \approx c$), this interval is proportional to its free path between these collisions, so that the new ν values should correspond to the α values in the spatial model. In view of this circumstance, the corresponding CTRF equations systems is

$$f(\mathbf{p}, t) = \int_0^t Q(t - t')F_{0\leftarrow 1}(\mathbf{p}, t')dt', \tag{8.16}$$

$$F_{0\leftarrow 1}(\mathbf{p},t)=\int w(\mathbf{p}\leftarrow\mathbf{p}')F_{1\leftarrow 0}(\mathbf{p}',t)d\mathbf{p}'+f_0(\mathbf{p})\delta(t), \tag{8.17}$$

$$F_{1\leftarrow 0}(\mathbf{p},t)=\int_0^t q(t-t')F_{0\leftarrow 1}(\mathbf{p},t')dt'. \tag{8.18}$$

The further transformation of the system is associated with the specification of the distributions $q(t)$ and $w(\mathbf{p}\leftarrow\mathbf{p}')$ distributions. The time is usually (one can say, always) taken in the exponential form $Q(t)=e^{-\mu t}$, $q(t)\equiv -dQ/dt=\mu e^{-\mu t}$ and therefore the master equation begins with the first-order time derivative $\partial f(\mathbf{p},t)/\partial t=\cdots$. This means that the process is assumed to be Markovian. However, the real distribution of the time intervals between collisions is unknown. For example, one can assume that it is of power law, $Q(t)\propto t^{-\alpha}$, rather than exponential. This is in agreement with the self-similar pattern of turbulent motions and with its power-type laws. The hypothesis of the fractal character of the interstellar medium [35] also provides power-law distributions. This concerns the behavior of magnetic field lines in interstellar space. They are usually represented as relatively smooth lines, which somewhere rest at the magnetic traps, intersect each other, sharply change their directions, and performing a "diffusion dance" in time. The leading centers of particles moving along spiral trajectories move along these lines. If these smooth sections become invisible and chaotic patterns of the structure become prevalent with the expansion of the field of view and the corresponding decrease in the scale, then this is an asymptotically homogeneous medium where the mean free paths can be simulated by a usual exponential function. If the expanding field of view includes increasing straight segments replacing the segments becoming small due to a decrease in the scale so that the structure remains (qualitatively) unchanged, this is a fractal structure. Under these conditions, the exponential distribution of mean free paths characteristic of a strongly mixing medium cannot be expected, but can not be completely rejected as well. The best compromise would be a family of distributions including both exponential and power-law distributions. Fortunately, such a family exists: it is a set of the functions

$$Q_\alpha(t)=E_\alpha(-\mu t^\alpha),\quad \alpha\in(0,1],$$

where $E_\alpha(z)=\sum_{n=0}^\infty z^n/\Gamma(\alpha n+1)$ are the Mittag-Leffler functions. The function $Q_\alpha(t)$ with $\alpha=1$ is a usual exponential and with $\alpha<1$ is a fractional exponential having a power-law asymptotic behavior $t^{-\alpha}$, $t\to\infty$. The corresponding density $q_\alpha(t)$ satisfies a fractional differential equation [73]; as a result, the entire system of Eqs. (8.16)–(8.18) has the fractional time-differential form

$${}_0\mathsf{D}_t^\alpha f(\mathbf{p},t)=\mu\mathsf{A}f(\mathbf{p},t)+f_0(\mathbf{p})\delta_\alpha(t). \tag{8.19}$$

Here

$$\mathsf{A}f(\mathbf{p},t)=\int w(\mathbf{p}\leftarrow\mathbf{p}')f(\mathbf{p}',t)d\mathbf{p}'-f(\mathbf{p},t) \tag{8.20}$$

is the acceleration integral with transitions density $w(\mathbf{p} \leftarrow \mathbf{p}')$. The time series of accelerating collisions forms a *fractional Poisson process of order* α [73], with $\alpha \to 1$ becoming an ordinary Poisson process which underlies classical kinetic equation [44]. Investigations performed in Uchaikin et al. [76] indicate a qualitatively new property of this process: the average number of collisions increases more slowly ($\propto t^{\alpha}$) than in the usual case ($\propto t$) and relative fluctuations of the number of collisions in the limit $t \to \infty$ do not disappear, but tend to a limiting distribution depending on α (some sort of KNO scaling).

The transition from the first time derivative to the fractional derivative of the order $\alpha < 1$ does not require solving a fractional differential equation; it is more convenient to use the relation between the solutions of Eq. (8.19) with $\alpha < 1$ and $\alpha = 1$ [73],

$$f_{\alpha}(\mathbf{p}, t) = (t/\alpha) \int_0^{\infty} f_1(\mathbf{p}, \tau) g_+(t\tau^{-1/\alpha}; \alpha) \tau^{-1/\alpha - 1} d\tau. \tag{8.21}$$

Here $g_+(t; \alpha)$ is the one-sided stable (in the Lévy sense) probability density determined by the Laplace transform

$$\int_0^{\infty} \exp(-\lambda t) g_+(t; \alpha) dt = \exp(-\lambda^{\alpha}),$$

and t, τ, λ are dimensionless variables. The Laplace transform of Eq. (8.21) in time with the use of the above formulas for the spectra provides the formula

$$N_{\alpha}(E; \tau) = N_1(E; \tau^{\alpha}). \tag{8.22}$$

It presents the effect of the fractal dimension of the fractional Poisson process of collisions $\alpha \in (0, 1]$ on the energy spectrum of cosmic rays: the spectrum $N_{\alpha}(E; \tau)$, formed by an ensemble of particles with the mean life time τ, which are accelerated according to the fractional Poisson law of the order $\alpha < 1$, coincides with the spectrum of particles that are accelerated by the usual Poisson process ($\alpha = 1$), but have the mean age τ (recall that the time is here dimensionless and the injection spectra $f_0(E)$ are the same in both problems). On the example of the Fermi spectrum, it is easy to see that the efficiency of acceleration decreases (the spectrum becomes steeper) with a decrease in the order of the process. The fractal character of the spatial distribution of accelerating regions naturally reduces the efficiency of acceleration.

8.3.3 *Fractional Fokker–Planck Equations*

Similar to the classical case, the transition from the kinetic equation to the Fokker–Planck equation is associated with the transformation of the collision integral to a differential form by expanding the integrand into a series in the momentum increment to the second order terms. There are two variants of such an expansion, which

provide slightly different equations (see, e.g., [44]). The first variant implies the smallness of the *absolute value of the change in the momentum* $|\Delta\mathbf{p}| = |\mathbf{p} - \mathbf{p}'|$, so that the momentum of the incident particle only slightly changes in the magnitude and direction in a single collision event (e.g., as in the case of the collision of a heavy particle with a light one). The second variant implies the smallness of the *change in the absolute value of the momentum* $\Delta|\mathbf{p}| = |\mathbf{p}| - |\mathbf{p}'|$, whereas the change in its direction is not small and can have a wide distribution up to the isotropic one (in the case of the collision of a light particle with a heavy one). Under the assumption of the isotropic scattering, the fractional differential generalization of the Fokker-Planck equation is obtained in the form

$$ {}_0\mathsf{D}_t^{\alpha} f(p,t) = \Delta_{\mathbf{p}}(K(p) f(p,t)) + f_0(p)\delta_{\alpha}(t), \tag{8.23} $$

where

$$ K(p) = (\mu/2) \int (\Delta\mathbf{p})^2 w(\Delta\mathbf{p};\, p) d\Delta\mathbf{p} $$

is the diffusivity in the momentum space. An energy analog of Eq. (8.23) ($\alpha = 1$) is well known in the physics of cosmic rays in the form (see Eq. (8.2) in Ginzburg and Syrovatskii [27])

$$ {}_0\mathsf{D}_t^{\alpha} n(E,t) = \frac{\partial[a_1(E)n(E,t)]}{\partial E} + \frac{\partial^2[a_2(E)n(E,t)]}{\partial E^2} + n_0(E)\delta_{\alpha}(t). \tag{8.24} $$

At the same time, Eq. (8.23) with diffusion term

$$ \Delta_{\mathbf{p}}(K(p) f(p,t)) = (\Delta_{\mathbf{p}} K(p)) f(p,t) + 2(\nabla_{\mathbf{p}} K(p))\nabla_{\mathbf{p}} f(p,t) + K(p)\Delta_{\mathbf{p}} f(p,t) $$

significantly differs from another diffusion type equation (Ginzburg and Syrovatsky [27])

$$ {}_0\mathsf{D}_t^{\alpha} f(p,t) = \nabla_{\mathbf{p}}(K(p)\nabla_{\mathbf{p}} f(p,t)) + f_0(p)\delta_{\alpha}(t). \tag{8.25} $$

The difference is due to the fact that Eq. (8.23) is derived in frame of the collision model, when the point presenting the particle instantaneously moves to another, possibly far, geometric point, violating the continuity of the trajectory in the momentum space, whereas the dynamic derivation of Eq. (8.25) implies that the trajectory in the momentum space is continuous and even differentiable. The classical versions (with $\alpha = 1$) of Eqs. (8.23)–(8.25) underlie the standard set of mathematical tools describing the fluctuation mechanisms of the acceleration of cosmic rays; their solutions are well known [6, 17, 27]. A common property of these models is the Gaussian character of the momentum distributions, which is due to the assumption that the second moment of the momentum acquired in an acceleration event, which enters into the diffusion coefficient, is finite and the acceleration rate is low. A change of the first time derivative in this equation to its fractional analog of the order $\alpha \in (0, 1)$ does not lead to an increase in the efficiency of acceleration; on the contrary, the

acceleration rate in the subdiffusion regime decreases further. It is also noteworthy that the statement of some authors that diffusion at $\alpha > 1$ is accelerated is erroneous. This statement is based on linguistic intuition ("if $\alpha < 1$ means subdiffusion and $\alpha = 1$ corresponds normal diffusion, then $\alpha > 1$ should mean super diffusion") and is erroneous, because the solution $f(\mathbf{p}, t)$ at $\alpha > 1$ loses its probability meaning: it is not positively definite in this case. Thus, the transition to the fractional time derivative (with the unchanged remaining, diffusion part of the equation) as a method for enhancing the high energy part of the spectrum is physically unpromising. At the same time, the parameter α presents the existence of possible correlations in the spatial distribution of the acceleration regions and can be useful in this respect (recall that $\alpha = 1$ corresponds to the uniform Poisson distribution of such regions, which do not correlate with each other). This parameter can be kept and the efficiency of the multiple acceleration mechanism can be increased only by modifying another operator of the equation, more precisely, by returning from the differential form of the acceleration operator, which describes a continuous slow collection of the energy, to the integral form describing acceleration as a sequence of events with large instantaneous (in terms of the considered "galactic" time scales) changes in the momenta. For this reason, it seems appropriate to change the momentum Laplacian to its fractional analog, because the fractional Laplacian contains a momentum integral operator with inverse power law kernel, which can ensure a high acceleration rate. Let us consider this variant in more detail.

The losses of the energy of a fast charged particle in a medium, which is described by equations similar to Eq. (8.24) (naturally, with $\alpha = 1$), cannot be larger than its initial energy and all of the moments of the energy loss are finite. In the problem of acceleration, there is no such definite limit of the energy increase; this fact is an additional reason to study the region with infinite dispersion, which attracts an increasing interest of researchers of anomalous diffusion processes. In this case, informative (in the asymptotic sense) results are obtained only when infinite dispersion is due to the power-law distributions:

$$\int_{|\Delta\mathbf{p}|>p} w(\Delta\mathbf{p}; \mathbf{p}')d\Delta\mathbf{p} \propto p^{-\gamma}, \quad p \to \infty. \tag{8.26}$$

If $\gamma > 2$, the second moment is finite and this corresponds to the classical diffusion region. If $\gamma < 2$, the second moment of the increment is infinite and this corresponds to the model of additive Lévy flights. In this case, the equations for the momentum and energy distributions $f(p, t)$ and $n(E, t)$, respectively, following from the asymptotic CTRW analysis,

$${}_0\mathsf{D}_t^{\alpha} f(p, t) = -K(-\Delta_{\mathbf{p}})^{\nu/2} f(p, t) + f_0(p)\delta_{\alpha}(t), \tag{8.27}$$

and

$$ {}_0\mathrm{D}_t^{\alpha} n(E,t) = \begin{cases} \partial^{\nu}[a_{\nu} n(E,t)]/\partial E^{\nu} + n_0(E)\delta_{\alpha}(t), & 0 < \nu < 1; \\ \partial[a_1 n(E,t)]/\partial E + \partial^{\nu}[a_{\nu} n(E,t)]/\partial E^{\nu} + n_0(E)\delta_{\alpha}(t), & 1 < \nu < 2. \end{cases} \tag{8.28} $$

seem to be reasonable. Here

$$ \nu = \begin{cases} \gamma, & \gamma \le 2; \\ 2, & \gamma > 2, \end{cases} $$

$\partial^{\nu}/\partial E^{\nu}$ is the symbol closer to the ordinary notation of the fractional differential operator and K, a_1, and a_{ν} are the constant coefficients. The constancy of these coefficients is very significant for the derivation of the equations. Fractional differential equations are usually derived with the use of integral transformations whose applicability requires the constancy of the coefficients. It would be incorrect to derive, e.g., Eq. (8.27) in such a manner and, then, to place the variable diffusion coefficient $K(p)$ in front of the fractional Laplacian (this is obvious even on the example of Eqs. (8.23)–(8.25) with the integer Laplacian).

Equation (8.28) is clearly similar to the νth term approximation of the expansion of a function that is zero at the reference point and it would seem that omitting the next (divergent!) term introduces an infinite error, but this is not the case. The reason is that the Taylor formula rather than the infinite Taylor series is used and the series expansion can be continued only until derivatives exist; after that, the remainder term, which is always finite, should be written. If the next derivative existed, its inclusion in the continued expansion would mean an approximation of the omitted term. If this derivative does not exist, it cannot be used to approximate the omitted term and one should return to the initial point, where the derivative of the preceding order is used for approximation. The diffusion packet, which is described by Eq. (8.27) and propagates from the origin of the momentum coordinates, has the form of the three-dimensional isotropic fractional stable distribution $\psi_3^{(\nu,\alpha)}$, $\nu \in (0, 2]$, $\alpha \in (0, 1]$, smearing proportionally to $t^{\alpha/\nu}$:

$$ f(p,t) = (Kt^{\alpha})^{-3/\nu} \psi_3^{(\nu,\alpha)}((Kt^{\alpha})^{-1/\nu} p). $$

Tails (or, more romantically, wings) of this distribution have a power-law form with the exponent ν. Physically, this means a peculiar leading effect: one of independent terms in the sum $\Delta\mathbf{p}_1 + \Delta\mathbf{p}_2 + \cdots + \Delta\mathbf{p}_n$ is always outstanding and compared in magnitude with the reminder. This leading effect disappears at $\nu = 2$, when the distribution becomes Gaussian (sub-Gaussian). As a result, the spectrum at $\nu < 2$ has the form

$$ N_1(E)dE \propto E^{-\nu-1} dE, $$

which is similar to the Fermi formula. The main difference is that the exponent ν in this case is independent of the age of the detected particles and is completely determined by the acceleration mechanism in an individual local event (collision). For this reason, the fractal character of the spatial distribution of accelerating regions also does not affect the slope of the resulting spectrum.

8.3.4 Integro-Fractionally-Differential Model

A drawback of the model with the fractional momentum Laplacian is that the increments of the momentum in the acceleration event are independent of the momentum of the particle involved in the interaction, whereas in the Fermi model and in its later variants, the increments of the energy (and, therefore, the momentum) are on average proportional to the energy (momentum) of the particle before the interaction. In this case, the energy of the accelerated particle is expressed in terms of the product of independent random variables, rather than their sum. This model is called *multiplicative walk* in order to distinguish it from the *additive walk* model considered above. The increment of the momentum in the multiplicative model is proportional (in the statistical case) to the absolute value of the momentum p' of the particle coming into interaction,

$$\Delta\mathbf{p} = \mathbf{p}'\mathbf{q}, \int_{|\Delta\mathbf{p}|>p} w(\Delta\mathbf{p};\mathbf{p}')d\Delta\mathbf{p} \propto (p/p')^{-\gamma}, \ p \to \infty. \tag{8.29}$$

Under the assumption that the distribution of the proportionality vector $\mathbf{q}$ is independent of $\mathbf{p}'$ and isotropic,

$$W(\mathbf{q};\mathbf{p}')d\mathbf{q} = (1/2)V(q)dqd\xi, \ \xi = \cos(\mathbf{q},\mathbf{p}),$$

kinetic Eq. (8.19) can be modified to the form

$$\begin{aligned} {}_0\mathsf{D}_t^{\alpha} f(p,t) = \mu \left\{ \int_{-1}^{1} \frac{d\xi}{2} \int_0^{\infty} V(q) \frac{f\left(p\Big/\sqrt{1+2\xi q+q^2},t\right)}{\left(\sqrt{1+2\xi q+q^2}\right)^3} dq - f(p,t) \right\} \\ + f_0(p)\delta_{\alpha}(t), \end{aligned} \tag{8.30}$$

representing a new model of a distributed reacceleration, more precisely, a new modification of the model proposed in Blanford and Ostriker [7] and Wandel et al. [84]. In order to make this model be closer to real processes of reacceleration, e.g., in the case of the intersection of shock fronts in the remnants of supernovae, we assume that (Wandel et al. [84])

$$V(q) = \gamma q^{-\gamma-1}, \quad \gamma > 1.$$

The resulting model can be called *multiplicative Lévy flights*.

Let us consider the equation for the spectral function in two extreme cases. In the first case, $\gamma > 2$, the second moment of the momentum increment proportional to E^2 exists, and this case corresponds to the classical diffusion with variable coefficients:

$${}_0\mathsf{D}_t^{\alpha} n(E,t) = \frac{\partial[a_1 E n(E,t)]}{\partial E} + \frac{\partial^2[a_2 E^2 n(E,t)]}{\partial E^2} + n_0(E)\delta_{\alpha}(t).$$

In the second case, we suppose $\gamma \ll 2$, so that only the term q^2 may be retained in the radicand in Eq. (8.30):

$$ {}_0\mathsf{D}_t^{\alpha} n(E,t) = \mu \left\{ \int_1^{\infty} \gamma q^{-\gamma-1} n(E/q,t)dq/q - n(E,t) \right\} + n_0(E)\delta_{\alpha}(t). \quad (8.31) $$

I do not claim that this approximation is very good, but namely this acceleration operator was used in (Wandel et al. [84]) for some calculations. Solving Eq. (8.31) with the method of the Mellin-Laplace transforms and using Eqs. (8.15) and (8.22), the following expression is obtained for the case of the mono-energetic source ($n_0(E) = \delta(E - E_0)$):

$$ N_{\alpha}(E;\tau) = \frac{\mu\tau^{\alpha}\gamma}{(1+\mu\tau^{\alpha})^2} \left(\frac{E}{E_0}\right)^{-1-\gamma/(1+\mu\tau^{\alpha})} \frac{1}{E_0}. \quad (8.32) $$

Although Eq. (8.32) was derived under the assumptions that strongly simplify the real situation and has a qualitative sense, it compactly presents the effect of *all three sources of fluctuation acceleration*: fluctuations of the age of the particle (parameter τ), fluctuations of the number of the acceleration events (α, μ), and fluctuations of the energy acquired in a single event (γ). Representing the scaling parameter μ in the form $\mu = \tau_{\mathrm{A}}^{\alpha}$, where τ_{A} is the characteristic time interval between the events of the acceleration of particles in the remnants of various supernovae (recall that τ is the mean lifetime with respect to nuclear collisions), the absolute value of the exponent of the integral spectrum can be written in the more clear form $\gamma' = 1 + \gamma/[1 + (\tau/\tau_{\mathrm{A}})^{\alpha}]$. At $\alpha = 1$, and $\mu\tau \gg 1$, we arrive at the Fermi formula ([23, 24]):

$$ N_1(E;\tau) \approx \frac{1}{a\tau} \left(\frac{E}{E_0}\right)^{-1-1/(a\tau)}, \quad a = \mu/\gamma. $$

Taking into account that all three above-considered mechanisms contribute to the acceleration (more precisely, reacceleration) of cosmic rays, this process can be analyzed using the integro-fractional-differential equation

$$ \begin{aligned} {}_0\mathsf{D}_t^{\alpha} f(p,t) &= \mu_0 \nabla_{\mathbf{p}}(K_0(p)\nabla_{\mathbf{p}} f(p,t)) \\ &+ \mu_1 \Delta_{\mathbf{p}}(K_1(p) f(p,t)) - \mu_2 K_2 (-\Delta_{\mathbf{p}})^{\nu/2} f(p,t) \\ &+ \mu_3 \left\{ \int_{-1}^{1} \frac{d\xi}{2} \int_0^{\infty} V(q) \frac{f\left(p/\sqrt{1+2\xi q+q^2}, t\right)}{\left(\sqrt{1+2\xi q+q^2}\right)^3} dq - f(p,t) \right\} \\ &+ f_0(p)\delta_{\alpha}(t). \end{aligned} \quad (8.33) $$

The terms with the coefficients μ_0, μ_1, μ_2, and μ_3 successively present the contributions from the processes of continuous acceleration (such as fluctuations of the momentum of the particle in turbulent plasma [17, 65]), collision accelerations with

a finite second moment (the model of walking magnetized clouds with a limited distribution of their velocities), additive Lévy-type collision accelerations given by Eq. (8.26) (the same with the power-law distribution of the velocities of the clouds), and multiplicative Lévy-type collision accelerations given by Eq. (8.29). Recall again that the last operator describing, in particular, acceleration at shock waves in the remnants of supernovae does not reduced to the fractional differential form, but holds in the integral form. Equation (8.33) should be supplemented by terms presenting energy losses, exist of particles from the Galaxy, and nuclear interactions (fragmentation).

However, only dominant terms can be retained in Eq. (8.33) applied to particular problems, as is usually done.

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Chapter 9
Closing Chapter



9.1 The Problem of Interpretation

Sometimes my students ask me: Which textbook do you recommend for learning your discipline (I give lectures in two disciplines: Mechanics and Probability). Usually, I answer: Each discipline is like a sculpture, and each textbook is a photocopy of it made from the author's point of view. For a superficial acquaintance, one good textbook may be sufficient, but if you wish to get a detailed knowledge of it, you must read a few textbooks represented various points of view. However, my book is not a textbook. I wanted to write a book which answers the question: What is the fractional derivative. In the process, I understand that it is answering the question: What they think is the fractional derivative? Of course, I could not present opinions of all of those who used or uses this calculus in practice. Nevertheless, feeling some responsibility for the degree of the representation completeness, I have collected in this chapter additional information and offer it to the reader with a few comments.

Since the very origin of integral and differential calculus, geometrical and physical interpretations played the most important roles (Table 9.1).

Due to the recurrent character of operations of higher integer orders n, these interpretations can be easily extended to an arbitrary integer n, and one obtains constructions such as "a slope of a tangent of a slope of a tangent …" and "an area of curvilinear trapezoid under plot of an area of curvilinear trapezoid …". However, the visibility and clearness disappears. Things are improved through introducing the terms "convexity", "concavity", "crookedness", and "acceleration", nevertheless the problem as a whole remains unsolved.

To reach better understanding of the fractional derivatives idea, one should trace ways of the fractional equations derivation.

This is an additional chapter, which has been included in the book not without hesitation. But finally, I've decided to dart a farewell glance at what we learned in this journey. . .

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Nonlinear Physical Science, https://doi.org/10.1007/978-981-96-0582-8_9

Table 9.1 Main classical operations and their interpretations

Operation	Geometrical interpretation	Physical interpretation
$\mathsf{D}_t f(t)$	Slope of a tangent	Velocity of particle with coordinate $x = f(t)$
${}_a\mathsf{I}_t f(t)$	Area of curvilinear trapezoid	Coordinate of point, work of force

Equations with a fractional time derivative were derived by Nigmatullin, who considered diffusion on fractal structures simulating porous and disordered media [80, 81].

One of the methods of deriving time- and space-fractional equations is based on the asymptotic transition from the integral equations of the Scher-Montroll random walk model [103] with the power-law waiting time distribution to fractional ones (for details, see [101]). It is known that the Scher-Montroll model describes the main features of dispersive transport.

Basing on the Langevin equation, Metzler and Klafter [77] developed a three-dimensional description of subdiffusion with the transition to macroscopic fractional differential dynamics. They analyzed a non-Markovian generalization of the Chapman–Kolmogorov equation for random processes with continuous time that is controlled by the waiting time distribution. The power-law distribution (with "heavy tails") of waiting times asymptotically leads to the Klein–Kramers equation from which fractional generalizations of the Rayleigh and Fokker–Planck equations are derived.

Fractional equations can be obtained from the trapping-emission balance equations of the multiple trapping model with the exponential density of localized states (for more details, see [10, 104, 130]).

The fractional differential equations are sometimes obtained by the simple change of the integer-order derivatives in the classical kinetic equations to the fractional-order derivatives and, only then, the physical justification of this change is attempted to be given (see, e.g., [9]). Some other authors give the following "justification" of fractional formulation: "Due to complexity and entanglement of trajectories of particle motion one can say that the gradient (coordinate derivative) becomes fractal and $j \sim \partial^{2\gamma} n/\partial t^{2\gamma}$".

Podlubny [91] notes that the problem of interpretation of fractional operators has existed for more than 300 years, that is as long as the operators themselves. In recent years, the interest to this problem has increased, and we are going to mention below some of the works, but in search of the meaning of fractional derivatives and integrals, the fact that the problem has not been solved even for integer orders is overlooked. I mean that we can not to interpret the operator D_t^n with the help of one phrase valid for all $n = \ldots -2, -1, 0, 1, 2$, and even for three values $n = -1, 0, 1$. Each integer order n requires its own geometrical and mechanical interpretation. We have a countable set of such values and the same number of corresponding interpretations. In the case of a real order ν any interval of ν contains infinite number of ... what? Velocities? Accelerations?

However, this does not prevent us enumerate the most common approaches to this problem.

9.2 Geometrical Interpretation

9.2.1 Shadows on a Fence

Following [91], we represent the left-sided Riemann–Liouville integral of order $\mu > 0$

$$ {}_0I_x^{\mu} f(x) = \frac{1}{\Gamma(\mu)} \int_0^x f(\xi)(x-\xi)^{\mu-1} d\xi $$

in the form of the Stieltjes integral

$$ {}_0I_x^{\mu} f(x) = \int_0^x f(\xi) dg_x(\xi), $$

where

$$ g_x(\xi) = \frac{1}{\Gamma(\mu+1)} \left[x^{\mu} - (x-\xi)^{\mu} \right]. \tag{9.1} $$

Let us take a three-dimensional Cartesian coordinate system, draw the graph g_x in the horizontal plane as a function of ξ, $\xi \in [0, x]$, and build up a trapezoid, formed by multiple segments of variable height $f(\xi)$, on the curvilinear basis. We project the "fence" obtained in such way on the vertical coordinate planes (Fig. 9.1). One can imagine that these projections are the shadows of the fence illuminated by the

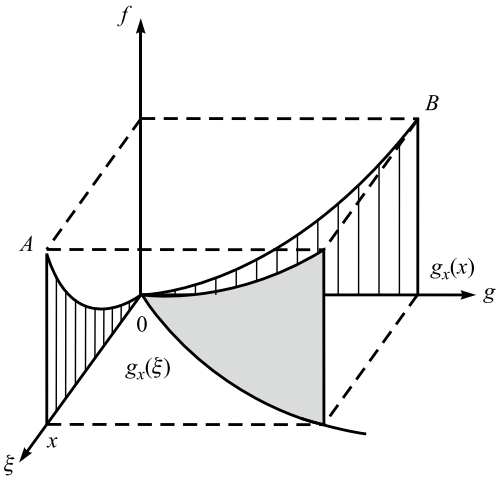


Fig. 9.1 To the geometrical interpretation of fractional integral

headlights of a distant car on the walls perpendicular to the rays. The area of the shadow OxA on the wall (ξ, f) is equal to an ordinary integral

$$ {}_0\mathsf{I}_x^1 f(x) = \int_0^x f(\xi)d\xi, $$

whereas the area of the shadow $Og_x(x)B$ on the other wall (g, f) is given by a fractional integral. The first assertion is a geometric interpretation of an ordinary integral, and the second relates to a fractional integral. The essential difference is that the shadow on the first wall does not change with change (e.g., increasing) of x, and the increasing of area occurs only due to the expansion of the integration domain. At the same time, the shadow falling on the second wall changes with x which contributes in addition to the change in the value of the fractional integral. This feature complicates the geometric interpretation both of the fractional integral and the fractional derivative.

The analogous interpretation of the right-sided integral and the corresponding fractional derivative is based on the right-sided Stieltjes integral

$$ {}_0\mathsf{I}_x^\mu f(x) = \int_x^b f(\xi)dh_x(\xi)d\xi $$

over the function

$$ h_x(\xi) = \frac{1}{\Gamma(\mu+1)}\left[x^\mu + (\xi - x)^\mu\right]. $$

In contrast to the previous case, all the points of the fence move when x changes (the end of the fence moves along the straight line $\xi = b$ in the plane (ξ, h), whereas in the first case the end was fixed at the point $\xi = 0$).

Let us present the one-dimensional Riesz operator as the sum of left-sided and right-sided Riemann–Liouville fractional derivatives:

$$ \begin{aligned} {}_0\mathsf{D}_x^{-\mu} f(x) &= \frac{1}{\Gamma(\mu)}\int_0^b f(\xi)|\xi - x|^{\mu-1}d\xi \\ &= \frac{1}{\Gamma(\mu)}\int_0^b f(\xi)(x-\xi)^{\mu-1}d\xi + \frac{1}{\Gamma(\mu)}\int_0^b f(\xi)(\xi - x)^{\mu-1}d\xi. \end{aligned} $$

It also can be presented in the form of the Stieltjes integral of $f(x)$ over the function

$$ r_x(\xi) = \frac{1}{\Gamma(\mu+1)}\left[x^\mu + \operatorname{sign}(x-\xi)(x-\xi)^\mu\right], $$

representing a form of a "fence" for this operator. It consists of two parts corresponding to the left-sided Riemann–Liouville integral $(g_x(\xi), (0 < \xi < x))$ and the right-sided one $(h_x(\xi),\ x < \xi < b)$. The area of the shadow on the first wall gives an ordinary definite integral with limits 0 and b:

$$ {}_a\mathsf{I}_b f(t) = \int_0^b f(\xi) d\xi. $$

Feller's operator can also be presented as a linear combination of the left-sided and right-sided Riemann–Liouville integrals of fractional order:

$$ \mathsf{D}_{u,v}^{-\mu} f(x) = u\ {}_a\mathsf{I}_x^{\mu} f(x) + v\ {}_x\mathsf{I}_b^{\mu}(x). $$

Now the fence has a gap at $\xi = x$. Its shadow on the first plane still corresponds to the usual integral of the first order, whereas the shadow on the other wall consists of two pieces that can be superimposed one on each other depending on the values of the coefficients a and b.

9.2.2 Tangent Vector and Gradient

Ben Adda [7, 8] developing a general concept of differentiability of order $\alpha \in (0, 1]$ for the function of several variables gives the following interpretation. Having introduced the directional derivative as

$$ \mathsf{D}_{\mathbf{u}}^{\alpha} f(\mathbf{x}) = \lim_{h \to 0} \frac{(T_\alpha f)(\mathbf{x} + h\mathbf{u}) - (T_\alpha f)(\mathbf{x}, \mathbf{x})}{h^\alpha} $$

(see Sect. 4.7.3.2 of Vol. I) with $\mathbf{u}$ being a d-dimensional unit vector and considering expression

$$ \sum_{j=1}^{d} \mathsf{D}_{\mathbf{e}_j}^{\alpha} f(\mathbf{x}) dx_j = \frac{1}{\Gamma(\alpha+1)} \frac{\partial^\alpha f}{\partial x_1^\alpha} dx_1^\alpha + \cdots + \frac{1}{\Gamma(\alpha+1)} \frac{\partial^\alpha f}{\partial x_d^\alpha} dx_d^\alpha $$

with Nishimoto's partial derivatives $\partial^\alpha f / \partial x_j^\alpha$ as a fractional differential

$$ d^\alpha f = \operatorname{grad}^\alpha f \cdot d\mathbf{l}^\alpha,\ d\mathbf{l}^\alpha = \{dx_1^\alpha, \ldots, dx_d^\alpha\}, $$

Ben Adda gives the following geometric interpretation of the fractional gradient of an order $\alpha \in (0, 1)$ in the three-dimensional space.

Let $f(x, y, z)$ be a function admitting $\operatorname{grad}^\alpha f$ at point $\mathbf{x}_0$, $\mathbf{x}$ be a point located on the surface S given by equation

$$ (T_\alpha f)(\mathbf{x}, \mathbf{x}_0) = C_0, $$

and $\mathbf{x}_0 + d\mathbf{l}$ be a point in a neighborhood of $\mathbf{x}$. Then the point $\mathbf{x}_0 + d\mathbf{l}^\alpha$ is not on S, but on a surface S' relating to another level C_0'. The displacement[1] $d^\alpha f$ is equal

[1] Strictly speaking, it is not a displacement in the case of fractional α.

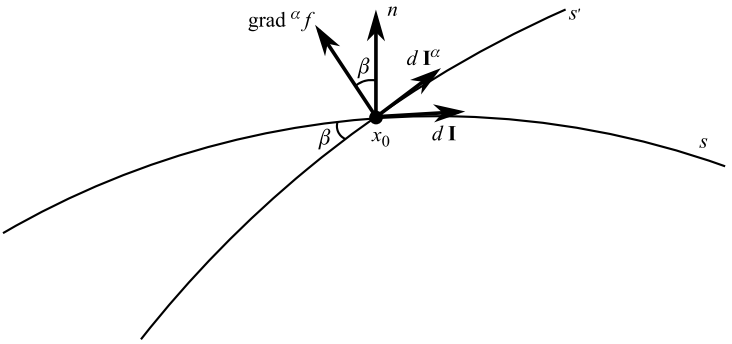


Fig. 9.2 The tangent vector in the level surface S

to zero in the level surface S, then the vector $d\mathbf{l}^{\alpha}$ is perpendicular to the gradient vector $\text{grad}^{\alpha} f$. This means that a tangent vector (in the level surface S) at point $\mathbf{x}_0$ is not normal to $\text{grad}^{\alpha} f$; its normal component is given by $\sin\beta|\text{grad}^{\alpha} f|$ where $\beta = \arccos|C_0' - C_0|/|d\mathbf{l}^{\alpha}|$ (see Fig. 9.2).

Albu and Opriş [3] present the fractional differential calculus on a differential manifold and consider the principal geometric structures on the fractional tangent bundle of order K is considered. On this tangent bundle are defined some objects and structures having a geometrical character. Fractional Lagrangeans on that bundle are also considered and the Euler–Lagrange equations are derived.

9.2.3 Fractals and Fractional Derivatives

9.2.3.1 Rutman Versus Nigmatullin

In 1992 Nigmatullin proposed to interpret the fractional integration as the integration over the fractal Cantor set. Presenting the evolution of the state of a physical system using the hereditary equation

$$J(t) = \int_0^t K(t,\tau) f(\tau) d\tau,$$

he suggested that the memory kernel $K(t,\tau)$ be nonzero only on the fractal Cantor set, and gave the following explanation.

"Only a part of states 'survive' during the evolution, and let the other part be lost in the evolution process. It means, that they are irreversibly lost and become unavailable for the system. The Cantor set is designed in such a way that the inaccessibility of these states provides automatically." Further, the author declares the intention "to show that the Cantor set in the limit $N \to \infty$ (N is the number of a decomposition

stage) converges to a fractional integral, and its order indicates the proportion of the remaining states and coincides with the fractal dimension of the set."

To construct the Cantor set of an arbitrary fractal dimension, one takes the time interval $(0, t)$ and removes the middle part, so that two intervals of length ξt ($\xi < 1/2$) remain. The hereditary kernel is assumed to be constant in these intervals and zero outside, and renormalized, in such a way that the integral of it is equal to 1. At the next stage, each of the remaining segments is exposed to the same procedure, etc. Assuming that the memory kernel is nonzero only at the points of the Cantor set, and using the Laplace transformation, R. Nigmatullin obtains the limiting ratio, allowing to interpret it as the Laplace transformation of the integral of fractional order equal to the fractal dimension of the Cantor set $\nu = \ln 2/\ln(1/\xi)$.

Rutman [99, 100] criticized conclusion and claims that "a direct link between fractional calculus and fractals is not established". Tatom [119] also expressed opinion that there is no simple one-to-one relationship between fractional calculus and fractals appears to exist. Fractal can be generated without the use of fractional calculus, and for each operation involving fractional calculus, there is not necessarily a corresponding fractal process. However, given a fractal process, fractional calculus can be used to generate related fractals with precise control over the fractal dimension (as we saw it in case of the fractional Brownian motion).

Nigmatullin acknowledged that Rutman's criticism had had foundations and forced him to revise his calculations, but he did not refuse his idea of the connection of fractional derivatives with fractals [74, 84]. In this new version, the logic of reasoning is as follows.

9.2.3.2 Integrating over the Cantor Set

Let us assume that some physical variable $J(t)$ is connected with some other one, represented by the smooth function $f(t)$ by the hereditary relationship

$$J(t) = \int_0^t K(t-\tau) f(\tau) d\tau \equiv K \star f(t),$$

where the memory kernel $K(t)$ is defined by the expression

$$K(t) = [1(t) - 1(t-T)]/T, \ t > 0,$$

where $1(t)$ is the Heaviside step function. The kernel $K_{T,\nu}^{(N)}(t)$, corresponding to the Nth stage of the construction of the Cantor set with the fractal dimension $\nu = \ln 2/\ln(1/\xi)$ on the segment $[0, T]$, is determined by the recurrent relation

$$K_{T,\nu}^{(N)}(t) = \left[K_{\xi T,\nu}^{(N-1)}(t) + K_{\xi T,\nu}^{(N-1)}(t - (1-\xi)T)\right]/2, \quad K_{T,\nu}^{(0)}(t) \equiv K(t).$$

Here $\xi \in [0, 1]$ is the scale parameter characterizing the compression ratio of the set when it passes on to the next stage. The height of the Cantor "comb" at the Nth stage of construction in accordance with the normalization is equal to $1/(2\xi)^N T$. The Laplace transformation of the recurrent relation leads to the following result:

$$\widehat{K}_{T,\nu}^{(N)}(\lambda) = \frac{1 + \exp(-\lambda T(1-\xi))}{2} \widehat{K}_{\xi T,\nu}^{(N-1)}(\lambda) = \frac{1 - \exp(-\lambda T \xi^n)}{\lambda T \xi^n} Q_N(\lambda T(1-\xi))$$

where

$$Q_N(z) = 2^{-N} \prod_{n=0}^{N-1} \left[1 + \exp(-z\xi^n)\right].$$

When $N \to \infty$,

$$\widehat{J}_N(\lambda) \to \widehat{J}(\lambda) = K_\nu[\lambda T(1-\xi)]\widehat{f}(\lambda), \quad K_\nu(z) = \lim_{N\to\infty} Q_N(z) = \frac{\pi_\nu(\ln z)}{z^\nu},$$

where $\pi_\nu(\ln z)$ is the periodic function with the period $\ln \xi$:

$$\pi_\nu(\ln z \pm \ln \xi) = \pi_\nu(\ln z).$$

This function was absent in the original version of Nigmatullin's calculations. To cope with the additional dependence, it was proposed to average the result over the period, in other words to replace

$$J(t) = K_\nu \star f(t),$$

by expression

$$J(t) = \langle K_\nu \rangle \star f(t),$$

leading to the fractional integral

$$J(t) = \frac{C(\nu)}{T^\nu \Gamma(\nu)} \int_0^t (t-\tau)^{\nu-1} f(\tau) d\tau,$$

with

$$C(\nu) = \int_{-1/2}^{1/2} \pi_\nu(\ln z + x \ln \xi) dx.$$

A further development of these ideas, based on the expansion of the periodic function π_ν into a Fourier series

$$\pi_\nu\left(\frac{\ln z}{\ln \xi}\right) = \sum_{n=-\infty}^{\infty} \exp\left(\frac{\ln z}{\ln \xi}\right),$$

led to the expression

$$K_\nu(z) = \sum_{n=-\infty}^{\infty} C_n \exp\left[(-\nu + i\Omega_n)\ln z\right].$$

Assuming further that the series can be approximated by the sum of three components,

$$\sum_{n=-\infty}^{\infty} C_n \exp\left[(-\nu + i\Omega_n)\ln z\right] \cong C_0 z^{-\nu} + A_n z^{-\nu+i\langle\Omega\rangle} + A_n^* z^{-\nu-i\langle\Omega\rangle},$$

$$\langle\Omega\rangle = 2\pi\langle n\rangle/\ln\xi,$$

the last two of which disappear after averaging, authors arrive at the previous result again.

Similar results were obtained by Ren et al. [96] and Yu et al. [141]. Interpretation remained practically the same. The physical system, which can be described in terms of fractional derivatives, should be characterized by the presence of slow (of a diffusion type, like in the comb or percolation models) channels creating the phenomenon of residual memory. The processes with residual memory correspond to the energy principle formulated by Jonscher for relaxation in dielectrics [59]. To this class, the authors refer transport in percolation clusters, fractal trees, porous systems, and similar heterogeneous structures.

9.2.3.3 Cookie-Cutter Support

Yu et al. [141] gave an interpretation of the fractional integral in terms of the generalized *cookie-cutter sets*. For any given $T \in (0, \infty)$, denote $E_0 = [0, T]$ and $\{\varphi_{n,j}(x) : E_0 \mapsto E_0,\ j = 1, 2, \ldots, K_n < \infty\}_{n=1,2,3,\ldots}$ is a family of functions satisfying:

1. $\varphi_{n,j} : E_0 \mapsto \varphi_{n,j}(E_0)$ is one-to-one mapping and $\mathrm{Int}(\varphi_{n,i}(E_0)) \cap \mathrm{Int}(\varphi_{n,j}(E_0)) = \emptyset$ $(i \neq j)$ for any n and $1 \leqslant i, j \leqslant K_n$.
2. For all n, the mapping $S_n : \cup_{j=1}^{K_n} \varphi_{n,j}(E_0) \mapsto E_0$, defined by $S_n|_{\varphi_{n,j}(E_0)} = \varphi_{n,j}^{-1}$ is $C^{1+\gamma}$-differentiable.

Then $S_n,\ n = 1, 2, 3, \ldots$ is called a sequence of *cookie-cutter maps*. Let $I_n = 1, 2, \ldots, K_n$, $\Lambda_n = I_1 \times I_2 \times \cdots \times I_n$

$$E_{j_1 j_2 \ldots j_n} = \varphi_{1,j_1} \circ \varphi_{2,j_2} \circ \cdots \circ \varphi_{n,j_n}(E_0), \quad E(n) = \cup_{j_1 j_2 \ldots j_n \in \Lambda_n} E_{j_1 j_2 \ldots j_n}.$$

Then $E_T = \cap_{n \geqslant 1} E(n)$ is called the *generalized cookie-cutter* set.

Basing their construction of the memory function upon this concept, the authors define the memory measure $d\mu(\tau)$ on E_T as a weak limit of

$$d\mu_n(\tau) = \sum_{j_1 \dots j_n \in \Lambda_n} P_{1,j_1} \cdots P_{n,j_n} 1(\tau \in E_{j_1 \dots j_n}) d\tau / |E_{j_1 \dots j_n}|,$$

where $P_{n,j} \geqslant 0, \quad \sum_{j=1}^{K_n} P_{n,j} = 1$. For any $A \subset E_0$

$$\int_A d\mu_n(\tau) = \sum_{j=1}^{K_n} P_{n,j} \int_A d\mu_{n-1} \circ \varphi_{n,j}^{-1}(\tau).$$

If $K_n \to K$, $P_{n,j} \to P_j$, and $\varphi_{n,j} \to \varphi_j$ as $n \to \infty$,

$$\int_A d\mu(\tau) = \sum_{j=1}^{K} P_j \int_A d\mu_{n-1} \circ \varphi_j^{-1}(\tau).$$

Applying the Laplace transform to the integral

$$J(t) = \int_0^t f(t-\tau) d\mu(\tau),$$

and assuming $\varphi_1(x) = \xi_1 x, \ 0 < \xi < 1$, they obtain

$$\hat{m}(\lambda) \equiv \int_0^\infty e^{-\lambda\tau} d\mu(\tau) \cong A\lambda^{-\nu}, \ \mathrm{Re}\lambda \to 0,$$

and

$$J(t) \cong A \ {}_0\mathsf{D}_t^{-\nu} f(t),$$

with A being constant and $\nu = \ln P_1 / \ln \xi_1$.

9.2.3.4 Other Points of View

Moshrefi-Torbati and Hammond [79] interpreted time-fractional operators on the base four concepts. These are fractal geometry, linear filters, construction of a Cantor set, and physical realization of fractional operators. It is concluded there that fractional operators may be grouped as filters with partial memory that occupies an intermediate position between filters with complete memory and those with no memory. Speaking in physical terms, the fractional order of a fractional integral is an indication of the remaining or preserved energy of a signal passing through such system, and the fractional order of a derivative reflects the rate at which a portion of the energy has been lost. To demonstrate a physical example of such a system, these authors refer to the article [86], authors of which have shown that the fractional integrating and differentiating can be interpreted in terms of signal propagation through

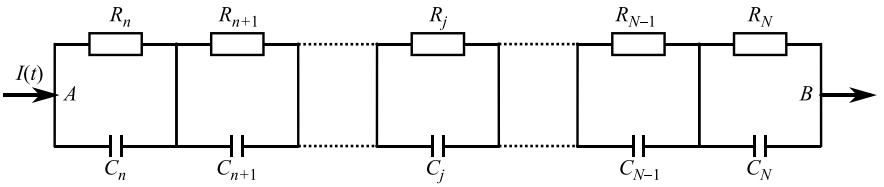


Fig. 9.3 The structure of a domino ladder. $R_j = R_0/g^j$, $C_j = C_0/G^j$

a simple circuit consisting of a chain of resistors and a chain of capacitors, interconnected as shown in Fig. 9.3, and called the *domino ladder network*. Passing through the circuit, the current $I(t)$ generates a potential difference $U_{AB}(t)$ between points A and B having the form

$$U_{AB}(t) = \int_0^t I(t-t') \sum_{j=-n}^{N} e^{-t'/\tau_j} \frac{dt'}{C_j}, \quad \tau_j = R_j C_j.$$

If the adjacent resistors and capacitors differ in value by constant factors g and G respectively, forming two geometric progressions, then in the limit $g \to 1$ and $G \to 1$ as $n \to \infty$ and $N \to \infty$, $U_{AB}(t)$ is given by

$$U_{AB}(t) = K_\nu \, {}_0\mathsf{D}_t^{\nu-1} I(t),$$

where

$$K_\nu = \frac{\pi \operatorname{cosec}(\pi\nu) R_0^\nu}{\ln(Gg) C_0^{1-\nu}}.$$

Nevertheless, as Stanislavsky and Weron [108] considered, the question about the connection between fractals and fractional calculus remained open. The point is that the exposition of Le Mehaute et al. [74] does not contain the exact expression for the kernel of convolution arising in the above-mentioned averaging procedure. Instead, the authors give its estimations which are not sufficient for ultimate conclusions. The exact analytical expression of the Cantor set for convolution of a smooth function with the normalized density of the Cantor set obtained by Stanislavsky and Weron [108] demonstrates that the self-similar kernel of this convolution cannot explicitly be reduced to any product of a power and a log-periodic function as suggested before. They used a functional equation for the kernel of convolution whose solution provide to be a more complicated function then the product of power and log-periodic functions. Thus, the procedure of averaging a smooth function over the Cantor set gives a very coarse approximation of the fractional integral and does not offer any advantages. The authors write: "This result rather shows that no direct relation between the calculus of fractional order and the fractal geometry is yet found."

In spite of this criticism, the "linguistic" idea of the relationship of fractal structures and fractional derivatives continues to live. The paper [76] builds on extension of continuum thermomechanics to fractal porous media by using the Tarasov (2005) approach. In its framework, the mass $m(V)$ contained in a three-dimensional region V is represented by the integral

$$m(V) = \int_V \rho(\mathbf{r}) c_3(D, r) dV,$$

where

$$c_3(D, r) = \frac{2^{3-D}\Gamma(3/2)}{\Gamma(D/2)} r^{D-3}.$$

When $\rho(\mathbf{r}) = \rho_0 = \text{const}$ and V is a sphere of radius R,

$$m(V) = \rho_0 \int_V c_3(D, r) dV = \rho_0 \left[\frac{4\pi 2^{3-D}\Gamma(3/2)}{D\Gamma(D/2)} \right] R^D.$$

The authors note that $c_3(D, r)$ solely depends on the scalar distance r, which in turn confines the formulations to isotropic fractals. It is hard to accept this statement. The point is that if this function is centered at a fixed point taken as an origin of coordinates, it doesn't keep this properties at displacement the origin. Thus, we meet merely the regular medium with the density smoothly decaying in all directions from the given origin, whereas the perfect fractal should possess this property relative to any of its point. Moreover, passing to large scales doesn't smooth over inhomogeneities as one could think looking at Fig. 3 in the cited work, because fractals are selfsimilar structures and cannot be homogenized at all (See Fig. 9.4.).

It may seem that the concept of the limited fractality (prefractality) allows this contradiction, but in reality it is not so: If a medium looks homogeneous at some scale, we must use operators of integer orders with constant coefficients.

Tarasov [113] writes in his article: "In the general case, the fractal media cannot be considered as continuous media. There are points and domains that do not filled of the medium particles. These domains can be called the porous. The fractal media can be considered as continuous media for the scales much more than mean value of the pore size. In this case, we can use the integration and differentiation." The first sentence is quite right, but the third is not. The generic property of any fractal is *its self-similarity at all scales*. If one finds such a large scale, on which the structure of the system can be neglected, then the system is not a fractal anymore and should be described in terms of usual derivatives. Certainly, the regular medium can also be inhomogeneous, but this inhomogeneity is of quite different kind: It does not possess translational invariance in all directions. In other words, one is not able to imagine a media, the mass density of which decays around each point in all directions according to the same law. For this reason, the mathematical scheme developed in works (Tarasov, 2005) seems nothing more than abstract mathematical exercises. As for direct testing of validity of the homogenization of a fractal, it seems quite unnecessary. In addition

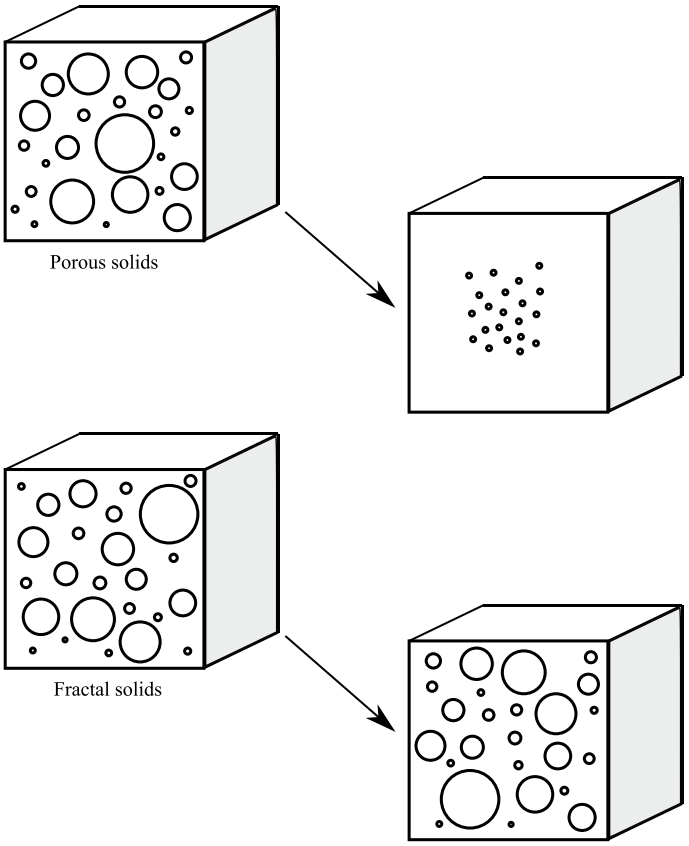


Fig. 9.4 What we see with increasing scales looking at a porous solid (the upper panel) and at fractal solids (the lower panel)

to abovesaid conceptual reasons, there exist direct evidences that a fractal can not be replaced by a regular homogeneous medium. Thus, the diffusion on fractals is *at all scales qualitatively different* from the diffusion in a homogeneous medium (see Sect. 3.7 and Ref. therein of Vol. I).

At the same time, it is worth to stress that everything said above relates to deterministic (non-random) systems. In random systems and processes, possessing the property of self-similarity, the link "fractal-fractional" is realized in a natural way (see Chap. 3 and Ref. therein of Vol. I and also Sect. 9.4).

9.3 Fractal and Other Derivatives

Discussions on fractals and fractional derivatives inspired some modifications of derivatives different from fractional ones. Some of them we touch in this subsection.

9.3.1 Fractal Derivative

Chen [21] defines and applies to anomalous diffusion problem [22] derivatives $\partial f(t)/\partial t^{\alpha}$ named him the *fractal derivatives*.[2] Obviously, this is nothing but a change of independent variable, and it is quite natural to get the solution to the equation with derivative df/dt^{α} in the form of the stretched exponent:

$$\frac{df}{dt^{\alpha}} = -\mu f(t) \quad \Rightarrow \quad f(t) = f(0)e^{-\mu t^{\alpha}}$$

instead of the case with the R-L derivative leading to the Mittag-Leffler function [22].

9.3.2 New Fractal Derivative

He [46] has introduced another type of non-standard derivative named it the *new fractal derivative* for engineering application. It is more closely and above all more clearly associated with fractality of the medium under consideration. Let us look at Fig. 9.5, illustrating such a media.

If we assume the smallest measure is l_0 and any discontinuity less than it is ignored, then the distance Δx between points A and B can be approximated by kl_0^{α}, where k is a constant and α is the fractal dimension. He defines his derivative via relation

$$\frac{Df(x)}{Dx^{\alpha}} = \lim_{\Delta x \to l_0} \frac{f(A) - f(B)}{\Delta x}.$$

9.3.3 Generalized Fractional Laplaian

Continuing to improve mathematical methods of image processing and developing tools for solving more complicated problems such as denoising, deconvoluting, and reconstructing *vector fields*, Tafti and Unser [112] approached this problem on the

[2] Observe that this term is used in Saichev and Woyczinzky [101] for fractional derivatives.

basis of generalizing the Laplace operator in two senses: making it of fractional orders and reweighting solenoidal and irrotational components.

There are considered certain convolution operators which act on vector fields $\mathbf{f}$ and can be written in terms of the inverse Fourier integral as

$$\mathsf{U} : \mathbf{f}(\mathbf{x}) \mapsto (2\pi)^{-d} \int_{\mathbb{R}^d} e^{i\mathbf{k}\cdot\mathbf{x}} \widetilde{\mathsf{U}}(\mathbf{k}) \widetilde{\mathbf{f}}(\mathbf{k}) d\mathbf{k},$$

where $\widetilde{\mathsf{U}}$ is the matrix-valued Fourier transform of the operator U. Such operators appear in equations of the form

$$\mathsf{U}\mathbf{B}_H = \mathbf{W},$$

modeling statistically self-similar (homogeneous) and rotation-invariant (isotropic) vector fields. The above properties impose the following conditions on the Fourier expression of U:

$$\widetilde{\mathsf{U}}(\mathsf{K}\mathbf{k}) = \mathsf{K}\widetilde{\mathsf{U}}(\mathbf{k})\mathsf{K}^{\mathrm{T}} \tag{9.2}$$

(rotational invariance), and

$$\widetilde{\mathsf{U}}(\sigma\mathbf{k}) = \sigma^{2\gamma}\widetilde{\mathsf{U}}(\mathbf{k}) \tag{9.3}$$

(homogeneity of degree 2γ).[3]

As a result, the vector convolution operator U satisfying the conditions (9.2)–(9.3) was represented in the form

$$\mathsf{U} = (-)^{\gamma} \left[e^{\xi_{\mathrm{irr}}} \mathsf{E} + e^{\xi_{\mathrm{sol}}} \mathsf{E}' \right].$$

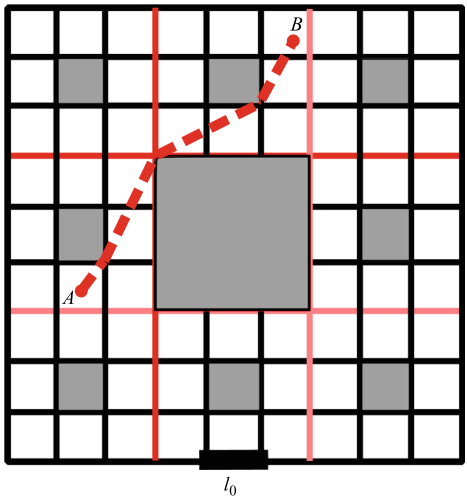


Fig. 9.5 For definition of the new fractal derivative

[3] $\mathbf{W}(\mathbf{x})$ is the vector white noise and γ relates to the Hurst exponent, by $H = 2\gamma - d/2$.

The penultimate operator E and its complement $\mathsf{E}' = \mathsf{I} - \mathsf{E}$ project a vector field onto its curl-free and divergence-free components, respectively:

$$\operatorname{div} \mathsf{E}' = 0, \quad \operatorname{curl} \mathsf{E} = 0,$$

$$\mathsf{E} \operatorname{grad} = \operatorname{grad}, \quad \mathsf{E} \operatorname{curl} = 0.$$

For this reason, Tafti and Unser [111] consider U as some sort of Laplacian operator

$$(-)^{\gamma}_{\xi} = (-)^{\gamma} \left[e^{\xi_{\mathrm{irr}}} \mathsf{E} + e^{\xi_{\mathrm{sol}}} \mathsf{E}' \right]$$

with $\xi = (\xi_{\mathrm{irr}}, \xi_{\mathrm{sol}}) \in \mathbb{C}^2$. In view of the properties of the operator E, this factorization means that the operator $(-)^{\gamma}_{\xi}$ combines a coordinatewise fractional Laplacian with a reweighting of the curl- and divergence-free components of the operand.

9.3.4 Fractional Derivatives in q-Calculus

In the theory of the quantum or q-calculus [38, 61], the derivative of a function $f(x)$ and the integral are defined by

$$D_q f(x) = \frac{f(x) - f(qx)}{x - qx}, \quad x \neq 0,$$

and

$${}_0D^{-1}_{q,x} f(x) = \int_0^x f(t) d_q t = x(1-q) \sum_{k=0}^{\infty} f(xq^k) q^k, \quad 0 \leqslant |q| < 1,$$

respectively. Al-Salam [5] and Agarwal [1] introduced several types of fractional q-integrals and q-derivatives. Rajković et al. [94] and Ortigueira [87] continue investigation in this direction.

9.3.5 Fuzzy Fractional Operators

By an ordered fuzzy number A they mean an ordered pair of continuous functions $f, g : [0, 1] \mapsto \mathbb{R}$. In the model of fuzzy real numbers developed by Kosiński [68], it is possible to utilize the fuzzy arithmetic in a simple way and to construct an Abelian group of fuzzy numbers, and then an algebra. Moreover, combining this concept with

fractional differential and integral operators yields *fuzzy fractional operators* each of which produces again a pair of functions related to fuzzy exponents.

9.4 Probabilistic Interpretation

9.4.1 Probabilistic View on the G-L Derivative

Analyzing the Grünwald–Letnikov derivative of order $\alpha \in (0, 1)$ written in the form

$$ {}_0\mathsf{d}_t^{\alpha} x(t) = \lim_{h \to \infty} \left[h^{-\alpha} \sum_{j=0}^{\infty} \gamma(\alpha, j) x(t - jh) \right], $$

$$ \gamma(\alpha, j) = (-1)^j \frac{\Gamma(\alpha + 1)}{j! \Gamma(\alpha - j + 1)}, $$

and taking notice of the equalities

$$ \gamma(\alpha, 0) = 1 $$

and

$$ -\sum_{j=1}^{\infty} \gamma(\alpha, j) = 1, $$

Tenreiro Machado [120] has come to the conclusion that the expression $-\sum_{j=1}^{\infty} \gamma(\alpha, j) x(t - jh)$ with $\alpha \in (0, 1)$ can be viewed as an expected value of the random variable X with the probability distribution

$$ \mathsf{P}(X = x(jh)) = |\gamma(\alpha, j)|, \quad j = 1, 2, 3, \ldots, $$

so, speaking in terms of signal transfer problems, the Grünwald–Letnikov definition gets the slope θ of a triangle composed by the "present" sample of the signal $x(0)$ and the arithmetic mean of the "past/future". As $h \to 0$, the slope $\theta \to {}_0\mathsf{D}_t^{\alpha} x(t)$. When $\alpha = 1$, the "past/future" has probability 1 for the sample near the "present" and zero for the rest of the "past/future", and when $\alpha = 0$ all the "past/future" has probability zero. We see that the integer orders correspond to a deterministic perspective whereas the fractional ones relate to a random process.

9.4.2 Stochastic Interpretation of R-L Integral

Interpretation of fractional equations in terms of stochastic processes is discussed in many works. The connection of the fractional integral with stable distributions was investigated by Stanislavsky [107]. He has shown that the equations with fractional time derivative describe evolution of a physical system whose time degree of freedom becomes stochastic and presents the sum of random time intervals with α-stable distribution.

A simple interpretation of the fractional R-L integral follows from the fractional Poisson process [16, 125, 128]. Because the mean rate of pulses counted to the past direction (from the initial moment t to $a < t$),

$$n_{-}(t-t') = \frac{(t-t')^{\nu-1}}{\Gamma(\nu)},$$

the integral is

$${}_{a}\mathsf{D}_{t}^{-\nu} f(t) = \frac{1}{\Gamma(\nu)} \int_{a}^{t} (t-t')^{\nu-1} f(t') dt' = \left\langle \sum_{j \in J(a,t)} f(t - T^{(j)}) \right\rangle,$$

where $T^{(j)} = T_1 + \cdots + T_j$, T_i are independent random variables with the common pdf

$$\psi_{\nu}(t) = t^{\nu-1} E_{\nu,\nu}(-t^{\nu}),$$

and $J(a,t)$ the subset of natural numbers $\{j : T^{(j)} < (t-a)\}$. One can say that this fractional integral is an integral over the random fractal dust averaged over all its realizations (see, for details, Sects. 3.5 and 6.3.5 of Vol. I).

9.4.3 Fractional Powers of Operators

Fractional powers of closed linear operators were first constructed by Bochner [11] and subsequently Feller [33], for the Laplacian operator. These constructions depend in an essential way on the fact that the Laplacian generates a semigroup.

Let A_{α}, $\alpha \in (0,1]$, be an infinitesimal generating operator of the continuous semigroup $\mathsf{T}_{\alpha}(t)$,

$$\mathsf{T}_{\alpha}(t) = \exp\{t\mathsf{A}_{\alpha}\}, \quad t \geqslant 0,$$

and

$$\mathsf{A}_{\alpha} = -(-\mathsf{A}_1)^{\alpha},$$

then the semigroups T_{α} and $\mathbf{T}_1$ are linked with each other via relation

$$\mathsf{T}_\alpha(t) = \int_0^\infty \mathsf{T}_1(\tau) g_+\left(\frac{\tau}{t^{1/\alpha}};\alpha\right)\frac{d\tau}{t^{1/\alpha}}.$$

From the formal point of view these identities look almost obvious,

$$\begin{aligned}\int_0^\infty \mathsf{T}_1(\tau) g_+\left(\frac{\tau}{t^{1/\alpha}};\alpha\right)\frac{d\tau}{t^{1/\alpha}} &= \int_0^\infty \mathsf{T}_1(zt^{1/\alpha}) g_+(z;\alpha)dz\\ &= \int_0^\infty \exp\left\{zt^{1/\alpha}\mathsf{A}_1\right\} g_+(z;\alpha)dz\\ &= \int_0^\infty \exp\left\{-z\left[t^{1/\alpha}(-\mathsf{A}_1)\right]\right\} g_+(z;\alpha)dz\\ &= \exp\left\{-\left[t^{1/\alpha}(-\mathsf{A}_1)\right]^\alpha\right\}\\ &= \exp\{-t(-\mathsf{A}_1)^\alpha\}\\ &= \exp\{t\mathsf{A}_\alpha\} = \mathsf{T}_\alpha(t).\end{aligned}$$

So is the expression of the fractional operator $(-\mathsf{A}_1)^{-\alpha}$ through the T_α:

$$(-\mathsf{A}_1)^{-\alpha} = \int_0^\infty \exp\{-t(-\mathsf{A}_1)^\alpha\}\,dt = \int_0^\infty \mathsf{T}_\alpha(t)dt.$$

For more substantial proof see Yosida [140].

Thus, according to Sect. 6.3.3.1 of Vol. I,

$$\mathsf{T}_\alpha(t) = \left\langle \mathsf{T}_1(t^{1/\alpha}S_+(\alpha))f\right\rangle$$

and

$$\begin{aligned}(-\mathsf{A})^{-\alpha}f &= \int_0^\infty\left[\int_0^\infty \exp\left\{-z[t^{1/\alpha}(-\mathsf{A})]\right\}dt\right] f g_+(z;\alpha)dz\\ &= \left\langle\left[\int_0^\infty \exp\left\{-S_+(\alpha)[t^{1/\alpha}(-\mathsf{A})]\right\}dt\right] f\right\rangle.\end{aligned}$$

Let us apply this approach to the Havriliak–Negami operator

$$[1 +_{-\infty} \mathsf{D}_t^\alpha]^\beta f(t) = \delta(t).$$

The semigroup generated by the infinitesimal operator $[1 +_{-\infty} \mathsf{D}_t^\alpha]^\beta$, can be found as

$$\widehat{\mathsf{T}}_{\alpha,\beta}(t)f = \left\langle \exp\left(-t^{1/\beta}S_+(\beta)\right) f\left(z - \left[t^{1/\beta}S_\beta\right]^{1/\alpha} S_+(\beta)(\alpha)\right)\right\rangle.$$

The inverse operator can be calculated with the help of an averaging procedure

$$[1 +_{-\infty} \mathsf{D}_t^\alpha]^{-\beta} f = \left\langle\, \beta[S_+(\beta)]^{-\beta} E^{\beta-1} f\left(z - S_+(\alpha)E^{1/\alpha}\right)\right\rangle,$$

where E is independent of S_+, exponentially distributed random variable with the unit mean value.

9.5 Classical Mechanic Interpretation

9.5.1 A Car with a Fractional Speedometer

Let us imagine a moving car on a highway. The driver measures the velocity $V(t)$ with a speedometer and on the time τ with his own watch. However, the watch does not show the right time: its reading is connected with the true time θ by the relation $\theta = g(\tau)$. Consequently, the driver time interval $d\tau$ corresponds to the true interval of the duration $d\theta = dg(\tau)$. Leaving point A at midnight ($\tau_A = 0$) and arrived at the point B at time $\tau_B = t$, the driver will compute the length of the path as the integral

$$S_{AB} = \int_0^t V(\tau)d\tau.$$

The correct path is expressed as a Stieltjes integral with the integrating function $g(\tau)$:

$$L_{AB} = \int_0^t V(\tau)dg(\tau).$$

Involving the function (9.1) as $g(\tau)$, we express this distance through the fractional integral

$$L_{AB} = {}_0\mathrm{I}_t^{\mu} V(t).$$

The time-derivative of this expression is equal to the speed of motion u calculated with the driver time. It is linked with the true speed as:

$$u(t) = \frac{1}{\Gamma(\mu)} \frac{d}{dt} \int_0^t \frac{V(\tau)d\tau}{(t-\tau)^{1-\mu}}.$$

Note that if the driver has repaired his watch, then μ will become equal to 1 and both these speeds will coincide: $u(t) = V(t)$.

Another example, easily interpreted in terms of fractional derivatives is processes with loss of memory in collisions has been proposed in Yu et al. [141]. Representing the Newton equation of motion in the integral form

$$\Delta \mathbf{v} = \frac{1}{m} \int_0^t \mathbf{F}(\mathbf{r}, \mathbf{v}, \tau)d\tau,$$

authors write: “If the interaction of the particle with a self-similar or generalized self-similar fractal structure has a collision nature, then the force can be presented

in the form

$$\mathbf{F}(\mathbf{r}, \mathbf{v}, \tau) = \mathbf{F}(\mathbf{r}, \mathbf{v}, \tau) \sum_k 1(t_k < \tau < t_{k+1})\rho_k,$$

where $1(t_k < \tau < t_{k+1})$ is the interval indicator (t_k, t_{k+1}), and ρ_k is the density of states. For the force acting only during a certain period of time

$$m\Delta\mathbf{v} = A[\Gamma(\nu)]^{-1} \int_0^t (t-\tau)^{\nu-1}\mathbf{F}(\mathbf{r}, \mathbf{v}, \tau)d\tau.$$

This equation can be used to describe both the Brownian motion and the loss in collisions", conclude the authors in the next to last section of their article. Note, however, that we are still far away from description of real processes with the help of this equation. First of all, we must make sure in the existence of such a fractal structure, establish the cause of its appearance and identify its key characteristics. But what is more important, the last equation assumes a special sort of averaging, namely, the averaging over the fractional Poisson flow (Sect. 3.5 of Vol. I).

From the other side, the more complex *the model* explaining the fractional derivative is, the less useful for *interpretation* of such operators, it is because the interpretation should facilitate intuitive applying. What are these "lost states of the system"? What is their physical meaning? Why are they necessarily characterized by a fractal distribution in time? How are they associated with completely different processes that allow a correct description in terms of fractional derivatives? These issues are more complicated than mathematical transformations, which we see in these works.

I should stress, however, that the most natural case of appearance of a fractional derivative in the simplest mechanical problem relates to the motion of a body in a viscous liquid (Sects. 7.3.1 and 7.3.2).

9.5.2 Dynamical Systems

Considering a fractional integral (and then a fractional derivative) from sufficiently well behaved function $f(t)$,

$${}_af^{(-\nu)}(t) = {}_a\mathsf{D}_t^{-\nu} f(t) = \frac{1}{\Gamma(\nu)} \int_a^t \frac{f(\tau)d\tau}{(t-\tau)^{1-\nu}},$$

Rutman [100] has stated the following relation between it and a causal dynamical model without memory and without fractal structures: fractional integral (derivative) can be represented as an infinite system of ordinary linear differential equation of the first order, with the coefficients depending on the independent variable,

$${}_af^{(-\nu)}(t) = C(t)z(t), \quad \frac{dz(t)}{dt} = A(t)z(t) + Bf(t).$$

The dimensions of matrices are infinite, but after a proper parametrization, the model can be approximately represented by a system of a finite order. This system is linear, stable and causal, with the approximation error controlled by the order of the system.

The following development of this idea can be found in articles [2, 142, 143], and others.

Tenreiro Machado and Galhano [121] accepting fractional calculus as a mathematical paradigm that describes the dynamics of hereditary systems or reflects an average of a large population of micro elements suggested analyzing the statistical dynamics of a system composed of a large number of micromechanical elements with backlash and impacts.

Each elemental component (See Fig. 9.6) consists of two masses m_1 and m_2 with coordinates x_1 and x_2 respectively, having a backlash h and subjected to the action of the random force

$$f(t) = f_{\max}\xi(t),$$

where $\xi(t) \in [-1, 1]$ is a random number generator. The motion of the masses m_1 and m_2 were calculated through the Runge-Kutta algorithm with a fixed time step and the transfer functions

$$H_k(i\omega) = \frac{\mathscr{F}\{\dot{x}_k(t)\}(\omega)}{\mathscr{F}\{f(t)\}(\omega)}, \quad k = 1, 2.$$

Results of simulations performed in this work have shown that the transfer functions $H_k(i\omega)$ can be approximated by $c_k\omega^{-\alpha_k}$ with $\alpha_1 = \alpha_2 = 1$. This reveals an integer-order dynamics for both types of the micromechanical elements. However, the simulation of the global dynamic evaluation of the multicomponent system transfer function

$$\langle H_k(i\omega)\rangle = \frac{1}{n}\sum_{j=1}^{n}[H_k(i\omega)]_j$$

with different levels of the exciting force $f_{\max}$ demonstrates an integer-order dynamics for m_1 but a fractional dynamics for m_2 ($\alpha_2 = 1,17$ for $f_{\max} =$

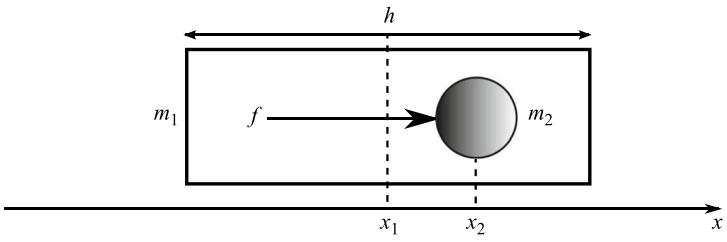


Fig. 9.6 A micromechanical element

{10, 20, 30, 40, 50} of dimensionless units). The authors conclude that, while individual dynamics of each element has an integer-order nature, the global dynamics reveal the existence of both integer and fractional dynamics.

9.5.3 Coarse-Grained-Time Dynamics

Considering the difference between time evolution operators $\mathsf{T}_\alpha(t)$ and $\mathsf{T}_1(t)$,

$$\mathsf{T}_\alpha(t) f(t_0) = \int_0^\infty \mathsf{T}_1(\tau) f(t_0) g_+\left(\frac{\tau}{t^{1/\alpha}}; \alpha\right) \frac{d\tau}{t^{1/\alpha}},$$

Hilfer concludes that basic issues, such as irreversibility, translation symmetry, or the meaning of stationarity are inevitably involved when proposing fractional dynamics, and advances as a basic postulate that *all time evolutions of physical systems are irreversible*. The concept of reversible behaviour is an idealization applicable only to a system being isolated from its past history and its environment. For an open system, it is not the case. In the framework of the Hilfer paradigm, a time evolution operator $\mathsf{T}_\alpha(t)$ formally introduced should be obtainable from a *coarse graining procedure*. A precise description of this procedure is in Hilfer [55]. The key idea lies in combining a time averaging

$$f(t) \mapsto \bar{f}_s(t) = \frac{1}{s} \int_{t-s}^{t} f(t')dt'$$

in the limit $t, s \to \infty$ with an appropriate rescaling of both variables.

9.5.4 Gradient Systems

Tarasov [117] uses the fractional forms of exterior calculus suggested in Cottrill-Shepherd and Naber [24, 25] for generalization of a gradient system described by the equation

$$\frac{dx_j}{dt} = -\mathsf{D}_{x_j} V(x_i), \quad i, j = 1, 2, \ldots, d,$$

to its fractional counterpart

$$\frac{dx_j}{dt} = -{}_0\mathsf{D}^\alpha_{x_j} V(x_i).$$

As an example, Tarasov considers the system

$$\frac{dx}{dt} = -\,{}_0\mathsf{D}^\alpha_x V(x, y), \quad \frac{dy}{dt} = -\,{}_0\mathsf{D}^\alpha_y V(x, y)$$

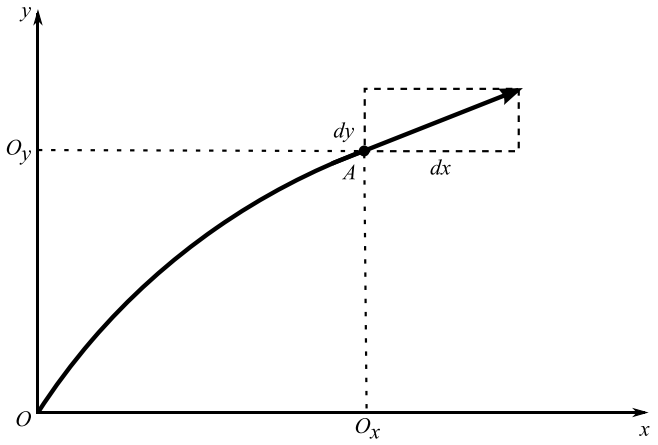


Fig. 9.7 Two-dimensional motion of a fractional gradient system

with a given field $V(x, y)$. Interpreting it as equations of motion of some body and writing in an explicit form,

$$\frac{dx}{dt} = -\frac{1}{\Gamma(1-\alpha)}\frac{\partial}{\partial x}\int_0^x \frac{V(x', y)dx'}{(x-x')^\alpha}, \quad \frac{dy}{dt} = -\frac{1}{\Gamma(1-\alpha)}\frac{\partial}{\partial y}\int_0^y \frac{V(x, y')dy'}{(y-y')^\alpha},$$

we see that the contours of integration in these equations is not connected to a real trajectory of the body under consideration (Fig. 9.7), and this deprives them of the physical sense.

There is also considered the simplest fractional Hamiltonian system,

$$\frac{dq}{dt} = {}_0\mathsf{D}_p^\alpha H, \quad \frac{dp}{dt} = -{}_0\mathsf{D}_q^\alpha H,$$

of a fractional order $\alpha \in (0, 1)$ with Hamiltonian

$$H(q, p) = ap^2 + bq^2.$$

The fractional differential form

$$\begin{aligned} d^\alpha H &= (dp)^\alpha\, {}_0\mathsf{D}_p^\alpha H + (dq)^\alpha\, {}_0\mathsf{D}_q^\alpha H \\ &= (dq)^\alpha \frac{1}{\Gamma(1-\alpha)}\frac{\partial}{\partial p}\int_0^p \frac{H(q, p')dp'}{(p-p')^\alpha} + (dp)^\alpha \frac{1}{\Gamma(1-\alpha)}\frac{\partial}{\partial q}\int_0^q \frac{H(q', p)dq'}{(q-q')^\alpha}. \end{aligned}$$

reveals the same inconsistency of a real trajectory and the integration contour which leads to the erroneous opinion on existence of stationary states of this system.

9.5.5 Chaos Kinetics

Grigolini et al. [42] note that one of the main tenets of the current literature on the derivation of statistical mechanics from dynamics rests on the conviction that randomness can be generated from Hamiltonian dynamics either as a consequence of the action of infinitely many degrees of freedom or as a consequence of chaos. The Hamiltonian

$$H = \sum_{i=-N}^{N} \left(\frac{p_i^2}{2m_i} \right) + \frac{k}{2}(q_{i+1} - q_i)^2,$$

where $m_0 = M$ and $m_i = m$ if $i \neq 0$, is an example of the first kind, and the sense of the Van Hove limit in that case is that the very short and the very large time scales are beyond the range of observation. Thus, the observer's limitation generates the randomness phenomena which in turn leads to fractional operators in asymptotical expressions. Grigolini et al. give an example of the second kind showing that the fractional operators can be generated by Hamiltonian system trajectory of which is almost of jumping (in a one-dimensional system—dichotomic) kind. This phenomenon can be interpreted as a manifestation rather of chaos than of statistics. This way to a deeper understanding of the essence of fractional calculus has been paved by Zaslavsky with coworkers and represented in his book [144] devoted to chaos arising in Hamiltonian systems. Zaslavsky pays a special attention to fractional kinetic as an approximate way to describe an ensemble of trajectories or particles, while neglecting some details of dynamics. In his opinion, the origin of fractional kinetics has two causes: first, the existence of singular zones in phase space, that create a set of sticky domains (traps), and second, the set is of a fractal or multifractal structure, generally speaking, in time and in phase space (configuration space) simultaneously.

To validate this approach, Zaslavsky refers to the *KAM* (*Kolmogorov-Arnold-Moser*) *theory* consequence stating that Hamiltonian systems with fully developed chaos demonstrate, as a rule, exponential decay laws, whereas the systems with a mixed phase space containing islands of regular motion usually have more slow algebraic decays due to the effect of stickiness of trajectories to the boundaries of such islands. This fact being complemented by assumption about self-similar (fractal) distribution of the traps over the phase space leads to bifractional kinetic equation. This model seems to be more physical than the hypothesis assigning the fractal properties directly to the real space and time. Let us look at this logic more attentively.

Consider a small domain $\Delta\Gamma$ in the phase space of a Hamiltonian system with finite dynamics and the point x_0 as an initial point for the trajectory starting in A. According to Poincaré-Zermelo theorem, this trajectory returns back to $\Delta\Gamma$ after some time t_1 and will continue to return repeatedly and infinitely many times at intervals $t_2, t_3, \ldots, t_k, \ldots$. For non-periodic dynamics, these cycles called the *Poincaré cycles* can be considered as random events, and recurrence times t_k as independent realizations of a random variable T. Zaslavsky indicated at existing some domains called the *stochastic layer traps* where the system can spend an arbitrary long time(the

layers are *sticky*). Some simple examples like the Sinai billiard and Bunimovich (stadium) billiard show that the random waiting time T in such traps may have a power-like tail. These layers bound the islands forming a singular part of the phase space. Moreover, there was shown that the set of islands is a fractal object: the boundary of the island is sticky, with that of the sub-islands being more sticky and so forth. As Zaslavsky claimed, this leads to self-similar patterns in time and phase space and makes it possible to apply the *renormalization group* approach. If the dynamics possesses the rescaling properties

$$\mathsf{R}: \ \Delta l \to \lambda_l \Delta l, \ \Delta t \to \lambda_t \Delta t.$$

the kinetic equation must have the form

$$\left(\frac{\partial}{\partial t}\right)^{\beta} P(x,t) = K \left(\frac{\partial}{\partial |x|}\right)^{\alpha} (AP(x,t)).$$

This conclusion does not seem to be unlikely but it hardly can be taken as a proof.

9.5.6 Continuum Mechanics

Two applications of fractional calculus to continuum mechanics are developing (see [18] and References therein). One of them is based on the local fractional derivative

$${}_x f^{(\nu)}(x) = \lim_{\xi \downarrow x} {}_x\mathsf{D}^{\nu}_{\xi}[f(\xi) - f(x)], \quad 0 < \nu \leqslant 1.$$

The following local fractional Taylor expansion of a fractal function $f(x)$ takes place:

$$f(x) - f(x_0) \sim \frac{{}_{x_0} f^{(\nu)}(x_0)}{\Gamma(\nu+1)} (x - x_0)^{\nu}, \quad x \to x_0, \quad 0 < \nu < 1. \tag{9.4}$$

Recall that the right-hand side of the equation is non-trivial and finite only if ν is equal to the critical order α of the function. When $\nu = \alpha = 1$, Eq. (9.4) is the equation of a tangent. Just as in the case $\nu = 1$ all the curves passing through the same point with the same first derivatives have the same tangent, in the case $\nu = \alpha \neq 1$ the curves with the same critical orders and the same fld form a class of equivalence modeled by $(x - x_0)^{\alpha}$. Carpinteri et al. [18] find it as a way to generalize the the geometric interpretation of derivatives in terms of "tangents".

This approach is addressed the solids where the deformation is localized on a fractal subset. On this assumption, the displacement field is represented by devil staircase-function. Let $1_C(x)$ be the indicator function of the Cantor set upon [0, 1]. Then, the solution of

$${}_x f^{(\nu)}(x) = 1_C(x)$$

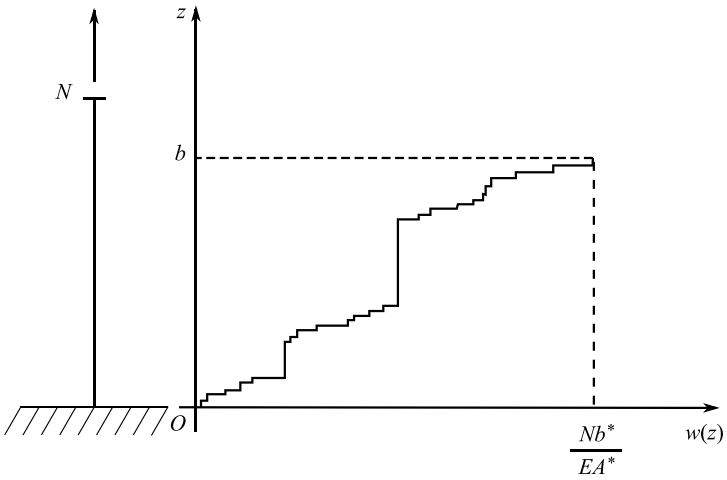


Fig. 9.8 Fractal bar subjected to an axial load (the left panel) and its stress-strain diagram (the right panel)

when $\nu = \alpha_C$ is expressed through the devil's staircase $S_C(x)$ as

$$f(x) = {}_x[1_C(x)]^{(-\nu)} = \frac{S_C(x)}{\Gamma(1+\nu)}.$$

Applying this forms to the stress-strain interrelation which models the displacement field $w(z)$ in a fractal bar subjected to an axial load (Fig. 9.8). The authors notice that the displacement increases less than linearly with the bar length, as occurs with classical elastic bodies.

Another approach considered in the cited work concerns to deformation of an elastic bar with nonlocal interactions between its elements. The force $F_j = f_j A \Delta z_j$ acting on a volume $\Delta V_j = A \Delta z_j$ of the bar is composed of the contact term ΔN_j and the long-range terms Q_j:

$$F_j = N_j + Q_j = N_j + \sum_{k=1}^{j-1} Q_{jk} - \sum_{k=j+1}^{m} Q_{jk}.$$

Using the approximation

$$Q_{jk} = \frac{\text{sign}(z_k - z_j) E c_\alpha \alpha [w(z_k) - w(z_j)]}{A\Gamma(1-\alpha)|z_k - z_j|^{1+\alpha}} \Delta V_j \Delta V_k, \quad 0 \leqslant \alpha \leqslant 1,$$

with c_α being a constant of the dimensions $[L]^{\alpha-2}$ and A the cross-section area of the bar, and passing to the limit as $\Delta z \to 0$, Carpinteri et al. arrive at the equation with derivatives of the Marchaud type

$$\frac{d^2w}{dz^2} - \frac{c_\alpha \alpha}{\Gamma(1-\alpha)}\left[\int_0^z \frac{w(z)-w(\zeta)}{(z-\zeta)^{1+\alpha}}d\zeta + \int_z^b \frac{w(z)-w(\zeta)}{(\zeta-z)^{1+\alpha}}d\zeta\right] = -\frac{f(z)}{E}.$$

In conclusion, the authors note that the nonlocal model represents an influence of internal forces.

9.5.7 Viscoelasticity

Let us consider an example of physical interpretation of the fractional differential model of high-elasticity polymers, given in Slonimsky [105], which clearly traced the indecomposability of the process into the sum of elastic and viscous components. The most important feature distinguishing high-elasticity deformation from the deformation of ordinary elastic bodies "such evident totality of relaxation phenomena, that the application of Hooke's law or any other relationship between stresses and strains, not taking into account time modes of mechanical interactions, is impossible even in rough approximation". In many works, deformation delay was associated with the presence of internal friction, and models of elastic bodies with internal friction and viscous body with elasticity were proposed. However, for a quantitative or, at least, semi-quantitative description of deformation of high-elasticity bodies, the construction of very complex mechanical models, consisting of a large number of different springs and different dampers, is necessary. The molecular dynamics of amorphous polymers, according to which relaxation processes are associated with slow processes of rearrangement of long and flexible molecules and coils (clusters) of these molecules, is manifested in changes of strains lagging behind changes of stress. Due to a very large length of a chain molecule, a simultaneous motion of all its parts (segments) at deformation is impossible, because the energy necessary for overcoming the intermolecular interaction greatly exceeds the energy of chemical bonds. The motion of this molecule is the result of the consecutive motion of its segments possible due to the flexibility of a high-elasticity molecule. Having the laws of motion of segments, one can understand the laws of motion of chain molecules, changes of their shape and, in the end, the laws of deformation of high-elasticity bodies. According to G. L. Slonimsky, a new way of determining the laws of deformation of high-elasticity bodies is to abandon the idea of high elasticity as a summation of elasticity and internal friction. The reason for such refusal is seen to be in strong distinction between elasticity of ordinary bodies and high elasticity of ones in low values of elastic module (2–3 orders of magnitude below normal ones), in huge quantities of reversible deformations (up to 1000%, and above), and in reverse signs of thermal effects of deformation and temperature module of elasticity. G. L. Slonimsky explains this distinction by the fact that high elasticity is conditioned by only a change of shape of flexible long chain molecules of polymer substances without a change of their energy of interaction (or their mobile parts-segments). At isothermal high elasticity deformation, the internal energy of a polymer body does not change, and all the work of deformation turns into heat, therefore the resistance of the body

of deformation is a result of entropy decreasing only. Thus, high elasticity deformation should be considered as an independent type of reversible deformation, and we should not try to divide it into elastic and viscous components. G. L. Slonimsky considers the fractional derivative as a mathematical symbol of such independence. The fractional derivative combines many features of elastic and plastic deformation, but it is not their combination.

Chatterjee [19] believes that the fractional behaviour of many viscoelasic materials can may arise from microstructural mechanisms of energy dissipations in disordered media. For linear processes in disordered systems a theoretical model suggested by Huber [57] led to the kinetic law

$$\langle I(t)\rangle = \exp\left\{-\int_0^\infty [1-\exp(-\lambda t)]\rho(\lambda)d\lambda\right\},$$

where $\langle I(t)\rangle$ is the average relaxation (survival) function of the process and $\rho(\lambda)d\lambda$ is the average number of relaxation channels with an individual relaxation rate between λ and $\lambda + d\lambda$. If $\rho(\lambda) \propto \lambda^{-\alpha-1}$, then $\langle I(t)\rangle \sim \exp[-(\mu t)^\alpha]$. For independent rate processes with static disorder and a self-similar distribution of reaction channels, a nonlinear analogue of the stretched exponential takes place. For nonlinear processes the tail of the averaged kinetic curve is self-similar and obeys the scaling law with a negative exponent [136]. This leads to fractional operators.

9.5.8 *Turbulence*

Multifractality for functions was introduced by Parisi and Frisch [88] to interpret experimental results on fully developed turbulence [6]. Frisch et al. [35] show that multifractality is connected to the tail behavior of fractional derivatives.

A homogeneous random function $u(\mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^d$ is called multifractal if (for some real interval $0 < h_{\min} \leqslant h \leqslant h_{\max} < 1$) there is a continuum of sets S_h of fractal dimension $D(h)$ such that, when $\mathbf{x} \in S_h$, the function u has a Hölder-like behavior with exponent $0 < h < 1$. In terms of increments this is expressed as

$$|\delta u(\mathbf{x},\mathbf{r})| \equiv |u(\mathbf{x}) - u(\mathbf{x}+\mathbf{r})| \propto |\mathbf{r}|^h, \quad \mathbf{r} \to 0, \quad \mathbf{x} \in S_h.$$

If one takes $D(h)$ to be the covering dimension (see [34], Sect. 8.5.1 and references therein) and then "thicken" the set S_h by covering it with balls of small radius ρ into the set S_h^ρ, then

$$\mathsf{P}(\mathbf{x} \in S_h^\rho) \propto \rho^{F(h)}, \quad \rho \to 0,$$

where $F(h) \equiv d - D(h)$.

When $u(x)$ is a homogeneous multifractal function, the moments of its increments, called *structure functions*, are defined as

$$S_p(\mathbf{r}) \equiv \langle(|\delta u(\mathbf{r})|^p)\rangle, \quad p \geqslant 0.$$

Considering the fractional derivative in the one-dimensional case,

$$\mathbb{D}^\alpha u(x) = \frac{\Gamma(\alpha+1)}{\pi} \sin\left(\frac{\alpha\pi}{2}\right) \text{p.v.} \int \frac{u(x) - u(x+r)}{|r|^{\alpha+1}} dr, \tag{9.5}$$

Frisch and Matsumoto [36] note that, at a point x_0 at which $u(x)$ has Hölder exponent $h > \alpha$ the fractional derivative is finite (provided the function u is, say, bounded). If $\alpha \geqslant h$ the fractional derivative will generally be infinite, but in the neighborhood of such a point

$$|\mathbb{D}^\alpha u(x)| \sim B|x - x_0|^{h-\alpha}, \quad x \to x_0.$$

Since $h - \alpha < 0$, the fractional derivative becomes very large near x. This is the case of a monofractal non-oscillatory singularity. In a multifractal case,

$$\mathsf{P}(|\mathbb{D}^\alpha u(x)| > \xi) \propto \xi^{-F(h)/(\alpha-h)}.$$

Nevertheless, the authors failed to establish on this basis a closer link between the turbulence physics and fractional derivatives. Later, Frisch et al. [35] answering to anonymous referee write that the idea of the multi-fractal analysis is based on the fact that the dynamics of a system determines a variety of scales. In the case of the Feigenbaum attractor studied in the article the scales depend on a symbolic location in a system of partition, but in physical systems like homogeneous turbulence, such partitions are difficult to define rigorously, authors conclude.

Equation (9.5) returns us to the Monin equation for the turbulent diffusion (Sect. 8.2.2). Now, we cannot but admit that it does not quite satisfy us: the motion of a tracer dropped into this flow looks somewhat strange; it stays during some time at some place, then suddenly jumps in a straight line from this place to another, which may be rather far from it. This waiting-jumping process is continued, and the direction of each jump is random, isotropically distributed, and independent of the previous one. But when $\alpha \to 2$, jump lengths and time intervals become shorter and shorter, so that the limit trajectory ($\alpha = 2$) is Brownian, everywhere continuous but nowhere differentiable. A possible way to improve the fractional model of the turbulence lies through introducing a finite velocity of motion and correlations between consecutive jumps.

9.5.9 *Plasma*

Kobelev et al. [64] offer the following way to take into account multifractal properties of the structure under investigation:

- to write out an appropriate differential equation for the corresponding regular medium;
- to replace the integer-order time-derivatives by its generalized fractional counterpart with the fractional index $\nu = 1 + d(\mathbf{r}, t)$,

$$ {}_0\mathsf{D}_t^{\nu(\cdot)} f(t) = \left(\frac{d}{dt}\right)^n \int_0^t \frac{f(\tau)d\tau}{\Gamma(n - \nu(\tau))(t-\tau)^{\nu(\tau)-n+1}}, $$

  where $d(\mathbf{r}, t)$ is the local fractional dimension of the multifractal system;
- to write out equations determining d using the system characteristics (interaction potentials and distribution functions).

Applied to the kinetics of the Coulomb plasma considered by the authors of this paper, their approach leads to the system of equations derived from the Vlasov equations with the Landau and Klimontovich dissipative integrals ($J_a^{(\nu)}$ and $J_a^{(\mathbf{r})}$ respectively):

$$ \left[{}_0\mathsf{D}_t^{\nu_a(\cdot)} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + e_a \mathbf{E}(\mathbf{r}, t) \cdot \frac{\partial}{\partial \mathbf{p}} \right] f_a(\mathbf{r}, \mathbf{p}, t) = J_a^{(\nu)}(\mathbf{r}, \mathbf{p}, t) + J_a^{(\mathbf{r})}(\mathbf{r}, \mathbf{p}, t), $$

$$ \nu_a(\mathbf{r}, t) = 1 + d_a(\mathbf{r}, t), $$

$$ \frac{\partial d_a}{\partial t} = \frac{\partial}{\partial P_j} \left[D_0(\mathbf{r}, t) \frac{\partial d_a}{\partial P_j} \right] + (\varphi_a - \varphi_{a0}) d_a + k_1 d_a^2, $$

where $d_a(\mathbf{r}, t)$ is the order parameter determined by varying the Ginzburg–Landau-type functional of the fractal dimension for the system free energy. For parameters P_i one can choose time, pressure, coordinates, temperature, etc.

Because these authors do not solve the equations that they derive, some inconsistency between the differential operators in the left-hand side of the first equation remains unnoticed. The point is that $\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + e_a \mathbf{E}(\mathbf{r}, t) \cdot \frac{\partial}{\partial \mathbf{p}}$ is a part of the total differential operator

$$ \frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + e_a \mathbf{E}(\mathbf{r}, t) \cdot \frac{\partial}{\partial \mathbf{p}}, $$

describing continuous motion of the particle with the given velocity $\mathbf{v}$ in a regular medium and corresponding to exponentially distributed paths between collision. Consideration of the fractal structures involves first of all a power-like distribution, relating to the fractional operator

$$ \left(\frac{d}{dt}\right)^\nu = \left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \frac{\partial}{\partial \mathbf{r}} + e_a \mathbf{E}(\mathbf{r}, t) \cdot \frac{\partial}{\partial \mathbf{p}}\right)^\nu. $$

Fractional equation with an operator of such kind is considered in our work [133].

Observe that the differential operator of a non-integer order surely contains an integral operator, which describes jump-like change of the corresponding independent variable. Thus, it acts on a pdf with respect to time, we observe a time-jump whereas the other variables (say, **r** and **p**) remains the same. In other words, time is flying but the particle remains motionless (they say, it is trapped). When the fractional operator acts on **r**, this relates to an instantaneous flight of the particle from one point to another being probably at large distance from the initial one. This is admissible if the particle rests in traps between the successive flights and durations of flights are negligible in comparison with those of trapping. Finally, if it concerns the momentum or velocity, we observe the collision process. Evidently, these considerations impose certain constraints on models governed by the fractional equations derived in Kobelev et al. [63–66], etc.: jump-like motion can not always model a real physical motion.

9.6 Quantum Mechanic Interpretations

9.6.1 Feynman Path Integrals

Embedding fractional derivatives into quantum mechanics, Laskin [73] refers to the *Feynman path integral* formulation based on the Brownian motion. The latter was historically the first example of the fractal in physics. Measuring the length of a Brownian path sample by using a straight line segment of a given length Δx and repeating the procedure with a smaller segment, one obtains

$$l(\Delta x) \propto (\Delta x)^{1-D_{\mathrm{B}}},$$

where $D_{\mathrm{B}} = 2$ is the fractal dimension of the Brownian path. The symmetric Lévy motion propagator

$$p_{\mathrm{L}}(x, t; x_0, t_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{-ik(x-x_0)} \exp\left[-|\sigma k|^{\alpha}(t - t_0)\right], \quad \sigma = \mathrm{const}, \quad 0 < \alpha < 2,$$

implies that

$$(x - x_0) \sim \sigma (t - t_0)^{1/\alpha},$$

and correspondingly

$$l(\Delta x) \sim \sigma (\Delta x)^{1-D_{\mathrm{L}}}$$

with $D_{\mathrm{L}} = \alpha$, $\alpha \in (1, 2)$. Using the formula for the length increments, $|\Delta x| = |x_j - x_{j-1}| \sim (\hbar^{\alpha-1} D_{\alpha})^{1/\alpha}$, in construction of the fractional path integral measure, Laskin [73] obtains the Lévy extensions of the Feynman propagator for a free particle as

$$K_{\mathrm{LF}}(x,t;x_0,t_0)=\frac{1}{2\pi\hbar}\int_{-\infty}^{\infty}dpe^{-ip(x-x_0)/\hbar}\exp\left[-i|\sigma p|^{\alpha}(t-t_0)/\hbar\right],\quad 0<\alpha<2.$$

By involving potential $V(x)$ into the path integral and computing an infinitesimal increment of the wave function,

$$\psi(x,t+\Delta t)-\psi(x,t)=\int_{-\infty}^{\infty}\left[K_{\mathrm{LF}}(x,t+\Delta t;x_0,t_0)-K_{\mathrm{LF}}(x,t;x_0,t_0)\right]\psi(x_0,t_0),$$

Laskin arrives at the space-fractional Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t}=\left[(\sigma\hbar)^{\alpha}(-)^{\alpha/2}+V(x)\right]\psi(x,t).$$

However, the question about the reasons causing such a "Zitterbewegung" of microparticles remains still open. Moreover, the situation is aggravated because of appearance of an additional parameter α with continuum of possible values. When ascribing such movement to the influence of the physical vacuum, the fractional equation can be validated only by admission of existence of a whole family of vacua with different properties.

9.6.2 *Lippmann–Schwinger Equation*

In [27, 28], the fractionalized *Lippmann–Schwinger equation*

$$\phi(\mathbf{r})=e^{i\mathbf{k}\cdot\mathbf{r}}-\frac{\kappa^{(2-\alpha)/\alpha}}{2\alpha D_{\alpha}(\pi\hbar)^2}\int\frac{\pi e^{i\kappa^{1/\alpha}|\mathbf{r}-\mathbf{r}_0|/\hbar}-\mathscr{G}_{\alpha}(|\mathbf{r}-\mathbf{r}_0|)}{|\mathbf{r}-\mathbf{r}_0|}V(\mathbf{r}_0)\phi(\mathbf{r}_0)d^3\mathbf{r}_0,$$

where $\kappa=(\hbar k)^{\alpha}$ and

$$\mathscr{G}_{\alpha}(r)=\sin(2\pi/\alpha).\mathscr{I}_1(kr)+\cos(2\pi/\alpha).\mathscr{I}_2(kr),$$

is used to study the fractional scattering problem.[4] In particular, the first Born approximation takes the form

$$\phi(\mathbf{r})\approx e^{i\mathbf{k}\cdot\mathbf{r}}-\frac{\kappa^{(2-\alpha)/\alpha}}{2\alpha D_{\alpha}\pi\hbar^2 r}e^{ikr}\int e^{-i\mathbf{q}\cdot\mathbf{r}_0}V(\mathbf{r}_0)d\mathbf{r}_0,\quad \mathbf{q}=k\mathbf{r}/r.$$

Bulgac et al. [15], Kusnezov et al. [70–72] investigated the motion of a particle coupled to a chaotic environment quantum mechanically and showed that the evolution a wave packet is described by a reduced density matrices which behaved as a Lévy motion. It is worth to stress that the fractional character of the equation appears only

[4] Observe that when $\alpha=2$, $\mathscr{G}_2=0$ and this equation reduces to the standard form.

in the macroscopic limit. From the general point of view on fractional operators as manifestation of complexity, this looks rather logical.

9.6.3 Time-Fractional Evolution Operator

Considering the time-fractional Schrödinger and Heisenberg equations, these authors recalculated some earlier results (see Chap. 5) and established the following basic properties of the time fractional evolution operator (TFEO)

$$\mathsf{U}_\nu(t,t_0)=\begin{cases} E_\nu\left\{[-i(t-t_0)]^\nu \mathsf{H}\right\}, & 0<\nu<1;\\ E_\nu\left\{[-i(t-t_0)]^\nu \mathsf{H}\right\}+w(t-t_0)E_{\nu,2}\left\{[-i(t-t_0)]^\nu \mathsf{H}\right\}, & 1<\nu<2.\end{cases}$$

- Hermitian conjugate of the TFEO is equal to its complex conjugate: $\mathsf{U}_\nu^+(t,t_0)=\mathsf{U}_\nu^*(t,t_0)$.
- $\mathsf{U}_\nu(t_0+0,t_0)=1$.
- In contrast to the standard case, the TFEO does not satisfy the composition law: $\mathsf{U}_\nu(t_2,t_0)\neq \mathsf{U}_\nu(t_2,t_1)\mathsf{U}_\nu(t_1,t_0), \quad \nu\neq 1$.
- TFEO is not a unitary operator: $\mathsf{U}_\nu^+(t,t_0)\mathsf{U}_\nu(t,t_0)\neq 1$.
- $\lim\limits_{t\to\infty}\mathsf{U}_\nu^+(t,t_0)\mathsf{U}_\nu(t,t_0)=\nu^{-2}$.
- A TFEO commutes with the Hamiltonian operator as well as its Hermitian conjugate: $\mathsf{U}_\nu(t,t_0)\mathsf{H}-\mathsf{H}\mathsf{U}_\nu(t,t_0)=0, \quad \mathsf{U}_\nu^+(t,t_0)\mathsf{H}-\mathsf{H}\mathsf{U}_\nu^+(t,t_0)=0$.

Ertic et al. [32] referred also to two specific examples where realistic physical systems have fractal attributes in space and time. When the cosmic microwave background radiation energy density and blackbody temperature are calculated with the help of fractional calculus, one gets compatible results with the observed and experimentally measured results, since the physical events are taking place in fractal space. The second example is the Bose–Einstein condensation of dilute atomic gases where the transition temperature is calculated in the frame of fractional calculus, i.e., by taking into account the fractality of the space. In this case, the calculated results again fit well with the experimentally measured data.

9.6.4 A Role of Environment

Chen [20], considering the frequency-dependent energy dissipation of acoustic wave propagation through soft matter, claimed fractional quantum-energy, and quantum-momentum relationships

$$E=\hat{h}\omega^{\gamma/2}, \quad p=\hat{h}k^{\gamma/2}, \quad \gamma\in[0,2],$$

which become normal quantum mechanical relations, when $\gamma = 2$. Chen explains this conjunction in the following way: "Traditionally, the phonon is mostly involved in the study of the atomic lattice of the crystal, being the quanta of lattice vibration energy, and is intimately related to the physical structures of materials. However, there are no lattice structures (at most fractal lattice) in the mesoscopic macromolecules of soft matter (e.g., porous media and amorphous solids), and it is thus difficult to apply the phonon concept here". To derive the fractional Schrödinger equation, he uses the quantum plane wave equation

$$\Psi(x,t) = Ae^{i(kx-\omega t)},$$

and the classical energy-momentum interrelation

$$E = \frac{p^2}{2m} + V(x).$$

As a result, the fractional counterpart of the Schrödinger equation is written as

$$e^{i\pi\gamma/4}\hat{h}\,{}_0\mathsf{D}_t^{\gamma/2}\Psi = \left[\frac{\hat{h}}{2m}(-)^{\gamma/2} + V(x)\right]\Psi.$$

However, the following fact casts doubt on validity of this conclusion: fractionalization of the Planck formula is explained by an influence of inhomogeneities in the soft matter, whereas the quantum plane wave is related to a homogeneous medium.

Recall that an open quantum-mechanical system surrounded by a bath is described by the general master equation

$$\mathsf{D}_t\rho(t) = -\frac{i}{\hbar}\left[\mathsf{H}(t),\rho(t)\right] + \int_0^t \mathsf{K}(t,t')\rho(t')dt'.$$

Since the bath hidden variables change in time, the conditioned system state will evolve in time also, and this time there is a meaningful relation between the conditioned system state at different times [37].

Tofighi [122] determines fractional quantum mechanics as a theory of quantum phenomena in fractal environments. Referring to the review [77] the author supposes that with the advent of many new material such as polymers,glasses, liquid crystals, and biopolymers one would expect that the motion of a particle in such complex systems is different from that of a simple system. One of manifestations of the difference is seen in an anomalous energy-frequency relation $E \propto \omega^\alpha, \quad 0 < \omega < 1$ called fractional Planck quantum energy relation. This relation leads to the fractional Schrödinger equation of the form

$$i^\alpha \hbar_\alpha\, {}_0^\alpha\mathsf{D}_t\psi = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t).$$

Again, it is hard to agree with this argument: observing a small region in a regular medium during a short time interval, we should see the Schrödinger equation in its standard form, whereas in a nonregular at all scales medium, say in a fractal one, we have no legitimate grounds for using the plane de Broglie wave as a step for deriving this equation [15, 70–72] and others.

West [138] suggested to interpret the fractional character of equation for the free particle Lévy propagator,

$$\frac{\partial K_L(x-x_0,t-t_0)}{\partial t}+iC(-)^{\alpha/2}K_L(x-x_0,t-t_0)=\delta(x-x_0)\delta(t-t_0),$$

as a result of long-range correlations in the vacuum which vanish when α becomes 2. No physical reasoning was involved in the derivation of this equation, so this conclusion seems somewhat artificial.

9.6.5 Standard Learning Tasks

Guo and Xu [45] have considered fractional versions of some standard learning tasks in quantum mechanics. Thus, the wave function of a free particle governed by the Shrödinger equation with the fractional Laplacian,

$$i\hbar\frac{\partial\psi}{\partial t}=D\left(-\hbar^2\right)^{\alpha/2}\psi(x,t)$$

is obtained in the form

$$\begin{aligned}\psi(x,t)&=\frac{A}{2\pi\hbar}\int_{-\infty}^{\infty}e^{ipx/\hbar}\exp\left[-iD|p|^{\alpha}t/\hbar\right]dp\\&=\frac{A}{\alpha|x|}H_{22}^{11}\left[\frac{1}{\hbar}\left(\frac{\hbar}{iDt}\right)^{1/\alpha}|x|\,\middle|\,\begin{matrix}(1,1/\alpha)&(1,1/2)\\(1,1)&(1,1/2)\end{matrix}\right].\end{aligned}$$

Considering a particle in a potential well

$$V(x)=\begin{cases}0, & \text{for } 0<x<l;\\ \infty, & \text{otherwise,}\end{cases}$$

the authors separate the spatial and temporal variables and write the solution in the traditional form

$$\psi_n(x,t)=e^{-(i/\hbar)E_nt}\phi_n(x),$$

where

$$\phi_n(x)=\begin{cases}A_n\sin\left(\frac{n\pi x}{l}\right), & \text{for } 0<x<l;\\ 0, & \text{otherwise.}\end{cases}\tag{9.6}$$

The only difference is in the energy spectrum:

$$E_n = D_\alpha \left(\frac{\hbar n \pi}{l} \right)^\alpha$$

instead of

$$E_n = D_2 \left(\frac{\hbar n \pi}{l} \right)^2$$

with $D_2 = \frac{1}{2m}$.

Deriving from the space-fractional Schrödinger equation its Lippmann–Schwinger counterpart, Dong [28] refers to the preceding paper [27] where the many-body quantum system governed by fractional Schrödinger equation is studied by using the density matrix approach. There are considered the Thomas-Fermi model, calculated the the quantum pressure of electron gas, and showed the validity of the Hohenberg-Kohn theorems.

As in the standard approach, Dong [27] divides the space into many small cells, each of volume $\Delta V = l^3$ and assumes that the wave function of each electron obeys the fractionalized Schrödinger equation

$$D_\alpha (-\hbar^2)^{\alpha/2} \psi(\mathbf{r}) = E \psi(\mathbf{r}) \tag{9.7}$$

with the boundary conditions $\psi(\mathbf{r}) = 0$, when $x = l$, or $y = l$, or $z = l$, or $x = y = z = 0$. Then, the author says it is easy to prove that $\psi(\mathbf{r}) = Ce^{i\mathbf{k}\cdot\mathbf{r}}$ with $\mathbf{k} = \{k_x, k_y, k_z\}$ is a basic solution to Eq. (9.7), satisfying

$$E = D_\alpha \hbar^\alpha |\mathbf{k}|^\alpha.$$

However, it must be admitted that there are at least two reasons for denying this claim. The first of them: the Hamiltonian $\mathsf{H}_\alpha(x, y, z) = D_\alpha(-\hbar^2)^{\alpha/2}$ can be represented as the sum $\mathsf{H}_\alpha(x) + \mathsf{H}_\alpha(y) + \mathsf{H}_\alpha(z)$ only in case $\alpha = 2$. For other values of α it is impossible, and consequently, the separation of variables is impossible as well. The second objection: the solution to the equation

$$D_\alpha \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right)^{\alpha/2} \psi(x) = E \psi(x) \tag{9.8}$$

depends not only on ψ at boundary points $x = 0$ and $x = l$, but also on the values of ψ outside the boundaries. Jeng et al. [58] write: "Eq. (9.8) is only valid if the function operated on is $e^{ipx/\hbar}$ everywhere; it is not a local equation that can be applied just in a restricted region. Because the quantum Riesz fractional derivative is a nonlocal operator, the wave function in the well knows about the wave function and potential outside the well." And further: "It may seem counterintuitive that Eq. (9.6) is not the correct ground state. The standard ($\alpha = 2$) Schrödinger equation for an

infinite potential well is equivalent to the Schrödinger equation on an interval with the Dirichlet boundary conditions $\psi(0) = \psi(l) = 0$. By raising that Hamiltonian to the power $\alpha/2$, we get a plausible fractional Laplacian and Eq. (9.6) is indeed a solution. However, this is not the Riesz fractional derivative. In other words, the fractional Schrödinger equation for an infinite potential well is not equivalent to the fractional Schrödinger equation on an interval. At this point, we do not know what the true solutions are for values of α other than 0, 2, 4. In [146] they found numerical solutions for the ground state. The solutions depend on α and differ from the simple sine wave solution in Eq. (9.6)."

9.6.6 *Fractional Laplacian in a Bounded Domain*

The boundary conditions for the first passage time problem,

$$\frac{\partial p(x,t)}{\partial t} = \sigma^{\alpha} \frac{\partial^{\alpha} p(x,t)}{\partial |x|^{\alpha}},$$

associated with two absorbing boundaries at $x = 0$ and $x = L$ are now nonlocal; reading

$$p(x,t) = 0 \quad \text{for} \quad x \leqslant 0, \quad \text{and} \quad p(x,t) = 0 \quad \text{for} \quad x \geqslant L.$$

Due to discontinuous character of trajectories of Lévy processes Dybiec et al. [30, 2008] note that the usual form of boundary conditions, i.e.,

$$p(x,t) = 0 \quad \text{for} \quad x = 0, \quad \text{and} \quad p(x,t) = 0 \quad \text{for} \quad x = L,$$

incorrectly employed in the literature [13, 14, 40, 41] is expected to lead to erroneous results; such a boundary condition does not account for the fact that the process can skip the location of the boundary without hitting it. The corresponding MFPT can be numerically integrated, yielding

$$MFPT = -\int_0^{\infty} dt\, t \int_0^{L} \frac{\partial p(x,t)}{\partial t} dx = \int_0^{L} \left[\int_0^{\infty} p(x,t) dt \right] dx.$$

Numerical results were compared with the theoretical findings with point-like boundary conditions,

$$T = \frac{4}{\pi \sigma^{\alpha}} \left(\frac{L}{\pi} \right)^{\alpha} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^{1+\alpha}}.$$

The numerical results for $\alpha < 2$ systematically exceed the theory result of Gitterman [40].

When the domain of variables where the fractional Riesz operator acts is translation invariant, say, unbounded homogeneous or periodic, it has a simple definition in terms of its Fourier transform

$$\mathscr{F}\{-(-)^{\alpha/2} f(\mathbf{x})\}(\mathbf{k}) = -|\mathbf{k}|^{\alpha} \widetilde{f}(\mathbf{k}), \quad \mathbf{x}, \mathbf{k} \in \mathbb{R}^d.$$

It may also be computed by using the matrix representation based on eigenfunctions $\psi_{\mathbf{k}}(\mathbf{x}; \alpha)$ of the fractional Laplacian

$$-(-)^{\alpha/2} \psi_{\mathbf{k}}(\mathbf{x}; \alpha) = \lambda_{\mathbf{k}}(\alpha) \psi_{\mathbf{k}}(\mathbf{x}; \alpha),$$

with eigenvalues $\lambda_{\mathbf{k}}(\alpha)$ connected to decay constants associated with corresponding $\psi_{\mathbf{k}}(\mathbf{x}; \alpha)$. This way can also be used for bounded domains if α takes an even value, because the operator $(-)^{\alpha/2}$ becomes local and it is enough to give the conditions only on the boundary surface (see, for example [146]).

However, in case of a bounded domain or aperiodic boundary conditions, the Fourier representation is not applicable to the fractional operator, the set of eigenfunctions and the corresponding spectrum of eigenvalues are awkward for computing under different boundary conditions. The statement of such a problem should be accompanied with a specification of the desired function values throughout an outer region. So, we have to return to the integral representation of the operator. The random flight interpretation can help in specifying the conditions but some subtle points such as distinction between first-passage and first-arrival times or between free and reflecting boundary conditions appear [146]. Zoia et al. [146] have investigated the matrix representation of the one-dimensional fractional Laplacian and solved numerically in connection to the first-passage problem (the Lévy-flights under absorbing boundary conditions) and to the long-ranged interfaces with no constraints at the ends (the free boundary conditions).

Krepysheva and Di Pietro [69] analyze the symmetric Lévy flights restricted to a semi-infinite domain by a reflective barrier. They show that the introduction of the boundary condition induces a modification in the kernel of the nonlocal operator:

$$\begin{aligned} -(-)^{\alpha/2} f(x,t) &= -\frac{1}{2\cos(\alpha\pi/2)\Gamma(2-\alpha)} \\ &\quad \times \frac{\partial^2}{\partial x^2} \int_0^{\infty} |x-\xi|^{1-\alpha} f(\xi,t) d\xi, \\ \mapsto -(-)^{\alpha/2}_{\text{refl}} f(x,t) &= -\frac{1}{2\cos(\alpha\pi/2)\Gamma(2-\alpha)} \\ &\quad \times \frac{\partial^2}{\partial x^2} \int_0^{\infty} \left[|x-\xi|^{1-\alpha} + (x+\xi)^{1-\alpha} \right] f(\xi,t) d\xi, \, 1 < \alpha < 2. \end{aligned}$$

The operators $-(-)^{\alpha/2}$ and $-(-)^{\alpha/2}_{\text{refl}}$ differ in the kernels, but the difference becomes small when $x + \xi$ is large. Nevertheless, omitting the term $(x + \xi)^{1-\alpha}$ we would get a decreasing integral with respect to x, whereas the total amount of the diffusing matter should be preserved.

Rafeiro and Samko [93] introduce a version of the fractional Laplacian for a bounded domain as a generalization of the Marchaud formula for one-dimensional fractional derivatives on an interval (a, b), $-\infty < a < b \leqslant \infty$, to the multidimensional case of functions defined on a region $G \subset \mathbb{R}^d$:

$$\mathbb{D}^{\alpha}_{G} f(\mathbf{x}) = C(\alpha) \left[a_G(\mathbf{x}) f(\mathbf{x}) + \int_G \frac{f(\mathbf{x}) - f(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|^{d+\alpha}} d\mathbf{y} \right], \qquad \mathbf{x} \in G \subset \mathbb{R}^d,$$

where $\alpha \in (0, 1)$,

$$C(\alpha) = \frac{\alpha 2^{\alpha-1} \Gamma[(d+\alpha)/2]}{\pi^{d/2} \Gamma(1-\alpha/2)}$$

and

$$a_G(\mathbf{x}) = \int_{\mathbb{R}^d \setminus G} \frac{d\mathbf{y}}{|\mathbf{x} - \mathbf{y}|^{d+\alpha}}.$$

In other words, this is the Riesz fractional derivative of the zero continuation of $f(\mathbf{x})$ from G to the whole space $\mathbb{R}^d$.

Guan and Ma [43], investigating the reflected symmetric α-stable processes, gave the name *regional fractional Laplacian* to the limit

$$-(-)^{\alpha/2}_{G} f(\mathbf{x}) \equiv \lim_{\varepsilon \downarrow 0} C(\alpha) \int_{G,\ |\mathbf{x}-\mathbf{y}|>\varepsilon} \frac{f(\mathbf{x}) - f(\mathbf{y})}{|\mathbf{y} - \mathbf{x}|^{d+\alpha}} d\mathbf{y},$$

provided it exists.

For more detail, the reader can be referred to the articles [12, 26, 56, 106, 110]. Better understanding of the Laplacian in a bounded domain can be achieved on the base of the non-local operator theory [29, 44, 135].

9.6.7 Application to Nuclear Physics Problems

Applications of fractional calculus to nuclear problem physics have been done by R. Herrmann. Using the mappings

$$x_i \mapsto \bar{x}(x_i) = \mathrm{sign} x_i |x_i|^{\alpha}$$

and

$${}^{\alpha}_{0}\mathsf{D}_{x_i} \mapsto \bar{D}_i = \mathrm{sign} x_i \, {}^{\alpha}_{0}\mathsf{D}_{|x_i|}$$

for definition of the set of conjugated operators on an Euclidean space for N particles in space coordinate representation,

$$\mathsf{P}_{\mu} = \{\mathsf{P}_0, \mathsf{P}_i\} = \left\{ i\hbar \mathsf{D}_t, -(\hbar/mc)^{\alpha} mc \bar{\mathsf{D}}_i \right\},$$

and

$$\mathsf{X}_\mu = \{\mathsf{X}_0, \mathsf{X}_i\} = \left\{ t, \frac{(\hbar/mc)^{1-\alpha}}{\Gamma(\alpha+1)} \bar{x}(x_i) \right\},$$

($i = 1, 2, \ldots, 3N$, with N being the number of particles), the author arrives at the following commutator rules on a function set $\{x^{n\alpha}\}$:

$$\left[\mathsf{X}_i, \mathsf{X}_j\right] = 0, \quad \left[\mathsf{P}_i, \mathsf{P}_j\right] = 0,$$

$$\left[\mathsf{X}_i, \mathsf{P}_j\right] = -i\hbar c(n,\alpha)\delta_{ij},$$

where

$$c(n,\alpha) = \frac{\Gamma(1+n\alpha)}{\Gamma(1+(n-1)\alpha)} - \frac{\Gamma(1+(n+1)\alpha)}{\Gamma(1+n\alpha)}.$$

Then, Herrmann defines in a usual way the generators of infinitesimal rotations in the (i, j)-plane

$$\mathsf{L}_{ij} = \mathsf{X}_i\mathsf{P}_j - \mathsf{X}_j\mathsf{P}_i = -i\hbar \left(\frac{\bar{x}(x_i)}{\Gamma(\alpha+1)} \bar{D}_j - \frac{\bar{x}(x_j)}{\Gamma(\alpha+1)} \bar{D}_i \right)$$

and applies it to investigation of the charmonium spectrum. In the frame of this approach, the charmonium spectrum may be understood quantitatively in the framework of (space) fractional Schrodinger equation with $\alpha \approx 2/3$.

The author establishes as well, that measured energies of rotational nuclear states may be reproduced by the fractional symmetric rigid rotor model [48] and emphasizes the close relation between fractional calculus and q-deformad Lie algebras and discusses the principle of local gauge invariance for fractional fields [50, 51]. In [50] a nuclear single particle Hamiltonian is constructed from fractional rotational groups $\mathrm{SO}^\alpha(3)$ and then applied to calculations of the microscopic part of the total nuclear energy within macroscopic-microscopic approach. The model reproduces magic numbers and for spherical nuclei and can be extended for deformed nuclei. Furthemore, the ground state properties of nuclei are correctly reproduced. The author sees advantages of this model, compared to phenomenological shell and relativistic mean field models, are that 1) the predicted magic numbers are not the results of a fit with a phenomenological ls-term, 2) there are no potential-terms or parametrized Skyrme-forces involved, and 3) all results are represented in analytical form.

The similar approach has been used in Herrmann [51] for obtaining electronic magic numbers in metal clusters. It has been demonstrated, that experimental data may be described assuming a sudden change in the fractional derivative parameter α which is interpreted as a second order phase transition in the region of cluster size with $200 \leqslant N \leqslant 300$. Herrmann has developed the four realizations of higher

dimensional fractional rotation groups and claims that they may successfully be connected to the four fundamental interaction types in nature.

A short review of these and other applications of fractional calculus to nuclear physics can be found in Rozmej and Bandrowski [97]. As to the interested reader, I would recommend the book [52].

9.7 Concluding Remarks

9.7.1 Hidden Variables

Analyzing various examples in order to understand a general nature of heredity we are often faced phenomena with *hidden variables*. The initial system described with a full set of variables is generally nonhereditary. As soon as we exclude ("hide") some variables from consideration, there appears a member reflecting the prehistory of the reduced system—the system becomes hereditary.

Let us see a simple mechanical example. A piston of mass m_1 is placed in a horizontal tube, opened along the edges, with the friction coefficient η along its borders. The piston is connected by a spring of length l and elasticity $k > (\eta/2)^2$ with a ball of mass m_2, moving through the tube without friction (Fig. 9.9) Force $F(t)$ is applied to the ball at time $t = 0$. We deal with a nonhereditary system with two degrees of freedom, which is described by the differential equation

$$m_1\ddot{x}_1 = -\eta\dot{x}_1 + k(x_2 - x_1 - l),$$

$$m_2\ddot{x}_2 = F(t) - k(x_2 - x_1 - l).$$

We supplement this system of equations with the following conditions

$$x_1(0) = 0, \quad \dot{x}_1(0) = 0, \quad x_2(0) = l, \quad \dot{x}_2(0) = 0,$$

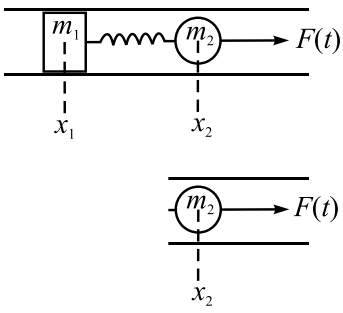


Fig. 9.9 The visual example of a hidden variable (x_1)

supposing that at the initial time the system is motionless and the piston is at the origin.

Resolving the equation regarding x_1 on the assumption that $x_2(t)$ is known yields

$$x_1(t) = \int_0^t K(\tau)[x_2(t-\tau) - l]d\tau,$$

$$K(\tau) = \frac{k}{m_1\omega_1}\exp\left(-\frac{\eta\tau}{2m_1}\right)\sin(\omega_1\tau), \quad \omega_1 = \sqrt{\frac{k}{m_1} - \left(\frac{\eta}{2m_1}\right)^2}.$$

Substituting the result into the second equation, we get the hereditary equation for the system, reduced to the single degree of freedom:

$$m_2\ddot{x}_2 + kx_2 = k\int_0^t K(\tau)x_2(t-\tau)d\tau + F_2(t), \tag{9.9}$$

where

$$F_2(t) = F(t) + kl\left[1 - \int_0^t K(\tau)d\tau\right].$$

To make this more evident, let us imagine that we cover with a screen the part of the tube containing the piston and the spring with a screen, so we can see only the ball, the motion of which satisfies Eq. (9.9). The ball's motion prehistory $x_2(t-\tau)$ affects on its behavior at time t via invisible (hidden) variable $x_1(t)$.

Another example we consider here concerns to the diffusion problem. The passage from the normal diffusion equation

$$\left[\frac{\partial}{\partial t} - \left(\frac{\partial}{\partial x}\right)^2\right] f(x,t) = 0$$

to the "semi-diffusion" equation

$$\left[A\left(\frac{\partial}{\partial t}\right)^{1/2} + B\frac{\partial}{\partial x}\right] f(x,t) = 0$$

may suggest to involve the matrix representation which leads to the conception of internal degrees of freedom of a system the state of which is described by spinors. This idea is developed in the article [134]. Interpreting f as a two-component column-vector,

$$f(x,t) = \begin{pmatrix} \varphi(x,t) \\ \chi(x,t) \end{pmatrix},$$

and A and B as 2×2 matrices satisfying the conditions

$$A^2 = I, \quad B^2 = -I, \quad AB + BA = 0,$$

the author obtains two realization of the algebra in terms of real matrices associated them to the Pauli matrices:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

As a result, two solutions are found,

$$f_1(x,t) = \begin{pmatrix} \varphi(x.t) \\ \varphi(x,t) \end{pmatrix} \quad \text{and} \quad f_2(x,t) = \begin{pmatrix} \varphi(x.t) \\ -\varphi(x,t) \end{pmatrix},$$

where $\varphi(x,t)$ is the solution of the normal diffusion equation. These solutions represent two possible probability distributions depending not only on the space-time coordinates, but also on the internal dichotomic degree of freedom.[5]

9.7.2 Complexity

There is a point of view that the simple systems (including decomposable into simple) are described by derivatives of integer order, whereas the complex ones (not decomposable into simple) are described by fractional derivatives. At first glance, there is a grain of truth, especially if you consider self-similar complex systems such as fractals. Indeed, dividing the simple inhomogeneous non-stationary medium into elementary volumes, considered during elementary times, we turn to homogeneous stationary elements, deviations of which from properties of real elements have the same infinitesimal order, or its integer power (area, volume). In the case of the inhomogeneous self-similar medium, dividing it into elements is simply a reproduction of the original medium: nothing becomes easier. A part of fractal is similar to the whole one. But fractional operators are based on the continuity concept, whereas the fractal structure is discontinuous at all scales (we can not pass on to the asymptotics, in which the fractal is seen as continuous). Only by averaging over an ensemble of fractals we can turn to continuity, hence the systems and processes must be random (stochastic). These are the ideas we started the book. At its beginning, we talked about the latent variables as a cause of heredity.

[5] I did not dare to discuss the use of fractional equations in cosmology without seeing any physical reasons for their introduction. Although it may be that the presence of hidden variables say relating to dark matter could provide such a basis.

The further development of this idea can be found in [90]. An additional requirement of self-similarity makes this hereditarity power law, and hereditary integrals turn into fractional ones. The rate of change of fractional integrals or of fractional integrals of velocities is the Riemann–Liouville or Gerasimov-Caputo fractional derivatives, respectively. Power-tail distributions produce α-stable statistics, which has simulated of stochastic approach to fractional calculus.

9.7.3 Finishing the Book

History teaches us that every step in improving our tools increases the freedom to choose the path of further development of science, culture, and other human activities. The successive discoveries the negative, rational, real, and complex numbers and the related theory of limits, not only created the modern arsenal of mathematical tools, they have changed the view of the world. In the arguments about the nature and the world, in the construction of the Babylon towers uniting humanity, a man has become less dogmatic and more free. The extension of natural numbers of operation (differentiation and integration) to real ones proposed by luminaries of mathematical thought Riemann and Liouville continued flow of the liberating ideas. These ideas turns out to be closely connected to such modern and important conceptions as non-Gaussian Lévy stable statistics, fractals, non-locality, and heredity. As consider the authors of the remarkable philosophical article [4], the main cause of it lies in the complexity of the world. A man is not a position to adequately map it, which gives rise the philosophical theory of reduction, forming the base for physicists' hypothesis on existence of hidden variables. As shown above, this hypothesis together with self-similarity paradigm leads to fractional operations. But many physicists and philosophers strongly criticize the reductionism...

Be that as it may, the fractional calculus is becoming more and more widely used in various fields of the science gradually changing their images. I tried to show this process in this book, and the reader may conclude how much I succeeded in doing this.

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Appendix A
Some Special Functions

A.1 Gamma Function and Binomial Coefficients

A.1.1 Gamma Function

The *Gamma function* generalizes the concept of the factorial function

$$n! = 1 \cdot 2 \cdot \ \ldots \ \cdot n, \quad n = 1,\ 2,\ 3,\ \ldots$$

to noninteger negative and even complex values of the argument. The point of departure for such extension is the integral

$$n! = \int_0^\infty e^{-x} x^n dx.$$

For a real $z = x > 0$, the Gamma function $\Gamma(x)$ is defined by means of the Euler integral of the second kind:

$$\Gamma(x) = \int_0^\infty e^{-\xi} \xi^{x-1} d\xi.$$

Note that $\Gamma(n+1) = n!$ for $n = 1,\ 2,\ 3,\ \ldots$.

This integral function is extended by analytic continuation from the real arguments x to all complex numbers z except the non-positive integers $z = -n,\ n = 0, 1, 2, \ldots$ (where the function has simple poles with residues $(-1)^n/n!$), yielding the meromorphic function we call the *Gamma function*.

Basic relations. Following relations take place for the Gamma function:

$$\Gamma(z+1) = z\Gamma(z),$$

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$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z},$$

$$\Gamma(1/2+z)\Gamma(1/2-z) = \frac{\pi}{\cos \pi z},$$

$$\Gamma(nz) = (2\pi)^{(1-n)/2} n^{nz-1/2} \prod_{k=0}^{n-1} \Gamma(z+k/n).$$

Asymptotic formulas. Saddle point approximation provides the following asymptotic formula called Stirling's formula:

$$\Gamma(z) \sim \sqrt{2\pi} z^{z-1/2} e^{-z}, \quad z \to \infty, \quad |\arg z| < \pi.$$

Some more asymptotic relations take place

$$\Gamma(z+a)/\Gamma(z+b) \sim z^{a-b}, \quad |z| \to \infty,$$

under condition that z do not coincide with points $z = -a,\ -a-1,\ -a-2,\ \ldots$ and $z = -b,\ -b-1,\ -b-2, \ldots$.

A.1.2 Three Integrals

Let us come back to the initial formula

$$\Gamma(x) = \int_0^\infty e^{-\xi} \xi^{x-1} d\xi, \quad x > 0.$$

Rotating the ray of integration to the imaginary axis,

$$\int_0^\infty e^{-i\xi} \xi^{x-1} d\xi = e^{-i\pi x/2} \Gamma(x),$$

and separating the real part from the imaginary, we obtain

$$\int_0^\infty \xi^{x-1} \cos \xi d\xi = \Gamma(x) \cos(\pi x/2)$$

and

$$\int_0^\infty e^{-i\xi} \xi^{x-1} d\xi = \Gamma(x) \sin(\pi x/2).$$

According to the principle of analytical continuation, these formulas remain also valid after replacement x by z for all points z where the integrals converge uniformly:

$$\int_0^\infty e^{-i\xi}\xi^{z-1}d\xi = e^{-i\pi z/2}\Gamma(z), \quad 0 < \mathrm{Re}\, z < 1,$$

$$\int_0^\infty \xi^{z-1}\cos\xi d\xi = \Gamma(z)\cos(\pi z/2), \quad 0 < \mathrm{Re}\, z < 1,$$

$$\int_0^\infty e^{-i\xi}\xi^{z-1}d\xi = \Gamma(z)\sin(\pi z/2), \quad -1 < \mathrm{Re}\, z < 1.$$

A.1.3 Binomial Coefficients

The binomial coefficients

$$\binom{m}{n} = \frac{m!}{n!(m-n)!} = \frac{m(m-1)\cdots(m-n+1)}{n!}, \quad n \leqslant m$$

admit the generalization to arbitrary non-integer and even complex $\mu(\neq -1, -2, \ldots)$ and ν:

$$\binom{\mu}{\nu} = \frac{\Gamma(\mu+1)}{\Gamma(\nu+1)\Gamma(\mu-\nu+1)} = \frac{\sin[(\nu-\mu)\pi]}{\pi}\frac{\Gamma(\mu+1)\Gamma(\nu-\mu)}{\Gamma(\nu+1)}.$$

For an integer $\nu = n$ and non-integer μ

$$\binom{\mu}{n} = \frac{\mu(\mu-1)\cdots(\mu-n+1)}{n!} = \frac{(-1)^{n-1}\mu\Gamma(n-\mu)}{n!\Gamma(1-\mu)} = O(n^{-\mu-1}), \quad n \to \infty,$$

and the following relation takes place:

$$\sum_{k=0}^{j}\binom{\mu}{k}\binom{\nu}{j-k} = \binom{\mu+\nu}{j}.$$

For integer $\nu = n$ and $\mu = m$ they vanish if $n > m$ and coincide with ordinary binomial coefficients if $n \leqslant m$.

A.2 Mittag-Leffler Functions

A.2.1 Mittag-Leffler Functions $E_\alpha(z)$, $E_{\alpha,\beta}(z)$

The *Mittag-Leffler function* $E_\alpha(z)$ with $\alpha > 0$ is defined by the series representation

$$E_\alpha(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\alpha n + 1)}$$

valid in the whole complex plane [13]. It provides a simple generalization of the exponential function because of the substitution of $n!$ with $\Gamma(\alpha n + 1)$. Particular cases, from which elementary functions are recovered, are

$$E_1(z) = e^z,$$

$$E_2(z^2) = \cosh(z),$$

$$E_2(-z^2) = \cos(z),$$

$$E_{1/2}(\pm\sqrt{z}) = e^z[1 + \operatorname{erf}(\pm\sqrt{z})] = e^z \operatorname{erfc}(\mp\sqrt{z}),$$

where erf (erfc) denotes the error (complementary error) function defined as

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-\zeta^2} d\zeta, \quad \operatorname{erfc}(z) = 1 - \operatorname{erf}(z).$$

By $\sqrt{z}$ one means the principal value of the square root of z in the complex plane cut along the negative real semiaxis.

Many of the important properties of function $E_\alpha(z)$ follow from its integral representation

$$E_\alpha(z) = \frac{1}{2\pi i} \int_{\mathrm{Ha}} \frac{\zeta^{\alpha-1} e^\zeta}{\zeta^\alpha - z} d\zeta, \quad \alpha > 0, \quad z \in \mathbb{C},$$

where the path of integration Ha (the *Hankel path*) is a loop which starts and ends at $-\infty$ and encircles the disk $|\zeta| \leqslant |z|^{1/2}$ in the positive sense.

For many purposes, the following properties of this function on the negative real axis are useful. The main of them are as follows.

(1) For $0 \leqslant \alpha \leqslant 1$, the function $E_\alpha(-x)$ is positive and completely monotonic, i.e.,

$$(-1)^n \frac{d^n}{dx^n} E_\alpha(-x) \geqslant 0, \quad x \geqslant 0, \quad 0 \leqslant \alpha \leqslant 1.$$

(2) For $1 < \alpha < 2$, the function $E_\alpha(-x)$ exhibits an add number of zeros, while $E_2(-x) = \cos(\sqrt{x})$ exhibits an infinite number of zeros.

(3) For $0 < \alpha < 1$ and $1 < \alpha < 2$, the following asymptotical expansion is valid

$$E_\alpha(-x) \sim -\sum_{n=1}^{\infty} \frac{(-x)^{-n}}{\Gamma(1-\alpha n)}, \quad x \to \infty, \quad 0 < \alpha < 1, \quad 1 < \alpha < 2.$$

(4) The Laplace transform of the function is

$$\mathscr{L}\{E_\alpha(-(\omega t)^\alpha)\}(\lambda) = \frac{\lambda^{\alpha-1}}{\lambda^\alpha + \omega^\alpha}.$$

(5) The duplication formula takes place

$$E_\alpha(z) = (1/2)[E_{\alpha/2}(\sqrt{z}) + E_{\alpha/2}(-\sqrt{z})].$$

(6) The integral duplication formula

$$\int_0^\infty \exp(-x^2/(4t)) E_\alpha(x^\alpha) dx = \sqrt{\pi t} E_{\alpha/2}(t^{\alpha/2}), \quad t > 0.$$

The *two-parameter Mittag-Leffler function* is obtained by replacing the additive constant 1 in the argument of the Gamma function by an arbitrary complex parameter β

$$E_{\alpha,\beta}(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\alpha n + \beta)}, \quad \alpha > 0, \ \beta \in \mathbb{C}.$$

Its integral representation has the form

$$E_{\alpha,\beta}(z) = \frac{1}{2\pi i} \int_{\mathrm{Ha}} \frac{\zeta^{\alpha-\beta} e^\zeta}{\zeta^\alpha - z} d\zeta, \quad \alpha, \ \beta > 0, \quad z \in \mathbb{C},$$

and asymptotical behaviour is determined via the expression

$$E_{\alpha,\beta}(z) \sim -\sum_{n=1}^{\infty} \frac{z^{-n}}{\Gamma(\beta - \alpha n)},$$

valid for $|\arg| < (1 - \alpha/2)\pi$ and $z \to \infty$.

The following general functional relations take place:

$$E_{\alpha,\beta}(z) = \frac{1}{\Gamma(\beta)} + z E_{\alpha,\alpha+\beta}(z),$$

$$E_{\alpha,\beta}(z) = \beta E_{\alpha,\beta+1}(z) + \alpha z \frac{d}{dz} E_{\alpha,\beta+1}(z),$$

$$\frac{d^n}{dz^n}[z^{\beta-1} E_{\alpha,\beta}(z^\alpha)] = z^{\beta-n-1} E_{\alpha,\beta-n}(z^\alpha), \quad n = 1, 2, \ldots .$$

Particular simple cases are

$$\begin{aligned} E_{\alpha,1}(z) &= E_\alpha(z), \\ E_{1,1}(z) &= E_1(z) = e^z, \\ E_{1,2}(z) &= \frac{e^z - 1}{z}, \\ E_{2,2}(z) &= \frac{\sinh(\sqrt{z})}{\sqrt{z}}. \end{aligned}$$

Some other functions linked to the Mittag-Leffler function:
hyperbolic function of order n

$$h_r(z, n) = \sum_{k=0}^{\infty} \frac{z^{nk+r-1}}{(nk + r - 1)!} = z^{r-1} E_{n,r}(z^n), \quad r = 1, 2, \ldots;$$

trigonometric function of order n

$$h_r(z, n) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{nk+r-1}}{(nk + r - 1)!} = z^{r-1} E_{n,r}(-z^n);$$

Robotnov's function

$$R_\alpha(\beta, z) = z^\alpha \sum_{k=0}^{\infty} \frac{\beta^k z^{k(\alpha+1)}}{\Gamma((1 + \alpha)(k + 1))} = z^\alpha E_{\alpha+1,\alpha+1}\left(\beta z^{\alpha+1}\right).$$

A.2.2 The Miller–Ross Functions

The *Miller–Ross function*

$$E_x(\nu, a) = x^\nu E_{1,\nu+1}(ax) = x^\nu \sum_{n=0}^{\infty} \frac{(ax)^n}{\Gamma(\nu + n + 1)}$$

and connected with it functions $C_x(\nu, a)$ and $S_x(\nu, a)$ (see below) play an important role in fractional calculus [12]. It has the integral representation

$$E_x(\nu, a) = \frac{1}{\Gamma(\nu)} \int_0^x e^{a(x-\xi)} \xi^{\nu-1} d\xi, \quad \text{Re } \nu > 0,$$

obeys the differential equation

$$D_x E_x(\nu, a) - a E_x(\nu, a) = \frac{x^{\nu-1}}{\Gamma(\nu)}, \quad \text{Re } \nu > 0,$$

and is tied with two-parameter Mittag-Leffler function and incomplete gamma function via relation:

$$E_x(\nu, a) = x^{\nu} E_{1,1+\nu}(ax) = a^{-\nu} e^{ax} \gamma(\nu, ax)/\Gamma(-a).$$

Special cases of the function:

$$\begin{aligned}
E_x(0, a) &= e^{ax}, \\
E_0(\nu, a) &= 0, \quad \text{Re}\nu > 0, \\
E_x(-1, a) &= a E_x(0, a), \\
E_x(-m, a) &= a^m E_x(0, a), \quad m = 0,\ 1,\ 2,\ \ldots, \\
E_x(1, a) &= \frac{E_x(0, a) - 1}{a}, \\
E_x(1/2, a) &= a^{-1/2} e^{ax} \text{Erf}[(ax)^{1/2}], \\
E_x(-1/2, a) &= a E_x(1/2, a) + (\pi x)^{-1/2}, \\
E_x(\nu, 0) &= \frac{x^{\nu}}{\Gamma(\nu + 1)}.
\end{aligned}$$

Recursion relations:

$$E_x(\nu, a) = a^m E_x(\nu + m, a) + \sum_{n=0}^{m-1} \frac{a^n x^{\nu+n}}{\Gamma(\nu + n + 1)}, \quad m = 0,\ 1,\ 2,\ \ldots,$$

$$\begin{aligned}
E_x(\nu, a) - E_x(\nu, b) &= a^m E_x(\nu + m, a) - b^m E_x(\nu + m, b) \\
&\quad + \sum_{n=1}^{m-1} \frac{(a^n - b^n) x^{\nu+n}}{\Gamma(\nu + n + 1)}, \quad m = 0, 1, \ldots .
\end{aligned}$$

Derivatives:

$$\mathsf{D}_x^m E_x(\nu, a) = E_x(\nu - m, a) = a^m E_x(\nu, a) + \sum_{n=0}^{m-1} \frac{a^n x^{\nu+n-m}}{\gamma(\nu + n + 1 - m)}, \quad m = 0, 1, 2, \ldots ,$$

$$\mathsf{D}_x^m[x^\mu E_x(\nu,a)] = \sum_{n=0}^{m}\binom{m}{n}\frac{\Gamma(\mu+1)}{\Gamma(\mu-n+1)}x^{\mu-n}E_x(\nu+n-m,a),\quad m=0,\ 1,\ 2,\ \ldots.$$

Integrals:

$$\int_0^x E_\xi(\nu,a)\xi^\mu d\xi = \Gamma(\mu+1)E_x(\nu+\mu+1,a),\quad \mathrm{Re}\,\mu,\ \mathrm{Re}\,\nu > -1,$$
$${}_0\mathsf{D}_x^{-m}E_x(\nu,a) = E_x(\nu+m,a),\quad m=0,\ 1,\ 2,\ \ldots,\quad \mathrm{Re}\,\nu > -1.$$

The Laplace transform:

$$\mathscr{L}\{E_x(\nu,a)\}(\lambda) = \frac{1}{\lambda^\nu(\lambda-a)},\quad \mathrm{Re}\nu > -1.$$

On multiplying of $E_x(\nu,a)$ by $x^{-\nu}$, one obtains a function of the product ax:

$$x^{-\nu}E_x(\nu,a) = \mathscr{E}(\nu,ax).$$

Special cases:

$$\mathscr{E}(\nu,0) = \frac{1}{\Gamma(\nu+1)},$$
$$\mathscr{E}(0,x) = e^x,$$
$$\mathscr{E}(1,x) = x^{-1}(e^x-1),$$
$$\mathscr{E}(2,x) = x^{-1}[\mathscr{E}(1,x)-1],$$

and so on.

The recursion formula:

$$x^m\mathscr{E}(\nu+m,x) = \mathscr{E}(\nu,x) - \sum_{n=0}^{m-1}\frac{x^n}{\Gamma(\nu+n+1)}.$$

For numerical calculations, the following asymptotical expansion is useful [12]:

$$\mathscr{E}(\nu,x) - x^{-\nu}e^x \sim -\left[\frac{1}{x\Gamma(\nu)} + \frac{1}{x^2\Gamma(\nu-1)} + \frac{1}{x^3\Gamma(\nu-2)} + \cdots\right]$$

as x increases without limit.

Another but close to this designation is used in [4]:

$$e_\nu(x;\lambda) = E_\nu(-\lambda x^\nu) = \sum_{n=0}^{\infty}\frac{(-\lambda x^\nu)^n}{\Gamma(n\nu+1)},\quad x>0,\quad \nu>0,\quad \lambda\in\mathbb{C}.$$

A.2.3 Functions $C_x(\nu, a)$ and $S_x(\nu, a)$

Replacing a by ia in the series for $E_x(\nu, a)$, one can obtain

$$E_x(\nu, ia) = C_x(\nu, a) + iS_x(\nu, a),$$

where

$$C_x(\nu, a) = x^\nu \sum_{n \text{ even}}^{\infty} \frac{(-1)^{n/2}(ax)^n}{\Gamma(\nu + n + 1)}$$

and

$$S_x(\nu, a) = x^\nu \sum_{n \text{ odd}}^{\infty} \frac{(-1)^{(n-1)/2}(ax)^n}{\Gamma(\nu + n + 1)}.$$

The functions obey the differential equation ($y = S_x(\nu, a)$ or $aC_x(\nu + 1, a)$):

$$\mathsf{D}_x^2 y + a^2 y = \frac{ax^{\nu-1}}{\Gamma(\nu)}, \quad \nu > 0.$$

They are expressed through the hypergeometrical function as follows

$$\begin{aligned} C_x(\nu, a) &= 2^{-1}x^\nu[{}_1F_1(1; 1 + \nu; iax) + {}_1F_1(1; 1 + \nu; -iax)]/\Gamma(1 + \nu) \\ &= a^{-\nu}[\cos(ax - \nu\pi/2) + (ax)^{\nu-2}/\Gamma(\nu - 1) - (ax)^{\nu-4}/\Gamma(\nu - 3) + \cdots] \\ S_x(\nu, a) &= (2i)^{-1}x^\nu[{}_1F_1(1; 1 + \nu; iax) - {}_1F_1(1; 1 + \nu; -iax)]/\Gamma(1 + \nu) \\ &= a^{-\nu}[\sin(ax - \nu\pi/2) + (ax)^{\nu-1}/\Gamma(\nu) - (ax)^{\nu-3}/\Gamma(\nu - 2) + \cdots]. \end{aligned}$$

Special cases of the functions:

$$\begin{aligned} C_0(\nu, a) &= 0, \quad \mathrm{Re}\nu > 0, \\ S_0(\nu, a) &= 0, \quad \mathrm{Re}\nu > -1, \\ C_x(\nu, 0) &= \frac{x^\nu}{\Gamma(\nu + 1)}, \\ S_x(\nu, 0) &= 0, \\ C_x(-1/2, a) &= 1/\sqrt{\pi x} - aS_x(1/2, a), \\ S_x(-1/2, a) &= aC_x(1/2, a), \\ C_x(1, a) &= (1/a)\sin(ax), \\ S_x(1, a) &= (2/a)\sin^2(ax/2), \\ C_x(-m, a) &= (-1)^{m/2}a^m\cos(ax), \quad m = 0,\ 2,\ 4,\ \ldots, \\ S_x(-m, a) &= (-1)^{m/2}a^m\sin(ax), \quad m = 0,\ 2,\ 4,\ \ldots, \\ C_x(-m, a) &= (-1)^{(m+1)/2}a^m\sin(ax), \quad m = 1,\ 3,\ 5,\ \ldots, \end{aligned}$$

$$S_x(-m,a) = (-1)^{(m-1)/2} a^m \cos(ax), \quad m = 1,\ 3,\ 5,\ \ldots\ .$$

Recursion relations:

$$C_x(\nu-1,a) = -aS_x(\nu,a) + \frac{x^{\nu-1}}{\Gamma(\nu)},$$
$$S_x(\nu-1,a) = aC_x(\nu,a),$$

$$C_x(\nu-1,a) + a^2 C_x(\nu+1,a) = \frac{x^{\nu-1}}{\Gamma(\nu)},$$
$$S_x(\nu-1,a) + a^2 S_x(\nu+1,a) = \frac{ax^{\nu}}{\Gamma(\nu+1)}.$$

Derivatives:

$$\mathsf{D}_x^m C_x(\nu,a) = C_x(\nu-m,a), \quad m = 0,\ 1,\ 2,\ \ldots\ ,$$
$$\mathsf{D}_x^m S_x(\nu,a) = S_x(\nu-m,a), \quad m = 0,\ 1,\ 2,\ \ldots\ ,$$
$$\mathsf{D}_x[x^\mu C_x(\nu,a)] = x^\mu C_x(\nu-1,a) + \mu x^{\mu-1} C_x(\nu,a),$$
$$\mathsf{D}_x[x^\mu S_x(\nu,a)] = x^\mu S_x(\nu-1,a) + \mu x^{\mu-1} S_x(\nu,a).$$

Integrals:

$$\int_0^x C_\xi(\nu,a)d\xi = C_x(\nu+1,a), \quad \mathrm{Re}\nu > -1,$$

$$\int_0^x S_\xi(\nu,a)d\xi = S_x(\nu+1,a), \quad \mathrm{Re}\nu > -1.$$

Laplace transforms:

$$\mathscr{L}\{C_x(\nu,a)\}(\lambda) = \frac{\lambda}{\lambda^\nu(\lambda^2+a^2)}, \quad \mathrm{Re}\nu > -1,$$

$$\mathscr{L}\{S_x(\nu,a)\}(\lambda) = \frac{a}{\lambda^\nu(\lambda^2+a^2)}, \quad \mathrm{Re}\nu > -2.$$

As for the function $E_x(\nu,a)$, the modified functions

$$\mathscr{C}(\nu,ax) = x^{-\nu} C_x(\nu,a),$$
$$\mathscr{S}(\nu,ax) = x^{-\nu} S_x(\nu,a)$$

are introduced and tabulated in [12].

A.2.4 The Wright Function

The *Wright function* is defined by the series [20]

$$W_{\alpha,\beta}(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!\Gamma(\alpha n + \beta)} \quad z \in \mathbb{C}.$$

Its integral representation is

$$W_{\alpha,\beta}(z) = \frac{1}{2\pi i} \int_{\mathrm{Ha}} \exp(\zeta + z\zeta^{-\alpha})\zeta^{-\beta} d\zeta, \quad \alpha > -1,$$

where Ha denotes the Hankel path in the complex ζ-plane with the cut along the line $\arg \zeta = \pi$ starting from $-\infty$ on the real axis, passing round the origin in a counterclockwise direction and returning to $-\infty$, thus enclosing the cut.

Some special cases:

$$W_{0,\beta}(z) = \frac{e^z}{\Gamma(\beta)},$$
$$(z/2)^{\nu} W_{1,\nu+1}(\mp z^2/4) = \begin{cases} J_{\nu}(z), \\ I_{\nu}(z). \end{cases}$$

In the case $\alpha = -\gamma \in (-1, 0)$ there is the following asymptotic expansion, valid in a suitable sector about the negative real axis

$$W_{-\gamma,\beta}(z) = Y^{1/2-\beta} e^{-Y} \left[\sum_{n=0}^{m-1} A_n Y^{-n} + O(|Y|_m) \right], \quad as \quad z \to \infty,$$

with $Y = (1-\gamma)(-\gamma^{\gamma} z)^{1/(1-\gamma)}$ and A_n being certain real numbers.

Recursion relation:

$$\alpha z W_{\alpha,\alpha+\beta}(z) = W_{\alpha,\beta-1}(z) + (1-\beta) W_{\alpha,\beta}(z).$$

Differential equation:

$$\frac{d}{dz} W_{\alpha,\beta}(z) = W_{\alpha,\alpha+\beta}(z).$$

The Laplace transform:

$$\mathscr{L}\{W_{\alpha,\beta}(\pm x)\}(\lambda) = (1/\lambda) E_{\alpha,\alpha}(\pm(1/\lambda)), \quad \alpha > 0,$$

$$\mathscr{L}\{W_{\alpha,\beta}(-x)\}(\lambda) = E_{-\alpha,\beta-\alpha}(-\lambda), \quad -1 < \alpha < 0.$$

A.2.5 The Mainardi Functions

The important cases of the Wright function, namely, the functions

$$F(z;\alpha) = W_{-\alpha,0}(-z)$$

and

$$M(z;\alpha) = W_{-\alpha,1-\alpha}(-z)$$

in case $0 < \alpha < 1$ have been considered in details by R. Gorenflo, F. Mainardi and coworkers [2, 7], and called *Mainardi functions*.

The functions are tied with each other via relation

$$F(z;\alpha) = \alpha z M(z;\alpha)$$

and have the series representations

$$F(z;\alpha) = \sum_{n=1}^{\infty} \frac{(-z)^n}{n!\Gamma(-\alpha n)} = -\frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-z)^n}{n!} \Gamma(\alpha n + 1) \sin(\alpha n \pi)$$

and

$$M(z;\alpha) = \sum_{n=1}^{\infty} \frac{(-z)^n}{n!\Gamma(-\alpha n + (1-\alpha))} = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-z)^{(n-1)}}{(n-1)!} \Gamma(\alpha n) \sin(\alpha n \pi).$$

Special cases:

$$M(z;1/2) = 1/\sqrt{\pi}\, \exp(-z^2/4),$$

and

$$M(z;1/3) = 3^{3/2} \mathrm{Ai}(z/3^{1/3}),$$

where Ai denotes the Airy function.

Moments and asymptotic expression in $\mathbb{R}^+$:

$$\int_0^{\infty} M(x;\alpha) x^n dx = \frac{\Gamma(n+1)}{\Gamma(\alpha n + 1)}, \quad n = 0,\ 1,\ 2,\ \ldots,$$

$$M(x/\alpha;\alpha) \sim \frac{z^{(\alpha-1/2)/(1-\alpha)}}{\sqrt{2\pi(1-\alpha)}} \exp(-(1/\alpha - 1) z^{1/(1-\alpha)}), \quad z \to \infty.$$

The Laplace transforms of the functions are of the form:

$$\mathscr{L}\{F(x^{\alpha};\alpha)\}(\lambda) = \exp(-\lambda^{\alpha}), \quad 0 < \alpha < 1,$$

$$\mathscr{L}\{M(x;\alpha)\}(\lambda) = E_\alpha(-\lambda), \qquad 0 < \alpha < 1,$$

$$\mathscr{L}\{x^{-1}F(x^{-\alpha};\alpha)\}(\lambda) = \mathsf{L}\{\alpha x^{-\alpha-1}M(x^{-\alpha};\alpha)\}(\lambda) = \exp(-\lambda^\alpha), \quad 0 < \alpha < 1.$$

A.3 The Fox Functions

A.3.1 Definition

The *Fox function* or *H-function*, also called the *generalized G-function* or *generalized Mellin-Barnes function*, represents a rich class of functions which contains functions such as Meijer's G-function, hypergeometric functions, Wright's hypergeometric series, Bessel functions, Mittag-Leffler functions, etc., as special cases (see for details [11]).

Let m, n, p, and q be integer numbers such that $0 \leqslant n \leqslant p$ and $1 \leqslant m \leqslant q$. The Fox function of the order (m, n, p, q) is defined by the Mellin-Barnes type integral

$$H_{pq}^{mn}(z) \equiv H_{pq}^{mn}\left[z \,\middle|\, \begin{matrix}(a_1,\alpha_1)\ldots(a_p,\alpha_p)\\(b_1,\beta_1)\ldots(b_q,\beta_q)\end{matrix}\right] = \frac{1}{2\pi i}\int_L K_{pq}^{mn}(s)z^s ds, \tag{A.1}$$

where

$$z^s = \exp\{s\ln|z| + i\arg z\}$$

and $\arg z$ is not necessarily the principal value. Further,

$$K_{pq}^{mn}(s) = \frac{A(s)B(s)}{C(s)D(s)},$$

where

$$A(s) = \prod_{j=1}^{m}\Gamma(b_j - \beta_j s), \quad B(s) = \prod_{j=1}^{n}\Gamma(1 - a_j + \alpha_j s),$$

$$C(s) = \prod_{j=m+1}^{q}\Gamma(1 - b_j + \beta_j s), \quad D(s) = \prod_{j=n+1}^{p}\Gamma(a_j - \alpha_j s).$$

The empty products are interpreted as unity: if $n = 0$ then $B = 1$, if $m = q$ then $C = 1$, if $n = p$ then $D = 1$.

In a general case

$$K_{pq}^{mn}(z) = \frac{[\Gamma(b_1 - \beta_1 s)\cdots\Gamma(b_m - \beta_m s)][\Gamma(1 - a_1 + \alpha_1 s)\cdots\Gamma(1 - a_n + \alpha_n s)]}{[\Gamma(1 - b_{m+1} + \beta_{m+1} s)\cdots\Gamma(1 - b_q + \beta_q s)][\Gamma(a_{n+1} - \alpha_{n+1} s)\cdots\Gamma(a_p - \alpha_p s)]}.$$

The parameters $\alpha_1, \ldots, \alpha_p$ and $\beta_1, \ldots, \beta_q$ are positive numbers while the parameters $a_1, \ldots, a_p$ and $b_1, \ldots, b_q$ are complex numbers. The parameters are restricted by condition

$$P(A) \cap P(B) = \phi$$

where ϕ denotes the empty set,

$$P(A) = \{s : s = (b_j + k)/\beta_j \equiv s_{j,k} | j = 1, \ldots, m;\ k = 0, 1, 2, \ldots\}$$

and

$$P(B) = \{s : s = (a_j - 1 - k)/\alpha_j \equiv \sigma_{j,k} | j = 1, \ldots, n;\ k = 0, 1, 2, \ldots\}$$

are sets of poles of functions $A(z)$ and $B(z)$ respectively. In other words, there must be no poles of $A(z)$ coinciding with ones of $B(z)$

$$\alpha_j(b_l + \nu) \neq \beta_l(a_j - 1 - \nu'),$$

where

$$\nu, \nu' = 0, 1, 2, \ldots, \quad l = 1, 2, \ldots, m, \quad j = 1, 2, \ldots, n.$$

Contour of integrating L runs from $\infty - ic$ to $\infty + ic$ where $c > |\mathrm{Im}\ b_j|/\beta_j,\ j = 1, 2, \ldots, m$. It separates the poles of the functions in such a way that the poles of $A(s)$ lie to the right and the poles of $B(s)$ lie to the left of the contour.

A.3.2 Some Properties

Let $\mu = \sum_{j=1}^{q} \beta_j - \sum_{j=1}^{p} \alpha_j$ and $\beta = \prod_{j=1}^{p} \alpha_j^{\alpha_j} / \prod_{j=1}^{q} \beta_j^{\beta_j}$.

Property 1. The Fox function is an analytic function of z (i) for every $z \neq 0$ if $\mu > 0$ and (ii) for $0 < |z| < \beta^{-1}$ if $\mu = 0$.

Property 2. In general, the Fox function is multiple valued due to the factor z^s in the integral representation but it is single valued on the Riemann surface of $\ln z$.

Property 3. The Fox function is symmetric in pairs $(a_1, \alpha_1), \ldots, (a_n, \alpha_n)$, likewise $(a_{n+1}, \alpha_{n+1}), \ldots, (a_p, \alpha_p)$; in $(b_1, \beta_1), \ldots, (b_m, \beta_m)$ and in $(b_{m+1}, \beta_{m+1}), \ldots, (b_q, \beta_q)$.

Property 4. If one of the pairs $(a_j, \alpha_j),\ j = 1, \ldots, n$, say (a_p, α_p), is equal to one of the pairs (b_j, β_j), say (b_q, β_q), then the Fox function reduces to one of the lower order, and p, q, and n decrease by unity:

$$H_{pq}^{mn}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_{q-1},\beta_{q-1})(a_p,\alpha_p)\end{matrix}\right.\right]=H_{p-1,q-1}^{m,n-1}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_{p-1},\alpha_{p-1})\\(b_1,\beta_1)\cdots(b_{q-1},\beta_{q-1})\end{matrix}\right.\right]$$

provided $n \geqslant 1$ and $q > m$.

Property 5. There exists an important property

$$H_{pq}^{mn}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right]=H_{qp}^{nm}\left[\frac{1}{z}\left|\begin{matrix}(1-b_1,\beta_1)\cdots(1-b_q,\beta_q)\\(1-a_1,\alpha_1)\cdots(1-a_p,\alpha_p)\end{matrix}\right.\right],$$

which enables us to transform a Fox function with $\mu=\sum_{j=1}^{q}\beta_j-\sum_{j=1}^{p}\alpha_j>0$ and $\arg z$ to one with $\mu<0$ and $\arg(1/z)$ and vice versa.

Property 6. For any positive γ

$$H_{pq}^{mn}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right]=\gamma H_{pq}^{mn}\left[z^{\gamma}\left|\begin{matrix}(a_1,\gamma\alpha_1)\cdots(a_p,\gamma\alpha_p)\\(b_1,\gamma\beta_1)\cdots(b_q,\gamma\beta_q)\end{matrix}\right.\right].$$

Property 7. For any complex σ

$$z^{\sigma}H_{pq}^{mn}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right]=\gamma H_{pq}^{mn}\left[z^{\gamma}\left|\begin{matrix}(a_1+\sigma\alpha_1,\alpha_1)\cdots(a_p+\sigma\alpha_p,\alpha_p)\\(b_1+\sigma\beta_1,\beta_1)\cdots(b_q+\sigma\beta_q,\beta_q)\end{matrix}\right.\right].$$

Property 8. For any admissible $p \leqslant q$, positive γ and complex r

$$H_{p+1,q+1}^{m,n+1}\left[z\left|\begin{matrix}(0,\gamma)(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_q,\beta_q)(r,\gamma)\end{matrix}\right.\right]$$

$$=(-1)^r H_{p+1,q+1}^{m+1,n}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)(0,\gamma)\\(r,\gamma)(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right].$$

Property 9. For the same parameters $p \leqslant q$, γ and r

$$H_{p+1,q+1}^{m+1,n}\left[z\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)(1-r,\gamma)\\(1,\gamma)(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right]$$

$$=(-1)^r H_{p+1,q+1}^{m+1,n}\left[z\left|\begin{matrix}(1-r,\gamma)(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\(b_1,\beta_1)\cdots(b_q,\beta_q)(1,\gamma)\end{matrix}\right.\right].$$

Property 10. For complex c and d, natural k and positive γ

$$\frac{d^k}{dz^k}H_{pq}^{mn}((cz+d)^{\gamma})=\frac{(-c)^k}{(cz+d)^{\gamma}}H_{p+1,q+1}^{m+1,n}\left[(cz+d)^{\gamma}\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)(0,\gamma)\\(k,\gamma)(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right],$$

$$\frac{d^k}{dz^k}H_{pq}^{mn}((cz+d)^{-\gamma})=\frac{c^k}{(cz+d)^{\gamma}}H_{p+1,q+1}^{m,n+1}\left[\frac{1}{(cz+d)^{\gamma}}\left|\begin{matrix}(a_1,\alpha_1)\cdots(a_p,\alpha_p)(1-k,\gamma)\\(1,\gamma)(b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right.\right].$$

Property 11. Fractional-order Riemann-Liouville integral:

$$ {}_0\mathrm{I}_z^{\mu} H_{pq}^{mn}(z) = z^{\mu} H_{p+1,q+1}^{m,n+1}\left[z \,\middle|\, \begin{matrix} (0,1)(a_1,\alpha_1)\cdots(a_p,\alpha_p) \\ (b_1,\beta_1)\cdots(b_q,\beta_q)(-\mu,1) \end{matrix} \right]. $$

The formula is valid if $\min\{\mathrm{Re}\,(b_j/\beta_j) \mid 1 \leqslant j \leqslant m\} > 0$.

The fractional-order Riemann-Liouville derivative is obtained from this formula by analytic continuation to $\mu = -\nu,\ \nu > 0$.

Property 12. The fractional differintegral of the product $z^{\alpha} H_{pq}^{mn}((az)^{\beta})$ for an arbitrary ν, positive α and β under condition $\alpha + \beta\ \min\{\mathrm{Re}\,(b_j/\beta_j) \mid 1 \leqslant j \leqslant m\} > -1$:

$$ {}_0\mathrm{D}_z^{\nu}\left\{ z^{\alpha} H_{pq}^{mn}\left((az)^{\beta}\right)\right\} = z^{\alpha-\nu} H_{p+1,q+1}^{m,n+1}\left[(az)^{\beta} \,\middle|\, \begin{matrix} (-\alpha,\beta)(a_1,\alpha_1)\cdots(a_p,\alpha_p) \\ (b_1,\beta_1)\cdots(b_q,\beta_q)(\nu-\alpha,\beta) \end{matrix} \right]. $$

Property 13. The theorem of residues enables one to express the Fox function as the infinite series

$$ H_{pq}^{mn}(z) = \sum_{i=1}^{m} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\beta_i} C_{ik} z^{s_{ik}}, $$

where

$$ s_{ik} = (b_i + k)/\beta_i $$

and

$$ C_{ik} = \frac{\prod_{j=1, j\neq l}^{m} \Gamma(b_j - \beta_j s_{ik}) \prod_{r=1}^{n} \Gamma(1 - a_r + \alpha_r s_{ik})}{\prod_{u=m+1}^{q} \Gamma(1 - b_u + \beta_u s_{ik}) \prod_{v=n+1}^{p} \Gamma(a_v - \alpha_v s_{ik})}. $$

This series can be used for the calculation of special values of the Fox functions and to derive their asymptotic behavior for $z \to \infty$.

Property 14. The Laplace transform

$$ \widehat{H}_{pq}^{mn}(\lambda) \equiv \mathscr{L}\{H_{pq}^{mn}(z)\}(\lambda) = \frac{1}{\lambda} H_{q,p+1}^{n+1,m}\left[\lambda \,\middle|\, \begin{matrix} (1-b_1,\beta_1)\cdots(1-b_q,\beta_q) \\ (1,1)(1-a_1,\alpha_1)\cdots(1-a_p,\alpha_p) \end{matrix} \right] $$

for $\mu \in [0, 1]$, and

$$ \widehat{H}_{pq}^{mn}(\lambda) = \frac{1}{\lambda} H_{p+1,q}^{m,n+1}\left[\frac{1}{\lambda} \,\middle|\, \begin{matrix} (0,1)(a_1,\alpha_1)\cdots(a_p,\alpha_p) \\ (b_1,\beta_1)\cdots(b_q,\beta_q) \end{matrix} \right] $$

for $\mu \geqslant 1$ respectively.

On the other hand, if the Laplace image of some function is of the form

$$ \widehat{F}(\lambda) = H_{pq}^{mn}\left[z \,\middle|\, \begin{matrix} (a_1,\alpha_1)\cdots(a_p,\alpha_p) \\ (b_1,\beta_1)\cdots(b_q,\beta_q) \end{matrix} \right], $$

then it reverse transform reads

$$F(x) = \mathsf{L}^{-1}\{\widehat{F}(\lambda)\}(x) = \frac{1}{x} H^{n,m}_{q,p+1}\left[x \,\middle|\, \begin{matrix}(1-b_1,\beta_1)\cdots(1-b_q,\beta_q)\\ (1-a_1,\alpha_1)\cdots(1-a_p,\alpha_p)(1,1)\end{matrix}\right]$$

for $\mu \in [0,1]$, and

$$F(x) = \frac{1}{x} H^{m,n}_{p+1,q}\left[\frac{1}{x} \,\middle|\, \begin{matrix}(0,1)(a_1,\alpha_1)\cdots(a_p,\alpha_p)(0,1)\\ (b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right]$$

for $\mu \geqslant 1$ respectively. These relations are valid if

$$\sum_{j=1}^{m} b_j - \sum_{j=m+1}^{q} b_j - \sum_{j=1}^{p} a_j > 0$$

and

$$\max \operatorname{Re}\left\{\frac{a_j-1}{\alpha_j};\ 1 \leqslant j \leqslant n\right\} < \min \operatorname{Re}\left\{\frac{b_j}{\beta_j};\ 1 \leqslant j \leqslant m\right\}.$$

Property 15. A more general formula is valid under additional condition $\alpha + \beta \min\{\operatorname{Re}(b_j/\beta_j) \mid 1 \leqslant j \leqslant m\} > -1$:

$$\mathscr{L}\{x^{\alpha} H^{mn}_{pq}((ax)^{\beta})\}(\lambda) = \frac{1}{\lambda^{\alpha+1}} H^{m,n+1}_{p+1,q}\left[\left(\frac{a}{\lambda}\right)^{\beta} \,\middle|\, \begin{matrix}(-\alpha,\beta)(a_1,\alpha_1)\cdots(a_p,\alpha_p)\\ (b_1,\beta_1)\cdots(b_q,\beta_q)\end{matrix}\right].$$

A.3.3 Some Special Cases

The beta type distribution (up to norming—here and below):

$$H^{10}_{12}\left[z \,\middle|\, \begin{matrix}(a,1)\\ (b,1)\end{matrix}\right] = \frac{1}{\Gamma(a-b)} z^{b}(1-z)^{a-b-1}, \quad |z| < 1.$$

The Lévy type distribution:

$$H^{10}_{01}\left[\frac{1}{z} \,\middle|\, \begin{matrix}--\\ (b,\beta)\end{matrix}\right] = \beta^{-1} z^{-b/\beta} \exp(-z^{-1/\beta}).$$

The Cauchy type distribution:

$$H^{11}_{11}\left[z^2 \,\middle|\, \begin{matrix}(1-\nu,1)\\ (0,1)\end{matrix}\right] = \frac{\Gamma(\nu)}{(1+z^2)^{\nu}} = \frac{1}{(1+z^2)^{\nu}}.$$

The ordinary Bessel function of the first kind:

$$H_{02}^{10}\left[\frac{z^2}{4}\middle|\left(\frac{a+\nu}{2},1\right)\left(\frac{a-\nu}{2},1\right)\right]=\left(\frac{z}{2}\right)^a J_\nu(z).$$

The modified Bessel function of the second kind:

$$H_{02}^{20}\left[\frac{z^2}{4}\middle|\left(\frac{a-\nu}{2},1\right)\left(\frac{a+\nu}{2},1\right)\right]=2^{1-a}z^a K_\nu(z).$$

The Whittaker function:

$$H_{12}^{20}\left[z\middle|\begin{matrix}(a-\lambda+1,1)\\(a+\mu+1/2,1)(a-\mu+1/2,1)\end{matrix}\right]=z^a e^{-z/2}W_{\lambda\mu}(z).$$

The generalized Mittag-Leffler function:

$$H_{12}^{11}\left[z\middle|\begin{matrix}(0,1)\\(0,1)(1-\beta,\alpha)\end{matrix}\right]=E_{\alpha,\beta}(-z).$$

The hypergeometric function:

$$H_{22}^{12}\left[z\middle|\begin{matrix}(1-\lambda,1)(1-\mu,1)\\(0,1)(1-\nu,1)\end{matrix}\right]=\frac{\Gamma(\lambda)\Gamma(\mu)}{\Gamma(\nu)}\,{}_2F_1(\lambda,\mu;\nu;-z).$$

The Meijer *G*-function:

$$H_{pq}^{mn}\left[z\middle|\begin{matrix}(a_1,1)\cdots(a_p,1)\\(b_1,1)\cdots(b_q,1)\end{matrix}\right]=G_{pq}^{mn}\left[z\middle|\begin{matrix}a_1,\ldots,a_p\\b_1,\ldots,b_q\end{matrix}\right].$$

A.4 Fractional Stable Distributions

A.4.1 Introduction

This class of distributions appeared in [7] (univariate distributions) and [15] (multivariate distributions) was investigated in work [1, 6] where it was named *fractional stable distributions*(FSD's). The reason of such term is that any random variable with distribution belonging to this class (FSRV) is equal in distribution to the ratio of two independent stable random variables (SRV's):

$$X(\alpha,\nu,\theta)\stackrel{d}{=}S(\alpha,\theta)/[S_+(\nu)]^{\nu/\alpha}. \tag{A.2}$$

Here α is the characteristic exponent of SRV $S(\alpha, \theta)$, θ is its skewness parameter (we consider as in the previous issue standard strictly stable laws when the scale parameter is 1 and the shift parameter is 0) and $S_+(\nu)$ is the *subordinator*, that is the stable random variable $S_+ \equiv S(\nu, 1)$, $\beta < 1$, with PDF $g_+(x; \nu)$ and characteristic function

$$\langle \exp\{-\lambda S_+(\nu)\rangle = \exp(-\lambda^\nu). \tag{A.3}$$

Equation (A.2) rewritten in terms probability density functions takes the form

$$q(x; \alpha, \nu, \theta) = \int_0^\infty g(xt^{\nu/\alpha}; \alpha, \theta) g_+(y; \nu) t^{\beta/\alpha} dt. \tag{A.4}$$

A.4.2 Characteristic Function

P. Lévy, who defined and described the class of stable laws, introduced the term the *second characteristics*

$$\psi(k; \alpha, \theta) = \ln \widetilde{g}(k; \alpha, \theta)$$

so that characteristic functions are expressed through the second characteristics via relation

$$\begin{gathered} \widetilde{g}(k; \alpha, \theta) = \exp\{\psi(k; \alpha, \theta)\}, \\ \psi(k; \alpha, \theta) = -|k|^\alpha \exp\{-i\alpha\theta(\pi/2)\text{sign } k\}. \end{gathered} \tag{A.5}$$

The most important for what follows property of the second characteristics is that in the strictly stable case they are homogeneous in Euler's sense, i.e.,

$$\psi(ak; \alpha, \theta) = a^\alpha \psi(k; \alpha, \theta) \tag{A.6}$$

for any positive a.

Going back to Eq. (A.4), multiplying it by $\exp(-ikx)dx$, integrating over the whole real axis, changing the order of integrations and using the second characteristics one can represent the characteristic function in the following form:

$$\widetilde{q}(k; \alpha, \nu, \theta) = \int_{-\infty}^\infty e^{ikx} q(x; \alpha, \nu, \theta) dx = \int_0^\infty \exp\{\psi(kt^{-\nu/\alpha}; \alpha, \theta)\} g_+(t; \nu) dt.$$

Taking into account property (A.6)

$$\widetilde{q}(k; \alpha, \nu, \theta) = \int_0^\infty \exp\{t^{-\nu} \psi(k; \alpha, \theta)\} g_+(t; \nu) dt.$$

and expanding the exponential function in the power series we obtain:

$$\widetilde{q}(k;\alpha,\nu,\theta)=\sum_{n=0}^{\infty}\frac{[\psi(k;\alpha,\theta)]^n}{n!}\int_0^{\infty}t^{-n\nu}g_+(t;\nu)dt.$$

The integrals present negative order moments of subordinator $S_+(\nu)$. It is known (and can be derived from Eq. (A.3)) that

$$\langle[S_+(\nu)]^{-\beta}\rangle=\int_0^{\infty}t^{-\beta}g_+(t;\nu)dt=\frac{\Gamma(1+\beta/\nu)}{\Gamma(1+\beta)},\tag{A.7}$$

consequently

$$\widetilde{q}(k;\alpha,\nu,\theta)=\sum_{n=0}^{\infty}\frac{[\psi(k;\alpha,\theta)]^n}{\Gamma(1+n\nu)}.$$

Comparing this expression with the series representation of Mittag-Leffler function

$$E_\nu(x)=\sum_{n=0}^{\infty}\frac{x^n}{\Gamma(1+n\nu)}$$

we obtain the final result in the form:

$$\widetilde{q}(k;\alpha,\nu,\theta)=E_\nu(\psi(k;\alpha,\theta)).\tag{A.8}$$

A.4.3 Inverse Power Series Representation

It is known that stable densities can be represented in the form of power series asymptotic for $\alpha<1$ and convergent for $\alpha>1$:

$$g(x;\alpha,\theta)=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n!}\frac{\Gamma(n/\alpha+1)}{\Gamma(n\rho)\Gamma(1-n\rho)}x^{n-1},$$

or in the form of inverse power series convergent for $\alpha<1$ and asymptotic for $\alpha>1$:

$$g(x;\alpha,\theta)=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n!}\frac{\Gamma(n\alpha+1)}{\Gamma(n\alpha\rho)\Gamma(1-n\alpha\rho)}x^{-n\alpha-1},\ x>0,$$

where $\rho=(1+\theta)/2$.

On substituting the second series into Eq. (A.4) we get

$$q(x;\alpha,\nu,\theta)=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n!}\frac{\Gamma(n\alpha+1)}{\Gamma(n\rho)\Gamma(1-n\rho)}\int_0^{\infty}(xt^{\nu/\alpha})^{-n\alpha-1}t^{\nu/\alpha}g_+(t;\nu)dt$$

$$=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n!}\frac{\Gamma(n\alpha+1)}{\Gamma(n\alpha\rho)\Gamma(1-n\alpha\rho)}x^{-n\alpha-1}\int_0^{\infty}t^{-n\nu}g_+(t;\nu)dt,\quad x>0.$$

Using here Eq. (A.7) we obtain the corresponding series for FSD densities:

$$q(x;\alpha,\nu,\theta)=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{\Gamma(n\alpha+1)}{\Gamma(n\alpha\rho)\Gamma(1-n\alpha\rho)\Gamma(1+n\nu)}x^{-n\alpha-1},\quad x>0.$$

For the fractional stable distribution function we have:

$$Q(x;\alpha,\nu,\theta)=1-\alpha^{-1}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}\frac{\Gamma(n\alpha+1)}{\Gamma(n\alpha\rho)\Gamma(1-n\alpha\rho)\Gamma(1+n\nu)}x^{-n\alpha},\quad x>0.$$

A.4.4 Integral Representation

Both convergent and asymptotic expansions are convenient tools of numerical analysis in the cases where the number of terms required to guarantee a reasonable accuracy is not very large. Otherwise one should prefer the integral representation of the distributions. From the computational viewpoint, the definite integral can be treated as the limit of a sequence of integral sums, i.e., as a series too; the existence of various schemes of numerical integration offers considerable scope for further improvements.

It is clear that the presence of oscillating integrand plagues the computation. To avoid this trouble, the inversion formula has to be transformed into an integral of non-oscillating function.

Let

$$g(x;\alpha,\beta)=g(-x;\alpha,-\beta)=\pi^{-1}\mathrm{Re}\int_0^{\infty}e^{ikx}\widetilde{g}(k;\alpha,-\beta)dk.$$

Omitting a special case $\alpha=1$, we assume that $x>0$ if $\alpha\neq 1$ and start with formula

$$\psi(k;\alpha,-\theta)=-|k|^{\alpha}\exp\{i\alpha\theta(\pi/2)\mathrm{sign}\,k\}.$$

The function $\widetilde{g}(k;\alpha,-\theta)=\exp\{\psi(k;\alpha,-\theta)\}$ allows the analytic continuation $\widetilde{g}^{+}(z;\alpha,\theta)$ from the positive semi-axis to the complex plane with the cut along the ray $\arg z=-3\pi/4$.

Considering the integral

$$J = \int_L e^{ixz} g^+(z; \alpha, -\theta) dz \equiv \int_L e^{-W(z,x)} dz$$

along the contour L which starts from zero and goes to infinity so that the $W(z, x)$ takes only real values, after some manipulations we obtain for the stable density

$$g(x; \alpha, \theta) = \frac{\alpha |x|^{1/(\alpha-1)}}{\pi |1-\alpha|} \int_{-\theta\pi/2}^{\pi/2} \exp\{-|x|^{\alpha/(\alpha-1)} U(\phi; \alpha, \theta)\} U(\phi; \alpha, \theta) d\phi,$$

where

$$U(\phi; \alpha, \theta) = \Big[\frac{\sin(\alpha\phi + \alpha\theta\pi/2)}{\cos\phi}\Big]^{\alpha/(1-\alpha)} \frac{\cos\Big((\alpha-1)\phi + \alpha\theta\pi/2\Big)}{\cos\phi}.$$

A similar expression takes place for the stable cumulative function:

$$G(x; \alpha, \theta) = \int_{-\infty}^{x} g(x'; \alpha, \theta) dx' = 1 - \int_{x}^{\infty} g(x'; \alpha, \theta) dx'.$$

For $x > 0$

$$G(x; \alpha, \theta) = (1-\theta)/2 + \pi^{-1} \int_{-\theta\pi/2}^{\pi/2} \exp\{-V(x, \phi; \alpha, \theta)\} d\phi$$

if $\alpha < 1$, and

$$G(x; \alpha, \theta) = 1 - \pi^{-1} \int_{-\theta\pi/2}^{\pi/2} \exp\{-V(x, \phi; \alpha, \theta)\} d\phi$$

if $\alpha > 1$. Here

$$V(x, \phi; \alpha, \theta) = x^{\alpha/(\alpha-1)} U(\phi; \alpha, \theta).$$

It is easy to derive the integral representation for fractional stable cumulative function. By definition,

$$\begin{aligned} Q(x; \alpha, \nu, \theta) &= \int_{-\infty}^{x} q(x'; \alpha, \nu, \theta) dx' \\ &= \int_{-\infty}^{x} dx' \int_0^{\infty} dt g_+(t; \nu) g(x' t^{\nu/\alpha}; \alpha, \theta) t^{\nu/\alpha} \\ &= \int_0^{\infty} G(x t^{\nu/\alpha}; \alpha, \theta) g_+(t; \nu) dt. \end{aligned}$$

Performing evident manipulations

$$G(xt^{\nu/\alpha};\alpha,\theta)=(1-\theta)/2+\pi^{-1}\int_{-\theta\pi/2}^{\pi/2}\exp\{-t^{\nu/(\alpha-1)}V(x,\phi;\alpha,\theta)\}d\phi,$$

we obtain the function for $\alpha<1$ in the form

$$Q(x;\alpha,\nu,\theta)=\frac{1-\theta}{2}$$
$$+\pi^{-1}\int_{-\theta\pi/2}^{\pi/2}d\phi\int_0^\infty\exp\{-t^{\nu/(\alpha-1)}V(x,\phi;\alpha,\theta)\}g_+(t;\nu)dt,\ x>0.$$

The inside integral can be calculated after the series expansion of the exponential function with the use of the subordinator moments (A.7):

$$\int_0^\infty\exp\{-t^{\nu/(\alpha-1)}V(x,\phi;\alpha,\theta)\}g_+(t;\nu)dt$$
$$=\sum_{n=0}^\infty\frac{[-V(x,\phi);\alpha,\theta]^n}{n!}\int_0^\infty g_+(t;\nu)t^{-n\nu/(1-\alpha)}dt$$
$$=\sum_{n=0}^\infty\frac{[-V(x,\phi;\alpha,\theta)]^n}{n!}\frac{\Gamma(1+n/(1-\alpha))}{\Gamma(1+n\nu/(1-\alpha))}.$$

As a result we have got (for $\alpha<1$)

$$Q(x;\alpha,\nu,\theta)$$
$$=\frac{1-\theta}{2}+\pi^{-1}\int_{-\theta\pi/2}^{\pi/2}\sum_{n=0}^\infty\frac{[-V(x,\phi;\alpha,\theta)]^n}{n!}\frac{\Gamma(1+n/(1-\alpha))}{\Gamma(1+n\nu/(1-\alpha))}d\phi,\ x>0.$$

This formula can be rewritten in terms of angular moments

$$\langle U^n(\alpha,\theta)\rangle=\frac{1}{(1+\theta)\pi/2}\int_{-\theta\pi/2}^{\pi/2}U^n(\phi;\alpha,\theta)d\phi,$$

and looks as follows:

$$Q(x;\alpha,\nu,\theta)=\frac{1-\theta}{2}+\frac{1+\theta}{2}\sum_{n=0}^\infty\frac{(-1)^n}{n!}\frac{\Gamma(1+n/(1-\alpha))}{\Gamma(1+n\nu/(1-\alpha))}\langle U^n(\alpha,\theta)\rangle x^{n\alpha/(\alpha-1)}.$$

A.4.5 Fox Function Representation

Fox function representation of FSD's is based on the Mellin transformation

$$\bar{q}(s;\alpha,\nu,\theta)=\int_0^\infty q(x;\alpha,\nu,\theta)x^s dx,$$
$$q(x;\alpha,\nu,\theta)=\frac{1}{2\pi i}\int_{\mathbb{C}}\bar{q}(s;\alpha,\nu,\theta)x^{-s-1}ds,\quad x>0.$$

Among integral transformations, the Mellin transformation is the most convenient tool for investigation of FSD because any FSD has the form of the Mellin convolution (A.4). Its Mellin transform is nothing but the product of Mellin transforms of stable distributions under convolution. Indeed, multiplying both sides of Eq. (A.4) by $x^s dx$ and integrating over the positive semi-axis yields

$$\begin{aligned}\bar{q}(s;\alpha,\nu,\theta)&=\int_0^\infty dx x^s\int_0^\infty g(xy^{\nu/\alpha};\alpha,\theta)g_+(y;\nu)y^{\nu/\alpha}dy\\&=\int_0^\infty dy g_+(y;\nu)y^{\nu/\alpha}\int_0^\infty g(xy^{\nu/\alpha};\alpha,\theta)x^s dx\\&=\int_0^\infty dy g_+(y;\nu)y^{-s\nu/\alpha}\int_0^\infty g(z;\alpha,\theta)z^s dz\\&=\bar{g}(s;\alpha,\theta)\bar{g}_+(-s\nu/\alpha;\nu).\end{aligned}\tag{A.9}$$

It is known that Mellin transforms of stable densities are of the form [14]

$$\bar{g}(s;\alpha,\theta)=\frac{\Gamma(s)\Gamma(1-\alpha' s)}{\Gamma(\rho s)\Gamma(1-\rho s)}=\rho\frac{\Gamma(1+s)\Gamma(1-\alpha' s)}{\Gamma(1+\rho s)\Gamma(1-\rho s)}$$

where $\alpha'=1/\alpha$ and $\rho=(1+\theta)/2$. Taking into account that

$$\bar{g}_+(s;\nu)\equiv\bar{g}(s;\nu,1)=\frac{\Gamma(1-s/\nu)}{\Gamma(1-s)},$$

we obtain for (A.9)

$$\bar{q}(s;\alpha,\nu,\theta)=\rho\frac{\Gamma(1+s)\Gamma(1-\alpha' s)\Gamma(1+\alpha' s)}{\Gamma(1+\rho s)\Gamma(1-\rho s)\Gamma(1+\alpha'\nu s)}.$$

If we put $-s-1$ in place of s we prepare the result obtained above to its representation in the form of the Fox function:

$$q(x;\alpha,\nu,\theta)=\frac{\alpha'}{2\pi i}\int_{\mathbb{C}}h(s;\alpha,\nu,\theta)x^s ds,\quad x>0.$$

where

$$\begin{aligned} h(s;\alpha,\nu,\theta) &= \alpha\bar{q}(-s-1;\alpha,\nu,\theta) \\ &= \frac{\rho\Gamma(-s)\Gamma(1+\alpha'+\alpha's)\Gamma(1-\alpha'-\alpha's)}{\alpha'\Gamma(1-\rho-\rho s)\Gamma(1+\rho+\rho s)\Gamma(1-\alpha'\nu-\alpha'\nu s)} \\ &= \frac{\Gamma(-s)\Gamma(\alpha'+\alpha's)\Gamma(1-\alpha'-\alpha's)}{\Gamma(1-\rho-\rho s)\Gamma(\rho+\rho s)\Gamma(1-\alpha'\nu-\alpha'\nu s)}. \end{aligned} \tag{A.10}$$

The case $\alpha > \nu$. Represent $h(s;\alpha,\nu,\theta)$ in the standard form connected with Fox function:

$$h(s;\alpha,\nu,\theta) = \frac{A(s)B(s)}{C(s)D(s)} \tag{A.11}$$

where

$$\begin{aligned} A(s) &= \Gamma(b_1-\nu_1 s)\Gamma(b_2-\nu_2 s), \quad B(s) = \Gamma(1-a_1+\alpha_1 s), \\ C(s) &= \Gamma(1-b_3+\nu_3 s), \quad D(s) = \Gamma(a_2-\alpha_2 s)\Gamma(a_3-\alpha_3 s). \end{aligned}$$

Thus $m=2$, $n=1$, $p=3$, and $q=3$. Continuation of comparison of (A.10) with (A.11) yields

$$a_1 = 1-\alpha', \quad \alpha_1 = \alpha'; \quad a_2 = 1-\rho, \quad \alpha_2 = \rho; \quad a_3 = 1-\alpha'\nu, \quad \alpha_3 = \alpha'\nu;$$

$$b_1 = 0, \quad \nu_1 = 1; \quad b_2 = 1-\alpha', \quad \nu_2 = \alpha'; \quad b_3 = 1-\rho, \quad \nu_3 = \rho.$$

As one can see from here,

$$\mu = \sum_{j=1}^{q}\nu_j - \sum_{j=1}^{p}\alpha_j = 1+\alpha'+\rho-\alpha'-\rho-\alpha'\nu = 1-\alpha'\nu$$

and when $\alpha > \nu$ the integral (A.1) along the contour C separating the poles of $A(s)$ and of $B(s)$ can be expressed through the Fox function

$$H_{33}^{21}\left(x \,\middle|\, \begin{matrix}(a_1,\alpha_1) & (a_2,\alpha_2) & (a_3,\alpha_3)\\ (b_1,\nu_1) & (b_2,\nu_2) & (b_3,\nu_3)\end{matrix}\right)$$

with parameters $\{a,\alpha,b,\nu\}$ given above. As a result we have the following formula for FSD density ($x>0$ and $\alpha>\nu$):

$$q(x;\alpha,\nu,\theta) = \alpha' H_{33}^{21}\left(x \,\middle|\, \begin{matrix}(1-\alpha',\alpha') & (1-\rho,\rho) & (1-\alpha'\nu,\alpha'\nu)\\ (0,1) & (1-\alpha',\alpha') & (1-\rho,\rho)\end{matrix}\right).$$

The case $\alpha < \nu$. In order to consider the opposite case $\alpha < \nu$, we replace x by $1/x$ and s by $-s-2$ in Eq. (A.10). After some typical manipulations with gamma-functions, we get:

$$q(1/x;\alpha,\nu,\theta)=\frac{x^2\alpha'}{2\pi i}\int_{\mathbb{C}}\frac{\Gamma(2+s)\Gamma(-\alpha'-\alpha's)\Gamma(1+\alpha'+\alpha's)}{\Gamma(-\rho-\rho s)\Gamma(1+\rho+\rho s)\Gamma(1+\alpha'\nu+\alpha'\nu s)}x^s ds.$$

Using again the standard form of the Mellin transform of the Fox function we find in this case:

$$A(s)=\Gamma(-\alpha'-\alpha's),\quad B(s)=\Gamma(2+s)\Gamma(1+\alpha'+\alpha's),$$
$$C(s)=\Gamma(1+\rho+\rho s)\Gamma(1+\alpha'\nu+\alpha'\nu s),\quad D(s)=\Gamma(-\rho-\rho s).$$

Now $m=1,\ n=2,\ p=q=3$. Comparison with Eq. (A.1) yields

$$a_1=-1,\quad \alpha_1=1;\quad a_2=-\alpha',\quad \alpha_2=\alpha';\quad a_3=-\rho,\quad \alpha_3=\rho;$$

$$b_1=0,\quad \nu_1=1;\quad b_2=1-\alpha',\quad \nu_2=\alpha';\quad b_3=1-\rho,\quad \nu_3=\rho.$$

The condition $\mu=\nu\alpha'-1>1$ is valid if $\alpha<\nu$ and for this case and $x>0$ we have

$$q(x;\alpha,\nu,\theta)=x^{-2}\alpha' H_{33}^{12}\left(1/x\,\middle|\,\begin{matrix}(-1,1) & (-\alpha',\alpha') & (-\rho,\rho)\\ (-\alpha',\alpha') & (-\rho,\rho) & (-\alpha'\nu,\alpha'\nu)\end{matrix}\right).$$

There exist a few cases when some terms in the function $h(s;...)$ cancel each other and the order of H-function becomes lower. This is the stable subfamily of FSD ($\nu=1$), the one-sided subfamily of FSD ($\alpha<1,\quad \theta=1$), and the one-sided stable subfamily ($\alpha<1,\quad \theta=1,\quad \nu=1$). The correspondent densities are of the form:

$$q(x;\alpha,1,\theta)\equiv g(\alpha,\theta)=\alpha' H_{22}^{11}\left(x\,\middle|\,\begin{matrix}(1-\alpha',\alpha'), & (1-\rho,\rho)\\ (0,1), & (1-\rho,\rho)\end{matrix}\right),\quad \alpha>1; \tag{A.12}$$

$$q(x;\alpha,1,\theta)\equiv g(x;\alpha,\theta)=x^{-2}\alpha' H_{22}^{11}\left(x^{-1}\,\middle|\,\begin{matrix}(-1,1), & (-\rho,\rho)\\ (-\alpha',\alpha'), & (-\rho,\rho)\end{matrix}\right),\quad \alpha<1; \tag{A.13}$$

$$q(x;\alpha,\nu,1)=\alpha' H_{22}^{11}\left(x\,\middle|\,\begin{matrix}(1-\alpha',\alpha'), & (1-\alpha'\nu,\alpha'\nu)\\ (1-\alpha',\alpha'), & (0,1)\end{matrix}\right),\quad \nu<\alpha<1;$$

$$q(x;\alpha,\nu,1)=x^{-2}\alpha' H_{22}^{11}\left(x^{-1}\,\middle|\,\begin{matrix}(-\alpha',\alpha'), & (-1,1)\\ (-\alpha',\alpha'), & (1+\alpha'\nu,\alpha')\end{matrix}\right),\quad \alpha<\nu<1;$$

$$q(x;\alpha,1,1)\equiv g(x;\alpha,1)=x^{-2}\alpha' H_{11}^{10}\left(x^{-1}\,\middle|\,\begin{matrix}(-1,1)\\ (-\alpha',\alpha')\end{matrix}\right),\quad \alpha<1. \tag{A.14}$$

Graphs of some distribution densities are presented in Fig. A.1.

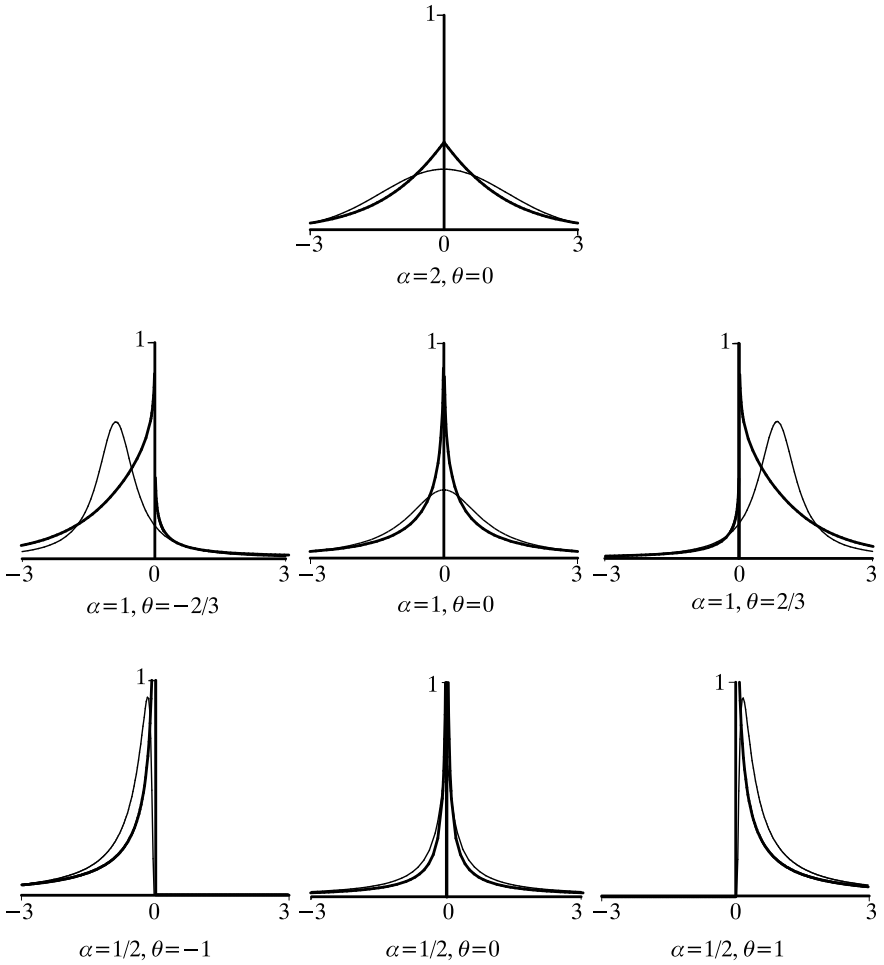


Fig. A.1 Graphs of some stable densities (thin lines) and fractional stable densities (thick lines)

A.4.6 Multivariate Fractional Stable Densities

Here we bring some properties of multivariate FSDs $\Psi_d^{(\alpha,\nu)}(\mathbf{x};\,\mu_o) \equiv \Psi_d^{(\alpha,\nu)}(|\mathbf{x}|)$.

1. The densities $\Psi_{d+2}^{(\alpha,\nu)}(r)$ and $\Psi_d^{(\alpha,\nu)}(r)$ are linked via the relation

$$\Psi_{d+2}^{(\alpha,\nu)}(r) = -\frac{1}{2\pi r}\frac{d\Psi_d^{(\alpha,\nu)}(r)}{dr}.$$

   The densities $\Psi_{d+1}^{(\alpha,\nu)}(r)$ and $\Psi_d^{(\alpha,\nu)}(r)$ are linked via fractional derivative of the order $1/2$ (Fig. A.2).

2. Similarly to the normal case, the projection of diffusible d-dimensional vector $X(t)$ on a d'-dimensional subspace ($d' < d$) diffuses according to a d'-dimensional law with the same parameters α and ν.

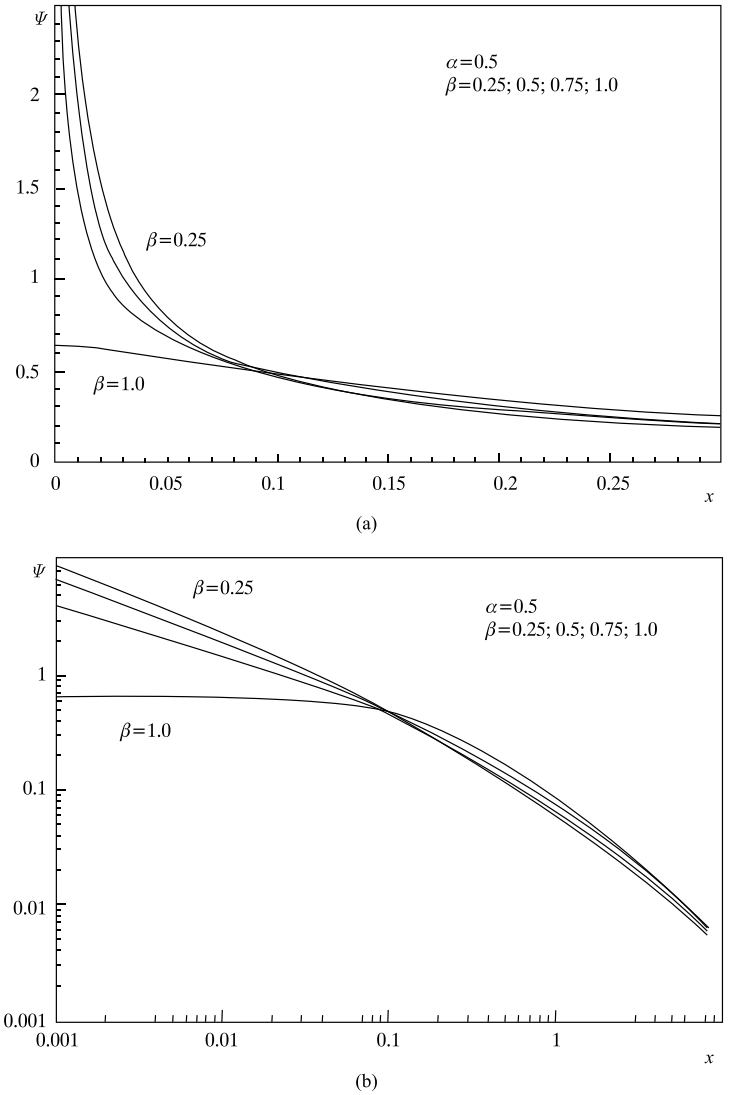
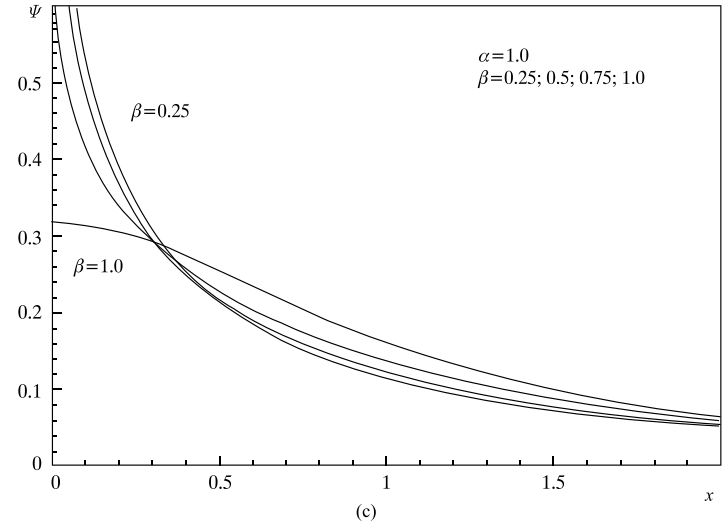
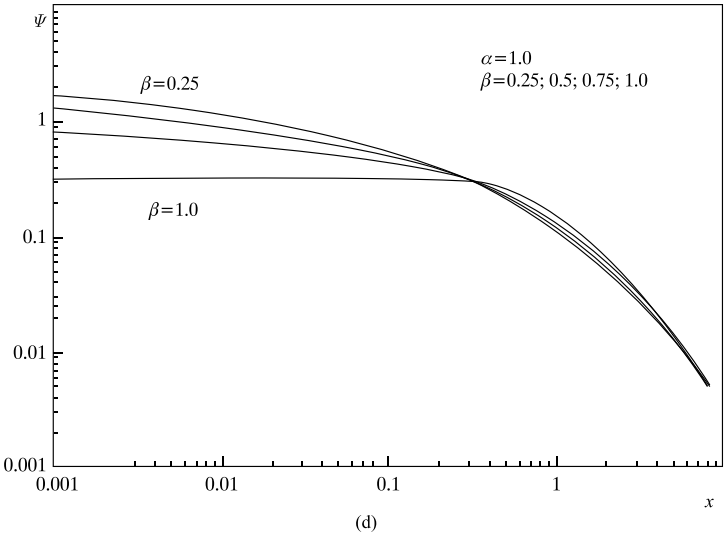


Fig. A.2 Graphs of one-dimensional fractionally stable densities with the indicated parameters

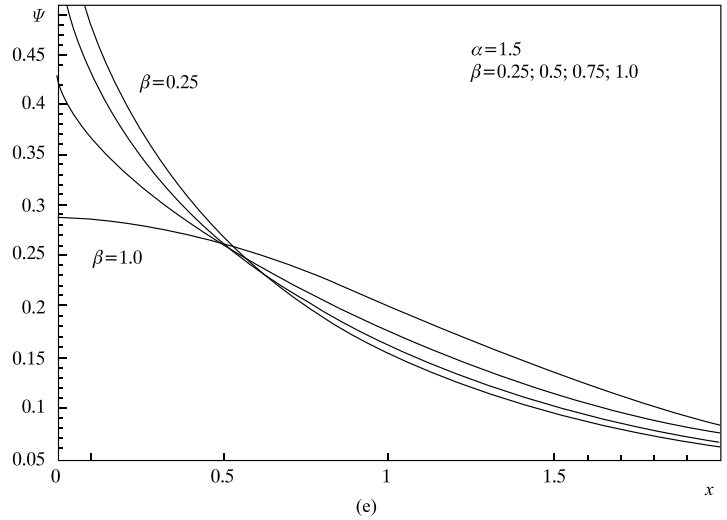


(c)

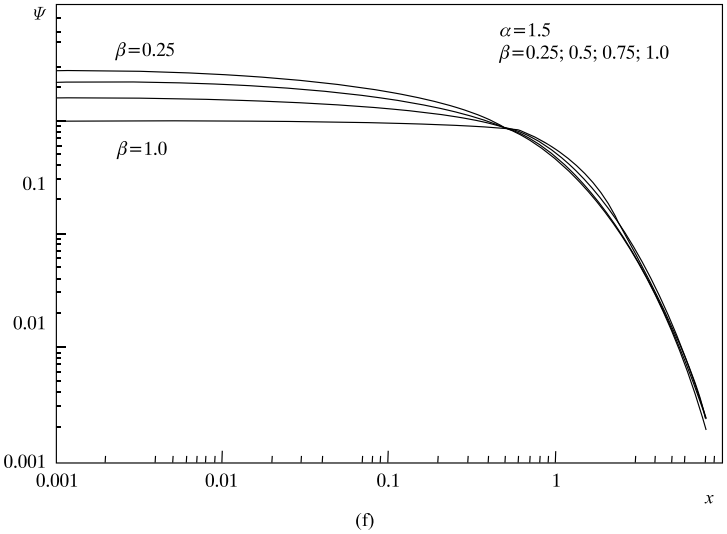


(d)

Fig. A.2 (continued)

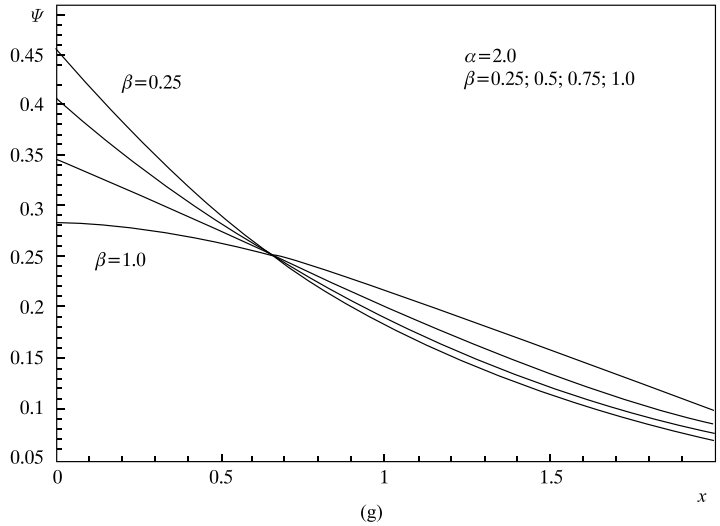


(e)

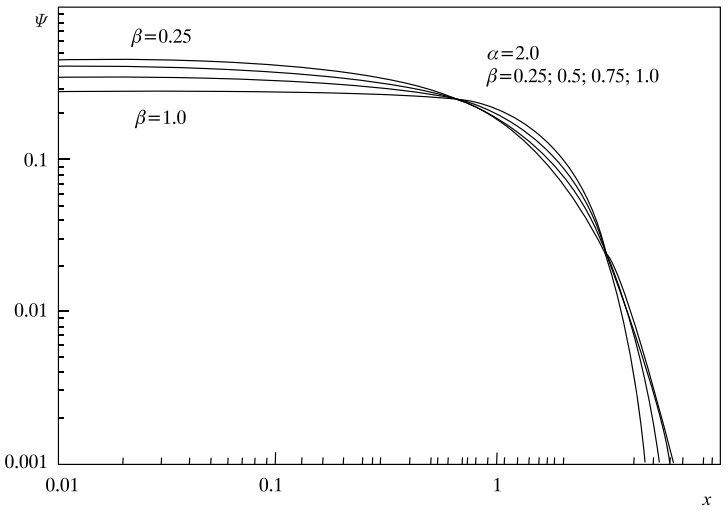


(f)

Fig. A.2 (continued)

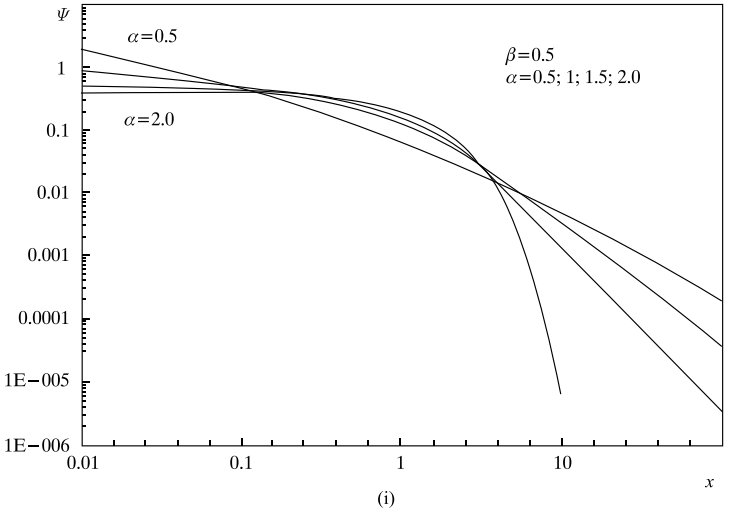


(g)

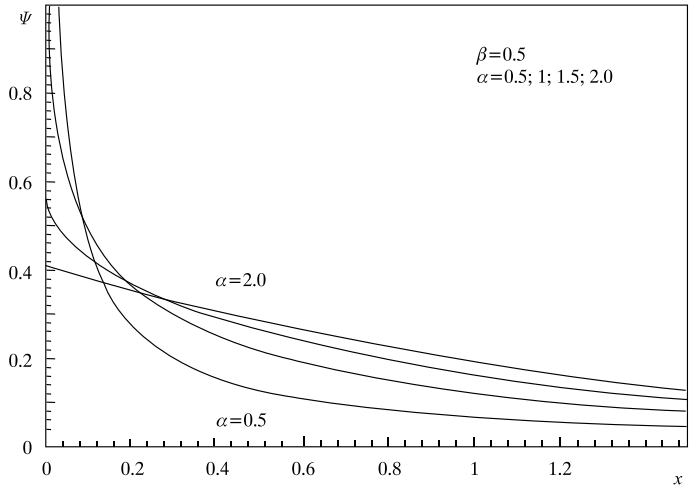


(h)

Fig. A.2 (continued)

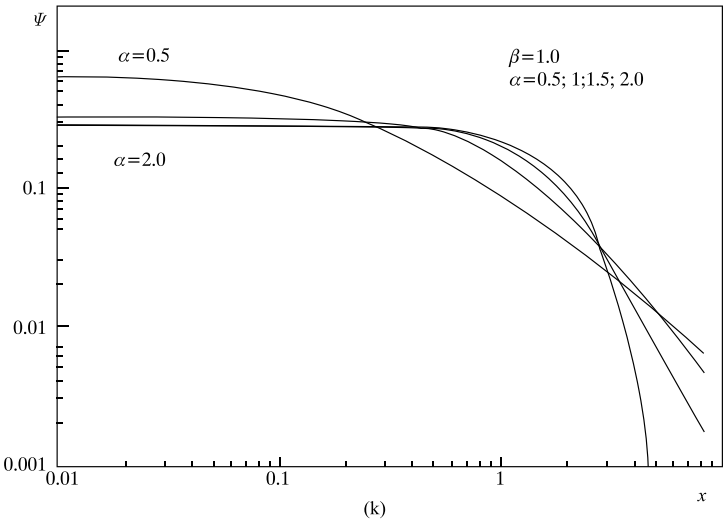


(i)

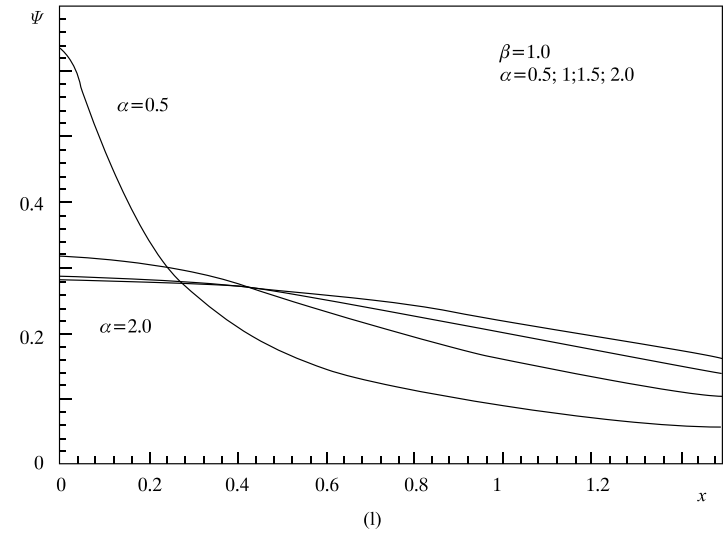


(j)

Fig. A.2 (continued)



(k)



(l)

Fig. A.2 (continued)

3. In contrast to the normal case, different coordinates $X_1(t), \ldots, X_d(t)$ of a particle performing anomalous diffusion (with $\alpha \neq 2$ or $\nu \neq 1$) are not independent of each other.
4. The FSD $\Psi_d^{(\alpha,\nu)}(r)$ is a decreasing function of r and its maximal value $\Psi_d^{(\alpha,\nu)}(0)$ is finite only if $d < \alpha$:

$$\Psi_d^{(\alpha,\nu)}(0) = \frac{\Gamma(1+d/\alpha)\Gamma(1-d/\alpha)}{(4\pi)^{d/2}\Gamma(1+d/2)\Gamma(1-d\nu/\alpha)}.$$

In particular,

$$\Psi_1^{(\alpha,\nu)}(0) = \frac{\csc(\pi/\alpha)}{\alpha\Gamma(1-\nu/\alpha)}.$$

5. In the case $\nu = 1$, the FSD becomes the isotropic LF distribution:

$$\Psi_d^{(\alpha,1)}(r) = \rho_d(r;\alpha)$$

$$\Psi_d^{(\alpha,1)}(r) = \rho_d(r;\alpha) = (2\pi)^{-d/2}\int_0^\infty e^{-s^\alpha} J_{d/2-1}(rs)(rs)^{1-d/2}s^{d-1}ds.$$

6. On combining this distribution with subordinator $g_+(t;\nu)$, one can obtain the relation

$$\Psi_d^{(\alpha,\nu)}(r) = \int_0^\infty \Psi_d^{(\alpha,1)}\left(r\tau^{\nu/\alpha};\alpha\right) g_+(\tau;\nu)d\tau.$$

7. If $\alpha = 2$ and $\nu < 1$, then

$$\Psi_1^{(2,\nu)}(0) = \Big[2\Gamma(1-\nu/2)\Big]^{-1},$$

$$\Psi_2^{(2,\nu)}(r) \sim \Big[2\pi\,\Gamma(1-\nu)\Big]^{-1}|\ln r|, \quad r \to 0,$$

and for $d \geqslant 3$

$$\Psi_d^{(2,\nu)}(r) \sim (4\pi)^{-d/2}\Big[\Gamma(d/2-1)\big/\Gamma(1-\nu)\Big](r/2)^{-(d-2)}, \quad r \to 0.$$

8. At large distances

$$\begin{aligned}\Psi_d^{(2,\nu)}(r) \sim\ & (4\pi)^{-d/2}(2-\nu)^{-1/2}\nu^{[(d+1)\nu/2-1]/(2-\nu)} \\ & \times (r/2)^{-d(1-\nu)/(2-\nu)}\exp\{-(2-\nu)\nu^{\nu/(2-\nu)}(r/2)^{2/(2-\nu)}\}.\end{aligned}$$

9. In the one-dimensional case with $\alpha = 2$

$$\Psi_1^{(2,\nu)}(r) = \nu^{-1}r^{-1-2/\nu}g_+\left(r^{-2/\nu};\nu/2\right).$$

Setting $\nu = 1$, we obtain

$$\Psi_1^{(2,1)}(r) = \frac{1}{\sqrt{4\pi}}e^{-r^2/4},$$

and in the case $\nu = 2/3$

$$\Psi_1^{(2,2/3)}(r) = \frac{1}{2\pi}\sqrt{r}K_{1/3}(2r^{3/2}/\sqrt{27}).$$

10. In the case $\alpha = 1$, $\nu = 1/2$ the FSD of all dimensions are expressed in terms of the incomplete gamma function:

$$\Psi_d^{(1,1/2)}(r) = \frac{2}{\sqrt{\pi}} \frac{\Gamma\Big((d+1)/2\Big)}{(4\pi)^{(d+1)/2}} e^{r^2/4}\Gamma\Big(1-(d+1)/2, r^2/4\Big).$$

For odd dimensions

$$\Psi_d^{(1,1/2)}(r) = \frac{2}{\sqrt{\pi}} \frac{\Gamma\Big((d+1)/2\Big)}{(4\pi)^{(d+1)/2}} \left(\frac{r^2}{4}\right)^{\mu} e^{r^2/4} E_{(d+1)/2}(r^2/4),$$

where $\mu = 1 - (d+1)/2$.

11. In the case $\alpha = 1, \quad \nu = 1$

$$\Psi_d^{(1,1)}(r) = \Gamma((d+1)/2)\left[\pi(1+r^2)\right]^{-(d+1)/2}.$$

12. The Mellin transform of the FSD is of the form

$$\begin{aligned}\tilde{\Psi}_d^{(\alpha,\nu)}(s) &\equiv \int_0^\infty \Psi_d^{(\alpha,\nu)}(r) r^{s-1} dr \\ &= \frac{2^s \Gamma(1-(d-s)/\alpha)\Gamma(s/2)\Gamma((d-s)/\alpha)}{\alpha(4\pi)^{d/2}\Gamma(1-(d-s)\nu/\alpha)\Gamma((d-s)/2)}.\end{aligned}$$

Therefore, $\Psi_d^{(\alpha,\nu)}(r)$ is expressed through the Fox H-function as follows:

$$\begin{aligned}&\Psi_d^{(\alpha,\nu)}(r) = (\nu/2)(4\pi)^{-d/2}\left(\frac{2}{r}\right)^{d+\alpha} \\ &\quad \times H_{32}^{12}\left(\left(\frac{2}{r}\right)^{\nu}\left|\begin{matrix}(-1,1/\alpha)\,(1-(\alpha+d)/2,\nu/2)\,(1-\alpha/2,1/2) \\ (0,1/\alpha)\,(-1,1/\alpha)\end{matrix}\right.\right), \quad \alpha < 1; \\ &\Psi_d^{(\alpha,\nu)}(r) = (\alpha r\sqrt{\pi})^{-d} H_{23}^{21}\left(\frac{r}{2}\left|\begin{matrix}(1,1/\alpha)\,(1,\nu/\alpha) \\ (1,1/\alpha)\,(d/2,1/2)\,(1,1/2)\end{matrix}\right.\right), \quad 1 \leqslant \alpha < 2; \\ &\Psi_d^{(2,\nu)}(r) = (2r\sqrt{\pi})^{-d} H_{12}^{20}\left(\frac{r}{2}\left|\begin{matrix}(1,\nu/2) \\ (1,1/2)\,(d/2,1/2)\end{matrix}\right.\right).\end{aligned}$$

13. When $\alpha = 2$ moments of all orders exist and are of the form

$$\langle |X(t)|^{2n}\rangle = \frac{\Gamma(n+d/2)\Gamma(n+1)}{\Gamma(d/2)\Gamma(n\nu+1)}\Big(4Dt^{\nu}\Big)^n.$$

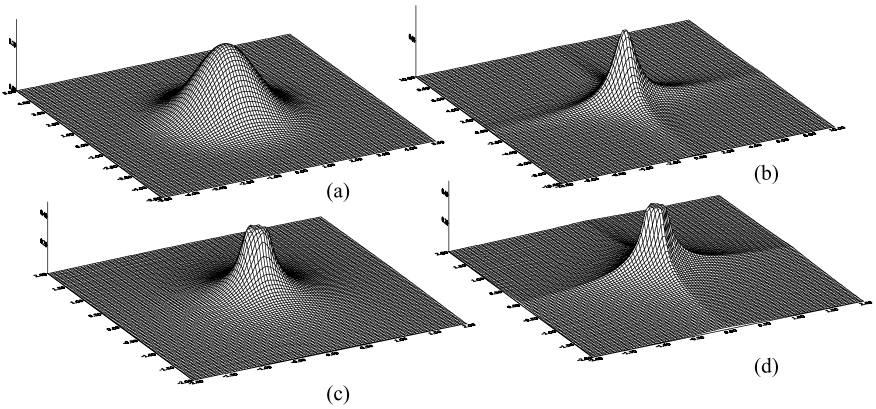


Fig. A.3 The densities $q_2(x_1, x_2; \alpha, \nu, \Gamma)$: **a** $\alpha = 2,\ \nu = 1,\ \Gamma = \Gamma_0$; **b** $\alpha = 1,\ \nu = 1,\ \Gamma = \Gamma_1$; **c** $\alpha = 1,\ \nu = 1/2,\ \Gamma = \Gamma_0$; **d** $\alpha = 1,\ \nu = 1/2,\ \Gamma = \Gamma_1$

Figure A.3 shows two-variable isotropic and anisotropic densities. Numerical values of isotropic densities $\Psi_d^{(\alpha,\nu)}(r) = q_d(\mathbf{x}; \alpha, \nu, \Gamma_0)$ for $d = 1$ and 2 are given in Tables B.1 and B.2.

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Appendix B
Fractional Stable Densities

See Tables B.1 and B.2.

Table B.1 One-dimensional FSD $2\Psi_1^{(\alpha,\nu)}(r)$

r	$\alpha = 0.25$				$\alpha = 0.5$				$\alpha = 0.75$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	0.5826	0.5980	0.6563	0.7994	0.9514	0.9212	0.9072	0.9535	1.0916	1.0027	0.8952	0.7337
0.2	0.3042	0.3178	0.3550	0.4315	0.5449	0.5494	0.5811	0.6826	0.6995	0.6781	0.6667	0.6733
0.3	0.2065	0.2177	0.2447	0.2954	0.3846	0.3960	0.4317	0.5194	0.5201	0.5203	0.5386	0.5988
0.4	0.1564	0.1657	0.1869	0.2241	0.2971	0.3100	0.3432	0.4140	0.4137	0.4225	0.4513	0.5253
0.5	0.1258	0.1338	0.1512	0.1802	0.2417	0.2545	0.2841	0.3412	0.3423	0.3548	0.3866	0.4591
0.6	0.1052	0.1122	0.1269	0.1504	0.2033	0.2155	0.2416	0.2883	0.2908	0.3048	0.3364	0.4021
0.7	0.0904	0.0966	0.1092	0.1289	0.1751	0.1865	0.2096	0.2483	0.2519	0.2662	0.2962	0.3537
0.8	0.0792	0.0848	0.0959	0.1127	0.1535	0.1641	0.1847	0.2171	0.2214	0.2354	0.2632	0.3128
0.9	0.0704	0.0755	0.0854	0.0999	0.1364	0.1463	0.1646	0.1922	0.1968	0.2104	0.2359	0.2783
1.0	0.0634	0.0681	0.0769	0.0897	0.1226	0.1318	0.1482	0.1719	0.1766	0.1896	0.2128	0.2490
1.1	0.0576	0.0619	0.0699	0.0813	0.1112	0.1197	0.1346	0.1551	0.1598	0.1720	0.1931	0.2241
1.2	0.0528	0.0568	0.0641	0.0743	0.1016	0.1096	0.1230	0.1410	0.1455	0.1570	0.1761	0.2027
1.3	0.0487	0.0524	0.0591	0.0684	0.0934	0.1009	0.1131	0.1289	0.1332	0.1441	0.1614	0.1842
1.4	0.0452	0.0487	0.0549	0.0633	0.0864	0.0934	0.1046	0.1186	0.1227	0.1329	0.1486	0.1682
1.5	0.0421	0.0454	0.0512	0.0589	0.0802	0.0868	0.0971	0.1096	0.1134	0.1230	0.1372	0.1541

(continued)

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Table B.1 (continued)

r	$\alpha = 0.25$				$\alpha = 0.5$				$\alpha = 0.75$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1	0.25	0.5	0.75	1
1.6	0.0395	0.0426	0.0479	0.0551	0.0749	0.0811	0.0905	0.1017	0.1053	0.1143	0.1272	0.1419
1.7	0.0371	0.0400	0.0451	0.0517	0.0701	0.0760	0.0847	0.0948	0.0981	0.1065	0.1183	0.1310
1.8	0.0350	0.0378	0.0425	0.0486	0.0659	0.0714	0.0795	0.0886	0.0916	0.0996	0.1103	0.1214
1.9	0.0331	0.0358	0.0402	0.0459	0.0621	0.0673	0.0749	0.0831	0.0859	0.0934	0.1032	0.1128
2.0	0.0314	0.0340	0.0382	0.0435	0.0587	0.0637	0.0707	0.0782	0.0807	0.0878	0.0967	0.1051
2.1	0.0299	0.0323	0.0363	0.0413	0.0556	0.0603	0.0669	0.0737	0.0760	0.0827	0.0909	0.0983
2.2	0.0285	0.0308	0.0346	0.0393	0.0528	0.0573	0.0634	0.0697	0.0717	0.0780	0.0856	0.0920
2.3	0.0273	0.0295	0.0330	0.0375	0.0502	0.0545	0.0603	0.0661	0.0679	0.0738	0.0808	0.0864
2.4	0.0261	0.0282	0.0316	0.0358	0.0479	0.0520	0.0574	0.0627	0.0643	0.0699	0.0763	0.0813
2.5	0.0250	0.0271	0.0303	0.0343	0.0457	0.0496	0.0547	0.0596	0.0611	0.0664	0.0723	0.0767
2.6	0.0240	0.0260	0.0291	0.0329	0.0437	0.0475	0.0523	0.0568	0.0581	0.0631	0.0686	0.0724
2.7	0.0231	0.0250	0.0280	0.0316	0.0419	0.0455	0.0500	0.0542	0.0553	0.0601	0.0651	0.0685
2.8	0.0223	0.0241	0.0269	0.0304	0.0402	0.0436	0.0479	0.0518	0.0527	0.0573	0.0620	0.0650
2.9	0.0215	0.0232	0.0260	0.0292	0.0386	0.0419	0.0460	0.0496	0.0504	0.0547	0.0590	0.0617
3.0	0.0207	0.0224	0.0251	0.0282	0.0371	0.0403	0.0442	0.0476	0.0482	0.0523	0.0563	0.0586
3.1	0.0200	0.0217	0.0242	0.0272	0.0357	0.0388	0.0425	0.0456	0.0461	0.0500	0.0538	0.0558
3.2	0.0194	0.0210	0.0234	0.0263	0.0345	0.0374	0.0409	0.0439	0.0442	0.0479	0.0514	0.0532
3.3	0.0188	0.0203	0.0227	0.0254	0.0332	0.0361	0.0394	0.0422	0.0424	0.0460	0.0492	0.0508
3.4	0.0182	0.0197	0.0220	0.0246	0.0321	0.0348	0.0380	0.0406	0.0407	0.0441	0.0472	0.0486
3.5	0.0177	0.0191	0.0213	0.0238	0.0310	0.0337	0.0367	0.0391	0.0392	0.0424	0.0453	0.0465
3.6	0.0172	0.0186	0.0207	0.0231	0.0300	0.0326	0.0355	0.0378	0.0377	0.0408	0.0435	0.0445
3.7	0.0167	0.0181	0.0201	0.0224	0.0291	0.0315	0.0343	0.0365	0.0363	0.0393	0.0418	0.0427
3.8	0.0162	0.0176	0.0196	0.0218	0.0282	0.0306	0.0332	0.0352	0.0350	0.0378	0.0402	0.0410
3.9	0.0158	0.0171	0.0190	0.0212	0.0273	0.0296	0.0322	0.0341	0.0338	0.0365	0.0387	0.0393
4.0	0.0154	0.0166	0.0185	0.0206	0.0265	0.0287	0.0312	0.0330	0.0326	0.0352	0.0373	0.0378
4.1	0.0150	0.0162	0.0180	0.0201	0.0258	0.0279	0.0303	0.0320	0.0315	0.0340	0.0359	0.0364
4.2	0.0146	0.0158	0.0176	0.0195	0.0250	0.0271	0.0294	0.0310	0.0305	0.0328	0.0347	0.0351
4.3	0.0142	0.0154	0.0172	0.0191	0.0243	0.0264	0.0285	0.0301	0.0295	0.0318	0.0335	0.0338
4.4	0.0139	0.0151	0.0167	0.0186	0.0237	0.0256	0.0277	0.0292	0.0285	0.0307	0.0324	0.0326
4.5	0.0136	0.0147	0.0163	0.0181	0.0231	0.0250	0.0270	0.0283	0.0276	0.0298	0.0313	0.0315
4.6	0.0133	0.0144	0.0160	0.0177	0.0225	0.0243	0.0262	0.0275	0.0268	0.0288	0.0303	0.0304
4.7	0.0130	0.0141	0.0156	0.0173	0.0219	0.0237	0.0255	0.0268	0.0260	0.0279	0.0293	0.0294
4.8	0.0127	0.0138	0.0153	0.0169	0.0213	0.0231	0.0249	0.0261	0.0252	0.0271	0.0284	0.0284
4.9	0.0124	0.0135	0.0149	0.0165	0.0208	0.0225	0.0243	0.0254	0.0245	0.0263	0.0275	0.0275
5.0	0.0122	0.0132	0.0146	0.0161	0.0203	0.0220	0.0236	0.0247	0.0238	0.0255	0.0267	0.0266

Table B.1 (continued)

r	$\alpha = 1$				$\alpha = 1.25$				$\alpha = 1.5$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	1.0916	0.9756	0.8315	0.6309	1.0308	0.9148	0.7724	0.5897	0.9581	0.8522	0.7266	0.5724
0.2	0.7783	0.7323	0.6823	0.6127	0.8042	0.7454	0.6754	0.5804	0.8020	0.7381	0.6629	0.5661
0.3	0.6110	0.5948	0.5862	0.5845	0.6638	0.6349	0.6047	0.5654	0.6906	0.6528	0.6109	0.5559
0.4	0.5021	0.5010	0.5134	0.5492	0.5635	0.5524	0.5466	0.5453	0.6036	0.5835	0.5648	0.5418
0.5	0.4242	0.4313	0.4546	0.5096	0.4868	0.4868	0.4965	0.5211	0.5326	0.5248	0.5227	0.5245
0.6	0.3652	0.3769	0.4055	0.4684	0.4259	0.4328	0.4522	0.4936	0.4732	0.4741	0.4838	0.5042
0.7	0.3190	0.3331	0.3637	0.4275	0.3760	0.3874	0.4126	0.4640	0.4227	0.4297	0.4475	0.4815
0.8	0.2817	0.2969	0.3276	0.3883	0.3345	0.3485	0.3769	0.4330	0.3792	0.3904	0.4137	0.4568
0.9	0.2510	0.2666	0.2963	0.3518	0.2995	0.3149	0.3446	0.4017	0.3415	0.3554	0.3821	0.4309
1.0	0.2254	0.2408	0.2689	0.3184	0.2695	0.2856	0.3154	0.3707	0.3084	0.3241	0.3526	0.4040
1.1	0.2036	0.2186	0.2448	0.2881	0.2436	0.2599	0.2888	0.3406	0.2794	0.2960	0.3251	0.3768
1.2	0.1849	0.1994	0.2235	0.2609	0.2211	0.2372	0.2647	0.3119	0.2537	0.2707	0.2995	0.3497
1.3	0.1688	0.1826	0.2046	0.2366	0.2013	0.2170	0.2428	0.2848	0.2308	0.2478	0.2758	0.3230
1.4	0.1547	0.1678	0.1879	0.2150	0.1839	0.1990	0.2229	0.2596	0.2105	0.2272	0.2537	0.2971
1.5	0.1424	0.1547	0.1729	0.1958	0.1686	0.1829	0.2049	0.2364	0.1923	0.2084	0.2333	0.2723
1.6	0.1314	0.1430	0.1595	0.1788	0.1549	0.1684	0.1884	0.2151	0.1760	0.1914	0.2144	0.2487
1.7	0.1217	0.1326	0.1474	0.1636	0.1426	0.1554	0.1735	0.1957	0.1613	0.1760	0.1970	0.2265
1.8	0.1131	0.1232	0.1366	0.1501	0.1317	0.1436	0.1599	0.1782	0.1481	0.1619	0.1810	0.2059
1.9	0.1053	0.1148	0.1268	0.1380	0.1218	0.1330	0.1475	0.1623	0.1362	0.1471	0.1663	0.1867
2.0	0.0983	0.1072	0.1180	0.1272	0.1129	0.1233	0.1363	0.1480	0.1254	0.1375	0.1527	0.1691
2.1	0.0920	0.1002	0.1100	0.1176	0.1049	0.1145	0.1260	0.1352	0.1157	0.1268	0.1403	0.1530
2.2	0.0862	0.0940	0.1027	0.1089	0.0976	0.1065	0.1167	0.1236	0.1068	0.1171	0.1289	0.1384
2.3	0.0810	0.0882	0.0961	0.1011	0.0910	0.0993	0.1082	0.1133	0.0987	0.1082	0.1184	0.1251
2.4	0.0762	0.0830	0.0901	0.0941	0.0849	0.0926	0.1004	0.1040	0.0914	0.1001	0.1089	0.1131
2.5	0.0718	0.0782	0.0845	0.0877	0.0794	0.0865	0.0933	0.0956	0.0847	0.0927	0.1002	0.1023
2.6	0.0678	0.0738	0.0795	0.0820	0.0744	0.0809	0.0869	0.0880	0.0786	0.0859	0.0922	0.0927
2.7	0.0641	0.0697	0.0748	0.0767	0.0697	0.0758	0.0809	0.0813	0.0730	0.0796	0.0849	0.0840
2.8	0.0607	0.0659	0.0706	0.0719	0.0655	0.0711	0.0755	0.0751	0.0679	0.0739	0.0782	0.0762
2.9	0.0576	0.0625	0.0666	0.0676	0.0616	0.0667	0.0705	0.0696	0.0632	0.0687	0.0721	0.0693
3.0	0.0547	0.0593	0.0630	0.0636	0.0579	0.0627	0.0660	0.0646	0.0589	0.0639	0.0666	0.0631
3.1	0.0520	0.0563	0.0597	0.0599	0.0546	0.0590	0.0618	0.0601	0.0549	0.0594	0.0615	0.0575
3.2	0.0495	0.0535	0.0566	0.0566	0.0515	0.0556	0.0579	0.0560	0.0513	0.0554	0.0569	0.0525
3.3	0.0471	0.0509	0.0537	0.0535	0.0487	0.0524	0.0544	0.0522	0.0479	0.0516	0.0526	0.0481
3.4	0.0450	0.0485	0.0510	0.0506	0.0460	0.0495	0.0511	0.0488	0.0449	0.0482	0.0488	0.0441
3.5	0.0429	0.0463	0.0485	0.0480	0.0436	0.0468	0.0481	0.0457	0.0420	0.0450	0.0452	0.0405
3.6	0.0410	0.0442	0.0462	0.0455	0.0413	0.0443	0.0453	0.0428	0.0394	0.0421	0.0420	0.0373
3.7	0.0393	0.0423	0.0440	0.0433	0.0392	0.0419	0.0427	0.0402	0.0369	0.0394	0.0390	0.0344
3.8	0.0376	0.0404	0.0420	0.0412	0.0372	0.0397	0.0404	0.0378	0.0347	0.0369	0.0363	0.0318
3.9	0.0360	0.0387	0.0401	0.0392	0.0353	0.0377	0.0382	0.0356	0.0326	0.0346	0.0339	0.0295
4.0	0.0345	0.0371	0.0384	0.0374	0.0336	0.0358	0.0361	0.0336	0.0307	0.0324	0.0316	0.0274
4.1	0.0332	0.0356	0.0367	0.0357	0.0320	0.0340	0.0342	0.0318	0.0289	0.0304	0.0295	0.0255
4.2	0.0319	0.0341	0.0352	0.0341	0.0305	0.0323	0.0324	0.0300	0.0272	0.0286	0.0276	0.0237
4.3	0.0306	0.0328	0.0337	0.0326	0.0291	0.0308	0.0308	0.0284	0.0257	0.0269	0.0258	0.0221
4.4	0.0295	0.0315	0.0323	0.0312	0.0277	0.0293	0.0292	0.0269	0.0242	0.0253	0.0242	0.0207
4.5	0.0284	0.0303	0.0310	0.0299	0.0265	0.0280	0.0278	0.0256	0.0229	0.0239	0.0227	0.0194
4.6	0.0273	0.0292	0.0298	0.0287	0.0253	0.0267	0.0265	0.0243	0.0217	0.0225	0.0213	0.0182
4.7	0.0263	0.0281	0.0287	0.0275	0.0242	0.0255	0.0252	0.0231	0.0205	0.0213	0.0200	0.0170
4.8	0.0254	0.0271	0.0276	0.0264	0.0232	0.0244	0.0240	0.0220	0.0194	0.0201	0.0189	0.0160
4.9	0.0245	0.0261	0.0266	0.0254	0.0222	0.0233	0.0229	0.0210	0.0184	0.0190	0.0178	0.0151
5.0	0.0237	0.0252	0.0256	0.0245	0.0213	0.0223	0.0219	0.0200	0.0174	0.0180	0.0168	0.0142

Table B.1 (continued)

r	$\alpha = 1.75$				$\alpha = 2$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	0.8927	0.8016	0.6929	0.5654	0.8398	0.7610	0.6687	0.5628
0.2	0.7870	0.7244	0.6504	0.5604	0.7679	0.7086	0.6398	0.5586
0.3	0.7010	0.6592	0.6115	0.5523	0.7020	0.6591	0.6102	0.5516
0.4	0.6279	0.6019	0.5745	0.5412	0.6415	0.6122	0.5803	0.5421
0.5	0.5645	0.5507	0.5391	0.5272	0.5861	0.5679	0.5503	0.5300
0.6	0.5090	0.5044	0.5050	0.5106	0.5354	0.5262	0.5203	0.5156
0.7	0.4599	0.4622	0.4722	0.4918	0.4889	0.4870	0.4906	0.4991
0.8	0.4162	0.4238	0.4407	0.4711	0.4463	0.4502	0.4613	0.4808
0.9	0.3773	0.3886	0.4105	0.4487	0.4073	0.4156	0.4326	0.4608
1.0	0.3424	0.3564	0.3816	0.4251	0.3716	0.3833	0.4046	0.4394
1.1	0.3112	0.3269	0.3541	0.4006	0.3390	0.3532	0.3775	0.4169
1.2	0.2830	0.2998	0.3280	0.3756	0.3091	0.3250	0.3512	0.3936
1.3	0.2577	0.2750	0.3032	0.3503	0.2818	0.2988	0.3260	0.3698
1.4	0.2348	0.2522	0.2799	0.3252	0.2569	0.2744	0.3019	0.3456
1.5	0.2141	0.2313	0.2579	0.3004	0.2341	0.2517	0.2788	0.3215
1.6	0.1954	0.2121	0.2373	0.2762	0.2132	0.2307	0.2569	0.2975
1.7	0.1784	0.1945	0.2180	0.2529	0.1942	0.2113	0.2362	0.2739
1.8	0.1631	0.1783	0.2000	0.2306	0.1768	0.1933	0.2166	0.2510
1.9	0.1491	0.1635	0.1832	0.2094	0.1610	0.1766	0.1982	0.2288
2.0	0.1365	0.1499	0.1676	0.1895	0.1465	0.1613	0.1809	0.2076
2.1	0.1250	0.1375	0.1532	0.1709	0.1333	0.1471	0.1648	0.1873
2.2	0.1145	0.1261	0.1399	0.1536	0.1213	0.1341	0.1498	0.1682
2.3	0.1050	0.1156	0.1277	0.1377	0.1103	0.1221	0.1359	0.1503
2.4	0.0963	0.1060	0.1164	0.1231	0.1003	0.1111	0.1230	0.1337
2.5	0.0884	0.0973	0.1060	0.1099	0.0912	0.1010	0.1111	0.1183
2.6	0.0812	0.0892	0.0965	0.0978	0.0829	0.0917	0.1001	0.1041
2.7	0.0747	0.0819	0.0878	0.0870	0.0753	0.0832	0.0900	0.0912
2.8	0.0687	0.0751	0.0798	0.0773	0.0684	0.0755	0.0808	0.0795
2.9	0.0632	0.0690	0.0726	0.0686	0.0622	0.0684	0.0724	0.0689
3.0	0.0582	0.0633	0.0660	0.0608	0.0565	0.0619	0.0647	0.0595
3.1	0.0536	0.0582	0.0600	0.0540	0.0513	0.0560	0.0577	0.0511
3.2	0.0494	0.0535	0.0545	0.0479	0.0465	0.0506	0.0514	0.0436
3.3	0.0456	0.0491	0.0495	0.0425	0.0422	0.0457	0.0457	0.0371
3.4	0.0421	0.0452	0.0450	0.0378	0.0383	0.0413	0.0405	0.0314
3.5	0.0389	0.0416	0.0410	0.0337	0.0348	0.0372	0.0358	0.0264
3.6	0.0360	0.0383	0.0373	0.0300	0.0315	0.0336	0.0317	0.0221
3.7	0.0333	0.0352	0.0339	0.0268	0.0286	0.0302	0.0279	0.0184
3.8	0.0308	0.0325	0.0309	0.0240	0.0259	0.0272	0.0246	0.0153
3.9	0.0285	0.0299	0.0282	0.0216	0.0235	0.0245	0.0216	0.0126
4.0	0.0264	0.0276	0.0257	0.0194	0.0213	0.0220	0.0190	0.0103
4.1	0.0245	0.0255	0.0235	0.0175	0.0193	0.0198	0.0166	0.0084
4.2	0.0228	0.0235	0.0214	0.0158	0.0175	0.0177	0.0145	0.0069
4.3	0.0211	0.0218	0.0196	0.0144	0.0159	0.0159	0.0126	0.0055
4.4	0.0196	0.0201	0.0180	0.0131	0.0144	0.0143	0.0110	0.0045
4.5	0.0183	0.0186	0.0165	0.0119	0.0130	0.0128	0.0096	0.0036
4.6	0.0170	0.0173	0.0151	0.0109	0.0118	0.0114	0.0083	0.0028
4.7	0.0158	0.0160	0.0139	0.0100	0.0107	0.0102	0.0072	0.0023
4.8	0.0148	0.0148	0.0128	0.0092	0.0097	0.0091	0.0062	0.0018
4.9	0.0138	0.0138	0.0118	0.0085	0.0087	0.0082	0.0054	0.0014
5.0	0.0129	0.0128	0.0109	0.0079	0.0079	0.0073	0.0046	0.0011

Table B.2 Two-dimensional FSD $2\pi r\Psi_2^{(\alpha,\nu)}(r)$

r	$\alpha = 0.25$				$\alpha = 0.5$				$\alpha = 0.75$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	0.5558	0.5593	0.5996	0.7316	0.8119	0.7443	0.6577	0.5497	0.7916	0.6640	0.4858	0.1882
0.2	0.2949	0.3034	0.3347	0.4124	0.4927	0.4765	0.4727	0.5253	0.5694	0.5129	0.4394	0.3189
0.3	0.2022	0.2103	0.2347	0.2884	0.3600	0.3584	0.3757	0.4506	0.4537	0.4274	0.4022	0.3842
0.4	0.1542	0.1614	0.1813	0.2218	0.2850	0.2892	0.3127	0.3843	0.3790	0.3684	0.3685	0.4035
0.5	0.1246	0.1312	0.1479	0.1801	0.2362	0.2429	0.2676	0.3310	0.3256	0.3240	0.3378	0.3967
0.6	0.1046	0.1106	0.1250	0.1515	0.2017	0.2095	0.2336	0.2887	0.2852	0.2889	0.3100	0.3773
0.7	0.0902	0.0956	0.1082	0.1306	0.1759	0.1842	0.2069	0.2547	0.2533	0.2603	0.2850	0.3525
0.8	0.0793	0.0842	0.0954	0.1147	0.1559	0.1642	0.1855	0.2270	0.2275	0.2364	0.2626	0.3265
0.9	0.0707	0.0753	0.0853	0.1022	0.1399	0.1481	0.1678	0.2041	0.2061	0.2161	0.2425	0.3012
1.0	0.0638	0.0680	0.0771	0.0921	0.1267	0.1347	0.1530	0.1849	0.1880	0.1986	0.2244	0.2774
1.1	0.0581	0.0621	0.0703	0.0838	0.1158	0.1235	0.1404	0.1687	0.1725	0.1834	0.2082	0.2557
1.2	0.0533	0.0570	0.0647	0.0768	0.1065	0.1139	0.1296	0.1548	0.1592	0.1701	0.1936	0.2359
1.3	0.0493	0.0528	0.0598	0.0708	0.0985	0.1057	0.1202	0.1427	0.1475	0.1583	0.1804	0.2180
1.4	0.0458	0.0491	0.0557	0.0657	0.0916	0.0985	0.1119	0.1322	0.1372	0.1478	0.1686	0.2019
1.5	0.0428	0.0459	0.0520	0.0613	0.0855	0.0921	0.1047	0.1230	0.1281	0.1384	0.1578	0.1874
1.6	0.0401	0.0431	0.0488	0.0574	0.0802	0.0865	0.0982	0.1148	0.1200	0.1299	0.1481	0.1744
1.7	0.0378	0.0406	0.0460	0.0539	0.0754	0.0814	0.0924	0.1076	0.1127	0.1223	0.1392	0.1626
1.8	0.0357	0.0384	0.0435	0.0509	0.0712	0.0769	0.0872	0.1011	0.1061	0.1154	0.1311	0.1519
1.9	0.0338	0.0364	0.0412	0.0481	0.0673	0.0729	0.0825	0.0952	0.1002	0.1090	0.1237	0.1423
2.0	0.0321	0.0346	0.0391	0.0456	0.0639	0.0692	0.0782	0.0899	0.0948	0.1033	0.1169	0.1335
2.1	0.0306	0.0329	0.0373	0.0434	0.0607	0.0658	0.0743	0.0851	0.0898	0.0980	0.1106	0.1255
2.2	0.0292	0.0314	0.0356	0.0413	0.0578	0.0627	0.0707	0.0807	0.0853	0.0931	0.1049	0.1182
2.3	0.0279	0.0301	0.0340	0.0395	0.0552	0.0599	0.0675	0.0767	0.0811	0.0886	0.0996	0.1116
2.4	0.0268	0.0288	0.0326	0.0378	0.0528	0.0573	0.0644	0.0731	0.0773	0.0844	0.0947	0.1055
2.5	0.0257	0.0277	0.0313	0.0362	0.0505	0.0549	0.0617	0.0697	0.0737	0.0806	0.0902	0.0999
2.6	0.0247	0.0266	0.0301	0.0347	0.0484	0.0526	0.0591	0.0666	0.0704	0.0770	0.0860	0.0947
2.7	0.0238	0.0256	0.0290	0.0334	0.0465	0.0505	0.0567	0.0637	0.0674	0.0736	0.0820	0.0899
2.8	0.0229	0.0247	0.0279	0.0321	0.0447	0.0486	0.0545	0.0610	0.0645	0.0705	0.0784	0.0855
2.9	0.0221	0.0239	0.0269	0.0310	0.0431	0.0468	0.0524	0.0585	0.0619	0.0676	0.0750	0.0815
3.0	0.0214	0.0231	0.0260	0.0299	0.0415	0.0451	0.0504	0.0562	0.0594	0.0649	0.0718	0.0777
3.1	0.0207	0.0223	0.0252	0.0289	0.0400	0.0435	0.0486	0.0540	0.0570	0.0623	0.0688	0.0741
3.2	0.0200	0.0216	0.0244	0.0279	0.0387	0.0421	0.0469	0.0520	0.0549	0.0599	0.0661	0.0709
3.3	0.0194	0.0209	0.0236	0.0270	0.0374	0.0407	0.0453	0.0501	0.0528	0.0577	0.0634	0.0678
3.4	0.0188	0.0203	0.0229	0.0262	0.0362	0.0393	0.0438	0.0483	0.0509	0.0556	0.0610	0.0650
3.5	0.0183	0.0197	0.0222	0.0254	0.0350	0.0381	0.0423	0.0466	0.0491	0.0536	0.0587	0.0623
3.6	0.0177	0.0192	0.0216	0.0246	0.0339	0.0369	0.0410	0.0451	0.0474	0.0517	0.0565	0.0598
3.7	0.0172	0.0186	0.0210	0.0239	0.0329	0.0358	0.0397	0.0436	0.0457	0.0499	0.0544	0.0574
3.8	0.0168	0.0181	0.0204	0.0232	0.0320	0.0348	0.0385	0.0422	0.0442	0.0482	0.0525	0.0552
3.9	0.0163	0.0177	0.0199	0.0226	0.0310	0.0338	0.0374	0.0408	0.0428	0.0466	0.0507	0.0531
4.0	0.0159	0.0172	0.0194	0.0220	0.0302	0.0328	0.0363	0.0396	0.0414	0.0451	0.0489	0.0512
4.1	0.0155	0.0168	0.0189	0.0214	0.0293	0.0319	0.0353	0.0384	0.0401	0.0436	0.0473	0.0493
4.2	0.0151	0.0164	0.0184	0.0209	0.0285	0.0311	0.0343	0.0372	0.0388	0.0423	0.0457	0.0476
4.3	0.0148	0.0160	0.0180	0.0204	0.0278	0.0302	0.0333	0.0362	0.0377	0.0410	0.0442	0.0459
4.4	0.0144	0.0156	0.0176	0.0199	0.0271	0.0295	0.0324	0.0351	0.0365	0.0397	0.0428	0.0443
4.5	0.0141	0.0153	0.0172	0.0194	0.0264	0.0287	0.0316	0.0342	0.0355	0.0385	0.0415	0.0429
4.6	0.0138	0.0149	0.0168	0.0189	0.0257	0.0280	0.0308	0.0332	0.0344	0.0374	0.0402	0.0415
4.7	0.0135	0.0146	0.0164	0.0185	0.0251	0.0273	0.0300	0.0323	0.0335	0.0363	0.0390	0.0401
4.8	0.0132	0.0143	0.0160	0.0181	0.0245	0.0267	0.0293	0.0315	0.0325	0.0353	0.0378	0.0389
4.9	0.0129	0.0140	0.0157	0.0177	0.0239	0.0260	0.0285	0.0307	0.0316	0.0343	0.0367	0.0376
5.0	0.0127	0.0137	0.0154	0.0173	0.0234	0.0254	0.0279	0.0299	0.0308	0.0334	0.0357	0.0365

Table B.2 (continued)

r	$\alpha = 1$				$\alpha = 1.25$				$\alpha = 1.5$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	0.6501	0.5176	0.3414	0.0985	0.4943	0.3836	0.2438	0.0710	0.3668	0.2824	0.1794	0.0593
0.2	0.5562	0.4746	0.3643	0.1885	0.4976	0.4116	0.2982	0.1391	0.4273	0.3478	0.2470	0.1170
0.3	0.4868	0.4363	0.3712	0.2636	0.4768	0.4138	0.3297	0.2018	0.4453	0.3781	0.2918	0.1719
0.4	0.4321	0.4021	0.3687	0.3201	0.4497	0.4055	0.3479	0.2570	0.4446	0.3912	0.3229	0.2226
0.5	0.3873	0.3714	0.3602	0.3578	0.4213	0.3921	0.3568	0.3031	0.4345	0.3941	0.3441	0.2681
0.6	0.3499	0.3436	0.3479	0.3783	0.3935	0.3760	0.3591	0.3393	0.4192	0.3905	0.3575	0.3075
0.7	0.3180	0.3185	0.3333	0.3849	0.3671	0.3586	0.3563	0.3655	0.4011	0.3825	0.3647	0.3402
0.8	0.2905	0.2958	0.3175	0.3810	0.3422	0.3408	0.3498	0.3821	0.3817	0.3716	0.3668	0.3660
0.9	0.2666	0.2751	0.3011	0.3697	0.3190	0.3229	0.3405	0.3900	0.3618	0.3587	0.3649	0.3848
1.0	0.2456	0.2563	0.2847	0.3537	0.2975	0.3054	0.3291	0.3903	0.3419	0.3446	0.3596	0.3969
1.1	0.2270	0.2392	0.2686	0.3349	0.2776	0.2885	0.3164	0.3843	0.3225	0.3298	0.3517	0.4026
1.2	0.2105	0.2235	0.2530	0.3150	0.2593	0.2722	0.3028	0.3734	0.3037	0.3145	0.3417	0.4025
1.3	0.1957	0.2092	0.2381	0.2948	0.2423	0.2566	0.2886	0.3588	0.2856	0.2992	0.3301	0.3973
1.4	0.1824	0.1961	0.2239	0.2750	0.2266	0.2419	0.2743	0.3417	0.2684	0.2840	0.3173	0.3878
1.5	0.1704	0.1840	0.2105	0.2561	0.2121	0.2279	0.2600	0.3231	0.2521	0.2691	0.3037	0.3748
1.6	0.1596	0.1730	0.1980	0.2383	0.1988	0.2147	0.2460	0.3037	0.2367	0.2546	0.2895	0.3590
1.7	0.1497	0.1628	0.1863	0.2217	0.1864	0.2023	0.2323	0.2842	0.2221	0.2405	0.2751	0.3412
1.8	0.1407	0.1534	0.1753	0.2063	0.1749	0.1907	0.2190	0.2650	0.2084	0.2269	0.2607	0.3220
1.9	0.1325	0.1447	0.1651	0.1921	0.1644	0.1797	0.2064	0.2465	0.1956	0.2140	0.2464	0.3021
2.0	0.1249	0.1367	0.1555	0.1790	0.1546	0.1694	0.1943	0.2289	0.1835	0.2016	0.2323	0.2820
2.1	0.1180	0.1292	0.1467	0.1670	0.1455	0.1598	0.1828	0.2123	0.1722	0.1898	0.2186	0.2620
2.2	0.1116	0.1224	0.1384	0.1560	0.1370	0.1507	0.1719	0.1968	0.1617	0.1786	0.2053	0.2425
2.3	0.1057	0.1160	0.1308	0.1459	0.1292	0.1423	0.1617	0.1824	0.1518	0.1680	0.1926	0.2238
2.4	0.1002	0.1100	0.1236	0.1366	0.1219	0.1343	0.1520	0.1690	0.1425	0.1580	0.1804	0.2060
2.5	0.0952	0.1045	0.1170	0.1281	0.1151	0.1269	0.1430	0.1567	0.1339	0.1486	0.1688	0.1892
2.6	0.0905	0.0993	0.1108	0.1203	0.1088	0.1200	0.1345	0.1455	0.1258	0.1397	0.1578	0.1736
2.7	0.0861	0.0945	0.1050	0.1132	0.1030	0.1135	0.1265	0.1351	0.1183	0.1313	0.1475	0.1591
2.8	0.0820	0.0900	0.0997	0.1066	0.0975	0.1074	0.1191	0.1256	0.1112	0.1234	0.1377	0.1457
2.9	0.0782	0.0858	0.0947	0.1005	0.0924	0.1017	0.1122	0.1169	0.1047	0.1160	0.1285	0.1335
3.0	0.0747	0.0819	0.0900	0.0949	0.0876	0.0963	0.1057	0.1090	0.0985	0.1091	0.1200	0.1223
3.1	0.0713	0.0782	0.0856	0.0897	0.0831	0.0913	0.0996	0.1017	0.0927	0.1026	0.1119	0.1121
3.2	0.0682	0.0747	0.0816	0.0849	0.0790	0.0866	0.0940	0.0950	0.0874	0.0965	0.1044	0.1028
3.3	0.0653	0.0715	0.0777	0.0805	0.0750	0.0822	0.0887	0.0889	0.0823	0.0907	0.0975	0.0944
3.4	0.0626	0.0684	0.0742	0.0764	0.0714	0.0781	0.0838	0.0833	0.0776	0.0854	0.0910	0.0868
3.5	0.0600	0.0655	0.0708	0.0726	0.0679	0.0742	0.0793	0.0781	0.0733	0.0804	0.0849	0.0799
3.6	0.0576	0.0628	0.0677	0.0690	0.0647	0.0706	0.0750	0.0734	0.0691	0.0757	0.0793	0.0736
3.7	0.0553	0.0603	0.0647	0.0657	0.0617	0.0672	0.0710	0.0690	0.0653	0.0713	0.0741	0.0680
3.8	0.0531	0.0579	0.0619	0.0626	0.0588	0.0640	0.0673	0.0650	0.0617	0.0672	0.0693	0.0628
3.9	0.0511	0.0556	0.0593	0.0598	0.0561	0.0609	0.0638	0.0613	0.0583	0.0634	0.0648	0.0582
4.0	0.0491	0.0534	0.0569	0.0571	0.0536	0.0581	0.0605	0.0578	0.0552	0.0598	0.0607	0.0540
4.1	0.0473	0.0514	0.0546	0.0546	0.0512	0.0554	0.0575	0.0547	0.0522	0.0564	0.0569	0.0501
4.2	0.0456	0.0495	0.0524	0.0522	0.0490	0.0529	0.0547	0.0517	0.0494	0.0533	0.0533	0.0466
4.3	0.0439	0.0476	0.0503	0.0500	0.0469	0.0505	0.0520	0.0490	0.0468	0.0503	0.0500	0.0435
4.4	0.0424	0.0459	0.0484	0.0479	0.0449	0.0483	0.0495	0.0465	0.0444	0.0476	0.0469	0.0406
4.5	0.0409	0.0443	0.0465	0.0459	0.0430	0.0462	0.0472	0.0441	0.0421	0.0450	0.0441	0.0379
4.6	0.0395	0.0427	0.0448	0.0441	0.0412	0.0442	0.0450	0.0419	0.0400	0.0426	0.0415	0.0355
4.7	0.0382	0.0412	0.0431	0.0424	0.0395	0.0423	0.0429	0.0399	0.0380	0.0403	0.0390	0.0333
4.8	0.0369	0.0398	0.0416	0.0407	0.0379	0.0405	0.0410	0.0380	0.0361	0.0382	0.0368	0.0313
4.9	0.0357	0.0385	0.0401	0.0392	0.0364	0.0388	0.0391	0.0362	0.0343	0.0362	0.0347	0.0294
5.0	0.0345	0.0372	0.0387	0.0377	0.0350	0.0373	0.0374	0.0345	0.0326	0.0343	0.0327	0.0277

Table B.2 (continued)

r	$\alpha = 1.75$				$\alpha = 2$			
	0.25	0.5	0.75	1	0.25	0.5	0.75	1
0.1	0.2733	0.2124	0.1375	0.0533	0.2091	0.1644	0.1102	0.0499
0.2	0.3619	0.2941	0.2090	0.1056	0.3073	0.2506	0.1810	0.0990
0.3	0.4056	0.3417	0.2607	0.1560	0.3664	0.3081	0.2360	0.1467
0.4	0.4260	0.3703	0.2996	0.2036	0.4021	0.3476	0.2797	0.1922
0.5	0.4325	0.3866	0.3289	0.2475	0.4222	0.3742	0.3144	0.2349
0.6	0.4301	0.3942	0.3503	0.2871	0.4313	0.3911	0.3417	0.2742
0.7	0.4218	0.3954	0.3652	0.3219	0.4325	0.4004	0.3624	0.3097
0.8	0.4095	0.3919	0.3745	0.3514	0.4278	0.4039	0.3774	0.3409
0.9	0.3946	0.3848	0.3790	0.3755	0.4188	0.4026	0.3872	0.3675
1.0	0.3781	0.3750	0.3795	0.3939	0.4067	0.3976	0.3927	0.3894
1.1	0.3607	0.3633	0.3765	0.4067	0.3923	0.3897	0.3941	0.4064
1.2	0.3429	0.3502	0.3705	0.4141	0.3765	0.3794	0.3921	0.4186
1.3	0.3249	0.3361	0.3622	0.4165	0.3596	0.3673	0.3871	0.4260
1.4	0.3072	0.3214	0.3519	0.4141	0.3422	0.3539	0.3795	0.4288
1.5	0.2898	0.3064	0.3400	0.4075	0.3245	0.3395	0.3697	0.4273
1.6	0.2729	0.2912	0.3268	0.3971	0.3068	0.3244	0.3581	0.4218
1.7	0.2567	0.2761	0.3128	0.3836	0.2894	0.3089	0.3450	0.4127
1.8	0.2411	0.2612	0.2981	0.3676	0.2723	0.2932	0.3308	0.4004
1.9	0.2262	0.2467	0.2830	0.3494	0.2557	0.2774	0.3157	0.3853
2.0	0.2120	0.2325	0.2678	0.3298	0.2396	0.2619	0.2999	0.3679
2.1	0.1986	0.2188	0.2526	0.3093	0.2242	0.2465	0.2838	0.3486
2.2	0.1859	0.2056	0.2375	0.2882	0.2094	0.2315	0.2675	0.3280
2.3	0.1739	0.1929	0.2227	0.2670	0.1954	0.2170	0.2512	0.3064
2.4	0.1626	0.1808	0.2083	0.2461	0.1820	0.2029	0.2350	0.2843
2.5	0.1519	0.1693	0.1944	0.2257	0.1694	0.1894	0.2191	0.2620
2.6	0.1419	0.1584	0.1810	0.2060	0.1574	0.1765	0.2036	0.2399
2.7	0.1326	0.1480	0.1683	0.1874	0.1461	0.1641	0.1886	0.2182
2.8	0.1238	0.1382	0.1561	0.1698	0.1355	0.1524	0.1742	0.1972
2.9	0.1156	0.1290	0.1446	0.1534	0.1256	0.1412	0.1603	0.1771
3.0	0.1079	0.1203	0.1337	0.1382	0.1162	0.1307	0.1472	0.1581
3.1	0.1007	0.1122	0.1235	0.1242	0.1075	0.1208	0.1347	0.1403
3.2	0.0940	0.1045	0.1139	0.1115	0.0994	0.1115	0.1230	0.1237
3.3	0.0878	0.0973	0.1050	0.0999	0.0918	0.1027	0.1120	0.1084
3.4	0.0820	0.0906	0.0967	0.0894	0.0847	0.0945	0.1017	0.0945
3.5	0.0765	0.0844	0.0890	0.0800	0.0781	0.0869	0.0921	0.0819
3.6	0.0714	0.0785	0.0818	0.0716	0.0720	0.0798	0.0832	0.0705
3.7	0.0667	0.0730	0.0752	0.0641	0.0663	0.0731	0.0750	0.0604
3.8	0.0623	0.0680	0.0691	0.0574	0.0610	0.0670	0.0674	0.0514
3.9	0.0582	0.0632	0.0634	0.0515	0.0561	0.0613	0.0604	0.0435
4.0	0.0544	0.0588	0.0583	0.0463	0.0516	0.0560	0.0541	0.0366
4.1	0.0508	0.0547	0.0535	0.0416	0.0474	0.0511	0.0483	0.0307
4.2	0.0475	0.0509	0.0492	0.0375	0.0436	0.0466	0.0430	0.0255
4.3	0.0444	0.0474	0.0452	0.0339	0.0400	0.0424	0.0382	0.0211
4.4	0.0416	0.0441	0.0415	0.0306	0.0367	0.0386	0.0339	0.0174
4.5	0.0389	0.0410	0.0382	0.0278	0.0337	0.0351	0.0300	0.0142
4.6	0.0364	0.0382	0.0351	0.0253	0.0309	0.0319	0.0265	0.0116
4.7	0.0341	0.0356	0.0323	0.0231	0.0283	0.0289	0.0233	0.0094
4.8	0.0319	0.0332	0.0298	0.0211	0.0259	0.0262	0.0205	0.0076
4.9	0.0299	0.0309	0.0275	0.0193	0.0237	0.0237	0.0180	0.0061
5.0	0.0280	0.0288	0.0253	0.0177	0.0217	0.0215	0.0158	0.0048

Appendix C
Fractional Operators: Symbols and Formulas

Table C.1 Fractional operators ($\mu,\ \nu > 0,\ n < \nu < n+1,\ -\infty < q < \infty$)

Name	Symbol	Formula
Riemann-Liouville integrals	${}_a\mathsf{D}_x^{-\mu} f(x),\ {}_a f^{(-\mu)}(x)$	$\dfrac{1}{\Gamma(\mu)} \displaystyle\int_a^x \frac{f(\xi)d\xi}{(x-\xi)^{1-\mu}},\ a < x$
	${}_x\mathsf{D}_b^{-\mu} f(x),\ f_b^{(-\mu)}(x)$	$\dfrac{1}{\Gamma(\mu)} \displaystyle\int_x^b \frac{f(\xi)d\xi}{(\xi-x)^{1-\mu}},\ x < b$
Riesz potential	${}_a\mathbb{D}_b^{-\mu} f(x),\ {}_a f_b^{(-\mu)}(x)$	$\dfrac{1}{2\Gamma(\mu)\cos(\mu\pi/2)} \displaystyle\int_{-\infty}^{\infty} \frac{f(\xi)d\xi}{\lvert x-\xi\rvert^{1-\mu}},\ \mu < 1$
	${}_{-\infty}\mathbb{D}_{\infty}^{-\mu} = (-_1)^{-\mu/2}$	
Conjugate Riesz potential	${}_a^{\circ}\mathbb{D}_b^{-\mu} f(x)$	$\dfrac{1}{2\Gamma(\mu)\sin(\mu\pi/2)} \displaystyle\int_{-\infty}^{\infty} \frac{\operatorname{sign}(x-\xi) f(\xi)d\xi}{\lvert x-\xi\rvert^{1-\mu}},\ \mu < 1$
d-Dimensional Riesz potential	$(-_d)^{-\mu/2} f(\mathbf{x})$	$\dfrac{\Gamma((d-\mu)/2)}{2^{\mu}\pi^{d/2}\Gamma(\mu/2)} \displaystyle\int_{\mathbb{R}^d} \frac{f(\mathbf{y})d\mathbf{y}}{\lvert\mathbf{x}-\mathbf{y}\rvert^{d-\mu}},\ \mu \neq d+2k, k = 0, 1, \ldots$
Feller potential	$\mathbb{D}_{u,v}^{-\mu} f(x)$	$\dfrac{A_{-\mu}^{-1/2}}{2\Gamma(\mu)} \displaystyle\int_{-\infty}^{\infty} \frac{u+v+(u-v)\operatorname{sign}(x-\xi)}{\lvert x-\xi\rvert^{1-\mu}} f(\xi)d\xi,\ \mu < 1$ $A_{-\mu} = (u+v)^2 \cos^2 \dfrac{\mu\pi}{2} + (u-v)^2 \sin^2 \dfrac{\mu\pi}{2}$

(continued)

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Table C.1 (continued)

Name	Symbol	Formula
Riemann-Liouville derivatives	${}_a\mathsf{D}_x^\nu f(x),\ {}_a f^{(\nu)}(x)$	$\frac{1}{\Gamma(n-\nu)}\left(\frac{d}{dx}\right)^n \int_a^x \frac{f(\xi)d\xi}{(x-\xi)^{\nu-n+1}},\ x>a$
	${}_x\mathsf{D}_b^\nu f(x),\ f_b^{(\nu)}(x)$	$\frac{(-1)^n}{\Gamma(n-\nu)}\left(\frac{d}{dx}\right)^n \int_x^b \frac{f(\xi)d\xi}{(\xi-x)^{\nu-n+1}},\ x<b$
Gerasimov-Caputo derivative	${}_a^\nu\mathsf{D}_x f(x),\ {}_a^\nu f(x)$	$\frac{1}{\Gamma(n-\nu)}\int_a^x \frac{f^{(n)}(\xi)d\xi}{(x-\xi)^{\nu-n+1}},\ x>a$
	${}_x^\nu\mathsf{D}_b f(x),\ {}^\nu f_b(x)$	$\frac{1}{\Gamma(n-\nu)}\int_x^b \frac{f^{(n)}(\xi)d\xi}{(\xi-x)^{\nu-n+1}},\ x<b$
Marchaud derivatives	${}^\nu\mathsf{D}_\pm f(x),\ {}^\nu f_\pm(x)$	$\frac{(\pm1)^{[\nu]}\{\nu\}}{\Gamma(1-\{\nu\})}\int_0^\infty \frac{f^{([\nu])}(x)-f^{([\nu])}(x\mp\xi)}{\xi^{1+\{\nu\}}}d\xi$
	$\mathsf{D}_\pm^\nu f(x),\ f_\pm^{(\nu)}(x)$	$\frac{1}{C_\nu}\int_0^\infty \sum_{k=0}^n (-1)^k\binom{n}{k} f(x\mp h\xi)\frac{d\xi}{\xi^{1+\nu}}$
		$C_\nu=\Gamma(-\nu)\sum_{k=0}^n (-1)^k\binom{n}{k}k^\nu$
Grünwald-Letnikov derivatives	$\mathsf{d}_\pm^q f(x),\ \mathrm{f}_\pm^{(q)}(x)$	$(\pm1)^q \lim_{h\to0}\frac{1}{h^q}\sum_{k=0}^\infty (-1)^k\binom{q}{k} f(x\mp kh)$
	${}_a\mathsf{d}_x^q f(x),\ {}_a\mathrm{f}^{(q)}(x)$	$\lim_{N\to\infty}\frac{1}{h_N^q}\sum_{k=0}^{N-1}(-1)^k\binom{q}{k} f(x-kh_N),\ h_N=\frac{x-a}{N}$
Riesz derivative	${}_a\mathbb{D}_b^\nu f(x),\ {}_a f_b^{(\nu)}(x)$	$-\frac{{}_a\mathbb{D}_x^\nu f(x)+{}_x\mathbb{D}_b^\nu f(x)}{2\cos(\nu\pi/2)}$
	$\mathbb{D}_x^\nu,\ \frac{\partial^\nu}{\partial\lvert x\rvert^\nu},\ (-\Delta_1)^{\nu/2}$	$-\frac{-{}_{-\infty}\mathbb{D}_x^\nu f(x)+{}_x\mathbb{D}_\infty^\nu f(x)}{2\cos(\nu\pi/2)}$
d-Dimensional Riesz derivative	$(-_d)^{\nu/2} f(\mathbf{x})$	$\frac{1}{K_d(\nu)}\int_{\mathbb{R}^d}\sum_{k=0}^n (-1)^k\binom{n}{k} f(\mathbf{x}-k\mathbf{h})\lvert\mathbf{y}\rvert^{-d-\nu}d\mathbf{y}$ $K_d(\nu)=\frac{\pi^{d/2}\Gamma(1-\nu/2)}{2^{\nu-1}\nu\Gamma((d+\nu)/2)}\sum_{k=0}^n (-1)^{k-1}\binom{n}{k}k^\nu$
Feller derivative	$\mathbb{D}_{u,\nu}^\nu f(x)$	$\frac{A_\nu^{-1/2}}{2\Gamma(-\nu)}\int_{-\infty}^\infty \frac{u+\nu+(u-\nu)\mathrm{sign}(x-\xi)}{\lvert x-\xi\rvert^{1+\nu}}[f(\xi)-$ $f(x)]d\xi$

Table C.2 Laplace transforms of fractional derivatives ($\nu > 0$, $n < \nu < n+1$)

Derivative	Transform
${}_0\mathsf{D}_x^{-\nu} f(x)$	$\lambda^{-\mu}\widehat{f}(\lambda)$
${}_0\mathsf{D}_x^{\nu} f(x)$	$\lambda^{\nu}\widehat{f}(\lambda) - \sum_{k=0}^{n-1}\lambda^k\, {}_0f_{0+}^{(\nu-k-1)}, \quad n-1<\nu<n$
${}_0^{\nu}\mathsf{D}_x f(x)$	$\lambda^{\nu}\widehat{f}(\lambda) - \sum_{k=0}^{n-1}\lambda^{\nu-k-1} f^{(k)}(0+),\ n-1<\nu\leqslant n$
$\mathsf{D}_+^{\nu} f(x),\ \mathsf{d}_+^{\nu} f(x)$	$\lambda^{\nu}\widehat{f}(\lambda)$

Table C.3 Mellin transforms of fractional derivatives ($\nu > 0$, $n < \nu < n+1$)

Derivative	Transform
${}_0\mathsf{D}_x^{-\nu} f(x)$	$\dfrac{\Gamma(1-s-\nu)}{\Gamma(1-s)}\overline{f}(s+\nu)$
${}_0\mathsf{D}_x^{\nu} f(x)$	$\dfrac{\Gamma(1-s+\nu)}{\Gamma(1-s)}\overline{f}(s-\nu) + \sum\limits_{k=0}^{n-1}\dfrac{\Gamma(1-s+k)}{\Gamma(1-s)}\left[\,{}_0f_x^{(\nu-k-1)}x^{s-k-1}\right]_0^{\infty}$
${}_0^{\nu}\mathsf{D}_x f(x)$	$\dfrac{\Gamma(1-s+\nu)}{\Gamma(1-s)}\overline{f}(s-\nu) + \sum\limits_{k=0}^{n-1}\dfrac{\Gamma(\nu-k-s)}{\Gamma(1-s)}\left[f^{(k)}(x)x^{s-\nu+k}\right]_0^{\infty}$
$\mathsf{D}_+^{\nu} f(x)$	$\dfrac{\Gamma(1-s+\nu)}{\Gamma(1-s)}\overline{f}(s-\nu)$
${}_0\mathsf{d}_x^{\nu} f(x)$	$\dfrac{\Gamma(1-s+\nu)}{\Gamma(1-s)}\overline{f}(s-\nu)$

Table C.4 Fourier transforms of fractional derivatives

Derivative	Fourier transform
${}_{-\infty}\mathsf{D}_x^{-\nu} f(x)$	$(-ik)^{-\nu}\widetilde{f}(k) = \lvert k\rvert^{-\nu}\exp\{i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad 0<\nu<1$
${}_0\mathsf{D}_x^{-\nu} f(x)$	$(-ik)^{-\nu}\widetilde{f}(k) = \lvert k\rvert^{-\nu}\exp\{i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad 0<\nu<1$
${}_x\mathsf{D}_{\infty}^{-\nu} f(x)$	$(ik)^{-\nu}\widetilde{f}(k) = \lvert k\rvert^{-\nu}\exp\{-i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad 0<\nu<1$
$\mathsf{R}^{\nu} f(x)$	$\lvert k\rvert^{-\nu}\widetilde{f}(k)$
$\overset{\circ}{\mathsf{R}}{}^{\nu} f(x)$	$i\ \mathrm{sign}(k)\lvert k\rvert^{-\nu}\widetilde{f}(k)$
$\mathbb{D}_{u,v}^{-\nu} f(x)$	$A_{-\nu}^{-1/2}\cos(\nu\pi/2)[(u+v)+i(u-v)\tan(\nu\pi/2)\mathrm{sign}(k)]\lvert k\rvert^{-\nu}\widetilde{f}(k), \quad 0<\nu<1$
	$A_{-\nu} = \cos^2(\nu\pi/2)[(u+v)^2+(u-v)^2\tan^2(\nu\pi/2)]$
${}_{-\infty}\mathsf{D}_x^{\nu} f(x)$	$(-ik)^{\nu}\widetilde{f}(k) = \lvert k\rvert^{\nu}\exp\{-i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad \nu\geqslant 0$
${}_x\mathsf{D}_{\infty}^{\nu} f(x)$	$(ik)^{\nu}\widetilde{f}(k) = \lvert k\rvert^{\nu}\exp\{i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad \nu\geqslant 0$
$\mathsf{D}_+^{\nu} f(x)$	$(-ik)^{\nu}\widetilde{f}(k) = \lvert k\rvert^{\nu}\exp\{-i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad \nu\geqslant 0$
$\mathsf{D}_-^{\nu} f(x)$	$(ik)^{\nu}\widetilde{f}(k) = \lvert k\rvert^{\nu}\exp\{i\nu(\pi/2)\mathrm{sign}(k)\}\widetilde{f}(k), \quad \nu\geqslant 0$
$\Delta_h^{\nu} f(x)$	$(1-e^{ikh})^{\nu}\widetilde{f}(k)$
$\mathbb{D}^{\nu} f(x)$	$\lvert k\rvert^{\nu}\widetilde{f}(k)$
${}^{\circ}\mathbb{D}^{\nu} f(x)$	$-i\ \mathrm{sign}(k)\ \lvert k\rvert^{\nu}$
$\mathbb{D}_{u,v}^{\nu} f(x)$	$A^{-1/2}\cos(\nu\pi/2)[(u+v)-i(u-v)\tan(\nu\pi/2)\mathrm{sign}(k)]\lvert k\rvert^{\nu}\widetilde{f}(k), \quad 0<\nu<1$
	$A = \cos^2(\nu\pi/2)[(u+v)^2+(u-v)^2\tan^2(\nu\pi/2)]$

Table C.5 Riemann-Liouville fractional derivatives ${}_0f^{(\nu)}(x)$ of elementary functions ($x > 0$, $-\infty < \nu < \infty$)

$f(x)$	${}_0f^{(\nu)}(x) = {}_0\mathsf{D}_x^{\nu} f(x)$
$x^{\mu},\ \mu > -1$	$x^{\mu-\nu}\Gamma(\mu+1)/\Gamma(\mu+1-\nu)$
$x^{\nu}/(1-x)$	$(1-x)^{-\nu-1}\Gamma(\nu+1)$
$(a+bx)^{\nu-1}$	$\Gamma(\nu)\sin(\pi\nu)/(a+bx)(a/x)^{\nu}$
$(1-x)^{\nu-1}$	$x^{-\nu}/[(1-x)\Gamma(1-\nu)],\ \lvert\nu\rvert < 1$
$e^{\mu x}$	$E_x(-\nu,\mu)$
$xe^{\mu x}$	$xE_x(-\nu,\mu) + \nu E_x(1-\nu,\mu)$
$x^{\nu-1}\ln(x+a)$	$\Gamma(\nu)[1-(a/(a+x))^{\nu}]/x$
$\ln x$	$x^{-\nu}(\ln x + \psi(1) - \psi(1-\nu))/\Gamma(1-\nu)$
$x^{\mu}\ln x$	$x^{\mu-\nu}[\ln x + \psi(\mu+1) - \psi(\mu+1-\nu)]\Gamma(\mu+1)/\Gamma(\mu+1-\nu),\ \mu > -1$
$\sin\mu x$	$S_x(-\nu,\mu)$
$\cos\mu x$	$C_x(-\nu,\mu)$
$x\sin\mu x$	$xS_x(-\nu,\mu) + \nu S_x(1-\nu,\mu)$
$x\cos\mu x$	$xC_x(-\nu,\mu) + \nu C_x(1-\nu,\mu)$
$\sin^2\mu x$	$\mu S_x(1-\nu,2\mu)$
$\cos^2\mu x$	$(1/2)C_x(-\nu,2\mu) + x^{-\nu}/[2\Gamma(1-\nu)]$
$\sin\sqrt{x}$	$(\sqrt{\pi}/2)(2\sqrt{x})^{1/2-\nu}J_{1/2-\nu}(\sqrt{x})$
$\sinh\sqrt{x}$	$(\sqrt{\pi}/2)(2\sqrt{x})^{1/2-\nu}I_{1/2-\nu}(\sqrt{x})$
$(1/\sqrt{x})\cos\sqrt{x}$	$\sqrt{\pi}(2\sqrt{x})^{-1/2-\nu}J_{-1/2-\nu}(\sqrt{x})$
$(1/\sqrt{x})\cosh\sqrt{x}$	$\sqrt{\pi}(2\sqrt{x})^{-1/2-\nu}I_{-1/2-\nu}(\sqrt{x})$

Table C.6 Gerasimov-Caputo fractional derivatives of some functions ($\nu > 0, \lambda > 0, \beta > n, n < \nu < n+1$)

${}^{\nu}\mathsf{D}$	$f(x)$	${}^{\nu}\mathsf{D}f(x)$
${}_{-\infty}^{\nu}\mathsf{D}_x$	$e^{\lambda x}(x)$	$\lambda^{\nu}e^{\lambda x}(x)$
${}_{x}^{\nu}\mathsf{D}_{\infty}$	$e^{-\lambda x}$	$\lambda^{\nu}e^{-\lambda x}$
${}_{a}^{\nu}\mathsf{D}_x$	$E_{\nu}[\lambda(x-a)^{\nu}]$	$\lambda E_{\nu}[\lambda(x-a)^{\nu}]$
${}_{x}^{\nu}\mathsf{D}_{\infty}$	$x^{\nu-1}E_{\nu}[\lambda x^{-\nu}]$	$(1/x)E_{\nu,1-\nu}[\lambda x^{-\nu}]$
${}_{a}^{\nu}\mathsf{D}_x$	$(x-a)^{\beta-1}$	$\frac{\Gamma(\beta)}{\Gamma(\beta-\alpha)}(x-a)^{\beta-1}$
${}_{x}^{\nu}\mathsf{D}_b$	$(b-x)^{\beta-1}$	$\frac{\Gamma(\beta)}{\Gamma(\beta-\alpha)}(b-x)^{\beta-1}$

Table C.7 Riemann-Liouville derivatives ${}_0f^{(\nu)}(x)$ of some special functions

$f(x)$	${}_0f^{(\nu)}(x) = {}_0\mathsf{D}_x^{\nu} f(x)$
$1(x-a) \equiv \Phi_1(x-a)$	$\frac{(x-a)_+^{-\nu}}{\Gamma(1-\nu)} \equiv \Phi_{1-\nu}(x-a)$
$\delta(x-a) \equiv \Phi_0(x-a)$	$\frac{(x-a)_+^{-\nu-1}}{\Gamma(-\nu)} = \Phi_{-\nu}(x-a)$
$\Phi_{\nu-n}(x-a)$	$\delta^{(n)}(x-a)$
$\Phi_{\mu}(x-a)$	$\Phi_{\mu-\nu}(x-a)$
$E_x(\mu,a)$	$E_x(\mu-\nu,a)$
$xE_x(\mu,a)$	$xE_x(\mu-\nu,a)+\nu E_x(\mu-\nu+1,a), \quad \mu>-2$
$C_x(\mu,a)$	$C_x(\mu-\nu,a), \quad \mu>-1$
$xC_x(\mu,a)$	$xC_x(\mu-\nu,a)+\nu C_x(\mu-\nu+1,a), \quad \mu>-2$
$S_x(\mu,a)$	$S_x(\mu-\nu,a), \quad \mu>-2$
$xS_x(\mu,a)$	$xS_x(\mu-\nu,a)+\nu S_x(\mu-\nu+1,a), \quad \mu>-3$
$x^{\nu-1}E_{\nu,\nu}(-\mu x^{\nu})$	$-\lambda x^{\nu-1}E_{\nu,\nu}(-\mu x^{\nu})$

Table C.8 Riemann-Liouville semiintegrals ${}_0f^{(-1/2)}(x)$ of elementary functions ($x>0$)

$f(x)$	${}_0f^{(-1/2)}(x) = {}_0\mathsf{D}_x^{-\nu_2} f(x)$
$(a-x)^{-1/2}$	$1/\sqrt{\pi}\ln[(\sqrt{a}+\sqrt{x})/(\sqrt{a}-\sqrt{x})], \quad x<a$
$e^{\mu x}$	$(1/\sqrt{\mu})e^{\mu x}\operatorname{erf}\sqrt{\mu x}, \quad \mu>0$
$\cos\mu x$	$\sqrt{2/\mu}[C(y)\cos\mu x + S(y)\sin\mu x], \quad y=\sqrt{2\mu x/\pi}, \quad \mu>0$
$\sin\mu x$	$\sqrt{2/\mu}[C(y)\sin\mu x - S(y)\cos\mu x], \quad y=\sqrt{2\mu x/\pi}, \quad \mu>0$
$\ln x$	$2\sqrt{x/\pi}[\ln 4x - 2]$

Table C.9 Riemann-Liouville fractional derivatives ${}_{-\infty}f^{(\nu)}(x)$ of elementary functions ($-\infty < x < \infty, -\infty < \nu < \infty$)

$f(x)$	${}_{-\infty}f^{(\nu)}(x) = {}_{-\infty}\mathsf{D}_x^{\nu} f(x)$
$H(x)$	$x^{-\nu}/\Gamma(1-\nu)$
$(b-ax)^{\mu},\ a>0$	$a^{\nu}(b-ax)^{\mu-\nu}\Gamma(\nu-\mu), ax<b, \mu-\nu<1$
$(x-a)_+^{\mu},\ \mu>-1$	$(x-a)_+^{\mu-\nu}$
$e^{\mu x},\ \mu>0$	$\mu^{\nu}e^{\mu x}$
$\sin(\mu x),\ \mu>0$	$\mu^{\nu}\sin(\mu x+\nu\pi/2), \nu>-1$
$\cos(\mu x),\ \mu>0$	$\mu^{\nu}\cos(\mu x+\nu\pi/2), \nu>-1$
$e^{\mu x}\sin(\lambda x),\ \mu>0,\ \lambda>0$	$(\mu^2+\lambda^2)^{\nu/2}e^{\mu x}\sin(\lambda x+\nu\arctan(\lambda/\mu))$
$e^{\mu x}\cos(\lambda x),\ \mu>0,\ \lambda>0$	$(\mu^2+\lambda^2)^{\nu/2}e^{\mu x}\cos(\lambda x+\nu\arctan(\lambda/\mu))$

Table C.10 Riemann-Liouville fractional derivatives $f_{\infty}^{(\nu)}(x)$ of elementary functions ($x > 0$, $-\infty < \nu < \infty$)

$f(x)$	$f_{\infty}^{(\nu)}(x) = {}_x\mathsf{D}_{\infty}^{\nu} f(x)$
$x^{\mu},\ x > 0$	$x^{\mu-\nu}\Gamma(\nu-\mu)/\Gamma(-\mu),\ \mu < \nu,\ x > 0$
$(ax+b)^{\mu}$	$a^{\nu}(ax+b)^{\mu-\nu}\Gamma(\nu-\mu)/\Gamma(-\mu),\ \mu < \nu$
$e^{-\mu x},\ \mu > 0$	$\mu^{\nu} e^{-\mu x}$
$\sin(\mu x),\ \mu > 0$	$\mu^{\nu}\sin(\mu x - \nu\pi/2),\ \nu > -1$
$\cos(\mu x),\ \mu > 0$	$\mu^{\nu}\cos(\mu x - \nu\pi/2),\ \nu > -1$
$e^{-\mu x}\sin(\lambda x),\ \mu > 0,\ \lambda > 0$	$(\mu^2+\lambda^2)^{\nu/2} e^{-\mu x}\sin(\lambda x - \nu\arctan(\lambda/\mu))$
$e^{-\mu x}\cos(\lambda x),\ \mu > 0,\ \lambda > 0$	$(\mu^2+\lambda^2)^{\nu/2} e^{-\mu x}\cos(\lambda x - \nu\arctan(\lambda/\mu))$

Table C.11 Riemann-Liouville fractional derivatives ${}_a f^{(\nu)}(x)$ of elementary functions ($x > a$, $-\infty < \nu < \infty$)

$f(x)$	${}_a f^{(\nu)}(x) = {}_a\mathsf{D}_x^{\nu} f(x)$
$(x-a)^{\mu-1},\ \mu > 0$	$(x-a)^{\mu-\nu-1}\Gamma(\mu)/\Gamma(\mu-\nu)$
$(b-x)^{\mu}$	$(b-x)^{\mu-\nu}\mathrm{B}_{(x-a)/(b-a)}(-\nu, \mu-\nu)/\Gamma(-\nu)$
$(x-a)^{\mu-1}(b-x)^{\lambda-1},\ a < x < b,$ $\mu > 0$	$(b-a)^{\lambda-1}(x-a)^{\mu-\nu-1} \times {}_2F_1(1-\lambda, \mu; \mu-\nu; \frac{x-a}{b-a})\frac{\Gamma(\mu)}{\Gamma(\mu-\nu)}$
$(x-a)^{\mu-1}/(b-x)^{\mu-\nu}$	$\left[(b-a)^{\nu}\frac{\Gamma(\mu)}{\Gamma(\mu-\nu)}\right](x-a)^{\mu-\nu-1}/(b-x)^{\mu},\ a < x < b$
$(x-a)^{\mu-1}/(x-c)^{\mu-\nu}$	$\left[(a-c)^{\nu}\frac{\Gamma(\mu)}{\Gamma(\mu-\nu)}\right](x-a)^{\mu-\nu-1}/(x-c)^{\mu},\ c < a < x$
$\exp(\mu x)$	$(x-a)^{-\nu}\exp(\mu x)\gamma^*(-\nu, \mu(x-a))$
$(x-a)^{\mu-1}\ln(x-a),\ \mu > 0$	$\frac{\Gamma(\mu)}{\Gamma(\mu-\nu)}(x-a)^{\mu-\nu-1}[\psi(\mu) - \psi(\mu-\nu) + \ln(x-a)]$

Table C.12 Some functions of Riemann's fractional operator ${}_0\mathsf{D}_x^{\nu} \equiv \mathsf{D}^{\nu}$

$F(\mathsf{D}^{\nu})f(x)$	Explicit expression
$(1+\lambda\mathsf{D}^{\nu})^{-1}f(x)$	$\sum_{j=0}^{\infty}(-1)^j\lambda^j\mathsf{D}^{j\nu}f(x)$
$\sqrt{\mathsf{D}+\delta}\,f(x)$	$\sum_{j=0}^{\infty}\binom{1/2}{j}\delta^j\mathsf{D}^{1/2-j}f(x)$
$\sqrt{\mathsf{D}+\delta}\,f(x)$	$e^{-\delta t}\mathsf{D}^{1/2}e^{\delta t}f(x)$
$(\mathsf{D}+\delta)^{\nu}f(x)$	$e^{-\delta t}\mathsf{D}^{\nu}e^{\delta t}f(x)$
$\sqrt{\tau\mathsf{D}^2+\mathsf{D}}f(x)$	$\sqrt{\tau}\sum_{j=0}^{\infty}\binom{1/2}{j}\tau^{-j}\mathsf{D}^{1-j}f(x)$
$\sqrt{\tau\mathsf{D}^2+\mathsf{D}}f(x)$	$\sqrt{\tau}\,\mathsf{D}^{1/2}e^{-x/\tau}\,\mathsf{D}^{1/2}e^{x/\tau}f(x)$
$\sqrt{\tau\mathsf{D}^2+\mathsf{D}}f(x)$	$\sqrt{\tau}e^{-x/\tau}\mathsf{D}^{1/2}e^{x/\tau}\mathsf{D}^{1/2}f(x)$
$\sqrt{\mathsf{D}-c\mathsf{D}^{-1}}f(x)$	$\sum_{j=0}^{\infty}a_j\mathsf{D}^{1/2-2j}f(x),\ a_j=(-1)^j\binom{1/2}{j}c^j$
$\sqrt{\mathsf{D}+b-c\mathsf{D}^{-1}}f(x)$	$\sum_{j=0}^{\infty}a_j\mathsf{D}^{1/2-j}f(x),\ a_0=1, a_1=b/2,$
	$a_2=-(a_1^2+c),\ a_j=(-1/2)\sum_{k=1}^{j-1}a_{j-k}a_k\ (j>3)$
$e^{-\gamma x}\sqrt{\mathsf{D}-\gamma}e^{\gamma x}f(x)$	$\mathsf{D}^{1/2}f(x)$
$e^{-\gamma x}\sqrt{\gamma-\gamma^2\mathsf{D}^{-1}}e^{\gamma x}f(x)$	$\sqrt{\gamma}e^{-\gamma x}\mathsf{D}^{-1/2}e^{\gamma x}\mathsf{D}^{1/2}f(x)$
$e^{-\lambda\mathsf{D}^{1/2}}f(x)$	$\mathsf{D}\int_0^x \operatorname{erfc}\left(\frac{\lambda}{2\sqrt{x-\xi}}\right)f(\xi)d\xi$
$\mathsf{D}e^{-\lambda\mathsf{D}^{1/2}}f(x)$	$-\mathsf{D}^{1/2}e^{-\lambda\mathsf{D}^{1/2}}f(x)$
$\mathsf{D}^{\nu}e^{-\lambda\mathsf{D}^{1/2}}f(x)$	$e^{-\lambda\mathsf{D}^{1/2}}\mathsf{D}^{\nu}f(x),\ \nu<1$
$\mathsf{D}^{-1/2}e^{-\lambda\mathsf{D}^{1/2}}f(x)$	$\int_x^{\infty}\left[e^{-\lambda\mathsf{D}^{1/2}}f(\xi)\right]d\xi$

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